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INTERNATIONAL SERIES OF MONOGRAPHS IN PURE AND APPLIED BIOLOGY

Division: **ZOOLOGY**

General Editor: G. A. Kerkut

Volume 25

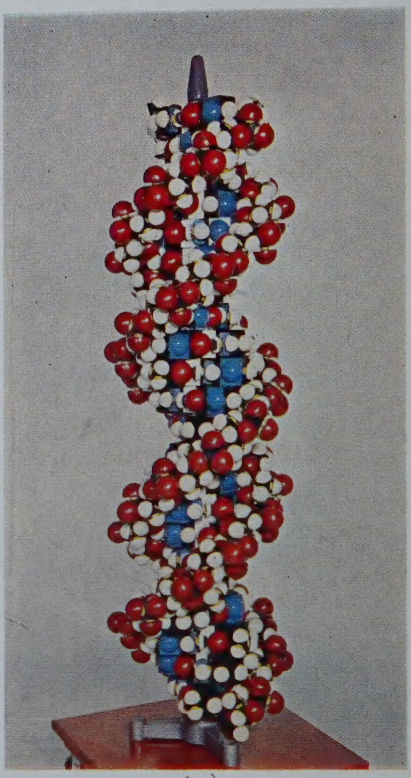
THE UNIQUENESS OF BIOLOGICAL MATERIALS

TABLE 1. PERIODIC TABLE OF THE ELEMENTS

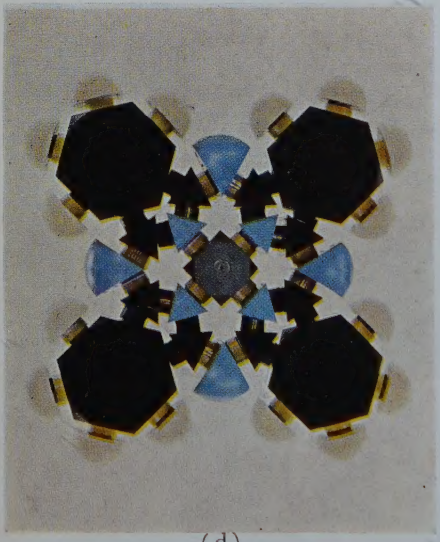
Group Period	0	I	II	III	IV	V	VI	VII	VIII Transitional
1		Hydrogen H (1) 1·01							
2	Helium He (2) 4·00	Lithium Li (3) 6·94	Beryllium Be (4) 9·01	Boron B (5) 10·82	Carbon C (6) 12·01	Nitrogen N (7) 14·01	Oxygen O (8) 16·00	Fluorine F (9) 19·00	
3	Neon Ne (10) 20·18	Sodium Na (11) 22·99	Magnesium Mg (12) 24·32	Aluminium Al (13) 26·98	Silicon Si (14) 28·09	Phosphorus P (15) 30·98	Sulphur S (16) 32·07	Chlorine Cl (17) 35·46	
4 Even series	Argon A (18) 39·94	Potassium K (19) 39·10	Calcium Ca (20) 40·08	Scandium Sc (21) 44·96	Titanium Ti (22) 47·90	Vanadium V (23) 50·95	Chromium Cr (24) 52·01	Manganese Mn (25) 54·91	Iron Fe (26) 55·85 Cobalt Co (27) 58·94 Nickel Ni (28) 58·71
Odd series		Copper Cu (29) 63·54	Zinc Zn (30) 65·38	Gallium Ga (31) 69·72	Germanium Ge (32) 72·60	Arsenic As (33) 74·91	Selenium Se (34) 78·96	Bromine Br (35) 79·92	
5 Even series	Krypton Kr (36) 83·80	Rubidium Rb (37) 85·44	Strontium Sr (38) 87·63	Yttrium Yt (39) 88·92	Zirconium Zr (40) 91·22	Niobium (Cb) Nb (41) 92·91	Molybdenum Mo (42) 95·95	Technetium Tc (43) 99 app.	Ruthenium Ru (44) 101·1 Rhodium Rh (45) 102·9 Palladium Pd (46) 106·4
Odd series		Silver Ag (47) 107·9	Cadmium Cd (48) 112·4	Indium In (49) 114·8	Tin Sn (50) 118·7	Antimony Sb (51) 121·8	Tellurium Te (52) 127·6	Iodine I (53) 126·9	
6 Even series	Xenon Xe (54) 131·3	Caesium Cs (55) 132·9	Barium Ba (56) 137·4	Lanthanum La (57) 138·9 Rare earths (58–71) Thallium Tl (81) 204·4	Haffnium Hf (72) 178·5	Tantalum Ta (73) 181·0	Tungsten W (74) 183·9	Rhenium Re (75) 186·2	Osmium Os (76) 190·2 Iridium Ir (77) 192·2 Platinum Pt (78) 195·1
Odd series		Gold Au (79) 197·0	Mercury Hg (80) 200·6		Lead Pb (82) 207·2	Bismuth Bi (83) 209·0	Polonium Po (84) 210	Astatine At (85) 210app.	
7	Radon Rd (86) 222	Francium Fr (87) 223 app.	Radium Ra (88) 226·1	Actinium Ac (89) 227 app.	Thorium Th (90) 232·1	Proactinium Pa (91) 231 app.	Uranium U (92) 238·1 Actinides (93–102)		

(a)

(b)



(c)



(d)

Atomic scale models of:

(a) a portion of the molecular chain of cellulose

(b) a portion of a polypeptide molecular chain, in the α–helix conformation (side chains trimmed)

(c) a portion of the double helical chain of the molecule of deoxyribonucleic acid (DNA)

(d) the molecule of nickel phthalocyanine, an analogue of the porphyrans

Colour code: black/carbon, white/hydrogen, red/oxygen, and blue/nitrogen

(*Courtesy of Messrs. Griffin and George*)

THE UNIQUENESS OF BIOLOGICAL MATERIALS

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University Lecturer in Zoology, Oxford

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Life is Unique: who will deny?
  In what respects there's less accord,
For who is certain how unique,
Or what, indeed, 'unique' implies?
Just 'single'—or the *non-pareil*?
On Earth? Or in the Universe?
Unique on Earth, no doubt, with but
One dawn, one Evolution;
Uniquely different from all else,
  In litho-, hydro-, and atmosphere;
Composed of carbon, hydrogen,
O,N,P,S,—and elements
In micro-, or in trace-, amounts;
In bodies organised, which true
To type perpetuate themselves.
But through what powers can Life do this?
Natural or Supernatural?
By properties inherent in
The molecules of which it's made?
  Built, as it was, selectively
Of those collectively endowed
With self-perpetuation as
Their tendency most-probable;
Or by extraneous, imposed power—
A *Deus ex machina* force,
Outside the laws of Nature and
The terms of reference of Science?
This is the problem: this indeed.

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PREFACE

THERE were two main reasons for attempting this book. The first was an interest in the philosophical question: Is the uniqueness of life inherent in the material of living organisms? The second was more mundane and practical and arose from the difficulty of persuading pupils that an essay on the properties of biological materials, carbohydrates, lipids, proteins and nucleic acids is as interesting as it is essential. In order to 'sell' this, the title of the essay was progressively changed through "The biological significance of the carbohydrates, etc." and other intermediaries to the title of the present work. It became evident that at each step the subject lost in amenability what it gained in interest, and in penetration, and that the relevant but often scattered information in the literature should be collected for the student by someone with more time and with a more long-standing conviction of its value.

In attempting to do this I am very conscious that there are many better qualified. However, the need for it is felt most acutely by those who have been trained against a background of evolutionary theory concerning the selected and selective value of biological systems. At the same time it is hoped that those who are more interested in the proximate, efficient causes of biochemical processes will, nevertheless, find the approach useful and stimulating.

The second motive is, of course, rooted in the first. The question whether, and in what ways, biological materials are unique is one of the most compelling of biological interests today. Whatever our view of the origin of life we believe that it has unique properties. The materialists among us will say that the uniqueness lies entirely in the physical and chemical properties of the materials, and it is of crucial importance to decide eventually if they are correct. We are rapidly approaching the time when it will be possible to give something like a definite answer, based on established fact.

In view of the practical motive, above, the resulting essay was often little more than an orthodox survey, if only because this is an essential background for anything more ambitious, and few had time to take the

subject further. It may fairly be contended, therefore, that this book likewise needs that background, but it would be wasteful of space and time to include more than is absolutely essential, when there are so many admirable standard works available. These may usefully be consulted as a prelude or as a sequel to the present work, or as a reference to hand. An attempt has been made to sketch just as much of the background biochemistry as is necessary to appreciate the uniqueness of the materials and their properties, without dominating, or distracting attention from, this foreground.

In addition to the many standard textbooks which have provided material for this venture, there are a number of books more specifically focused on this aspect of biochemistry, and I hope that their help has been adequately acknowledged in the text. It is invidious to discriminate but my heaviest single debts are to Sexton (1953), Jessop on p. 532 (1949, 1961), Oparin (1957), Florkin (1960) and Oncley (1959).

The work has outgrown the bounds of a weekly essay, and the advice of Bacon to Coke is very apposite, that one "should not so much labour what to speak as to find what to leave unspoken . . .". At present, however, it is not always easy to decide what can safely be left unspoken. There is much which may prove important; the reader can no doubt select what seems so at present.

A.E.N.

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My sincere thanks are here recorded to the following for permission to reproduce their published illustrations as the figures indicated: Professor S. W. Fox for Fig. 56; Professor L. C. Pauling and Dr. R. B. Corey, and the Publishers of the Proceedings of the National Academy of Sciences, Washington, for Fig. 33c; the same authors, for Figs. 29, I, and 30; Dr. A. Rich and the American Institute of Physics, New York, for Fig. 32, A and B; Professor H. D. Springall and Butterworths Publications, Ltd., London, for Fig. 33, A and B; Professors E. S. West and W. R. Todd, and the Macmillan Publishing Company of New York, for Fig. 18; and Messrs. Griffin and George for the Courtauld atomic models shown in the frontispiece.

To Dr. Gerald Kerkut I owe much of the incentive to write this book and many valuable suggestions subsequently.

I am also greatly indebted to Mr. J. S. Haywood for photographing some of the figures.

ABBREVIATIONS USED

Å	Ångström unit (10^{-7}mm)
ACh	Acetylcholine
ACTH	Adrenocorticotropic hormone of pituitary
ADH	Antidiuretic hormone of pituitary
ADP	Adenosine diphosphate
AMP	Adenosine monophosphate, adenylic acid
APGH	Somatotropic hormone (anterior pituitary growth hormone)
aspNH_2	Asparagine
atm	Atmosphere
ATP	Adenosine triphosphate
BAL	British anti-lewisite
1C, 2C, etc.	First, second, etc., carbon atom in molecule, according to conventional numbering
C_1, C_2, etc.	Compound or unit with 1, 2, etc., carbon atoms in molecule
CDP, CTP	Cytidine di- and triphosphate
C-even, C-odd	Compounds with an even or odd number, respectively, of carbon atoms in the molecule
CF	Citrovitrorum factor
CoA	Coenzyme A
COHb	Carboxyhaemoglobin
CoQ	Coenzyme Q
c.t.i.	Chemotherapeutic index
D-, L-	Optically active compound based on D- and L-glycerose respectively
d.c.	'Direct' current
DCA	Dicarboxylic acid
$\triangle F^\circ$	Change in free energy
DNA	Deoxyribonucleic acid
DNP	Dinitrophenol
DPA	Dipicolinic acid
$E'_\circ$	Oxidation–reduction potential (difference in potential between aqueous system, when half oxidised, and potential of standard hydrogen electrode)
e	Electron

EDTA	Ethylenediaminetetracetic acid
e.s.u.	Electrostatic unit
FAD	Flavin-adenine dinucleotide
FMN	Flavin mononucleotide
FSH	Follicle-stimulating hormone of pituitary
GA	Glutamic acid
g-atom	Atomic weight in grams
g cal	Small calorie (cal)
g-equivalent	Gram-equivalent (equivalent weight in grams)
$gluNH_2$	Glutamine
g mol	Gram-molecule (molecular weight in grams, mole)
GMP, GDP, GTP	Guanine mono-, di- and triphosphate
GSH	Glutathione
Hb	Haemoglobin
HOpro, HOlys, etc.,	Hydroxyproline, hydroxylysine, etc.
ICSH	Interstitial cell-stimulating hormone of pituitary
IMP, ITP	Inosine mono- and triphosphate
L-	Optically active compound based on L-glycerose
α-KG	α-Ketoglutaric acid
k.m.f.	Keratin-myosin-fibrinogen group of proteins
M	Molar (molecular weight in one litre of solution)
mμ	10^{-6}mm
mol., mole	Molecular weight in grams
mV	Millivolt
N	Normal (equivalent weight in one litre of solution)
N	Organic compounds substituted on the nitrogen atom
n	Straight-chain isomer ('normal' isomer)
NAD, NADH	Nicotinamide-adenine dinucleotide
NADP, NADPH	Nicotinamide-adenine dinucleotide phosphate
NMN	Nicotinamide mononucleotide
NMP, NDP, NTP	Nucleotide mono-, di- and triphosphate
OAA	Oxaloacetic acid
o, *m*, *p*	*Ortho*, *meta* and *para* disubstituted aromatic compounds
-P or -Ⓟ	Phosphate or phosphoryl radical (not elemental phosphorus)
P	Phosphate radical with high group transfer-potential
PABA	*p*-Aminobenzoic acid
PBG	Polybenzolyglutamate
PGA	Pteroylglutamic acid
pH	Log. reciprocal of hydrogen ion concentration, in gram-ions per litre
ϕ	Phenyl radical (substituted benzene)
P_i	Inorganic phosphate
pK	Dissociation constant (pH at stage of half-ionisation)
PP	Pyrophosphate unit

ppm	Parts per million
PRPP	5′-Phosphoribosyl-1′-pyrophosphate
Q_{10}	van't Hoff's temperature coefficient (factor of speeding for a 10° rise in temperature)
R or Ⓡ	Unspecified organic radical
rH	Log. reciprocal of hydrogen atom concentration (atmospheres) in equilibrium with system
RNA	Ribonucleic acid
TCA	Tricarboxylic acid
TMP, etc.	Thymidine monophosphate, etc.
TMV	Tobacco mosaic virus
UMP, UDP, UTP	Uridine mono-, di- and triphosphate
V	Volt
V.P.	Vapour pressure
w/w	Weight for weight
μ	10^{-3}mm
ϕ	Phenyl radical (substituted benzene)
$\sim$	Bond of high group transfer potential (also cycles of alternating current)
$\approx$	Approximately equal

Abbreviations for amino acids: First three letters of name

For symbols of chemical elements see Table 1, frontispiece.

CHAPTER 1

INTRODUCTION

THE aim of this book is not to demonstrate the uniqueness of life as a process, or of living organisms as the mechanism of the process: this has long been manifest, to laymen as well as to biologists, whatever their opinion on the nature and source of the uniqueness. The present interest is essentially in this further problem—the basis for the uniqueness of life—to what extent it is due to the intrinsic uniqueness of the materials which go to build *bioplasm*, and discrete organisms. In this context 'materials' are the elements used (Table 1, *frontispiece*), and those of their compounds which are used; for convenience they may be set in contradistinction to the 'organisation' which directs their collective properties into the process of living. This distinction is justified since it is theoretically possible that there is nothing innately unique about the materials: uniqueness might be a purely *emergent* property at the level of the collective organisation. Either possibility would be consistent with a 'natural' theory of the nature and origination of life, with natural selection as the 'final' cause, so that the problem is biological, and not purely metaphysical.

In order to investigate the uniqueness of the materials it is necessary to have a clear idea of the whole nature of biological uniqueness since if the innate properties of the materials are outstanding in ways irrelevant to life they have no further interest. In fact it may prove (p. 4) that virtually any outstanding property is relevant and useful, but it is important to recognise the principle, and it is important to keep in mind throughout the essential nature of life. It is not the only self-perpetuating system in the universe, or even on Earth, and it is not even the only one with a dynamic type of perpetuation (Needham, 1959). Its uniqueness may lie not merely in its chemical nature—as opposed to the mechanical perpetuation of such systems as atoms and solar systems—but in the complexity of the chemical system which self-perpetuates under this banner of life.

Mechanical perpetuation such as that of the solar system seems fairly easy to understand: the intrinsic momentum of the planets is exactly balanced against the gravitational pull of the central Sun—and, with the

almost perfect efficiency of the motion, this system will persist for a very long time. Chemical systems, however, are changing materially, and would soon reach some static end-state if there were not exactly restorative reactions. Under any particular set of conditions, however, only one set of reactions are 'probable' or spontaneous, and it is not evident how a chemical system, however complex, can be dynamically self-perpetuating on earth, or anywhere else. A one-way trend is also in accord with the laws of entropy, and even atoms and solar systems may be running down slowly; chemical systems, using the much less efficient energy-transfers of molecular bonds, run down much more quickly, but this is now realised to be no calamity on earth because it is an open system with a boundless replenishment of solar energy (von Bertalanffy, 1949; Oparin, 1957a). It is this which also provides the necessary condition for a perpetual system, since it has become balanced against the spontaneous entropy of the energy-flow: in daylight autotrophes such as plants 'fix' and store solar energy in chemical form while elsewhere, and at other times, the stores are degraded, and material returns to base. The living system might be compared with the plastic balls poised on the fountains of the fairground shooting gallery, the intrinsic weight of its entropy balanced on the upthrust of solar energy-flow. As engineers discovered, however, rotary motion is more efficient than reciprocation, and biological material returns to base by another route; like the mechanical analogues mentioned above, life consists of cyclic processes, as many ancient philosophers fully appreciated, and as Ezekiel realised in his mystic, perhaps deliberately obscurantist, fashion:

> . . . and their appearance and their work
> Was, as it were, a wheel within a wheel. . . .
> For the spirit of the living creature was in the wheels.
> (*Ezek.* i, 16, 20)

The odds against simple reciprocation being perpetual are familiar to those who frequent the shooting gallery, whereas the die is loaded in favour of cyclic return, by some route, however devious: unless material is extraordinarily stable it is likely to be caught up in some such cycle. The uniqueness of life, therefore, is in the speed and discreteness of its cyclic systems, which are very relevant to its complexity. In this connection it is worth a moment's digression to consider a simpler, spontaneous natural cycle which has the further virtues, first of being actually essential for life on land, and, secondly, of showing that life is not the only dynamic perpetuating system on earth which involves intricate discrete bodies. This is the water cycle: evaporation, mainly from the sea, produces clouds which precipitate as rain and snow, partly on the land, and the

water returns to the sea via streams and rivers. The latter may be compared with living organisms to the extent that they maintain a very constant form, although the material flowing through them is continuously and rapidly changing. For the individual stream and organism the flow is unidirectional and we may tend to forget the larger cycle of which it is part; it is a typical steady-state system, with inflow exactly balanced by outflow. Another amusing simple analogy is between the discrete form of clouds and that of living organisms. The limiting membrane is absent and clouds are much more transient than the amoebae they superficially resemble, but the mere formation of such discrete bodies, of such mobile materials, in such a mobile medium, is a very significant phenomenon. In fluid media much more permanent bodies can be expected to form spontaneously. Clouds are liquid-in-gas colloids (p. 225) in fact. The cloud-cover is as perpetual as life, on earth, but another great difference is that clouds form and disappear spontaneously whereas living organisms now always originate from other organisms of the same genetical type. Individuals, like clouds, are mortal however, ultimately, because of the same laws of entropic chance, and it is only the system as a whole which has achieved immortality.

For this reason the emphasis should be on the living system and the *biomass* as an entity, rather than on the structure or the metabolism of a particular organism. The material flows through the system as a whole every few thousand years at most (Needham, 1959), and this is the cycle into which biological elements and compounds have been caught up. They flow through minor cycles, like eddies in a stream, the metabolic cycles in particular (Fig. 8, p. 65), and their properties may be most relevant to these minor cycles, but these should always be considered as components of the main biological cycle.

> With Earth's first Clay They did the Last Man's knead
> E'en then of the Last Harvest sow'd the Seed:
> Yea, the first Morning of Creation wrote
> What the Last Dawn of Reckoning shall read.
>
> (OMAR KHAYYÁM: *Rubáiyát*)

The *general* course of the material through the system is more important than the particular species or individual through which it flows. It could be argued that the individual and species are as unimportant as the form of a particular stream or cloud for the water cycle, but this is not true with the same force, because the water cycle, again, is only a simple physical one. The properties of water are not affected by the form of cloud or stream but the fate of chemicals may vary considerably according to the chemical composition of the body through which they flow.

This in fact is the basis for evolution. All the biological elements and most of their compounds have been available since the dawn of life, and their properties are immutable. It is in the course they have been induced to take in specific bodies that the possibility of evolutionary change has rested. Evolutionary change is genetic, permanently codified (until the next change) in the organisational instructions of the individual. Genetic specificity and evolution are very relevant to the theme of this book, and will be considered when the intrinsic properties of the materials have been surveyed. It may be remarked in passing that evolution, a type of change which appears novel and non-reiterative (perhaps because it is part of a cycle too protracted to have been detected as an entity), also is not peculiar to living organisms, but applies to the universe as a whole. Biological evolution however is unique once more in its complexity, as well as in the very purposive or *teleonomic* behaviour (Pittendrigh, 1958) which it has produced in its vehicle, the living organism. How far is this inherent in the ultimate biological materials, which not only flow through the organism but collectively *are* the living organism during that time? The individual molecule flows but its type is a permanent component of the organism.

From these considerations it is possible to form some idea of the kind of unique properties to look for in a survey of biological constituents. Any properties which facilitate the trapping of solar energy and its use in directed ways are relevant, as well as any which speed the return of materials to base. Also relevant are any which help to segregate individual parts or the whole of the living system from the rest of the world, since any contamination must tend to defeat the advantages of uniqueness. This would seem to demand outstanding properties and never average ones, and the more pertinent question might be a further corollary: Were all elements and compounds with outstanding properties drawn into the biological system? Is life precisely this? It is outstanding enough among terrestrial systems for this to be true, and its complexity is consistent with everything which could contribute to a self-perpetuating, dynamic, chemical system actually having been incorporated. If life originated spontaneously this is in fact precisely what would be anticipated: some materials were virtually unaffected by solar energy-inflow while others were extremely sensitive. The two would evolve a sharp 'interface' as a river wears a sharp bank in the stationary rocks. Amenable peripheral material is drawn into the stream and its bed becomes ever deeper and sharper. As applied to the chemical system this might imply that living organisms should be found to have depleted the Earth of certain elements, and this is precisely the implication of Table 2.

If the abundance of the elements in living organisms is plotted graphi-

TABLE 2. ELEMENTAL COMPOSITION OF LIVING ORGANISMS, COMPARED WITH THAT OF THEIR ENVIRONMENT

Element	Relative abundance of the elements				
	As a percentage by weight of				In g-atoms per 100 g of living organisms
	Lithosphere (outer 24 miles)	Hydrosphere	Atmosphere	Living organisms	
Oxygen	47·33	85·79	23·02	63·00	3·94
Hydrogen	0·22	10·67	0·02	10·00	10·00
Carbon	0·19	0·01	0·01	20·00	1·67
Nitrogen	0·02	0·02	75·53	2·50	0·18
Calcium	3·47	0·05	—	2·45	0·06
Phosphorus	0·12	0·000004	—	1·10	0·036
Chlorine	0·23	2·07	—	0·16	0·0045
Sulphur	0·12	0·05	—	0·14	0·0044
Potassium	2·46	0·04	—	0·11	0·0028
Sodium	2·46	1·14	—	0·10	0·0043
Magnesium	2·24	0·14	—	0·07	0·0029
Iron	4·50	0·002	—	0·01	0·00018
Manganese	0·08	—	—	0·003	—
Fluorine	0·03	0·00014	—	0·0004	—
Bromine	—	0·01	—	0·0003	—
Copper	0·01	—	—	0·0002	—
Iodine	—	0·006	—	0·0001	—
Zinc	0·004	0·00001	—	0·0001	—
Cobalt	0·003	—	—	0·00004	—
Nickel	0·03	—	—	0·00004	—
Titanium	0·46	—	—	0·00005	—
Barium	0·08	0·00001	—	—	—
Strontium	0·02	—	—	—	—
Chromium	0·06	—	—	—	—
Vanadium	0·04	—	—	—	—
Aluminium	7·85	—	—	—	—
Silicon	27·74	—	—	—	—
Argon	—	—	1·40	—	—
Krypton	—	—	0·01	—	—
Xenon	—	—	0·005	—	—

cally (Fig. 1) against their abundance in the lithosphere (the logarithmic scale is necessary because of the wide range of the figures), they fall into two groups, in fact; calcium, oxygen and a number of microconstituents show a direct, and most of the others the inverse, correlation. The physiological ions (Chapter 25), other than calcium, might be said to fall on both lines. Very few lie far from one or the other. It will be noted that

the slopes of the two lines fitted are very different from 45°, so that the regressions are logarithmic and not linear; conceivably this has some fundamental significance.

Apart from oxygen, calcium, and a few elements of minor importance, then, there is a rather clear negative correlation between the abundance of an element in the lithosphere and that in living organisms. The maximal thickness of sedimentary rocks is about fourteen miles (Darwin, 1884),

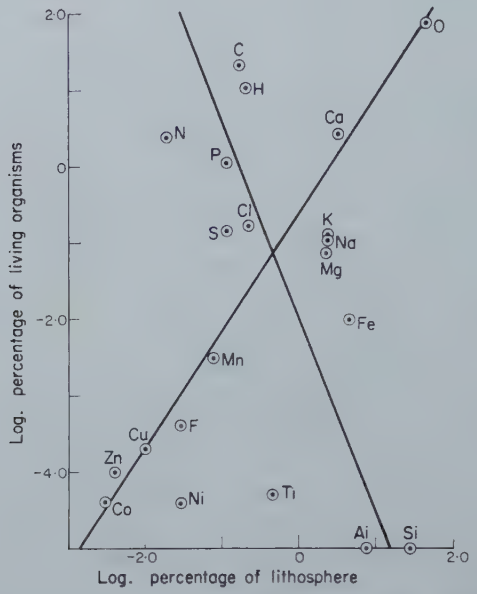

FIG. 1 Abundance of the elements in the material of living organisms as a function of their abundance in the lithosphere, both as percentages, plotted on a logarithmic scale for convenience. Some elements show a direct, and most others an inverse, correlation. The lines fitted have slopes very different from 45°, so that the relations are not simple linear ones.

and the analysis for the outer ten miles alone is very similar to that given in Table 2, so that it is a feasible idea that the lithosphere, and the abiosphere *in toto*, has been depleted of the elements most useful for biological purposes. In the universe as a whole the elements carbon, hydrogen, oxygen, and nitrogen are the most common, so that the rarity of carbon, hydrogen, and nitrogen in the Earth's abiosphere is certainly curious. Unfortunately for this view, however, these elements would still be rare if the whole of the biosphere, together with coal and other biological deposits, were distributed once more uniformly throughout. The biomass today is only about 1/300 of the mass of the atmosphere, only 1/69100

of the hydrosphere (Florkin, 1960), and perhaps no more than $1/10^7$ of the mass of that part of the lithosphere which has been exposed to exchange, so that none of the elements in living organisms would greatly enrich the abiosphere—or dilute its more abundant elements. The Earth's poverty in these precious elements remains a mystery (p. 507): they can scarcely have been depleted, on this scale, by insects lost to space, or by any other biological cause. The alternative possibility therefore emerges that elements were exploited biologically precisely because they were already rare and exceptional—unique quantitatively. There are objections to this also, because of oxygen and calcium, on the one hand, and, on the other hand, because of the many rare elements which have no biological significance. There is the further possibility that the biological elements have largely been depleted on earth by some non-biological agency, but perhaps for similar reasons to those which make them biologically unique.

In their distribution on Earth, therefore, the biological elements show their first, and collective, unique property and there is some reason to think of them collectively, in the kind of evolution postulated above. Collective uniqueness seems self-contradictory and perhaps savours of the abuse 'unique' often receives in trade circles, but it is justified in the interests of economy in terminology. At the same time it is a reminder that a number of different grades and shades of uniqueness should be recognised. Properties which distinguish all biological elements from all others represent the lowest grade. The next, which is rather less important, distinguishes particular groups of elements among the biological members. Next and most important is the uniqueness of particular elements. Finally particular groups of compounds of an element and even particular compounds can be uniquely significant. By variation in shade is meant that in each grade it may or may not be possible to say that the uniqueness has biological significance. Sometimes it is possible to say that it has biological value, and even that it is indispensable, but not yet to explain exactly how. Equally there are some instances of manifest importance biologically without any clear evidence of uniqueness. These are the problems of the subject, but they are not insuperable obstacles, and it is possible to test the main theme in some detail.

The plan of the ensuing treatment is to consider the relevant elements (Table 1) in turn, approximately in decreasing order of importance—at any rate in so far as one *can* rank elements which are all indispensable. Hydrogen and oxygen will be considered first because of the importance of water as the medium for bioplasm, and then carbon, the foundation of the main constituents of this bioplasm. Its compounds with hydrogen and oxygen must be classified and considered in detail, and the introduction

of the next important element, nitrogen, adds a large number of further groups of biological compound. The remaining elements by contrast add few new classes of biological compound, and collectively require only six chapters. Having completed this analytical survey an attempt will be made to draw general conclusions and to deal with biological elements and compounds collectively, and more synthetically. The final step in this direction is to return to the question of uniqueness in relation to evolution.

Love was the most ancient of all gods. . . . There was also another Cupid, or Love, the youngest son of the gods. . . . Love seems to be the appetite or incentive of the primitive matter, the natural motion or moving principle of the original corpuscles or atoms. . . . Cupid is a perpetual child; for compounds are larger things and have their period of age, but the first seeds or atoms of bodies are small and remain in a perpetual infant state.

(FRANCIS BACON, *Cupid or Atoms*)

CHAPTER 2

HYDROGEN AND OXYGEN

HYDROGEN and oxygen are present in nearly every biological compound, so that there would be some justification for considering them first even if water were not so outstandingly important as the dispersion medium in which carbon compounds have come to life. Even so, water is so inseparable from every manifestation of life that it seems best to deal with its unique biological properties before turning to its elements themselves. There are more g-atoms of hydrogen and oxygen in bioplasm than of any other element (Table 2) because water is the major constituent of all organisms. Even in insects, almost the most desiccated of animals, there is 46 per cent and in jellyfish there is as much as 96 per cent. Life is a system of carbon compounds in aqueous medium and it would seem justifiable to modify those famous and powerful lines:

> And the spirit of God moved on the face of the waters,
> And God said 'Let there be *Life*'. (*Genesis:* I, 2)

> . . . Proteus denotes matter. . . . He is represented as the servant of Neptune because the various operations and modifications of matter are wrought in a fluid state. The herd of Proteus seems to be no other than the several kinds of animals, plants and minerals in which matter appears to diffuse and spend itself.
>
> (FRANCIS BACON, *Proteus or Matter*)

2.1 The Uniqueness of Water

Only systems based on a fluid medium could display the properties which we should accept as Life. Gaseous systems are too volatile and lack the powers of spontaneously segregating sub-systems with genetic continuity: clouds (p. 3), with their nebulous segregations, are unusual among gaseous systems. At the other extreme, solid systems are too rigid and inert. Many of them have a considerable degree of perpetuation, but usually of a static type, contrasting with the steady-state, dynamic type of a living organism. The molecules of a solid are closely and often very regularly packed, so that movements are very limited, above the level of

electrons. The molecules of a liquid, by contrast, although they make contacts on all sides and readily pass over to crystalline and other solid-state conditions, preserve a high degree of mobility in the normal 'bulk phase'. Molecules of a solute in the liquid are sufficiently mobile at low concentrations for the gas laws to define their behaviour, and they present as much surface for chemical reaction as gases.

In gases the increase in entropy, that is the tendency towards a uniform distribution of energy, is very rapid but in liquids it is slow enough for local differences to be maintained by a modest inflow and for steady-states to be set up. All gases mix freely with any other, but not all liquids. Some form discontinuities or interfaces, with solid-state properties, where they meet another liquid, and complex polyphasic systems are readily formed which further retard the increase in entropy and increase the potentialities for steady-state perpetuation.

On many counts water is the uniquely suitable liquid in these and in other ways essential for life. It is particularly fortunate that water is so abundant on earth. It has been shown (Henderson, 1913; Firsoff, 1962) that conceivable alternative living systems could be based on liquid ammonia (p. 153) or on other liquids (p. 505), but all are inferior to water in most respects, in their liquid states. None are serious alternatives under terrestrial conditions. It is not enough to have suitable physical and chemical properties: the liquid must be abundant and distributed all over the surface of the globe. For the evolution of a living system its constituent materials must have maximal facilities for transport, interaction, segregation, etc. The terrestrial hydrosphere is ideal for this. It is in fact one of the many examples (p. 508) of the fitness of the environment for life (Henderson, 1913, 1924), which have led to the idea that our brand of life is a rather parochial one, adapted to the particular local conditions. At the same time there is no question of the absolute superiority of water as a biological medium.

It is probably highly significant that the body temperature of homoiothermic vertebrates is around 37·5°C, and that large insects warm up to this temperature before taking flight. This is precisely the temperature at which the specific heat of water is minimal, so that its molecules are then most easily mobilised per unit of energy supplied. Compost and damp hay tend to this temperature, under bacterial action, though in large artificial masses the reactions may then get out of control and the temperature may soar to firing point. The aqueous solution of some salts has a minimal volume at this temperature, and for instance sodium carbonate is most soluble here, indicating a maximal affinity for the water at this point; other properties of water and of its solutions also show turning points near the same value.

Water has a uniquely high surface tension (Table 3) exceeded by few substances other than liquid selenium, and this at a very much higher temperature. Water, therefore, is ideal for the formation of discrete living bodies, with stable limiting membranes. Air–water interfaces are less important, perhaps, than those between water and lipids, which likewise have high values. Other biologically useful consequences of the high tension are that materials which can lower the tension, surface-active materials, tend to accumulate at the surface, and also to orientate

TABLE 3. SURFACE TENSION OF LIQUIDS (dynes/cm)

Liquid	Temperature (°C)	Surface tension	Liquid	Temperature (°C)	Surface tension
Selenium A	217	92·4	Chloroform A	20	27·1
Water A	20	72·5	Cyclohexane A	20	25·5
Glycerol A	20	63·4	Ammonia V	11·1	23·4
Sulphuric acid A, V	20	55·1	Ethanol V	20	22·8
Quinoline A	20	45·0	Acetaldehyde V	20	21·2
Aniline V	20	42·9	Chlorine V	20	18·4
Phenol A, V	20	40·9	Oxygen V	−183	13·2
Benzene A	20	28·9	Argon V	−188	13·2
Naphthalene A, V	127	28·8	Carbon dioxide V	20	1·16
Acetic acid V	20	27·8	Helium V	−270	0·24

A = against air, V = against own vapour

there. Most of the biologically important carbon compounds have this property, which promotes their aggregation and concentration, as well as the formation of organised membranes. From a very dilute solution of protein virtually all of the molecules accumulate in the superficial 1 mm or so in 24 hours (Bateman, 1945). Another way in which the high surface tension of water is biologically valuable is in maintaining a thin film of solution over soil particles, so permitting a rich growth of micro-organisms, and also the continuous absorption by plant roots, without those anoxic conditions which accompany waterlogging.

The high surface tension is one manifestation of a very high affinity between water molecules, which has other important consequences. Fortunately it does not lead to an excessively high viscosity (Table 4), since the molecules are relatively small and compact. Many organic liquids have a higher viscosity and some values are greater by many orders of magnitude. Because of its low viscosity water permits the rapid

TABLE 4. VISCOSITY OF LIQUIDS (centipoises)

Liquid	Temperature (°C)	Viscosity	Liquid	Temperature (°C)	Viscosity
Glucose	22	$9{\cdot}1\times10^{15}$	Acetic acid	15	1·31
Glass (soda)	575	11×10^{12}	Water	20	1·00
Pitch	15	$1{\cdot}3\times10^{10}$	Cyclohexane	20	0·66
Glycerol	20	1490	Benzene	20	0·65
Olive oil	10	138	Chloroform	20	0·58
Sulphuric acid	20	25·4	Acetone	15	0·34
Tristearin	75	18·5	Air	−192	0·17
Creosote	20	12·0	Hydrogen	−253	0·01

diffusion of solutes (Danielli, 1952), and this is one factor which contributes to its great solvent action on other materials (p. 16). This is also partly the reason for its good lubricant action.

Water also has the virtue of not being excessively dense (Table 5), its

TABLE 5. DENSITY OF REPRESENTATIVE ELEMENTS AND COMPOUNDS (g/ml)

Substance	State	Density	Substance	State	Density
Alkyl borines		0·7–0·9	Gelatin	solid	1·27
Nitrogen	liquid	0·81	Brick	solid	1·4–2·3
Ammonia	liquid	0·82	Magnesium		
Benzene	liquid	0·88	acetate	solid	1·42
Silicon–carbon			Nitric acid	liquid	1·50
compounds		< 1·0	$Na_2CO_3.7H_2O$	solid	1·51
Water	liquid (4°C)	1·00	Chlorine	liquid (−33·6°C)	1·51
Organic			Bone	solid	1·7–2·0
compounds		*ca.* 1·0	Sulphuric acid	liquid	1·84
Metallo-organic			Sulphur		
compounds		> 1·0	(monoclinic)	solid	1·96
Oxygen	liquid	1·14	Caustic soda	solid	2·13
Cyanic acid	liquid	1·14	Diamond	solid	3·01–3·52
Hydroxylamine	liquid	1·20	Tungsten boride	solid	17·2

density lying around the average for organic liquids. It is very much lighter than most metals and their salts; by convention it has unit density and on that scale tungsten boride, WB, has a value of 17·17. Most inorganic liquids also have a higher density than water.

Its specific heat (Table 6), which is roughly proportional to density,

also is not outstandingly high and in fact is low compared with that of most other polar compounds. It is higher than that of almost all organic compounds but lower than that of many inorganic compounds. Its molecules are therefore relatively easily mobilised. At the same time the value is high enough for large masses of water to act as valuable stores of heat, with important consequences for individual organisms and for life in general. The Gulf Stream is a well-known instance.

Table 6. Specific Heats of Representative Materials (cal/g)

Substance	Temperature (°C)	Specific heat	Substance	Temperature (°C)	Specific heat
Potassium dichromate	397	0·034	Isoamyl valerate	20	0·46
Glycerol	−250	0·047	Olive oil	6·6	0·47
Phthalic acid	20	0·232	Acetic acid	0	0·49
Chloroform	20	0·234	Aniline	20	0·52
Succinic acid	0	0·248	Acetone	20	0·53
Glucose (solid)	20	0·275	Glycol	40	0·53
Benzoic acid (solid)	20	0·287	Hydrogen peroxide	0	0·55
Urea	20	0·320	Water	20	1·01
Sulphur dioxide	20	0·327	Ammonia	20	1·13
Glycerol	0	0·330	Hydrogen (gas)	0	3·4
Sulphuric acid	10	0·339	Mercury	− 39 to 100	6·68
Benzene	20	0·406	Caustic soda (fused)	400	6·7
Palmitic acid	20	0·430	Sodium (fused)	200	7·3
Pyridine	21	0·431	Potassium chloride (fused)	800	8·8

Related to the high surface tension is a high heat of vaporisation (Table 7), greater than that of any other known substance. This is invaluable for animals living on dry land in two important ways: it economises water-loss and it cools them maximally per unit loss. The vapour pressure of water (Table 8) is correspondingly low. The heat of fusion (Table 9) of water also is high, though it is exceeded by that of some polar inorganic materials, including ammonia and sodium fluoride. The high surface tension of water tends to depress the heat of fusion somewhat since the liquid is able to take up a form of minimal surface, but much more significantly it is also depressed because unlike most solids ice shrinks in volume when it melts. This is perhaps why ice melts at a temperature 15°C lower than would be expected on other grounds, a

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feature attributed to its *internal pressure*. But for this 15° grace, most of the earth's water would be permanently ice.

Ice is also unusual in being less dense than water. This is because there is a space in the centre of each polygonal cell of its molecular lattice (Klotz, 1962); if the molecules were close-packed the density would be

Table 7. Latent Heat of Vaporisation of Liquids (cal/g)

Liquid	Temperature (°C)	Latent heat of vapn.	Liquid	Temperature (°C)	Latent heat of vapn.
Water	100	538	Acetic acid	118	96·8
Sulphur	316	362	Benzene	80	95·5
Ammonia	−33·4	327	Ethyl ether	34·6	89·3
Ammonia	0	302	Carbon dioxide	−60	87·2
Ethanol	78·3	204	Naphthalene	218	75·5
Hydrogen cyanide	20	210	Mercury	358	68·0
Glycol	197	191	n-Decane	160	60·2
Phosphorus	287	130	Chloroform	61	58·0
Formic acid	101	120	Boron trichloride	10	38·2
Methyl formate	33	111	Argon	−186	37·6
Hydrogen	−253	108	Carbon dioxide	20	35·1
Pyridine	114	107	Helium	−269	6·0

Table 8. Vapour Pressures of Representative Materials (mm Hg) (for gases the boiling point (V.P. = 760 mm) is given)

Material	Temperature (°C)	Vapour pressure	Material	Temperature (°C)	Vapour pressure
Mercury	20	0·001	Carbon tetrachloride	23	100
Phosphorus, yellow (solid)	76·6	1·0	Molybdenum hexafluoride	17·2	400
Chlorobenzene	20	8·8	Ammonia	−33·6	760
Carbon	3946	10·0	Nitrogen tetroxide	21·0	760
Beryllium borohydride (solid)	28·1	10·0	Nitrogen	−196	760
Arsenic trichloride	23·5	10·0	Hydrogen cyanide	25·9	760
Water	20	17·5	Hydrogen fluoride	19·7	760
Boron tribromide	14	40·0	Chlorine trifluoride	11·5	760
Tin tetrachloride	35·2	40·0	Carbon dioxide	−78·2	760
Germanium tetrachloride	27·5	100·0	Carbon dioxide	22·4	45600

as high as 1·6. The specific heat of ice also is lower than that of water, of course. Because of its lower density ice should float up to the surface as it forms, and build a slush there. Owing to another anomaly of water, however, namely a density which is maximal at 4°C, and not 0°C, the coldest water is already *at* the surface and freezes as a solid mass *in situ*.

Table 9. Heats of Fusion of Representative Solids (cal/g)

Solid	Melting point (°C)	Heat of fusion	Solid	Melting point (°C)	Heat of fusion
Sodium fluoride	992	186	Palmitic acid	55	39·2
Sodium chloride	804	124	Anthracene	216·6	38·7
Ammonia	−75	108	*p*-Aminobenzoic acid	188·5	36·5
Potassium fluoride	860	108	Benzoic acid	121·8	33·9
Water	0	79·7	Sulphuric acid	10·4	24·0
Nitrogen pentoxide	29·5	76·7	Aniline	−7·0	21·0
Hydrogen peroxide	−1·7	74·1	Gallium	3·0	19·2
Potassium chloride	772	74·1	Methanol	−97	16·4
Cobalt	1495	62·0	Bromine	−7·3	16·2
Formic acid	80	58·9	Hydrogen	−259	14·0
Quinol	172·3	58·8	Hydrogen chloride	−114	13·9
Stearic acid	64	47·6	Sulphur	119	13·2
Glycerol	18	47·5	Iodine	113·7	11·7
Carbon dioxide	−56·2	45·3	Argon	−190	6·7
Acetic acid	16·6	44·7	Nitrogen	−210	6·1
			Phosphorus	44·2	5·0
			Oxygen	−219	3·0

This transparent crust insulates the water below, retarding its further freezing, minimising stirring and safeguarding the life of animals and plants below. Ice is not a good conductor of heat, so that the latent heat of freezing at its lower surface will be absorbed by the water below and not by the air. Except in shallow waters, therefore, freezing reaches a steady-state limit.

If ice were heavier than water and settled to the bottom of lakes and of the polar oceans, it would thaw extremely slowly and would long ago have locked up most of the water of the planet and denied its further use to living organisms. Water itself also is not a very good conductor of heat, and convection would have little melting value if the water were entirely above the ice. It is possible, by heating a test-tube of ice near the top, to melt it and even to boil the water locally without thawing that at the bottom (Bayliss, 1960).

The need for expansion on freezing causes water to supercool rather readily, even down to −72°C (Kraus, 1947). This again may play an important biological role. It will also superheat to 178°C, under certain conditions (Cook, 1956).

Although water has a low vapour pressure the atmosphere can hold more of the vapour than of any other gas, largely due to the formation of the colloidal dispersion, cloud; it also readily supersaturates with the vapour. As much as 30 per cent of the solar energy reaching the earth may be used in raising water for cloud and rain. The energy is partly recovered in rain, as heat and as activated molecules; plants grow far better after rain than after watering (even with water which has not been chlorinated). Considerable amounts of valuable minerals circulate in the water (Newbould, 1958).

The high intermolecular attractive forces in water are reflected also in its relatively high boiling and freezing points. But for this it would be gaseous at the Earth's surface temperatures. Its uniqueness in this respect is shown by comparing these values with those of the hydrides of neighbouring elements, as well as with those of neon which follows fluorine in the periodic system and has the same electronic configuration as the hydrides (Edsall and Wyman, 1958):

°C	H_4C	H_3N	H_2O	HF	Ne
Boiling point	−161	−33	+100	+19	−246
Freezing point	−184	−78	0	−92	−249

The values for water make a very sharp peak; at the same time it is noteworthy that the values show a regular gradient from those of this hydride, in both directions; water is outstanding, not quite peculiar.

Water is outstanding among solvents for the insensitivity of its boiling and freezing points to the presence of solutes (Table 10). This does not imply that it is a poor solvent: on the contrary it is the best solvent known. It implies that solutes in general disturb the architecture of water very little: this accommodates a great variety of forms of solute molecule, many of them becoming surrounded by a normally packed layer of water molecules (Klotz, 1962). It is one expression of the great affinity of water for so many solutes, in fact.

The affinity of water for solutes is shown by the fact that they, ions in particular, increase the surface tension of the water, and also by their high energies of hydration. It is also shown by the fact that the volume of the solution is less than that of the water plus that of the undissolved

solute; the molecular packing is closer than in either pure substance. The affinity is shown further by a chemical interaction of water with most solutes.

There are few substances which are not appreciably soluble in water,

TABLE 10. DEPRESSION OF FREEZING POINT AND ELEVATION OF BOILING POINT OF SOLVENTS BY ONE GRAM MOLECULE OF SOLUTE

A. CRYOSCOPIC CONSTANT (MOLECULAR DEPRESSION OF FREEZING POINT)

Solvent	Normal freezing point	Molecular depression (°C)
Water	0	1·85
Acetic acid	17·0	3·9
Stearic acid	67·0	4·5
Benzene	5·5	5·12
Urethane	49·0	5·14
Naphthalene	80·0	6·9
Phenol	39·0	7·4
Ethylene dibromide	9·3	11·8
Camphor	175·0	40·0

B. MOLECULAR ELEVATION OF BOILING POINT

Solvent	Normal boiling point	Molecular elevation (°C)
Water	100·0	0·51
Methanol	64·7	0·83
Ethanol	78·3	1·22
Ethyl ether	35·4	2·02
Benzene	80·2	2·53
Aniline	184·4	3·52
Chloroform	61·2	3·63
Carbon tetrachloride	76·8	5·03
Bromobenzene	156·0	6·26

and many are highly soluble. Even the inert gases and hydrocarbons form a kind of solution or hydrate with water (Buswell and Rodebush, 1956; Klotz, 1962). Some of these hydrates are crystalline solids melting at temperatures well above 0°C, showing that the hydrocarbon increases the steric organisation of the water molecules. This is shown also by an increased heat capacity; more heat is required to raise the temperature by 1°C, i.e. to break down the molecular order. A considerable fraction of the molecules of amino acids, lipids and other biological materials is

hydrocarbon so that these hydrates and their properties may be important *in vivo*. The bound water may be a fundamental part of the living fabric.

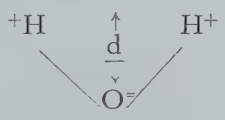
Few inorganic salts are soluble in any solvent except water (Bayliss, 1959), so that it is an indispensable go-between for organic and inorganic materials, both of which play a vital role in living systems. The solvent power of water for inorganic salts depends largely on its ability to ionise these solutes, but it also has a great affinity for, and solvent action on, non-ionising molecules such as sugars. The aqueous solution of these compounds depends on the affinity of the oxygen atom of the water, through its two lone pairs of electrons, for almost any other atom, including the hydrogen atoms of other water molecules. This explains the uniformly high affinity of water for so many solutes. As a member of Group VI of the periodic system (Table 1) oxygen has six electrons in its outer, valency, shell; two are shared, one with each of the hydrogen atoms of the molecule of water, leaving two lone but very active pairs, and it is these which bond the water molecules themselves into such a compact, quasi-crystalline fabric and have such a strong affinity for solute molecules. Each of the lone pairs can be regarded as making a potential chemical bond with other molecules, each oxygen atom having effectively four bonds.

There is an approximate solid angle of 105° between each of these bonds and its neighbours and they are directed towards the corners of an imaginary tetrahedron, therefore, just like the bonds of a carbon atom (p. 30). The tetrahedral arrangement is indefinitely extensive in space, as a quasi-crystalline lattice (Jacobson, 1954), and since the bond-lengths in water are not very different from those of carbon (Table 39, p. 467), a hybrid lattice is possible with little restriction on the type or amount of the carbon compound, provided it will bond freely with the water. As a liquid, water will suffer considerable distortion without destroying the essential structure. Both bond-angles and bond-lengths can vary considerably (Mason, 1957). The degree of association into crystalline arrays by water molecules themselves varies considerably; in steam virtually all the molecules are free, and the degree of association increases progressively as the temperature of water is lowered. The individuals rapidly change partners and this contributes to the good lubricant action of water. The size and details of the association vary also with solute and other conditions, and further emphasise the great versatility of water. Even ice has three distinct crystalline forms, in different temperature ranges (Mason, 1957); the familiar hexagonal form of snowflakes is the natural form just below freezing point.

The other hydrides in the table above have more or fewer lone pairs

of electrons than water and it appears to be the balance between the lone pairs and the shared electrons which gives water its outstanding position in that table of values. Ammonia, with only one lone pair, is satisfied with one other ammonia molecule and so does not form a continuous fabric, while methane has no lone pairs and is self-sufficient. Each hydrogen fluoride molecule has three pairs but its neighbours have only one hydrogen atom to share so that no stable fabric is possible with hydrogen fluoride alone, notwithstanding that fluorine forms stronger secondary bonds with available hydrogen atoms than any other element. The strength of this secondary, so-called *hydrogen-bond*, decreases in the order F > O > N > C. It is a partial or potential valency bond and that of O . . . H has a heat of formation of 5 kcal/g-equivalent of reagents, of the same order as that of the typical electrovalency bond, such as that between H^+ and OH^- themselves. In fact the structure H—O . . . H resonates freely with H . . . O—H, and both bonds are electrovalent by turns. The hydrogen bond therefore is very strong, particularly in quantity, as in all aqueous solutions and in many biological molecules; this is illustrated by the fact that when the hydrogen of the hydroxyl group of tyrosine becomes involved in a hydrogen bond it behaves as though it were actually separated from the oxygen, and the hydroxyl ionised as O^- and H^+ (Anfinsen, 1959). The absorption spectrum in fact shows the appropriate shift to a longer wavelength, with more intense absorption. The biological importance of the hydrogen bond will become abundantly clear.

Through the partial sharing of electrons between neighbouring water molecules the polarity of the molecules is increased; the hydrogen atoms become slightly positively charged and the oxygens negatively. Since the molecule is V-shaped, with the oxygen at the apex the of the V, it forms a *dipole*, thus:



Its *dipole moment*, the product of the magnitude of the charges and the distance, *d*, is 1·8 Debye units or $1{\cdot}8 \times 10^{-18}$ electrostatic units (e.s.u.). This is less than that of quite a number of polar organic substances, and considerably less than that of amino acids (Table 11), and inorganic salts, but it is high enough to have considerable electrostatic and electrodynamic activity. The force necessary to rotate the molecule is proportional to the dipole moment, so that water molecules are relatively easily orientated in an electrical field, and facilitate electrolysis.

The easy rotation into a polarised position gives water a high *dielectric*

Table 11. Dipole Moments of Representative Molecules in Debye Units (electrostatic units $\times 10^{18}$)

Substance	Temperature (°C)	Dipole moment
CO_2, N_2, C_2H_4, C_2H_6, CCl_4, Br_2, and most non-polar substances		0
Hydrogen chloride	−72–216	1·05
Ethyl ether	25	1·15
Ammonia	1–184	1·47
Ethanol	25	1·7
Water (steam)	150–210	1·85
Hydrofluoric acid	27–97	1·91
Ethylene glycol	144–233	2·28
Hydrogen cyanide	25	3·0
Potassium chloride	676	8·0
Urea	25	8·6
Glycine	25	15 app.
Diglycine	25	25 app.
ϵ-Aminocaproic acid	25	29·0
Gliadin	25	190
β-Lactoglobulin	25	750
Edestin	25	1400

constant (Table 12); in an electric field it therefore acts as a condenser with a high capacity, holding a large number of + and − charges orientated in a field at a low potential difference between the poles. Very few other materials have a higher dielectric constant than water. It is an ideal medium for holding charges polarised but only slightly separated, a kind of electrical buffering. There are a number of ways in which this contributes to the kind of steady-state conditions typical of biological systems. Water is readily converted from an insulating to a conducting system by traces of electrolytes; capacity and conductivity can be infinitely graded according to the amount and nature of solutes present. Solute ions can form charged layers at interfaces and associated lipids (p. 78) can contribute condenser properties of the alternative type, of high potential and low capacity.

The high dielectric constant of water is associated with its great power of dissociating electrolyte molecules into their constituent ions. By mass-action effect this promotes further solution of the solid and accounts for the particularly high solubility of electrolytes in water. Dissociation involves spatial separation of the ions so that water, although itself an insulator, is largely responsible for the high electrical conductivity of solutions of electrolytes.

Water itself ionises to some extent: in pure water $1/10^7$ of the molecules are ionised, the free H^+ ion rapidly associating with a water molecule, so that the main ionic species are $H_2O.H^+$ and ^-OH. For most purposes, however, it is adequate to represent the cation as H^+. Each ion has a brief life of 1 pico-sec (10^{-12} sec), but each is capable of inducing ionisation in a neighbouring molecule, and there can be a proton shift throughout the system and an electron shift in the opposite direction, a semiconduction

TABLE 12. DIELECTRIC CONSTANTS OF REPRESENTATIVE MATERIALS (FACTOR OF REDUCTION IN FORCE BETWEEN UNIT CHARGES, COMPARED WITH VACUUM AS 1·0)

Substance	Dielectric constant	Substance	Dielectric constant
Hydrogen (1 atm.)	1·0003	Sulphur dioxide (liquid)	13·8
Air (1 atm.)	1·0006	Ammonia	17·8
Carbon dioxide (50 atm.)	1·60	Ethanol	26·0
Octane	1·96	Methanol	35·0
Paraffin wax	2·3	Methyl cyanide	39·0
Nitrogen tetroxide	2·5	Thallium chloride	47·0
Ebonite	3·15	Water	81·0
Sulphur	3·52	Formamide	109·0
Plate glass	4·67	Hydrogen cyanide	116·0
Mica	6·64	Glycine	137·0
Casein	6·1–6·8	Rutile (parallel to optic axis)	170·0
Glass	8·45		

or 'brigade' effect (Klotz, 1962). This effect improves with the degree of order in the system; it is up to 1000 times as great in ice as in water, but, as already seen, organic molecules can greatly increase the crystallinity of their aqueous medium. This may be relevant to the possibility of action between molecules which are in contact only indirectly, through the medium (Rothen, 1956).

Thermal conductivity is generally proportional to that of electricity, and in water it is greater than that of most liquids, but less than that of many solids (Table 13). Curiously enough it is not much less than that of ice, perhaps because of the more open texture of the latter (p. 14); it is considerably greater than that of snow because of the air trapped in the flakes. Another physical virtue of water is its transparency to all but the shortest radiation from the sun. Primitive living organisms tend to be equally transparent so that photo-activation can occur throughout

the body. Trapping pigments become common in the higher organisms, either to use the light energy or to protect against it, or again for camouflage.

To return to the ionisation of water: although this is so slight it is greatly increased by the presence of electrolytes, an effect which is mutual, therefore. The *ionisation constant* of water, the product of the concentrations of its two ions, remains constant at 10^{-14} ($10^{-7} \times 10^{-7}$), so that any

Table 13. Thermal Conductivity of Representative Materials
(g cal/sec cm² for temperature gradient of 1°C/cm)

Material	Conductivity	Material	Conductivity
Cotton wool	0·00004	Snow	0·00051
Eiderdown	0·000046	Water (20°C)	0·0014
Inorganic gases		Brick	0·0015
(excl. water)	0·00004–0·0003	Ice	0·002–0·005
Organic compounds		Glass	0·0025
in general	0·0002–0·0007	Earth's crust	0·004
Cork	0·00015	Quartz	0·016–0·030
Wood	0·0005	Brass	0·26
Glycerol	0·0007	Copper	1·00

other ion which reacts with, and removes, either one allows more ionisation of water. If the concentration of ^-OH is depressed to 10^{-13} then that of H^+ rises by further ionisation to 10^{-1}. It will be recalled that the negative logarithm of the concentration of H^+ ions is known as the pH, that of pure water being 7·0, and that of an N/10 hydrochloric acid solution 1·0. Factors which affect the ionisation of water have powerful effects on biological systems; life is impossible in a medium more than a fraction of a unit removed from pH 7·0, though some acidity (excess of H^+ ions) is usual inside cells. Living organisms possess a number of buffering materials which will mop up excesses of either H^+ or ^-OH, and so keep the pH constant within narrow limits. Sodium bicarbonate is one of the simplest and most important: in the presence of excess acid it is converted to carbonic acid, a weak acid, largely un-ionised, while in the presence of alkalis (excess ^-OH) it is converted to sodium carbonate and water. Even more aesthetically pleasing are the double-headed buffers, such as the amino acids (p. 167), which deal with H^+ ions at one site and ^-OH ions at another (Fig. 25), according to requirement.

The critical importance of pH indicates that it is the ions of water

which really concern bioplasm: by comparison most other ions are of minor importance, except in so far as they depress the concentration of the water ion of the same charge. It is no doubt significant that H^+ and ^-OH, in that order, are the two most mobile ions in aqueous media. It is understandable that H^+ is the most rapid since it is absolutely the smallest ion, but there are other cations and anions smaller than ^-OH. No doubt Klotz's brigade effect is important here: it gives the solvent ions a unique advantage.

The importance of pH to living organisms is partly that they themselves produce many acids and bases, some of them strongly ionising, for specific uses in the body. Extraneous acids and bases seriously interfere with these normal agents and so the body also produces large amounts of weak acids and bases to act as buffers where necessary. In addition to the strong neuro-mediator bases, and other organic bases and acids to be considered, living organisms can segregate high concentrations of mineral acids and bases. The oxyntic cells of the mammalian stomach secrete 0·2 per cent hydrochloric acid and the snail *Dolium* secretes 0·5 per cent hydrochloric acid + 4 per cent sulphuric acid. Some ascidians have as much as 9 per cent sulphuric acid in their blood cells. It is known that gastric hydrochloric acid production depends on the initial production of the weakly ionising acid, carbonic acid, followed by an exchange of the HCO_3^- for the Cl^- anion. Some organisms can secrete a strong alkali.

Water is unique also in its great chemical reactivity. Virtually no reaction is rapid in the complete absence of water and very few are not significantly catalysed by it: it appears in fact to be the universal catalyst. This is not surprising, perhaps, for reactions in aqueous media, in view of the extent to which solution depends on interaction between solute and water molecules. Hydrolysis, and dehydration or condensation, are two very common reactions in aqueous media, both in the laboratory and *in vivo*. In other cases water merely activates the primary reagents, by causing their ionisation, etc.

However, it is often an equally good catalyst for reactions in gaseous media, and between solids. The universal role is probably connected with the fact that every reaction is an oxidation–reduction reaction and that oxygen and hydrogen are the most powerful elements for this purpose. They unite to form water with the liberation of 34 kcal/mole, compared with 7·0, 2·3, 1·6, for the oxidation of carbon, sulphur and iron respectively (Fearon, 1949). The H^+ ion of water can be reduced to H and the ^-OH oxidised to OH, O or H_2O_2, and this commonly occurs in the metabolism of living organisms. *Redox* reactions are as common in aqueous media as in gaseous and lipid media, and often involve water

itself, through its ions. They are consequently sensitive to pH and interact with the hydrolytic type of activity. It is not surprising, therefore, that so many reactions are dependent on water.

It is in the chemical activities of water that the properties of its individual elements are most evident, and these will now be considered.

2.2 Hydrogen

As the lightest element hydrogen is certainly in a unique position and it is not merely incidental that there are more g-atoms of this than of any other element in biological material (Table 2). This is still true of the dry matter: there are roughly two hydrogen atoms per carbon atom in aliphatic compounds and the bulk of biological compounds are aliphatic (p. 127). All the common biological elements have light atoms and the abundance of hydrogen is in keeping with an approximately inverse relationship between abundance and atomic weight.

Something of the properties of hydrogen has already been seen in its most important compound, water. It is the most mobile ion and also the most strongly electropositive. Free hydrogen is the most mobile gas and soon escapes from our atmosphere. The high oxygen content there is probably an index of the amount of hydrogen, from the breakdown of water, which has been lost in this way. Since the atmosphere has 300 times the weight of the biomass, there is 60 times as much free oxygen and almost 50 times as much in the hydrosphere as in living organisms. However, there is 10^4 times as much carbon locked up in coal, etc., as in the latter, so that the free oxygen could have come from carbon dioxide now fixed in living and fossil forms. Hydrogen can penetrate the crystal lattice of silica glass (Barrer, 1934), whereas other gases such as oxygen and nitrogen can only pass between the crystallites (Davson and Danielli, 1952). A thin glass membrane therefore can be used to measure the H^+ ion potential difference between solutions on either side of it, and this is the most common method of measuring pH. Hydrogen is a unique gas also in being 'occluded' by metals—that is, absorbed.

Hydrogen conducts heat five times as rapidly as air; it also has a specific heat of 3·4 at 0°C, which is unusually high for a gas. The first property is consistent with the great mobility of its molecules and the second with the large number of molecules per unit weight. These and other unique properties therefore are due to its position as the lightest element. They may all be biologically significant but since free hydrogen is of trivial importance it is the unique chemical properties of an element in this position which attract most attention.

The hydrogen atom has only a single orbital electron; it therefore

readily donates this to other atoms, to achieve a stable state, as the H^+ ion. Equally readily it acquires a share in a second electron, however, since this completes the small first electronic shell and also produces a stable state. No other element can be so simply and symmetrically ambivalent, in this way. It behaves in both ways simultaneously and this is why it forms the extra, hydrogen bonds. By electron-donation it forms a typical electrovalent bond, but by sharing a second electron it usually forms a covalent bond, having about ten times as much energy of formation and correspondingly great stability. This is the bond formed with carbon. In strong acids, such as hydrochloric acid, hydrogen is entirely in the ionic form, while in water it can behave in both ways, ionising slightly or considerably according to the chemical environment. This can be regarded as another aspect of the uniqueness of water. The ambivalence of hydrogen itself is further stressed by the fact that it forms hydrides with the lighter electropositive, metallic elements, and here behaves as an anion, monopolising the second electron.

Hydrogen is very active chemically since in general activity is inversely proportional to atomic size. In this connection it cannot be too soon or too much emphasised that all the biologically important elements are relatively light and active. Hydrogen reacts explosively with oxygen on heating and very readily with the halogens, sulphur, nitrogen, phosphorus and carbon—in fact with all the major biological elements. It is with the physiological metals (Chapter 25) that it most readily forms hydrides.

Biologically its compounds with carbon are second in importance only to water. They have particularly strong covalent bonds (Witzemann, 1945), the symmetrical distribution of the four around the carbon atom adding to the general stability. Even when one or two of the carbon bonds are held by other elements, the methyl, $-CH_3$, and methylene, $= CH_2$, radicals remain stable units. The former is regularly transferred as a unit in metabolism. Hydrogen atoms bonded to carbon no longer have much affinity for water and hydrocarbons dissolve sparingly and endothermically (Rodebush and Masterman, 1954). The interesting hydrates (p. 17) of hydrocarbons are based on physical rather than chemical properties. As soon as other elements are introduced into the molecule, however, the carbon compound becomes more soluble in water, and most biological compounds are of this type. Hydrogen bonded directly to carbon does not readily form hydrogen bonds with other elements: in the paraffins, therefore, two very reactive light elements have formed a stable unreactive type of compound, an excellent foundation for biological material, to be leavened as required by smaller proportions of other elements.

Since hydrogen is basically univalent and in hydrocarbon form does

not form additional hydrogen bonds, it effectively seals off the carbon, and perhaps its most important significance in carbon combination is its tendency to limit polymerisation of carbon to one dimension as contrasted with the solid three-dimensional fabric of elemental diamond. It could be regarded as a plasticising or 'shortening' agent. Without its help carbon could not form molecules of potential biological value (Henderson, 1913; Conger, 1949).

Because of its small size the hydrogen atom has minimal effect on the size and shape of any molecule in which it occurs. This is important in biological molecules, which contain so much hydrogen. At the same time it contributes considerably to the energy content of the molecule per unit volume and mass.

2.3 Oxygen

By weight oxygen is the most abundant element, both on earth generally and in the bodies of living organisms. It shares with calcium the distinction of being abundant in both organisms and environment, in contrast to most biological elements (Table 2). Unlike calcium it is abundant in the hydro- and atmospheres as well as in the lithosphere. It constitutes about half of the Earth's crust (Fearon, 1949), and has probably never been much less, so that some qualification is necessary in any theory that conditions were at one time generally reducing (Oparin, 1957a; Needham, 1959). Today there is rather more oxygen in the surface than the deeper layers, possibly owing to the loss of hydrogen from water (p. 24), but water is not a reducing compound and has always been abundant. About two-thirds of the weight of living organisms is oxygen, but this is largely in the forms of water and mineral skeletons. In the dry, organic fraction the number of g-atoms is less than those of hydrogen and carbon. This is not to minimise the importance of oxygen, however, in organic combination as well as in water.

It occurs mainly as carbonyl, $\rangle C{=}O$, and carbinol, $-COH$, compounds in living organisms, formed by the progressive reduction of the stable carbon compound, carbon dioxide—today the precursor of all biological carbon compounds, and the form to which heterotrophes return most of these (p. 35). Today the average oxygen atom runs through this cycle in about 2000 years (Wald, 1954), a relatively short time which owes much to the uniquely good fortune that both carbon dioxide and oxygen are gaseous, and diffuse rapidly and freely. Carbon dioxide is also very soluble in water and oxygen reasonably soluble. It would be difficult to overemphasise the biological importance of these properties.

Carbon dioxide dissolves in water to form carbonic acid, $\begin{matrix}HO\\HO\end{matrix}\!\!\diagdown\!\!\diagup C{=}O$, which therefore already has the two main oxygen groups of organic compounds. The hydroxyl is more strongly acid than it is in water, because the hydrogen can resonate between its present O and the $=O$, a situation even more favourable in the strong oxyacids such as sulphuric and nitric. Organic derivatives, represented by the general formula, $\begin{matrix}R\\HO\end{matrix}\!\!\diagdown\!\!\diagup C{=}O$, also in general are stronger acids, since there is not a second hydroxyl to interfere with the resonance. The name, oxygen or acid-former, refers to this property of $HO{-}\underset{|}{X}{=}O$ compounds in general. In organic compounds the OH and $=O$ groups show greater variety of properties than in simple compounds and this might be regarded as a simple example of emergence (p. 1), of properties which depend on the association with other atoms and radicals. This is not true only of oxygen, but is a general phenomenon. The carbinol or alcohol group, C—OH, without an $=O$ on the same carbon atom, behaves as a weak anion, ^{-}OH, but when attached to an aromatic carbon ring (p. 131) it becomes strongly acid, $-C-O^{-}H^{+}$, again because of a resonant shift of the hydrogen. However, when an alanine side-chain (p. 175) is substituted in the *para*-position to this hydroxyl (p. 131), it becomes strongly alkaline, the p*K*, that is the pH at which it is half ionised, shifting from 6·5 to 10·0.

The hydroxyl group confers outstandingly high 'molecular cohesive energy' (Cohn and Edsall, 1943), a property shared by other oxygen-containing groups such as $O{=}C{-}NH_2$. It is particularly evident in the stickiness of sugars, gums and mucilages, with large numbers of hydroxyl groups in the molecule (Warson, 1957), and probably reflects the general readiness of oxygen to form extra, covalent and strong, bonds with many elements. Also two hydroxyl groups readily condense with the elimination of water, because of their acid–base ambivalence, and polyhydroxy molecules therefore can form extensive reticula in this way. In addition the strong hydrogen-bonding capacity of oxygen may be involved.

The further properties of organic oxygen radicals will be considered in appropriate contexts. The ambivalence of hydroxyl is further illustrated by the ease with which it binds inorganic ions of both kinds, for instance metals and such anions as phosphate and sulphate. This ambivalence is, of course, already inherent in the properties of water.

Oxygen is outstanding among non-metals and gases in being significantly paramagnetic, attracting and concentrating the lines of force in a magnetic field. Most other gases are slightly diamagnetic, repelling the

lines of force (Szent-Gyorgyi, 1955). Liquid oxygen is much more strongly paramagnetic. The property is due to unpaired electrons with unopposed magnetic spins, and indicates that molecular oxygen readily assumes the more active atomic form, with two unpaired electrons. This unusual readiness to dissociate adds to the reactivity of oxygen and is biologically important. It largely explains the strong tendency of oxygen to share an electron of the atom of various other elements and form strong covalent bonds with them rather than with another oxygen. This is reflected in the instability of hydrogen peroxide, HO—OH. Oxygen, therefore, behaves as the most strongly electrophilic of all elements with the exception of fluorine.

An unusual property of oxygen is that at high temperatures it is absorbed by a number of metals—platinum, palladium, silver, gold—rather as hydrogen is 'occluded' (p. 24), but in this case the gas is released on cooling.

Allen (1896), in an article remarkably penetrating for its day, maintained that oxygen-transfer occurs in every significant biological reaction, and that it is the medium of exchange between the other three major biological elements, hydrogen, carbon and nitrogen. No doubt this follows largely from the fact that carbon dioxide is the biosynthetic starting point, but it is still a reflection of the uniqueness of oxygen among biological elements.

CHAPTER 3

THE UNIQUENESS OF CARBON

Bioplasm is mainly a dispersion of carbon compounds in water and in dried organisms there are more g-atoms of carbon than of any other element. This was already well appreciated when the study of carbon compounds began under the title of organic chemistry. A question of some theoretical interest is whether organic compounds ever occur in nature except through the activity of living organisms and, if so, whether we should consider them potentially alive. Sir James Jeans, with characteristic emphasis, suggested that life was as inherent to carbon as magnetism is to iron and in a sense no doubt he was right, for magnetism is a general property of atoms which happens to be particularly well developed in iron.

The uniqueness of carbon lies primarily in its versatility, in the almost unlimited number and variety of compounds it can form under natural and mild laboratory conditions. The number already known from living organisms is greater than the total number formed by any known means from any other one element, though it is still a very small fraction of the total number of carbon compounds synthesised in the laboratory (Beilstein, 1918 *ff.*) and a still smaller fraction of the theoretically conceivable number. It would perhaps be surprising if something like our biological system had not evolved from among this number and variety: Nägeli used to say that it would be nothing short of a miracle (Holmes, 1948).

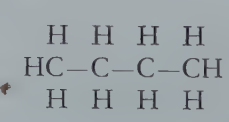
The versatility of carbon depends mainly on its position in Group IV, the central group of the periodic system; it is therefore ambivalent, neither strongly electropositive nor strongly electronegative, but forming compounds readily with other elements of both types from all groups of the system. Moreover, with all of these its bond is covalent, each partner contributing one electron, so that the hydrocarbons in some respects differ very little from the compounds with the halogens at the other extreme of the periodic table. Carbon equally readily forms compounds with a great variety of radicals, that is stable groups of other atoms, such as $-OH$, $-NH_2$, $-H_2PO_4$. The strong absorptive powers

of charcoal, a porous form of carbon prepared from wood, for both electropositive and electronegative dyes, and for other materials, is probably a further manifestation of this same ambivalent affinity of the element. It also permits the ready combination of carbon atoms with each other, and this leads to the most uniquely important result biologically, the potentially unlimited number of carbon compounds, of any conceivable molecular size and form.

The four bonds of each carbon atom are directed towards the corners of an imaginary tetrahedron, with the carbon as its centre, so that by bonding with other carbon atoms an indefinitely extended three-dimensional fabric is possible, similar to that of water (p. 18); consequently solutes and medium can readily form an orderly continuum (Jacobson, 1954). Moreover, large polymer molecules contain more energy per unit volume than smaller molecules (Firsoff, 1962). The basis of bioplasm is essentially a fabric of this kind, a continuum of carbon *macromolecules* in water.

As already indicated the bonds are covalent and stable: their heat of formation is 50 kcal/mole or more. They are therefore not formed rapidly without catalytic activation. On the other hand the heat required to activate hydrocarbons for oxidation is 58 kcal/mole so that they are very stable once formed (McNab *et al.*, 1952), and this is the basis for a good deal of the stability of carbon compounds in general. At the same time carbon compounds are notable for lability as much as for their stability. Few remain unchanged when heated above 300°C, and most are gaseous at that temperature, if not already decomposed. As in so many other respects, carbon seems to have the best of both worlds, in fact, combining stability with lability, momentum with inertia. Most organic compounds are metastable (Haldane, 1954b), that is to say they are not in complete equilibrium with their environmental conditions and are easily induced to react further.

As already noted also (p. 26), other elements, particularly hydrogen which has only a single valency bond, can limit the size of carbon polymers, giving the molecule discrete form and specific properties. The aliphatic hydrocarbons, structurally the simplest of carbon polymers, are mostly single chains of carbon atoms, with all other bonds held by hydrogen atoms, e.g. n–butane:



Because of the tetrahedral form of each carbon unit the chain in fact must be either spirally coiled or zigzag, and this is true of linear carbon com-

pounds in general. Some molecules are two-dimensional rings, or systems of rings, in some cases lying fairly neatly in one plane. Extensive solid, three-dimensional carbon molecules are relatively uncommon in living organisms, however probable in principle, and three-dimensional fabrics are built mainly from linear molecules—in which every atom is potentially in a position to react chemically. In fact the linear molecule seems to to be the most probable type also in laboratory syntheses.

Some of these linear polymers can build three-dimensional fabrics, as will be seen, by a variety of cross-links. These are always more sparsely and irregularly distributed than in simple crystalline solids, which in a

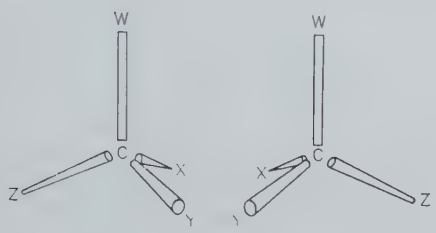


FIG. 1A Average position of the four bonds of a carbon atom to show the mirror-image relationship between the two optical isomers of any compound having four different groups, W, X, Y and Z, bonded to the same C-atom.

sense also are polymers (Stone and Graham, 1962), and this contributes to the plasticity and elasticity of organic polymers, already based on the zigzag or spiral form of the individual chains. It should be emphasised that a number of other elements can form the necessary chains for fabrics of this kind (Stone and Graham, 1962), but in nearly every case more than one element is required in the backbone, and none has both the pliability of the carbon chain and its versatility for cross-linking.

Because of the tetrahedral form of the bonded carbon atom the resulting molecule is asymmetrical if all four bonding groups are different, and any molecule containing at least one such carbon atom is asymmetrical. If any two bonding groups are identical there is a plane of symmetry. This steric asymmetry is shown also by compounds of other atoms with the tetrahedral disposition of bonds (p. 153), but these are rare and their compounds few. It is very common in carbon compounds and is biologically very important. When such asymmetric compounds are produced in the laboratory there is an almost equal number of the two possible, mirror-image forms of the molecule (Fig. 1A), a *racemic* mixture as it is called. In living organisms, however, usually an almost pure

preparation of one of the two *enantiomers* is produced and this pure asymmetry is genetically perpetuated. Such a pure enantiomer preparation can be recognised by its action in rotating a plane-polarised beam of light passed through its solution. This purity heightens the specificity of the material, and biological systems have become specialised to react only to the natural isomer. L–hyoscyamine (henbane) the natural product, based on the *laevo*-rotary (L—) parental compound, constricts the pupil of the eye but the *dextro*-rotary (D–) isomer has no action. The natural L–adrenalin has twelve times the pressor activity of the D–form, and L–nicotine is fatal in concentrations at which the D–form causes only shivering. Conventionally the parental compound is glycerose (p. 41), the simplest molecule which could be asymmetric, and it may be noted that while the L–series is the natural one in the examples quoted, and in amino acids (p. 161) and other groups, in the carbohydrates it is the D–series.

It would seem that optical purity must be an emergent property at the biological level, but there does appear to be a slight preponderance of one enantiomer even in laboratory reactions (Oparin, 1957a; Keosian, 1961) and it is conceivable that under certain conditions the difference would spontaneously increase in an open system (Langenbeck, 1935). Pure isomers certainly tend to racemise in the laboratory and probably *in vivo* also, but Langenbeck believes that the racemate is the thermodynamically unstable condition. There is some evidence that the degree of optical purity is higher in high latitudes than near the equator (Klabunovskii and Patrikeev, 1959), which would support the idea that a preponderance of one enantiomer is due to automatic natural factors. Morphological asymmetry is said to increase in parallel, but there are many organisms with morphological symmetry yet the same high optical purity of their molecules as in the most asymmetrical. This is one of the most fascinating properties of carbon compounds, and has great, but still somewhat obscure, biological significance.

The ambivalency of carbon explains not only its unique ability to form a large number of compounds but also the mildness of the properties of these compounds. Not many are violently reactive and these few, for instance nitrocompounds, rarely occur in living organisms. The biological acids and bases in general are of moderate strength only. Some carbon compounds are endothermic and some exothermic, so that there are considerable possibilities of spontaneous formation of a mixture of the two types. In large molecules one group may react with another in the same molecule: it is this which makes nitroglycerin so violently explosive, however. The heat of formation of carbon compounds from their immediate precursors is rarely very great, so that, once the initial

steps of carbon dioxide reduction have been effected, a large variety of compounds form relatively spontaneously. They would not be highly improbable even in a closed system.

The majority of carbon compounds are more unstable, on the one hand, than the fully oxidised forms, such as carbon dioxide, and also than the fully reduced forms, such as the hydrocarbons, at the other extreme (Urey, 1952). Both ethane, C_2H_6, and carbon dioxide are formed exothermically from their elements. The oxidation of methanol, CH_3OH, and of glucose (p. 40) to carbon dioxide and water is exergonic, and so is their reduction to methane and water. Biological materials therefore are, as it were, lightly poised between two stable extremes, relatively spontaneously changing in either direction under slight shifts in redox potential, but having effective 'stops' at the extremes.

The chemical versatility of carbon is well illustrated also by the so-called chain-reactions of hydrocarbons, when oxidised catalytically at low temperature (Pringle, 1954). If RH represents the molecule of such a hydrocarbon, then light and other catalysts promote reactions of the type:

$$RH \rightarrow R— + H—$$

and

$$O_2 + RH \rightarrow ROO— + H—$$

the free radicals R— and ROO— spontaneously react further in such ways as:

$$R— + O_2 \rightarrow ROO—$$

$$ROO— + RH \rightarrow ROOH + R—$$

$$ROO— + RH \rightarrow ROOR + H— \rightarrow RO— + RO—$$

There is a net increase in free radicals and the process can be self-accelerating. In other cases the number reaches a steady-state which has been maintained for long periods experimentally (Newitt and Gardner, 1936). This type of process is particularly significant as a potential property of living systems, which do operate at relatively low temperatures. No other element can produce such a variety of free radicals, and consequent variety of alternative pathways of reaction. Most organic reactions give a large range of products (Gilbert, 1950) and in biological metabolism there is often more than one. The impression is that metabolic reactions have each been naturally selected from a wide range of alternatives which collectively could make any conceivable pathway possible—under appropriate conditions. The reaction selected is not always the most probable one in the laboratory, i.e. the one which gives most product, and this is another emergent property of the biological organisation, under selection. The side reactions which still occur may be a major cause of 'wear-and-tear', i.e. of inefficiency, at the molecular level and of *metabolic turnover*.

The low-temperature oxidations of hydrocarbons are very sensitive to external factors, such as radiation and temperature. Increase in temperature actually retards some of the reactions, a response which demonstrates not only the unique versatility of carbon compounds but also their capacity for relatively rapid reaction at low temperatures. Given a suitable medium (p. 505) carbon-based life should be possible on planets much colder than earth.

The most important biological radicals of carbon are carboxyl $-C(O)OH$, carbonyl $=CO$, carbinol $-CH_2OH$, methyl $-CH_3$, acetyl $CH_3C(=O)-$, guanidinium $-HN-C(=NH)NH_2$, and there are also important radicals not containing carbon itself, $-OH$, $=O$, $-NH_2$, $-SH$, $-OP(=O)(OH)_2$ and $-P(=O)(OH)_2$ in particular.

Many further examples of the ambivalence and versatility of carbon could be given, and indeed most of the book will be concerned with just this. For all but the simplest of carbon compounds there are several, often many, isomers, compounds with the same empirical or elemental formula, quite apart from optical and other steric (geometric) isomers. For instance even the relatively small $C_{10}H_{16}O$ is the empirical formula of 120 different organic compounds. Many of these are quite unrelated in significant structure and properties, and this simple isomerism is in fact largely of academic interest, in contrast to stereoisomerism.

Until recently (p. 475) diamond was the hardest substance known, and most of its competitors for this distinction are carbides. Carbon, in fact, has also a number of unique physical properties. Diamond, and forms of the neighbouring elements, boron (p. 473) and beryllium (Mendelsohn, 1949), have anomalously low atomic heats (specific heat $\times$ atomic weight) at ordinary temperatures, 2·4 compared with the 6·3 cal/g-atom for other elements. This reflects the great regularity of packing of the carbon atoms in diamond so that its state of entropy is much lower than that of most other elements. This is another aspect of the stability of the element —and of its compounds.

Carbon is also versatile in its physical properties. Whereas diamond is hard and transparent, graphite is soft, black and opaque. This is associated with a difference in crystal form. The atoms of diamond have the typical tetrahedral arrangement of bonds, and the resulting three-dimensional lattice is equally strong in all directions. This would be the texture of molecules of carbon compounds, also, if they had carbon polymerised extensively in three dimensions (p. 31). In graphite, by contrast, three bonds of each carbon atom are distorted so as to lie in one plane, and neighbouring atoms also lie in the same plane. Their fourth bond is directed perpendicularly to this plane, to link with an atom of the neighbouring layer. When polar gases, water vapour in particular, penetrate

between the layers they slide readily on each other (Braithwaite, 1956) and give graphite its lubricant property. This structure transmits no light and so appears black; when it has been well spread or polished it shines because much of the light is reflected. Diamond scintillates because of the great amount of light reflected and refracted from its many artificially cut facets.

In charcoal there is probably a third arrangement, the carbon atoms tending to form rings of six, as they do also in many compounds. Charcoal is prepared from wood so that its structure probably depends on that of cellulose (p. 54). Lampblack is amorphous carbon. Under high pressure carbon forms diamond and graphite, and also threads, which recall the structure of some silicates (p. 468).

Similar physical contrasts are seen among the compounds of carbon: carbon dioxide, methane and others are gaseous at ordinary temperatures, ethyl alcohol, ether, acetone and the light paraffins are liquid, while the higher paraffins and many other compounds are solid. Order increases and entropy decreases with molecular size, therefore. In addition, properties are affected by other elements in the molecule. Those, like oxygen and nitrogen, which form polar groups increase the order, and depress the boiling and melting points. They also increase the solubility in water, as already seen, and collectively carbon compounds show all grades of solubility from highly insoluble to extremely soluble. All those liquid carbon compounds which are immiscible with water are miscible with each other and also dissolve most of those carbon solid compounds which are insoluble in water. They are therefore collectively known as lipids (p. 76), and form a distinct lipid phase in various situations in living organisms. Biological activity makes considerable use of the interplay between lipid and aqueous phases (Green, 1962).

This is not the place to single out particular compounds or groups, but carbon dioxide does demand this distinction since it is not usually regarded as an organic compound (as it should be), and certainly falls in a group of its own, with its two hydroxyl groups on the same carbon atom in the hydrated form, carbonic acid. As already indicated it is the final stable product of all carbon compounds when they are fully oxidised —whether by burning or by biological respiration. The good fortune of its being gaseous (p. 26) might be emphasised since it is in fact one of the very few gaseous oxides at ordinary temperatures (water vaporises more than most others). It is probably the limiting factor in the turnover of biological material, since its concentration in the air remains very constant at 0·04 per cent. Even so the amount of free carbon dioxide in the biosphere is ten times the whole biomass (Blum, 1955; Lederberg and Cowie, 1958).

It is the instability of the two hydroxyl groups on the same carbon atom which makes carbonic acid such a weak and harmless acid. In spite of the high solubility of carbon dioxide in water the equilibrium concentration of the acid is low. Substitution of one or both H^+ ions by a metal greatly stabilises the molecule, however, and the acid is a good buffer for alkalis. Most salts except those of the alkali metals (p. 416) are rather insoluble, so that calcium carbonate is ideal for skeleton formation, since it is so easily available *in vivo*.

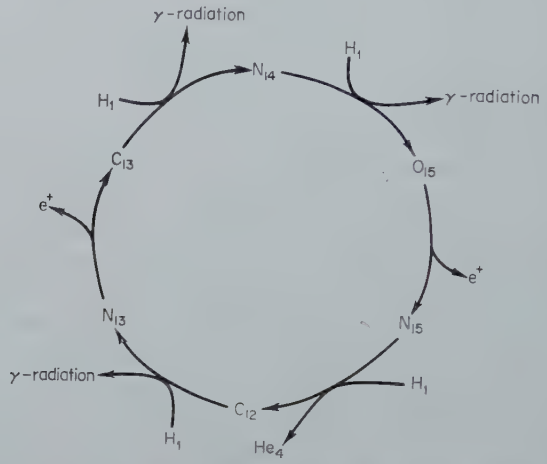


FIG. 2 The Energy Production Cycle of the Sun (Simon, 1949). To show the key roles of carbon, hydrogen, oxygen and nitrogen.

There are other unique properties of carbon which probably are relevant to its biological role, though at present it is not always clear in what way. For instance carbon appears to be the first stable element formed when the cooling of a star permits the transmutation of hydrogen to heavier elements (Jeans, 1943; Oparin, 1957a). Moreover, carbon can act as a catalyst for the further transmutation of hydrogen to helium (Simon, 1949), being itself transmuted in the process, through isotopic forms of nitrogen and oxygen back to carbon and helium. Conceivably the biological importance of hydrogen, oxygen and nitrogen is not unrelated to this interesting cycle (Fig. 2).

Among non-metals carbon has an outstandingly high electrical conductivity, electrons passing relatively easily from one atom to another. This is probably a manifestation of its orderly crystalline structure and perhaps also of its amphoteric lability. Carbon is also the element with the highest melting point, 3500°C, another expression of its strong molecular structure. Its boiling point, at 4200°C, is lower than that of tungsten,

at 5900°C, however. There are even very few compounds with a higher melting point than elemental carbon, and again these are carbides.

3.1 Other Elements of Group IV

With four electrons in their outermost valency shell these elements have a family resemblance to carbon, but the differences which result from a larger atomic size, with more electronic orbitals, are great enough to leave carbon unique, biologically, and this is equally true for elements in other groups. Differences essentially quantitative in principle very often become strongly qualitative in expression. Silicon, the second member of Group IV, like carbon forms almost equally strong bonds with hydrogen, on the one hand, and oxygen and chlorine, on the other. It is ambivalent, therefore, and the bonds are almost as strong as those of carbon, but in sharp contrast to carbon it forms much weaker bonds with its identity (Edsall and Wyman, 1958):

Bond-strength with: (kcal/equivalent)	Self	Hydrogen	Oxygen	Chlorine
Carbon	81·6	98·8	81·5	78·0
Silicon	42·5	75·1	89·3	85·8

It may be noted that the N—N, O—O, and S—S bonds also are weaker than C—C. Silicon therefore does not polymerise as carbon does, to form such a varied range of compounds. It differs so much in consequence that a separate consideration in another context (p. 465) is preferable.

The higher members of the group become progressively more metallic in properties, their electropositive property therefore predominating, and they have a low affinity for other electropositive elements such as hydrogen. Tin and lead do polymerise to some extent but they form no significant series of compounds based on this. Through their increasing atomic weight their compounds are necessarily less volatile, reactive and versatile. Because their valency electrons are further from the nucleus of the atom their binding energy is less than in carbon; they do not share electrons so readily and do not form compounds with so little energy of formation (Kavanau, 1947).

Some coals have a high content of germanium, another element in this group (Table 1). This is believed to be a secondary 'enrichment' process, the germanium replacing some of the carbon of the original organic

deposit (Pirie, 1957; Hallam and Payne, 1958). It does not imply that some organisms once used germanium in place of carbon, though it is probably due to properties which the two elements have in common. Germanium is the first member of the Ge—Sn—Pb sub-group. Titanium, the first member of the 'odd' sub-group, will be considered later (p. 453) since it has biological significance in its own right. Zirconium, its next higher analogue, acts as an antigen (p. 270) to animals and may form a complex with their proteins (Burnet, 1959), but this is not the only antigenic inorganic element.

Of the many known carbon compounds relatively few are really common and functionally active in living organisms, and the number has probably decreased during the course of evolution (Lwoff, 1944). It is as important to know why others have not been exploited as why these have, but the latter is an easier problem since we usually know the purpose for which they are useful. These compounds will now be considered, and where possible the disadvantages of those not exploited.

CHAPTER 4

CARBOHYDRATES

She found me roots of relish sweet
And honey wild and manna-dew . . .
(J. KEATS, *La Belle Dame Sans Merci*)

ORGANIC chemists tend to think of the aliphatic hydrocarbons, that is open chain carbon compounds with only a single bond between neighbouring carbon atoms and all other bonds held by hydrogen, as the simplest organic compounds, and so deal with them first. Since biosynthesis in most autotrophes begins at the other extreme, however, from fully oxidised carbon dioxide, the carbohydrates represent the first stable stage and the first group of biological compounds to be considered. Their simplest members, the monosaccharides or *monose* sugars, are C_2 to C_9, aliphatic molecules, with every carbon but one in the carbinol, $\begin{array}{c} | \\ H—C—OH \\ | \end{array}$, state. The other is in the aldehyde, $\begin{array}{c} | \\ H—C=O \end{array}$, state if it is terminal in the chain, or in the ketone, $>C=O$, state if it is not; monoses therefore are *aldoses* or *ketoses*. The *hexoses*, glucose (Fig. 6, I) and fructose (Fig. 6, II), are the best-known aldose and ketose respectively. Conventionally the carbon atoms are numbered from the aldehyde end; where there is a keto group it is usually at the 2C, i.e. it is the result of isomerisation of . . . $HC(OH)—C(=O)H$ to $—C(=O)—C(OH)H_2$.

4.1 Monoses and their Derivatives

The empirical formula of the monoses is $C_nH_{2n}O_n$ or $C_n(H_2O)_n$ which indicates how the name carbohydrate arose. The emphasis on water has a good deal of justification, as will be seen. The one possible biose (Fig. 6, III), glycolaldehyde, occurs in living organisms, but is of minor importance: the most common are the tri-, pent- and hexoses, though one tetrose and one heptose function in the metabolic cycles of photosynthesis (Fig. 3) and the gluconic shunt (Fig. 5). Glycolaldehyde and the

ketotriose, dihydroxyacetone (Fig. 6, IV), are the only monoses without an asymmetric carbon atom (p. 3). In fact every carbon atom of the open chain glucose (Fig. 6, I), except the two end members, is optically active; the resultant rotation of plane-polarised light by a pure biological isomer is an algebraical summation of the contributions of the individual asymmetric atoms. The amount of optical isomerism among monoses is obviously very great; there are eight isomers of (Fig. 6, I) and in addition

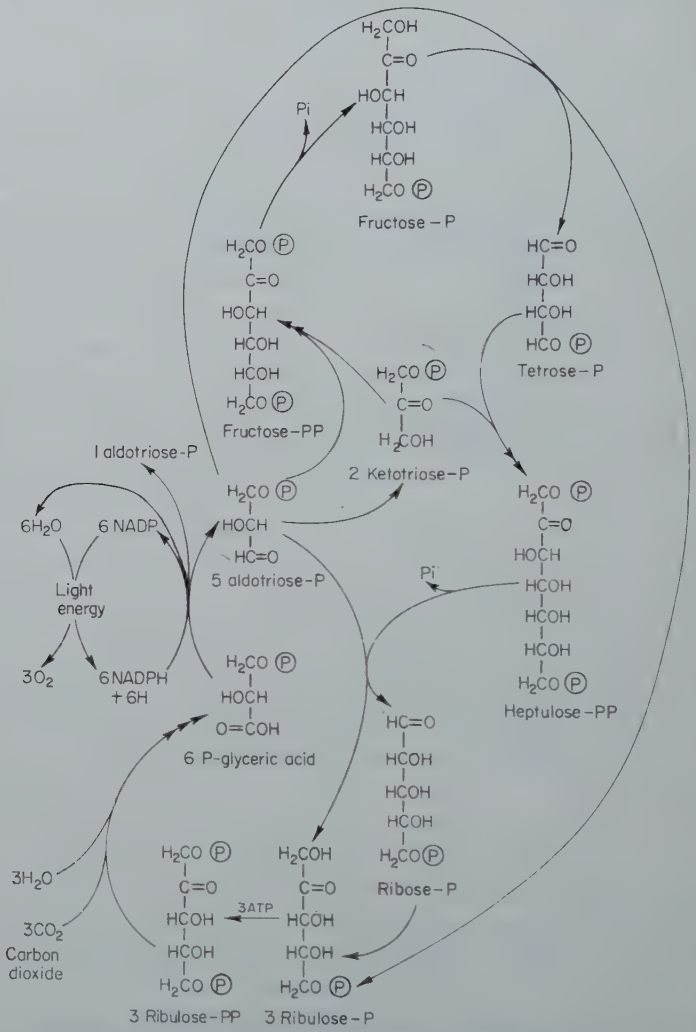


FIG. 3 Simplified Scheme showing the Unit of Photosynthesis. One triose molecule is produced in each complete revolution of the cycle.

this undergoes ring closure (p. 42) which makes 1C also asymmetrical and doubles the number of isomers. As already noted (p. 32) the aldotriose, glyceraldehyde (Fig. 6, V), is conventionally regarded as the structural 'parent' of all optically active compounds, since only its 2C is asymmetric. The isomer which is *dextro*-rotary is the natural (biological) one and in one plane this is represented as in Fig. 6, V; also as already noted all the natural monoses are derived from this, as Fig. 3 shows, and so they are all called D–forms, though fructose and others on balance are *laevo*-rotary.

Both of the possible trioses (Figs. 6, IV and V) play a key part in metabolism (Figs. 3, 4, and 5) and the following pentoses occur in living organisms: ribose, deoxyribose, ribulose, xylose, xylulose, and arabinose. The common biological hexoses are glucose, fructose, galactose, mannose, and fucose. Glucose is outstandingly the most important of the hexoses, with ribose and deoxyribose among pentoses. There appears to be a rather low specificity implied in this abundance of monoses, but some of them are intermediaries in pathways leading to others, and to other classes of metabolites; in general there is not very much overlap in function: no two are simple alternatives.

The way in which carbon dioxide is initially 'fixed' by autotrophes such as green plants, and reduced as far as the monose stage, is indicated in Fig. 3; for further details the standard textbooks should be consulted. It will be seen that with one exception the substrates and intermediaries in the process are all monoses, so that the process has very much an autocatalytic quality. The pathways are cyclic and can operate indefinitely so long as the motive force of solar energy, and carbon dioxide, are available, and of course water which is the initial donor of hydrogen for reduction. With each complete revolution of the main cycle three molecules of carbon dioxide are converted to one of glyceraldehyde. Two of the carbon atoms have been reduced to the carbinol and one to the aldehyde stage (Calvin, 1962). The role of chlorophyll, water and other agents is now well understood (Calvin, 1959–62). The phosphorylated condition of each intermediary also is catalytically important (p. 51).

All living organisms, and not merely autotrophes, can fix carbon dioxide, and do so by the same system of reactions (Calvin, 1962). Heterotrophes differ in obtaining the necessary energy indirectly from organic food, and not directly from the Sun. Given energy, therefore, their biosynthetic powers are not necessarily very much inferior to those of autotrophes.

Infra-red absorption studies indicate the existence of aldehydes on Mars (Colthup, 1961). This is important because there is independent

evidence, from the colour change (p. 508), for a seasonal cycle of vegetation on this planet.

The condensation of two triose molecules to form hexose, and of triose with tetrose to form heptose, is effectively the familiar aldol condensation of the laboratory, where potassium carbonate is usually used as the catalyst. *In vivo* carbon dioxide or its derivative conceivably may act in this capacity. The relative spontaneity of this aldol reaction raises the question why the condensations do not continue indefinitely to produce macromolecules. The monoses do go on to form macromolecules (p. 54) but these are polysaccharides, that is to say monose units linked indirectly through an oxygen atom. The explanation is that once the monose reaches a certain chain-length the aldehyde group can condense reversibly, through an hydroxyl group, with another carbon atom of the same molecule, forming a stable ring structure. The solid angle between neighbouring bonds of carbon, around 109°, dictates the tendency to form a ring and also explains why the pentose and hexose rings are the most stable, causing least distortion of the bond-angle; the six-membered ring opens them somewhat and the five-membered ring closes them a little. This ring closure and the stability of the rings formed by the pentoses and hexoses account for their relative abundance among natural monoses.

The ring in fact is closed through an oxygen atom but, as indicated by the tetrahedral structure of water (p. 18), this has bond-angles of around the same value. Consequently, aldotetroses also can, and do, form ring-structures. In pentoses and hexoses the terminal (ω–) carbon is excluded from the ring, and in hexoses sometimes 5C also, so that the rings are either amylene oxide (pyranose) or butylene oxide (furanose) derivatives (Figs. 6, VI, VII). The two additional asymmetric forms of glucose caused by ring closure, the α– and β–forms, also are illustrated by these two compounds. Both are D–sugars because the 5C has the same symmetry as the one asymmetric atom of D–glyceraldehyde. In solution the α– and β–forms are in dynamic equilibrium through the open chain form, and either pure isomer spontaneously racemises, as shown by a steady change in optical rotation. This is perhaps why the cyclic form retains the reactivity and other potential aldehyde properties at 1C, which therefore is known as a *hemi-acetal* and not a carbinol. This is just one of the many examples of the versatility conferred by dynamic lability.

It is evident that the tetrahedral bonding of carbon prevents the pyranose ring from being quite planar and this applies to many other aliphatic, or *alicyclic*, rings. Unsaturated rings, by contrast (that is rings with double bonds between carbon atoms —C=C—), are constrained into a more planar form. The pyranose ring, however, assumes the form

of either a 'boat' (Fig. 6, VIII) or a 'chair' (Fig. 6, IX); as in Figs. 6, VI and VII, the heavy line marks the edge of the ring facing the observer, to suggest a simple three-dimensional effect. The chair is the more common; most of the substituent groups lie in the main plane of the seat so that there is some approximation to coplanarity of the whole molecule.

The bonds of the oxygen atom are shorter than those of carbon (Table 39), so that the heterocyclic pyranose ring has somewhat smaller angles and therefore is more stable than the hexane, or 6-carbon, homocyclic ring. The furanose ring will be less stable than pentane, however, since the bond-angles will be further closed and biological furanose compounds usually need to be stabilised by phosphorylation (p. 355). Even so pentoses exist more extensively in the open chain state than the hexoses. The stability of the hexoses no doubt accounts for their outstanding biological role. In crystalline form, or in solutions too concentrated to be attacked by living organisms, they decompose very slowly. On the other hand, once activated they oxidise rapidly, even explosively (Oparin, 1957a), and are ideal fuels for ready use; they again have the best of both worlds.

Use as 'current coin' for respiration in fact is the main biological use of the carbohydrates, and there are special fuel-storage forms among the polysaccharides, or polymerised monoses. Aldohexoses such as glucose are more useful forms than ketoses since they have the usual strong reducing action of aldehydes. However, fructose is easily broken down to aldotriose, and in fact it is a normal intermediary in the glycolytic sequence (Fig. 4) which is the universal initial pathway of biological oxidation of the hexoses. It will be seen that for every molecule of glucose oxidised as far as pyruvic acid there is a net synthesis of two molecules of ATP (p. 404) from ADP; this synthesis is the universal method of transferring energy for use in the body. About 36 kcal/g-molecule is released in this sequence, up to the lactic stage, and perhaps as much as 22 kcal of this is usefully stored in ATP, an efficiency of 60 per cent or so. This is fairly representative of the efficiencies of biological processes and compares very favourably with those of laboratory reactions and of machines. The total energy released when one mole of glucose is completely oxidised to carbon dioxide and water is 686 kcal so that most of this is still locked up in pyruvic acid, and is released in a further pathway, the Krebs cycle (Fig. 8, p. 65). The unique value of the glycolytic sequence is that it releases energy without an external supply of oxygen, by intramolecular changes—a mild version of what happens in some explosive molecules (Read, 1942). The virtue of this carbohydrate halting-stage in biosynthesis, with considerable oxygen still in the molecule, is

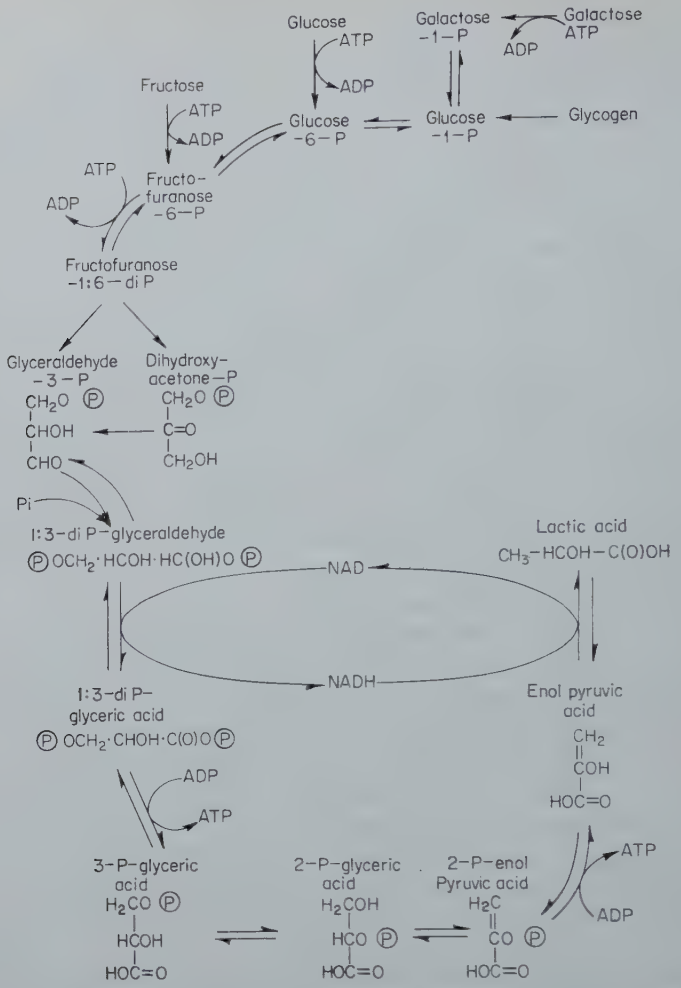


FIG. 4 The Glycolytic Sequence of Respiratory reactions, also largely reversible for use as a pathway of biosynthesis. The final product, pyruvic acid, feeds into the Krebs Cycle for further oxidation if oxygen is available. Otherwise it is reduced to lactic acid so as to regenerate the dehydrogenase coenzymes.

therefore obvious, for there are situations in which energy is required but free oxygen is not available. Provided the acid products are not allowed to accumulate, glycolysis may continue for long periods. Another proviso is that the coenzyme NADH must be continuously reoxidised by the continuation of the sequence to lactic acid, so that it does not yield as much energy as it would if halted at the pyruvic stage. It is a mean respiratory pathway, therefore, and most living organisms have a so-called Pasteur

effect mechanism: when oxygen is available it allows NADH to be oxidised via the electron-transfer sequence (Fig. 20) and pyruvate passes straight on to the Krebs cycle for complete oxidation. The switch may be very largely controlled by mass action, i.e. automatically.

It is well known that yeasts glycolyse to ethanol, CH_3CH_2OH, instead of to lactic acid, so that there is some latitude in this type of pathway; only the terminal steps differ from the more common glycolysis, but there is a rather better yield of energy, 50 kcal/mole of glucose. Ethanol also has the virtue of volatility, so that under most conditions yeasts can get rid of it rapidly enough to ferment indefinitely. This is, of course, very wasteful, and probably there has been selection for the lactic pathway in animals with a good circulatory system: in mammals the lactic is removed to the liver where oxygen is abundant and here 20 per cent is completely oxidised to re-synthesise the rest to glucose. The complete balance sheet therefore is that 137 kcal are expended for the gain of 22 kcal in ATP; the process is inefficient and living organisms appear to use it only when necessary.

There is another alternative to the early steps of the glycolytic sequence —the gluconic acid 'shunt' (Fig. 5). It also demands no free oxygen, and it leads to the complete oxidation of one C per hexose molecule. This high yield may be less important than the value of the shunt as a biosynthetic pathway for ribose, which is further used in the synthesis of nucleic acids (p. 355). The main coenzyme is NADP, and not NAD as in glycolysis.

Sugars also have the biological virtue of being very soluble in water owing to their many hydroxyl groups, and the molecules are small enough to diffuse rapidly. With the further ability to polymerise into relatively insoluble storage forms, such as starch in plants and glycogen in animals, they again have the best of both extremes. Also because of their many hydroxyl groups they are non-volatile, like the hydrophil inorganic salts—and like some other organic substances. In general, volatility is a disadvantage biologically—i.e. within the body. Sugars might be said to have tamed both the volatility and the acidity of carbon dioxide.

Most of the properties of the sugars in fact depend on the many hydroxyl groups—including the characteristic sweetness. No doubt the taste has become attractive because of their biological value, and has no more absolute significance, and it is interesting, therefore, that a number of relatively useless organic molecules, such as saccharin, taste as sweet or even sweeter. In fact the laboratory-synthesised compound, 1–propoxy–2–amino–4–nitrobenzene, is 5600 times as sweet as sucrose. It seems therefore that polar groups other than hydroxyl also taste sweet.

For some purposes sugars can be regarded simply as polysubstituted waters, retaining many of the properties of the water, such as hydrogen-bonding. Hydrogen bonds probably account for the stickiness of most soluble carbohydrates in adequate concentration, when the molecules

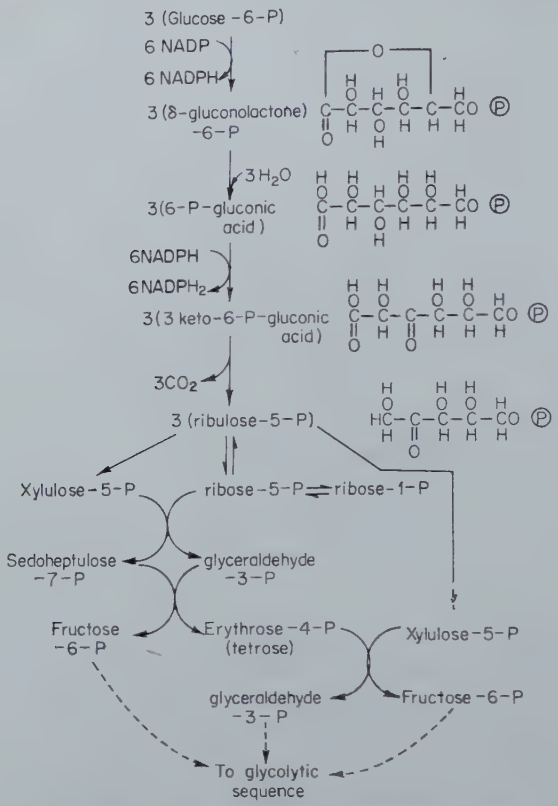
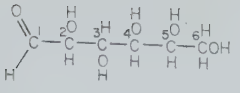
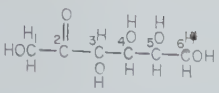


FIG. 5 The gluconic acid shunt-pathway, an alternative pathway for the incipient oxidation of the hexoses and a biosynthetic pathway for ribose. Other products feed into the glycolytic pathway. Note resemblances to photosynthesis cycle (Fig. 3).

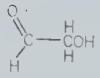
will be hydrogen-bonded to each other, directly or through water molecules. At lower concentrations relatively few water molecules will be bonded between two saccharide molecules and the solution becomes more lubricant. The addition to carbohydrate of strongly hydrophil groups, such as sulphuric acid, increases the size of the water shell round each molecule and the resulting mucins are among the best biological lubricants. In the muco-polysaccharides of the connective tissue, and in agar,



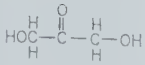
I
Glucose



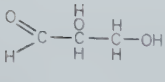
II
Fructose



III
Glycolaldehyde or
Glycollic aldehyde



IV
Dihydroxyacetone



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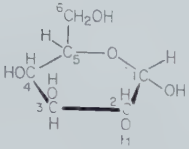
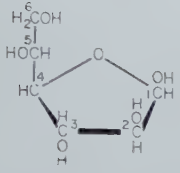
D-glyceraldehyde



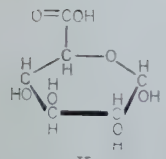
VI
α-D-glucopyranose



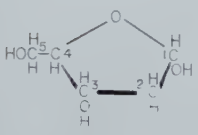
VII
β-D-glucofuranose

VIII
"Boat" form

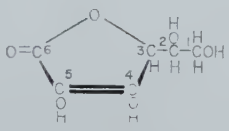
IX
"Chair" form
of pyranose ring



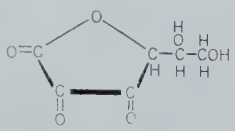
X
Glucuronic acid



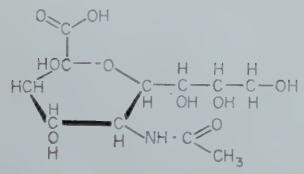
XI
Deoxyribose



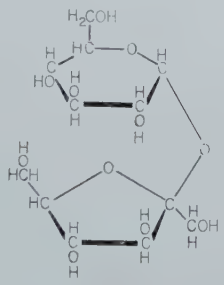
XII
Ascorbic acid (vitamin C)



XIII
Dehydroascorbic acid



XIV
Sialic acid



XV
Sucrose (cane sugar)

FIG. 6 The more important monose sugars and their derivatives.

the water-storing property is more important than lubrication. Even the more insoluble polysaccharides have a high affinity for water, which may help in the eventual hydrolysis of starch and glycogen. That of cellulose, in plant cell-walls, may assist the transport of water to and from the cell. Both starch and cellulose are used in partition chromatography because they hold what amounts to a stationary water-phase which will partition solutes against a flowing phase.

Their affinity for water makes saccharides also good anti-desiccants and good anti-freeze agents (Billingham, 1955), without the disadvantages of electrolytes (as the alternative). These allow the water to freeze and so they concentrate up to dangerous levels. In the presence of certain other materials sugars rather readily supersaturate water, as in the syrups, which are highly resistant to desiccation and to freezing. It is probably significant that temperate and polar plants store carbohydrate, whereas tropical members store oils which are hydrophobe and scarcely affect the aqueous phase. Sugars even protect cells against damage by irradiation (Wood, 1959), which acts mainly on the intracellular water, ionising it and forming a great deal of hydrogen peroxide.

In a sense, no doubt, the osmotic action of solutes depends on an affinity for water. Sugars are biologically important for osmotic purposes not because their osmotic power per mole is different from that of other molecules, however, but as in the case of the anti-freeze function because they can replace electrolytes. In this instance they do not introduce the complication of Donnan and electrical differentials, as electrolytes do. Further, the amount of sugar can be rapidly changed from inside by metabolism.

In the fixation of carbon dioxide into sugars the $=O$ is separated from all hydroxyl groups (p. 164) and the acidity disappears. The sugar hydroxyl groups behave like other alcoholic hydroxyl radicals, and form esters and ethers, i.e. they can form salt-like compounds with acids and bases. Ether formation is probably the more common and the resulting compounds are known as glycosides, but the reaction is usually with another hydroxyl compound and it is not always easy to say which is acting as acid and which as base, even where the other reagent is a strong acid, such as phosphoric. Monoses form glycosides with other saccharides and this is also the basis for indefinite polymerisation, since each residue has a number of hydroxyl groups available. In fact condensation occurs most readily between the hemi-acetal group of one and an ordinary hydroxyl group of another residue, so that simple linear polymers are most usual.

Monoses and disaccharides form salts with calcium, strontium and potassium. They behave as such weak acids that these salts are decomposed by carbon dioxide, and this provides a very convenient method of

commercial purification, since calcium and strontium are precipitated as insoluble carbonates.

Natural glycosides with *aglycone* partners include the nucleosides (p. 355) and cerebrosides (p. 294), plant pigments such as alizarin and the anthocyanins, the pigments of aphids, tannins and phlorrhizin. The drugs digitalin and strophanthin also are glycosides. Salicin from the willow is a glycoside of salicylic alcohol while amygdalin, from almonds, is the glucoside of an aromatic cyanide. Frequently the glycose moiety stabilises the aglycone component, protecting it from destruction or alternatively protecting the body from its effects. Precursors of active reagents are kept inactive until the right moment by this kind of conjugation. An example is the phenolic material used to harden the ootheca of the cockroach (Brunet and Kent, 1955). Indican is a naturally occurring, colourless glycosidic precursor of indigo, the colour of which appears dramatically when indican is hydrolysed.

Insects use glucose to *detoxicate* phenolic and other poisons (Smith, 1955), for excretion from the body. This is therefore a variant of the protective function. Occasionally the glucoside is more toxic than the original, but this is true of the products of other detoxicating agents (Williams, 1959), and is the occasional, unavoidable metabolic misfortune. Vertebrates use instead glucuronic acid (Fig. 6, X); this is effectively glucose oxidised to the carboxyl stage at 6C so that it forms closely analogous compounds, as some other monose derivatives do. There are other detoxicating agents but none so versatile as these. Glucuronic acid (Table 42) forms esters with acid groups, ethers with alcoholic hydroxyl groups, and combines with —SH, —NH_2, and a variety of alkyl (aliphatic) and aryl (p. 127) radicals. Indeed it seems surprising that normal metabolites with these groups are immune to attack. Glucuronic acid uses mainly its hemi-acetal group for the purpose, as glucose does, and the rather strong carboxylic acid group remains free.

Most of these functions are uniquely restricted to the hexoses. As already noticed, the other monoses occur mainly as intermediaries in the pathways of photosynthesis (Fig. 3), glycolysis (Fig. 4) and gluconic acid (Fig. 5). The pentoses have some additional functions but they overlap those of the hexoses relatively little. Some micro-organisms can oxidise them in respiration but mammals tend to excrete them unchanged. Large amounts of these reducing sugars in the urine, coming from proprietary soft drinks, have on occasion given a false alarm of diabetes. The most important biological pentoses are the riboses, as components of nucleic acids (p. 355). Very rarely other sugars play the same role in a nucleotide, but this only serves to emphasise the unique role of the riboses. Deoxyribose, the form in DNA (p. 347), as its name implies is unusual in having

the hydroxyl on 2C replaced by hydrogen (Fig. 6, XI); two deoxyhexoses, L–fucose and L–rhamnose, also occur biologically but they are reduced at the 6–position, are stable and have different, but still rather obscure, functions. Both riboses are also unusual in being very unstable (Baldwin, 1953, p. 352), and depend on their nucleotide combination for their stabilisation. In this form therefore they combine their reactivity with the compensatory virtue of stability. Other pentoses are more stable and occur mainly in polymers called pentosans, in plants. The pentoses have a higher affinity for water than the hexoses, and pentosans are contained in gums and mucilages, combined with various other monoses and their derivatives. As already noted, hexoses hold water to this extent only when combined with other hydrophil molecules.

A number of monose derivatives are of biological importance. Glucuronic acid (Fig. 6, X), in addition to its function in detoxication, is an intermediary in the synthesis of vitamin C from glucose. Gluconic acid, which is glucose carboxylated at the 1C position, is an intermediary in pentose synthesis, and in a respiratory pathway (Fig. 5). Saccharic acids, carboxylated at both 1C and 6C, also occur biologically. Although more oxidised than the monoses they are synthesised from them *in vivo*, and represent our first example of retrogressive or oxidative synthesis; this is by no means uncommon.

Ascorbic acid (Fig. 6, XII), or vitamin C, is another example. It could be regarded as further oxidised, since it is formed from glucuronic acid by removal of hydrogen from 4C and 5C, with the formation of the double bond. The term *lactone* is given to compounds ring-closed in this way through a carboxyl carbon, which loses its acidity in the process. The acidity of ascorbic therefore depends on the 4 and 5 hydroxyl groups which as in those on an aromatic ring (p. 131) become acidic through resonance between =C·OH and —C=O. The C=O already at 6C can resonate in the same way, and in fact there is some approach to a *conjugate* system of double bonds (p. 116) which further increases the activity of the molecule. Resonance activity is more usually associated with redox than acid–base activity, and indeed ascorbic acid is a metabolic redox agent of great importance. It becomes oxidised to dehydroascorbic acid (Fig. 6, XIII), which is very closely comparable to the aromatic quinones. Indeed ascorbic acid will reduce aromatic quinones to the phenolic state. It is outstanding among carbohydrate and other aliphatic derivatives in this: although two steps more oxidised than glucose it is a much stronger reducing agent, and illustrates the almost unlimited possibilities among carbon compounds. The metabolic importance of ascorbic acid has been known since the days of the first long sea voyages and their scourge of scurvy, resulting from a dietary deficiency of the vitamin. Its redox

potential, E_o', is + 0·080 V, very near the middle of the metabolic range (Table 31).

Another modification of the monose molecule which is important biologically is the replacement of one hydroxyl group, usually that on 2C, by the amino group, —NH_2. This is therefore a substituted ammonia and is basic in reaction, though only weakly (pK=7·8). Once more the one derived from glucose is the most common: glucosamine occurs in the A, B, O and Le^a blood group antigens, and in further derivatives. The ammonia group commonly becomes acetylated, and acetyl-glucosamines are components of the hyaluronic acid of the ground substance of connective tissue and of cartilage, of mucin, of the anticoagulant, heparin, and of pepsin, and other materials. In most of these it is the glucose derivative, again, but in cartilage and connective tissues there is galactosamine. Acetyl glucosamine can still polymerise, as the parent monose does, and the skeletal material, chitin, is so produced.

This acetylation is a condensation and not a salt-formation so that it is not strictly correct to say that the product has the trinity of components of some other biological compounds (p. 355), viz. acid + base + neutral moiety. The bond is analogous to the peptide bond (p. 168), strong and covalent, the ionic properties quenched in the water molecule removed:

$$R_i\,C(O)OH + H_2N{\cdot}R_2 \rightarrow R_i\,C(O){\cdot}HN{\cdot}R_2 + H_2O$$

The NH_2 in fact acts as an acid and the acetic as a base: hence the distinction *acetyl* and not acetate. The acyl-amino group retains an ability to form hydrogen bonds and these may play the same important part in skeleton building as in protein materials (Chapter 15).

Acetylation reduces the water-affinity of saccharides, and some of the commercial products of cellulose, acetylated in all three available positions, are particularly insoluble and make very good, hard-wearing fabrics. They are also much less inflammable than the parent saccharides, an index of low reactivity which also is valuable in skeletal fabrics. By contrast the sulphuryl derivatives increase the water-affinity, since one hydroxyl of the sulphuric acid is still free. In chondroitin sulphate, the unit in the structure of the ground substance of cartilage, the acetyl galactosamine residues are sulphurylated at 1C, and the combined properties may help to account for the interesting hydrated yet solid properties of cartilage.

Structurally the phosphoryl compounds of the monoses resemble those of sulphuric acid but they have a very different and much more general function, connected with the ideal resonating property of the phosphoric acid molecule (p. 403) and its consequent activating action on its conjugates. There is virtually no metabolic reaction of monoses and their derivatives which does not involve their phosphorylation (Table 35).

In principle, to judge by the ribose of RNA, every sugar hydroxyl group can be conjugated in this way and the phosphorus group is readily shifted from one to another, sometimes resonating between them (Todd, 1954). Also, as already indicated, the link is sometimes phosphoryl and sometimes phosphate in type. When the phosphate is transferred to some other combination the amount of energy involved may vary between 3 and 16 kcal/mole (p. 402) so that altogether these conjugates are extremely active and versatile.

The hexose phosphates are actually stronger acids than the free phosphoric acid, and their salts of calcium and magnesium are more soluble than the corresponding inorganic forms. Biologically this is no doubt a valuable property.

Of the remaining biological derivatives of the monoses one of the most interesting is sialic or neuraminic acid, also known as gynaminic, haemataminic, lactaminic and serolactaminic acid (Fig. 6, XIV). It is an aminated nononic acid, usually acetylated on the amino group, as shown (Heimer and Meyer, 1956). Its carboxyl group has a p*K* as acid as pH 2·7. It is an important physiological agent, as its many names indicate, and is also a component of gangliosides, the complex glycolipids of the nervous system, and of the carbohydrate moiety of some of the glycoproteins.

In the glycolipids monose derivatives are involved but in some at least of the glycoproteins the carbohydrate is in polysaccharide form. The properties of these complexes are not all known in detail, but in glycolipids the carbohydrate depresses the lipid-solubility (Fearon, 1949) and this helps to stabilise them as constituents of membranes. Reciprocally the lipid depresses the water solubility of the saccharide, and the complex is most soluble in the intermediate solvents, the weakly polar alcohols and acetone.

4.2 Oligosaccharides

There are a number of disaccharides in living organisms (Table 14) and a disaccharide unit seems to play a part in the construction of the polysaccharides. Thus starch is broken down to maltose, and cellulose to cellobiose. These monose-monosides are in general more stable than the free monoses. Sucrose or cane sugar is non-reducing because the glucose residue is bonded to the fructose by its aldehyde group, and trehalose because the two glucose residues are united face to face through both aldehyde groups. Trehalose is also non-permeant and in insects it is formed from glucose as it is absorbed from the gut; consequently the concentration-gradient of glucose is maintained favourable for spontaneous absorption (Treherne, 1958). Sucrose remains however very soluble and sweet, and it is interesting that it should be the storage form

in cane, beet, etc. Probably its significance is in fact osmotic, or more generally colligative.

Unlike glucose the disaccharides have a limited distribution in most cases. Sucrose (Fig. 6, XV) seems to be synthesised only by plants and

Table 14. Disaccharides and Trisaccharides

Disaccharides	
Maltose	4-D-glucopyranosyl-α-D-glucopyranoside
Trehalose	1-α-D-glucopyranosyl-α-D-glucopyranoside
Cellobiose	4-D-glucopyranosyl-β-D-glucopyranoside
Gentiobiose	6-D-glucopyranosyl-β-D-glucopyranoside
Sucrose	1-α-D-glucopyranosyl-β-D-fructofuranoside
Turanose	3-D-fructofuranosyl-α-D-glucopyranoside
Lactose	4-D-glucopyranosyl-β-D-galactopyranoside
Mellibiose	6-D-glucopyranosyl-β-D-glucopyranoside

Trisaccharides	Components
Melezitose	glucose, fructose, glucose
Gentianose	fructose, glucose, glucose
Raffinose	fructose, glucose, galactose
Mannotriose	galactose, galactose, glucose
Robinose	galactose, rhamnose, rhamnose
Rhaminose	galactose, rhamnose, rhamnose

if injected into the body of animals it is excreted unchanged. Lactose is the sugar of mammalian milk, and occurs nowhere else in their bodies, except occasionally in the urine of nursing mothers; some very special significance for the infant may be suspected and indeed it is a good anti-rachitic agent, promoting the absorption of calcium and phosphorus from the gut (Brody, 1945, p. 802). It is also a convenient source of galactose for connective tissue, cartilage and the cerebrosides, all of which are being synthesised very rapidly at this stage. It is a β–glycoside and mammals have few enzymes opening this type of link, in contrast to the many attacking the α–glycoside bond, but they do possess a lactase; digestion is probably assisted also by the *Lactobacillus* symbiotic in the infant gut. Lactase occurs also in almond seeds, fungi and baker's (but not brewer's) yeast, so that the sugar itself probably has a wider distribution. It is one of the least soluble and least sweet of sugars.

A few trisaccharides occur sporadically in living organisms (Table 14).

There are very few pure tetrasaccharides, such as stachyose, D–galactosyl–D–galactosyl–D–glucosyl–D–fructose. However, in human milk there is an interesting hetero-tetrasaccharide: D–galactosyl–*N*–acetyl–D–glucosaminyl–D–galactosyl–D–glucose, which appears to be an essential growth factor for the intestinal *Lactobacillus* (Barker and Stacey, 1957); clearly the bacterium is specially adapted to this diet, in this ecological situation.

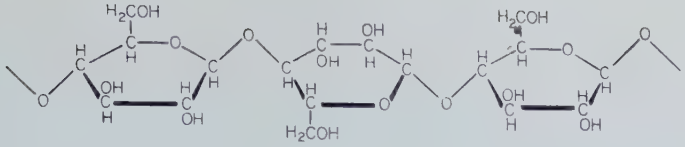
There are not many oligosaccharides of biological importance and they are rather restricted in distribution. By contrast there are a number of important polysaccharides, a situation very similar to that among peptides (p. 203). There are in fact biological 'nodes' in the various series of organic compounds; the many intermediates have no unique properties but for the most part only such as are shown more strongly by nodal members, and they alone have been biologically exploited.

4.3 Polysaccharides

The two main biological functions of the polysaccharides have already been noticed, fuel-storage and skeleton-construction. The most common fuel-stores are starch (Fig. 7, II) and inulin in plants, and glycogen and galactogen in animals; inulin and galactogen are rather rare and merely emphasise the unique value of the main form in each kingdom, both polyglucosides. Inulin is a polyfructoside and galactogen, as its name indicates, a polygalactoside. In plants the main skeletal material is cellulose (Fig. 7, I), a glucosan, that is a linear polymer once more of glucose. It also occurs in the test of tunicates and in the connective tissue of mammals (Hall, 1958). The corresponding galactosans, mannans and xylans, also occur in plant skeletal materials. In animals chitin is more common, a simple linear polymer of acetyl–β–glucosamine (p. 51).

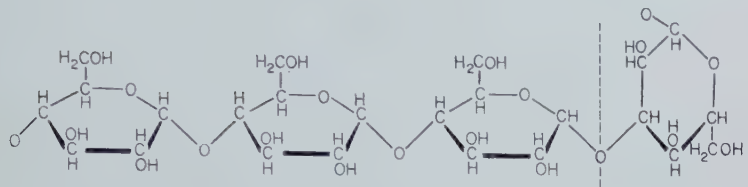
In the capsular polysaccharide of *Pneumococcus*, types II and III, the repeat-unit is a hetero-dimer, D–glucosyl–D–glucuronic acid. In type I the dimer is a more unusual one, acetyl–D–glucosamine and D–galactosamine. Other strains have other variants on the hetero-dimer theme. This complexity and variety is unusual in carbohydrates and it is perhaps significant that these are among the few saccharides found to have antigenic properties. The anthrax bacillus has a hexose-hexosamine dimer, with alkaline properties, contrasted with the acid reaction of most. Hyaluronic acid, a skeletal component of the ground substance of the connective tissue, of synovial fluid and of the vitreous humour, has a dimer of *N*–acetyl–β–glucosamine, and glucuronic acid, while the chondroitin of cartilage (III) has *N*–acetyl–galactosamine–sulphate and glucuronic acid (p. 50).

Plants produce a number of accessory skeletal materials such as the cement, *pectin*, and various gums. Pectin is a fairly simple polymer of



I Cellulose

4-D-glucopyranose-β-D-glucopyranoside links

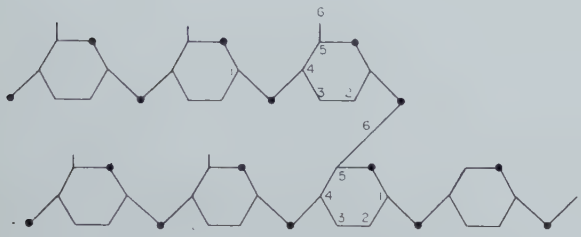


IIa

Amylose (starch)

4:1α-glucopyranoside links

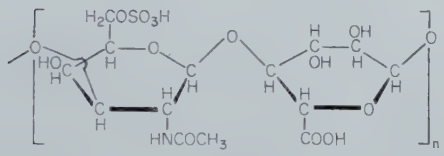
Occasional 1:3 β-link



IIb

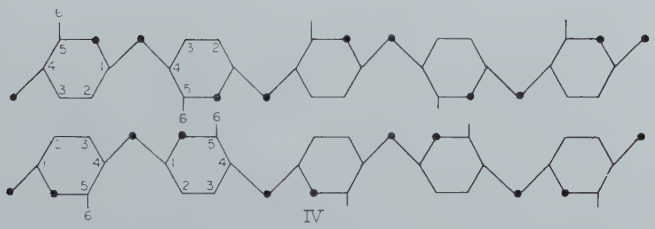
Amylopectin (starch and glycogen)

1:4 α links with branching by 1:6 α links



III

Chondroitin sulphate



IV

Antiparallel arrangement of cellulose molecules

FIG. 7 The structure of representative polysaccharides.

methylated galacturonic acid, interspersed with a few residues of galactose and arabinose; it is exploited commercially to make jams set, a property which survives the long boiling. Some fruits themselves contain enough for the purpose. Agar jelly from seaweed has nine D–galactose residues to one of L–galactose–sulphuric acid. Alginic acid from other Algae is a poly–D–mannonic acid. Gum arabic from acacia contains galactose, arabinose, rhamnose and glucuronic acid. At present the significance of each monose derivative is not clear, but each polymer has its own unique properties and no doubt this depends on the particular monomers. In these materials the carbohydrates have really 'gone to town'.

Although the other functions of polysaccharides may seem quite unrelated to the two main ones, there is a connection, which has already been seen in relation to hydration (p. 45), and the general structure is similar to that of the skeletal polymers. Mucin, the lubricant, is a polymer of *N*–acetyl–glucosamine–sulphate and glucuronic acid. The great difference in physical properties between it and chondroitin therefore seems to depend on the difference between glucosamine and galactosamine sulphates. However, the mucus of the snail appears to be based on D- and L–galactosamines, so that there is more in this problem. In passing, the racemic mixture in this mucus is interesting: as an external secretion perhaps the optical impurity is not deleterious.

Heparin, the anti-coagulant, has a rather similar mechanical function to mucin, liquefying rather than solidifying, and is similar in composition. The difference is that each acetyl glucosamine residue has two and not one sulphuric acid group and in addition there is a further sulphuric acid group on alternate glucuronic residues; the complete repeat-unit is therefore a tetramer. The high sulphuric content gives the high affinity for water, and may be the essential factor in anti-coagulation.

The jelly surrounding the eggs of echinoderms has important functions in addition to mechanical protection. It contains a number of sugar residues, varying in the different species, as the bacterial polysaccharides do, and having a very definite antigenic function (Runnstrom, 1952). They agglutinate the sperm of foreign species in such a way as to inactivate them, and the sperm of the same species in such as way as to assist fertilisation. The A, B, O and Le^a antigens of man are very similar heteropolysaccharides; for instance A contains L–fucose, D–galactose, D–mannose and D–glucosamine (Barker and Stacey, 1957). Pectin is able to protect red blood cells against the influenza virus, acting as a deputy for the normal antigen of the A, B, O group: perhaps this is the rationale for the 'apple-a-day' maxim, since apples are very rich in pectin. These antigens are directed against proteins and like most polysaccharides are normally associated with indigenous protein.

In polymerisation the hemiacetal group of the one monose condenses usually with the 4C or 6C hydroxyl group of the next residue, but occasionally with that of 3C (Fig. 7, II). Ketoses use the ketone group, which is usually at 2C, and the bond demands at least one aldehyde or ketone group; they can form linear polymers of unlimited size and the macromolecules of cellulose probably contain 4700–5000 residues (Bell, 1960) in virtually all cases. Hyaluronic acid has a molar weight of $2–5 \times 10^5$, which is equivalent to 1000–3000 residues. This is the first example of a really large biological polymer and it is noteworthy that it is not an unbroken chain of carbon atoms but a string of carbon-based monomers united through an atom other than carbon—in the present case oxygen. Paraldehyde and paraformaldehyde by contrast are unbroken carbon polymers and they polymerise spontaneously whereas 4 kcal/g-equivalent is required to forge the bonds in glycogen. The alkyl aldehydes also polymerise by addition and not by condensation.

The monoses have the important additional facility of forming branched polymers (Fig. 7, IIb), with the proviso that each residue can be *active* partner in only one link, using its aldehyde group. The polyhydroxy alcohols (p. 72) are quite unable to polymerise in this way. In general the storage-polysaccharides have exploited the potentiality for branching while the skeletal types remain strictly linear: it is not possible to weave with tangled threads. However the plant gums and mucilages have branched molecules and this probably contributes in a simple mechanical way to their water-holding power. However the water-affinity of the pentoses is no doubt the main factor, since the branched hexosans are not so hydrophil. In the storage-hexosans branches perhaps provide extra points of attack when it comes to mobilisation and they may also facilitate the formation of granules as the ideal form for storage. The resulting tuft seems to be less viscous than the same mass in the form of a single tangled thread (Oncley, 1959 or 1959b).

The solubility of polysaccharides decreases with their molar weight, the smaller polymers being still fully soluble, larger ones colloidal and the largest completely insoluble, notwithstanding the large number of hydroxyl groups still free. This is a valuable property in skeletal materials, permitting initial synthesis and orientation in solution, but complete insolubility once completed. Cellulose is remarkably insoluble, considering its simple glucoside nature, and in the laboratory special reagents, such as Schweitzer's, are necessary to dissolve it. In chitin, as already noted, solubility is further depressed by acetylation of the residues. Commercial acetylation of cellulose destroys even its hygroscopic affinity for water and the derivative can be used for chromatographic techniques in which a water-phase is undesirable. Cellulose acetate becomes soluble

in acetone, nitrobenzene and chloroform, lipid-soluble that is; when the residues are completely acetylated (p. 51) it is no longer soluble in the more hydrophil organic solvents, such as acetone. The analogous trinitrocellulose, however, being substituted with a more polar group, *is* soluble in the polar organic solvents—acetone, alcohol, ether and ethyl acetate. Glycogen and starch are more water-soluble than cellulose because α–glucose polymerises as a spiral thread whereas β–glucose forms flat ribbons (Oncley, 1959b).

Cellulose is soluble in acidified zinc chloride, acetic anhydride and concentrated sulphuric acid, and in sodium hydroxide followed by carbon disulphide, as well as in Schweitzer's reagent which is essentially alkaline cupric hydroxide. This is due to the partial hydrolysis of the cellulose to hemicellulose and in part to the hydrating effect of these ionic hydroxy-compounds. On diluting with water, however, the hemicellulose tends to precipitate so that there is probably also a salting-in effect (p. 223). The cellulose-'xanthate' produced by sodium hydroxide and carbon disulphide may be considerably hydrated, judging by its high viscosity. It is also another virtually non-inflammable polysaccharide. In appropriate concentration (70–80 per cent) sulphuric acid swells paper just enough to convert it to translucent parchment, which will act as a dialysing membrane permeable to small molecules, just as the nitro-cellulose, prepared membranes. These are just a few of the many potentialities of the polysaccharides; a number of them no doubt are exploited *in vivo*.

Acids hydrolyse polysaccharides to monomers; *in vivo* of course they are hydrolysed enzymatically, at a virtually neutral reaction. Both acids and alkalis swell cellulose and the latter absorbs both acidic and basic dyes readily. Its hydroxyl groups can be substituted by both acidic and basic ionic groups and the product can then be used technically as a suitable material for ion-exchange chromatography (Long, 1961). Two well-known types of basic ion-exchange resins are 'ecteola' and 'deae'. These can absorb protein and nucleic acid equal to one-fifth of their own weight and release it again on *eluting* with a medium of slightly different pH or ionic strength. This powerful ion-binding property plays a biologically significant part in the function of the egg-jelly of echinoderms.

As further examples of the versatility of the polysaccharides there are the melting properties of agar and of methyl cellulose. Agar does not melt until heated to 95–100°C but on cooling it does not set until the temperature is down to 35°C. Technically this is very convenient since the sterilised agar can be mixed with nutrients, cooled, inoculated with bacteria before or after setting, as desired, and poured into a suitable vessel for setting. Methyl cellulose has the unusual property of setting

on heating and liquefying on cooling (Bull, 1943; Sexton, 1953). It is not known to be a common derivative *in vivo*, but conceivably it could be used to buffer the effect of temperature on orthodox materials if mixed with them.

The mechanical strength of polysaccharides as skeleton materials depends first on the glycoside bond, and their rigidity and toughness on the parallel alignment of these bonds, in the virtually straight molecules (Bell, 1960). This bonding is supplemented by hydrogen bonds between molecules, directed at right angles to their axes and parallel to the planes of the glucose residues. In addition, neighbouring molecules are anti-parallel (Fig. 7, IV), the 1C ends of their glucose residues pointing in opposite directions; this allows the closest approach between the molecules, and the non-specific attractive forces of London–van der Waals therefore are relatively strong. The micelles, or aggregates of molecules, are typically ribbons about 150–300 Å wide and 10 μ or so long. In the plant cell-wall they lie in the plane of the wall but run a spiral course, which increases strength against forces which may come from any direction and at the same time leaves maximal flexibility for growth in length; growth simply increases the pitch of the spirals and opens the lozenge-shaped spaces between two countercoursing systems of ribbons. As a long and strong thread the cellulose molecule is intrinsically ideal for skeleton formation, and living organisms clearly have exploited it to build a variety of textures at higher levels of magnitude.

In the 'wood', or xylem, of plants whole cells become virtually fusiform 'fibres' of cellulose, the micelles following a high-pitched spiral as in commercial cables, designed for pulling strength, with some resilience: twisted threads are stronger than a bundle of straight ones because they bind together as they are pulled. The wood cells interdigitate at their ends with further relays of cells, and so build the macroscopic skeletons of plants of all sizes. It is effectively interdigitation of this kind which permits the spinning of strong commercial threads from the long cellular fibres of the cotton boll. Lateral adhesions are very important for this, and it is significant that the strength of flax increases 20 per cent on wetting (Colhoun, 1958), which improves lateral adhesion. The swelling of the fibres and increased rigidity of a wet rope is common observation: the improved adhesion is partly mechanical, non-specific, and partly no doubt due to the formation of more hydrogen bonds. The water-affinity of cellulose therefore is valuable in skeletal structures, also.

The cellulose fibres of flax can bear a tension of 50–100 kg/mm cross-section (Colhoun, 1958), which is more than half that of steel wire (170 kg/mm^2). Silk bears 65 kg but wool only 21. The manufactured thread, nylon, at 75 kg, compares with an average flax.

Around less specialised plant cells the cellulose must withstand strains in all directions, mainly associated with the varying hydrostatic pressure in the cell. The micelles follow a spiral of low pitch relative to the main axis of the cell, giving flexibility for growth, as already indicated. A number of layers of micelles are laid down; within each layer all are parallel but there is a large angle between the preferred orientation in neighbouring layers. The strength of this texture is well known from a comparison of plywood with ordinary wood of the same thickness. The device has been exploited also in protein fabrics (p. 484). There is not space here to deal in full with the fascinating details of this exploitation and these may be found in the works of Preston (1952), Frey-Wyssling (1957), Picken (1960) and others.

The higher grades of organisation of such materials are in any case probably beyond our terms of reference (p. 477). It is far from certain, and indeed improbable, that organisation at these levels is an automatic consequence of the intrinsic properties of the molecules and equally certain that it depends on what may be called systemic organisation, for want of a better understanding of its exact nature. It is a subject amenable to investigation, however, and it would be possible to ask the question: at what level does systemic organisation take over? Is the anti-parallel orientation of the neighbouring cellulose molecules spontaneous or is this already systemically dictated? Is even the linear molecule of cellulose, rather than a branched polysaccharide, systemically determined? Cellulose fibres are often laid down with their axes at right angles to the direction of growth of the cell, whereas chitin fibres are usually parallel to the line of growth. The deployment of pectin, lignin and other accessory materials also seems to demand systemic organisation.

Skeletal material can scarcely fail to be also protective, but in addition there are a number of specifically protective polysaccharides. Dry, loosely woven cellulose fibres form good mechanical protection, through their resilience, but they are even more useful as heat insulators through trapping still air, non-conducting and non-convecting. This is the significance of the cotton boll. The fibres are usually also waxed and hydrofuge.

Polysaccharides also protect in more chemical ways, at the molecular level. They stabilise other biological materials with which they are conjugated. The chitin–protein association in the exoskeleton of arthropods is harder and more insoluble than either component alone: the action is in fact reciprocal. The polysaccharide, chondroitin moiety of cartilage contributes to its rigidity and it stabilises the protein of tendon (Sylven, 1956). As little as 0·2–0·8 per cent of hyaluronic acid is enough to give almost complete immobilisation of the protein component of the ground substance of connective tissue. Biochemical stability is illustrated by the

intestinal mucins which presumably withstand a whole range of carbohydrases and proteases. The precise nature of the protection is not yet clear in all cases; in the arthropod skeleton the chitin molecules may lie across the protein threads, at right angles, the —NHCO— pseudopeptide group of the acetylglucosamine residues being hydrogen-bonded with the peptide groups of the protein.

A striking example of stabilisation by polysaccharides is in the commercial preparation of ersatz creams, based on agar. Another of biological importance is the laboratory formation of coacervates (p. 233), small bodies with a size range up to that of small cells, and with a range of *eobiological* properties. Gum arabic, araban, agar and soluble starch have been most used for this purpose, and they may conceivably have some related function *in vivo*. Phosphorylation greatly improves the ability of polysaccharides to form coacervates, probably by converting them to ionic species. In the membranes of cells and organelles these ionic polysaccharides may help to control selective permeation (Kent and Whitehouse, 1955).

In all, therefore, this first group of stable carbon compounds to be formed in biosynthesis has been put to a variety of uses. These all follow from a small number of intrinsic, unique properties.

CHAPTER 5

TERNARY COMPOUNDS MORE FULLY OXIDISED THAN CARBOHYDRATES

SOME of these have already been considered (p. 50) and the reason for considering them after the carbohydrates themselves was given—that in biosynthesis they are formed after the carbohydrates, and from them. Some of them are merely intermediaries in the pathways of oxidation of the carbohydrates, in respiration, but others are synthesised in their own right for useful purposes. Since the empirical formula of monose carbohydrates is $C_nH_{2n}O_n$ each carbon is associated with two less oxygen atoms than carbonic acid, or the equivalent of four hydrogen atoms, and there are many possible intermediaries when n is a large number. Of course in a C_n monose there are $2(n-1)$ bonds held by other carbon atoms, but when opened each pair of these could be oxidised to 2(—COH) just as much as if each carbon were free; consequently one oxygen molecule is required to oxidise each carbon of any monose back to carbon dioxide:

$$C_nH_{2n}O_n + nO_2 \rightarrow nCO_2 + nH_2O$$

Any ternary compound (i.e. containing only carbon, hydrogen, and oxygen) which has an oxygen/carbon ratio higher than in carbohydrates or a lower hydrogen/carbon ratio, or both, then qualifies for the present group. Some are mononic, monuronic and saccharic acids, collectively classified as polyhydroxycarboxylic acids, whereas others are more easily classified according to their metabolic function.

> Sweets of today tomorrow sour.
> (LESLIE COULSON, *But a Short Time to Live*)

5.1 Polyhydroxycarboxylic Acids

Carbonic acid itself (p. 35) could be considered the first member of all the series in this group, since its structure is $\begin{matrix}HO\\HO\end{matrix}\!\!\diagdown\!\!\diagup C{=}O$. The next member of the mononic series is glycollic, or hydroxyacetic, $CH_2OH{\cdot}C(=O)OH$, occurring in unripe grapes and other fruits. The next member,

glyceric acid, $CH_2OH{\cdot}CHOH{\cdot}C(=O)OH$, is an intermediary in the photosynthetic cycle (Fig. 3) as well as in the glycolytic sequence (Fig. 4). This means that there is an oxidation step in the former, also: this is probably one aspect of its spontaneity. Gluconic acid (p. 50) is the only other common member.

The lowest member of the uronic series is glyoxylic acid or glyoxal, $HC(=O){\cdot}C(=O)OH$. It is a member of a number of metabolic pathways (Kornberg, 1961), which will be considered later. Glucuronic acid seems to be the only other common member (p. 50).

In a sense carbonic might be considered the lowest member also of the saccharic acids, since it is dibasic and both hydroxyl groups act as carboxylic acid groups—with a common carbonyl group. More strictly the first member is oxalic, $HO(O=)C{\cdot}C(=O)OH$, a potent poison in quantity but present in rhubarb, sorrel and elsewhere. It is poisonous for a number of reasons: like some other dicarboxylic acids it may act as a metabolic antagonist of more useful analogues; it forms insoluble calcium salts and so may excite tissues (p. 427) and prevent blood-clotting, and it is a strong reducing agent. Because of the proximity of the two carboxyl groups both tend to oxidise to carbon dioxide rather readily; this may be a good reason why it is not an important normal metabolite: in sorrel, rhubarb, etc., its biological significance may be protective.

The next member, tartronic, $HOOC{\cdot}CHOH{\cdot}COOH$, is formed in the oxidation of the C_4 member, tartaric, as well as of lactic, and of malonic acid, $HOOC{\cdot}CH_2{\cdot}COOH$. It is also formed when two molecules of glyoxylic acid condense and partially oxidise (p. 68): in fact it is the semi-aldehyde which is actually formed in this case, and it is then reduced to glyceric acid. Tartaric acid is present in unripe grapes and other fruit and is historically famous because of Pasteur's work on its optical isomers. Pasteur noticed that in a racemic mixture two mirror-image crystals separated out; he isolated these under the microscope, dissolved each separately and proved that they were the two pure isomers. This is perhaps the justification for saying that the mixture will resolve spontaneously (p. 32). The presence of this and other acids in unripe fruit may be a biological protection since the 'purpose' of fruits is defeated if they are eaten before the seeds are ripe. In the ripe fruit they are supplemented or replaced by sugar, and this again is adaptive: it does not imply that the acids are necessary precursors of the sugars—rather the contrary.

Tartaric is specially interesting as an optically active compound since its two asymmetric carbon atoms have precisely the same four groups attached; consequently in one position (mesotartaric) their rotations are equal and opposite and the solution appears inactive. This is known as 'internal compensation'. Pasteur isolated the two non-compensated forms.

5.2 Compounds of the Respiratory Pathways

These include glyoxylic, glyceric and gluconic acids but for the most part they belong to other classes, not yet considered. Some can be regarded as merely dehydrated polyhydroxy acids, so that there is no further increase in oxygen/hydrogen: if one carbon acquires two hydrogen groups and its neighbour two hydroxyl groups the latter carbon can be dehydrated to C=O. Alternatively an HCOH unit can be dehydrogenated to C=O and in this case the oxygen/hydrogen ratio *is* further increased. Some of the compounds of these pathways, for instance lactic and acetic acids, have the same empirical formula as the monoses, having one carbon as much more oxidised as another is reduced: indeed lactic is formed essentially by shifting hydrogen and oxygen within the triose molecule, in this way. Collectively, therefore, the present group range from the carbohydrate level to levels more oxidised than even the saccharic acids (Table 15), although some individual carbon atoms may be fully reduced. In the glycolytic sequence only glyceric and pyruvic acids are more oxidised than monoses and in the gluconic shunt only the three

Table 15. Ternary Compounds more Fully Oxidised than Monoses

Compound	Empirical formula	Difference from corresponding monose, $C_nH_{2n}O_n$	Total difference in H-units
Acetic acid	$C_2H_4O_2$	none	0
Glycollic acid	$C_2H_4O_3$	$+O$	2
Glyoxal	$C_2H_2O_3$	$-2H+O$	4
Glyoxylic acid	$C_2H_4O_4$	$+O_2$	4
Lactic acid	$C_3H_6O_3$	none	0
Glyceric acid	$C_3H_6O_4$	$+O$	2
Pyruvic acid	$C_3H_4O_3$	$-2H$	2
Tartronic acid	$C_3H_4O_5$	$-2H+O_2$	6
Succinic acid	$C_4H_6O_4$	$-2H$	2
Fumaric acid	$C_4H_4O_4$	$-4H$	4
Malic acid	$C_4H_6O_5$	$-2H+O$	4
Oxaloacetic acid	$C_4H_4O_5$	$-4H+O$	6
Tartaric acid	$C_4H_6O_6$	$-2H+O_2$	6
α–Keto glutaric acid	$C_5H_6O_5$	$-4H$	4
Gluconic acid	$C_6H_{12}O_7$	$+O$	2
Glucuronic acid	$C_6H_{10}O_7$	$-2H+O$	4
Saccharic acid	$C_6H_{10}O_8$	$-2H+O_2$	6
Aconitic acid	$C_6H_6O_6$	$-6H$	6
Citric acid	$C_6H_8O_7$	$-4H+O$	6
Oxalosuccinic acid	$C_6H_6O_7$	$-6H+O$	8

gluconic derivatives themselves, but in the pathways of *terminal* respiration, i.e. those which complete the oxidation, naturally most are at these levels. However, since fully oxidised carbon and hydrogen units are split off as they are formed the level of the remainder remains intermediate.

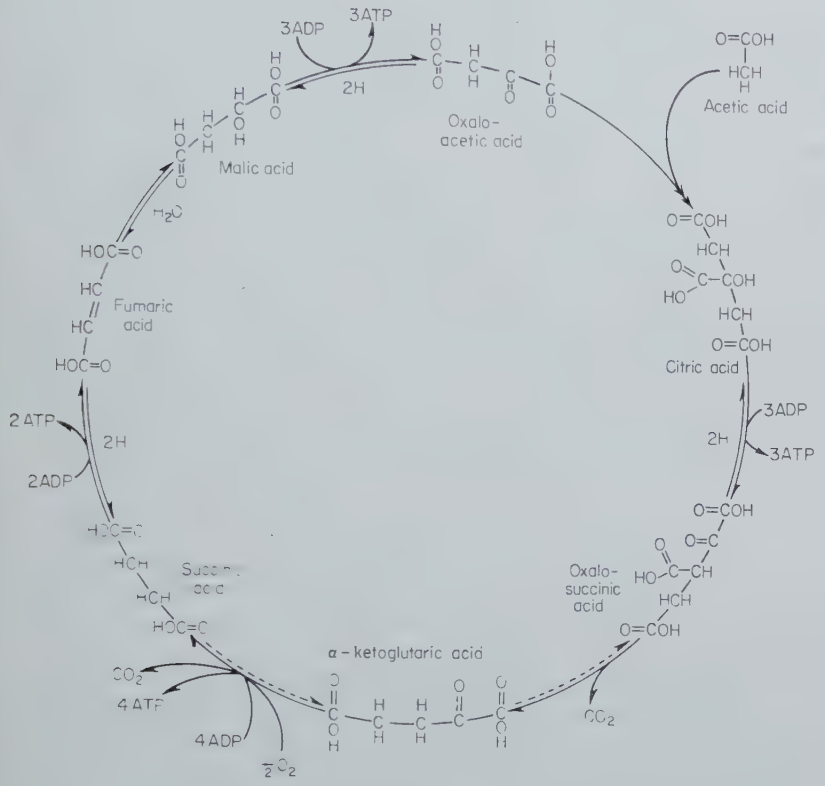


FIG. 8 The Krebs cycle or tricarboxylic acid (TCA) cycle of respiratory reactions, also largely reversible for use in biosynthesis. In each revolution of the cycle one molecule of acetate is completely broken down to carbon dioxide and water.

The most important terminal pathway is the Krebs or TCA cycle (Fig. 8); the DCA cycle (Fig. 9) is a variant of part of this, and is much more restricted in occurrence, to yeasts and bacteria. Not only carbohydrates but also fatty acids, the glycogenic and ketogenic amino acids, and in fact virtually every respiratory substrate, is fed into this cycle for final oxidation. Acetic is obviously the common intermediary for these various substrates and since the oxaloacetic (OAA) is recovered at the

completion of each revolution of the cycle this involves the complete oxidation of one acetic molecule each time, or at least (p. 67) its equivalent. The efficiency of such cyclically operating pathways needs no emphasis; certain steps are irreversible so that the cycle revolves potentially spontaneously and indefinitely so long as substrate is available. There is of course the usual wear and tear and so the OAA must be continuously 'topped up' by carboxylation of pyruvate. In any case there can be considerable permanent loss of members from the cycle because it can also take part in biosynthesis, a number of definitive metabolites being produced in offset pathways.

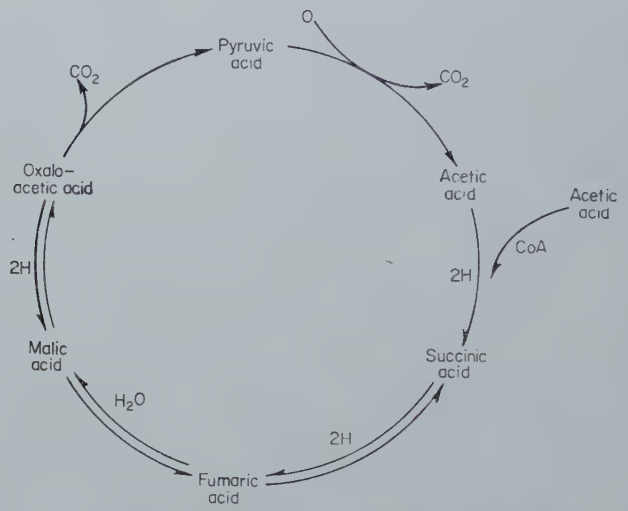


Fig. 9 Dicarboxylic acid (DCA) cycle of respiratory reactions, occurring in some bacteria and yeasts—an alternative cycle for appropriate purposes.

This is probably one reason why there are so many members and steps in the cycle, but in any case the number is scarcely excessive, since reactions typically proceed in steps of a single electronic change. The two carbon atoms are split off in two separate stages, and a preliminary softening up by intramolecular rearrangement and by partial oxidation (dehydrogenation) is necessary to permit this. Further oxidation steps bring the C_4 residue back to the OAA stage. Four oxidation steps in all, spaced round the cycle, yield 3, 4, 2, and 3 ATP molecules respectively, from ADP and inorganic phosphate (P_i), the most useful aspect of the whole sequence. The large number of steps also improves the possibilities of controlling the pathway, and to some extent the ability to isolate and recognise the intermediaries in the laboratory is a reflection of the amenability of the cycle to being checked at these points.

For our main theme the crucial question is whether these intermediaries are unique, the only ones, or at least the best, for oxidising acetic acid so as to yield energy for biological use. *A priori* it might seem that acetic could be oxidised without condensation in this way with OAA. However it will be noticed (Fig. 8) that the acetic does not become attached to the terminal carbon of OAA and that in fact it is the two terminal carbon atoms of the OAA which are oxidised in the next cycle, the acetic being oxidised piecemeal in the succeeding *two* cycles. Conjugation with another molecule is therefore essential. OAA is not the only suitable conjugant and indeed in the DCA cycle (Fig. 9) a second molecule of acetic is used. Again it is the two terminal carbon atoms of the conjugate which are oxidised away. In the DCA cycle the decarboxylation steps are the two final ones and this may be the kind of factor which determines the use of this as an alternative to the TCA cycle. In any case, within each cycle it seems that the relatively large number of intermediaries is necessary and that each is uniquely essential for that pathway. We have already seen something of the reasons why the glycolytic, gluconic and terminal pathways to some extent can behave as alternatives (p. 45). There are

Table 16. Alternative Metabolic Pathways in Micro-organisms

Substrate available	Anabolic pathway	Catabolic pathway
Acetate	1. Glyoxylic cycle 2. Tricarboxylic cycle	Tricarboxylic acid cycle
Glycollate	1. Reversed glycolytic cycle 2. Tricarboxylic cycle	Dicarboxylic acid cycle
Lactate	1. $C_1 + C_3$ condensation 2. Tricarboxylic cycle	Tricarboxylic acid cycle

some other alternative pathways (Table 16) not perhaps of very widespread use, but important because something is known about the conditions which determine their use (Kornberg, 1961).

One of these is the glyoxylic cycle of *Escherichia coli* (glyoxylic acid, $CH(OH)_2 \cdot COOH$, is the hydrate of glyoxal (p. 63)). This comes into operation when carbon compounds such as acetate alone are available as nutrients. Isocitric acid, an intermediary of the Krebs cycle not shown in Fig. 8, then breaks down to succinic and glyoxylic and the latter condenses with acetic to form malate, another member of the Krebs cycle. This therefore permits a rapid increase in the amount of material in the

cycle and it can be used for both respiration and biosyntheses. If the carbon substrate is as oxidised as glycollic acid (p. 62) it is itself oxidised to glyoxylic, which forms malate as before, but this is oxidised in the DCA path (Fig. 9), liberating acetic to re-cycle. To obtain a net synthesis, two glyoxylic residues condense and are decarboxylated via tartronic semi-aldehyde, with reduction, to glyceric acid; from this carbohydrates on the one hand, and acetic on the other, can be synthesised. From the extra acetic, synthesis in the TCA cycle and other pathways is possible. With lactate as a source of energy, condensation with a C_1 unit permits synthesis of new material for the Krebs cycle and other pathways, while the Krebs cycle itself can also continue to function as the main respiratory cycle (Table 16).

These examples help to explain why there are more alternative respiratory cycles than might have been anticipated, particularly since all substrates can funnel through acetic if necessary. The alternatives are necessary in the first place because of vagaries in the supply of nutrients, which can be very specialised even in the wild, but more significantly, perhaps, because each pathway must also be used for certain syntheses and cannot do both functions at the same time. This is certainly true when the synthesis requires endergonic steps in the main pathway, but not all syntheses need to be of this type in fact. The TCA cycle in principle could produce glutamic and aspartic acids and alanine (p. 162) when revolving in the respiratory direction, since all lie on offset-pathways. However, whichever way it revolves the products of synthesis drain off the vital respiratory intermediaries: since, as already seen, they are not exhausted in the operation of the respiratory cycle any switch to synthesis is like suddenly taking catalysts to become substrates. During such time, therefore, some other pathway must be found to supply respiratory energy. This may be a main reason why embryos and some other systems respire glycolytically even when oxygen is abundant.

When the DCA cycle is in operation half of the TCA cycle could be used in syntheses; carbon dioxide could be fixed in α–ketoglutaric and citric thus synthesised as an end in itself. The step from α–ketoglutaric to succinic is irreversible so that there is minimal danger of any flow from DCA cycle into this path. Fixation of carbon dioxide in this way is known to occur. Large concentrations of citric acid are built up in citrus fruits and other plant products, in semen and elsewhere, so that specific syntheses of this kind are essential. The large number of these metabolites required for specific purposes is adequate justification for the generous provision of alternative respiratory pathways to permit synthesis in the others.

Some further idea of the variety is given by the many products of

glycolysis in animals collectively (Table 17); many of these animals are parasites, or specialised in other ways, and probably illustrate special restrictions in diet and oxygen supply and in disposal of waste rather than problems of synthesis. Moreover many of the products belong to groups

TABLE 17. RARER PRODUCTS OF GLYCOLYSIS IN ANIMALS

Product	Produced by
Acetic acid	*Fasciola*, *Litomosoides* (Nematoda), *Trichomonas*, some yeasts
Glycerol	Trypanosomes
Pyruvic acid	Trypanosomes
Propionic acid	*Fasciola*, *Heterakis*
Succinic acid	*Hymenolepis*, Nematodes, *Trichomonas*, Trypanosomes, *Tetrahymena*, other Protozoa
DL–2–methyl butyric acid	*Ascaris*
cis–2–methyl crotonic acid	*Ascaris*
n–valeric acid	*Ascaris*

of materials not yet considered. However, they do give some idea of the potential variety in metabolic pathways as a background against which the orthodox glycolytic and Krebs pathways stand out as unique in importance.

Because of the variety of minor respiratory pathways, and of pathways of synthesis intermeshing with them, some of the members of the present group have great biological importance and appear as 'nodal' molecules on which a number of pathways converge, or from which they diverge. Pyruvic acid is the most outstanding (Fig. 10). Fumaric acid is one of the

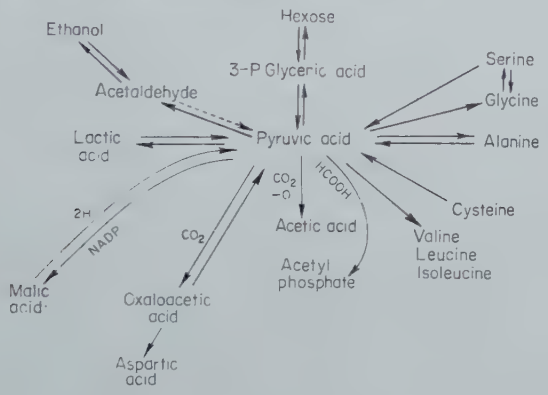


FIG. 10 The pyruvic acid 'node' of metabolism.

products of oxidation of phenolic compounds (West and Todd, 1957, p. 1095) and of the ornithine cycle for urea production (Fig. 26), in addition to its role in the TCA and DCA cycles. Succinic acid is interesting because it is a dicarboxylic fatty acid and behaves very much like the fatty acids. It is a reminder that the lower members of such series are much more fully oxidised and polar in properties than the higher members, and in fact succinic is highly water-soluble. It is also a strong reducing agent and so an ideal respiratory substrate; it has two fully hydrogenated carbons, although on balance it is more oxidised than tetrose.

The occurrence of many of these acids in fruits is very interesting. They are attractive to animals and probably assist their metabolism. It seems probable that they have been evolved to attract seed-dispersing animals and not for any value to the seeds. The seeds of many pass through the animal gut without harm and so are able to benefit from the fruit being eaten.

CHAPTER 6

TERNARY COMPOUNDS OF INTERMEDIATE GRADES OF REDUCTION

Most of the remaining groups of biological materials are more reduced than carbohydrates and contain correspondingly more energy per unit weight. This is in fact a relevant factor and the fatty acids (p. 87), the most important form of storage-energy, are as fully reduced as subsequent mobilisation-needs allow. Moreover the number of biological groups of compounds with intermediate grades of reduction is rather limited: if reduction is an advantage there is every reason for 'going the whole hog'. However most of the remaining biological groups are active in other capacities than simply as a source of energy and they are highly reduced because this is a stable stage whereas many of the intermediaries are unstable, as they are in so many pathways. Therefore, biological austerity is partly due to innate properties of the materials.

Reduction depresses the solubility in water but most of the important metabolites still to be considered contain nitrogen or other polar elements in the molecule; they are therefore mostly water-soluble although extensively reduced. It is difficult to visualise a range of biological agents combining the contrasting virtues of stability and activity in aqueous media without the co-operation of these other elements.

Carbohydrates are abundant not only because they represent a stable stage but also because they can be used for anaerobic respiration (p. 43). They are also very easily mobilised chemically and physically. Where bulk is not a great disadvantage, and where dry storage is feasible, carbohydrates remain the main stores, for instance in plants, which have no locomotor problems. A further reason for this is that reduction is not 100 per cent efficient so that it is most economical to store carbohydrates. External, solar energy is trapped only in the one initial process of carbon dioxide fixation: all subsequent reductions must be powered by chemical energy so fixed, and there is every reason for economy.

A digression is perhaps justified here. It is sometimes maintained that some living organisms have other external sources of energy, since they

build their fabric and reserves by the oxidation of various simple molecules such as ferrous iron, hydrogen sulphide, sulphite, thiosulphate, and even hydrogen, not to mention the organic materials methane and other hydrocarbons. However, with the trivial exception of some volcanic materials all oxidisable materials exposed on the surface of the Earth today are of relatively recent biological origin (Smith, 1953). Otherwise they would have oxidised long ago. In a sense, therefore, these 'chemo-autotrophes' are as heterotrophic as ourselves. All true autotrophes are photoautotrophes, although they do not all use water as the photoreductant and do not always release oxygen therefore (Calvin, 1962).

6.1 Polyhydroxy Alcohols

This is virtually the only group of ternary compounds in the intermediate grades of reduction which has much biological importance and they are very near the carbohydrate stable state itself. The aldehyde, $HC(=O)$ group has been reduced one step to carbinol, $HCOH$, and there is evidence that they are actually synthesised from monoses in this way. They may be called *monitols*, therefore, and their relative abundances resemble those of the monoses; there is one C_3 member, glycerol, one pentitol, and three hexitols: $H_2COH(CHOH)_4HOCH_2$, widely distributed, as well as a C_4 member, erythritol, in lichens and Algae.

Glycerol is an indispensable constituent of the glycerides of all organisms, triesters of glycerol with fatty acids. Erythritol also will form esters with fatty acids (Bateman, 1945), and there is the interesting problem why in practice only glycerol has been exploited. All of the hydroxyl groups can be esterified in the C_6 cyclic member, inositol. Ribitol has only one known biological function, as analogue of ribose in one group of nucleotides, the flavin group (p. 356); this proves important enough, because of the key position of the flavo-protein enzymes. The three hexitols are not so widely distributed and have no generally important function. All occur in the sap of trees and to some extent in fungi, and they may be simply suitable stable derivatives of the hexoses for transport purposes. They are the basis of the biblical manna (Heb. *Man hu*: what is this?), and are very sweet in taste to man; curiously enough blowflies, which are very fond of sugar, are repelled by all three: dulcitol, mannitol and sorbitol (Hodgson, 1961).

Although the monitols are formed by reduction of the already strongly reducing aldehydes they are themselves non-reducing and this may be the

most important respect in which they are more stable than the monoses. Also they are not fermented by yeast and this may have been subject to natural selection. It is possible that they are formed from monose by a Cannizzaro reaction, the mononic acid being formed in stoichiometric amount.

$$2R{-}HC{=}O \xrightarrow{H_2O} R{-}H_2COH + R{\cdot}HOC{=}O$$

Having lost the aldehyde group, even the hexitols do not close to a ring structure, and this difference from the monoses also may have biological significance.

6.2 Cyclitols

Ring-closed hexitols do occur, however, the inositols, and one of these, *meso*-inositol, myoinositol or *i*-inositol, is a vitamin for many animals (Fig. 11, I). It is probably formed from hexose by direct ring closure between C_1 and C_6, by aldol condensation—an addition reaction and not a dehydration as in the formation of pyran and furan rings (p. 42). The empirical formula remains unchanged, therefore, but the reducing property is lost and the inositols are even more stable than the open chain hexitols. They retain the sweet taste.

Since the carbon atoms can all be regarded as having identical groups on two of their bonds—the remainder of the ring in fact—they are optically inactive. However, one of the eight possible isomers does exist in two optically active enantiomorphic forms. This is because the molecule *as a whole* can be divided into two non-identical halves wherever the

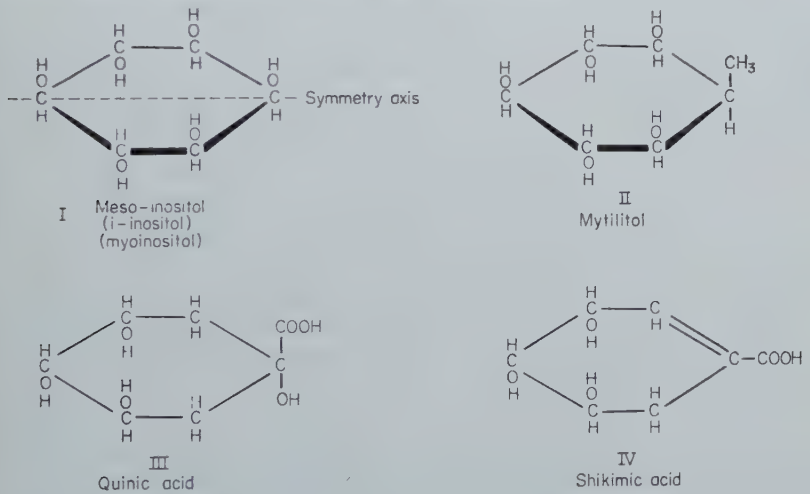


Fig. 11 Monitols and related compounds of biological significance.

line of bisection: all the other isomers have a plane of symmetry. This is a new manifestation of optical activity, therefore. The various isomers of inositol also illustrate a type of geometric stereoisomerism which has not yet been mentioned specifically but is very common. It is a type which results whenever there are rings in a molecule. If two particular substituents on a ring, A and B, are on the same side of the ring the properties are significantly different from those of the molecule having them on opposite sides, and the two forms therefore are known as *cis* and *trans* isomers respectively. In some cases the ring structure is different as viewed from its two faces and then the number of geometric isomers is further doubled, but the inositol ring is identical from either aspect and there are only eight isomers, of which one has two optical forms, as already noted. The furone and pyrone rings are not identical from the two sides.

Meso-inositol is one of the inactive forms and so is scyllitol, from elasmobranch fishes and plants. Scyllitol has the distinction of being the all-*trans* isomer, i.e. the hydroxyl groups lie alternately on either side of the ring with perfect regularity. The optically active form (but in the racemic state!) occurs in blackberries and mistletoe and in the outer layers of some seeds; it cannot be compared in biological importance with *meso*-inositol. It is not yet clear in what way this form is unique since its biological properties seem to be those of any inositol. It forms esters with acids, particularly with phosphoric, and in plants it commonly occurs as the hexaphosphate, *phytin*, possibly a store of phosphorus. It esterifies on every carbon, therefore, like the open chain monitols. In phytin the phosphate groups further form salts with magnesium and calcium, and no doubt increase the solubility of these cations (p. 410). Phospho-inositol forms compounds with diglycerides, analogous to the phosphatides (p. 295), so that inositol is analogous to the nitrogen base of the latter. Like the phosphatide, lecithin, inositol is 'lipotropic', that is it mobilises fats for metabolism, but while lecithin mobilises glycerides in particular, inositol affects mainly lipids of the steroid type (p. 118). The two do not seriously overlap in function, therefore.

This hexahydroxycyclohexane also has neurological significance and its analogue, α–hexachlorocyclohexane, or gammexane, is a powerful insecticide. Inositol protects against its anti-metabolic action.

Cyclic pentitols are not known from living organisms and other rings would be unstable (p. 42).

6.3 Cyclitol Derivatives

Mytilitol, from the muscle of the mussel, *Mytilus*, is a methyl inositol (Fig. 11, II), and has the appearance of being a derivative of scyllitol.

Quinic acid (Fig. 11, III), from cinchona bark and other plants, also the source of quinine (p. 339), has the appearance of a mytilitol in which the methyl side chain has been oxidised to carboxyl. Its state of oxidation therefore is the same as that of the inositols since there has been reduction at two carbons. In this connection it may be noted that the cyclitols, unlike the open chain monitols, are only in the same state of reduction as the carbohydrates, and might have been considered with them. Other cyclitol derivatives of biological importance in general are more fully oxidised, and provide further examples of retrogressive synthesis (p. 50), since there is good reason to believe that they are derived *in vivo* from cyclitols, and ultimately from hexose. Shikimic acid (Fig. 11, IV) occurs naturally in star anise and may be a universal intermediary in the synthesis of aromatic compounds (p. 187). It is a dehydrated quinic acid, still at the same level of oxidation, therefore, but other derivatives have more double bonds in the ring, i.e. are more aromatic, through dehydrogenation. All are therefore characterised by the large numbers of hydroxyl groups persisting; examples are gallic acid (p. 128), phenolic enough to act as a tanning agent, tetrahydroxybenzoquinone, and other pigments of fungi. Further derivatives will be considered in appropriate contexts.

In all, the number of intermediaries in the pathway of *reduction* of hexoses is seen to be small. Glycerol and ribitol would seem to be the only strict members of any great biological importance.

CHAPTER 7

FATTY ACIDS AND RELATED LIPIDS

As already indicated (p. 71), once the reduction of carbohydrate is begun there is no stable stage short of full reduction. This is perhaps best appreciated by starting from the other extreme, that is from the paraffins; their stability and inertness disappear once a single carbon atom is partially oxidised (Bayliss, 1959, p. 338). In point of fact nearly all biological lipids have this minimal degree of oxidation, and their molecules are a compromise between the contrasting virtues of stability and chemical reactivity. Others are reactive because they are partially *unsaturated*, that is they have C=C double bonds in the molecule, and these also are more oxidised than the paraffins, in the sense that they have less hydrogen. The presence of oxygen also confers some degree of polarity and water-solubility on the molecule as contrasted with the hydrocarbons, which are so insoluble that they form a kind of negative hydrate (p. 25). At the same time biological lipids are sufficiently insoluble to form a separate phase in the living cell and this has great biological value. This is of course the necessary qualification for classification as 'lipid', the ability to build a fluid phase immiscible with water. This includes not only substances which are liquid between 0° and 100°C, but solids and gases which are soluble in these: a large group of water-insoluble carbon compounds are in fact freely intermiscible and there are therefore only two fluid media in living organisms, aqueous and comprehensive lipid. There are of course biological materials, such as the melanins and some of the proteins, which are very sparingly soluble in either, under physiological conditions, but most metabolites are physically, and often chemically, mobile in one or both of these two media.

Biological lipids are almost entirely binary (C and H) and ternary (C, H, and O) compounds; virtually all elements except hydrogen and the halogens tend to make carbon compounds water-soluble—particularly when the molecule is small, so that the ratio of hydrogen to the other element is low. Oxygen appears to be outstanding in its ability to occupy a single site, or very few sites on a large, otherwise hydrocarbon, molecule. The fatty acids, $CH_3{\cdot}(CH_2)_n{\cdot}COOH$, which are the essential

components of the fats and oils stored by animals and plants, are the best example, and most of what follows is particularly relevant to them, and their near relatives: other biological lipids will be considered in later chapters, but they also show many of the following general properties.

Because they form a separate medium the physical properties of lipids, just as in the case of water (p. 10), are often as important as their chemical functions. For instance many organisms make use of the fact that carbon is lighter than oxygen, so that most lipids are lighter than water. Oil drops give buoyancy to some Protozoa, siphonophores, embryos of Crustacea and fishes, and other plankton, and lipids contribute to the buoyancy of many seeds, of whales, etc. In adult marine fish the fat content of the body is fairly closely related to the height in the water at which they normally swim.

Lipid also forms a good cushion and shock-absorber, and the viscera of mammals are usually well surrounded by fat; the subcutaneous layer has another function but even so this may be an important auxiliary one. In all cases the lipid is also, if not primarily, a reserve fuel (p. 87), so that the cushioning is a gratuitous asset; consequently it might be biologically desirable to use it in this way even if it were a poorer agent than water. In fact, however, it has advantages for some purposes: the high viscosity of lipids reduces their rate of flow under shearing forces, and there will be less displacement of the organs of the body during locomotion, etc., than if water were the packing fluid. It is of course essential that the lipid should be fluid at body temperature, and this is found to be true. Suet is hard at room temperature, and a useless cushion, but deep in the body of the sheep it is just fluid, and its viscosity is therefore maximal for the fluid state. Poikilotherms have fats of lower melting point, mainly due to a larger percentage of unsaturated fatty acids, oleic in particular. Since their body temperature is very near that of the environment the amount of unsaturated fat increases with latitude, within the same species. It is therefore unfortunate that plants, on which they feed, tend to store oil most abundantly in low latitudes, and sugars in colder latitudes (p. 48); however, animals very readily convert carbohydrate to lipid. It appears that the melting point of the body fat of homoiotherms varies also with distance from the surface, those of the subcutaneous layer having the lowest melting point: everywhere they are just fluid, and their viscosity maximal. In man the temperature gradient is very considerable, the skin being as cool as 24°C, compared with the maximum of 37°C.

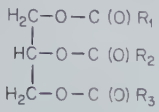
The main significance of the subcutaneous lipid of homoiotherms is in heat-insulation, another physical function. It is perhaps not easy to persuade the adipose person that this is true since his cutaneous thermo-

receptors are as sensitive as those of the leaner kine, and it would be even more difficult to persuade him that the insulation works also in the opposite direction, at times when metabolic heat is causing more discomfort than the Sun without, but it is significant that the subcutaneous lipid layer is thickest in mammals such as man and the pig, which have little hair for insulation. The extreme examples are the Cetacea, with a layer of blubber up to fifteen inches thick, which enables them to thrive in polar waters. This thick layer contrasts with the relatively thin pelage of terrestrial polar mammals, subjected to temperatures much below 0°C (Irving, 1951), and no doubt lipid is less efficient than a stationary layer of air, trapped in the hair. The latter is impracticable under water, and it is possible that man and the pig reduced their pelage because of aquatic habits (Hardy, 1960). There is, however, the alternative possibility that the reduction has the advantage of facilitating heat-loss in tropical latitudes. This appears to be the reason for the loss of hair by the elephant (contrast the arctic mammoth); elephants have their reserve fuel-fat deep among the viscera where it least impedes heat-loss. In passing it may be noted that the blubber-layer of the Cetacea also improves the streamlining of the body and also that hair does not necessarily increase water-resistance; that of the seal probably has the positive virtue of reducing turbulence. As a heat insulator, again, lipid is more effective as a fluid than as a solid; conduction is the critical factor since in any case the lipid is intracellular, and convection therefore very limited.

Lipids are even more valuable as electrical insulators; most of the molecule is hydrocarbon, non-ionising and bearing only weak electrostatic charges, except under special conditions. Most electrical activity in living organisms depends on ions rather than on electrons and the insulating properties of such lipid structures as the myelin sheath of nerve fibres are very great. They have a high resistance, and lipid therefore also acts as a good dielectric medium for condensers of high potential and low capacity; this type is believed to play its part in the generation of the sudden large changes in electrical potential which accompany the propagation of a nerve impulse. For such condensers a low dielectric constant is required (Table 12), and that of paraffin is only twice that of a vacuum, as compared with eighty times for water.

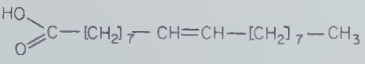
Insulation against ionic movement is part of the general incompatibility of lipids and water. The outer surfaces of leaves and of the bodies of many animals are waxed to prevent waterlogging or water-loss, or both, i.e. to direct water-transport through the correct channels. That of insects is directed mainly against loss but that of aquatic Crustacea presumably against ingress. There is a lipid component of the wall of all cells and this controls not only water-transport but also that of solutes.

In this way lipid controls the whole rate of metabolic and physiological activity. The control of transport acts differentially on different solutes; to some extent, at least, this is due to the lipid components of the membrane, so that these partly control the whole pattern of activity in the body.



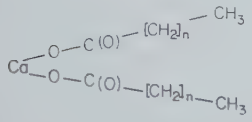
I

Triglyceride
(generic formula)



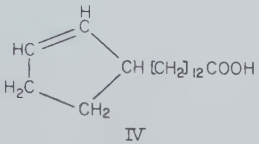
II

Oleic acid



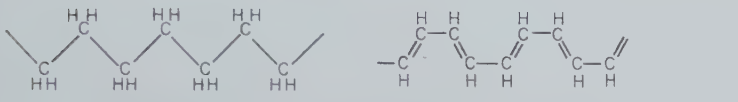
III

Wedge theory of structure
of calcium soap



IV

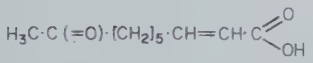
Chaulmoogric acid



V

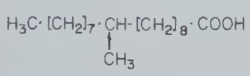
Saturated chain

Unsaturated chain (all-*trans* configuration)



VI

9-ketodec-2-enoic acid
(Queen-substance of bees)



VII

10-methylstearic acid
(tuberculostearic acid)

FIG. 12 Fatty acids and related Lipids (*continued on p. 80*).

The control is not merely that of lipid barriers at strategic points of an aqueous system. It is becoming increasingly clear (Green, 1962) that a good deal of metabolism occurs in lipid medium, in particular the terminal steps of aerobic respiration, involving the cytochrome system (Fig. 20), so that the control is partly promoter. Both media are essential for complete metabolism, and transport between them is as important as that from one aqueous phase to another through a mere membrane of lipid. It so happens, however, that the metabolically active lipid media are mere membranes in thickness, and so they have a maximal area of contact with interleaved aqueous media; this is well seen in the mitochondria,

which contain the enzymes of the terminal steps of respiration, and in plant chloroplasts which are carrying out essentially the reverse steps—of reducing carbon dioxide and oxidising water (Fig. 3). Iron and magnesium porphyrans (p. 315), the prosthetic groups of the cytochrome

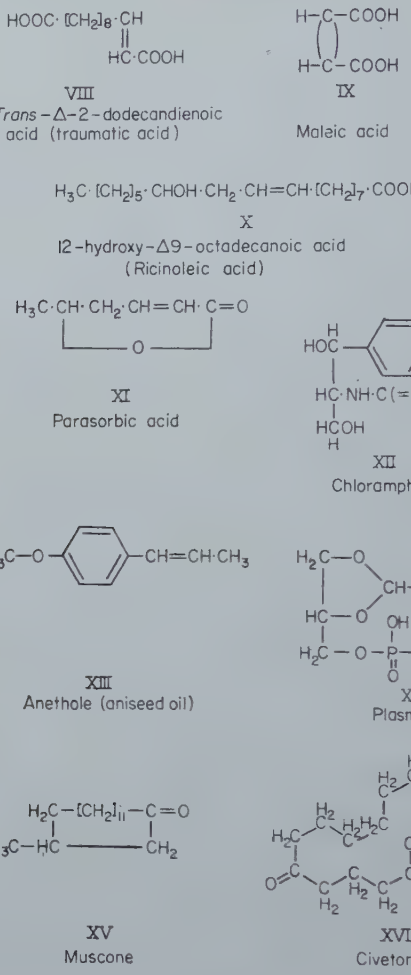


FIG. 12 (*continued*).

enzymes and of chlorophyll, are, like most metallo-organic chelate complexes, quite soluble in lipids (Albert, 1950), and this is true also of a number of the heterocyclic nitrogen bases of other coenzymes. Of the two chlorophylls normally present together in chloroplasts, one is in the lipid and the other in the aqueous phase (Glass, 1961). Similarly the

dehydrogenases which catalyse the proximate steps of respiration are in the aqueous phase so that traffic between the two phases must be continuous, not merely occasional.

One of the outstanding features of narcotics is that their potency is related to their lipid-solubility and it was thought that they acted by dissolving in, and affecting the permeability of, the cell membranes. It also appeared, however, that they affected respiration and the two effects can now be seen as one. At the same time it must be admitted that other explanations of narcosis are still envisaged; Klotz (1962), for instance, suggests that they may act by forming the 'negative' hydrates already mentioned in connection with the hydrocarbons. These effectively inactivate or 'freeze' water, and so narcotics may act essentially on this phase.

There are further possibilities of biologically significant activity in lipid medium; for instance it has been suggested that in lipid media natural magnetic fields and circularly polarised light might be adequate to determine the synthesis of optically active materials, i.e. the preferential synthesis of the one isomer of asymmetrical molecules (p. 31). Szent-Gyorgyi (1960, p. 95) suggests that the free energy of hydrolysis of ATP should be greater in lipid than in aqueous media, in much the same way as an insulating medium, such as air, improves the production of an electric spark. It might also be combined with the condenser action of lipids.

Since most of the metabolically active lipid seems to be in the form of thin, paucimolecular layers, activity at lipid–aqueous interfaces is probably more relevant than that in the bulk phase of lipid droplets, or even of the smaller emulsified droplets which occur in many cells and body-fluids. Much work has been done on the properties of these interfacial membranes, and of simple laboratory models, mainly with an interest in transport across membranes. It is not possible to deal with this in detail here (Davson and Danielli, 1952; Harris, 1960), but one or two observations bearing on chemical activity at such interfaces are significant. If oleic acid (Fig. 12, II), a fatty acid with one double bond in the middle of the molecule, is deposited as a monomolecular layer, in contact with water containing an oxidising agent, then on compressing the layer in its own plane some of the molecules are forced down into the water, exposing the double bond to the oxidising agent. A reaction begins, but ceases immediately the pressure is removed and the molecules return in plane with the rest. Protection and exposure in this way might well play an important part in biological processes. Haemin (p. 320) will catalyse the oxidation of linoleic acid (Table 18) by free oxygen so long as it is at the interface of an emulsion in water, but this ceases if the emulsion is converted into a homogeneous fluid (Haurowitz and Schwerin, 1940). Enzymes

in membranes sometimes become active simply on stretching the membrane (Bradfield, 1950). Fatty acids are oxidised as fuel mainly by β–oxidation, which involves nibbling C_2 (acetic) units progressively from one end, the polar —COOH end, which is the one presented to the aqueous phase at any interface. The enzymes are in the aqueous phase in living cells, where they are in a position to nibble and no more. As the molecule is shortened, and the size ratio of hydrophobe to hydrophil portions decreases, it eventually passes into aqueous solution. These are just a few of the many types of activity possible at interfaces.

Interfaces may be defined as organised molecular layers at the junction between two media. They occur at air–water junctions as well as at lipid–aqueous boundaries and much work has been based on layers and membranes deposited here; with due precautions the results may be extrapolated to aqueous–lipid interfaces and membranes. There is always an asymmetry across any boundary and this is usually adequate to polarise molecules near the boundary. Lipids such as paraffins and fatty acids, with long chain molecules, form a monomolecular layer in which all the molecules are parallel and orientated normal to the surface. This arrangement is the foundation of the lipid component of all biological membranes. If a very small amount of some lipids is spread, by suitable techniques, on a large water surface, the individual molecules are separate, irregularly orientated and effectively 'gaseous', and if compressed in the plane of the surface (by drawing tight a loop of thread floated on the surface) they pass through a 'liquid' state to the densely packed palisade, which has the rigid crystallinity of a solid. The lipids in biological membranes are in this state, although in the bulk they are fluid at that temperature, and it is important to note that the thermal death point of animals proves to be precisely the temperature at which their membrane-lipids 'melt', or disorganise.

The experiment of compression of a thin film shows that the proximity of other lipid molecules is an important condition for forming crystalline membranes and that the asymmetry of forces across the boundary is not alone sufficient. The chains are held in orientation by the London–van der Waals attractions of neighbouring molecules and this is effective only within a certain range of chain-lengths, namely C_{16} to C_{36} or so; this is precisely the range of the common biological fatty acids (Table 18, p. 89). Beyond this length the molecules topple and tangle, while shorter chains have insufficient attraction. If the lipid is also polar, like the fatty acids, that is if it has a hydrophil group in the molecule, then members shorter than C_{16} also tend to pass into aqueous solution too readily to form stable membranes. The commonest fatty acids are C_{16} and C_{18}, near the lower limit of stability, and there are at least two teleo-

nomic reasons for this: first that rapid permeation demands a minimal thickness and secondly that lipids protect proteins, with which they are associated, in the biological membranes, most effectively at a length of C_{12} —and even at $C_{7\text{-}8}$ if the protein is concentrated (Putnam, 1953a).

It seems clear that a good crystalline monolayer will form in every interface between water and discrete drops of lipid, but most of the active lipid is sheer membrane (p. 79) and its formation in the mitochondria and chloroplasts, and elsewhere within the cell, is less easily explained. It is known that any lipid in aqueous solution tends to accumulate at the surface, since by so doing it reduces surface tension (Höber *et al.*, 1945), and this might help to explain the origin of lipid in the cell-membrane, though not that of intracellular membranes. Lipids also migrate spontaneously to regions of strongest ionic gradient, and form a monolayer across this (Bernal, 1954), and this may be relevant to the formation of intracellular membranes though it is not easy to see how the dense system of intracytoplasmic membranes in some cells could be formed simply by this mechanism. A further, major difficulty is that all the membranes, around and within the cell, are not monolayers but bimolecular lipid layers, together with protein and sometimes other components. The hydrophobe ends of the lipid molecules are all orientated towards the other lipid layer and the hydrophil ends everywhere face outwards into the aqueous phase. Gorter and Grendel (1925) first demonstrated the bimolecular lipid layer, by chemical methods, in the wall of the erythrocyte, and this has been confirmed more generally by electronmicroscopy. A *unit membrane* of universal occurrence (Fig. 13) is now recognised, consisting mainly of the double lipid layer, with a layer of protein on each surface.

The lipid component could be regarded as a drop of lipid, flattened until it is nothing but an upper and a lower interface, but mechanically this would be the least probable and least stable form, and could occur spontaneously only with the help of spreading agents which lower the surface tension of the lipid itself, and of stabilising agents improving the rigidity once the bimolecular layer is formed. Inorganic ions such as sodium and potassium act as spreaders for the fatty acids, and the resulting soaps spontaneously form bimolecular and thicker layers, with hydrophobe and hydrophil surfaces always facing their like (Smith, 1951). Most organic molecules tend to decrease the surface tension of water (Cohn and Edsall, 1943, p. 174), and reciprocally most hydrophil materials will depress that of lipids. This may be the significance of some of the other components of biological membranes, including the phosphatides (p. 294), which are particularly hydrophil lipids and are abundant in all biological membranes. With an optimal amount of hydrophil material

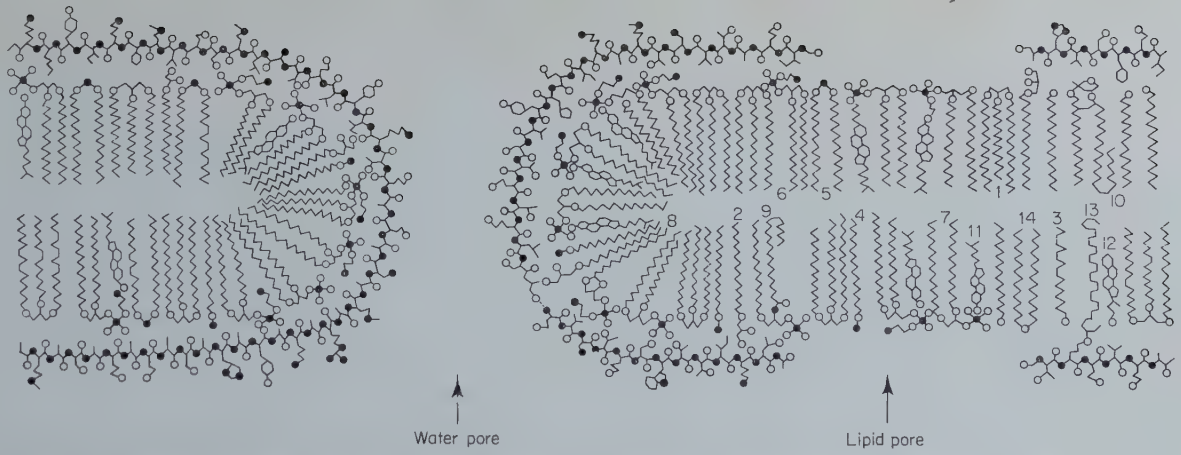


FIG. 13 Schematic section through a small portion of a biological unit-membrane, consisting of two monomolecular layers of lipid, with the hydrophobe ends of all molecules directed in towards the other layer and the hydrophil ends out towards a stabilising layer of protein, also of monomolecular thickness, covering each face of the lipid bilayer. The hydrophobe side-chains of the protein molecules are directed towards the lipid, and most of the hydrophil side-chains out towards the aqueous medium. The two protein monolayers are in continuity through 'water-pores', facilitating the transport of water-soluble materials across the membrane. In surface view the protein molecules form a meshwork, exposing naked lipid patches ('lipid pores') in places, facilitating the permeation of lipid materials. At these places there must be a high proportion of self-stable lipids, such as the phosphatides. These are probably also abundant round the water pores where molecules with a broader hydrophil than hydrophobe end are required. A number of types of biological lipid which could, and probably do, contribute to the structure of membranes (not necessarily all in any one membrane) are represented, viz. 1, triglycerides; 2, monoglycerides; 3, free fatty acids; 4, Na-soaps; 5, Ca-soaps; 6, phosphatides; 7, phosphatidic acids; 8, plasmalogen; 9, sphingomyelin; 10, cerebrosides (cerasine); 11, cholesterol; 12, bile acids; 13, carotinoids; 14, aliphatic esters. Several possible alternative configurations of some are shown. Free ends of neighbouring protein molecules are held by hydrogen-bonds. A carotoprotein bond is indicated and hydroxy amino acids are shown associated with the phosphatide phosphorus, but other possible lipoprotein bonds are not specified. For clarity the =O groups of the peptide bond units are represented perpendicular to the plane of the membrane but they are actually in the plane, hydrogen-bonded to neighbouring NH groups. Clear circles represent polar groups containing oxygen; unmarked positions are C atoms, while atoms of all other elements except hydrogen are black discs. For clarity hydrogen is not represented.

in the lipid, and of organic solutes in the aqueous, phase there seems no reason why greatly extended interfacial membranes should not form spontaneously and be quite stable.

The protein layers have a powerful stabilising action on the formed lipid bi-layer, the hydrophobe side chains of constituent amino acids (p. 16) being directed into the lipid layers and the hydrophil side chains into the water. It is known that the protein main chains lie in the plane of the membrane but little is yet known of the way they are arranged within that plane. They may form a meshwork which leaves some bare patches of lipid, and elsewhere they may line 'water-pores' through the membrane so that both aqueous and lipid solutes have their own preferential pathways through barrier membranes (Fig. 13). Another important stabilising agent is the calcium ion, which is relatively highly concentrated in the cell membrane (p. 422), and may play an important part in the formation of 'surface precipitation membranes' which form spontaneously over wounds in the cell membrane (Heilbrunn, 1952).

The strength of lipid layers is increased also by other types of lipid (Ries, 1961), and this may be another reason for the heterogeneity found in the lipid component of biological membranes. It is well known that even a simple mixture of paraffin wax and stearic acid (Table 18) is less brittle than either alone (Read, 1935, p. 265). Great variation in length of molecule, however, makes for instability and it is significant that the more important lipid components of biological membranes, fatty acids, phosphatides, phosphatidic acids and sterols (Booij and de Jong, 1956), have much the same effective length.

A further important property of lipids which is relevant to the formation and function of bi-layers is that of lubrication and detergency. Many solid carbon compounds, such as paraffin and other waxes, fats, etc., as well as elemental graphite, are good lubricants. The essential property is the tendency to form layers which shear readily on each other, because they are held only by the weak van der Waals forces, while adhering more firmly to the two objects between which movement is desired. The typical bi-layer is ideal because the hydrophil ends of the molecules readily form the firm bonds required, while the hydrocarbon chains of the two layers slide on each other like the pile of a cat's pelage. At the same time it must be admitted that the apolar paraffins are better lubricants than might be anticipated: possibly they are anchored to the bearing surfaces by a cement of their negative hydrates (p. 76). It may be for the same reason that they form molecular layers in the first instance so much better than expected, and develop electrical potentials as great as 200 mV across a multilayer construction (Beament, 1961). Emulsions, because of their large surface, and of the spherical form of the droplets, make very good

lubricants, with rolling added to shearing motion, but most lipids are good, since they spontaneously form their orientated monolayers on any exposed surface.

Detergency is a lubrication between soil and the soiled surface, and the classical detergents, the soaps, are good lubricants—as most of us have proved on occasion! Once both surfaces are coated with the detergent water flows easily between them and rinses off the soil particles. Lipid contaminants are emulsified and the droplets then act as ball-bearings for the solid contaminants. The extent to which lubrication and detergency by lipids is significant *in vivo* is not very clear. Soaps possibly, are formed in the more alkaline region of the intestine, but this organ actually uses the mucins (p. 56) as a lubricant. Here the sliding is between hydrophil surfaces and it will be noticed that the same must be true for the soaps and other water-soluble detergents and lubricants. The hydrophobe surfaces must adhere to the bearing surfaces and water flows between hydrophil surfaces.

The great advantage of synthetic detergents such as the alkyl 'sulphonates', in which sulphuric acid replaces the carboxylic acid of the soaps, is that the alkali salts of this strong acid are stable in acid media, where the detergent action of the soaps is destroyed. The alkyl sulphonates confirm previous findings about the optimal length of molecule for monolayer formation, the shorter and longer members of the series being poor detergents.

As already noted, there is an asymmetry potential across a monolayer of paraffin, and across one of a polar lipid, such as the fatty acids, this can be augmented by the intrinsic electrical polarisation of the molecule. The polar group shows the negative charge of the oxygen ion while the complementary positive charge on the hydrogen ion tends to be distributed, by *transmission effect*, throughout the rest of the molecule. The hydrophobe end therefore is positive to the hydrophil end. Such potentials across monolayers can be used for continuous transport or other work across them, so long as the transported ions are not allowed to 'neutralise' the membrane potential (Beament, 1961), and they probably are so used in the insect exoskeleton, where lipid multilayers are built up, all with the same polarity. This cannot be the basis of the 'resting' potential across cell-walls and other structures with the unit-membrane construction, however, since the potentials of the two face-to-face lipid layers of each unit will be equal and opposite. There is little charge across the myelin sheath of nerves, which is built of many turns of a double fold of unit membranes.

The lipid of the active membranes of living organisms constitutes the classical *constant element* of Terroine, which is conserved during starvation —an important piece of teleonomy. There is still the main function of the

variable element to consider, that of reserve fuel. Apart from the compactness of fat, as a single large droplet in each fat cell, and the auxiliary functions of the gross layers, already considered, the great virtue of fat as a storage fuel is its high calorific value, 9·3 kcal/g compared with 4·2 kcal/g of sugar. This is due simply to the replacement of the heavy, non-combustible oxygen by hydrogen. For the same reason fat yields more 'metabolic' water per unit weight, a factor of importance where water is as precious as it is to insects. The stability and inertness of a lipid are other great virtues.

This storage fat and oil is almost entirely triglyceride, that is to say glycerol (p. 72) esterified with three fatty acid molecules per molecule. The resulting trident (Fig. 12, I) is a larger and more viscous molecule than the simple linear fatty acid. It is also a much better form for storage, being neutral in reaction ('neutral fats'), and less reactive, less polar and less water-soluble than either free acid or free alcohol. The change due to esterification is seen more dramatically in the case of ethyl acetate which is soluble only in seventeen volumes of water, whereas ethanol and acetic acid themselves are both freely miscible with water in all proportions. It is probable that fatty acids can build lipid layers with the glycerol 'clip' still binding them in trios, and this may be the reason (p. 72) why no higher polyhydric alcohol is used for the esterification. Phosphatides (p. 294) and inositides are formed by one fatty acid unit of the triglyceride being replaced respectively by a phosphoryl–nitrogen base unit and by an inositol derivative. They play a part in mobilising glycerides and other lipids, as well as in membrane-formation.

The fatty acids and their glyceride esters are the most important group of biological lipids and will be considered first, along with the related aldehydes, alcohols and esters. The acid has the most hydrophil and active polar group, and is considerably less volatile than the corresponding paraffin. This is another important virtue of partial oxidation (p. 71), particularly for storage purposes. This effect of the hydrophil, and particularly the ionic, character on volatility is not restricted to organic compounds: the strongly ionising inorganic salts are particularly non-volatile.

7.1 The Fatty Acids (Acylic Acids)

In vivo all the fatty acids are probably synthesised from acetic acid, $CH_3{\cdot}COOH$, which is the second member of the series itself, and the product of partial *oxidation* of the carbohydrates. The energy liberated when acetic is produced in this way for fatty acid synthesis no doubt can be well used in the subsequent stages, which involve the successive

addition of acetic units and the complete reduction of each carbonyl group, except the last, to methyl. In fact of course additional energy may be required, but it may be noted that since a C_2H_6 unit is ultimately required from each triose, $C_3H_6O_3$, there should be enough spare energy in the triose, if used efficiently. It is of course transferred by ATP (p. 404) and hydrogen by the pyridine coenzyme NADP (p. 334); ATP is formed directly and NADPH indirectly, via NADH, both in the glycolytic sequence (Fig. 3). While acetic is being thus used for fat synthesis of course the Krebs cycle (Fig. 8) is starved, and respiration must be mainly glycolytic.

The precise mode of synthesis of the longer chain fatty acids is still uncertain (Wakil, 1962) and may change at one stage of the elongation. It appears that what was once considered the main mechanism, namely a reversal of the β–oxidation by which the acids are known to be re-oxidised in respiration, applies only to the elongation beyond C_{16}, which is a trivial fraction. It is in effect a Claisen condensation, widely used in the laboratory, and so an efficient reaction. *In vivo* it is catalysed by coenzyme A (p. 356) which may be represented as HS·CoA, and activates the acylic acids by conjugation. Each elongation by C_2 therefore may be represented:

$$R_nCH_2CO{\cdot}SCoA + CH_3CO{\cdot}SCoA \rightarrow R_nCH_2CO{\cdot}CH_2CO{\cdot}SCoA +$$
$$HSCoA \xrightarrow[-O]{+2H} R_nCH_2CH_2CH_2CO{\cdot}SCoA$$

At shorter chain lengths carbon dioxide acts as a catalyst, in the form of bicarbonate. This condenses with acetyl–CoA to form malonyl–CoA, which condenses with the growing acyl chain, eliminates carbon dioxide once more, and is reduced as in previous mechanism. In fact it differs from this only in the malonyl intermediary stage and in some of the enzymes:

$$CO_2 + CH_3CO{\cdot}SCoA \xrightarrow[\text{Biotin}]{\text{ATP, } Mn^{2+}} CH_2(COOH){\cdot}CO{\cdot}SCoA$$
$$R_nCO{\cdot}SCoA + CH_2(COOH)\,CO{\cdot}SCoA \xrightarrow{2NADPH} R_n{\cdot}CH_2CH_2CO{\cdot}SCoA$$
$$+ CO_2 + HSCoA$$

Malonic acid readily breaks down to acetic and carbon dioxide in the laboratory but the teleonomy of its initial formation is curious unless the new carboxyl group is the one which condenses with the acyl chain and that bound to CoA is the one released as carbon dioxide. This might well facilitate the process and recalls the significance of the condensation of acetic with OAA in the Krebs cycle (p. 67). Malonic acid inhibits a critical step in the Krebs cycle and so promotes fatty acid synthesis in this way also.

Both methods explain why the C–even fatty acids and derivatives are very common and the C–odd members very rare *in vivo* (Table 18).

TABLE 18. THE MORE COMMON BIOLOGICAL FATTY ACIDS

A. SATURATED: $CH_3(CH_2)_nCOOH$

Number of C atoms	Acid	Occurrence
C–even Acids		
2	Acetic	Vinegar, oil of spindle tree, nettle sting
4	Butyric	Butter, rumen bacteria
6	Caproic	Goat's milk, palm oils
8	Caprylic	
10	Capric	
12	Lauric	Laurel, coconut, spermaceti
14	Myristic	Nutmeg, coconut
16	Palmitic	Palm oil, animal fats
18	Stearic	Mutton fat, plant oils
20	Arachidic	Peanut oil, rape oil, lard
22	Behenic	Ben oil
24	Lignoceric	Peanut oil
26	Cerotic	Beeswax, wool fat
28	Montanic	Beeswax, Montan wax
30	Mellisic	Beeswax
32	Dotriacontanoic	Lac wax
34	Tetratriacontanoic	Ghedda wax
36	Hexatriacontanoic	
38	Octatriacontanoic	
C–odd Acids		
1	Formic	Ant sting, nettle sting
3	Propionic	Rumen bacteria
5	Valeric	Valerian root
9	Pelargonic	

Propionic (C_3) is an exception but is formed from carbohydrate; other C–odd acids do occur (Shorland, 1952), sporadically, perhaps again by special pathways or by demethylation of a C–even member. Living organisms in fact have great difficulty in metabolising the C–odd molecules presented to them experimentally (Cosgrove and Swanson, 1952) and these prove good fungicides. In fact adult human beings secrete them, C_{11} in particular, on their hair and so are immune to ringworm (Ainsworth, 1958). Children have not yet acquired this power which therefore seems to be a recent special acquisition. The C–odd members produced by other organisms likewise may have a special antibiotic function. The fluorine derivatives of the C–odd members however are *less* toxic than those of the evens because they are not competing with

B. Unsaturated

Number of C atoms	Number of double bonds	Position of double bonds	Acid	Occurrence
4	1	2:3	Crotonic	Croton oil
5	1		Tiglic	
11	3		Undecylenic	Scalp oil (adult)
14	1	9:10	Myristoleic	Fish oils, butter
16	1	9:10	Palmitoleic	Butter, fats of vertebrates
18	1	9:10 *cis*	Oleic	Olive and other plant oils
18	1	9:10 *trans*	Elaidic	(not biological)
18	1	11:12	Vaccenic	Butter
18	2	9:10, 12:13	Linoleic	Plant oils, animal fat
18	3	9:10, 12:13, 15:16	Linolenic	Linseed oil, animal fat
18	4		Stearidonic	Cod liver oil
20	1	9:10	Gadoleic	Fish liver, brain lipids
20	4		Arachidonic	Butter, liver, P-lipids
22	1	13:14	Erucic	Rapeseed oil
22	5		Clupanodonic	P-lipids

normal metabolites. Formic acid, HCOOH, is particularly toxic since in addition to being odd it has the usual anomalous properties of the first member of most series. It occurs, for its corrosive value, in the stinging fluids of ants (whence the name), bees, nettles and the Mad Tree of Australia. Because of its hydrogen and =O on the same carbon it remains effectively an aldehyde, with all the additive reactivity of the group. Acetic usually occurs with it in sting-fluids: in concentration it also is corrosive, of course, but the occurrence probably indicates that it is also the source of formic, by de-methylation.

The C_{18} fatty acids are much the most abundant in plants and animals and the C_{16} members are second in importance. C–even members up to C_{38} have been discovered in biological material (Table 18) but they only serve to emphasise the unique exploitation of the $C_{16\text{-}18}$ node. Anything shorter does not build stable crystalline layers and anything longer is less soluble in water, protects proteins less well, has too high a melting point, and so on. There are in fact no universally abundant members between the node and acetic, either. Butyric and the capric acids are abundant in mammalian milk but this should be considered a sporadic use. In living organisms collectively the C_{20} to C_{24} acids are perhaps more common than the C_6 to C_{14}. The size range therefore compares with that

among the steroids (p. 121). These are synthesised by another pathway so that the reasons for the resemblance are probably certain common functional requirements. At the same time it seems that C_{18} marks some kind of node in biosynthesis, in the fatty acid line. This is perhaps borne out also by the fact that linoleic, one of the unsaturated C_{18} acids, can polymerise as a unit, a property largely responsible for the hardening of linseed oil in the setting of paint (Robertson, 1947). In passing it may be

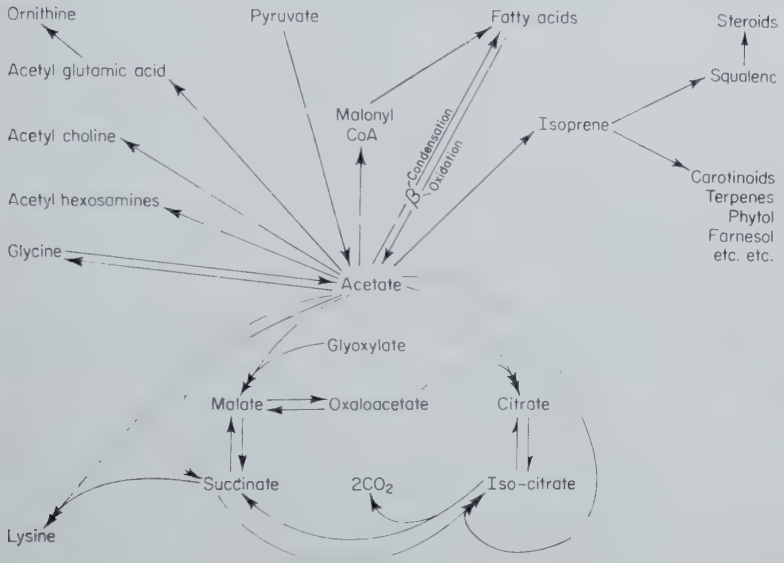


FIG. 14 Active acetate (acetyl coenzyme A) as a metabolic node.

noted that this kind of polymerisation is rare in biological lipids, in contrast to monoses, amino acids and nucleotides. However, it does occur in the other main lipid pathway, via isoprene (p. 107), and commercially one group of plastics is formed by the polymerisation of the unsaturated C_3 member, acrylic acid, $CH_2{=}CH{\cdot}COOH$. The C_{16} to C_{18} stage is the minimal length for stability in other series, also, and in aggregates (Nash, 1958), and this concerns synthesis as well as subsequent function (p. 82). That the precise chain-length is indeed critical for biological function is shown by the fact that oleic acid ceases to be a growth factor for *Corynebacterium* if the molecule is lengthened by C_2—though it can be shortened to C_{16} without serious loss of activity (Boughton and Pollock, 1952).

The unique biological virtue of acetic acid is that it lies, like pyruvic acid (p. 69), at an important metabolic node (Fig. 14), particularly between carbohydrates and lipids. It is miscible with water in all proportions,

because of its high polar/non-polar ratio, whereas propionic ('first fat') already can be salted out as an oil by adding calcium chloride or other dehydrating agents. Since formic is so anomalous, acetic is the only member which could be used for fat metabolism. It is at the same level of oxidation as the carbohydrates (Table 15) which facilitates their interconversion.

The alkali salts of the fatty acids are soluble even up to C_{18}. Palmitic (C_{16}) soaps are soluble and surface active even in sea water and are used to make 'marine soap'. This is a good indication of the great difference in properties which can result from one C_2 unit, more or less. The greater solubility of the salts is of course due to their greater ionisation. Calcium soaps and the salts of other polyvalent metals are insoluble because they form aggregates with the apolar ends of the acyl residues directed outwards and the polar groups towards the centre. This follows from the wedge shape of the molecule (Fig. 12, III), and it explains why calcium tends to cause the formation of water-in-oil emulsions while the monovalent cations induce the oil-in-water type.

Most relevant properties grade regularly along the series of fatty acids, and along such *homologous* series in general. Up to C_{10} the fatty acids are liquids of increasing oiliness, while beyond this point they become greases and eventually waxy solids; the paraffins show the same trend but the acids are less volatile than the corresponding hydrocarbon because of their hydrophil group. The melting point is depressed, however, by each double bond which is formed, so that it is possible to have glycerides fluid at all normal environmental temperatures and yet with molecules long enough to form stable membranes.

Like other aliphatic hydroxy compounds the fatty acids form anhydrides, one molecule acting as a base and the other as an acid:

$$2RC(=O)OH \rightarrow RC(=O)O(O=)CR$$

The bond formed is much more labile than that between the glucose residues in glycogen and it is not difficult to understand why the fatty acids do not even form dimers. The lability of the bond is related to the large amount of energy liberated when the anhydride is hydrolysed back to acyl acid. It has in fact an energy-transfer value as high as that of some phosphate compounds (p. 402) and might be represented RCO—O~CR. In order to harness this energy acyl phosphates $RCOO\sim PO(OH)_2$ are formed *in vivo*, and the acetic member, *acetyl phosphate*, plays an important part in metabolism. As will be seen (p. 404), ~P compounds are much more stable kinetically than the acyl anhydrides. Another stable form, perhaps of more general importance, is the acyl–CoA compound already seen, which may be represented $RCO\sim SCoA$, the usual form of

active acylate. It would be difficult to overemphasise the importance of this in metabolism.

In contrast to the high specificity of fatty acids used for particular metabolic purposes it does not matter how long a molecule of storage fat is, or whether it is saturated or not, provided it has the correct melting point. In consequence there is a very strong tendency for stored fat to resemble in composition the source of food of an animal. The coconut crab stores a fat very similar to coconut oil, with a low melting point, and there are many other recorded examples. Unrelated animals living under the same conditions may have fats more alike than near relatives living under different conditions. The fat of insects is more similar to that of mammals than it is to that of Crustacea. In these cases, of course, the exigencies of the environment rather than the type of food may determine the type of fat, and in this event it is functional and not fortuitous. Individual living organisms do not always produce a narrow range, either; the C_4 to C_{18} saturated acids are all present in mammalian milk and the C_8 to C_{16} members in palm seed oils. Of course these are stores for the young so that they probably strengthen the evidence that only storage fats are relatively variable in composition.

7.1.1 Unsaturated Fatty Acids

These are usually labelled *enoic* to distinguish them from the *anoic*, fully saturated series. C_{18} members are very common, abundant and metabolically important; otherwise they occur rather sporadically but in a considerable variety of chain-lengths (Table 18). The C_{18} group act as vitamins for mammals, and other animals tested, and this indicates that their significance may be chemical, depending on the usual reactivity associated with the double bonds. The enoic acids in fact rather readily form addition compounds, particularly with the halogens, so that the 'iodine number' is used to measure the number of double bonds in the molecule. It may be either oxidised or reduced at these positions and also broken. Most of those with more than one double bond have them in alternate neighbour positions so that they form a conjugate system which can resonate as a unit, i.e.

$$\ldots\text{—CH=CH—CH=CH—}\ldots \leftrightharpoons \ldots\text{=CH—CH=CH—CH=}\ldots$$

This extension of resonance increases activity proportionately (p. 116).

The double bond can be regarded as a ring (p. 74) of minimal size, since it prevents rotation about it, and introduces the *cis–trans* type of isomerism. Since the double bond can occur in principle at any position except next to the carboxyl group there are 31 possible mono-enoic C_{18} acids (West and Todd, 1957, p. 152) and the number of polyenoic isomers

is even greater because of the *cis–trans* innovations with each additional double bond. Biological activity is rather rigidly restricted to particular isomers: the 6:7 and 11:12 isomers of oleic can deputise for it to some extent but not the 4:5 isomer (Boughton and Pollock, 1952). This is in contrast to the low specificity of storage fats.

The value of the enoic acids where low-melting-point fats are required, has been mentioned. They are also useful in spreading as a monolayer more readily than the anoic series: oleic spreads spontaneously but stearic requires a special solvent or spreader. The restriction on rotation within the molecular chain may be a disadvantage in membranes, though there is no definite evidence about this. The double bond also results in a deeper folding of the molecule (Fig. 12, V), so that the molecule is shorter, and is stouter in the relevant region, than the corresponding anoic acid. This again may be relevant to membrane-formation, though the increased variety of these features may be a positive advantage in some situations.

Among the rarer enoic acids, biologically, is the *queen substance* of bees, which is produced by the queen and inhibits queen-development in the workers. It is 9–keto–dec–2–enoic acid (Fig. 12, VI) (Callow and Johnston, 1960). The enoic bond is near enough to the carboxyl =O to show conjugate resonance with it. There are two natural ring-closed enoic acids, chaulmoogric (Fig. 12, IV) and its C_{16} homologue hydnocarpic, an interesting confirmation of the C_{16} to C_{18} node. Both are invaluable cures for that stubborn disease, leprosy, perhaps as analogues of the open chain fatty acids, though something more specific to the ring structure may be suspected.

Acrylic acid (p. 91), being C_3, is not a biological enoic acid but it is interesting in ways which may serve as models for biological materials. Its polymers, already mentioned, may be orientated by heating and bonded laterally to a polyvalent alcohol such as glycerol, and will then behave like a simple model of actomyosin, responding to changes in pH and other relevant conditions by changes in molecular dimensions (Kuhn, 1950).

Few acetylenic acids occur biologically (Walsh, 1961).

7.1.2 Branched Chain Fatty Acids

These are rare and usually have only methyl side chains. Nevertheless the change radically affects the molecule and many organisms cannot metabolise them by the usual β–oxidation; they treat them as toxic substances, conjugating them with a suitable detoxicant and excreting without further change (Williams, 1959). Organisms which produce them therefore may do so for their antimetabolic or toxic action. The tubercle bacillus produces tuberculostearic, or 10–methyl stearic, $C_{17}H_{35}COOH$ (Fig. 12, VII), and phthioic acid, $C_{25}H_{51}COOH$. Their toxicity may be due in part to their effect when incorporated into membranes. Isostearic acid

requires 50 per cent more space laterally than stearic, and its monolayer in consequence has only one-third the strength (Ries, 1961).

Isovaleric acid, $(CH_3)_2{:}CH_2{\cdot}CH_2{\cdot}COOH$, occurs in the roots of *Valeriana*, and in dolphins and nematode worms. It is notorious for its intensely rancid smell—perhaps a warning of its metabolic undesirability. It may be derived *in vivo* from the branched chain amino acids (p. 179), in which case it belongs to the isoprene pathway. In the products of this pathway methyl branches are usually very abundant, but the similarity of tuberculostearic to stearic and oleic leaves little doubt that some methyl branched lipids do come from the acyl pathway.

7.1.3 Dicarboxylic Fatty Acids

This series with the second carboxyl group at the far (ω) end of the molecule is a more important one, biologically. In fact the mono-carboxylic series are fairly easily oxidised in the ω–position and the resulting dicarboxylic acid then can be β–oxidised from both ends of the molecule. This may be the origin of those members which now have a special function in the body. For some reason (Baldwin, 1953) the C_8 to C_{12} dicarboxylics are not easily oxidised from either end, though those outside this range, on either side, are readily and completely catabolised. It is therefore significant that the dicarboxylic acids suberic (8), azelic (9), sebacic (10), undecandioic (11) and traumatic acid (Fig. 12, VIII) all occur biologically; when excreted they are excreted unchanged. However, pimelic (7) and adipic (6) also occur in nature, and glutaric (5) is formed in an offset pathway from the Krebs cycle. Succinic (4) is an essential member of this cycle (Fig. 8) and malonic is an intermediary in the main pathway of fatty acid synthesis (p. 88). Oxalic acid (p. 63) could be regarded as the first member of the series, so that up to C_{12} they are as fully represented in living organisms as the monocarboxylics, though only sporadically for the most part, and in small quantities. Succinic and malonic are probably the only ones with a general metabolic function, and even so there must be some restriction on malonic since it is a powerful inhibitor of the Krebs cycle through competitive antagonism of succinic. It is of course possible that this ensures that acetic shall not be drawn into the Krebs cycle when it is required for the synthesis of fatty acids (p. 88).

Traumatic acid is a mon-enoic member, with the carboxyl groups in the *trans* relation, while fumaric acid, a member of the Krebs cycle, is the corresponding C_4 acid. The *cis* isomer of this, maleic acid (Fig. 1, IX), also occurs naturally, in apples and elsewhere. This pair again show the high biological specificity to one geometric isomer, since maleic attracts the spermatozoa of echinoderms while fumaric does not (Rothschild, 1952); in fact it has less effect than some other, unrelated, substances.

The rate of hydrolysis of the two forms is very different (Sexton, 1953) because of the different accessibility of the double bond, depending on the positions of the surrounding groups. The two isomers differ also in the ease with which they are ejected from membranes (Schulman, 1951).

Traumatic acid, as the name implies, promotes wound-healing in plants (English *et al.*, 1938), and the activity is fairly specific to this acid; however the saturated dicarboxylics of about the same chain-length, e.g. sebacic, do have some activity. The degree of resemblance between the molecules of traumatic acid and the queen substance in bees is interesting (Fig. 12, VIII).

The dicarboxylic acids are much more soluble in water than their monocarboxyl counterparts, as might be expected. They are also stronger acids, quite apart from having twice the titratable acidity per molecule. The two carboxyl groups must affect the ionisation of each other, and they readily form internal anhydrides, particularly when this produces a 5- or 6-membered ring (succinic and glutaric). These acids therefore have functions distinct from those of the monocarboxylics.

7.1.4 Hydroxy Fatty Acids

The lowest members of this series are automatically members also of the more fully oxidised and very water-soluble group of hydroxy acids already considered in Chapter 6. They are of course molecules of an intermediate grade of reduction, but like the dicarboxylic series rarely have more than two polar groups in the molecule. They are essentially lipids, therefore, though polar enough not to be soluble in petroleum oils. The two polar groups tend to interact as in the dicarboxylic series, mutually increasing activity if near together and undergoing ring closure if at the optimal distance of four to five carbons apart. The inner anhydride ring compound is a lactone, with some resemblance to ascorbic acid (p. 50). They tend to inhibit cell division and growth—a fairly common property of lactones. Indeed there is some evidence that ascorbic acid itself inhibits cell division under certain conditions (Shapiro, 1948). Parasorbic acid (Fig. 12, XI), a laboratory-synthesised relative of natural lactones occurring in the mountain ash, and a member of the present group, is a powerful inhibitor of germination of seeds, and of the proliferation of mesenchyme cells *in vitro* (Medawar *et al.*, 1943). Like ascorbic acid it is an enoic form with the potentiality for conjugate resonance. The sorbic lactones are thought to form addition compounds with —SH or —NH_2 groups of key metabolites and block growth in that way. Ascorbic acid in fact controls the oxidation of the SH group of glutathione (p. 203). It is ironical that the sorbic acids should be its metabolic analogues since there is no etymological or other connection between the two suffices.

The best known of the naturally occurring hydroxy-fatty acids are given in Table 19. They are long chain members, occurring mainly in

Table 19. Biological Hydroxy-Fatty Acids of High Chain-length

Number of C atoms	Acid	Occurrence
Saturated		
16	Trihydroxypalmitic (aleuritic)	Lac insect
18	Dihydroxystearic	Castor oil
24	Cerebronic	Lipids of brain
24	Hydroxylignoceric	Cerebrosides
32	Lanoceric (dihydroxy-triacontanoic)	
Unsaturated		
18	Ricinoleic	Castor oil
24	Hydroxynervonic	Lipids of brain

nervous tissue and castor oil. Their significance is at present obscure, and so is that of the lower, lactone-forming members—at any rate to the organism producing them. Defence against predators is unlikely to explain the distribution of those of nerve tissue.

7.2 Esters of Fatty Acids

One example, the polyester with glycerol, has already been noticed and most of the biologically important esters are with alcohols, mono- or polyhydric. Ester-formation is the classical example of a balanced reaction, but the equilibrium lies 67 per cent in favour of the ester compound. In theory enzymes do not affect the position of equilibrium but in practice they do seem to move it further in favour of the ester, under appropriate conditions *in vivo*—as well as speeding the reaction in the usual way. Like so many organic reactions esterification is slow at ordinary temperature without catalytic help. The decreased solubility of the ester (p. 87) is probably the main factor favouring the forward reaction, and in the cells this may be assisted by segregating the more lipid-soluble ester across intracytoplasmic membranes.

The glycerides are by far the most abundant esters in living organisms and they are probably the ideal compounds of fatty acids for storage and other purposes. Any alcohol would tend to neutralise the acid and to

make it less water-soluble and less reactive, properties all valuable in storage material. The virtues of glycerol are that it is a small molecule and can esterify three acid molecules simultaneously. By binding them into bundles of three it accelerates their movement and membrane-formation. On the other hand the three acids for steric reasons do not pack quite parallel, and so a trihydroxy alcohol is the longest chain which could reasonably be used for the purpose.

There are also metabolic reasons why glycerol is ideal. It is produced as a by-product of the glycolytic pathway, which must be used extensively for respiration when acetic is required for fatty acid synthesis (p. 88). Reduction of dihydroxyacetone to glycerol could be used as a method of re-oxidising NADH, so as to continue the oxidation of the other triose, glyceraldehyde, on the way to acetic acid. With the latter required for fat synthesis it is not feasible to regenerate NAD by reducing pyruvic to lactic (Fig. 3), or via acetaldehyde to ethanol. A commercial method of inducing yeast to synthesise glycerol is based on this principle. Glycerol is fluid at ordinary temperatures and is an excellent solvent, so that it has a number of useful chemical properties.

A number of polybasic inorganic acids form mixed salts with more than one metal but glycerol is a polyacidic base which forms salts with three different acids; it is in fact unusual to find any two acids alike in one glyceride molecule, though in any sample of fat most of the triglyceride units may have the same composition. Inositol forms a hexaphosphate (p. 74) but it is not yet possible to say if glycerol is unique in usually forming hetero-esters. The fatty acids used rarely vary in length by more than a C_2 unit, so that the requirement for a uniform 'pile' in lipid membranes is not seriously contravened. The small spaces caused by the shorter acids may accommodate water or other small molecules. When phosphatides (p. 294) are used in the membrane there are larger gaps, since the phosphoryl-base unit is directed towards the aqueous phase. The phosphatides are a further example of the tendency of glycerol to form hetero-esters; in this case the hydrophil, phosphoryl-base unit greatly increases the hydrophilia of the whole molecule, i.e. among other virtues it promotes the mobilisation of the two fatty acids, when required. Collectively glycerides have great potentialities.

Esters with monohydric alcohols are quite common but have different and more restricted functions. There is a large group in which both acid and alcohol are short-chained; they are therefore volatile oils. Almost without exception they have strong and attractive odours, notwithstanding the unpleasantness of some of the free acids, such as butyric, valeric and caproic, and some of the free alcohols (amyl). Indeed the more unpleasant the acid the more attractive the ester. The reason for this,

chemical or biological, is not clear, but most fruits have a number of these esters and there is little doubt that they attract animals to eat the fruit and disperse the seeds. They are chemically more inert than the free acids but it is not clear why the latter are so unpleasant or why other volatile attractants should not have been used. To the taste the attraction is mainly the Krebs cycle acids (p. 65), and these are not volatile enough to attract from a distance; possibly the esters are the simplest volatile scents which could be exploited and in addition they are probably metabolised by the animal. They tend to be intoxicant and promote well-being. The most typical are ethyl and isoamyl acetates which smell like pears, ethyl butyrate (pineapple) and isoamyl isovalerate (apples). The last and some others imply that acid and alcohol may have a common precursor.

There is the same indication in the esters of the longer chain acids (Table 20). The three waxes shown are the best known; spermaceti is

Table 20. Components of Waxes: Similarity in Number of C–atoms

Wax	Number of C–atoms in components				
	Saturated acids	Unsaturated acids	Saturated alcohols	Unsaturated alcohols	Paraffins
Spermaceti	10–18	12–20	14–18	16–20	—
Beeswax	24–34	—	24–34	—	25–31
Carnuba wax	26–34	—	26–34	—	27

obtained from the oil in the anterodorsal cavities of the head of the cachalot, or sperm whale, while beeswax may come indirectly from the plant waxes, used for waterproofing the surfaces of leaves. All are mixtures of a range of esters, but the range of acids in each case is very similar to that of the alcohols. All are C–even and where there are unsaturated acids the alcohols also have double bonds. In addition, beeswax and plant waxes contain some hydrocarbons and these are always C–odd, strongly implying that they are formed by decarboxylation of the acids (or decarbinolation of the alcohol). The plasticising effect of paraffin admixture has been mentioned (p. 85). These esters with large molecules are less easily hydrolysed than the glycerol esters which would not be suitable for long-term waterproofing of external surfaces. Acid and alcohol tend to lie side by side, with the common hydrophil group at the bend of the 'hairpin', so that they readily form good molecular layers of the usual type. Layer-formation is improved by mechanical shearing, and highly polished hydrofuge surfaces can be produced by rubbing.

Lanolin, or wool fat, is another fatty acid ester of high molecular weight but in this case the alcohol is a sterol (p. 118). It is notable for the amount of water it will absorb without dissolving and this makes it invaluable as a base for ointments, keeping the skin moist and facilitating the inunction of both lipid and water-soluble materials. These functions may be equally useful to the sheep and other mammals where also a completely waterproof surface is not as vital as in insects and plants. Esterification makes the sterols themselves less soluble, and more slowly reacting, and the hormonal sterols are often sold in ester form for this reason, to increase their stability and prolong their action. Acids as short as propionic are effective.

7.3 Oligohydric Aliphatic Alcohols

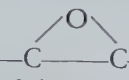
These are two steps more reduced than the fatty acids, but they are biologically more important than the aldehydes and will be considered first. In addition to those in ester form there are some free mono- and dihydroxy alcohols in living organisms, or at least they are extracted free. The more active and volatile lower members tend to be toxic and this contrasts with the biological value of the fatty acids on the one hand and of the polyhydroxy alcohols on the other (p. 72). The monohydroxy alcohols are in fact less toxic than the dihydroxy series, for instance propanol (C_3) than dihydroxy-propane. The antibiotic, chloramphenicol (Fig. 12, XII), is a substituted dihydroxy propane and is a powerful inhibitor of protein synthesis; it has, of course, other toxic groups. This toxicity of the alcohols helps to explain why compounds of intermediate grade of reduction are not common *in vivo* but the reason for the toxicity is not very clear; it may be not so much intrinsic as secondary to the exploitation of a metabolism based on other series—because these are the most useful on other grounds. The reason for not exploiting many of the series of intermediate grades of reduction has been considered, and it is possible that the monohydroxy alcohols are toxic because they are inferior alternatives to the fatty acids. The branched chain alkyl monols are less toxic than the corresponding straight chain alcohols, which may imply that the latter are active metabolic antagonists of the fatty acids.

The alcohols are known to depress the ionisation of the fatty acids and they dissolve, or occlude, in soaps. They are narcotics because they dissolve in the cell-membrane and depress its permeability or for the alternative reason (p. 81). If they were metabolic antagonists of the fatty acids, however, the C–odd members should be harmless, whereas amyl (C_5) is the most toxic. Moreover the fatty acids around that chain-length are not of wide biological use. In fact the Protozoa are able to utilise the

C–even alcohols as food, but not the C–odd members (Cosgrove and Swanson, 1952). Ethanol is very readily used as a source of energy, via acetic acid, and other members can be used in the same way, on a small scale. Ethanol even stimulates the normal metabolism of pyruvate (West and Todd, 1957, p. 990) and carbohydrate generally. It may reasonably be contended that the lower fatty acids are as objectionable and toxic as the alcohols and are attractive only as esters. In this light the alcohols are perhaps more liable to be competitors of glycerol than of the acids. The oligohydric alcohols in general therefore may be antagonists of the polyhydric series.

In dihydroxy alcohols toxicity increases with the distance between the two hydroxyl groups, at any rate in dihydroxy-butane (C_4). This could be connected with ring closure but it is also known that hydroxyl groups are more easily metabolised if they are close together (Williams, 1959).

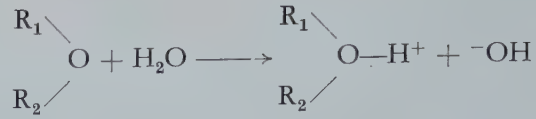
There is the possibility that the short chain alcohols may be toxic as analogues of water itself, and the resemblances of ethanol will be considered (p. 505). It is a good solvent, forms alcohol-of-crystallisation and forms molecular aggregates. It is interesting that water itself in high concentration can cause mild symptoms of intoxication.

Among the longer chain oligohydric alcohols occurring naturally are *astrol*, or batyl alcohol, $C_{23}H_{48}O_3$, in starfish (Bergmann, 1943), and *scymnol* from elasmobranch fishes. The latter is in fact a steroid but has a hydroxyl and an anhydride or epoxide group,  , in the side chain. It occurs as a bile salt, esterified with sulphuric acid, and so differs from most bile salts in that the steroid is the base and not the acid. From its methyl branches it is clear that this whole molecule is of isoprenoid origin, and this is true of most other open chain alcohols—among the carotenoids (p. 112) and in other groups. In all, therefore, the simple aliphatic oligohydric alcohols do not play a very important biological role, except in ester form.

7.4 Aliphatic Ethers ($R_1{\cdot}CH_2{-}O{-}CH_2{\cdot}R_2$)

These anhydrides of the monohydric alcohols can be compared—and contrasted—with the glycosides, except that having only the one hydroxyl group the dimer is the limit of polymerisation. They are less polar, more lipid-soluble and inflammable, and more volatile than the alcohols. They are strong narcotics and have no unique advantages for biological purposes. Perhaps few groups better illustrate biological unsuitability combined with laboratory value. They are apolar enough to be rather

unreactive chemically but the oxygen is still capable of becoming an oxonium ion (p. 152):



They do this best with strong acids such as hydrochloric acid and so can be regarded as conjugate bases.

There are few di-alkyl ethers in living organisms but alkyl-aromatic mixed ethers are common as essential oils, i.e. fragrant oils, of plants. Examples are anethole (Fig. 12, XIII), in aniseed oil, eugenol in clove oil and vanillin. Because of the acidity of the aromatic hydroxyl group, these oils behave more as esters than as ethers. The methyl glycosides are somewhat more like the ethers.

7.5 Alkyl Aldehydes and Ketones

These are the aldehydes and ketones corresponding to the fatty acids and monohydroxy alcohols. They are rarer than these series, in diametric contrast to the polyhydroxy aldehydes (Chapter 4), in relation to their alcohol and acid counterparts. The lower members are in fact more toxic than the lower fatty acids and alcohols, because of the great reactivity of the aldehyde group, which very readily forms addition compounds simultaneously with two other groups or molecules and so hobbles them both. The compounds are very stable, much too irreversible for biological purposes. It is this which makes formaldehyde such a good reagent for the fixation of the amino group of amino acids in the Sørensen method of estimation. Among the poisons used for defence by insects are a number of aldehydes (Roth and Eisner, 1962), n–hexanal, *trans*–2–hexenal, *trans*–2–heptenal, 2–octenal, 2–decenal, citral, citronellal and the dialdehyde of iridomyrmecin, iridodial, as well as some aromatic aldehydes. The second main group of insect poisons are ketones, but mostly aromatic (quinones).

Not all biological aldehydes of this group are poisons, particularly at the longer chain-lengths. The C_{16} member occurs in plasmalogens (Fig. 12, XIV), phosphatide derivatives in which the CHO group forms the usual double additive compound, known in this case as a *gem*-diether. The plasmalogens are very abundant in embryos (Brachet, 1950). *N*–decylic aldehyde, $C_9H_{19}CHO$, occurs in oils of orange, lemon grass and coriander. It occurs along with aldehydes and other derivatives of the isoprenoid pathway (p. 110). An open chain aldehyde may be the sub-

strate for light-production in some bacteria, which is interesting in connection with retinene (p. 114).

Alkyl ketones are even rarer *in vivo* than the aldehydes. The two best known are alicyclic ring compounds, *muscone* (Fig. 12, XV) from the musk rat and *civetone* (Fig. 12, XVI) from the civet cat. They have powerful, pleasant odours, possibly of aphrodisiac function. The size of the ring (Fig. 12, XV) is almost unique and it is suspected that this is incipiently folded into a potential steroid form (Fig. 12, XVI), although the structure is otherwise very similar to some of the derivatives of the C_{16} and C_{18} fatty acids. Some ketonic steroids do have a musky smell and the steroid relationship may have some relevance to the aphrodisiac effect of musk. It will be noticed that civetone is an enoic ketone, and that some of the insect poisons are enoic aldehydes.

CHAPTER 8

PARAFFINS (C_nH_{2n+2})

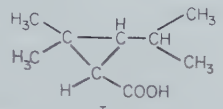
THE rarity of fully reduced aliphatic compounds in living organisms has already been emphasised, and the main biological reasons given, namely water-insolubility and low chemical reactivity in aqueous systems at ordinary temperatures. The rarity is at first a surprise to the organic chemist, starting his preparations from the wealth of naturally occurring hydrocarbons, and having evidence that these are of biological origin (Smith, 1953; Oparin, 1957a; Kropotkin, 1957). Petroleum deposits contain every paraffin from C_1 to C_{30}. Fish oils yield considerable amounts on simply heating under relatively anaerobic conditions (Read, 1935, p. 125), so that there is little doubt that they form fairly spontaneously from compounds such as the biological lipids, which are almost fully reduced. Their rarity in living organisms therefore is a biological-level phenomenon: work must be expended in keeping the lipids partially oxidised and active. It is, therefore, no surprise that some micro-organisms can catabolise paraffins where profitable (Zobell, 1950), or excrete them as by-products, but they rarely make use of them as agents in the body. Small amounts occur in waxes (p. 99) and may serve a useful plasticising function, and aliphatic hydrocarbons occur in sulphate-reducing bacteria (Smith, 1953), though here they may be by-products.

Although completely paraffinic molecules are rare it is possible that the hydrocarbon portions of various biological molecules may exploit certain paraffin properties to advantage (Buswell and Rodebush, 1956; Klotz, 1962). Hydrocarbons have a negative heat of solution (Rodebush and Masterman, 1954), that is to say heat is required to make spaces for their molecules, where they hold a rigid shell of water molecules protecting them from true solution. These phobic, repulsion hydrates may be as effective anti-freeze devices as the positive association of water with glycerol and sugars (p. 48), provided cooling is slow. However, these hydrates freeze at temperatures as high as 40°F and signs of frost-damage in plants have been detected at such temperatures, perhaps due to them. This is of course near the temperature at which water itself is densest but there seems no reason why this should cause signs of damage. Snows

formed of crystals of these hydrates sometimes block pipelines of 'natural gas', methane, at temperatures as high as 68°F. This kind of water presumably is part of the 'bound' water of proteins and other molecules.

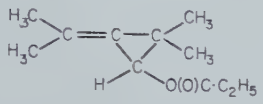
8.1 Cycloparaffins

These may be defined as alicyclic compounds with no double bonds in the ring. They are as rare as the open chain paraffins in living organisms, though common in some petroleum deposits, mainly as cyclopentanes and -hexanes. These are known as 'naphthenes', not to be confused with the naphthalenes (p. 135), and *not* enoic compounds in spite of the suffix. There is as much as fifty-eight parts per million (ppm) of naphthenes in some fish and lamellibranchs (Smith, 1953). As usual the C_5 and C_6 rings are the most stable and common, and it is therefore particularly interesting to find some biological cyclopropanes, such as chrysanthemum mono- and dicarboxylic acids (Fig. 15, I). Another is the sex-attractant substance



I

Chrysanthemum monocarboxylic acid



II

2,2-dimethyl-3-isopropylidene propionate

Fig. 15 Alicyclic compounds synthesised by living organisms.

of cockroaches (Jacobson *et al.*, 1963), 2,2,dimethyl–3–isopropylidene cyclopropyl propionate (Fig. 15, II). These are partially oxidised compounds, of course, and could have been considered among the acids and esters; pure cycloparaffins are as rare *in vivo* as open chain paraffins. The propane ring remains an unusual structure to find, in any case.

CHAPTER 9

OLEFINES AND TERNARY UNSATURATED COMPOUNDS

Open chain hydrocarbons with one or more unsaturated bonds are more common biologically than paraffins, and there are many ternary derivatives of these olefines. Acetylene derivatives, with triple bonds, $R_1C \equiv C{\cdot}R_2$, by contrast, are very rare in living organisms. They are probably too reactive to be effectively controlled. The olefines have the optimal degree of chemical reactivity, which depends on the readiness with which they resaturate the double bonds. Their formation from the paraffin or alkane state is endothermic and the energy absorbed facilitates resaturation:

$$C_2H_6 \xrightarrow{-2H} CH_2{=}CH_2 \quad -32{\cdot}2 \text{ kcal}$$

Linoleic acid oxidises spontaneously in air, but it is noteworthy that saturation by hydrogenation is equally spontaneous and exothermic. The acetylenic state is the most endothermic, since its formation from hydrogen and carbon, as well as by the desaturation of paraffins and olefines, is endothermic, and so it is not surprising that acetylenes are rare biologically. It is also clear why the natural accumulations of carbon are in three forms: carbon dioxide and carbonates, elemental carbon (coal) and paraffins.

In principle a carbon atom could be double-bonded to both of its neighbours, but in practice this is very rare, and is unknown biologically. This carbon atom would be elemental but in fact elemental carbon usually has only single bonds between atoms. The tendency in biological olefines, as in the aromatic series (Chapter 10), is to approach the condition of a regular alternation of single and double bonds in any sequence of carbon atoms, a condition of maximal conjugate resonance (p. 116). Molecules with this structure are readily activated and powerfully activate other molecules. The interest of this chapter is mainly in this double bond but it so frequently affects the properties of substituent groups on the chain that the ternary derivatives also will be considered. Biologically of course they constitute the larger percentage.

Olefines form addition compounds equally readily with hydrogen and

with the electronegative elements, as would be expected from the intrinsic properties of carbon (p. 29). The bias however is towards oxidation since this is more exothermic than hydrogenation and free oxygen is abundant. A number of them therefore are valuable antioxidants, maintaining other metabolites in a state of reduction.

The simplest olefine, the gas ethylene, $CH_2{=}CH_2$, plays a role in the ripening of apples. It is produced by the ripe fruit and added ethylene hastens ripening. The gas also induces growth and other tropistic responses in plants, and initiates root development (Sexton, 1953, p. 367). As little as one volume in 10^7 of air gives a positive effect, so that it may well be a normal biochemical agent. Other simple olefinic compounds have a similar but smaller effect. However, some saturated halogenated derivatives also have some activity of the same kind, for instance ethylene chlorhydrin, $CH_2Cl{\cdot}CH_2OH$ (also well known as one of the few solvents for melanin (p. 188)). The proximity of chlorine and hydroxyl makes the compound more reactive than paraffins or their simple halogen substituents, which are almost equally inert (p. 457). It may be noted in this context that this inertness is the main reason why alkyl halides and some other series are rare biologically.

Most of the biological olefines seem to be based on a C_5 unit which is either *isoprene* (Fig. 16, I) or a nearly related molecule. In general they are simple polymers of this unit, the only extensively polymerising biological lipid; in this, in its double-bond system, and in a strong tendency for the polymers to form ring structures, the isoprenoid pathway stands sharply distinct from that of the fatty acids. At the same time there do seem to be cases of convergence, i.e. very similar compounds produced by both pathways, and in any case the isoprenoid unit is synthesised from acetic acid (Kit, 1960). The pathway is probably:

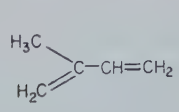
$$2CH_3COOH \longrightarrow CH_3CO{-}CH_2COOH \longrightarrow CH_3COCH_3 + CO_2$$

$$\begin{matrix} CH_3 \\ CH_3 \end{matrix}\!\!>\!CO + CH_3COOH \xrightarrow{-H_2O} \begin{matrix} CH_3 \\ CH_3 \end{matrix}\!\!>\!C{=}CH{\cdot}COOH \xrightarrow{-O_2}$$

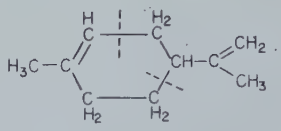
$$\begin{matrix} CH_3 \\ CH_2 \end{matrix}\!\!>\!C{-}CH{=}CH_2$$

Speaking teleonomically this is a way of obtaining unsaturated lipids from acetic acid. Isoprene, which does occur free in living organisms, may be the first stable stage and subsequent syntheses by polymerisation are perhaps not so very different from what happens in the fatty acid pathway, effectively a polymerisation of acetic units. Both pathways make use of C_1 side chains, i.e. bonds are formed with non-terminal carbons ('secondary bonds'), temporarily in the condensation of each malonyl

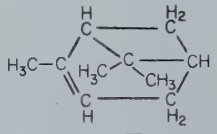
unit in the fatty acid pathway, but permanently in isoprene. The methyl side chain restricts resonance in the isoprene unit and this may be its useful significance, to stabilise the unit. By further polymerisation, however, there is unlimited scope for large resonating systems (p. 116).



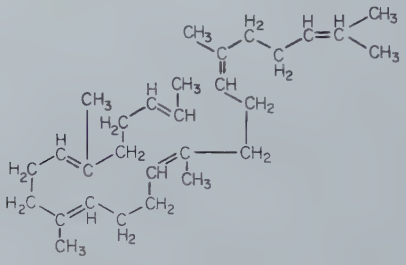
I
Isoprene



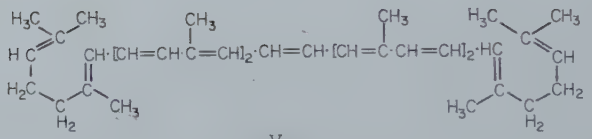
II
Limonene



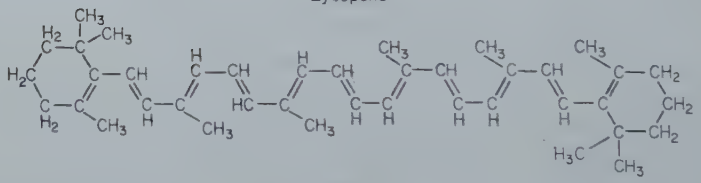
III
Pinene



IV
Squalene



V
Lycopene



VI
β-carotene

Fig. 16 Biological Olefines (*continued on pp. 111 and 113*).

Each polymerisation of course resolves one double bond and is no doubt spontaneous: in order to produce extensively resonating systems further desaturation is necessary after polymerisation. Ring closure similarly is spontaneous, produces a more stable structure, but does not yield aromatic rings without further desaturation. In fact the binary and ternary aromatic compounds are biosynthesised via another pathway

(p. 187), though some of the steroids and other isoprenoids approach full desaturation in some of their rings. The initial double bonds of isoprene arise from the carboxyl =O group, of course, by tautomeric shift.

The natural open chain polymers of isoprene are mostly large molecules: for stability in smaller units ring closure may be essential. There is a large group of dimers, the terpenes, the largest group of essential oils of plants, with drier, more aromatic odours than the aliphatic esters (p. 98). All are rather volatile, of course, distilling at temperatures below 200°C. *Limonene* (Fig. 16, II) is one of the best known: the conventional numbering does not bring out the isoprene construction, which has been indicated by the dotted lines. Even more widespread is *pinene* (Fig. 16, III), in conifers and in other plants, such as eucalyptus, fennel, carrot, coriander, rosemary, neroli, lavender and lemon. It is the main oil in turpentine and is further interesting for its *para*-bridged structure, a further double bond being resolved in forming this. Another terpene is *phellandrene* in *Eucalyptus* and fennel.

Tri-isoprenes, usually known as sesquiterpenes, occur in living organisms, and there are also tetramers, the diterpenes. One of the shortest open chain polymers is *squalene* (Fig. 16, IV), a hexamer, abundant in elasmobranchs, and believed to be the near precursor of the steroids (p. 118). As represented it implies the mode of multiple ring closure to form steroids, with a side chain. There is strong evidence that squalene *is* synthesised from acetic acid, via isoprene, in the body.

The carotinoid pigments, a very large group, synthesised by plants and secondarily exploited by animals, are octamers. *Lycopene* (Fig. 16, V) is structurally a suitable parent substance, though not necessarily the precursor of all the others *in vivo*. As shown, it implies the way in which hexene (*ionone*) rings are formed at each end of the molecule in many of the group, for instance in *carotene* (Fig. 16, VI), the best known of the group. Many of them are ternary derivatives (p. 112). They owe their strong colours to the very extensive double bonding (p. 116), which must be completed as indicated, after polymerisation.

Even higher polymers of isoprene are the natural rubbers, which can be broken down to isoprene by careful heating out of air. *In vivo* the rubber is in the form of latex, an oil-in-water emulsion, which readily breaks as the water evaporates, and leaves a resilient, solid seal to wounds in the rubber tree. There is further polymerisation on exposure and the long kinked molecules, with methyl side chains, entangle extensively. They are sufficiently kinked, and therefore resilient, to be drawn out collectively in any direction and to recoil on release, and this is the unique peculiarity of rubber. It is said to have rubbery elasticity, quite anomalous among elastic materials; the molecules of most materials become

disordered on stretching but those of rubber become more aligned and orderly and liberate heat. This is absorbed when the rubber contracts to its higher entropic state of disorder. Some of the commercially synthesised rubbers have in addition ring structures in the molecule, which will tend to improve tangling.

The methyl side chains of the isoprenoids merit more attention. They are rarely used in the bonds of polymerisation and of ring closure, and their function may be largely restricted to the special property of rubber and to the synthesis and stabilisation of the initial units. A methyl group does not occupy much space and need not interfere seriously with most functions. On the other hand in some compounds methyl groups are of vital importance, as will be seen, and their removal from the synthetic pseudo-steroid, stilboestrol (Fig. 17, VI), destroys its activity. The methyl group is cationoid, whereas the $=CH_2$ group is anionoid, so that the association between the two in the present group should promote appropriate types of chemical activity.

9.1 Oxy-isoprenoid Compounds

Since the oxygen of the acetic acid must be sacrificed to form the —C=C— double bonds of the isoprene it is probable that *in vivo* the ternary olefinic compounds are derived secondarily, from the hydrocarbons, by oxidative synthesis. The ready oxidation of the olefines has been stressed already; dilute potassium permanganate will oxidise terpenes all the way to glycols (Read, 1935, p. 613).

In the terpenes oxygen greatly increases the scent; the water-solubility increases in parallel, and this is a reminder that chemicals must dissolve in fluid on the olfactory epithelium in order to be smelled. Solution in alcohol enhances the scent because the ethanol itself is so hydrophil, and also helps to volatilise the oil. The oxygen probably stabilises the molecule, also, since a number of the oxy-terpenes are open-chained. *Geraniol* (Fig. 16, VII), from indian geranium and from attar of roses, is an example. Its ester with acetate is an essence in some species of *Eucalyptus*, and illustrates the wide distribution of most of these terpene essences. In ring-closed form it occurs as *terpineol*, in conifers. Its aldehyde, *geranial* or *citral*, is an oil of citrus fruits, lemon grass and some species of *Eucalyptus*. *Citronellal* (Fig. 16, VIII) differs from this only in having the double bond in the chain saturated; it occurs in the lemon-scented eucalyptus. The corresponding alcohol, *citronellol*, therefore has only one unsaturated bond in the molecule; it occurs in attar of roses. As already noted (p. 102) citral and citronellal are present in the poisons of some insects (Roth and Eisner, 1962). *Mint* and *camphor* (Fig. 16, IX) are two other terpenes

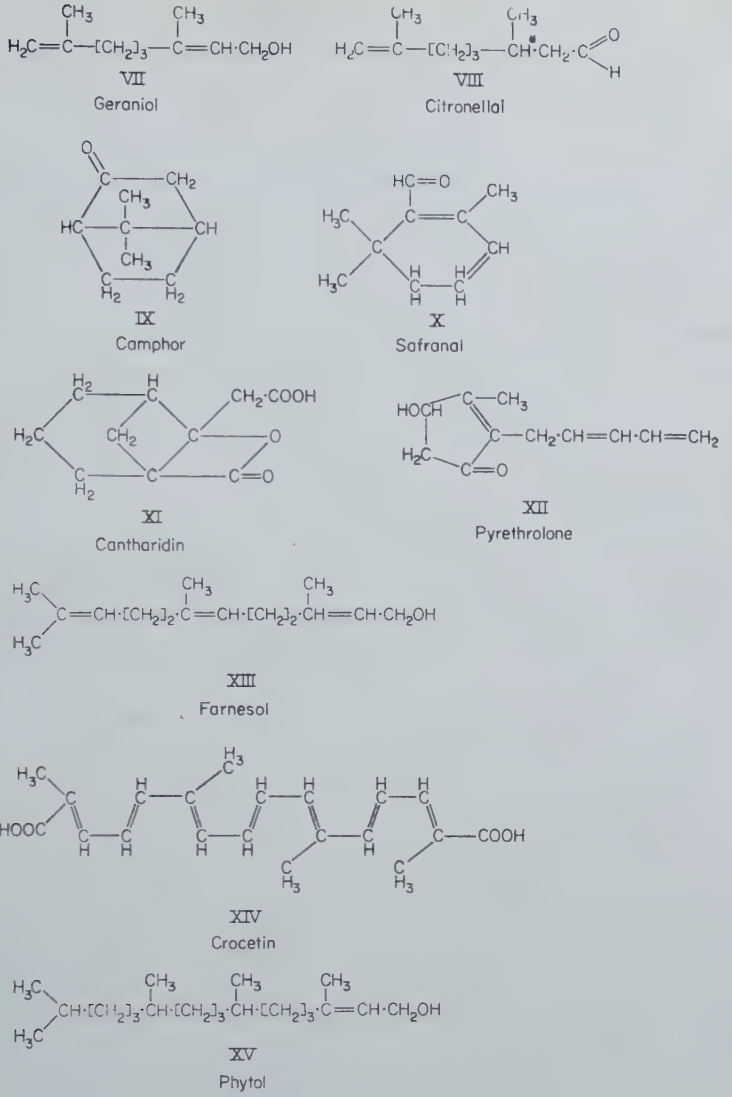


FIG. 16 (*continued*).

with distinctive smells. This has been traced to the —C—C(=O)—CH_2 grouping (Haagen-Smit, 1952), and it is interesting that as the size of the ring containing this group is increased the smell changes through cedar and musk to civet (p. 103).

The camphors and other oxy-terpenes have pharmacological, and in some cases physiological, significance. The camphors are *para*-bridged

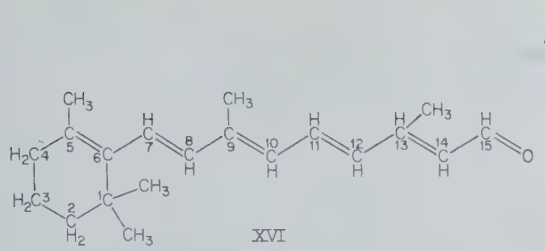
aldehydes and ketones, and *borneol* is the secondary alcohol corresponding to the ketone shown. *Methone* is another ketone and its alcohol, *menthol*, has mild antiseptic and anaesthetic properties. *Eucalyptole* or *cineole* is an internal ether, between two alcohol groups of the same molecule, and again is a mild antiseptic. *Safranal* (Fig. 16, X) is an aldehyde and appears to have male-determining properties in the Algae. *Cantharidin* (Fig.16, XI), with a good deal of oxidation in the side chains, two of which have formed a lactone ring, is the main principle of the poison of meloid beetles (Maass-Berlin, 1937).

The *pyrethrins*, first isolated from pyrethrum, are powerful insecticides and are structurally interesting because they have a pentene and not a hexene ring (Fig. 16, XII). *Pyrethrolone* is perhaps the parent substance, and is a C_{11} compound, like chrysanthemum dicarboxylic acid (p. 105). Pyrethrins cause neuromuscular paralysis. They are inactivated if the side chain is saturated; in addition the hydroxyl group must be esterified and the ketone group is essential for activity.

There are a number of oxidised sesquiterpenes of biological importance. One is *farnesol* (Fig. 16, XIII), which has many of the actions of the insect hormone, *neotenin*, and may be identical with it (Schmialek, 1961). Including the steroids and the algal termones there are a number of hormones of isoprenoid origin. *Crocetin* (Fig. 16, XIV), or saffron, the yellow pigment of the crocus, is one of the algal sex-determinants, and is a diterpene with a complete system of conjugated double bonds.

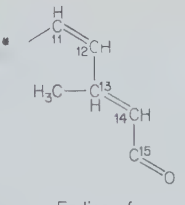
By contrast, *phytol* (Fig. 16, XV), the long side chain on the porphyrin ring of chlorophyll (p. 320), is a diterpene alcohol with an almost fully saturated chain. *Vitamin A* is a similar diterpene with a terminal alcohol group, but with the other end of the molecule closed to a hexene ring. It is known to be formed by the exact fission of the carotene molecule (p. 109), with hydroxylation of the groups so exposed. As is well known, vitamin A controls the morphogenesis of epidermis and bone, and the general resistance to infection in mammals. Isoprene derivatives therefore act as hormones, and vitamins, and in other useful capacities to be considered, as well as in chemical defence.

One important function of vitamin A is as a source of its aldehyde, *retinene*, the photosensitive pigment of the retina of vertebrates and many invertebrates. It is normally conjugated with a protein, *opsin*, to form the complete active pigment, *rhodopsin*. Retinene is the 11-*cis* stereo-isomer of this aldehyde, and only this isomer of the large theoretical number is active (Fig. 16, XVI). This is particularly interesting because the 11-*cis* form is a sterically hindered one, and so is one of the least 'probable' thermodynamically. However, once formed the 11-*cis* form is kinetically very stable, to all agents except light, and this is no doubt ideal. The all-

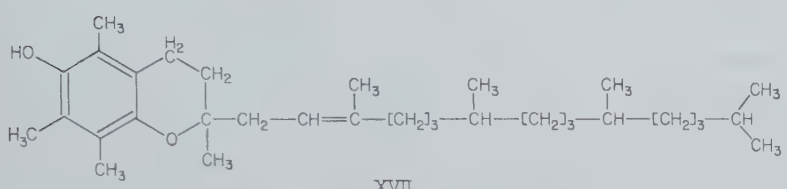


XVI

All-*trans* retinene 1

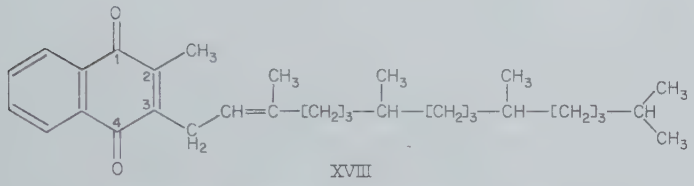


Ending of 11-*cis* retinene 1



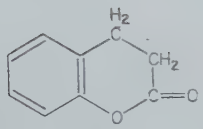
XVII

α-tocopherol (vitamin E_1)



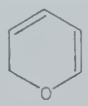
XVIII

2-methyl,-3-phytyl-1, 4-naphthoquinone (vitamin K)



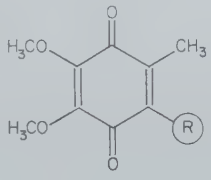
XIX

Coumarin



XX

Pyrane ring



XXI

Co-enzyme Q (ubiquinone)

FIG. 16 (*continued*).

trans form is the most probable configuration and the 11-*cis* isomer reverts to this when it is activated by light. All-*trans* shows complete resonance (p. 116), which is suppressed in 11-*cis* (Finar, 1959), so that the *cis-trans* change may play an essential part in the actual photochemical event. It is all-*trans* vitamin A which is stored in the liver, but the 11-*cis* (neo *b*) alcohol is formed in the retina ready for oxidation to the corresponding aldehyde. The retina can also isomerise all-*trans* retinene to the 11-*cis* form.

A number of steps in the rhodopsin cycle are known to be spontaneous. The conjugation between retinene and opsin is exergonic and the energy

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in fact promotes the oxidation of the alcohol to the aldehyde. It also removes the products of this reaction and accelerates it in that way (Glass, 1961, p. 901). No doubt there is significance also in the fact that photodecomposition affects other points in the chain and not the terminal very reactive aldehyde group itself. There are parallels to this in the biochemistry of the sugars and of other metabolites. Possibly also significant here is the role of a long chain aldehyde in light-*production* by some bacteria (p. 103).

Marine and terrestrial vertebrates have the retinene shown but freshwater fishes and amphibia have *retinene*$_2$, with an extra double bond in the ring, in a position to extend the resonance ranges. The complete pigment, *porphyropsin*, absorbs light maximally at 522 mμ therefore, compared with 500 mμ for rhodopsin. These differences may be adaptive in media with slightly different light-transmission maxima. It is known that the absorption maximum of the pigment of deep-sea fishes is at a shorter wavelength than in surface-living forms, and this is correlated with the greater penetrating power of the short wavelength rays. Here, therefore, there has been much exploitation of differences in molecular detail for specific biological purposes; of the possibilities it is not always the most probable which is used and systemic means of circumventing this are implied.

Most of the oxy-carotinoids are octaisoprenes, like carotene (p. 109) and lycopene, and the molecule is usually in the all-*trans* form. There are other smaller polymers of biological importance, but it seems preferable to consider the carotinoids first because of their relative uniformity of structure and simple dimer relationship to vitamin A. As already indicated they form a large group of brightly coloured pigments, often integumental but also associated with germ cells (Goodwin, 1952) and other growing tissues. This is the main group of fat-soluble pigments in living organisms. The colours range from scarlet through orange to yellow, though some are violet or blue. Conjugation with protein tends to produce the latter colours, which turn orange on denaturing the protein.

There are so many coloured substances of biological importance that a word about their significance is appropriate here; the carotinoids illustrate most of the relevant points. Substances are coloured if they absorb sufficiently strongly and selectively a particular, localised band of wavelengths in the visible spectrum: their colour is the residual spectrum which they transmit, dominated by the complementary colour to that absorbed. The subjective phenomenon of colour vision is a physiological one, of course, in which the visible spectrum is not an open band but a closed circle, with infrared apposed to ultraviolet; on this circle complementary colours occur at opposite ends of each diameter. Consequently, as absorption shifts from the shortest visible waves in the violet through

the spectrum to the longest red wavelengths, the transmitted colour seen changes from yellow through red to purple (red+violet), and so round through violet and blue to green. This is rather quaintly known as a deepening of the colour, and has no implication of an increasing saturation or of intensity.

If a molecule absorbs electromagnetic radiation this is because radiation of that wavelength and quantum-energy value enhances electronic oscillations and other motions in the molecule. The radiation which does so is that 'tuned' to the motions in question. The motion of most importance chemically and biologically is the raising of an electron from the highest occupied orbital to an unoccupied orbital, since this activates the atom concerned, i.e. makes it unstable, and leads to chemical reaction as the electron shifts to a more stable situation, in the same or in some other molecule. Electron jumps of this kind need the help of energy of a high quantum value and short wavelength. Some require ultraviolet radiation but biological reactions depend mainly on covalent bonds, requiring 40–90 kcal/mole for their rupture, and this is appropriate to activation by rays of 320–711 mμ. Valency bonds require 20–100 kcal/mole corresponding to activation by 'light' of 280–1430 mμ. The centre of this, and virtually the whole of the range for covalent bonds, is in the visible spectrum (380–790 mμ), and so most of the useful activating substances are coloured. In practice they may be activated by chemical or other forms of energy and not by light, but their colour remains a useful label and a rather precise measure of the quantum value necessary for their activation.

Those activated only by light of shorter wavelength than the visible range can be detected in the same way with an ultraviolet spectrophotometer, but appear colourless to the eye unless they also absorb in the visible range. Examples of this group are the aromatic amino acids, the nucleic bases and some others. Absorption of the low energy of infrared radiation keeps us warm and speeds reactions, but does not absolutely initiate many that are biologically important.

For activation by radiation the molecules must already have electronic motions of near threshold value for the orbital transition, and one of the first requirements for this is molecular asymmetry (Finar, 1959, I, p. 741). It is notable that in fact virtually all pigments, and most other metabolites, have some asymmetry of the molecule. Asymmetry confers dipole properties (p. 19) and the intensity of light-absorption depends on the amount of change possible in this electric dipole of the molecule—on the *transition dipole* as it is called.

The second type of property which is valuable in the molecule is one which decreases the difference in energy levels between the resting, or *ground-state*, of the relevant electron and its active state, since this permits

the use of external energy of lower quantum value and longer wavelength, more abundant in the part of the solar spectrum reaching the biosphere. Reactions which otherwise would need ultraviolet for activation can then be initiated by visible light. Resonance is a property of this kind: it is a spontaneous oscillation between alternative electronic and bonding states in a molecule, and is tantamount to a sub-threshold activation of the relevant electrons. Resonance is particularly well developed in molecules with double bonds, which readily exchange the double bond state with a neighbouring single bond, accompanied by a shift of atoms between the groups bonded:

$$H_3C{-}CH{=}CH{-}CH{=}CH{-}HC{=}O \rightleftharpoons H_2C{=}CH{-}CH{=}CH{-}CH{=}HCOH$$

The double bonded groupings most important in living organisms, in decreasing order of potency for resonance phenomena, are: N=O, C=S, N=N, C=O, C=N, C=C. Compounds with an N=O bond are most easily activated and those with C=C least so, though much more easily than single bonded structures. If there is more than one of these double bonds in a molecule their effects may summate, particularly if they form a conjugated series of regularly alternating double and single bonds, as shown above, and resonate as a single unit. For this purpose the C=C bonds are evidently uniquely suitable, despite their relatively low individual resonance potency, since the other types rarely occur in series. A single —CH=CH— can be activated only by the very short ultraviolet radiation but six in series begin to absorb in the far violet, transmitting yellow, and there is the usual *bathochromic* shift with increasing numbers of conjugated double bonds. This probably explains why the conjugation of an orange carotinoid with protein shifts the colour round to blue, the C=O groups of the peptide being conjugated in series with the C=C system. The bathochromic shift also accompanies the addition of each ring of the polycyclic aromatic compounds (Chapter 10): benzene is colourless, naphthalenes mostly yellow, anthracenes red and the aphins (p. 136) often green, or even black, which is the end of the bathochromic series. Black pigments absorb strongly in the infrared, in fact, and this accounts for their heat-absorbing power. If the melanins (p. 188) are poly-indols they probably are at least tri- or tetramers.

The energy requirement for activation is minimal when the charge on the molecule is in a terminal position. Vitamin A is therefore ideal; many of the other carotinoids have their polar groups on the ionone rings, i.e. effectively terminal. Molecules with the maximal number of atoms in a single plane show greatest resonance so that the ionone rings further contribute to this. The *trans*-isomers are more coplanar than the *cis*-forms and are more strongly resonant, as already seen for retinene.

Resonance is enhanced and colour 'deepened' by *auxochromes*, that is radicals with a key atom having a lone pair of electrons. Auxochromes therefore facilitate reactions which otherwise would demand energy of higher quantal value.

Isoprenoid compounds, and carotinoids in particular, are unique as the only extensively resonant, open chain, binary and ternary biological compounds. Most others are polycyclic and contain nitrogen in addition. It should be stressed again that to produce the complete conjugated system of most carotinoids there must be considerable further desaturation after the polymerisation. Virtues of the carotinoids for resonance-activation include the ability to switch resonance on and off, by the *cis–trans* change, to activate at a wide range of energy levels by varying the degree of polymerisation and of desaturation, and to activate over considerable spatial distances by the resonant shift of electrons or protons throughout the length of the molecule.

There are always carotinoids associated with chlorophyll and they may play an essential role in photosynthesis by passing on the light energy initially absorbed to the main pigment. It is a most surprising fact that chlorophyll itself absorbs maximally at 680 mμ, in the red (transmitting the complementary green), which is far from the blue-green wavelengths most abundant in sunlight reaching the earth. It is also far from the wavelength most effective in activating water or carbon dioxide (Bayliss, 1959), the two key molecules in the process (p. 40). It is therefore thought that energy is absorbed in the blue-green by the carotinoids and passed on as energy corresponding to 680 mμ light, to chlorophyll. Porphyrins such as chlorophyll are very stable (p. 310), and are ideal for holding and storing energy once it is trapped (Wald, 1959), but it is curious that a porphyrin capable of efficient initial absorption should not have been evolved: the two-stage transfer no doubt provides an extra point for controlling the process.

Other biologically important isoprene derivatives include vitamins E, *α–tocopherol*, the fertility vitamin, and K, *naphthoquinone*, the blood-clotting factor (Fig. 16, XVII, XVIII). The side chain of vitamin K is the same as in chlorophyll and that of E is a shorter but similarly saturated one, in contrast to the opposite tendency in the carotinoids. Their ring structures, however, tend to be more fully unsaturated. Only one ring of E is fully unsaturated and so it absorbs and is activated by only light of short wavelength 290–300 mμ. Both rings of K are fully unsaturated and resonance is further increased by the quinonoid C=O groups so that it is coloured and absorbs at a number of wavelengths between 240 and 330 mμ. Its physiological action is due almost entirely to the ring structure, since a synthetic analogue without any side chain, *menadione*, is equally

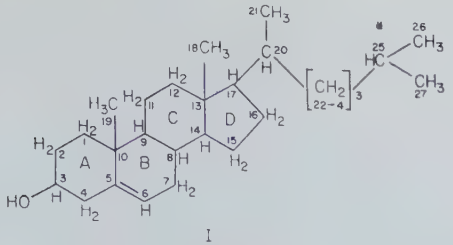
effective. The side chain therefore may be an inert 'handle' or holdfast, for some purpose. The fully unsaturated rings have the properties of aromatic rings (Chapter 10), although they are almost certainly of common, isoprene origin with the side chains. The hydroxyl of the unsaturated ring of vitamin E has alcoholic properties and not the acidity of phenols (p. 131), so that the double bonds may partially resonate to the other ring.

In sweet clover there is an antagonist to vitamin K which inhibits blood-clotting in animals feeding on the plant. This is the notorious *coumarin* (Fig. 16, XIX). Like menadione it has no side chain, but in addition it is based on the *chromane* ring structure of vitamin E rather than on the naphthalene system of K. Some of the quinone derivatives of vitamin E themselves have some haemorrhagic activity (Sexton, 1953, p. 200), while coumarin antagonises also some other naphthyl derivatives such as naphthyl acetic acid, the plant hormone (p. 136). The *pyrane* ring (Fig. 16, XX) in fact is a chemical analogue of benzene so that these two vitamins are chemically quite close analogues. It should be noted in passing that vitamin E is a further isoprene concerned with reproduction.

Another isoprenoid which is very relevant here is coenzyme Q or ubiquinone (Green, 1962). It consists of the benzoquinone derivative (Fig. 16, XXI) with a side chain, R, of 6 – 10 isoprene units according to species. It is reversibly reducible to the diquinol state and is a link in the electron-transfer sequence of terminal oxidation (Fig. 20, p. 133). In *Mycobacterium phlei* a vitamin K derivative acts as an alternative to CoQ in this sequence, so that the quinone structure and probably the side chain are the significant components. The side chain of CoQ is of course more unsaturated than that of vitamin K.

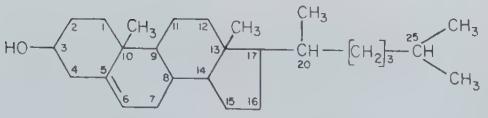
9.2 Steroids

There is now little doubt that the steroids are synthesised *in vivo* from isoprenoid intermediaries in the way indicated (p. 109). The resulting *cyclo*pentanophenanthrene condensed ring system (Fig. 17, I), and certain other features, are common to all biological steroids: all have HO— or =O at position 3 and CH_{13} at 13, for instance. In other details, however, there is great variation, and the steroid plan has radiated very widely in the physiological chemistry of living organisms. There is a distinct hiatus between other isoprenoids and this relatively complex ring, also known as the *sterane* ring when it is in the fully saturated condition (*cyclo*pentanoperhydrophenanthrene). In fact ring closure of squalene leaves one double bond in ring B, and the further tendency is towards more desaturation, though never as far as in phenanthrene itself. The female gonadal hormone, oestrone, has ring A fully unsaturated and



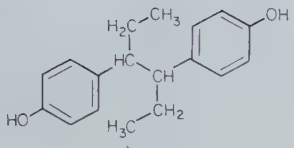
I

Cholesterol



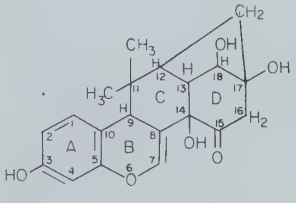
II

Cholesterol redrawn to show 3-17-25 axis



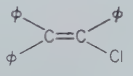
III

Stilboestrol



IV

Miroestrol



V

Triphenyl
ethylene chloride
(a pseudo-oestrogen)

FIG. 17 Steroids and related molecules.

equilenin from the mare has both A and B in this state. Rings C and D are never unsaturated so that as in vitamins E and K the resonance is restricted to the end of the molecule remote from the side chain.

The ring system of the steroids lies roughly in one plane except that the individual rings are in the chair (*trans*) or boat (*cis*) form already seen in the hexoses (p. 43). The bodies of rings A and B are coplanar and have the same *trans*-shape in one series of the steroids, the cholestane series, and the same *cis*-form in the other main series, the coprostanes or *allo*–cholestanes. Rings C and D are again coplanar, so far as the pentane ring permits, and have the same *trans*-form (Fig. 18). This is also identical in the two main

series, so that they differ in A and B only. The contiguous ends of the four rings are coplanar and perpendicular to the planes of their bodies, so that the molecules provide very geometric and specific constructional pieces. Like the pieces in children's constructional kits these will fit together in several distinct patterns, giving a number of crystal forms (Bateman, 1945). However the different steroids very readily form mixed crystals, since there are only the two types of piece, and this has set a severe technical problem in purification.

The molecule is asymmetrical in a number of ways, which may all have their biological significance. It is reminiscent of a key which will fit only special locks. However, an approximate axis can be recognised

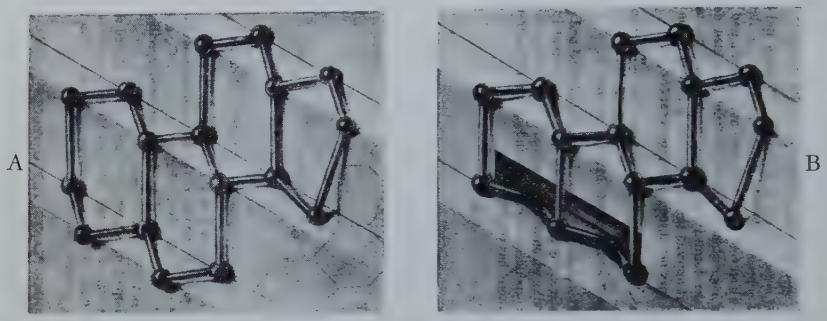


FIG. 18 Three-dimensional models of two main types of biological steroid. A, cholestane, and B, allocholestane (coprostane) series. (From *Textbook of Biochemistry*, by E. S. West and W. R. Todd, by courtesy of the authors and the Macmillan Co. of New York.)

(Fig. 17, II) running from the invariably present polar group at 3 to the end of the side chain, and the molecules of the sterol group, in particular, orientate rather readily at interfaces, with the 3 polar group in the aqueous and the side chain in the lipid phase. There are no fewer than eight asymmetric carbon atoms in the cholestanes, at positions 5, 8, 9, 10, 13, 14, and 17 in the ring system and at 20 in the side chain, so that there are 256 (2^8) possible optical isomers; the number is of course smaller, but still quite considerable, in the *cholestenes*, steroids with double bonds in the rings. In practice most of the biological steroids prove to be optically identical, further emphasising their radiation from a very restricted group of sterenes. Variation occurs only at positions 3 and 5. When the substituents at these positions are both on the same (β) side of the ring system as the CH_3 at 10, which is used as a reference, they are conventionally called *cis*-forms, and *trans*-forms when on the other, α, side. The α–side is above the plane of the paper in the representation of cholesterol (Fig. 17, I).

Although all are built on the same plan, several functional groups can

be recognised, based on minor structural differences. The sterols, including vitamin D, have a hydroxyl group at 3 and a long side chain at 17, while the bile acids have a shorter chain, ending in a carboxyl group. The aglycone moiety of digitalis and related cardiostimulatory ethers is a steroid with the side chain closed to a lactone ring (p. 50), while the surface-active plant saponins have the chain closed to an internal ether ring. Finally the steroid hormones have the side chain shortened to an hydroxyl or =O group. There has been a general evolutionary trend towards reduction in length of the side chain (Bergmann *et al.*, 1943), so that the hormones may be the most highly evolved steroids. The evolutionary trend makes the steroids of great taxonomic value: bacteria in general lack steroids of all kinds (Florkin, 1960, p. 41), plants and the lower invertebrates have members with longer side chains than in the higher Metazoa, the ophiuroids and echinoids have cholesterol whereas the other echinoderms have a different alcohol, stellasterol, and the gastropods have cholesterol whereas the Lamellibranchs never do. As for the glycerides (p. 93) the position is somewhat complicated by an influence of dietary steroids on the types present in animals: herbivores tend to have members with long side chains, approximating to those of their plant food, and marine animals have members with longer side chains than those of freshwater and terrestrial animals. This is a close parallel to the variations in glyceride-distribution and it is further significant that the total number of carbon atoms in the two groups of compound is very similar, ranging down from C_{30} or so but settling on C_{16} to C_{18} in the most highly evolved members; since the two pathways of synthesis are different the resemblance must be due to selection for similar functional properties, such as thickness of monolayer built by the molecule.

The side chain improves the adsorption of polycyclic molecules of this type on to other materials, so that the hormones presumably do not adsorb so well as cholesterol. A similar contrast may perhaps be anticipated between echinochrome (p. 137) and vitamin K andthis is possibly the kind of handle suggested for the side chains of vitamins K and E (p. 118).

The main structural features of some of the more important biological steroids are summarised in Table 21. The hydroxyl group on an alkane ring behaves as would be expected, like that of an alcohol, while it behaves as an acid when on an alkene ring. The =O substituent behaves as an aliphatic ketone since it is rarely found on a fully saturated ring and never in pairs, the essential feature of the aromatic quinones; at the same time it does have redox properties and so is intermediate between the two types of >C=O group. The significance of some of the other structural details is known. The carboxyl group of the bile acids, together with the number of hydroxyl groups, makes these acids and their esters

very surface-active and water-soluble. The saponins are sterol glucosides and again have the necessary groups for high surface-activity. The cardio-toxicity of the digitalis and strophanthus glycosides is associated, as might be anticipated, with the unsaturated lactone ring formed by the side chain of the steroid moiety (Fearon, 1949, p. 197). *Progesterone* and the adrenal cortex hormones have an =O group at 3, androgens have =O,—OH or —H, and oestrogens have —OH or —H. Androgens and oestrogens have no carbon side chain at 17 but a polar group instead, so

Table 21. The More Common Mammalian Steroids: Summary of Structural Features

Steroid	Total number of C–atoms	Number or position of double bonds	Nature of side chain at 17	Nature of groups at positions: 3	10	11	13	Other features
Cholesterol	27	1 (4 : 5)	C_8	H, HO	CH_3	H, H	CH_3	
Cholic acid	24		$-C_4-COOH$	H, HO	CH_3	H, H	CH_3	OH at 7 and 12
Deoxycortico-sterone	21	1 (4 : 5)	$-CO.CH_2OH$	=O	CH_3	H, H	CH_3	
Corticosterone	21	1 (4 : 5)	$-CO.CH_2OH$	=O	CH_3	H, HO	CH_3	
Progesterone	21	1 (4 : 5)	$-CO.CH_3$	=O	CH_3	H, H	CH_3	
Oestrogens	19	2 to 4	=O or —OH	H, OH	H	H, H	CH_3	
Androgens	19	0 or 1	=O or —OH	H, OH or = O	CH_3	H, H	CH_3	

that both ends of the molecular axis (Fig. 17, II) are polar. The significance of some of these features is still uncertain but it is known that the polar–apolar type form bimolecular 'cells' in their crystallisation pattern, whereas the bipolar type form single layers (Bateman, 1945, p. 156). Esters formed with small molecules orientate at interfaces in the same way as the glycerides but those with long chain partners have the latter folded back so that its hydrocarbon end faces the lipid phase, again as in similar fatty acid esters.

The differences between the various steroids give the impression that the steryl nucleus is invariable and that relatively minor variations in degree of saturation or in one or two substituent groups account for all differences in properties. The molecules might seem to be as unique as any in living organisms. However, in vitamin D the B-ring is open and the properties of the oestrogens are mimicked by much more simple, synthetic compounds such as stilboestrol (Fig. 17, III), here represented so as to suggest its possible geometrical resemblance to the natural steroids. If the two potential central rings were closed the system formed

would be *isosteric* with the sterene system, and there is in fact a natural compound of this kind (Barrington, 1963, p. 105), *miroestrol* (Fig. 17, IV), in the roots of the Thailand plant, *Pueraria*, and it has oestrogenic potency. It differs from the steroids proper not only in D being a C_6-ring but also in B being pyranose. Stilboestrol is actually more potent than the natural oestrogens and the related hexoestrol, with only saturated bonds in the chain-link, is even more potent. Some of the synthetic forms of vitamin D are likewise more potent than the natural form and it might therefore seem that natural biological compounds are not uniquely potent for their purpose after all. However, we need to know more about all the physiological ramifications of the effects of the synthetic substitutes before being certain of this. Compounds of the form shown in Figure 17, V, also mimic the oestrogens, and even *p*-propyl phenol has some oestrogenic activity. In some respects, therefore, the specificity of the steroids is rather low, and they are a group which merits a good deal more attention. Removal of the terminal carbon atoms from the side chains of the link in hexoestrol destroys its activity so that the pseudosterene ring system would seem to be essential (Jellinek, 1958), and efficient mimics are close mimics. At this level of molecular complexity, perhaps, we are concerned with the finer grades of specificity having quantitative, rather than fundamental qualitative, differences in action.

The sterols, that is the steroids with an alcoholic —OH substituent, are soluble in the apolar lipid solvents, but in alcohol only on heating, and they are virtually insoluble in water. It would therefore be surprising if their main function were not in relation to lipid media and lipid–aqueous interfaces. The more hydrophil steroids of course act as internuncials between the two phases, like the soaps and the phospholipids, and some are freely water-soluble. Like some of the modern detergents they are surface-active even in acid media, where soaps are ineffective. It has already been seen that the sterols have a suitable molecular size, shape and structure for incorporation into molecular films, and it is found that haemolytic substances are those which penetrate cholesterol films (Sexton, 1953, p. 72). It seems probable that the flattened molecule, set on end across the membrane, will restrict movement in the plane of the membrane more than the filiform fatty acid molecule, limiting it to the direction of the plane of the molecules, which naturally pack with these planes parallel. The steroids do in fact help to stabilise membranes in this way and Booij and deJong (1956) refer to them as the cement of the membrane. On the other hand flat molecules should move fairly easily in a direction perpendicular to the plane of the membrane and sterols do seem to penetrate more rapidly than would be expected from their general molecular size and complexity. Labelled steroids are found

to penetrate cells and accumulate on the intracellular particles (Waddington and Goodhart, 1949), and this may be one of the reasons for the carcinogenic activity of polycyclic hydrocarbons—including some steroids.

The biological action of steroids is found to be proportional to their effect on the surface potential of a monolayer of a protein such as gliadin, so that no doubt in living membranes they are associated with protein in much the same way as the glycerides (p. 83). They do not act merely as a stabilising cement but also control the transport of sodium, and therefore, indirectly, that of water. The mineralocorticoids of the adrenal cortex are outstanding for this purpose, depressing sodium-excretion and conserving its level in the body. Addison's disease, due to a deficiency of this group of hormones, is characterised by great muscular weakness, and sodium is known to have a stimulatory effect on the excitable tissues (p. 421). *Strophanthidin*, the steroid of one of the cardiac glycosides, and its relatives also have a powerful effect on sodium-transport (Berliner, 1959). *Cortisone*, another of the adrenal steroids, alleviates rheumatic stiffness, probably by improving water-elimination, and so reducing oedema, as well as inflammation. *Stigmasterol*, a plant sterol, also is said to have an anti-stiffness action. Oedema, or waterlogging of the tissues, is associated with high cholesterol in the plasma (Fearon, 1949, p. 190); cholesterol is known to have a high affinity for water, as in lanolin (p. 100), and since it is at the same time very insoluble in water it may form a repulsion hydrate typical of the hydrocarbons. As a membrane-component, therefore, it may act as a water carrier, readily releasing it on the far side of the membrane.

Transport of calcium, across the gut-wall in particular, is promoted by vitamin D and this is a main aspect of its anti-rachitic action. Calcium salts are insoluble in alkaline media so that acid media, and steroid detergents, seem essential for their absorption.

The stabilising action of the less soluble steroids, such as cholesterol, is undoubtedly one of their more unique biological properties, and in most cases it is applied at the molecular membranes. Haemolytic substances can be shown to disperse protein films if these are not stabilised by sterols (Sexton, 1953, p. 72), and protein coacervates also are stabilised by them (Machovschi, 1959). Sterols might be said to precipitate the membrane-protein and cholesterol can be shown to precipitate normal globulins (Makari, 1960). It is probable that the action is reciprocal, as in the case of the associations between proteins and nucleic acids (p. 370), polysaccharides (p. 60) and glycerides (p. 83). As much as 75 per cent of the oestrogen of the blood is associated with protein and this again may be for stabilisation, helping to prolong the action of this steroid.

Cholesterol protects even against the bile salts, saponins and other

detergent steroids (West and Todd, 1957, p. 930), as well as against the phospholipids, which have a similar lipotropic action (p. 297). It counteracts the haemolytic action of phospholipids (p. 296), perhaps by preventing their hydrolysis to the active form lysolecithin since it protects the body against the haemolytic action of the venoms of the cobra and rattlesnake, known to work by enzymatic release of lysolecithin. Cholesterol promotes the deposition of neutral fat in the liver, whereas the mobilisation of this is one of the main lipotropic actions of lecithin.

Sterols may be said to stabilise fatty acids by direct esterification as well as in this, possibly more indirect, way. As in the glycerides (p. 97) the ester-stabilisation is reciprocal, that is to say the ester is less soluble also than the free steroid. Acetic and propionic acids, in particular (p. 100), are used technically to prolong the action of the steroid hormones. The action on the hydrophil steroids also may be regarded as reciprocal in the sense that digitonin, the active steroid of foxglove, precipitates most other steroids. The more acidic steroids may form esters with the more alcoholic members.

Effete steroids are made soluble and excreted as the alcohol partners of esters with the strong acids, glucuronic (p. 49) and sulphuric. This is the clue also to the solubility of the relatively strong bile acids, which in turn can act as solvents for other steroids such as cholesterol. They also dissolve lipoprotein conjugates such as rhodopsin (p. 112) and mitochondrial membranes, as neither pure lipid nor pure aqueous solvents could. Glucuronic and sulphuric acids are used also for the detoxication and excretion of other alcohols and this is a significant reminder of the association between the two acids also in the mucopolysaccharides (p. 56). They may keep the polysaccharides in solution.

Some steroids make nucleic acids more soluble (Henry and Stacey, 1946). There may also be a more specific relationship here; the adenine ring system of nucleic acids fits very closely the α–side of rings C and D of a typical steroid, and electron transfer readily takes place between them in consequence (Szent-Gyorgyi, 1960; Engel, 1956). This is not necessarily related to the solvent action but it is certainly another important aspect of steroid function.

Some of the steroids are in fact very active electron-transfer agents and they do work in conjunction with nucleotide coenzymes. Oestrogens catalyse the oxidation of the NAD coenzymes by peroxidases (Williams-Ashman *et. al.*, 1959; Burzatta, 1961). The most important hydrogen-transfer controlled by steroids may be that between NAD itself and the triphosphate, NADP (Long, 1961, p. 352). The ketonic steroids are among the most active, a further indication that they are rather quinonic in nature. Stigmasterol alleviates the growth-inhibitory action on *Tetrahymena* of dinitrophenol (DNP), which is a notorious respiratory

inhibitor, uncoupling oxidation from the essential phosphorylation process. As usual, therefore (p. 192), the steroid group has both physical and chemical functions, the latter only very recently recognised. It is possible that redox action plays a wide role in the biological functions of steroids; for instance the fact that deoxycorticosterone, with hydrogen at position 11, has ten times the hormonal activity of corticosterone, which has hydroxyl in that position, may have a redox basis. The conspicuous association between the steroid hormones and the vitamin C–GSH redox system in the adrenal cortex is significant here. Most steroids are active in such small amounts that an enzymic function may be suspected, though clearly an agent which controls permeation through membranes should also show this quantitative feature of catalysts. Steroids promote the action of proteolytic enzymes in some way (Deborin, 1959), and tend to inhibit growth in consequence; this also may depend on a redox effect since sulphydryl reducing compounds depress, and oxidising metals promote, proteolysis.

Most steroids have too few double bonds in the molecule for resonance on a useful scale, but those with at least one ring fully unsaturated absorb strongly in the ultraviolet. Ring B is particularly sensitive to radiation and that is why it is opened, to form vitamin D (calciferol), on exposure of ergosterol to ultraviolet. The pseudosteroid stilbenes, such as stilboestrol (p. 122), have a more extensive system of conjugated double bonds, and absorb very strongly. However a resonant shift can only cross the link chain if the side chains are *not* ring-closed, and this may be why the stilbenes are not so highly resonant in practice as Fig. 17, III, would imply. Like the steroids, they absorb only in the ultraviolet, though they are fluorescent, which means that they re-emit in the visible range; the fluorescence is white and emission therefore is distributed over the visible range in much the same proportions as in sunlight. In consequence the compound 4:4–diamino stilbene–2:2–disulphonic acid is used commercially to enhance the 'whiter than white' efficiency of some fabric-cleansing detergents! In any case, like its steroid analogues it is itself a good acid detergent.

The steroids are thus a group with a unique structure, and with many highly specific variants in detail. Among lipids, and even among biological materials more generally, they are an outstanding group in practical importance and in theoretical interest. Much remains to be discovered about this. The isoprenoids more generally are a large, versatile and important group, in ways which for the most part contrast with the main functions of the glycerides. Chemically, and probably functionally also, they are intermediate between the aliphatic lipids and the aromatic group to be considered in the next chapter.

CHAPTER 10

AROMATIC HYDROCARBONS AND THEIR TERNARY COMPOUNDS

AROMATIC compounds are fully unsaturated ring structures (Fig. 19, I), with or without side chains, which may not be fully unsaturated. As already emphasised (p. 106), full unsaturation means alternating double and single bonds, and the chains of some carotinoids are in this state. They are not regarded as aromatic, however, since on ring closure some of the double bonds are resolved and the apparent saturation increases. The theoretical open chain precursors of aromatic rings therefore are acetylenic polymers, and benzene in fact is synthesised from acetylene itself in the laboratory. The acetylenes are very unstable, however, and many biological aromatic ring compounds are synthesised by an entirely different pathway (p. 187), involving desaturation after ring closure, as in the isoprene pathway.

Aromatic rings clearly represent the limit of useful desaturation since any further dehydrogenation would yield elemental carbon as the next stable stage; this is all too evident when benzene is allowed to burn. As usual 5- and 6-membered rings are the only stable ones and even cyclopentene is very rare. Benzene (Fig. 19, I) therefore is the parent substance for all significant binary and ternary aromatic compounds. The ring is usually represented as in Ia, CH being understood at each position; the substituents in the ring, to form heterocyclic compounds, and those in place of hydrogen as side chains, are always represented, however. Phenol therefore is II, and it is such an important simple substituent that all other substituents are known as phenyl derivatives, the phenyl radical, C_6H_5, being benzene less one hydrogen atom; for brevity it is often represented as ϕ, which is not only the appropriate Greek initial but can be taken to imply a ring with substituent groups. All aromatic radicals are *collectively* termed *aryl* as distinct from the *alkyl* radicals of the paraffins and the *alkene* radicals of the olefine series.

In living organisms there is a far greater number of aliphatic than of aromatic compounds, and in total bulk the disparity is even greater. One

reason for this no doubt is the greater value of open chain molecules for building fabrics, which play such an important part in the organisation of living organisms, not merely in simple mechanical ways but in spatially deploying enzymes (Green, 1962), and organising metabolism in general;

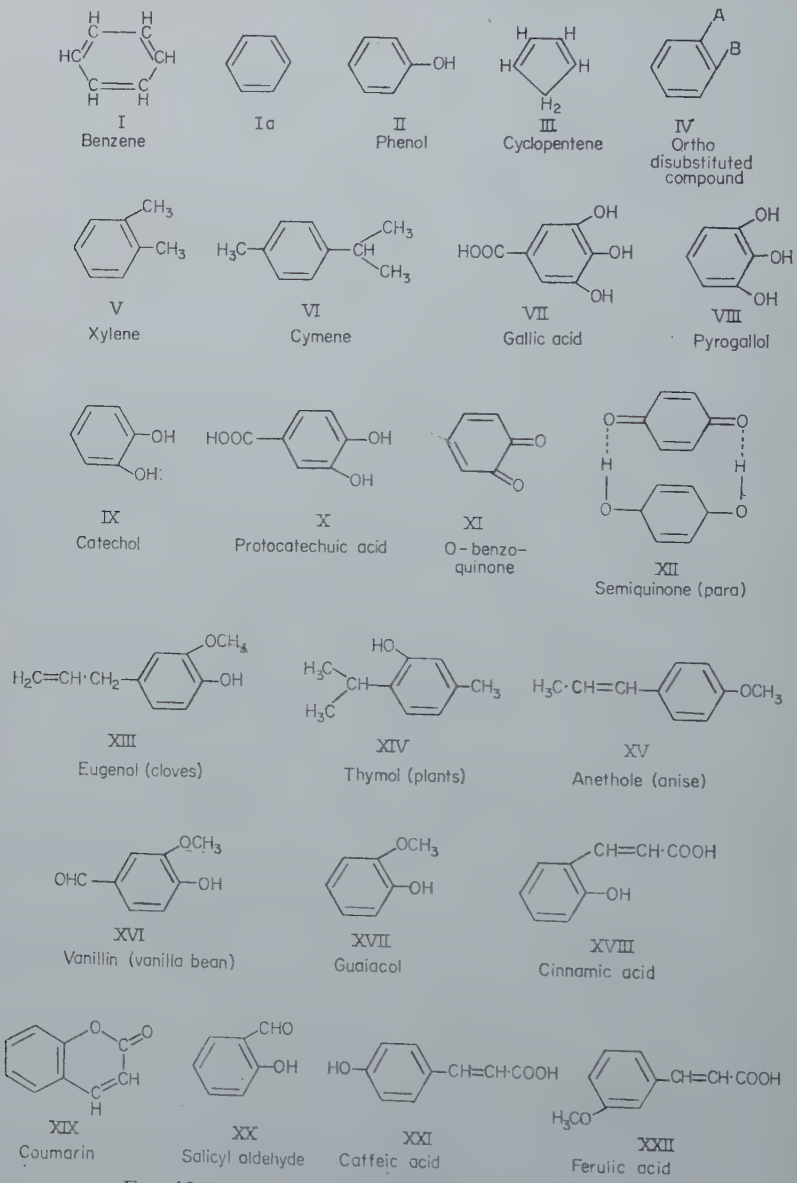


FIG. 19 Benzene derivatives of biological significance.

this is a biological, teleonomic reason. In addition, there is the fundamental chemical reason that synthesis starts with carbon dioxide, which polymerises to relatively saturated chain molecules; these do not spontaneously desaturate to the aromatic stage. To form benzene from three molecules of ethane requires 85 kcal/mole, so that aromatic compounds in general are not synthesised without good reason. It may be noted that even so the energy required is small compared with the 241 kcal/mole required to desaturate the ethane to three moles of acetylene; it is not surprising that the latter and its derivatives rarely occur in living organisms. From these figures it is clear that acetylene polymerises to benzene spontaneously, with the liberation of considerable energy:

$$3\ CH{\equiv}CH \rightarrow C_6H_6 + 156\ \text{kcal/mole}$$

In the laboratory, by contrast, a very large number of aromatic compounds have been synthesised, for a number of relevant reasons. In the first place the natural starting materials—coal, petroleum, etc.—contain considerable aromatic material, which shows that it may form spontaneously post mortem, to some extent, under appropriate conditions, and that we do not have to evoke supernatural causes to explain those aromatic materials which *are* formed *in vivo*. In the second place, aromatic compounds are much more reactive than the paraffins and therefore have interested chemists more. The products of reaction are more varied in properties than those of aliphatic reactions. They have more commercial uses, the dyestuffs industry being the classical, historical example; they are very useful in living organisms for the same reason and the biological importance is out of all proportion to their insignificant bulk. A number of them are coenzymes and other key reagents, potent in very small amounts. An interesting question here is whether the pleasant smell of many aromatic compounds, from which the group earned its name, is due to their biological desirability. Aromatic compounds are often brightly coloured, which facilitates the study of their reactions; this again is biologically relevant, since it is also a measure of their resonance, their reactivity and their ability to activate other molecules (p. 115).

The benzene ring in fact is fully resonant, as its structure shows (Fig. 19, I). With only three double bonds, however, it absorbs in the ultraviolet and is uncoloured, but additional rings, or resonant structures in side chains, shift the absorption to the visible range and increase the biological value. The biological value of benzene derivatives lies not merely in their reactivity, however, but in the invaluable admixture of stability. Benzene is much more stable and less reactive than the double bonding might imply (Read, 1935, p. 520). The molecule resonates powerfully between the two alternative arrangements of the double bonds

and the molecule can be regarded as gaining stability from the *resonance energy*, which can be measured as the difference between the actual energy required to synthesise the substance and that theoretically required to synthesise either resonant form of the molecule. In consequence of this effect, benzene is actually *more* stable than cyclohexane, which of course should be regarded as the result of ring closure by a C_6 chain with *one* double bond, and therefore not a simple paraffin. These intermediate stages of desaturation are as usual the least stable, and benzene in fact behaves more like a typical partially unsaturated compound if one of its double bonds is saturated. Once more, therefore, we have to do with a stable node among organic compounds. Most substituents on the benzene ring increase its activity and reduce its stability, and permit an infinite gradation in properties.

It is now evident why cyclopentenes are so rare, even in laboratory products. Two unsaturated bonds are the maximum possible in a C_5 ring (Fig. 19, III) and this leaves the molecule in the unstable state of partial saturation; complete resonance throughout the ring is impossible and there is much less resonance stabilisation. As will be seen, *heterocyclic* pentene rings are more stable and common since, like benzene, they are able to form a closed circuit of six π–electrons (Finar, 1959, p. 485).

No carbon atom with a double bond can be sterically asymmetric (p. 31) and benzene itself is optically inactive. Moreover, it has no geometric isomerism since there is only one substituent at each point on the ring and this lies almost in the plane of the ring. Resonance makes all six positions in the ring identical, so that it is identical when viewed from either side, and no extra series of isomers arise from that type of asymmetry. The constraint of the double bonds makes the ring much more coplanar than alicyclic rings, and it becomes a plane of symmetry. This greatly simplifies the chemistry of the aromatic compounds, and they have fewer of these 'trivial' variants than aliphatic series. Some isomerism does arise among derivatives, however, because properties vary with the position of two or more substituents on the ring. Two substituents, A and B, can be on neighbouring positions (Fig. 19, IV) or may be separated by one or two carbon atoms. These are known respectively as the *ortho* (*o*), *meta* (*m*) and *para* (*p*) positions; there is only one of each of these isomers because of the 'reversibility' of the ring.

The differences between these isomers depend on interaction between the substituent groups and between them individually and the ring, and examples of the effects will be seen. The interaction between ring and side chain is reciprocal, and one or two typical examples may be considered here. Because of its unsaturation the ring is electron-donating and this puts a more negative charge on the oxygen atom of a hydroxyl sub-

stituent than in open chain compounds. Consequently phenol, ϕ—OH, ionises as ϕ—O^- and H^+, and is a strong acid, 7 pH units more so than the aliphatic alcohols, in fact; it can still form ethers with other alcohols, however, and esters with strong acids. For most biological purposes phenol is much too strong an acid, and 'carbolic' has long been a popular antiseptic. Similarly the amino group, —NH_2, becomes much less basic than in aliphatic compounds. The ring affects radicals on carbon side chains also, but this decreases rapidly with distance, so that most side chains of any length behave as aliphatic structures. In benzoic acid, ϕ—COOH, however, the acidity is greatly enhanced by the ring and the dissociation constant, K, is 0·0060 as compared with 0·0018 for acetic acid.

Substituents affect the ring mainly by influencing its degree, or its pattern, of unsaturation, particularly well seen in the polyphenols. Substituents may affect each other via the ring, but they may also interact directly, by *steric hindrance*, as when groups *ortho* to carboxyl prevent it from forming esters. An effect via the ring is the depression of acidity of hydroxyl by an alanine side chain, in tyrosine (p. 186). Another of importance is the depression of acidity, mutually, in poly-hydroxy substituents. Their reducing power rises in proportion, assisted by the shift of double bonds from the ring to form quinone, ϕ=O, groups.

10.1 Monocyclic Biological Aromatic Compounds

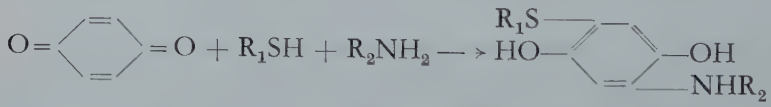
As usual the binary compounds are not very common and we can be sure that the reduction, if not the actual ring closure, of the benzenoid hydrocarbons in coal and natural oils is a post-mortem change. Small amounts exist free in plant oils, just as in the case of the paraffins, and *xylene*, or di-methyl benzene (Fig. 19, V), is obtained commercially by distilling wood. It induces the germination of seeds (Sexton, 1953, p. 388) and may be a normal physiological agent. Similarly *toluene*, or methyl benzene, as well as *bromobenzene*, induces the germination of sponge gemmules (Rasmont, 1959). *Cymene*, or *p*-methylisopropyl benzene (Fig. 19, VI), occurs in oil of eucalyptus and in other plants.

Its site of occurrence, the C_{10} size, and the general form of the molecule, very strongly imply an origin from terpenes and in fact cymene is readily synthesised from camphor (p. 111) in the laboratory, by dehydration. Many of the ternary benzenoids of living organisms also bear the stamp of terpenoid origin so that there are two biosynthetic pathways for biological benzenoids, via terpenes and via monoses and shikimic acid (p. 187). Derived from the latter is *gallic acid* (Fig. 19, VII), from gall 'nuts'; *pyrogallol*, the photographic developer (Fig. 19, VIII), is prepared from this, and is the best example of a powerfully reducing polyphenol.

It is possible that other polyphenols are derived via this pathway, either directly or via the phenolic amino acids (p. 186). *Catechol* (Fig. 19, IX) occurs in the secretion of acacia while certain derived diphenols, the *catechins*, give tea its attractive flavour (Harlow, 1956); it may be noted that the flavour therefore has no relation to the stimulant action, which depends on purines (p. 347).

Of the ternary derivatives of the phenolic amino acids *protocatechuic* acid (Fig. 19, X) is one of the most important; it is an intermediary in the degradative synthesis of quinones which tan or harden the arthropod exoskeleton (Pryor, 1940). The effective agent is probably *o*-benzoquinone (Fig. 19, XI), though the *p*-isomer is a possible alternative (Dennell, 1958); synthesis of the *m*-isomer is not possible since clearly this would necessitate either an almost saturated ring or a valency of more than four in one of its carbon atoms. The quinones are powerful oxidising agents because of their readiness to recover the hydrogen to restore the phenolic state. They will alternatively bond with any other electron-donor and since the activity of the two =O groups is linked, via the resonance shifts in the ring, the molecule binds together two others. This is the essential factor in producing insoluble proteins, the stability of the bonds also being a critical factor. Formation of quinone from diphenol adds one double bond to the molecule, but it adds two carbonyl groups which each have a much bigger resonance effect than C=C (p. 116) and so the quinones are coloured and their polymers may be as bathochromic as black. Darkening usually, but not always, accompanies tanning. The necessary length of resonance chains to produce a black colour may be obtained by peptide bond groups becoming in series with benzene rings, though it is also possible that the quinone molecules polymerise to some extent. They do interact rather readily, and the formation of interesting dimers, semiquinones or quinhydrones (Fig. 19, XII), as unstable intermediaries, plays a considerable part in their reactions.

Quinones can also form addition compounds at the *o*-positions to the =O groups:



Again protein macromolecules can be bound together by this process. The formation of large insoluble masses of course depends on each macromolecule having a number of receptor sites for quinones, so that it becomes bound to a large number of other polypeptide chains.

Parabenzoquinone is clearly the basis of coenzyme Q, or ubiquinone, a respiratory redox reagent of many groups of organisms, occupying a

position in the electron-transfer sequence shown in Fig. 20. This does not preclude the *p*-quinone from also taking a part in the tanning of the arthropod exoskeleton, though it calls for careful investigation. Polyphenol oxidases are common in plants and fungi and sometimes they catalyse irreversible reactions, with the formation of stable products, while in other cases they act as reversible, potentially or actually respiratory, metabolites.

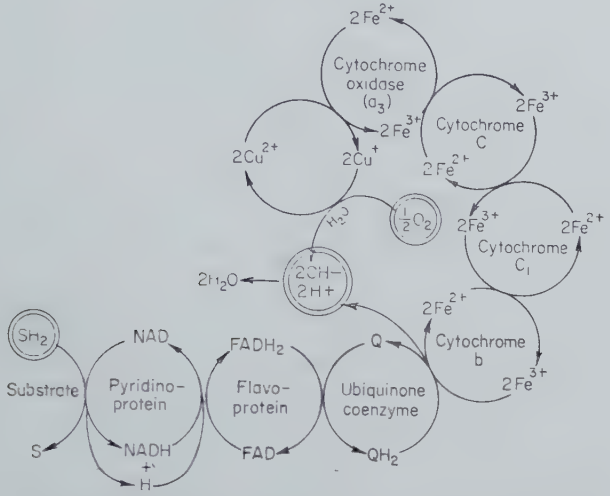


FIG. 20 The hydrogen- and electron-transfer sequence in respiration: schematic representation of theoretical example with maximal number of steps, showing the position of the ubiquinone cycle. Each step is operated by a cyclically regenerating coenzyme. The proximal four steps involve hydrogen-transfer and the terminal five electron-transfer. These five are represented as a loop so as to bring the ionised oxygen near to the ionised hydrogen, for the final reaction: as far as substrate atoms are concerned, the pathway short-circuits the loop, though this is absolutely indispensable for the continuous production of both ionisations.

Among the probable terpenoid derivatives are *eugenol* (Fig. 19, XIII), the essential oil of cloves, *thymol* (Fig. 19, XIV) from various plants, *anethole* (Fig. 19, XV) from anise, and *vanillin* (Fig. 19, XVI) from the vanilla bean. The last has a shortened side chain but the other substituents imply a kinship with eugenol, while *guaiacol* (Fig. 19, XVII), from beechwood, has lost the carbon side chain completely. Like the terpenoid essential oils themselves, this group have pharmacological actions, thymol having antiseptic properties and eugenol being a mild anaesthetic. *Cinnamic acid* (Fig. 19, XVIII) is another flavouring, like vanilla, and its ester with its own alcohol-analogue has been isolated from storax resin,

an interesting parallel to some of the aliphatic esters (p. 99). *Coumarin* (Fig. 19, XIX), the anticoagulant from sweet clover (p. 118), is very probably a lactone, ring-closed, cinnamic derivative. The completed ring system is that of vitamin E, and it has slightly the activity of this vitamin. *Salicyl aldehyde* (Fig. 19, XX), from oil of meadowsweet, probably also has a terpenoid origin, with shortening of the side chain; its derivatives include the analgesics of the aspirin (acetylsalicylic acid) type. By treatment with sodium acetate (Perkins reaction) it can be converted to *o*-hydroxycinnamic acid and to coumarin. In willow bark salicyl alcohol occurs as the glucoside, *salicin*; its methyl derivative is oil of wintergreen, an histological clearing agent. The salicylates and benzoates can act as respiratory substrates (Sexton, 1953, p. 278) and affect carbohydrate metabolism (Bornstein, 1952), while coumarin depresses the glycosuria due to alloxan (p. 350). *Caffeic acid* (Fig. 19, XXI), which has no relationship to caffein (p. 350) except in occurrence, and *ferulic acid* (Fig. 19, XXII), from the tomato, are interesting in inhibiting germination of seeds (Sexton, 1953, p. 380). Even *benzoyl alcohol* (Fig. 19, XXIII), present in balsam, may be a terpene derivative, and therefore also benzaldehyde, benzoic acid, and other simple benzene compounds. Benzaldehyde occurs as a glucoside, *amygdalin*, in oil of bitter almonds; its aldehyde group forms an addition compound simultaneously with the cyanide, CN, group which is interesting because both benzaldehyde and hydrocyanic acid have the odour of almonds. Have they been exploited to reinforce this odour? They do not smell alike merely because they are associated naturally in this glucoside, since we distinguish the odours of a number of components in some natural products.

The benzoquinones are clearly the most important ternary benzene derivatives biologically and add to the importance of the aromatic amino acid pathway (Fig. 27). The terpenoid line produces a number of interesting minor metabolites which functionally perhaps are more closely related to the parent terpenes than to the major phenolic group.

O blest unfabled Incense Tree,
That burns in glorious Araby,
With red scent chalicing the air, . . .
With amber tears and odorous sighs . . .
O, fast her amber blood doth flow
From the heart-wounded Incense Tree, . . .
(GEORGE DARLEY, *O Blest Unfabled Incense Tree*)

We have rose-candy, we have spikenard,
Mastic and terebinth and oil and spice, . . .
(JAMES ELROY FLECKER, *The Golden Journey to Samarkand*)

10.2 Polycyclic Aromatic Hydrocarbons and Ternary Compounds

Binary and ternary benzene derivatives with long side chains are not very common biologically and the only other group to be considered in this chapter are polycyclic compounds, again with short side chains for the most part. There are two possible types of polycyclic compound, chain-linked and condensed. The first is the least common biologically, though the stilbenes (p. 122) are a good example of two benzene rings thus linked. This type is perhaps more common among alicyclic compounds and a large number of the carotinoids are of this general form (Goodwin, 1952). Condensed rings share two carbon atoms, rarely more or less, with each neighbour. Naphthalene (Fig. 21, I), therefore, the

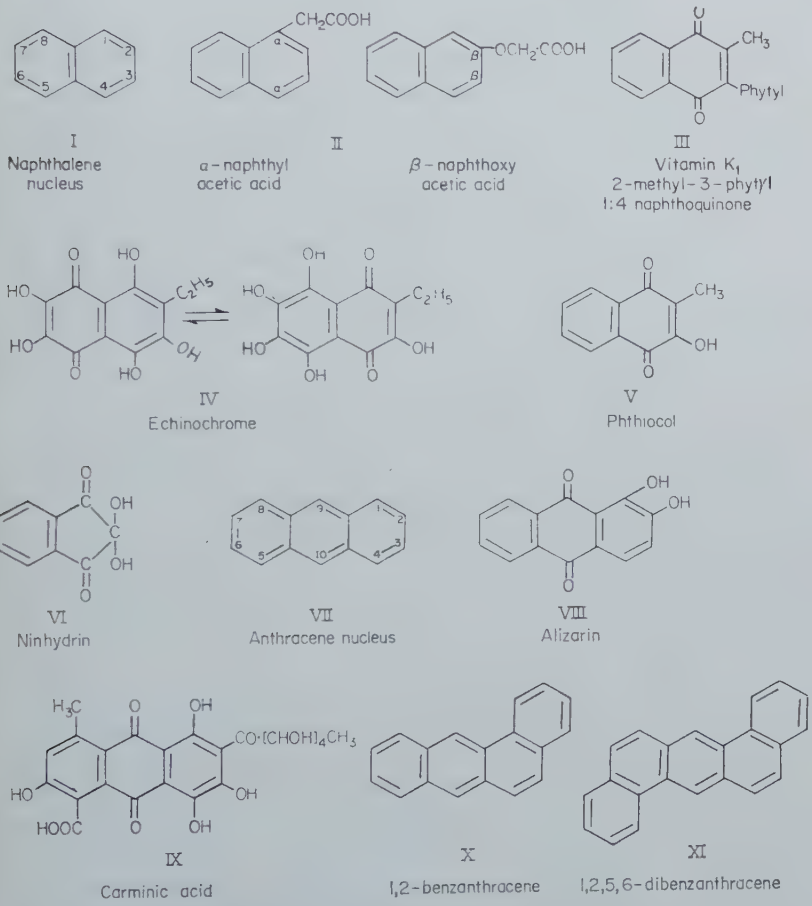


FIG. 21 Polycyclic hydrocarbons of biological significance (*continued on p. 140*).

simplest member, is a C_{10} compound and could not be synthesised by condensation of two benzene rings. The stability of the benzene ring would make this synthesis unlikely, quite apart from the problem of the two spare carbon atoms. Polycyclic compounds *are* formed from simpler aromatic ring compounds but by using side chains to close additional rings. This requires further desaturation to produce fully aromatic polycycles so that it would be wrong to think of them simply as polymerised benzene derivatives. Some contain a mixture of fully and partially desaturated rings.

It is in this group that some of the most complex biological monomers occur. The *aphins* (Fig. 21, XVIII), integumental pigments of aphids, are the outstanding example (Fox and Vevers, 1960). They have seven condensed homocyclic rings as well as two heterocyclics also condensed; the rings are not all maximally unsaturated. Such complex monomers are probably rare and it is no doubt true that, in nature as in the laboratory, the difficulties of synthesis increase with molecular size: to this extent life is opportunist. By contrast to the synthesis of large monomers, polymerisation tends to be rather generally spontaneous and large masses of varied open chain polymers are used in living organisms; aromatic compounds are certainly fewer in number and less in bulk, but they include the most complex monomers, that is molecules without a repeat-unit. It should be pointed out, however, that there are fewer carbon atoms in an aphin than in a carotinoid molecule, and that our subjective impression of complexity is enhanced by condensed ring formation. At the same time this impression probably estimates correctly the amount of work involved in the synthesis of the condensed ring system.

Whereas benzene might be said to have a molecule of complete radial symmetry, naphthalene (Fig. 21, I) has only biradial symmetry, but it is again identical from either face, so that positions 1, 4, 5, and 8 are all equivalent and so are 2, 3, 6, and 7. For monosubstituents, therefore, there are only two isomers, *ortho* or *meta*, to the two shared carbons; they are known as the α and β isomers (Fig. 21, II). Although it could not be synthesised from two benzene rings it is also abundant in coal extracts and in petroleums. In fact it is the most common aromatic hydrocarbon in coal tar distillates, though like the other aromatic series it does not occur in the tar produced by carbonisation of the coal at low temperature (450–550°C). They may be largely artefacts of the processing, therefore.

The free hydrocarbon is probably very rare *in vivo*; most of the biological derivatives are ternary or higher compounds, the binary series as usual being the last stage in biosynthesis. One of the most important biological derivatives is *α–naphthyl acetic acid* (Fig. 21, IIa), a plant growth hormone, or *auxin*; the compound *β–naphthoxyacetic acid* is even more

potent in stimulating organ-formation in plants (Sexton, 1953, p. 373), and the action on plant growth appears to be one of rather low specificity. A variety of mono- and bicyclic aromatic compounds, phenyl, acenaphthyl, fluoryl and indolic, also have significant effects; their common property seems to be little more than a fully resonant, oligocyclic nucleus. This may perhaps be compared with the relatively low specificity of the natural oestrogens (p. 123), among animal hormones. Again some of the compounds synthesised in the laboratory have proved more potent than the natural agents isolated; one of the most powerful is 2:4-*dichlorophenyl acetic acid*, popular as a lawn weed killer since it causes many weeds to outgrow their strength and die, while the grass thrives.

The ring structure of vitamin K (p. 117) is 1:4-*naphthoquinone* (Fig. 21, III). This is one of three common naphthoquinones, the 1:2, 1:4, and 2:6 isomers; there are six theoretically possible isomers of a disubstituted naphthalene. Both quinonoid groups are on the same ring and it resembles *p*-benzoquinone chemically and metabolically; in fact, as already noted (p. 118), it appears to act as an alternative to ubiquinone in some organisms. As an antihaemorrhagic factor it appears to promote the synthesis of prothrombase, the main enzyme in blood-clotting.

Echinochrome (Fig. 21, IV) is another biological 1:4-naphthoquinone, with the further unique feature of being a self-contained semiquinone (p. 132), and resonating between the two alternative forms. With this remarkable property it would be surprising if it did not have a metabolic redox function, and it reversibly reduces to the heptahydroxy condition (Baldwin, 1953, p. 184). In distribution, however, it is partly an integumental pigment of echinoids, and prompts the question why so many integumental pigments have redox properties; in many cases it is possible to say that a coloured molecule can scarcely fail to be sensitive to oxidation–reduction but that the change is not reversible in most integumental pigments, under physiological conditions. Echinochrome, however, could act as a physiological respiratory mediator and perhaps does so in the blood cells and eggs. In the skin it perhaps acts in conjunction with terminal enzyme systems receiving oxygen directly from the external medium.

Echinochrome owes its red colour to the extra resonance between the two quinone positions, since 1:4-naphthoquinone itself is yellow. The large number of hydroxyl substituents is unique among biological aromatic compounds and recalls the carbohydrates and related alcohols. A number of the carbohydrate hydroxyl groups are retained in shikimic acid and its immediate derivatives (Fig. 27) and it may indicate that echinochrome is synthesised via this pathway. The point is of some importance since the pigment has considerable resemblance to the ring system

of vitamin K for which an isoprenoid origin seemed possible (p. 118) in common with the side chain. However the phytyl side chain *is* attached to a moiety of quite a different synthetic pathway in chlorophyll, and the question remains open.

Phthiocol (Fig. 21, V), not to be confused with phthioic acid (p. 94), is a pigment of human tubercle bacilli and has slight vitamin K activity. This may tend to reduce the haemorrhage due to bacterial damage to the lung tissues of the host.

Ninhydrin, or triketohydrindene hydrate (Fig. 21, VI), an interesting compound used to detect amino acids by its colour reaction, is a quinone with a pentacyclic ring, which is as fully unsaturated as it could be in the quinonoid form. The nucleus as a whole is known as indane, because of its resemblance to indole, and could also be regarded as a homologue of the naphthalene nucleus. The unusual *gem*-diol form shown is more stable than in most organic compounds; it dehydrates to the triketo form, as the name indicates. Ninhydrin is not a biological compound, of course, and re-emphasises the rarity of biological cyclopentene compounds.

Anthracene (Fig. 21, VII), the simplest triple condensed homocyclic ring, also occurs in petroleum and in coal products, but not significantly in living organisms. Few of its ternary compounds, except the 9:10 quinones, are important either. Quinones, therefore, are outstanding among aromatic, ternary, biological compounds. The anthraquinones, with the extra resonance of the third ring, are bright red pigments. *Alizarin*, 1:2–dihydroxy–9:10–anthraquinone (Fig. 21, VIII), occurs in the root of the madder plant, and *carminic acid* (Fig. 21, IX) is another 9:10 quinone, from the carmine or cochineal pigment obtained from the female of the scale insect, *Coccus cacti*, L. It has several more substituents than alizarin, including a deoxyhexose. The vast amounts of the pigment accumulated by the insect probably implies that it is a by-product of the plant food. These quinones undergo a colour change when they are reduced and one, laccaic acid from the lac insect, also changes colour with pH and so can be used as an indicator of both types of change. In the darkness of the plant root, alizarin, if it has a useful function, is presumably activated by chemical means.

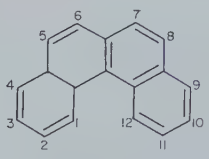
The 9:10 quinone is the most common isomer in the laboratory, as well as in nature, and indeed it is the most common product of oxidation of all anthracene derivatives (Finar, 1959, p. 700). For instance not only do other 9:10 substituents oxidise to this quinone but so do 5:8 di-substituents. The biological compound is the most probable one, therefore, but in this case we cannot be sure that it is more than a by-product of metabolism. Alizarin is present in the madder root as the di-glucoside, *ruberythric acid*, which by analogy with other glucosides (Brunet and

Kent, 1955; Smith, 1955) is probably an inactive form, and possibly a detoxicated form. However, it is difficult to believe that the redox property, and the ability to dye fabrics and tissues, are both biologically valueless. Dyeing depends on the formation of insoluble lakes with metal cations already mordanted on to the fabric or tissue, the bonding being via the hydroxyl groups at 1 and 2 and not via the quinone groups. The commercial Turkey red is an aluminium lake. Because of their greater resonance network the anthraquinones could activate where naphthoquinones would be ineffective.

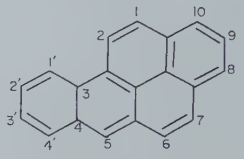
Since the simpler polycyclic aryl hydrocarbons are rare in living organisms it is fairly certain that those with larger ring systems are even rarer. It is therefore surprising, but probably very significant, that some of them are strongly carcinogenic, and that in general this property increases with molecular size. The explanation of this has a wider relevance than to growth alone. Like the molecules with auxin activity in plants they are rather diverse in structure: some of the most potent carcinogens are X–XIV, quadruple and quintuple condensed hydrocarbons of varied form. Among the compounds of any one of these series relatively few are very active, so that side chains also no doubt play a part. Some nitrogen-containing series also are active, but in the same sporadic way. If these are compounds not normally encountered, the degree of specificity seems very high, but if they are analogues of normal metabolites then it is rather low. All the most active carcinogens have a fully or very extensively resonant molecule, and no doubt electronic activation plays a major part in the phenomenon (Allison and Nash, 1963), but not all highly resonant polycyclic compounds are strong carcinogens. As a result of resonance in these large ring systems there is often a high density of electrons in the so-called *K*- or meso-region of the centre of the molecule, and it has been suggested that this is an essential property for carcinogenesis (Pullman, 1945; Coulson, 1953), but again the correlation is not absolute, either way. Nitrogen contributes to carcinogenicity most strongly as a —N=N— link between rings, so that the resonance–activation type of action is no doubt of primary importance and the problem is to find what other structural or behavioural features can make or mar this property.

Two main possibilities emerge, substituent groups and interaction with the biological materials. In general methyl groups as side chains increase carcinogenicity, because of the opportunities they provide for extending resonance, and for closing to form further rings. This may be the significance of the great potency of the 5:10 di-methyl substituent of anthracene. The stilbenes (p. 122) are among the carcinogens with the fewest rings in the molecule but they have potentially two more rings by

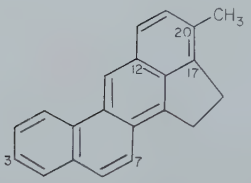
closure of short side chains. Some cyanide derivatives increase activity, perhaps both by adding a C_1 unit and through the —C≡N triple bond. Side chains rather generally increase the energy absorption of the molecule, as would be expected if they permit an extension of resonance. The hydroxyl and nitro groups as substituents do not increase, and may



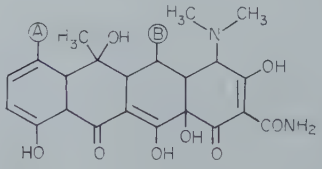
XII
Benzphenanthrene



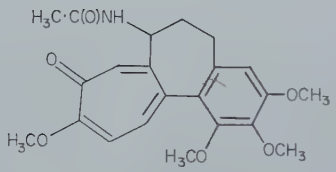
XIII
Benzpyrene



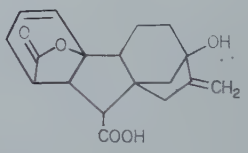
XIV
20-methylcholanthrene



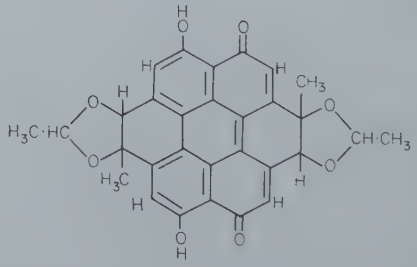
XV
Terramycin: A = H, B = OH
Aureomycin: A = Cl, B = H



XVI
Colchicine



XVII
Gibberellic acid
(Gibberellin X)



XVIII
Erythroaphin

FIG. 21 (*continued*).

even depress carcinogenic activity, although they are potentially capable of increasing resonance.

Carcinogenic activation of bioplasm may demand resonance between the carcinogenic molecule and, say, the proteins, and this may further limit the number of actual carcinogens. The energy absorbed by many carcinogens in order to produce the first excited π–singlet state of their molecule corresponds to light energy of a wavelength around 350 mμ, which is the region in which tryptophane, one of the amino acids (p. 189), emits energy most strongly and efficiently (Birks, 1961). Effective carcinogens therefore may be those which can draw off energy from proteins via their tryptophan residues. It is interesting that tryptophan is deficient in tissue which has become cancerous.

Oxidation–reduction effects are inevitably involved in the action of these highly resonating molecules (Lovelock, 1962), and neoplastic tissues are characterised by a changed respiratory pattern (Hieger, 1961). The excitation of the π–electrons of carcinogens is very similar to that in bioluminescent systems, which essentially involve oxidation. In addition carcinogens must act in lipid media or at lipid–aqueous interfaces; the extent to which respiration occurs in lipid media (Green, 1962) has largely removed the apparent paradox in this.

Polycyclic, largely unsaturated, molecules occur among antibiotics but perhaps there is nothing very specific about this, since collectively antibiotics seem to use every possible antimetabolic device in the armoury. Each one in fact usually has several strings to its bow and those in the present group, *terramycin* and *aureomycin* (Fig. 21, XV), are no exceptions. Primarily there may be semi-aldehyde resonance between their OH and =O groups; in addition there are relatively uncommon and therefore probably toxic groups such as the amide and the tertiary amine, and the linear chain of four rings also is rather unusual.

There are other biological polycyclic ternary compounds of mainly aromatic character, but varying in degree of saturation, in ring structure and other features. Here, no doubt, is a large field for future work, when simpler molecules are better understood. *Colchicine* (Fig. 21, XVI), from the autumn crocus, now notorious for its inhibitory action on the normal behaviour of the division-spindle of the cell, is a typical example. It has one benzene ring, a heptacyclic fully unsaturated *tropolone* ring and an almost saturated heptacyclic ring which nevertheless could conduct resonance between the other two. The large number of methoxy, ether groups also is striking, and the lack of free hydroxyl groups. The acetyl amino group recalls the carbohydrates. In its bizarre form, and large number of a relatively uncommon type of side chain, it resembles antibiotics, and possibly has this kind of significance in *Colchicum* itself.

The *gibberellins* (Fig. 21, XVII) have an even more bizarre molecule, which is alicyclic rather than aromatic, though the three main rings give it a resemblance to *benzfluorene* (p. 137) which like the gibberellins promotes the growth of plants. The gibberellins, in fact, may have some functional relationship to the auxins, the simplest compounds of this section of the chapter. This might be taken as a striking demonstration of the low functional specificity of polycyclic compounds, and there is at least a germ of truth in this. The contrast to the great specificity of some of the nitrogen-containing molecules to be considered is very sharp.

The aphins (Fig. 21, XVIII) are outstanding not only for the size of their condensed ring system, but also for its almost full unsaturation and for the degree of symmetry of the system. A line between the two ethane side chains divides the molecule into two mirror-image halves, except for the hydrogen and methyl groups on the dioxypentane rings. There is also the internal semiquinone property of echinochrome and terramycin, and further the aphins provide one of the many examples of body pigments with potential redox properties. Like echinochrome, the molecule has a modicum of asymmetry, necessary for maximal resonance (p. 115). Rather surprisingly, it is water-soluble in its natural state, and very sparingly soluble in lipids; this is partly due to its being conjugated as a glycoside, and partly, perhaps, to the number of polar groups round the periphery. It is noteworthy that oxygen atoms in ring structures behave as oxonium ions and add to the resonance of the nucleus, as well as to its water-solubility. Altogether this is a most remarkable molecule, and a difficulty for those who might consider the aphids degenerate. There is no known plant-product from which aphins could be synthesised without much elaboration; structure varies in details in the different species of aphid. It is important to know if they are isoprenoid in origin. If not they stand 'out on a limb' among biological aromatic compounds.

10.3 Chain-linked Cyclic Compounds

There are few homocyclic, aromatic compounds of this type with biological significance, whereas chain-linked heterocyclic rings, both aliphatic and aromatic, are not uncommon, for instance the polysaccharides, the flavones (p. 146) and the nucleosides (p. 355). One possible reason for this may be that condensed aromatic rings are automatically all coplanar, because the bonds common to two rings are double, and therefore rigid, through resonance; the lability of chain-links would neutralise the advantage, which is considerable, of this coplanarity. It is therefore interesting that *azobenzene*, ϕ—N=N—ϕ, is a potent carcinogen, since it is not in a position to close, as the analogous stilbenes may (p. 122), to a quad-

ruple condensed nucleus. The high resonance value of N=N (p. 116) may be the explanation, and a planar molecule may not be so critical for carcinogenesis as for some other activities (Szent-Gyorgyi, 1960). It is also interesting that both stilbenes and azo compounds are acaricidal.

In the laboratory it is possible to link as many as six benzene rings via an ethane link, for instance: $(\phi)_3\equiv C—C\equiv(\phi)_3$. The bond between the two link-carbons becomes so weak, however, that the radical $(\phi)_3\equiv C—$ can exist free—a tervalent carbon. It is of course very reactive and readily returns to the tetravalent state. The free tri-*di*phenyl methyl radical, $(\phi—\phi)_3\equiv C—$, also has been prepared, with benzene rings directly linked in pairs, without intermediary atoms; there is a limit to the number of aromatic rings which can be stably chain-linked, via the one chain, but there are endless theoretical possibilities of serial linkage and there must be other reasons why chain-linkage is rare biologically, in aromatic compounds.

CHAPTER 11

HETEROCYCLIC TERNARY COMPOUNDS

THESE are compounds with oxygen in the ring, whether alicyclic or aromatic. Several examples have been seen already, the monoses in particular, and apart from these it is not a very big group, but it does merit separate recognition before concluding the consideration of oxygen in biological materials. Technically the term heterocyclic ring is reserved for the more stable, unsaturated closed-chain structures (Finar, 1959, p. 711), but since the elements mainly concerned, oxygen, nitrogen and sulphur, are all capable of bringing resonance stability to the ring through extra double bonds, associated with their onium state, the distinction is rather artificial. Oxygen, as already noted (p. 142), and sulphur sometimes resemble nitrogen in acquiring double bonds in the ring by becoming onium ions. Certainly simple rings like ethylene oxide (Fig. 22, I) are easily opened, and also the lactone ring (Fig. 22, II), but the tetrahydrofuran and -pyran rings of the hexoses are relatively permanent in polysaccharides, so much so that different and specific enzymes are required for the α– and β–glycosides.

Even ethylene oxide, or epoxide, compounds are often stable, and the di-epoxides (III and IV) inhibit cell division (Loveless and Revell, 1949), so that they may prove practical anti-tumour agents (Sexton, 1953, p. 355). The monepoxides are inactive. The next member of this simple series is rare but the following two are the tetrahydrofuran and -pyran structures of the pentoses and hexoses. They are often said to be furan and pyran derivatives but by convention these should be the fully unsaturated rings (Figs. 22, V and VI), which do occur in some biological molecules.

The lactones also are a very distinct and significant group biologically, and some of the aliphatic members have already been considered (p. 96). They differ from the epoxide series in being essentially internal esters as distinct from internal ethers. *Protoanemonin* (Fig. 22, VIIa) is an antibacterial lactone with a partly unsaturated ring and therefore intermediate between the aliphatic type and the aromatic lactones, or *pyrones*. *Patulin* (Fig. 22, VIII) is a lactone antibiotic, with its second ring a

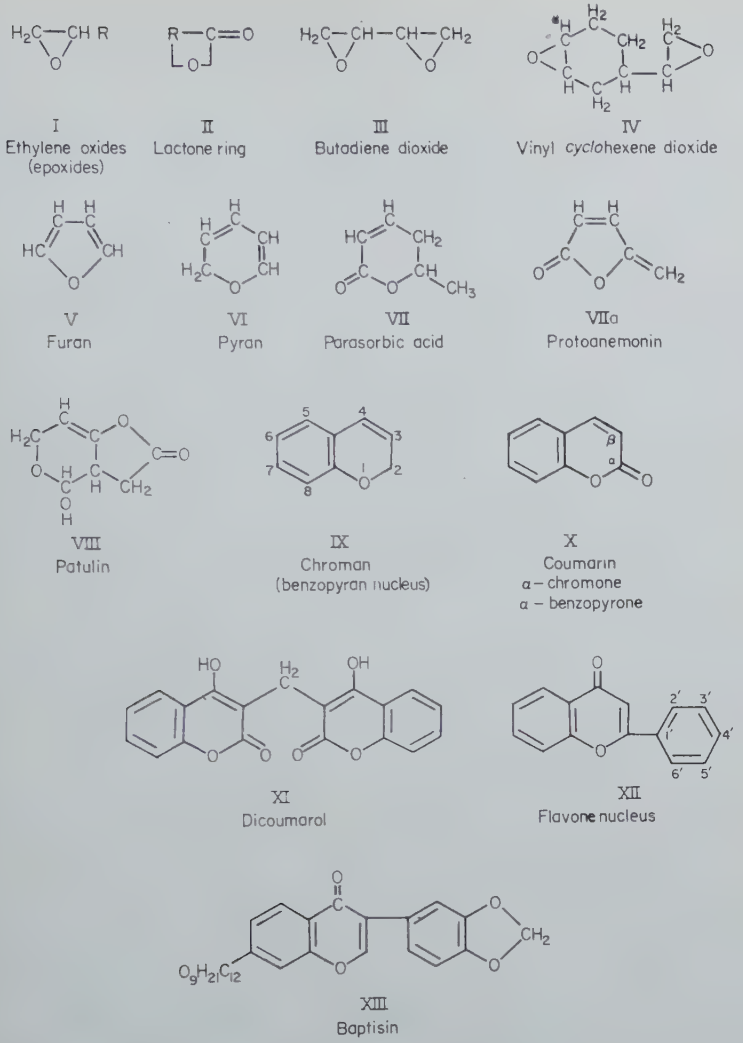


FIG. 22 Ternary cyclic compounds of biological significance (*continued on p.* 147).

partially saturated pyran. Lactones in fact tend to be rather generally inhibitory to growth (Alamercery *et al.*, 1951), and particularly (Fig. 22, VIIb) the partially unsaturated types (Medawar *et al.*, 1943). They antagonise the plant auxins.

A number of condensed oxy-heterocyclic systems occur in living organisms, and patulin could be taken as an example. Another simple type with metabolic value is the *chroman*, or *benzopyran*, nucleus (Fig.

22, IX), of vitamin E (p. 117); it is maximally unsaturated. More common is the *chromone* or *benzopyrone* nucleus (Fig. 22, X), the corresponding lactone. Coumarin, the toxic principle of sweet clover, is in fact a simple α–chromone, with the keto group in the *o*–position to the ring-oxygen. The active form may be a dimer (Fig. 22, XI), chain-linked and possibly capable of acting as an internal semiquinone by virtue of the symmetrically placed =O and —OH groups. It recalls dibromoindigotin (Fig. 28, X) in its general form and symmetry.

The γ–chromone, with the keto =O *para* to the ring-oxygen, occurs in a large group of plant pigments, the *flavones*, primarily pigments of the flowers, whereas carotinoids occur mainly in leaves and fruit. They are not all yellow as the name might imply but in fact also range through red and purple to blue. There is a great range in degree of resonance, therefore, and greens may be absent only for obvious reasons. Some of the *anthocyanidin* sub-group are based on the chromone nucleus alone, with the ring-oxygen a fully active oxonium cation, but most of them have this nucleus chain-linked to a benzene ring, forming the so-called *flavone* system (Fig. 22, XII). They are usually conjugated as glucosides, mainly via position 3; perhaps this is why they are not toxic like most lactones, and in some cases have other useful physiological functions. In fact one at least, *baptisin* (Fig. 22, XIII), does inhibit growth, and inhibits germination in seeds as powerfully as hexenolactone (Sexton, 1953, p. 367) and coumarin; the name is somewhat inappropriate, therefore. It has an additional dioxypentane ring, like those in the aphins.

The flavones with hydroxyl groups at 3′ and 4′ of the linked benzene ring are good reversible redox agents; they catalyse the reduction of metabolic hydrogen peroxide by ascorbic acid (p. 50), itself a lactone. *Citrin* or *vitamin P* is a flavone and may act in much the same way in animals, since it synergises with ascorbic acid, and improves both its action and its storage in the body. In fact it proves to be a mixture of two flavones, *hesperidin* and *eriodictyol*, the latter having the required 3′:4′ hydroxyl groups. Other flavones, such as *quercetin*, are believed to affect the reproductive behaviour of Algae, while *genistein*, from clover, is an anti-oestrogen; the pyrone ring can break to give a stilbene-like compound (Jellinek, 1958).

There are a few flavones in insects, probably derived from their plant food. *Xanthone* (Fig. 22, XIV), however, is an insect pigment consisting of a benzene ring condensed, and not merely linked, with the chromone nucleus; this is probably synthesised *in situ*, therefore.

Possibly related to the flavones is *rotenone*, a powerful paralysant for insects, occurring in derris root. It has five condensed rings (Fig. 22, XV), of which no fewer than three have an oxygen atom. One of them

can be either pyran or furan, partially saturated, and the other two are pyran and pyrone. The remaining rings are benzenoid. It is perhaps a flavone in which two more rings have been formed by closure of side chains, initially attached where glycose units are usually attached in the glycosides of flavones.

Two instances of a dioxy-heterocyclic ring have been seen already. In

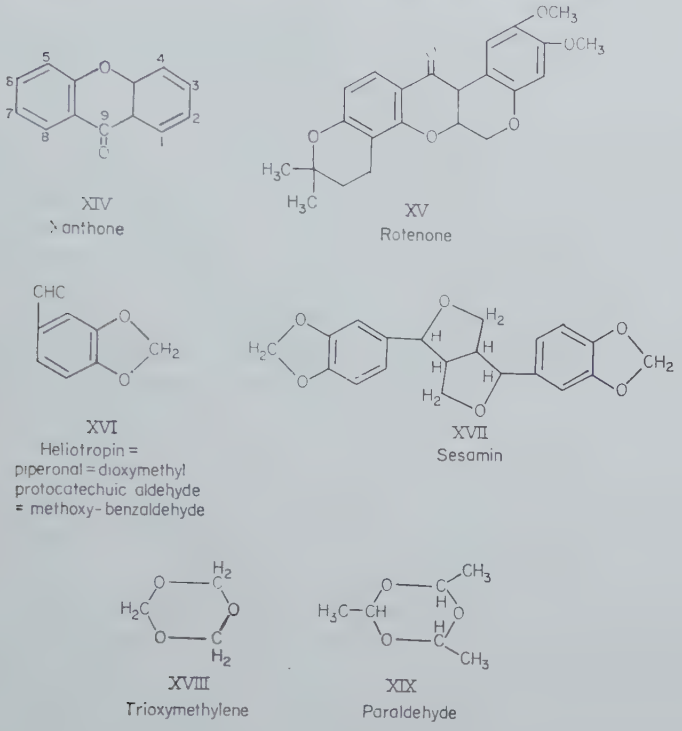


FIG. 22 (*continued*)

both cases they are part of the structure methoxybenzene (Fig. 22, XVI), which occurs free as *heliotropin* or *piperonal* (not to be confused with the *N*-heterocyclic piperidines). It can be synthesised by the oxidation of piperic acid from the alkaloid piperine (Chapter 18) or from a diphenol such as protocatechuic acid. A variety of piperine derivatives synergise with the pyrethrins (p. 112) as knock-down paralysants (Sexton, 1953, p. 331), and so does the more complex *sesamin* (XVII) which contains two piperonal units.

The heterocyclic X_6 ring with two oxygen atoms has been synthesised in the laboratory; *dioxan* is the fully saturated form and is useful as a

histological clearing agent since it has the unique property of being miscible with both water and apolar lipids such as xylene. In the laboratory the pentacyclic ring with as many as three oxgyen atoms has been synthesised, by treating olefinic double bonds with ozone. The compounds are therefore known as ozonides; they are explosively unstable, and this no doubt helps to explain why there is rarely more than one oxygen atom in biological rings.

Formaldehyde polymerises to trioxymethylene (Fig. 22, XVIII), among other compounds, and acetaldehyde to paraldehyde (Fig. 22, XIX), both having a trioxy-hexacyclic ring. This is unstable and the trimers rather readily revert to free monomers. It is not a ring of biological value, therefore.

It must not be concluded that oxy-cyclic rings are biologically unimportant, but only that most of the more outstanding of them have been, or will be, considered in other contexts. The hydrofuran and -pyran basis of the carbohydrates and many of their derivatives is of first importance, and ascorbic acid is a unique lactone with a quinonoid tendency in addition (p. 50). Vitamin E among chroman derivatives is metabolically unique, while the flavones form a large group, with more than aesthetic value.

CHAPTER 12

THE UNIQUENESS OF NITROGEN

FROM the preceding chapters it is clear that oxygen greatly increases the activity of carbon compounds, and this is at least equally true for nitrogen. Its contribution is unique, and it is not simply an alternative to oxygen; compounds containing both elements add properties shown by neither alone. Moreover these are not direct compounds between nitrogen and oxygen: organic nitrogen oxides are extremely rare. Both elements are strongly polar, tending to form ions in water; in virtually any form, singly or together, they increase the activity of carbon compounds in aqueous media.

Like carbon, hydrogen and oxygen, nitrogen is a unique element and no other can substitute for it except in very trivial ways. Like the other primary biological elements it is the lightest in its chemical group. Allen (1896) was already very impressed by the uniquely important properties of nitrogen, and called it the critical element. This is no doubt justified when we consider the outstanding importance of nitrogen compounds in living organisms: the amino acids, proteins, nucleotides, nucleic acids, B-vitamins, quaternary ammonium compounds and phosphatides in particular. Collectively these are certainly the most important 'agents' in the body, that is to say materials with a function other than merely as a source of energy. Allen considered that every vital phenomenon is due to some change in a nitrogen compound and indeed in the nitrogen atom of that compound; this may seem an exaggeration but when we consider how much activity in the body is enzymically promoted, and that all enzymes are nitrogen compounds, the claim seems justified. Nitrogen atoms are often the key to the enzyme's activity, as in the cytochromes, the pyridine dehydrogenases, and so on.

As an element nitrogen does not appear to be very reactive, since the free nitrogen gas constitutes roughly 80 per cent of the atmosphere, but this could be regarded as due to the great affinity of the element for itself. The nitrogen molecule is in consequence extremely stable and its heat of dissociation is as high as 210 kcal/mole, greater than that of hydrogen, oxygen, bromine, chlorine, iodine, sulphur, selenium, and tellurium. Its

oxides in fact are formed from nitrogen gas endothermically, and decompose again readily; this indeed is the main reason for the explosive properties of many organic nitrogen oxides, and for the rarity of this type of compound in living organisms. Most other elements form oxides exothermically and they are stable. The peculiarity of nitrogen is shown also in its greater affinity for hydrogen than for oxygen. Ammonia, NH_3, is formed from nitrogen exothermically, to this extent like the paraffins from carbon, but ammonia is more stable than the nitrogen oxides. Most biological nitrogen compounds in fact are ammonia derivatives, or 'substituted ammonias'. If oxygen were bound to nitrogen in an organic compound it should pass spontaneously to carbon and other atoms, and so it does in nitroglycerine and other explosives—with the encouragement of a little activation energy! Metabolism does not normally operate at such speeds. Dinitrophenol (p. 189) is relevant here.

The heat liberated when ammonia is formed from nitrogen gas is only 15 kcal/mole, and that absorbed in forming nitrous oxide from the gases is not much more; both reactions would be strongly exothermic starting from *atomic* nitrogen and any apparent anomalies therefore are associated with the one main feature, the great affinitity of nitrogen for nitrogen. However this does have, as a consequence, that a small shift of energy will permit the interchange of ammonium and nitrogen oxide compounds; the plant reducing nitrates and nitrites to amino compounds has not so far to go, energetically, as we sometimes suppose. Both groups of compound are also easily synthesised by 'fixing' atmospheric nitrogen; because of the great stability of the latter, in fact, Fogg (1955) suggests that the process should be called unfixing, a very pertinent suggestion, though it is important to stress equally that very little of the energy released when N_2 was formed from 2N needs to be returned in order to form the oxides, while a little more is released in forming ammonia. Biological nitrogen compounds are among its most stable forms, as are those of carbon, and their biosynthesis has a high degree of spontaneity. Nitrogen compounds are at the same time readily interconvertible yet stable, and various micro-organisms reduce the oxides while others oxidise amino compounds. It should be pointed out that the final stages of oxidation, to nitrite and nitrate, *are* exothermic, so that this last group obtains respiratory energy in that way. However, the main fact of the ready interconvertibility and small energy changes is valid, and large amounts of nitrogen oxides are formed in lightning discharges.

Nitrogen is a rare element on Earth (Table 2) since although constituting such a high percentage of the air it is absent from the hydro- and lithospheres. Its relative abundance in mobile gaseous form therefore is another of the unique strokes of fortune, like the mobility of carbon

dioxide, and of oxygen. There is some ammonia in volcanic gases, and a little ammonium-aluminium silicate in the Earth's crust, and some nitrate deposits of biological origin, but atmospheric nitrogen remains the main external source, if we may regard most of the nitrite and nitrate formed by bacterial decay as remaining within the biological system. Their solubility is an important safeguard against serious loss to the lithosphere. Otherwise all the oxgyen of the atmosphere might also have been lost in this way.

12.1 Ammonia

Ammonia, the basis of most biological nitrogen compounds, in turn has a number of unique features. As a member of Group V of the periodic system (Table 1) nitrogen has one pair and three unpaired electrons in its outer shell. The three become paired by covalent sharing of an electron, each with a hydrogen atom, forming bonds with much the same strength as the covalent bonds of methane. On account of its remaining *lone* pair of electrons, however, the nitrogen atom can form a dative or co-ordinate bond with a fourth hydrogen atom, that is to say a bond in which the nitrogen donates both electrons, so that the hydrogen is free to donate its single electron to an electrophilic group. The NH_4 therefore becomes a cation (Fig. 23, I), the basis for the *quaternary* ammonium compounds in general. In the figure the crosses indicate the electrons received from the initial three hydrogen atoms. NH_4^+ is a monovalent cation with properties similar to those of the alkali metals, though the base ammonium hydroxide is much weaker than sodium hydroxide and potassium hydroxide, and so is more useful for the relatively mild biological reactions. However, some of the organically substituted ammonias become very strong bases and are used as such *in vivo*—with safeguards. This is a good example of the influence of carbon on other biological elements, and of new properties emerging at biological levels. No ternary compound is at all strongly alkaline, as we should expect from the name 'oxygen', and amino bases are the biological alkalis *par excellence*.

Biological ammonia compounds retain the essential properties of the parent substance, since the covalent bonds are used for the C—N links, leaving the ionic bond still available. This would of course be impossible with cations such as Na^+ with only one valency bond, and it would also be impossible with divalent cations, such as Ca^{++}, because the second bond is also ionic and would not give a stable compound with carbon. In any case only monovalent cations form bases strong enough for useful purposes. The transitional elements, iron, copper, and others (Chapter 26), by using available electrons in the penultimate shell for binding

covalently with other elements, *are* able to form stable complexes, with their electrovalencies still free, but these are used for redox rather than for pH purposes.

It will be recalled that oxygen also is able to form a complex action corresponding to ammonium (p. 142). Although oxygen has two lone pairs of electrons only one of these is donated to an extra, electrovalent

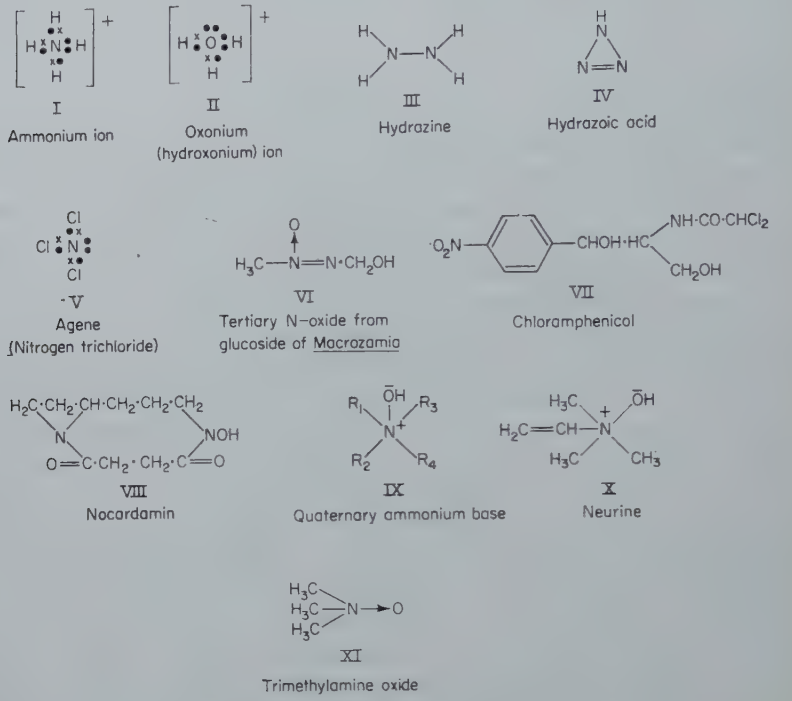


FIG. 23 Simple nitrogen compounds of biological importance.

hydrogen, so that the oxonium cation OH_3^+ also is monovalent (Fig. 23, II). Sulphur forms an analogous sulphonium cation, SH_3^+, although again in principle it could form a more complex ion; sulphur often displays a valency of 6 in fact (Chapter 23). These *onium* ions have properties very similar to those of ammonium, and are strong active bases by biological standards. However, as already stressed, oxygen bases are not stable and strong enough to deputise for NH_4^+.

Sulphonium is a much stronger base than OH_3^+ and it is responsible for a number of important biological reactions, in compounds such as thiamine (p. 397). It is unsuitable for many purposes, however, because the bonds of its other two hydrogen atoms are relatively weak, and tend

to become ionic, so that hydrogen sulphide, the *conjugate base*, is in fact a dibasic acid! Of course ammonia does act as a weak acid on occasion, as will be seen. Even as covalent bonds, S—H links are not very stable and the hydrogen is readily exchanged for other elements, making —SH compounds strongly reducing. This property is put to good use in living organisms but *ipso facto* it is an additional reason why NH_4^+ has become *the* biological base.

Another reason why ammonium has been uniquely exploited is that the fourth hydrogen makes the whole complex tetrahedral, like methane, and so it fits sterically the tetrahedral fabrics already dictated by carbon compounds and by water (p. 18). The reality of the tetrahedral structure is shown by the fact that when the four hydrogen atoms are substituted all by different groups the whole molecule becomes optically active, like those of carbon.

In its physical properties ammonia has some of the biological virtues of water. Its specific heat is in fact greater than 1·0, and its latent heat of fusion is 84 cal/g compared with the already high figure of 80 for water. The latent heat of vaporisation is lower than that of water, however, 332 cal/g compared with 539, and it is not such a good temperature buffer as liquid water. The viscosity of liquid ammonia is only one-quarter of that of water, so that it would not be so suitable for maintaining the mechanical stability of molecular structures, the segregation of reserve materials, and many other purposes. These properties are worth considering seriously since ammonia is liquid between —77·7° and —33·4°C, which is not a low temperature of the absolute scale. Such temperatures prevail on some of the planets of our own solar system and there are considerable amounts of liquid ammonia on some of them (Oparin, 1957a).

Firsoff (1962) has discussed in some detail the possibility of a form of life based on liquid ammonia in place of water and decides that a closely parallel type of system would be possible. Like water, liquid ammonia is a good solvent for many other substances; their relative solubilities are different, however, so that the ammonia-based life would differ in a number of details, quite apart from those associated with the different temperature range. The dipole moment of ammonia is 1·47 compared with 1·85 for water, so that it is a poorer solvent for polar types of molecule but a better one for non-polar and lipid substances. Among polar types the chlorides, so abundant in terrestrial living organisms, are mostly insoluble in ammonia, and so are the sulphates, sulphides, hydroxides and oxides. However, metals and some non-metallic elements, many of them good catalysts, are more soluble than in water. The alkali metals dissolve without any chemical reaction, forming interesting blue solutions; these are good demethylating agents (p. 178).

Just as in water, the solution of ionic and other polar molecules in ammonia depends on the slight ionisation of NH_3 into $^-NH_2$ and H^+; the latter tends to associate with another molecule of ammonia to form the ammonium ion, NH_4^+. The dissociation constant of ammonia is very low, however, $1{\cdot}9 \times 10^{-33}$ compared with 10^{-14} for water, and this explains its lower solvent power for polar materials and its lower electrical conductivity, though its solution of sodium chloride has a higher conductivity than aqueous sodium chloride. The dielectric constant of liquid ammonia is only one-quarter of that of water so that it separates charges in solutes less strongly—another manifestation of its poorer solvent power for polar molecules.

Hydrolysis has its parallel in ammonolysis, H^+ being added to the one product and $^-NH_2$ to the other. Ammonia also forms ammoniates corresponding to hydrates, so that it readily adds to the fabric of carbon compounds, as well as acting as a medium for them. It will also react in other ways with organic molecules to form analogues of the various series derived from water. Since hydrogen is common to both water and ammonia a hydrocarbon foundation could be common to the fabrics of bioplasms in aqueous and ammonious media. Among the more important radicals $—NH_2$ would replace $—OH$, and $=NH$ would replace $=O$. In addition $\equiv N$ is analogous to $=O$ and atmospheric nitrogen might be used in place of oxygen for respiratory purposes:

$$6CH_4 + 7N_2 \longrightarrow 8NH_3 + 3C_2N_2$$

where cyanogen is the analogue of carbon dioxide; unfortunately it is solid below −35°C and therefore could be returned to the biological cycle only via solution in the medium. However it may be true, as Firsoff suggests, that Jovians drink liquid ammonia and breathe atmospheric nitrogen!

The ammonia analogue of the amino acids (p. 161) is $R{\cdot}CH(NH_2){\cdot}C(:NH)NH_2$, the α–amino group being common to both, and an amidine group replacing the carboxyl group. Amidine compounds would be the analogues of the fatty acids and guanidine, $HN{:}C(NH_2)_2$, of urea. Peptides would be formed by the elimination of ammonia instead of water from the two condensing amino acids. The ammonia analogue of the carbohydrate monoses has actually been synthesised in the laboratory (p. 159). A number of these analogues and partial analogues, i.e. oxygen–nitrogen hybrids, are already components of our terrestrial bioplasm, so that the ammonia-based life need not be so very different from our own. Reciprocally most of the oxygen–water series of compounds are soluble in liquid ammonia. There is the possibility that life might evolve from water-based to ammonia-based or vice versa as the temperature changed

in the appropriate direction. If the temperature ranges of the two liquids were more nearly overlapping this might have to be taken very seriously. As it is, the gap between them might be bridged by materials capable of depressing the freezing point of water (p. 505). Replacement of oxygen and hydroxyl by their nitrogen analogues might not be so necessary in the actual fabric, since their compounds can be active in solution in ammonia at temperatures far below the freezing point of water; they could be as useful as nitrogen compounds are in our present water-based bioplasm. The nucleic acids (p. 360), which seem to be the essential genetic material of our present life, are also in a particularly favourable hybrid state. Protagonists of the theory that the Earth had a cold origin may be tempted to suggest that our life actually was ammonia-based at first and that the genetic material retains the stamp of this. Others might contend, with no less justification, that both nitrogen and oxygen groups of nucleic acids are essential to their vital function, in any conceivable medium.

Ammonia forms a dimer, hydrazine (Fig. 23, III), and a trimer, hydrazoic acid (Fig. 23, IV), and one or two other small polymers occur in living organisms. At low temperatures it forms polymers with as many as eight nitrogen atoms, but this is not to be compared with the faculty of polymerisation of carbon. Lead azide, the salt of hydrazoic acid, is very unstable and is used as a detonator. Such compounds as agene (Fig. 23, V) also are explosive and in fact nitrogen forms few stable compounds except with carbon and hydrogen. Agene, like most explosives, is very poisonous to living organisms (Read, 1942). The poisonous nature of these compounds, and indeed of ammonia itself, is not necessarily a serious argument against the possibility of an ammonia-based life since many simple carbon compounds are very poisonous, e.g. CO, RCN, C_2N_2 and HCHO; the most powerful poisons in fact are the nearest analogues to normal metabolites. In any case the latter might be different in an ammonious medium. Nitrogen *might* replace oxygen—but not carbon.

12.2 Organic Oxides of Nitrogen

The reason why oxygen directly bonded to nitrogen is very rare in living organisms has been indicated (p. 150); while the higher oxides are formed exergonically and are reasonably stable in inorganic form, they can be explosively unstable in the presence of carbon and hydrogen which have such a high affinity for oxygen. In trinitroglycerin, $C_3H_5N_3O_9$, there is enough oxygen to satisfy carbon and hydrogen by intramolecular rearrangement, and this is a main reason for its power as an explosive, but even where there is not enough, as in trinitrotoluene (TNT), the explosive

power is very great; the excess carbon is then deposited as soot, the process being much too rapid for oxidation by atmospheric oxygen. In passing it may be noted that this is a demonstration of the relative stability of elemental carbon. A mere mixture of elemental carbon and nitrates reacts explosively, as we know from the first gunpowders, and to bring the two, i.e. C and $—NO_2$, together in the same molecule is certainly playing with fire!

Most of the few biological nitrogen oxides are in fact antibiotics in which mildly explosive properties are an advantage, since the necessary activation energy keeps them safe until they reach a target organism. An example is Fig. 23, VI, found in a glucoside of *Macrozamia*, and another is *chloramphenicol* (Fig. 23, VII), an antibiotic which inhibits protein-synthesis. In chloramphenicol the halogen atoms contribute to the toxicity and we have the usual hydra-headed agent. *Nocardamin* (Fig. 23, VIII) also is an antibiotic; it has the oxime, $=N—OH$, group which is particularly rare because this is the most endergonic stage of nitrogen oxidation. The molecule is interesting also because of the degree of symmetry; as usual this is imperfect, but in this case there is rather restricted resonance even assuming that the nitrogen atoms become quaternary. Another nitroxy compound occurs in the defensive secretion of the bombardier beetle.

The compound, ammonium hydroxide, in theory has the ^-OH associated with the ionic H^+ of the molecule, but in effect the positive charge is transferred to the nitrogen atom, and quaternary organic nitrogen compounds are formed (Fig. 23, IX) in which all four hydrogen atoms are replaced by a carbon radicle, and the nitrogen remains cationic; in fact this group includes some of the strongest nitrogen bases. Occasionally the free base occurs, as shown, for instance in neurine (Fig. 23, X), and so there are nitroxy compounds among normal metabolites. Another, somewhat related, substance is trimethylamine oxide (Fig. 23, XI), an excretory product of elasmobranchs, teleosts and cephalopod molluscs; here the nitrogen donates its lone pair of electrons to complete the two half-filled orbitals of oxygen. The bond is dative, as in NH_4, but there is no tendency to ionise and the product is stable. Both here and in the quaternary hydroxides the oxygen is not so labile as in covalent nitrogen–oxygen links, and it is only the latter which are biologically dangerous as explosives. This is not to say that all toxic nitroxy compounds are necessarily of this type, since there are many causes of toxicity, but it is worth noting that the hydroxyl of hydroxylamine, H_2:N—OH, is covalently and not electrovalently linked, and similarly in the oximes.

The rarity of oxides of nitrogen in biological materials is due to biological, natural selection, causes since there is no great difficulty in synthesising those of any level of oxidation (p. 150). There is a thermo-

dynamic bias in favour of reduced nitrogen as of oxidised carbon, and this is reflected in the forms in which they are excreted. Urea and uric acid are of course a compromise between oxidised carbon and reduced nitrogen; on the other hand trimethylamine oxide could be said to have its nitrogen oxidised and its carbon atoms reduced, and living systems have little difficulty in preparing both elements in most levels of oxidation–reduction except the acetylene and nitrous oxide levels.

There appear to be considerable amounts of free gaseous oxides of nitrogen on Venus (Moore, 1957), Mars (Kiess *et al.*, 1960), Jupiter and Uranus. This may imply that there is little reduced carbon and nothing similar to our type of life. On the other hand there is much evidence for some form of life on Mars, at least, and the gaseous nitrogen oxides present an interesting problem.

12.3 Cyanides

Since in general the nitrogen is directly bonded to carbon in biological compounds, it is reasonable to glance at the cyanides, or nitriles, with the smallest organic nitrogen radical, —C≡N, as the possible starting point for biosynthesis. The great toxicity of hydrocyanic acid and organic cyanides, RCN, need not deter us unduly, for the reasons given, though this does raise the problem of a suitable, safe precursor. Some biologists have postulated the evolutionary origin of amino acids (p. 161) from alkyl cyanides by hydrolysis, which is spontaneous, but in fact the laboratory reaction is:

$$RCN + H_2O \rightarrow RC\begin{matrix}\diagup\!\!=\!\!\diagdown NH \\ \diagdown\!\!\diagup OH\end{matrix} \rightarrow RC\begin{matrix}\diagup NH_2 \\ \diagdown\!\!=\!\!\diagup O\end{matrix} \xrightarrow{H_2O} RC\begin{matrix}\diagup\!\!=\!\!\diagdown O \\ \diagdown\!\!\diagup OH\end{matrix} + NH_3$$

the intermediate compound having the amino group on the carboxyl carbon and not on the next, or α, carbon. Others have postulated a polymerisation of cyanide (Oparin, 1957a, p. 213), as the means of forming the peptide bond at one step, but this gives a main chain entirely of carbon atoms, with nitrogen only at the ends:

$$HC{\equiv}N + HC{\equiv}N \rightarrow N{\equiv}C{-}CH{=}NH \xrightarrow{HCN} NC{\equiv}\overset{\overset{NH}{\|}}{C}{-}CH{=}NH$$

$$\xrightarrow{HCN} N{\equiv}C{-}\overset{\overset{HN}{\|}}{C}{-}\overset{\overset{NH}{\|}}{C}{-}CH{=}NH \text{ etc.}$$

whereas the NH group is an integral part of each peptide link. This was intended as a theory of phylogenetic origin of peptides but there is no reason to think that this would differ essentially from the normal meta-

bolic origin, which has probably not changed radically since then (Needham, 1959). Plants today start with nitrates and reduce them to ammonia, or direct to amino compounds.

Although they may not be significant as precursors of biological nitrogen compounds cyanides do have biological importance. Their great toxicity is due mainly to their ability to form complex anions, the ferro- and ferricyanides (Fig. 38, XVII), analogous to the iron porphyrans (p. 315). The iron of the latter bonds with cyanide in preference to the normal oxygen, in the case of haemoglobin, or instead of undergoing the normal electron-shift, in the case of the cytochromes, and so aerobic animals die of asphyxia. In cyanocobalamin (p. 321), however, —CN is bonded in this way to cobalt, in porphyran combination, as a normal metabolite. Cyanide is a stable radical because the electronic orbitals of nitrogen are all filled and it is further stabilised by resonance between the forms H—C≡N and H—N=C. The radical is so firmly bound as a unit that the hydrogen of hydrocyanic acid ionises as a weak acid, the nitrogen in the second resonant form is effectively quaternary, $\overset{+}{H}N \equiv C^-$. ^-CN changes rather easily from an anion to a covalently bonded stable radical, which is the biologically important form. At the same time it may be significant that hydrocyanic acid has a higher dipole moment and even a higher dielectric constant than water (Tables 11, 12).

Another possibly physiological action is as a plant growth factor, in indoyl acetonitrile. On the other hand the '*lathyrus* factor', *β–aminopropionitrile*, $H_2N{\cdot}CH_2{\cdot}CH_2{\cdot}CN$, of the sweet pea is very toxic and there are some other cyanides probably used defensively. This may be true for amygdalin (p. 134) and the secretion of some millipedes contains free hydrocyanic acid (Jones, *et al.*, 1962).

12.4 Amino Compounds

In the broad sense virtually all biological nitrogen compounds are substituted ammonia or amino compounds. The most important of these will be considered in the following chapters. Most of them are fairly fully reduced compounds and so follow on rather naturally from the previous chapters but one group, the amino sugars, belong to more oxidised levels (p. 51). As in amino compounds in general their nitrogen is bonded directly to carbon, usually the 2C, and ammonia effectively behaves as an acid. At 2C the amino substituent protects the 1C aldehyde group, so that it no longer gives the Molisch or other aldose reactions. This may be biologically important, since sugars aminated at other positions, 3C or 6C, still behave as aldehydes. Another effect of amination

is that the glycosides formed are more stable; many other properties also are changed. Enzymes acting on the free hexose will not attack the hexosamine. The hexose reciprocally affects the amino group which now has a p*K* near neutrality and so is much less alkaline than ammonia.

Although the glycosides of the hexosamines are stable, the free hexosamine is very labile and reactive, especially in alkaline medium. Among the compounds formed is a dimer which is bonded via the amino group of both residues to form a C_6 ring, effectively a pyrazine ring (p. 346), therefore. This illustrates the ease with which *N*–heterocyclic rings are formed, and further examples of this will be seen later.

As already noted (p. 51) acetylation of the amino group stabilises it in alkaline medium, and this is at least part of its significance in chondrin, mucin and chitin. However, acetylation largely restores the sensitivity of the hexose to acid, against which the amino substituent is a protection, so that the free hexosamine is to be expected in acid and the acetylated derivative in alkaline media. Acetylation specifically at the 2C position is favoured by lipid media. The properties of galactosamine differ in detail from those of glucosamine: its acetyl compound oxidises more easily and this may be relevant to the differences in texture and other properties between chondrin and mucin.

The 2C–, or α–amino, hexoses are aldehyde analogues of the amino acids (p. 161), and when the aldehyde group is oxidised to carboxyl the resulting amino hexonic acid *is* a polyhydroxy amino acid. It shows the properties of the hydroxy amino acids such as serine (p. 177). However, these amino hexonic acids are rare *in vivo*; they are inferior to the natural amino acids for their specific purpose, no doubt, even if they are not frankly troublesome as competitive analogues.

The amino group of the glycosamines can be further substituted, as in most amino compounds, to form secondary, $(R_1R_2)=NH$, and tertiary, $(R_1R_2R_3)\equiv N$, amines, where R_1 is the hexose. These are not formed as readily as by a simple alkyl primary amine such as ethanolamine, $C_2H_5NH_2$. However, the tertiary base goes on to form quaternary compounds, $(R_1R_2R_3R_4)N^+A^-$, and these behave like the most physiologically active of the quaternary nitrogen compounds, *choline* (p. 289), though much more feebly than acetyl choline.

The monoses also form amino compounds at the 1C position, and the nucleosides (p. 355) are essentially compounds of this type. Occasional substituents at other positions also occur in living organisms and in the laboratory hexoses have been aminated at all six positions by long chain aliphatic amines (Danehy and Pigman, 1951). In the antibiotic *puromycin* (*achromycin*) there is a 3–amino–3–deoxy ribose, another example of the use of rare structures in this group of agents; in this same antibiotic there is

also a biologically abnormal purine, 6–dimethyl amino purine (Kent and Whitehouse, 1955), so that the molecule as a whole may be a nucleoside analogue. It is a metabolic antagonist of the nucleic acids.

Monoses are readily aminated also by other amino compounds; urea (p. 284), guanidine (p. 300), amino acids (p. 161), pyrimidine and purine bases (p. 347) and others. The amino acid compound depresses both the reducing action of the aldose and the amino group activity of the amino acid. It readily polymerises, so that it may be relevant to the formation of carbohydrate–protein associations in skeletal materials, if not to glycoproteins in general. It may be responsible for the aesthetically objectionable browning of sugars in some foods (Thompson, 1950).

Sialic or *neuraminic* acid (p. 52) is a more complex derivative of an acetyl monosamine. The polyhydroxy alcohols also form amino compounds, for instance in ribityl flavin (p. 380). A diamine occurs in the antibiotic, *neomycin A*.

CHAPTER 13

AMINO ACIDS

THERE is a startling paradox in the term 'amino acids' and no doubt they are the outstanding example of this unique property of organic molecules, of being able to bear groups ionising as bases alongside acidic groups, without the two reacting to form an internal salt. The molecule therefore behaves simultaneously as acid and base, and this gives it a power and versatility, coupled with a mildness of resultant pH, and a buffering power, which is unrivalled elsewhere in nature. It is significant that in variety, in bulk of products, and in functions, the amino acids are the most important of biological amino compounds.

As already stressed, amino compounds are the only strong bases capable of forming organic compounds and by good fortune the carboxylic acids approximately balance them in strength. Other acids are sometimes associated with amino bases but amino acid is taken to mean this special compound, the aminocarboxylic acid, $H_2N{\cdot}R{\cdot}COOH$. In fact it is usually taken to mean a much more restricted group, in which the rest of the molecule, R, is almost fully reduced, i.e. at the lipid level of biosynthesis, so excluding the aminohexonic acids (p. 159) and others at this general level. The use by biologists of amino acid, without qualification, for the lipid-level amino acids, alone, is justified by their overwhelmingly predominant role. Many biologists go further and use the term without qualification for the α–amino acids alone (Fig. 24, I), those with the amino group on the carbon next to the carboxyl, which of course must be in a terminal position of the molecule. It is members of this group which build the proteins, the main fabric of bioplasm, so that the restrictive emphasis is largely justified. In any case the number of other types of amino acid present in living organisms is small, for the usual reasons of unique exploitation and restricted stable states. Those of intermediate grades of reduction are rare because all compounds in these grades are rare. The amino group is usually introduced into a molecule already extensively reduced and stable.

The α–amino acids are conveniently regarded as consisting of an α–complex, $H_2N{\cdot}\overset{|}{C}H{\cdot}COOH$, and a side chain, R, which in the simplest

member, glycine, is simply a second hydrogen atom, attached to the α–carbon. In other amino acids, however, the side chain is relatively long and is the main chain as far as the individual amino acid is concerned. The term side chain becomes relevant when the amino acids are built up into peptides. The side chain of some of the amino acids is mainly a ring structure (Fig. 24).

There are twenty α–amino acids commonly found in the proteins of all living organisms (Fig. 24A); sporadically one or two more occur but this twenty is an impressive demonstration of the common kinship of

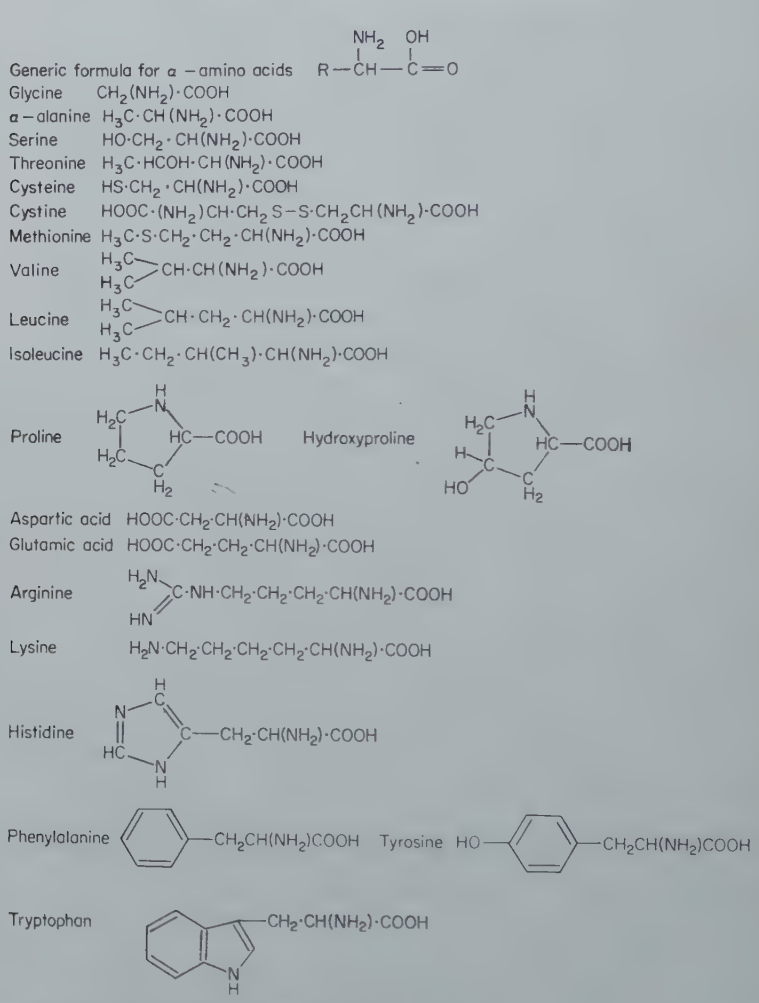


FIG. 24 The more important biological amino acids. A: Protein-building.

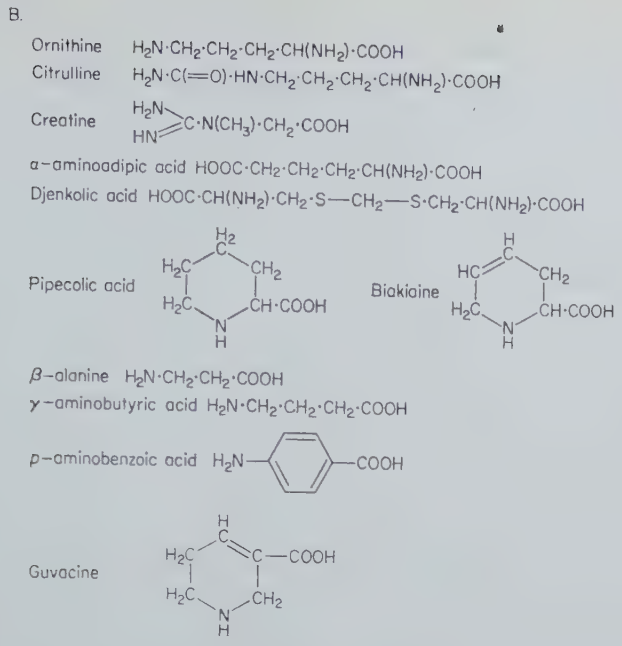


FIG. 24B: Non-protein amino acids.

all organisms, particularly in view of the high percentage of protein in the organic material of the body. One or two other α–amino acids, such as *ornithine* (p. 183) and *creatine* (p. 183), are important as free metabolites and may never be incorporated into protein. Even including these and rarer ones the total number of biological amino acids is as usual a relatively small fraction of that theoretically possible; by general biological standards, however, this is an impressively large number, as would probably be agreed from the evidence of preceding chapters. The groups still to be considered confirm the conclusion that there are more amino acids common to *all* organisms than compounds of any other one group. Moreover each amino acid is individually more unique and active in general metabolism than the members of perhaps any other group. They are the key biological materials, and we must consider first the common α–complex and then the specific properties of each side chain.

13.1 The α-Complex

Of course side chain and α–complex do interact, and in some cases this may be the most important feature of the molecule, but analytical convenience is not the only justification for considering them separately. To

some extent the α–complex is simply a device for stringing together a number of metabolically active side chains. Some amino acids have an extra amino group and others a second carboxyl but these have a different significance from those of the α–complex. For the understanding of the latter the few amino acids with their *only* amino group in some other position are most instructive. Three of these are biologically important, β–alanine, γ–aminobutyric acid, and *p*–aminobenzoic acid (Fig. 24).

The common amino acids could be regarded as aminated fatty acids and a striking feature therefore is the extent to which the amino group increases water-solubility and depresses solubility in lipids. Very few of the amino acids are insoluble in distilled water and none is insoluble in dilute acids. At the same time the close proximity of amino group and carboxyl at one end of the molecule gives it a strong polar–apolar morphology, in some of the amino acids, and this is biologically useful. With this structure chains much longer than those actually occurring would counteract the hydrophil effect of the α–complex. Even as it is, once the latter is involved in a peptide bond the balance is shifted, and side chains can be so orientated as to make parts of the protein molecule insoluble. Here, better than in any other case perhaps, we can see the importance of ionisation for water-solubility; amino acids are about 400 times as soluble as the isomeric hydroxy-amides, $R{\cdot}CH(OH){\cdot}C\begin{matrix}NH_2\\ \diagdown\!\!\!\!\diagup\\ O\end{matrix}$ (Cohn and Edsall, 1943). As a point of interest, their relative rates of permeation through membranes are in the same ratio, and this is less easily understood, since in general lipid-soluble molecules are found to permeate best, and ions most slowly.

The comparison with hydroxyamides reveals some further important details of the properties of the amino and carboxyl groups, individually and in proximity. On any but the carbonyl carbon, the hydroxyl ionises as poorly as that of the alcohols, but on the same carbon as $=O$ it ionises as $O{=}C{-}O^-$ and H^+, because of the resonance between the two structures $-C\begin{matrix}OH\\ \diagdown\!\!\!\!\diagup\\ O\end{matrix}$ and $-C\begin{matrix}O\\ \diagdown\!\!\!\!\diagup\\ OH\end{matrix}$. On any carbon but this one, $-NH_2$ ionises by the usual quaternary change: $-NH_2 + H_2O \rightarrow -NH_3^+ {}^-OH$, but on the carbonyl group, as in the amides, there is resonance between $-C\begin{matrix}NH_2\\ \diagdown\!\!\!\!\diagup\\ O\end{matrix}$ and $-C\begin{matrix}{}^+NH_2\\ \diagdown\!\!\!\!\diagup\\ O^-\end{matrix}$ and this does not labilise an anion, the group as a whole remaining externally neutral. Nitrogen is more nucleophilic than oxygen and holds its H^+, while oxygen is not nucleophilic enough to form oxonium $-{}^+OH_2$ at α, or at any other position in an open chain.

Any amino acid differs from the corresponding hydroxyamide in this way, but in addition there is an interaction between the two ionising groups which varies with position and is maximal for the α–relationship. By contrast to the α–amino acids the amino benzoic acids, in which the amino group can never be on the same carbon as the carboxyl group, ionise rather poorly (Cohn and Edsall, 1943), and the amino phenols scarcely at all, although the phenolic hydroxyl is a strongly acid group. The two groups mutually promote ionisation and this is maximal for close proximity; this is equally true for solubility, therefore, so that there is the somewhat paradoxical situation that the molecule is most soluble with its polar groups crowded at one end. Because of the low dissociation constant of water the ionised amino acid will exist mainly as the dipolar ion, $H_3^+N{\cdot}CHR{\cdot}COO^-$, also known as a *zwitterion*. Its existence is actually favoured by agents which depress the dissociation of water and it is the main 'species' even in 90 per cent ethanol. To some extent it could be considered to be an internal salt, although the proximity of the two ions prevents their actual union, and the amino acids have the general physical properties of inorganic salts, including a high melting point. Some amino acids will even sublime without melting. The actual values of their melting points are lower than those of many inorganic salts, of course, because of the rather different and labile nature of the zwitterion, and because of the extensive hydrocarbon portion of the molecule.

The two charges of opposite sign, separated by more than 4Å, give the molecule a high dipole moment (Table 11); this is actually maximal for the α–relationship since when the two groups are more widely separated in the chain they become in practice more closely associated through incipient ring closure promoted by the tendency to internal salt-formation. This in addition tends to diminish the number of free charges. The dipole moment of glycine is 15 Debye units compared with the value of 1·85 for water, which already has a value higher than that of most materials (Table 11). In peptides the two charges are separated by great distances and very high values of the dipole moment are recorded.

As in the case of water itself (p. 20), the high dipole moment facilitates the separation of ionic charges in the molecules of other solutes, promoting their ionisation and solution. Amino acids improve the solubility of salts of calcium, iron and other metals, and improve the ionic activity of such salts as the iodates of silver and lead. A number of them enhance the solubility of other amino acids, so that a mixture is usually more soluble than a pure amino acid; this is one reason why proteins are often as soluble as individual free amino acids, notwithstanding their large molecules and small number of free amino and carboxyl groups. However, this interaction between amino acids is not purely generic but is specific

to particular pairs and so concerns the side chains as well as the α–complex. Moreover, certain amino acids depress the solubility of others, and some enhance that of one but depress that of others; glycine enhances that of cystine (p. 177) but depresses that of asparagine (p. 181). This reveals a vista of interactions which must contribute greatly to the versatility of living systems and of proteins in particular.

Reciprocally the solubility of amino acids and proteins is very sensitive to the action of other solutes. Inorganic salts increase the solubility of most of them, with the exception of the ketogenic amino acids (p. 65) and those with a terminal, hydrophobe, methyl group. Rather naturally, the most strongly electrovalent metals have the most effective salts, the alkali and the alkaline earth metals in particular, which are the common salts of natural waters because of their own solubility. This is in fact an instance where living systems have used the naturally occurring potentialities (p. 4); their body fluids approximate in ionic ratios to those of sea water mainly for this reason, though the ratios have been changed in detail to meet specific requirements—even in marine organisms which cannot change the total osmolarity of their internal medium (Robertson, 1957).

The high melting point makes the amino acids relatively stable, and also the proteins. It is estimated that some of them would be stable for millions of years in the dry state (Abelson, 1957), and it is not surprising that spores of bacteria and other micro-organisms will survive for many years and germinate when re-moistened. Even the seeds of angiosperms such as the lotus may possibly germinate after several thousand years in a dormant state. Alanine and other amino acids have been identified in Cambrian fossils, and many laboratory-stored protein materials appear to retain their properties for many years (Keilin, 1953a).

Interaction between amino and carboxyl groups inevitably affects their strength as base and acid. The α–amino group increases the acidity of the carboxyl about fourfold, the pK being depressed from pH 4·8 to 4·3, presumably because RNH_3^+ reduces the ability of H^+ to suppress the ionisation. On the other hand the carboxyl *depresses* the alkalinity of the $^+NH_3$ by as much as 3 pH units, or 1000-fold. This can probably be interpreted as a depression of ionisation of $R{\cdot}NH_3OH$ by COO^- ions. In consequence of these interactions the isoelectric point of the α–amino acids, the pH at which the two groups are ionised to an equal extent, is not near neutrality, as might have been anticipated from the pK values of amines and fatty acids, but considerably on the acid side of neutrality. There are more large anions, $H_2N{\cdot}R{\cdot}COO^-$, than large cations, $H_3^+N{\cdot}R{\cdot}COOH$, in solution in physiological media, which are usually near neutrality, so that amino acids and proteins react with metals rather than with anions. The asymmetrical position of the isoelectric

point of course does not vitiate the term neutral amino acid for those with one amino and one carboxyl group, and no other acid or base groups; this merely implies that they carry an equal number of potential + and − ions and do not appreciably change the pH of distilled water. In aqueous solution the ^{-}OH ion suppresses the ionisation of RNH_3OH more than $^{+}OH_3$ does that of RCOOH, and the reaction remains neutral.

Neither group is as strongly ionising as mineral acids and bases and therefore the amino acids are good buffers, capable of mopping up any excess of free ions of either charge, to form on the one hand un-ionised acid or base and on the other hand water, together with a large ion which affects pH very little. Thus an added acid H^+A^- suppresses the ionisation of carboxyl and reacts with RNH_3OH to form $RNH_3^+A^- + H_2O$, while added alkali, $B^{+-}OH$, has the reciprocal effects. Within the buffering power of the amino acids, which depends on their concentration, the pH does not change; pH is a matter of the balance between the ions of water, H^+ and ^{-}OH specifically, and the ions RNH_3^+, A^-, $RCOO^-$ and B^+ do not affect it very much. In most amino acids the p*K* of the two groups is sufficiently far apart for them to act as buffers over distinct pH ranges, and the great advantage over simpler buffers is just this. If a neutral solution of an amino acid is titrated against an acid the curve is of the type typical for buffers (Fig. 25), i.e. it shows a plateau in pH over a considerable range of amount of acid added. If the neutral amino acid is titrated against alkali a second plateau is obtained at a different pH. The amino acid buffers at these two pH values, and in fact over a range centred on these two values, the plateau being not quite flat. A number of different amino acids collectively cover the whole of the biologically significant pH range.

Although monobasic, inorganic, weak acids and bases have only one buffering range a tribasic acid such as phosphoric has three, and is a very useful buffer for that reason. The second range, $BH_2PO_4 \rightleftharpoons B_2HPO_4$, centres on neutrality, pH 7·0, and therefore is particularly valuable *in vivo*; some of the simpler amino acids in fact have their two ranges some distance on either side of neutrality and so are less useful in this respect. However small changes in pH are less dangerous than further changes away from neutrality and the collective distribution of ranges among amino acids is a good one. Mammals will not tolerate a movement of the pH of their blood by more than 0·4 pH units from neutrality, and the phosphate buffers of the plasma are no doubt invaluable. It is well known that bicarbonate and some other acids and bases also play a part in buffering *in vivo*. Proteins are less good than amino acids, weight for weight, because peptide-formation removes the ionising power of the amino and carboxyl groups involved, but because of the amount of proteins they still constitute one of the major buffer systems of the body.

Probably more important than the ambivalent ionic properties of the amino acids is their ability to polymerise with their identity and with each other, by means of the *peptide bond*, —HN—C(=O)—, to form chain-molecules or peptides, of indefinite length, with molecular weights running up to several millions of Daltons. Each amino acid can form a peptide bond with each of its α–polar groups, and *in vivo* probably not

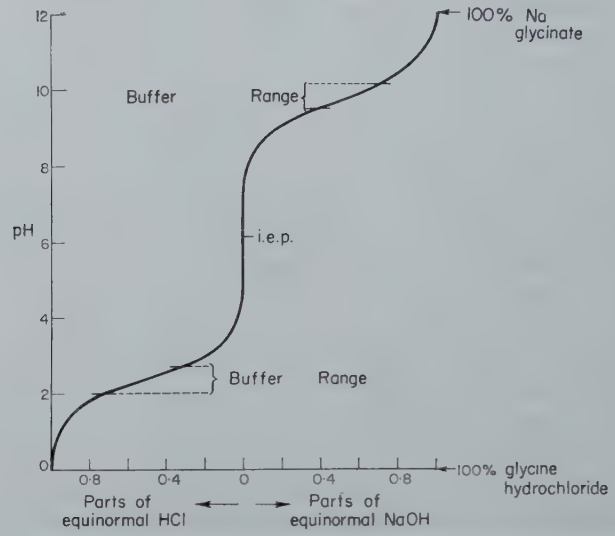


FIG. 25 The titration curve of glycine, as a typical amino acid, against equinormal hydrochloric acid and NaOH. The zwitterion acts as a buffer with both acid and alkali, but over widely separated pH ranges, in this case both far from pH 7·0. Collectively, however, the amino acids as buffers cover the whole of the significant pH range.

with amino or carboxyl groups in other positions, so that linear peptides are *de rigueur*. The linkage between amino groups of one and carboxyl of another amino acid is no doubt one manifestation of the ambivalence of the α–complex, but it is important to note that it is not a simple salt-formation as might have been anticipated. Electrovalent bonds of the type $R_1NH_3^+ \ldots {}^-OOC{\cdot}R_2$ in fact would not give the strength found in the peptide chain and absolutely essential for the functions it performs. The reaction is actually a condensation, with the elimination of water, but with the amino group effectively acting as acid and the carboxyl as base, forming a carbonyl amine:

$$HOOC{\cdot}R_1{\cdot}NH_3^+ + {}^-OOC{\cdot}R_2{\cdot}NH_3OH \rightarrow$$
$$HOOC{\cdot}R_1{\cdot}NH{-}OC{\cdot}R_2{\cdot}NH_3OH + H_2O$$

Both groups are therefore individually ambivalent and in principle

could form anhydrides with their identity if this were stable. The peptide bond is endothermic to the extent of 3500 cal/g equivalent, but this is a small value compared for instance with that for the acyl anhydrides (p. 92) and it is proportionately more stable, requiring enzymic help for hydrolysis at a reasonable rate. The atoms concerned in the actual bond, —HN—C(=O)—, are virtually coplanar and form a relatively rigid structure, properties which have biological value, as will be seen.

Peptides appear rarely to undergo ring closure between their two ends, which would stop their growth, and so they have a free amino group at one end and carboxyl at the other and remain large zwitterions of a kind, quite apart from polar ionising groups in the side chains. Their dipole moment is very great, as already indicated, and their solutions have a high dielectric constant, with a great power to hold charges separated, potentially ready for electrical work. Cyclic peptides *do* occur, but probably only in antibiotics, one of the many abnormalities which they exploit. *Gramicidin-S* is a cyclic pentapeptide.

Another situation in which the peptide lacks the zwitterion structure is when either proline or hydroxyproline (p. 179) are present. These are *imino* acids, the α–amino having condensed with the terminal carbon atom of the side chain (Fig. 24) and being no longer free. It is not available for peptide bond formation, either, and so these amino acids can normally only occupy a terminal position (but see p. 179). There is known to be a terminal proline residue in haemoglobin.

13.2 The Side Chains

There are two main questions about the side chains: what is their general function and why have this particular twenty been exclusively exploited? In answer to the first, the simplest possibility would be that they are merely 'handles' for manipulating the α–complex while it is forming a peptide bond. In this event, however, no variety would be necessary. Moreover the ease with which peptides, including homopeptides, are formed by glycine, which has effectively no side chain, shows that this is incorrect, and that a side chain is not necessary for the synthesis of the α–complex in the first place.

One general function of all side chains, though it may be quite incidental, is that they make the α–carbon asymmetric and the amino acid optically active. Glycine, which alone has two identical (H) groups on this carbon, is inactive. In passing it may be noted that the natural amino acids are usually regarded as being L–forms, structurally derivable from L–glycerose, whereas the natural carbohydrates are all based on D–glycerose. However the hydrogen atom is the only substituent common

to both glycerose and the α–carbon of the amino acids and asymmetry is lost between the carbohydrate and amino acid stages (in the α–keto intermediary) so that the kinship to L–glycerose is very tenuous. The actual rotation of the amino acids depends on the four substituents, of which only the side chain varies. This sometimes also contains extra asymmetrical carbon atoms so that their net rotation varies in magnitude and sometimes in sense. They are all pure isomers as usual, and apart from the occasional antibiotic all have the same arrangement of the four groups on the α–carbon.

Another possible indication of a general function is that all side chains are relatively short. That of lysine is virtually the longest, C_4 plus the ω–amino group. Some bacteria have its next higher homologue, diaminopimelic, but this is rare. The effective length of those with ring structures in the side chain is not greater than that of lysine. As already indicated, this ensures that the free amino acids are water-soluble and it may also ensure that all can co-operate on equal terms as side chains of a completed peptide. Further it avoids any serious danger of chemical interference (Waugh, 1959), or of tangling with each other and with the peptide chain itself. A number are known to be used for linking neighbouring peptide chains together and for this a uniform and preferably short length is optimal. Dipeptidases will attack bonds only between amino acids with short, natural side chains (Sexton, 1953, p. 67), but of course this may imply simply that they have been selected to deal only with the natural length and does not help to explain why this is short, and rather uniform. It contrasts sharply with the uniformly *long* molecules of the fatty acids and the carotinoids.

In diametric opposition to the idea that the side chains are mere handles for the α–complex is the possibility that the peptide chain is merely a convenient device for deploying chemically active side chains. This is equally easily refuted in the case of the skeletal proteins, at least, where the peptide chains are the whole basis of the fibrous fabric. On the other hand it might well be true for some enzymes, and partly true for most. In the final analysis it is difficult to avoid the conclusion that both parts of the molecule are independently important and that their relative importance varies in the different proteins. Proteins have physical or chemical functions, or both simultaneously, as will be seen, and a similar duality is found in the functions of many of the individual amino acids, though in this case it is not necessarily based on α–complex and side chain respectively. Multiplicity of exploitation in fact is inherent in biological evolution and is not restricted to amino acids and proteins.

However uncertain our conclusions about the *general* function of amino acid side chains there seems little doubt that each has a specific

function. Their variety alone provides strong *a priori* grounds for this and it is worth considering them in turn. Before doing so, however, there are several points which concern them collectively. Their uniqueness as a group, among biological groups of compounds, is seen very clearly by comparing them on the one hand with the fatty acids and on the other with the monose carbohydrates, the sterols or the nucleic bases. The former have a molecule with a similar plan, always with a carboxyl group at one end, but the rest of the molecule is rather monotonous and varies only in detail, in great contrast to the side chains of the amino acids. In the other set the general plan of the molecule, quite complex in some cases, is common to all members which vary only in details; again there is nothing corresponding to the variety of amino acid side chains. In the amino acids the α–complex is the common feature and side chains belong to a wide variety of biochemical groups. Indeed in some ways it would be easier to deal with them in the context of these various groups, some of which have not yet been introduced: in practice a compromise on this seems preferable, as it has been, perhaps, for life itself.

The great variety of side chains is reflected in their biosynthesis. For heterotrophes some amino acids are *non-essential*, i.e. they can be synthesised at home, while the *essential* group must be acquired in the food or deficiency diseases develop, such as *kwashiorkor* due to lack of tryptophan. Davis (1961) has shown that the non-essential amino acids are synthesised from the stage of nitrogen-free precursors by one to three enzymically catalysed steps (average 1·4), whereas the essential group need an average of 5·5, the actual number again varying between the individual members. The broad distinction is opportunist, therefore, but for the present purpose the main point is that there are rarely more than two common amino acids synthesised via any one pathway, and that a number of them involve three to ten steps even from the stage of a near precursor. For the α–complex the pattern of synthesis is rather uniform: the precursor is obtained in the form of an α–keto acid, R·C(=O)·COOH, and the amino group is *trans-aminated* from glutamic acid (p. 180) which in turn can be aminated by ammonia direct. For the common portion, therefore, a common mechanism has been evolved, even though this does come at the end of the pathways, which have been very variable up to this point.

This contrast heightens the distinction between side chain and α–complex, and together with the variety and length of individual pathways it also heightens the paradox that in a number of laboratories during the past twelve years a variety of amino acids—among other biological molecules—have been synthesised together by merely activating a mixture of simple precursors such as CO_2, H_2O, N_2, NH_3, HCHO, CN, etc. (Table 22). The means of activation clearly are more powerful than

TABLE 22. LABORATORY SYNTHESES OF AMINO ACIDS AND OTHER METABOLITES

Raw materials	Method	Products	Reference
H, NH_3, CH_4, H_2O	Electrical discharge	Amino acids, fatty acids hydroxy acids, polyhydroxy compounds	Miller, 1953
CO, NH_3, CH_4, H_2O	Electrical discharge	Various amino acids	Pavlovskaya and Passynski, 1957
CO, N_2, H_2, H_2O and other mixtures	Electrical discharge	Various amino acids	Abelson, 1957
Organic ash, NH_3, H_2S, CH_4	20kv, 50~/sec	Hydrocarbon polymers	Wilson, 1960
CO_2, H_2O	Cyclotron irradiation	Succinic acid, etc.	Garrison *et al.*, 1951
Ammonium acetate	Cyclotron irradiation	Glycine, aspartic acid, etc.	Hasselstrom *et al.*, 1957
HCHO, H_2O, KNO_3 and other mixtures	U.V. irradiation, $FeCl_3$ catalyst	Various amino acids	Bahadur, 1957
HCHO, NH_3	U.V. irradiation	Amino acids, heterocyclic compounds	Calvin 1956
NH_4OH (or NH_2OH), KNO_3, simple ternary compound	U.V. irradiation	Glycine, glutamic acid, β–alanine, γ–aminobutyric acid, norvaline, norleucine	Ferrari and Cultrera, 1961
C_2H_2, NH_3	Heat (300–400°C), oxide as catalyst	Succinic acid, pyrroles, pyridines	Oparin, 1957a
NH_3, malic acid or urea, malic	Heat (140°C)	Aspartic acid, peptides, ureidosuccinic acid	Fox *et al.*, 1955
HCHO, NH_2OH	Heat	Amino acids	Oró *et al.*, 1959
CN, NN_3, H_2O	30–100°C	Adenine, 4–aminoimidazole–5–carboxamide, glycinamide, glycine, alanine	Oró, 1961

Raw Materials	Method	Products	Reference
NH_2OH, N_2H_4, CH_3CHO	30–100°C	Amino acids, etc. Also aliphatic acids	Oró, 1961
HCN, NH_4OH	70°C	Aspartic acid, alanine, glycine	Oró and Kamat, 1961
NH_4CN	60°C	Amino acids, peptides, purines, fatty acids	Lowe *et al.*, 1963
HCHO	Room temp. $Ca(OH)_2$ as catalyst	Monose sugars	Oparin, 1957a
		C_3–C_6 sugars	von Mayer and Jaschke, 1960
HCHO	Alkalis as catalysts	Sugars	Pfeil and Muckert, 1961
HCN oligomer	Alkaline hydrolysis	Amino acids (glycine)	Wipperman, 1876

those available *in vivo* today but the important point is that once activated the simplest of precursors produce a variety of metabolites, including a number of amino acids, by spontaneous pathways. There is much the same evidence from natural photosynthesis: much the same variety of products appears within seconds of exposing leaves or chloroplast preparations to light (Calvin, 1961). The amino acids obtained to date in the laboratory activations are in fact among the simpler, non-essential group, but even so they represent a number of separate pathways and are a warning not to exaggerate the difficulties of amino acid synthesis. The main point for the present purpose is the variety and not the difficulty of the amino acid pathways of biosynthesis. Each side chain certainly needs separate consideration both in synthesis and in its subsequent functions.

The interaction between side chain and α–complex has been mentioned and in the free amino acids at least the two components are far from independent. Scarcely any property of the α–complex is identical in different amino acids. Free amino acids do play a considerable part in metabolism, and they are the precursors of various other metabolites, so that this kind of interaction may be important. It may also affect the formation of peptide bonds, making that between each particular pair of amino acids unique. There appears to be a distinct enzyme for the activation of each amino acid preparatory to peptidation, if not for the final step itself, and proteolytic enzymes also have a high

degree of specificity to particular bonds. Here the nature of the side chains on either side of the bond is shown to be critical, and in the well-analysed case of dipeptidase a constellation of features, embracing both side chains and peptide bond as a unit, seems to be necessary. Desulphuration of the side chain of cysteine (p. 177) and dehydroxylation of that of serine (p. 177) seem to expose the α–complex to deamination and this kind of interaction may be common in free amino acids. The cysteine NH should be protected once it is in peptide form, and this is perhaps one reason why as a reducing agent it is normally in tripeptide combination (p. 177) or in a protein. The phenyl side chain of phenyl alanine (p. 185) depresses the ionisation of the α–complex, so that only 50 per cent of the molecules exist in the zwitterion form, compared with over 99 per cent for the alkyl amino acids (Hawk *et al.*, 1947, p. 115). Reciprocally both the $^{+}NH_3$ and the carboxyl ^{+}H quench the fluorescence due to the benzene ring—probably putting the energy to better use.

13.3 The Individual Amino Acids

The simplest amino acid, *glycine* (Fig. 24), can be regarded as virtually a free α–complex, a very useful 'dummy' to be inserted, like an *em*, into a peptide chain wherever an active side chain would be troublesome. There is a very high percentage (44·4) of glycine in silk fibroin (p. 216), a protein designed to be spun easily, before it dries, into long and thin, but strong, threads; side chains would prevent the peptide chains sliding on each other. The α–complex, and therefore the whole glycine molecule, is virtually coplanar, and polyglycine forms flat ribbons which pack very closely together. This decreases the water-solubility of proteins such as silk (Cohn and Edsall, 1943, p. 195) and polyglycine is unique among amino acid homopolymers in becoming increasingly insoluble with chain-length. The free amino acid is of course one of the most soluble, because of the absence of any hydrocarbon portion, so that protein solubility properties are nicely graded according to the percentage of glycine. Because of the close packing of the residues its polymers are said to have a high crystal lattice energy.

Glycine has a number of unique properties for the reason usual in the first member of an organic series. Another is that it has a low *activity coefficient* (Cohn and Edsall, 1943), that is to say the electrical conductivity of ions of glycine, and various other properties, do not change very rapidly with concentration, because the ions do not interfere with each other very much as the concentration is increased. In this it resembles inorganic salts. Also like these it increases the surface tension of its aqueous solutions. At low temperatures it is even unique among all non-electrolytes

in decreasing the heat-capacity of aqueous solutions (Cohn and Edsall, 1943, p. 170), that is the heat required to raise the temperature of a standard volume by 1°C. Some of its properties are shared by one or two other amino acids; glycine and proline are the only two which do not, as the hydrocarbons do, absorb heat when they dissolve. *Alanine* shares with it the property of increasing the surface tension of water and also the interesting, unique ability to form *two* hydrogen bonds through its α–NH group, when this is already in peptide-linkage.

Alanine has a short, chemically inactive side chain so that it also is rather abundant in such proteins as silk (26·4 per cent). Its side chains are long enough to cause some adhesion between peptide chains, without which a coherent macroscopic thread could not be produced. The methyl side chain is long enough also to cause a much higher activity coefficient than that of glycine. Like glycine, alanine is common in many proteins, but whereas glycine also has a number of chemical functions in the body, alanine appears to be significant only as an inert structural unit. The biosynthetic origin of the amino acids is shown in Table 23 and the most important molecules synthesised from them in Table 24. Alanine is not

TABLE 23. BIOSYNTHESIS OF COMMON α–AMINO ACIDS: MAIN SOURCES

Amino acid	Main precursor(s)
Glycine	1. Serine
	2. Glucose → glyoxalate
Alanine	Pyruvate
Serine	P–glycerate → HO–pyruvate
Threonine	Aspartic acid → homoserine
Cysteine	1. Serine + methionine
	2. Sulphate
Methionine	Aspartic acid → homoserine
Valine, leucine	Glucose → pyruvate
Isoleucine	Aspartic acid → homoserine → α–ketobutyric
Proline	Glutamic acid, ornithine
Hydroxyproline	Proline
Aspartic acid	Oxaloacetic acid
Glutamic acid	α–Ketoglutaric acid
Arginine	Glutamic acid → ornithine
Lysine	1. α–Amino adipic acid
	2. Diaminopimelic acid
Histidine	Fundamental → imidazole–glycerol–phosphate
Phenylalanine	Glucose → shikimic acid → prephenic acid
Tyrosine	Phenylalanine
Tryptophan	Shikimic acid → anthranilic acid → indole–3–glycerol phosphate

a major or a direct precursor even of valine and leucine. Although alanine itself seems to be chemically unimportant, however, a large number of the other amino acids can be considered as substituted alanines, which perhaps implies that special side chain groups must be attached to the carbon

Table 24. Functions of Amino Acids in Further Biosyntheses

Amino acid	Other amino acids synthesised from it	Other metabolites synthesised from it, wholly or in part
Glycine	Creatine	Porphyrins, purines, riboflavin, glycocholic acid, hippuric acid, glutathione
Alanine	Valine, leucine	
Serine	Glycine, cysteine	Purines, ethanolamine, choline
Threonine	Serine, isoleucine, glycine	
Cysteine	Cystine	Taurine, thioethanolamine, mercapturic acid, sulphate
Methionine	Cysteine, isoleucine, creatine (methylation)	Choline (methylation)
Valine		Pantoic acid
Proline	Hydroxyproline, glutamic acid	
Aspartic acid	Isoleucine, threonine, methionine β-alanine	Pyrimidines, purines
Glutamic acid	Proline, ornithine, arginine, γ-aminobutyric acid (and transamination)	Glutathione
Arginine	Glutamic acid, creatine	Urea, octopine, agmatine
Lysine	Glutamic acid	Cadaverine
Histidine	Serine, glutamic acid	Histamine, purines, ergothioneine, carnosine, anserine
Phenylalanine	Tyrosine	
Tyrosine		Thyroxin, thyronine, tyramine, adrenalin, melanin
Tryptophan		Nicotinic acid, tryptamine, quinolinic acid, ommochromes, ?tyrian purple
Ornithine	Arginine	Ornithuric acid, putrescine

next to the α–complex. Alanine itself conceivably may serve to keep extraneous molecules at that distance.

In this light *β–alanine*, with an amino group on this carbon and none on the α–carbon, should perhaps be considered an entirely different kind of substance. It does not occur in proteins and has an entirely different origin (Table 24), but it occurs in a pseudopeptide linkage in carnosine,

anserine and pantothenic acid (p. 356), its only biological products of significance. It is much more soluble than α–alanine, presumably because the molecule is now polar at both ends. This is further important because not all amino acids become more soluble by the addition of a polar group at the far, or ω, end of the side chain.

Serine (Fig. 24) can be regarded as β–hydroxyalanine and *threonine* as β–methylhydroxyalanine. The hydroxyl group is as usual important for bonding and is probably the only one which will bind phosphate to protein. It depresses the activity coefficient of the molecule but makes it *less* soluble in water, as well as in lipids. They are among the most unstable of the amino acids and this is probably a clear demonstration that intermediaries between the poly–hydroxy compounds and the hydrocarbons are all unstable; even in protein combination they remain very sensitive. In the laboratory very violent reagents are necessary to phosphorylate them and the resulting esters also are very unstable, but their biologically synthesised phosphates are stable, an interesting example of the way biological mechanisms do circumvent chemical difficulties. Even so they remain examples of rather 'improbable' biological molecules.

Their hydroxyl group has been found essential for the function of some proteins. In addition serine is the precursor of other amino acids, and of other metabolites (Table 24), and threonine can be converted to serine, glycine and isoleucine; there is no doubt of their importance in metabolism. Although serine can be synthesised from threonine their main pathways of biosynthesis are separate (Table 23) and effectively each is a unique metabolite.

Cysteine (Fig. 24) is the sulphur-analogue of serine, and the —SH radical is a stronger acid than —OH. In practice, however, it behaves more often as a reducing agent than as an acid, since sulphur is less electrophilic than oxygen (p. 388), so that a hydrogen atom is released more readily than an H^+ ion. The sulphur is therefore able to make a strong covalent rather than an ionic bond and this is particularly important in reactions of the type $2RSH \rightarrow RSSR + H_2$. In this way two peptide chains, or two turns of one chain, are firmly bonded together and the strong insoluble keratins are formed, in hair, claws, feathers, etc., of vertebrates. The bond is so strong that it survives the hydrolysis of the protein and the dimer *cystine* is obtained (Fig. 24). In spite of this strength, however, cysteine is a good *reversible* reducing agent, in appropriate form. The best is the tripeptide, *glutathione* (GSH), which in fact is one of the few abundant, free oligopeptides with a major metabolic function. It acts in conjunction with the ascorbic acid redox system (p. 50). At the same time the SH group of the cysteine residues of many enzyme–proteins is an active and indispensable redox agent. The uniqueness of

this reducing group is that it is active in alkaline media where most others are fully oxidised. Moreover acid media protect it from oxidation so that it does not compete with those which are active in that pH range.

Provided hydrogen-acceptors are available the cysteine SH group will bond with a large number of organic and inorganic radicals. It plays a part in the detoxication of some phenolic poisons, forming so-called mercapturic acids. These owe their name to the fact that SH forms analogous compounds with mercury, and other metals; egg-albumen, with a high content of cysteine, therefore, is one of the best antidotes against mercury poisoning. By the same token, of course, HS-enzymes are very sensitive to traces of free salts of heavy metals. As an alternative to their own conjugation, cysteine and cystine promote the polymerisation of various small molecules such as methanol, formaldehyde, and even ammonium carbonate, under the influence of light (Gordy *et al.*, 1955).

The body of the molecule of cysteine is derived from serine and the sulphur group from methionine, the other sulphur-containing amino acid, by a somewhat circuitous route. Cysteine is seen to be the precursor of a number of further metabolites (Table 24) in addition to its function as an amino acid.

Structurally *methionine* is the methyl derivative of the C_4 homologue of cysteine, though it is not synthesised in this way *in vivo* (Table 23). The methyl substituent entirely changes the properties and methionine in function is as different from cysteine as from any other amino acid. It is primarily important as a methyl donor for the biosynthesis of such molecules as choline (p. 289); for the purpose it becomes activated as *S*-adenosyl methionine (Fig. 48, III), a conjugate with adenosine (p. 394) and not with the complete adenylic acid. In the laboratory violent reagents such as hydriodic acid, red phosphorus or sodium in liquid ammonia (p. 153) are necessary to remove methyl from methionine, so that this is another example of biological facilitation. Through the control of choline and lecithin-synthesis methionine controls the whole fat metabolism of the body.

Methionine therefore donates methyl to a number of recipients and sulphur to serine. It may give methyl, or a larger portion, to the synthesis of isoleucine (below), but no doubt transmethylation is its unique contribution to life. There is no apparent reason why this function should not operate while it is in protein combination, as the prosthetic group of the enzyme in fact. While re-emphasising its independence of cysteine, it is interesting to find, in certain organisms, a rare amino acid, *djenkolic acid*, which has structural characters of both methionine and cysteine (Fig. 24).

Valine, *leucine* and *isoleucine* (Fig. 24) can be regarded as alanines sub-

stituted by branched hydrocarbon chains. Structurally, and probably biosynthetically, they are related to the isoprene series of metabolites (p. 107), but lack one outstanding feature of that series, the double bonds in the carbon chain. It is interesting that no open chain amino acid of the twenty has a double bond in the chain. That of valine and the leucines is less reactive than the isoprene compounds. The hydrocarbon chain would be expected to make them relatively soluble in lipids but in fact they are most soluble in polar solvents. They are certainly soluble in polar organic solvents but leucine is more soluble in 70 per cent aqueous ethanol than in absolute alcohol, and valine is very soluble in water. It is more soluble than other amino acids of the same number of carbon atoms, so that the hydrophobe/hydrophil ratio probably depends on the maximal length of hydrocarbon chain, irrespective of branching.

The most important property of these branched chain amino acids is implied in their isoprene-like structure. They are major constituents of the protein *elastin*, occurring in the elastic fibres of connective tissues, in elastic cartilage and elsewhere, and there seems little doubt that their branched chains confer rubbery elasticity rather as the branched isoprene units do in natural rubber (p. 109). The percentage of valine is high in casein and egg-albumen, and this is probably an index of the importance of elastic proteins in the developing young. However, valine is rarely absent from any protein, and the possibility of a second main function must be envisaged. Valine seems to be the precursor of pantoic acid, a component of the vitamin pantothenic acid (p. 356).

It is doubtful if *norvaline* and *norleucine*, the unbranched isomers, exist in protein combination, though they are sometimes isolated as the free amino acids (Desnuelle, 1953). *Pipecolic acid* (Fig. 24) might be a ring-closed derivative of norleucine, and *biakiaine* a partially unsaturated derivative. A curious feature of leucine is that the natural L–form tastes bitter whereas the unnatural D–isomer is sweet.

In *proline* and *hydroxyproline* there is again an entirely different side chain. They could be considered as a ring-closed norvaline, and this is perhaps a good reason for doubting if the latter itself exists in protein combination. The pyrrolidine ring is the only fully saturated, alkane ring among the side chains of amino acids, the tendency being for rings to be unsaturated and open chains saturated. Proline is the next lower homologue of pipecolic acid, with the α–amino group in the ring. As already indicated (p. 169) this restricts its ability to form peptide bonds and proline often ends peptide chains. This is not an absolute restriction, however, and there is 17 per cent of proline and 15 per cent of hydroxyproline in collagen, the main protein of the white fibres of connective tissue. The ring does, however, prevent the free rotation of the atoms

about the bonds of the α–complex and this may contribute to the inextensibility of collagen fibres (Gross, 1961). The ring also causes a reversal of the sense of coiling of a peptide chain (p. 220), so that the molecule resembles the compensating coils used in industry (Rich, 1959a). Proline is abundant in elastin so that the pyrrolidine ring alone does not cause very great inextensibility, which may be due rather to the hydroxyl group of hydroxyproline forming bonds with neighbouring peptide chains (Piez and Gross, 1960). Other hydroxyamino acids, such as serine and threonine and the rarer hydroxyglutamic acid and hydroxylysine, may contribute to this type of bond. Hydroxyproline is absent from elastin, which is in favour of this view.

A most remarkable property of proline is its great solubility in water, as glue-makers appreciate. It is actually six times as soluble as glycine, although it has a C_3 side chain; it would seem that the alkane ring has a more favourable effect on solubility even than the branched chain of valine, which otherwise proline resembles in general solubility-properties. It has the same, but greater, solubility in ethanol and other polar organic solvents, and the *prolamins* (p. 209), with a high proline content, are unique among proteins in being soluble in 80 per cent alcohol. On the other hand *casein*, the main protein of milk, is insoluble in ethanol in spite of its 15 per cent proline. Another surprise is that hydroxyproline has only a quarter of the water-solubility of proline, so that a hydroxyl group on a ring depresses solubility, as it does at the ω–end of an open chain!

To continue the paradoxes, proline has an unusually low melting point among amino acids (153°C), whereas among inorganic salts the most hydrophil are usually those with highest melting points. The implication is that proline is poorly ionising, as the α–amino is involved in the ring, the solubility being similar to that of the sugars. Proline is in fact sweet, and shares this property with glycine and the other more soluble amino acids.

Both proline and hydroxyproline are abundant in seed-proteins as well as in casein, probably an indication of their importance in the proteins of growing organisms. Hydroxyproline is formed from proline itself and occurs in few proteins, but proline itself is rarely absent, so that again it may have other major functions.

The unique structural feature of the side chain of *aspartic* and *glutamic acids* is an extra carboxyl group at the ω–end, and they are therefore known as the dicarboxylic amino acids. Like valine and leucine they are serial homologues, but in this case C_4 and C_5 (Fig. 24). Unless we regard glycine and alanine as serial homologues these are the only two strict pairs. In the present instance there is no doubt that the two homologues are simple alternatives for certain purposes, but each has in addition functions peculiar to it, and this may prove true of valine and leucine also. Both

aspartic and glutamic acids form the amides of their ω–carboxyl, *asparagine* and *glutamine* respectively, and all are useful as stores and donors of the amino group, for the synthesis of other amino acids from their α–keto precursors, and of other amino compounds. The aspartic pair are predominant in plants and the glutamic pair in animals, so that there is no complete overlap in function. In animals α–ketoglutaric acid (α–KG), the precursor of glutamic acid, can be aminated by ammonia itself whereas oxaloacetic acid (OAA), the precursor of aspartic, must be aminated by transamination from glutamic, so that the latter is the dominant member for this reason.

The ω–carboxyl groups can form simple salt-linkages with the ω–amino of the diamino amino acids (below), and contribute to bonds between peptide chains, or they can form salts with other cations, for more purely chemical purposes. In addition they can form peptide bonds, for instance in glutathione, and pseudopeptide bonds, as in the two amides. In addition each is very active and important in a number of ways, not all clearly related to the ω–carboxyl structure. Both have α–keto precursors which are members of the Krebs cycle (p. 65) and there is evidence that glutamic acid plays some unique role in the respiration of nerve tissue; it affects the permeation of Na^+ and K^+ and so may be important in nerve conduction itself (Terner *et al.*, 1950). Glutamic acid also improves the uptake of iron across the gut wall, probably as a soluble glutamate. Both are precursors also of the side chain of other amino acids, aspartic of β–alanine, isoleucine, threonine and methionine and glutamic of proline, ornithine and arginine. In addition, aspartic plays a part in the synthesis of pyrimidines, purines (p. 347) and urea, while glutamic is the precursor of *γ–aminobutyric* acid (Fig. 24) and a component of GSH. Clearly each is indispensable and they are the two most common amino acids in proteins. Both may be essential for peptide-formation; in the laboratory (Fox, 1960) both, and particularly aspartic, promote the heteropolymerisation of the other amino acids.

Lysine is a diamino-counterpart of the dicarboxylic amino acids and in some organisms, at least, is synthesised via α–aminoadipic acid, the C_6 homologue of glutamic (Kit, 1960), which itself occurs in some fungi. The C_5 homologue of lysine, *ornithine* (Fig. 24), is an important metabolite but probably never occurs in protein combination, while α, γ-diamino butyric acid, the C_4 member, rarely occurs biologically. *Arginine*, which is another amino acid common in proteins, is the ω–amidine derivative of ornithine, i.e. guanidino–norvaline. It behaves in many ways similarly to lysine and so does *histidine*, another 'essential' amino acid of proteins, and structurally (but not biosynthetically) a ring-closed derivative of arginine. The three—lysine, arginine and histidine—are therefore known

collectively as the *hexone bases*, though in addition they have each its own unique function, as the very different endings of their side chains indicate. All three endings are unique among the twenty. Their main function in common is the construction of proteins with an alkaline reaction, to balance the acidity of nucleic acids (p. 360), of the haem group (p. 315) and of the dicarboxylic amino acids. They act as counterparts of the latter for other purposes also.

Lysine has the distinction of the longest side chain among common amino acids; nevertheless, although this is the optimal length for ring closure it does not close, probably because of repulsion between ω– and α–amino groups. A similar repulsion may operate between the two carboxyl groups of glutamic acid, but in forming the pyrrolidine ring its ω–carboxyl condenses with the α–amino. Lysine is very soluble, the ω–amino group increasing solubility as in β–alanine; by contrast the ω–carboxyl of the dicarboxylic acids, like hydroxyl, depresses solubility. This may be the reason why the diamino amino acids have longer chains than the dicarboxylics. The ω–amino is more alkaline than the α–amino, its p*K* being 10·53 compared with 8·95 for the latter. It is of course an amine group (p. 289), but does not show the powerful neurotropic action of the amines formed by α–decarboxylation of amino acids.

Lysine is particularly abundant in the *globins*, which are *histones* (p. 208), conjugated with the haem nucleus, in the haemoglobins, cytochromes and myoglobins of vertebrates. Invertebrate haemoglobins have more arginine and less lysine, so that the hexone bases do seem to be interchangeable to some extent. There is also a high percentage of histidine in the haemoblogin of gnathostomes but little in that of lampreys and of the invertebrates, or in the cytochromes. Lysine also occurs as a complex with the vitamin biotin (p. 397), which is a carboxylic acid (Wright *et al.*, 1950), and its alkaline ω–amino group is probably its one significant functional feature. It does occasionally form the hydroxy-derivative already mentioned, however, and this can bond with phosphate and other groups.

It is a lone amino acid in a number of ways; a certain amount of it may be converted to glutamic acid *in vivo*, but it gives rise to few products other than those mentioned; it is synthesised only by autotrophes, and by two different pathways—both peculiar to lysine, among the common metabolites (Table 23). It is quite indispensable but at present it is not clear why the other hexone bases cannot deputise for it. They both have a number of functions in addition to that in the alkaline proteins, and possibly lysine serves as the essential hexone base where these other functions might be adversely affected.

The other functions of arginine are related to its ω–guanidine group,

as would be expected. This is one of the strongest of biological bases (p. 300), and arginine, like lysine, is very water-soluble. Among the important properties of guanidine is the ability to form a bond of high transfer energy with phosphate (p. 405) and represented: R·HN·C (=NH)·HN~P(=O)$(OH)_2$. The phosphate is readily transferred, without loss of bond-energy, to ADP to form ATP, and arginine phosphate is therefore known as a phosphagen. It is particularly vital to the sustained action of the muscles of many invertebrates, as a store of ~P for the rapid regeneration of ATP from ADP. In the vertebrates, and in some invertebrates, another guanidino–amino acid, *creatine* (Fig. 24), serves as the phosphagen, and is synthesised from glycine, but it receives its guanidine group from arginine. Other guanidine derivatives act as phosphagens in other invertebrates, but it is not yet known if arginine is the universal biological source of the guanidine group.

The main other function of arginine is as a source of urea (p. 284) for excretory purposes (Fig. 26), the arginine being continuously regenerated in the *ornithine cycle*, another important metabolic cycle discovered largely by Krebs (Krebs and Henseleit, 1932). This in fact is the main metabolic significance of ornithine, but as its name implies it is abundant in birds which do not in general excrete urea! In birds ornithine is used to detoxicate phenolic poisons, in the same way as cysteine and glycine in other groups; it has the advantage of conjugating phenols via both of its amino groups. There are many features of interest in the ornithine cycle, not all relevant in the present context; one is the neutrality of urea, as contrasted with the alkalinity of guanidine. Another is the intermesh between the cycle and the Krebs TCA cycle (Fig. 26). More immediately relevant is the part played by both glutamic and aspartic acids in the process, the use of simple components for the synthesis, ammonia and carbon dioxide, and the further amino acid intermediary, *citrulline* (Fig. 24). As its name implies citrulline also occurs as a free amino acid in melons, where its function is uncertain, since plants do not form excretory urea; it is probably the intermediary in the normal pathway of synthesis of arginine from glutamic acid through ornithine.

In the synthesis of ornithine from glutamic acid the α–ammonia is acetylated while the ω–carboxyl is being reductively aminated; otherwise ring closure to pyrrolidine might occur. Ornithine itself is a component of the antibiotics, *tyrocidin* and *gramicidin*, one of the many examples of the use of unusual or special molecules for this purpose. Arginine is the probable precursor of *agmatine*, *octopine* and other guanidine derivatives occurring sporadically in invertebrates (Baldwin, 1953). It has a number of major and minor functions, therefore, for the most part chemical.

As already indicated, histidine (imidazolyl alanine) has the appearance

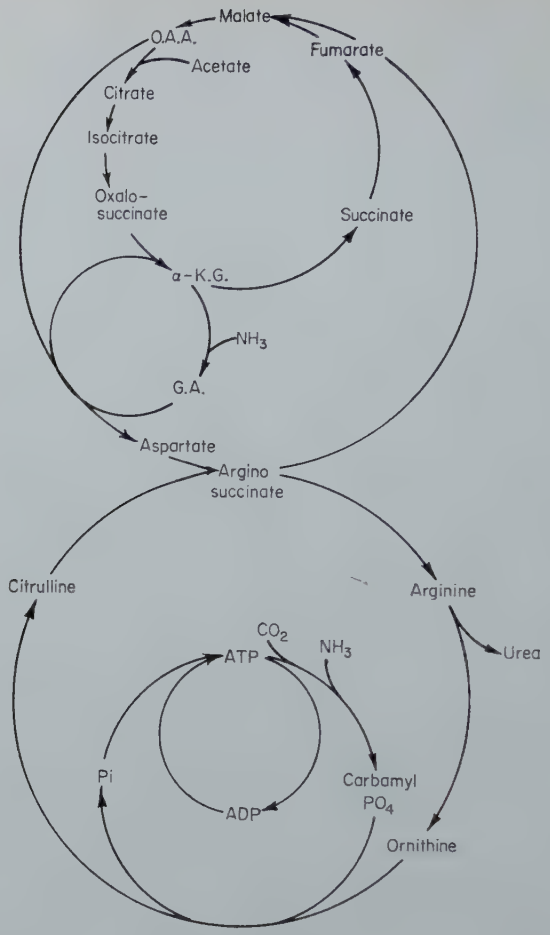


FIG. 26 The ornithine cycle.

of a structural derivative of arginine, and it is interesting that amino-guanidine inhibits the enzyme which normally destroys histamine, a derivative of histidine. At present, however, little is known about any but the final stages of synthesis; the general view is that its pathway is different from that of arginine and equally from that of the purines (p. 347), which also contain the imidazole ring in their molecule. If so this further illustrates the unexpected degree of independence of biosynthetic pathways. The ring is maximally unsaturated and with the further property of two nitrogen atoms to share in the resonance, it is more active than the benzene ring. It shows this in a wealth of activities which rather dwarf the hexone base function. The tertiary nitrogen of the ring

becomes a quaternary base with a p*K* near pH 6·0; together with the α–NH_3^+, with a p*K* of 9·17, it buffers effectively over much of the range, pH 5 to 9—the most useful part of the physiological range. In the dipeptide, histidinyl histidine, the p*K* of the two imidazole nitrogens comes to differ by 1 pH unit, an effect of the peptide bond, so that its effective buffering range is even greater. The buffering power of haemoglobin is related to its histidine content (Prosser and Brown, 1961, p. 231). In addition, histidine residues form the actual link between the globin and the haem nucleus in some if not all haem-proteins, so that even here the hexone-base function is not its only function.

As an activating molecule it is essential for the action of chymotrypsin (Oparin, 1957a, p. 245), and for the activation of D–amino acid oxidase (Edlbacher, 1946). It is itself inactivated by light (Anfinsen, 1959, p.132) and in ribonuclease this affects the whole enzyme molecule. Laboratory-prepared cobalt–histidine complexes (p. 328) act as oxygen-carriers (Milward, 1951). Imidazole will form an acetyl derivative with a bond transfer energy 2·8 kcal/mole greater than that of acetyl phosphate (p. 92) (Picken, 1960, p. 15); at 17·3 kcal it is one of the highest values in useful biological compounds. It is not surprising that histamine, the product of decarboxylating histidine, is a potent neurotropic agent; it mimics the action of the main parasympathetic mediator, acetylcholine (p. 290), presumably through the amine–nitrogen becoming quaternary, as it does in the interesting derivative, *ergothioneine* (p. 298). Pilocarpine, from the jaborandi plant, is another imidazole derivative with parasympathomimetic action. Histamine is released when tissues are wounded and causes some of the subsequent responses of the ancient tetrad: rubor (vasodilation), turgor (oedema), calor (temperature increase) and dolor (tenderness). It has now been shown to promote regenerative growth after wounding and other types of growth (Kahlson, 1962), so that it is probably a major physiological, and not merely a pathological, agent.

The imidazole ring, like that of tyrosine (p. 186), is substitutable by iodine (Roche *et al.*, 1952), and mono- and di-iodohistidines may occur *in vivo*; no other amino acids will form these compounds. In all, therefore, histidine is a very active and versatile metabolite. It acquires an SH substituent on the ring in ergothioneine (Fig. 37, XXI, p. 285), and in addition the α–complex is changed to the active betaine form (p. 298). This occurs in red blood cells and other tissues as well as in the rust fungus, ergot, and may have physiological significance. Histidine also exists as two dipeptides with β–alanine, *carnosine* or β–alanyl histidine, and the methyl derivative, *anserine* (Fig. 29, III).

Histidine can be regarded as imidazolyl–alanine; the benzine analogue, *phenylalanine* (Fig. 24), also is one of the common amino acids,

and so is its *p*-hydroxy derivative, *tyrosine*, in general the more active of the two. There is another benzenoid amino acid, *p-aminobenzoic acid* (p. 378), but it does not occur in proteins. The mono- and di-iodine and bromine derivatives of tyrosine do occur in protein combination, in the special skeletal proteins of sponges and Anthozoa. They, and the other derivatives of the two primary benzenoid amino acids, have other functions in addition, and this depends again mainly on the resonant activity of their ring. Although in itself not as powerfully resonant as the imidazole ring, certain substituents can make it active enough: the hydroxyl group of tyrosine and the amino group of *p*-aminobenzoic acid are examples. Both are in the *para* position to the alanine chain, and this is the most active position. The hydroxyl has in degree the acidic properties of aromatic hydroxyl substituents (p. 131) so that in a sense these are analogues of the open chain dicarboxylic and diamino amino acids. The tyrosine hydroxyl group reacts with a variety of reagents, and since the products are usually sufficiently resonant to be brightly coloured these reactions have become the basis of many tests for proteins, very few of which lack tyrosine. Apart from this, virtually the only characteristic groups useful for the purpose are the SH of cysteine, and the imidazole and indole rings. The *p*-hydroxyl group facilitates the substitution of iodine and bromine at the *ortho* positions to it and it reacts with the alanine chain of a second residue to form the dimer, *thyronine*, on which the thyroid hormones are based (Fig. 28).

Figure 27 gives an idea of the large number of important metabolites formed from phenylalanine or from tyrosine. With a few exceptions, probably including *tropic acid* (Fig. 28, I), all biological benzene derivatives are synthesised through these two amino acids; here more than in any other instance, perhaps, we have the impression that the side chain is the important component and that the α–complex and peptide-linkage are just convenient ways of carrying and deploying it. The origin of the two amino acids from glucose, via heptulose and shikimic acid (Fig. 27), has already been mentioned, together with some of the ternary intermediaries. *Tryptophan*, the remaining common amino acid, is synthesised in the same general pathway, and in addition another line of indole derivatives, including the black pigment, melanin, are synthesised from tyrosine. Apart from excretory products, normal and abnormal, most of the significant products of the benzenoid line are not direct derivatives of phenylalanine: *tyramine*, *adrenalin*, *thyroxin* and the other halogen derivatives, and *melanins*, all are tyrosine derivatives. There is no overlap in function between the two, perhaps because phenylalanine is really only a precursor of tyrosine. The latter seems as uniquely important as histidine.

Not only does it form iodine substituents, as histidine, but its amine,

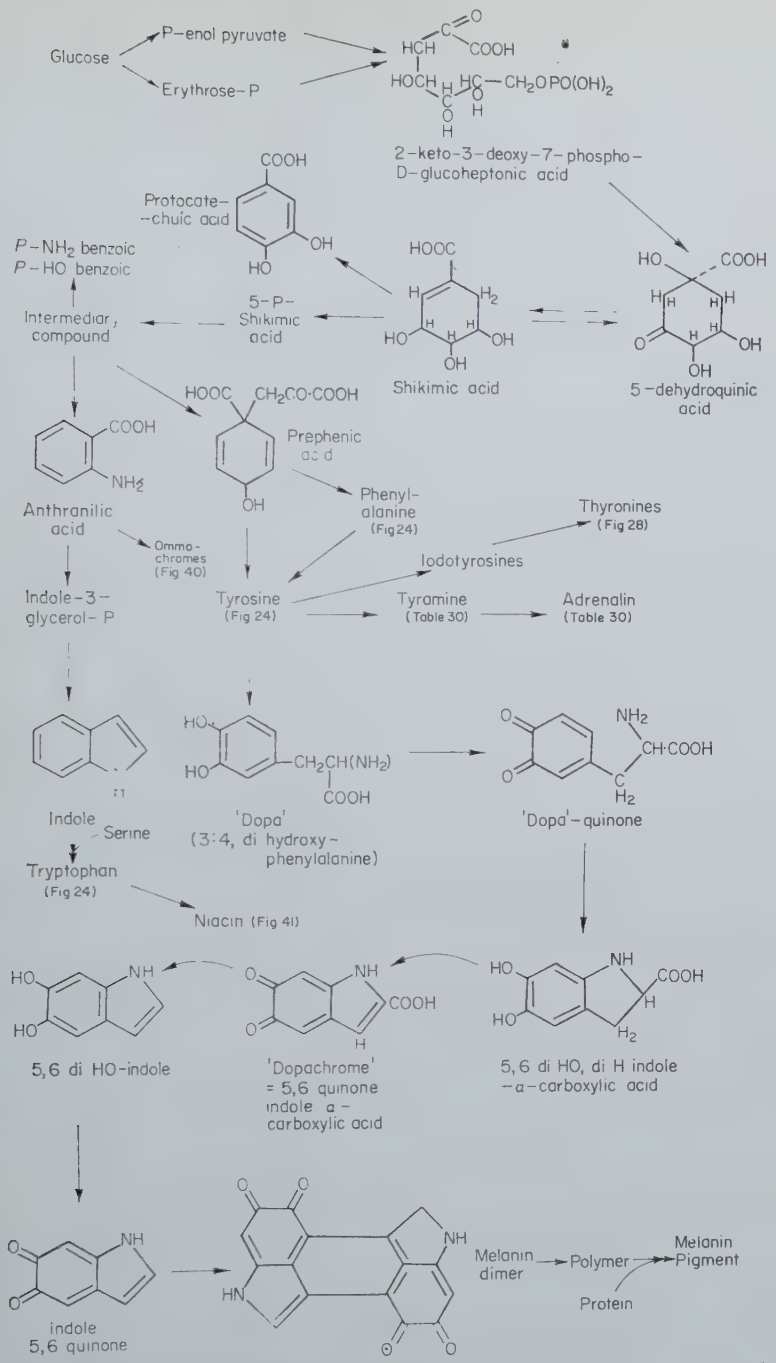


FIG. 27 Pathways of biosynthesis of biological phenolic and indolic compounds.

the product of decarboxylation, is another powerful neurotropic agent, and may be a physiological transmitter in the neuro-effector systems of some invertebrates. Corresponding amines of other amino acids (Table 30) have some neurotropic effect, but tyramine is one of the more potent. It is formed in the vertebrate gut but this is thought to be an incidental product of bacterial action. It is also synthesised in the kidney, under anoxic, acid conditions. Under aerobic, more alkaline, conditions tyrosine is deaminated, leaving the carboxylic acid. This rôle may help to control pH in the body and has been observed in the behaviour of other amino acids, in some organisms. Like most amines tyramine affects muscular tone and blood pressure, but as a normal pressor agent it has been superseded in the vertebrates and some invertebrates, at least, by *epinephrine* or *adrenaline* (Table 30), secreted by the postganglionic endings of the sympathetic system and by the adrenal medulla. Related dihydroxyphenols serve a similar function in insects (Wigglesworth, 1954). The second hydroxyl substituent on the ring is as usual *ortho* to the first and the methylation of the primary amine probably increases its strength in the usual way (p. 289). The poison of the nematocysts of coelenterates may be a phenolic compound associated with protein (Picken, 1957), and so may be related to these neurotropic derivatives of tyrosine. Tyramine itself seems to be a normal synaptic transmitter in sea anemones (Ross, 1945).

On standing in solution adrenalin oxidises to a reddish *adrenochrome* (not to be confused with adenochrome), similar to the red compound *dopachrome* (Fig. 27), formed as an intermediary in the synthesis of melanin. Adrenalin, *homogentisic acid* (a human pathological product) and other phenolic compounds oxidise to their own particular brand of melanin, in fact (West and Todd, 1957, p. 1100). The final, dark and very stable, insoluble and unreactive pigment is a high polymer of an indole quinone, conjugated with protein, and this synthesis, therefore, seems to be a spontaneous change in many phenolic compounds. It is probably a by-product of the hardening changes in the exoskeletal polyphenols of arthropods, and where necessary they are able to harden without darkening (Dennell, 1958). Melanins are unique in their resistance to chemical agents, light, etc., and are ideal integumental pigments. They do bleach to some extent in bright sunlight, and in hydrogen peroxide, and they show some redox change in colour. The darkest melanins absorb strongly at all wavelengths in the visible spectrum, as well as in the infrared. The latter provide heat energy and this is the value of dark pigments to the young frog tadpole and to some animals in high latitudes, or at high altitudes.

The substitution of halogens in organic compounds is a somewhat improbable process, but it is more spontaneous in aromatic than in aliphatic

compounds. Low temperatures are in its favour (Finar, 1959, p. 518). Chlorine is a more active halogen but only the bromo and iodo derivatives are common biologically. Iodination tends to be reversible but iodine is fixed readily and without catalytic help if there is already an amino or a hydroxyl group on the ring. It takes up the preferred positions *ortho* to the hydroxyl. Mono- and di-iodo tyrosines, as well as the corresponding bromo compounds, occur in the skeletal materials mentioned and in other tissues of invertebrates, but iodine compounds predominate (Gorbman, 1959) as they do in the gnathostome thyroid. Here the condensation of two di-iodo tyrosine residues gives thyroxin (Fig. 28) and loss of the iodine at position 5′ yields 3,5,3′-thyronine, about five times more active as a thyroid hormone than thyroxin (Barrington, 1963). The uniqueness of the iodo compounds is discussed later (p. 457); biological thyroid activity is shown by the analogues of the other halogens and by the nitro-analogue but in strongly descending order $I > Br > Cl > NO_2$, i.e. in inverse order of oxidising power. The uncoupling action, by dinitrophenol, of phosphorylation from respiratory oxidation reactions (p. 458) is best understood in this context. Thyroid hormone promotes oxidative phosphorylation but in excess it also can have this uncoupling action. The nitro-analogue of thyroxin has only 1/3000 of its effect on the metamorphosis of Amphibia (Glass, 1951, p. 895), and it seems that the normal responses are very precisely tuned to the mild oxidising action of the iodo compounds.

They are in fact much more precisely tuned than this. A good deal of work has been done in testing the various components and features of the molecule for thyroid activity and a surprising number of these must be just as they are. This is true of other metabolites which have been tested in detail, but thyroxin illustrates the point particularly well (Du Toit, 1952). The minimal requirements are two phenolic rings with an ether bridge between them, and halogens (or nitro group) substituted in the ortho positions to this on the distal ring, i.e. the one with the alanine side chain. There must be an active group in this side chain and finally an anionic group must be substituted on the proximal ring, either *ortho* or *para* to the bridge, but preferably *para*, and the hydroxyl group is probably the best for the purpose. It will be noticed that halogens on this ring are not essential and this helps to explain why tri-iodothyronine is more active even than thyroxin.

Tryptophan, or β–indole alanine (Fig. 28), owes its name to having been first recognised in a tryptic digest of protein. With a fully resonant condensed ring system, including one nitrogen atom, it is an even better activator, using light energy in the visible spectrum, than histidine, and its products are frequently brightly coloured. Indole derivatives of

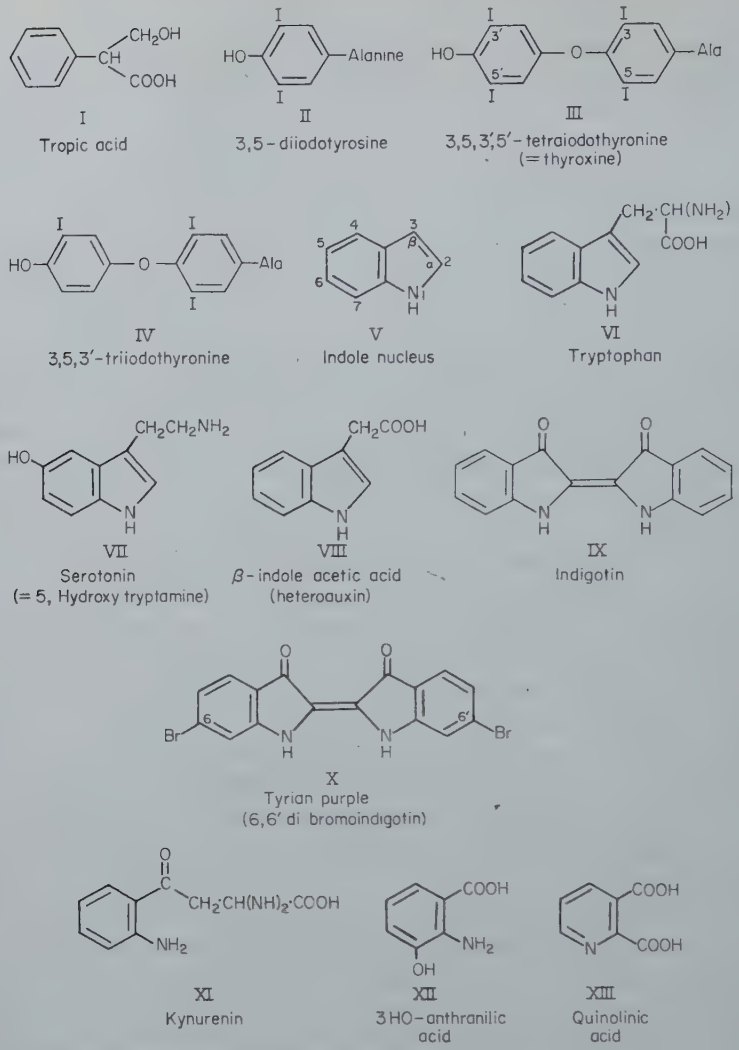


Fig. 28 Further derivatives of the phenylalanine, tyrosine and tryptophan pathways of biosynthesis.

biological importance are probably more numerous than those of either imidazole or benzene, and there is little doubt about the value of an indolic amino acid. What is less clear is why these three aromatic rings, alone among the number used in living organisms, should be in amino acid form whereas the others are conjugated with protein, if at all, in other ways. As in the case of tyrosine there is much to give the impression that

the α–complex is merely a way of manipulating an active molecule as side chain. A number of products become free of the α–complex but, as with the phenolic derivatives, virtually all funnel through a pathway of biosynthesis which passes first through one of these amino acids; as already noted (Fig. 27) some indoles are formed from tyrosine and not via tryptophan itself.

Tryptophan is never a very abundant amino acid in proteins (Table 27), but it is rarely absent. It is demanded in the diet by heterotrophes, and yet seems to be utilised rather wastefully (West and Todd, 1957, p. 1104). Again, destruction of the indolyl group by acid hydrolysis of proteins does not greatly affect their properties, as though it were a casual passenger in proteins, and is required only in small quantities even for other purposes. A fact which is possibly relevant here is that alanine can be substituted on the indole nucleus very easily, by comparison with the imidazole ring, for instance (Davis, 1961), so that it is convenient to prepare as an amino acid. The pathway from indole glycerol phosphate to tryptophan is a single step whereas four are required between imidazole glycerol phosphate and histidine. The reason is that a shift of electrons from the indole nitrogen to the glycerol chain labilises its attachment and allows serine to replace it *in toto*. In the imidazole counterpart, however, a shift of the double bond to the C–N link adjacent to the glycerol side chain actually stabilises this attachment, and the glycerol must be changed to alanine *in situ*, piecemeal. There is an opportunist reason, therefore, why indole should be manipulated as an amino acid, but by the same token there must be other positive reasons why imidazole is prepared in this form; such functions as bonding the haem nucleus to protein are probably adequate teleonomic reasons.

Also for teleonomic reasons, however, it seems unlikely that any radical is prepared in amino acid form and bound into protein unless it has some function in that combination. Ornithine is a good example of a metabolically active amino acid which is manipulated without the need for protein combination. The distribution of tryptophan in proteins is not indiscriminate; Gudernatsch (1934) considered that, along with the other aromatic amino acids, it was most abundant in differentiated functioning, as opposed to embryonic proteins, and apart from the high content in the seed-protein, *gliadin* (Table 27), this is probably true, and is consistent with the picture of the resonant ring system being able to activate a variety of processes. The high concentration in a seed-protein certainly presents a problem since these autotrophes can synthesise their own tryptophan as soon as they need specialised proteins. There is also a high concentration in *avidin*, a protein of egg-albumin, but this is known to be a very specialised protein for the chemical defence of the unincubated

egg—effectively a parental protein. There is little tryptophan in the embryonic *tropomyosin* but considerable amounts in functional *myosin* and *actin*. It is also abundant in another protein with a mechanical function, *fibrinogen*, and in *feather-keratin* with a static mechanical function. Curiously enough, cataract, an opacity of the lens, is due to deficiency of tryptophan. It is also concerned in some immunity reactions (Anderson, 1948; Miller and Goebel, 1949), and is abundant in a number of enzymes, lysozyme, chymotrypsinogen, lactic dehydrogenase and many others (Table 27). As with so many other amino acids, and some other metabolites, its functions can be divided into two groups, physical and chemical; of course functions can only be physical or chemical: the significant point is that few are absolute specialists, having functions in the one group only. This is particularly clear in the functions of the proteins, collectively. It is also interesting to see such paradoxes as the participation of tryptophan in blood-clotting, on the one hand, and in the lytic activity of lysozyme on the other.

Further unique properties of tryptophan and other indoles are illustrated by particular derivatives. The amine, *tryptamine*, is a powerful neurotropic agent, but in this it resembles the amine counterparts of other amino acids. It may be a physiological agent in some animals but in general it has been superseded by a hydroxy-derivative, *serotonin*, or 5-hydroxy tryptamine, sometimes called *enteramine*; this is a close parallel to tyramine and adrenalin. The study of serotonin has shown that the indole nucleus certainly does contribute to the properties of these amino derivatives (Szent-Gyorgyi, 1960). The flat, strongly resonant molecule is ideal for what is known as *charge-transfer*, or activation by single electron-shifts, as contrasted with the two-electron transfers of more orthodox reactions. Serotonin is a particularly active agent in charge-transfer, which would seem to underlie its unique role as a neuromediator: only amines with a large, flat ring of this kind *could* show its particular physiological properties. Other indole-derivatives with neurotropic effects include *reserpine*, *lysergic acid*, *harmine* and *yohimbine*, so that this is not a minor side-line of indolic activity. *Melatonin*, the recognised hormone of the pineal gland, is, like serotonin, a 5-hydroxy-derivative.

Probably fairly closely related to the neurotropic group functionally are the plant hormones of the *heteroauxin* group, which may be compared with the ternary auxins already considered. They show the same broad group-specificity, β–indole acetic (Fig. 28) being the most active, but most indole derivatives having *some* action. This is sometimes a metabolic antagonism of the action of those which actually promote growth. They do also show a variety of differential effects, some acting mainly on shoot or on root growth, others on germination of seeds, on fruit-setting, on

the epinastic response, and so on. Few of these compounds may be normally produced, physiological agents, but they illustrate a valuable way of analysing the function of those which are, and a similar technique could usefully be applied to many other instances. As a result of this type of analysis it is possible to list the essential requirements in an effective heterauxin molecule, as for thyroxin (p. 189): there must be a side chain with a carboxyl group, or its precursor, and this must be separated from the ring by at least one C—C bond, but by not more than two. A halogen at positions 4 or 6 in the ring increases the activity. The side chain must be at β and not at α. The coplanar, resonant ring is of course essential and there also must be an appropriate ratio of hydrophil to lipophil portions of the molecule.

The other biological indoles are mostly pigments, or otherwise more directly concerned with photo-activation. The best known, the indigo pigments, have a di-indole structure, with a double bond connecting the two moieties (Fig. 28), planar, rigid and maximally resonant. The transmission spectrum therefore is very bathochromic, from the blue of woad and commercial indigo (Fig. 28, IX) to the purple of Tyrian Purple, or dibromoindigotin (Fig. 28, X). Absorbing mainly towards the red end of the spectrum, woad has a warming effect, and so had more than aesthetic value to the earlier natives of Britain! The commercial Tyrian Purple comes from the hypobranchial (adrectal) gland of certain pectinibranch gastropods, *Murex* and related genera, and is still used to dye fabrics (Fox and Vevers, 1960). The iodine analogue presumably would not be formed unless there were an hydroxyl or an amino substituent on the ring, but this might reduce the resonance-possibilities.

The ommochrome pigments (p. 342), first isolated from the ommatidia of insect-eyes, and later from the meconium and integument (Butenandt *et al.*, 1954, 1960), and from the integument of cephalopods (Schwink, 1956) and isopods (Needham and Brunet, 1957), are tryptophan derivatives, again with the dimer structure, though differing in detail from the mode of formation in thyroxin, melanin, and indigo. The pyrrole ring of the monomer is first opened, to form *kynurenin* (Fig. 28, XI), and two of these then condense to form a tricyclic condensed ring, *phenoxazine*. The best-known ommochrome, *xanthommatin* (Fig. 40, XXIV) (p. 332), is a low polymer of this kind, with a sharp redox colour change and a general reactivity contrasting with the inertness of the melanins—which it sometimes mimics *in situ*. There are also higher polymers, however, the *ommins* which are more like the melanins in properties.

In *Cypridina* an indole derivative may be the source of its bioluminescence (McElroy and Glass, 1961), emitting light in response to chemical activation, therefore, in contrast to the reciprocal, and more familiar,

photo-activation of chemical reactions. Tryptophan itself is responsible for much of the fluorescence of the globular proteins, absorbing energy and re-emitting it as light of lower quantal value. If the energy were efficiently harnessed the fluorescence would be 'quenched' and this brings us back to the view that in proteins tryptophan is sometimes a casual passenger to some other functional destination.

A rather distinct product of tryptophan, alone sufficient to make it an indispensable metabolite, is nicotinic acid (p. 333), one of the B-vitamins and the unique component of the molecules of a group of dehydrogenase coenzymes. It may therefore be more than coincidence that there is a high tryptophan-content in the apoenzyme of lactic dehydrogenase. The pathway from tryptophan to 'niacin' itself is full of interest. It is common to the ommochrome pathway as far as kynurenine and then passes through 3-*hydroxyanthranilic acid* (Fig. 28, XII) and *quinolinic acid* (Fig. 28, XIII). The indole ring nitrogen is therefore introduced into the benzene ring after opening up the pyrrole ring. This seems to make light work of opening and remodelling cyclic structures. Moreover an indole is used to synthesise a ring which might have come more plausibly from a benzenoid compound direct, just as it might seem that melanin should be synthesised from indoles direct; no doubt there are reasons for this and it would be helpful to know if these are chemical or biological.

Tryptophan certainly seems to indicate that the functions of the amino acids become more numerous and varied with increasing complexity of the side chain—and less restricted to peptidation.

13.4 Interactions between Amino Acids

It is not possible to obtain a complete picture of the unique biological role of the amino acids from a consideration only of the individuals in isolation, and of a generic α–complex. Their most important interaction is in building peptides and proteins, of course, and this will be the subject of the next two chapters. The present need is for general principles, which help to understand the heteropolymerisation of natural peptides, on the one hand, and, on the other, why these particular 20 amino acids have been singled out for the purpose. The determination of the particular amino acid sequence in each protein is attributed to a genetic code but this does not explain the evolutionary processes which led to a set of amino acids which could be coded in this simple way. If their order were determined solely by a coding device external to them then we might ignore any kind of interaction occurring before peptidation, but from the evidence of the laboratory peptidations of Fox (1960) it is clear that aspartic and glutamic acids have a specific promoter effect on peptide forma-

tion by all other amino acids, and conceivably some of the other amino acids may be included in proteins purely for internal services of that kind. At any rate this offers a clue to the kind of factors which may have determined the particular personnel of the 20, among the many theoretical possibilities.

Glutamic acid also promotes the uptake of most other amino acids into the cell (Gale, 1953), and there are other interactions of this kind (Christensen *et al.*, 1948). The effects of the amino acids on the solubility and related physical properties of each other affect uptake and peptidation, as well as other functions. Then there is the interaction at an earlier stage still, in the synthesis of the amino acids themselves. Glutamic acid is of unique importance as the universal amino-donor for the amination of the α–keto precursors of other amino acids, with aspartic acid and alanine as first deputies. In addition, a number of the amino acids provide the body of the molecule for others (Table 24); the outstanding examples are serine, alanine, aspartic acid, glutamic acid and threonine. The more complex amino acids produce mainly metabolites of other groups.

Table 25. Recorded Antagonisms between Amino Acids

Between	And
α–Alanine	Glycine
β–Alanine	Asparagine
Serine	Threonine
Glutamic acid	Aspartic acid
Valine	Leucine
Isoleucine	Leucine
Isoleucine	Norleucine
Valine	Norvaline
Valine	Phenylalanine
Valine	Threonine, serine
Methionine	Norleucine
Lysine	Arginine

The issue is complicated by the occurrence of antagonistic interactions (Gladstone, 1939; Roblin, 1946) so that under some conditions particular pairs may be incompatible, or otherwise increase the amount of each other necessary for normal function (Table 25). In most cases the antagonism is seen to depend on resemblances in structure and properties, and illustrates the need to avoid, as far as possible, molecules which *could* duplicate a function but in practice compete, and so inhibit it. The amount of this *in vivo*, due to amino acid antagonisms, may not be serious

but there is evidence that it is not negligible, and that it can exacerbate any incipient tendency towards imbalance.

Interactions after incorporation into proteins are best left until a later stage (p. 213). It is clear that there is a good deal of interaction and interrelationship even prior to that stage.

13.5 Conclusions

Even if we did not yet know anything about proteins we should probably conclude that the amino acids constitute a group of substances quite unique among the metabolites considered so far. As indicated at the beginning of the chapter their side chains are far more varied than any part of the molecule of any previous group. This is not due entirely, or even mainly, to the introduction of nitrogen into the molecule, since the variety lies in the side chains and only four of these contain nitrogen. A new type of compound is involved, in which the feature which has been exploited is the substitution of a great variety of metabolically active radicals on a suitable adapter, the α–complex. The buffering and other properties of the latter are also relevant, of course, but the position is comparable to what would hold if the main importance of the carbohydrates was in the variety of groups substituted on the monose skeleton; in fact the acetyl–amino group is one of the few substituents in this case. The collective activities of the 20 amino acids of the proteins, together with a number of coenzymes still to be considered, are responsible for most of the vital activities. Apart from the nucleic acids they are in bulk the last major group of biological materials. There remain others, the B vitamins in particular, which are of critical importance, but for the most part as isolated members of their chemical group, and used in small, catalytic amounts. Amino acids, used in large quantities on the other hand, are more in the nature of substrates and this may have been a main factor determining exactly which radicals should be used in the amino acid form (p. 190).

By comparison with the B-vitamins and other nitrogen compounds to be considered, the amino acids are conspicuous for relative simplicity rather than for complexity of structure. As precursors of many other metabolites (Table 24) this is perhaps to be expected. They therefore combine great variety and quantitative abundance with relative simplicity, and this fairly summarises their unique position.

Their functions can be grouped under four main headings: (1) nonspecific functions of the α–complex, (2) side chain participation in peptidation, and in the properties of the resulting peptide, (3) specific metabolic and physiological functions of side chains, whether as free amino

acids or in protein-combination, and (4) functions as precursors of other metabolites. Examples under 1, 3, and 4 have been already numerous; more under 2 will be seen in the next two chapters.

There is one remaining feature of the amino acids collectively which is worth consideration and this is their tendency to fall into a simple grouping, which might be held to reduce their apparent variety. They tend to group first in pairs or threes, as the purines and pyrimidines do (p. 347), and these again in pairs, a simple mathematical pattern which might be relevant to genetic coding (Crick, 1963). One pair at least, phenylalanine and tyrosine, are probably related as precursor and product, but this is by no means universally true (Table 23). Glycine and alanine form the first pair, having a short inert side chain, and then there are two hydroxy-aliphatic amino acids, two dicarboxylic, two prolyl and two phenolic amino acids. Together with the valine–leucine-, diamino-, and sulphur-containing trios this leaves only tryptophan as an odd member. In some ways histidine is a partner to tryptophan, as already seen, in which case there are seven pairs and two trios. Eight of these nine groups again form pairs, glycine and alanine with the leucine group because they have purely hydrocarbon side chains, the hydroxyl pair with the sulphur group because of the analogy between OH and SH, the dicarboxylic pair with their diamino-counterparts, and the four aromatic ring compounds together. This makes four groups and the prolines are the only odd pair. There is a certain amount of irregularity and artificiality about this grouping but it seems more than fortuitous and any possible *natural* basis for genetic coding should receive sympathetic consideration. If phenylalanine is simply a precursor of tyrosine it is necessary to explain why it is also so abundant in proteins, and the metabolic antagonism detected between members of other pairs (Table 25) implies that there must be some compensating advantage also. It is clear that aspartic and glutamic acids have major properties in common and that it has been necessary to restrain the one in plants and the other in animals. There is not much serious overlap in function between related amino acids in living organisms, and this may be due largely to selection against overlap, avoiding the need to sacrifice a pattern necessary for some other purpose, such as coding.

CHAPTER 14

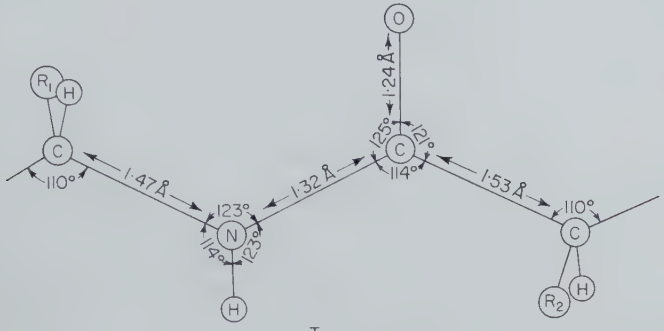
PEPTIDES

Any two or more amino acids, polymerised by the peptide type of bond, —HN—(O=)C—, between successive pairs, constitute a peptide, the size of which is in principle unlimited, and in practice often very great. Classification of peptides is to some extent based on natural properties, but is still partly arbitrary. The smaller polymers are called oligopeptides and the larger ones polypeptides, but any reasonably large *biological* peptide is called a protein, whereas no polypeptide synthesised in the laboratory, however large and heterogeneous, would be admitted as a protein. For the most part the proteins have a secondary and a tertiary grade of structure which can be disorganised by quite mild chemical and physical agents, completely destroying its normal biological function without necessarily breaking a single peptide bond. A typical peptide is little affected by such agents and this might well become the one criterion for distinguishing all peptides from all proteins. At present, however, there may be individual polymers wrongly classified, according to this criterion.

The present chapter will be confined to a consideration of the general properties of the peptide bond, of peptidation and of peptides, as exemplified by simple polymers of biological and laboratory origin. Most biological peptides are heteropolymers, usually with a high degree of heterogeneity, but a number of homopolymers, of glycine and of some other amino acids, have been synthesised in the laboratory. Nearly all are bonded through the α–complex only, though glutathione (GSH) is a natural oligopeptide with one bond involving an ω–carboxyl group, of glutamic acid. The rarity of ω–bonds is reflected in the rarity of branched peptides, but occasionally the two terminal amino acids of a chain join by their α–groups to form a cyclopeptide. Substances other than amino acids on occasion form —HN—(O=)C— bonds and it is sometimes convenient to call the product a pseudopeptide.

As already noted, the energy required to form a peptide bond is not great, a mere 3·5 kcal/gm equivalent, compared with the total energy content of the typical covalent bond, such as that between carbon and

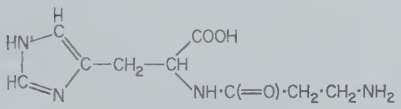
nitrogen, of 50 – 100 kcal. It is less than that associated with the hydrogen bond of oxygen (p. 19), and considerably less than the 5 kcal required for glucose polymerisation. Moreover it decreases with the size of the existing peptide making the bond, and may be spontaneous at the molecular sizes used *in vivo*.



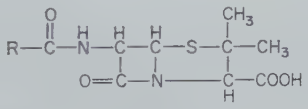
I

II
Ocytocin: Cys_I·Tyr·Ileu·Glu NH_2·Asp NH_2·Cys_I·Pro·Leu·Gly(NH_2)
Pitressin: Cys_I·Tyr·Phe·Glu NH_2·Asp NH_2·Cys_I·Pro·Arg·Gly(NH_2)
Bradykinin: Arg·Pro·Pro·Gly·Phe·Ser·Pro·Phe·Arg

III Anserine



IV Penicillin



In penicillin G, R = benzyl: C_6H_5·CH_2—
" X, R = *p* HO—benzyl: HOC_6H_4·CH_2—
" F, R = 2 pentenyl: CH_3—CH_2—CH=CH—CH_2—
" dihydro F, R = *n*-pentanyl: CH_3—$[CH_2]_4$—
" K, R = *n*-heptanyl: CH_3—$[CH_2]_6$—

FIG. 29 The peptide bond unit (after Pauling and Carey) and some oligopeptides.

The dimensions and form of the *peptide bond unit*, that is the complete unit which repeats regularly between successive α-carbons along the peptide chain, are very constant (Fig. 29, I) and not much affected by the different side chains. The length of the carbonyl, C=O, bond is greater than it would be if it were an isolated structure, and that of the adjacent C—N, peptide, bond is considerably shorter than the 1·47 Å of a typical

single C—N bond, so that there must be resonance between them and therefore they must also be virtually coplanar. The resonance does not produce a complete conjugate system along the peptide chain, but with hydrogen bonds and other links between neighbouring turns of the chain, and between a chain and its neighbours, extensive resonant networks may often develop (p. 213), and may play a large part in the unique behaviour of proteins.

Rotation is possible around the other two bonds of the peptide repeat unit, provided the side chains are appropriate, but the resonant form of the bond, —N=C(OH)—, dictates a basically *trans*-arrangement of the neighbouring side chains. In the simplest case, therefore, they approximate to a simple 180° rotation between successive residues, producing a flat zigzag ribbon, as in a simple laboratory polymer, polyglycine I (Rich, 1959a). In another form, polyglycine II, however, the angle is 120°, producing a cylindrical or so-called helical spiral, with three amino acid residues per turn of the screw (Fig. 32). These features are revealed mainly by X-ray diffraction methods.

It is interesting to find that simple industrial polymers, such as *polyethylene*, . . . —CH_2—CH_2—CH_2— . . . , forms ribbons like polyglycine I, while *teflon*, or polyfluoroethylene, . . . —CF_2—CF_2—CF_2— . . . , forms a spiral thread. The reason is that the sphere of influence, or van der Waals' radius, of the fluorine atom is greater than that of hydrogen, so that the equilibrium form, with the closest packing of the atoms, is a spiral, permitting each fluorine to fit into the space between neighbouring turns of the screw. Biological peptides usually form spiral threads, since they have side chains at least as bulky as the fluorine atom, but silk (p. 216) may be based on the polyglycine I plan and β–keratin appears to have a *pleated sheet* construction (Fig. 33). In this case the individual peptide chain zigzags in *two* planes at right angles to each other, the side chains lying in the plane of the major zigzag, projecting alternately in opposite directions.

The number of possible structures for the peptide chain, consistent with efficient function, is somewhat narrowly limited by the basic steric requirements already indicated, namely (1) the coplanarity of the peptide bond unit, (2) the relatively fixed bond-lengths and angles (Fig. 29, I), (3) the *trans*-arrangement of successive side chains and (4) the need of hydrogen-bonding for subsequent strength and stability (Pauling and Corey, 1953; Pauling *et al.*, 1951). The structure which combines these requirements with a minimum of restrictions concerning side chains and other features has been called the α–helix (Fig. 30), and this appears to be the actual form of many biological peptides. In it the plane of each peptide bond is tangential to the surface of the enveloping cylinder of the helix, and the side

chains all protrude outwards, normal to the surface. Hydrogen bonds are formed between each *n*th peptide group and each $(n + 3)$th. Since there are 3·6 residues per turn of the spiral these hydrogen bonds make only a slight angle with the longitudinal axis of the cylinder. The result is a very strong *secondary* structure to the molecule, binding in all the α–complexes and exposing all the side chains for further action. It is a closely packed structure, with no empty space down the axis, and it can accommodate

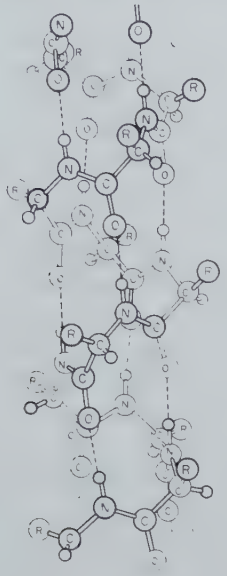


FIG. 30 The α-helix type of peptide molecule (by courtesy of Professors Linus Pauling and R. B. Corey), showing the secondary structure due to intramolecular hydrogen bonds, and other features described in the text.

the reversal in sense of coiling caused by proline residues, without disturbing the basic structure. This is a unique structure, ideal for many biological purposes.

Some other sterically feasible and also stable structures have virtues for particular purposes (Donahue, 1954), and actually occur in some natural proteins. All are helices, with the properties of springs, of cables, or both. Any regular repetition of an asymmetrical structure must produce a helix, and the few ribbons can be regarded as helices with a rotational period of 180°.

Fascinating and very instructive properties are shown by another laboratory model peptide: this is polybenzylglutamate, PBG, studied in the solvent mixture dichloroacetic acid–dichloroethane (Zimm, 1959).

This forms an organised, helical coil at high, and a random coil at low, temperatures, i.e. the molecule melts on cooling and sets on warming. This paradoxical behaviour implies that the orderly, solid form is the form of maximal and not minimal entropy in that system. The probable explanation is that at low temperatures all the hydrogen bonds of the peptide are held by solvent molecules while intramolecular bonds become dominant only at high temperatures. This illustrates the importance of considering the system as a whole when determining the state of entropy. The type of relationship may have its biological counterparts, and helps to explain the importance of the solvent as part of the fabric. It also shows how a macromolecule may increase its order and decrease its local entropy without affecting the validity of the second law of thermodynamics for the system as a whole, and under conditions which cause an increase in the entropy of the latter.

One biological possibility is that a thermo-solidifying peptide might be combined with a more orthodox thermo-melting one, so that the conjugate maintains a perfectly steady state over a large temperature range. The ratio of the two components might be genetically or epigenetically adjustable to fit the different temperature ranges in the various parts of the world.

The random coil of PBG at low temperature, and of other α–helices, can be orientated by stretching in a manner somewhat analogous to the micelles of rubber. This may be relevant also to the solidification of silk, as it is extruded in fluid form from the silk gland (Penn, 1949). There is a further surprise from α–helical structures, however, since further stretching may cause the helices to uncoil. Here again is a possible basis for homeostasis, a change in conditions towards either extreme, from the optimum, causing effectively the same change in organisation. It is possible to visualise that this could become the basis for a single mechanism capable of reversing the effect of conditions at either extreme on the body.

There are a number of further paradoxes of this kind in the physical properties of peptides, which emphasise their great versatility, individually and collectively. If a solution of collagen (p. 217) is heated the polypeptide chains shorten and coil into helices, like the synthetic PBG, and this form is stable, but if it is stretched and cooled the molecules snap back into the extended configuration (Penn, 1949). There are two stable states, therefore, under different conditions, with a state of instability in between. This situation is potentially valuable for organising regular cycles of change, as in the contraction and relaxation of muscle. This is not to imply that the present example is at all a close analogy to the muscle cycle, of course, though collagen does have properties relevant to the latter (Pryor, 1950).

In a concentrated, 20 per cent, solution in methylene chloride, PBG forms another very interesting structure, the liquid–crystal, separating out as spherical droplets, 300 mμ in diameter, composed of concentric layers 1·0–15 μ thick. The crystal is a form of closest packing, of maximal economy of space, and so may be a spontaneous formation in concentrated solutions. The structure is similar to some produced by proteins, and probably to some existing within living cells. Liquid–crystal *mesoforms* in fact play a large part in the protein structures of living organisms (p. 231), and have a wide range of properties.

A number of the properties of peptides have already been considered, in particular the effect of peptidation on α–complex and side chain, and the change in properties with size of peptide. Ionisation, buffering action, orientation and behaviour in an electric field, and solubility properties are among the most important. Some peptides become more soluble as they increase in size, but polyglycine and polyarginine become less soluble. Solution is often autocatalytic, and many peptides are more soluble in their own solutions than in water. Reciprocally dilution may precipitate them and so they avoid indefinite dispersion, a property which may have been of major importance to the primitive, eobiotic systems. It would not be surprising if heteropeptides collectively could produce a complete range in this and in almost every property.

Some of the ways in which side chains and α–complex interact have been noted, and also the effect of side chain on peptide form. Other examples are given by Tristram (1953). One very interesting case is that racemisation of the amino acids, that is of their α–complex, occurs more easily in peptide linkage than in the free amino acid. This is perhaps why the metabolic turnover of amino acids in protein is so rapid, and why D–amino acid oxidases are often more abundant than the L–oxidases.

Most of the general properties of peptides have been studied on laboratory models and on degradation-products of natural proteins, and the number of natural oligopeptides in fact is rather small. This is perhaps the now familiar phenomenon of stable nodes with few intermediaries. Of the few known, most are antibiotics, characterised by the exploitation of unusual structures. The only very common metabolic oligopeptide, the tripeptide GSH, glutamyl cysteinyl glycine, has one of its bonds in the abnormal ω–position (p. 198). The posterior pituitary hormones, oxytocin (*ocytocin*) and *pitressin*, and the hormone *bradykinin* (Fig. 29, II), released in allergic responses, are all nonapeptides, and are among the smallest of orthodox pattern. The two free dipeptides which occur sporadically, *carnosine* and *anserine* (Fig. 29, III) (p. 185), contain one amino acid which is not an α–type.

GSH is a good example of a unique metabolite, not only the sole

member of its class but also performing a function of almost incalculable importance. As already seen (p. 177) it is essentially an intracellular redox agent, by virtue of its cysteine residue, which depends on the other two amino acids for its stability. Cysteine is the central amino acid of the three and the tripeptide is the smallest unit which could give it the necessary protection. At the same time it is noteworthy that the other two amino acids, glutamic acid and glycine, are very active amino acids in a number of ways (Chapter 13), and both, like cysteine, have detoxication as a secondary function. Each belongs to a different one of the four quartettes of amino acids recognised in the previous chapter (p. 197).

It is believed that a growth factor for micro-organisms, *strepogenin*, obtained by hydrolysing casein, may exist as a discrete oligopeptide (Sexton, 1953, p. 230). Seryl–glycyl–glutamic acid shows some of the activity of strepogenin, which also counteracts the action of the analogue, seryl–glycyl–aspartic acid, or *lycomarasmin*, the cause of wilt in tomatoes. This pathological action of a normal α–peptide of L–amino acids is rather surprising and may be one reason for the rarity of natural oligopeptides. Antagonism between related amino acids is again in evidence, as also in the aspartic analogue of folic acid (p. 376). The tripeptide, leucyl–glycyl–glycine, activates pancreatic lipase (Prosser and Brown, 1961, p. 126) and the tripeptide size seems to be more common than other oligopeptides. If this were to form a cyclopeptide the ring would have the most stable, hexatomic size and there is in fact evidence that heterocyclic substances do inhibit tripeptidase (Long, 1961, p. 297), presumably as competitive substrates. The mere existence of specific tripeptidases is significant. There are of course also enzymes which act only on dipeptides but this is technically a special case because both amino acids forming the bond are at an end of the chain and so the enzyme must be doubly an *exopeptidase*; no other protease will act on a dipeptide and the requirements for its effective hydrolysis are very specific.

Very few other physiological oligopeptides are known at present, though a number have been demonstrated by chromatographic methods in tissues, body-fluids and urine (Dent, 1950; Sanger, 1950). Their number has probably decreased during the course of evolution (Synge, 1957), an instance of the general law of increase in austerity (Lwoff, 1944). Quite a number have been exploited as antibiotics, however, and they are perhaps the major group of this very varied type of biological agent. The peptide part contains a high proportion of D–amino acids (p. 169), among the usual variety of novel structural features in antibiotics, and a tendency towards the cyclopeptide form is perhaps another of these. Penicillin, one of the most powerful of antibiotics, in fact, is doubly ring-closed, by what would appear to be triple bonding between

a cysteine and a valine residue (Fig. 29, IV), but only one of the bonds is a peptide link, the carboxyl of the valine being free and the amino of the cysteine forming a pseudopeptide bond with one of a number of carboxylic acids. *Gramicidin–S* is a cyclic pentapeptide (Haurowitz, 1950), while some of the gramicidins are much larger peptides (Fox and Foster, 1957). The intensive study of the peptide antibiotics no doubt will be very rewarding, for the usual reason that the abnormal throws clearest light on the normal. It is already known that some of these antibiotics precipitate proteins or denature them in other ways, while others are strongly surface-active. Both carboxy and amino groups of the antibiotic may be used, and also labile —S—S— linkages, and probably others of the active radicals of amino acids. Antibiotics are known to affect one or more of the key processes controlled by these active groups, such as the redox activity of SH compounds. Other processes known to be affected by them collectively include amino acid assimilation, protein synthesis, transpeptidation, nucleic acid metabolism, and polysaccharide synthesis. There is no doubt a good reason why peptide antibiotics are the most potent and versatile; they reflect the breadth and depth of function of the normal proteins, now to be considered.

The classical triumph in the laboratory synthesis of polypeptides, by Emil Fischer in 1907, was an octadecapeptide, L–leu–tri-gly–L–leu–tri–gly–L–leu–nona-gly. This was not very heterogeneous and could only be regarded as a laboratory compound. Recently, however, the specific hetero nonapeptide, bradykinin II, has been synthesised (Boissonnas *et. al.*, 1960) and has the properties of the natural compound; indeed an error in determining the structure of the latter was corrected by the study of laboratory variants.

CHAPTER 15

PROTEINS

PROTEINS are natural large polypeptides which can be denatured, that is inactivated biologically, without breaking peptide bonds (p. 198). It is not always possible to say that this criterion is satisfied, however, and some biochemists would prefer to regard any large peptide of biological origin as a protein. It has been seen that there is usually a secondary structure, that is to say a secondary system of bonding, in peptide chains, stabilising them by linking successive turns of the helix, and a general, if not universal, feature of proteins is a further, *tertiary* system of bonding between peptide chains, or between successive turns of a higher order of folding within the one chain. These are characteristic respectively of the fibrous and globular main subdivisions of the proteins, though not absolutely restricted to either. It would be expected that the tertiary structure is the most labile and sensitive to damage, but not necessarily that biological function should depend upon it as much as denaturation seems to imply. There is little doubt that in fact all features of the molecule, primary, secondary and tertiary bonding, and the properties of the individual side chains, are concerned in the functioning of proteins collectively, but the probability is that their specific co-operation is disorganised if the tertiary structure is damaged. This controlling of specific function is just one of the many further potentialities which emerge between peptide and protein levels.

The absolute uniqueness of the proteins, collectively, in fact lies in their great versatility, the variety and complexity of their structure, and of their physical and chemical properties, and the consequent variety of functions they perform. Many of these functions themselves seem to be unique among natural phenomena, and the proteins are toweringly outstanding components of the unique biological system. Their scope and power are increased by the large number of *conjugated proteins*, a term used to mean conjugation with non-protein moieties. There is no limit to the amount of conjugation *between* proteins, but it is not always so easy to distinguish the individual members of the conjugate as it is in actomyosin, and most protein–protein conjugates at present rank as *simple*, but often very large,

protein molecules. No other type of metabolite forms conjugates with so many other types of moiety as the proteins, and it may well prove that few metabolites are biologically active unless conjugated with protein. It is known that some *prosthetic* groups, for instance the pyridine coenzymes, are very loosely bonded to protein, and that retinene makes and breaks its protein linkage during each cycle of activity, so that a labile, functional conjugation may be even more common than at present appears. The most important conjugated proteins in addition to the glyco– and lipo–proteins, and the melano– and other chromo–proteins, already noticed, are the nucleoproteins which are the most uniquely important of all biological materials, B-vitaminoproteins, porphyrinoproteins, phosphoproteins, further chromoproteins, and direct conjugates with iron, copper, vanadium, and other inorganic elements.

15.1 Classification of Proteins

Because of the great variety among proteins it is particularly necessary to have classificatory labels in order to discuss them at all, although inevitably the classification presupposes a knowledge of the structures and functions on which it is based. In fact, of course, the relevant aspects of structure are still very poorly understood and classification has grown up piecemeal and empirically, sometimes based on rather trivial properties. It is scarcely possible to design an ideal, dichotomous or oligotomous classification at present (Fig. 31); ideally a classification should be a précis of the properties of the sub-groups, and this is perhaps an exacting requirement for this particular group of materials. The properties actually used at present are molecular size and shape, and other structural features, amino acid composition, physical and chemical properties and, in the last resort, biological function.

The distinction between simple and conjugated proteins is useful, and within each of these the fibrous/globular dichotomy. Conjugated proteins are probably mostly globular, but the nucleoproteins are fibrous and the lipoproteins lamellar. Solubility and other differences follow the fibrous/globular dichotomy, and within the globular proteins solubility differences are useful. Among the fibrous proteins secondary and tertiary structural features help and in both groups the amino acid composition, to some extent. None of the distinctions is absolute, and it is necessary to define fibrous and globular proteins arbitrarily as those with an axial ratio greater or less than a particular value, for instance 15/1. In fact myosin has a ratio of 45–50/1 but is soluble and is usually regarded as a globulin (West and Todd, 1957, p. 385). Again the albumins, pseudoglobulins and globulins show a continuous spectrum of solubilities, as

tested by salting out (p. 223); this kind of continuous gradient makes classification more difficult, but not impossible. Figure 31 will serve as a framework for a more detailed study of structure and properties.

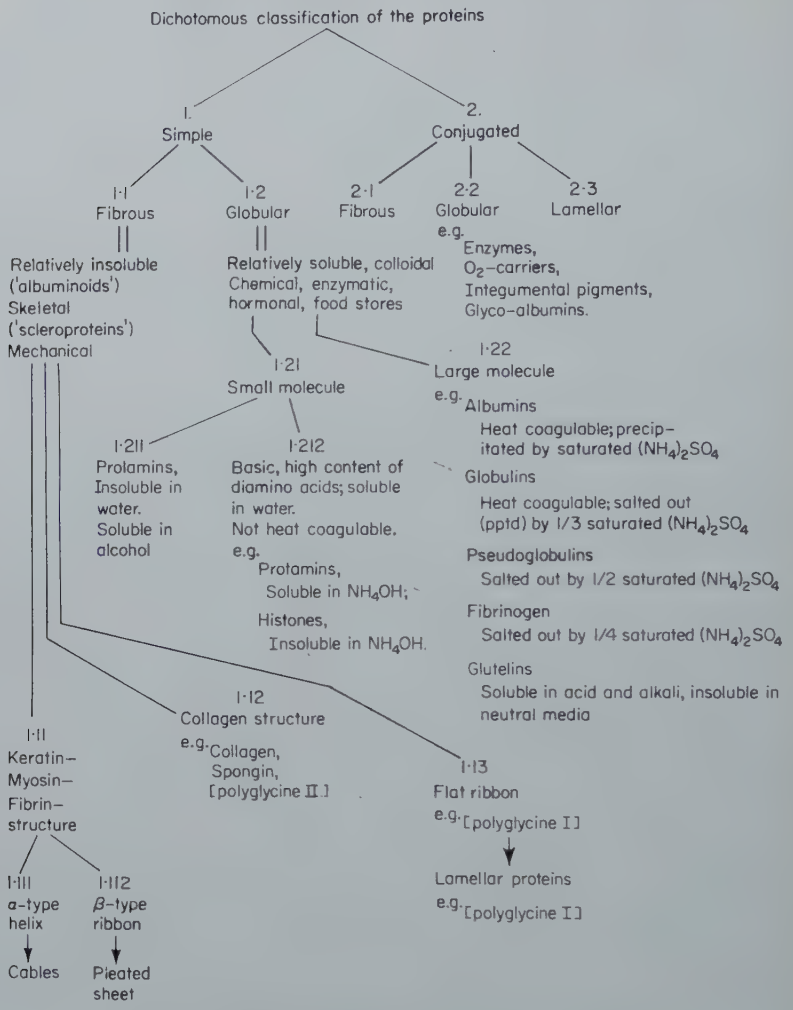


FIG. 31 Dichotomous classification of the proteins.

15.2 Molecular Size

The molecular weight of those proteins in which it has been possible to make reliable estimates varies very considerably, from a few thousand to 60 million Daltons (Table 26). Some of the fibrous proteins indeed may

TABLE 26. MOLECULAR WEIGHTS OF REPRESENTATIVE SELECTION OF PROTEINS

Protein	Type	Source	Molecular weight
Salmin	Protamin (from nucleo-protein)	Salmon sperm	5600
Cytochrome	Haemohistone	Most cells	15,600
Thymus histone	Nucleoprotein	Calf thymus	15,600
Myoglobin	Haemohistone	Striated muscle	17,200
Haemoglobin	Haemohistone	Blood corpuscles (gnathostomes)	68,000
Gliadin	Prolamin	Wheat	27,400
Zein	Fibrous prolamin	Maize	40,000
Ovalbumin	Glycoalbumin	Egg-white	45,000
Seralbumin	Albumin	Blood serum	69,000
Conalbumin	Albumin	Egg-white	87,000
Myogen	Albumin	Sarcoplasm	150,000
β–Lactoglobulin	Globulin	Milk	37,300
	Pseudoglobulin	Serum	50,000 to 100,000
	Globulin	Serum	167,000
Actin	Globulin	Myofibrils	70,000
Myosin	Globulin	Myofibrils	1,000,000
meromyosin L	Globulin	Myofibrils	96,000
meromyosin H	Globulin	Myofibrils	230,000
Haemocyanin	Globulin	*Homarus*	752,000
		Helix	6,600,000
Edestin	Globulin	Hempseed	309,000
Ribonuclease	Globulin	Widespread	15,000
Pepsin	Globulin	Stomach, mammals	34,400
Flavoprotein	Globulin	Yeast	82,000
Catalase	Globulin	Liver	250,000
Urease	Globulin	Jackbean	480,000
Thromboplastin	Globulin	Blood	170,000,000
Insulin	? Globulin	Islets, pancreas	48,000
ACTH	?	Anterior pituitary	20,000
APGH (somato-tropin)	?	Anterior pituitary	45,000
FSH	?	Anterior pituitary	67,000
Thyroglobulin	Glycoglobulin	Thyroid	650,000
Gelatin	Fibrous	Collagen	10,000 to 100,000
Fibroin	Fibrous	Silk	217,000
Fibrinogen	Fibrous (k.m.f.)*	Blood plasma	450,000
Feather keratin	Fibrous (k.m.f.)*	Feathers	850,000
Casein	P–protein	Milk	75,000 to 375,000
Bushy stunt virus	Nucleoprotein	Tomato	7,600,000
Tobacco mosaic virus (TMV)	Nucleoprotein	Tobacco	60,000,000

* k.m.f. = keratin–myosin–fibrinogen.

have an indefinite molecular size, ranging up into visible dimensions, but they are not easily obtained in solution without damage and it is not certain whether or no they have units which could be called molecules. When they are brought into solution, like gelatin from the native collagen, the particles are of variable size and so may be fortuitously broken fragments of a fabric in which the idea of a molecule is meaningless. Globular proteins usually have a definite unit, of consistent size and weight, which can be regarded as a molecule, but sometimes this is an aggregate of sub-units which might be better candidates for the term, when they are all alike and represent the *repeat* unit of the protein. The haemoglobin of gnathostomes consists of four identical portions, each with one iron–porphyran unit; however, each is rather irregular in shape, whereas the four fit together to make a good sphere (Perutz, 1960) and so the tetramer is usually regarded as the molecule. Myoglobin and the haemoglobin of the cyclostomes have no such aggregation and the same monomer, of weight 17,600 Daltons, is the molecule. It is sometimes convenient to recognise discrete molecular aggregates as *micelles*.

The protamins and histones rarely exist as independent molecules so that the smallest simple proteins are some of the hormones and enzymes. There is a wide range of sizes in nearly every functional group, with considerable overlap between groups. This might seem to deny any specific significance to molecular size but in fact it is probably always very nicely adapted to the function of the protein, the spread of sizes being due to the many factors which often control function. Most hormones are released from their cell of origin, travel in the blood stream and may enter the cells of a 'target' organ; a small molecule therefore is essential. This is also true for those enzymes, such as pepsin, which must be released from their cell of origin, and after action conceivably are resorbed. Lysozyme, the enzyme secreted in tears, has a molecular weight of only 15,000. The exceptionally large size of thyroglobulin among hormones may have precisely the converse significance, to keep the active principle within the thyroid follicles until required for free circulation; many endocrinologists believe that the circulating form is a much smaller molecule, perhaps free thyroxin or 3,5,3′-thyronine. A similar function may explain the large size of thromboplastin, the enzyme which initiates the series of reactions in blood-clotting; with a molecular weight of 170×10^6 it holds the size record among globular proteins, whereas thrombase, the enzyme which catalyses the next step in the process, has a weight of only 140,000. It is normally present in the plasma whereas thromboplastin is intracellular and is released only upon wounding. It is imperative that it should not leak out at other times and cause thrombi in the blood vessels, and the large size contributes to this safeguard.

Enzymes which are formed and function within the one cell also in general are large molecules, for instance phosphorylase has a molecular weight of 400,000 (Tristram, 1953); it is desirable that most enzymes of this type should be held strictly within bounds.

A large size is also useful for keeping globular molecules within the blood stream, and in general for preventing permeation through membranes. The molecules of the plasma-proteins are too large to pass through the healthy wall of Bowman's capsule in the nephron, and so are not lost from the body; also by remaining in the blood subsequently supplying the tubule of the nephron they provide a small osmotic component of value in the resorption of water from tubule into the blood stream. A similar mechanism operates in the tissues in general, which receive fluid with small-moleculed nutrients from the arterioles and may return some osmotic water on the venular side. No other specific function has yet been attributed to the blood-albumins (Fox and Foster, 1957, p. 338).

There are some blood-proteins, such as the haemocyanins (Table 26), and the haemoglobin of the earthworm (molecular weight $= 3 \times 10^6$), which have a far larger molecule than the average plasma-protein, so that their osmotic effect is insignificant. In fact they appear to be meeting a quite different biological requirement, that of maximal total mass combined with minimal viscosity. These are oxygen-carrier pigments, and the oxygen supply to the tissues depends on their mass, among other things, but the work of circulating the blood is minimised by having a fluid of low viscosity. Weight for weight a small number of large bodies produce a less viscous solution than a large number of small particles. The large molecule of earthworm haemoglobin contrasts with the molecular weight, 68,000, of gnathostome haemoglobin, and the even smaller molecule in cyclostomes; in the vertebrates it is highly concentrated and retained within the erythrocytes, which for the purpose can be regarded as aggregates even larger than the 29 mμ of the molecules of *Helix*–haemocyanin. The erythrocytes are flattened so that oxygen can diffuse readily to and from each molecule of haemoglobin, and they are plastic enough to flow easily through capillaries. Some invertebrates have their haemoglobin segregated in corpuscles and once more the molecular weight is low.

The small molecular size of the protamins, and also of those histones which are associated with nucleic acids, is somewhat surprising in view of the large molecular size of the nucleic acids themselves. They and the nucleoproteins are largely confined to the cells, and DNA (p. 355) largely to the nucleus, and the small protein molecules are sufficiently firmly bound to contribute to their total size. At the same time the small size of protein may facilitate masking and unmasking of the nucleic

acid moiety, piecemeal or totally. With only 15–30 amino acids per molecule of protamin this will cover only one turn or so of the nucleic acid helix, and should peel away easily.

For most of the fibrous proteins, serving a mechanical function, the larger the effective molecular size the better, since in general intramolecular are stronger than intermolecular bonds. However, fibrinogen, the precursor of fibrin, must be present in the normal blood and must be polymerised *in situ* to form an efficient clot, and so a small molecule is optimal. The size is well below that of the free carrier-pigments, but further reduction is limited by the need for an axial ratio suitable for subsequent fibre-formation yet with a stoutness obviating the risk of pseudothrombus formation, through fortuitous tangling in the confined spaces of the capillaries.

15.3 Molecular Shape

As in the case of fibrinogen, molecular shape also can be very important biologically. For molecules bowled along in the blood stream, or in other fluids, a sphere or an ellipsoid is the ideal shape, the sphere for rolling, the prolate ellipsoid for flowing and the oblate for spinning. Most of the plasma-proteins seem to be strongly prolate though with an axial ratio usually less than 5/1 except in the special case of fibrinogen. Beyond this value there is a danger of jamming, or at least of excessive viscosity in narrow tubes, when flow is not laminar. Under simpler conditions of flow even myosin, with an axial ratio of 45–50/1, flows quite well, with all the molecules lying parallel to the direction of flow; in consequence the solution shows birefringence of flow, which disappears as soon as the molecules are allowed to resume a random orientation.

Insulin has a spherical molecule and its sub-units are regular sections of this. As already indicated, the sub-units of the haemoglobin of gnathostomes by contrast are very irregular, but nevertheless combine to build a good sphere. This desirable shape therefore has been evolved in different ways, which may override the shape of sub-units. Since the haemoglobin of vertebrates is confined within cells, the spherical shape of the molecules may seem pointless, but the erythrocytes must pass through spaces small compared with their diameter, and they are greatly distorted in the process; a set of ball-bearings as contents is a great asset.

The shape of the units of the fibrous proteins and some of the features of the globular group are better considered along with other aspects of their structure.

15.4 The Structure of Protein Molecules

The structure of the molecule of vertebrate myoglobin is very similar to that of the 1/4 units of gnathostome haemoglobin (Kendrew, 1959; Perutz, 1960). In spite of its irregularity the details of the folding of the peptide chain are quite consistent and the haem nucleus fits into a specific position. Tertiary bonding must be equally specific and consistent. In the sub-units of insulin three —S—S— bridges between six SH groups play the main part in this, but in other globular proteins hydrogen bonds between the nitrogen and oxygen atoms of the peptide bond units may be the main links, as they are in the tertiary bonds of many fibrous proteins. It is probably an extensive and regular system of this kind which permits an electron-shift started at the haem nucleus of haemoglobin to affect distant carboxyl groups in the protein moiety, by providing a fully resonant meshwork. Reciprocally a change in an SH group of the protein may affect the oxygen dissociation balance of the haem (Najjar and Robinson, 1959). Some of the fibrous proteins also may possess this semiconductor property for the same structural reason. The persistence of life-like behaviour throughout the length of mammalian hairs may depend on this kind of property.

The peptide components of some globular proteins are in one of the more regular secondary forms described in Chapter 14, and the tertiary structure also may be regular or 'crystalline'. Some have molecules with crystalline and aperiodic portions. Myosin and a number of other globular proteins are based on peptides with the α–helix form (p. 200), but in general they are less crystalline than the fibrous group. Relatively few of the active side chain groups are detectable in globular proteins but this does not necessarily imply that most of them are actually involved in tertiary structural links; some of them may be masked for chemical purposes. In either case, however, they are often revealed by denaturation (p. 237) which breaks tertiary bonds in the first instance. The molecule therefore opens out into the fibrous form. At the same time it usually loses its biological activity (Putnam, 1953a), which may depend mainly on the group protected in the native molecule. Some globular proteins are resistant to denaturation and cytochrome *c* recovers much of its activity on cooling after heat-denaturation. The specific structure therefore may often develop as spontaneously as the simpler, orderly structures of inorganic crystals.

The laboratory peptide, polyglycine I (p. 200), spontaneously forms very regular crystalline structures resembling the structural fibrous group rather than the chemically acting, globular proteins. Its ribbon-like peptide chains build sheets (Fig. 32) by alignment in regular antiparallel

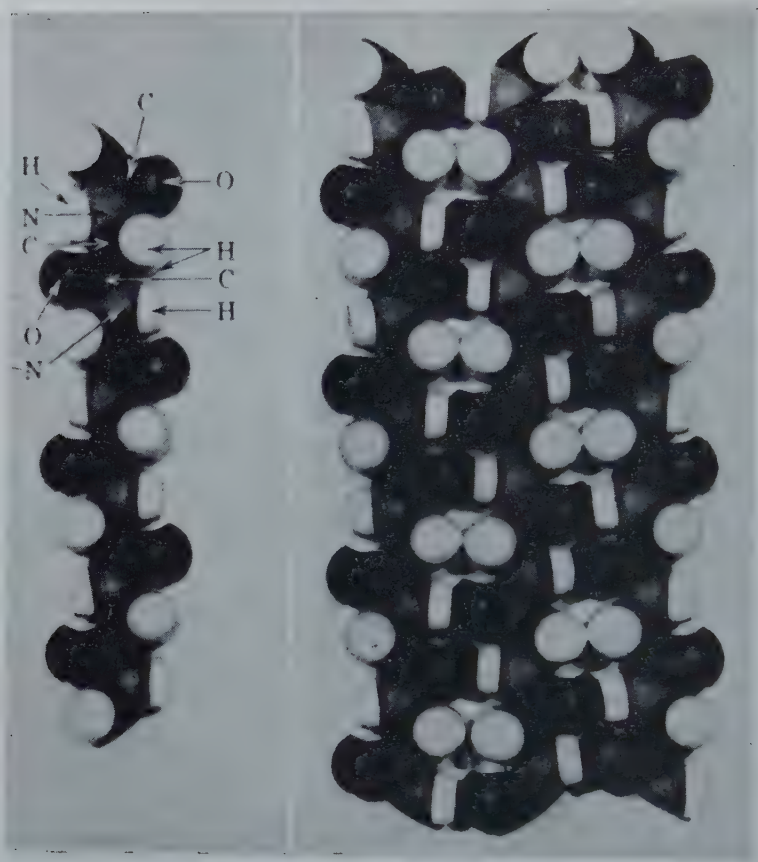


FIG. 32 A. *On Left:* Model of molecule of polyglycine I. The atoms of the four elements are differently coloured and labelled. The simple serrations of the successive peptide bond units (Fig. 29, 1) lie all in the same plane, with all hydrogen-bonding groups.

On Right: Sheet formed by hydrogen-bonded union of molecules of polyglycine I, lying all parallel but alternate ones with opposite orientations of NH_2 and COOH ends ('antiparallel'). The two α-hydrogen-atoms of each glycine residue fit neatly against those of a residue in the neighbouring peptide chains, on either side. The whole sheet is a virtually flat hexagonal mesh; each chain provides two sides of each hexagon and the other two are constituted by the hydrogen-bond unit N—H . . . O = C and the van der Waals-linked —CH_2 H_2C— unit.

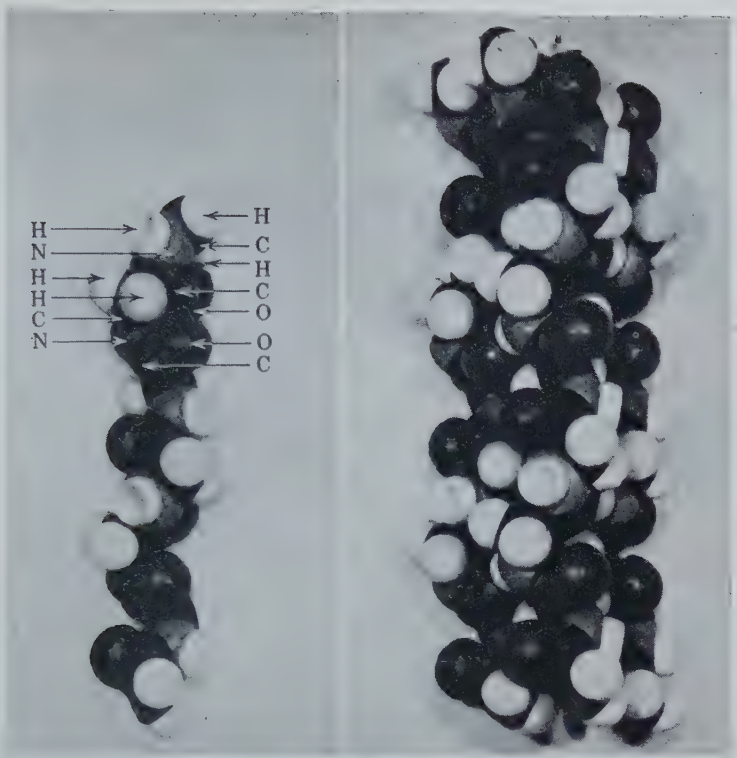


FIG. 32 B. *On Left:* Model of molecule of polyglycine II, a helix with three residues per turn; the H-bonding groups therefore are no longer all in one plane and the peptides form three dimensional bundles (right) rather than flat sheets. Three chains are shown but all the hydrogen-bonds are satisfied only when each is surrounded by six neighbours. [All from Rich (1959).] (From *Review of Modern Physics*, **31**, 50, by courtesy of the author and the American Institute of Physics.)

fashion, i.e. with the nitrogen terminal end of successive chains in opposite directions. The two hydrogen atoms of each α–carbon then fit snugly against those of the two neighbouring chains, in turn, while the peptide bond NH group fits against a C=O of one neighbour and the C=O group against an NH of the other. The hydrogen bonds between these peptide bond groups are the sole tertiary links in this case. Only polyglycine can form flat sheets of this kind, for the steric reasons already considered, but the zigzag type of heteropeptide chain (p. 200) can form pleated sheets (Fig. 33) through alignment with their major zigzags in parallel and in dress, the side chains now protruding out from both faces of the sheet, and enhancing the ridges. The chains therefore are turned 90° on their axes compared with those in the polyglycine sheets. This is

the actual structure of β–keratin, in which —S—S— and hydrogen bonds form the main tertiary links.

The crystalline parts of silk-fibroin also may be built on this plan, in which the chains are again in antiparallel array. Fibroin consists of 40 per cent glycine, and 40 per cent alanine and serine which have the same chain-length. Glycine therefore alternates regularly with one of the C_3

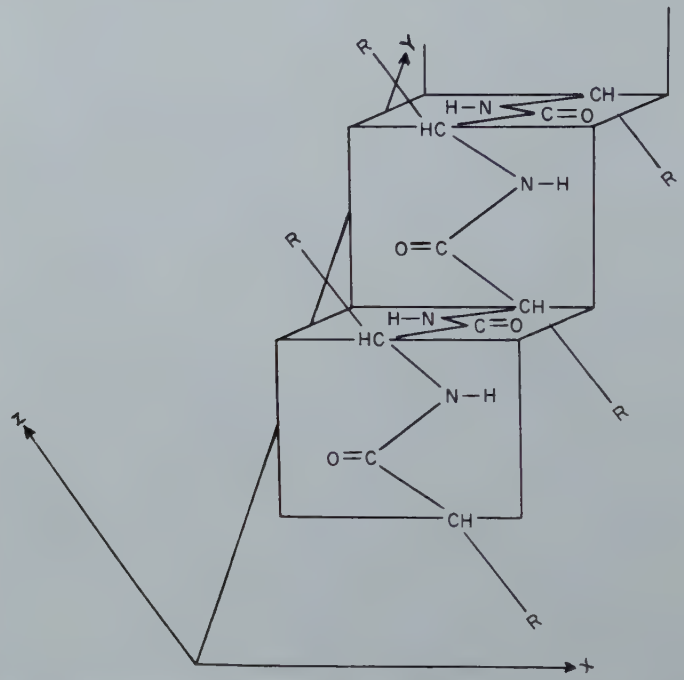


FIG. 33 A. Diagram of the β-type of natural heteropeptide, with two orders of serration, in planes mutually at right angles. There is one major serration per two amino acid residues, compared with the $1\frac{1}{2}$ per residue of the minor zigzags. The side chains are all in the plane of the major serrations, directed alternately in opposite directions. The hydrogen-bonding groups, as in polyglycine I, are all in the plane of the minor serrations.

amino acids so that one side of the sheet bears only hydrogen side chains and the other C_1 groups. The sheets themselves probably polymerise in their turn, in antifaced fashion, so that the hydrogen atoms of one face fit between those of the neighbouring sheet, and similarly for C_1 side chains across the alternating interfaces. This compact fitting of the pleated sheets, in addition to the hydrogen bonds weaving the chains into their sheets, gives great strength to silk-fibroin, with slight elasticity due to the corrugations. Spinning through the narrow orifices of the

silk glands promotes the parallel orientation of chains and their shearing into stable positions (Meyer and Jeanerat, 1939).

Collagen likewise contains a high percentage of glycine, but together with proline, hydroxyproline and other amino acids with bulky side chains, and so the peptide chains are helical, probably based on the polyglycine II structure. In pure polyglycine II each molecular helix is held by hydrogen bonds to its six nearest neighbours, giving an indefinitely

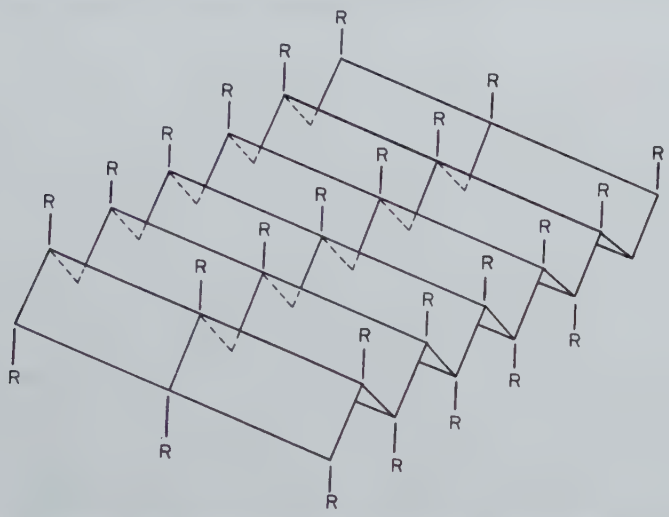


Fig. 33 b. Pleated sheet formed by hydrogen-bonding, in the plane of the minor serrations, between β-peptides in antiparallel array. The chains are in dress and the major serrations give the sheet its pleats. Both from Springall (1954), *Structural Chemistry of Proteins*, by courtesy of the author and Butterworths, London.

extensible array of very regular pattern. In collagen, however, three helices become co-coiled around a common fibre axis, so that their individual helical axes themselves run a spiral course, of rather high pitch, making a strong cable of the trio. The α–hydrogen atoms of the glycine residues of all three occupy and fill the limited, common axial space, while the longer side chains protrude normal to the surface, like the snags on barbed wire. There are no serious steric problems, therefore, and in principle any amino acid could be incorporated. Most of the side chains probably help to tie together neighbouring triplices, and contribute to the strength of collagen (Gross, 1961). The hydroxyl group of hydroxyproline, serine, threonine and possibly also hydroxyglutamic acid and hydroxylysine play the major part in this. Between such twiggy cables there is little scope for hydrogen bonds, which are limited to distances around 2·76 Å. This structure is only possible with primary helices which

conform to a simple pattern of exactly three amino acid residues per turn, and every third a glycine residue. There is in fact around 33 per cent of glycine in all collagens.

The α–helix (p. 200) is the basis of a number of the fibrous proteins, as well as some globular ones. In the former the individual peptide helices may lie in simple parallel orientation or they may be co-coiled in groups of three (Florkin, 1960). Another arrangement, usual in the α–form of hair-keratin, is a group of six helices co-coiled around a seventh, the AB_6 arrangement. When α–keratin is subjected to moist heat and stretched it

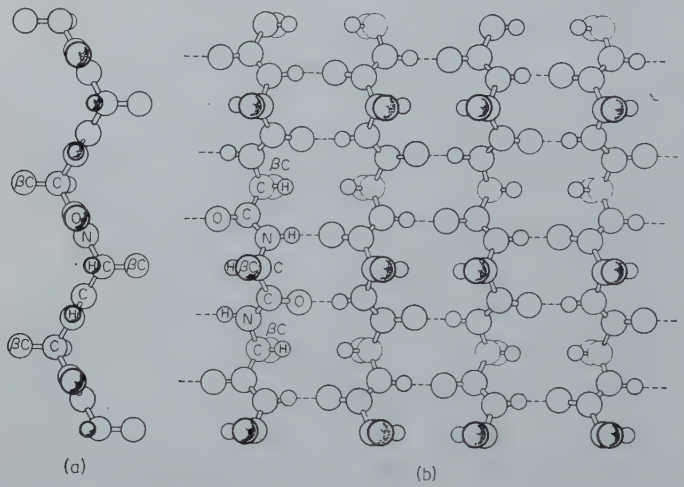


FIG. 33 c. Diagram of steric model of same: (a) single chain, (b) pleated sheet. (From Pauling and Corey (1951), *Proceedings of the National Academy of Sciences, Washington*, **37**, 729, by courtesy of the authors and the National Academy of Sciences.)

changes to the β–form, probably with the pleated-sheet structure already described. If so then the lability of structure to simple physical forces is remarkable. In any case it is certain that considerable changes in structure *are* caused in proteins such as hair-keratin by such treatment, and are the rationale for 'permanent waving' and for some of the treatments of wool.

Myosin is based on the α–helix and *in situ* it is a fibrous protein. The individual filaments, now visualised by high-power electron-micrographs (Huxley, 1960), are 100–140 Å in diameter, and may be small aggregates of AB_6 or similar cables. The electron micrographs show side chains apparently directed towards the thinner 40 Å filaments of the second protein of the myofibrils, *actin*. These once more form a ring of six, equally spaced around each stouter, myosin filament. According to the telescoping theory of muscular contraction (Hanson and Huxley, 1957; 1960) these cross-links are labile and are rapidly and repeatedly broken, to

be re-formed with receptor groups further along, as the one type of protein filament slides past the other. A considerable number of the links are by SH groups, an example of the lability of sulphur bonds (p. 390). There seems little doubt that the unique motor power of muscle and related effector proteins is based on inter- or intramolecular changes in bonding between peptides and that in muscle it is almost possible to relate events to this at all levels of magnitude. There is every reason for the intensive study devoted to it in recent years.

The two proteins of the myofibril can be extracted as soluble globulins by strong potassium chloride solutions. These are physically much more labile than the static skeletal proteins. The myosin molecule rather readily breaks down into four light and two heavy sub-units. If a solution of myosin is forced through a narrow aperture into distilled water, it forms solid fibres with the same X-ray diffraction-picture as the fibrils *in situ*, so that the natural structure is probably a spontaneous one. These artificial fibres can be made to shorten under the influence of ATP (p. 404) in low concentration and to lengthen again at higher concentrations, an interesting analogy to the response of the complete muscle fibril to ATP. Myosin in solution will also spontaneously unite with actin, again in the presence of ATP. There is a great increase in viscosity and in birefringence of flow, indicating an increase in fibrogenesis; ATP can also reverse this change. Thus various simple models of the normal contraction-cycle of the two proteins can be prepared in the laboratory.

Actin is further interesting because it rather readily and reversibly polymerises between a globular monomer, the globular form, and a fibrous form, resembling a string of beads. Globular actin has a molecular weight of 70,000: since that of myosin is nearly 10^6, and the myosin filament has about nine times the sectional area of the actin filament, there are about 16 globular actin units per myosin molecule, *in situ*. Each myosin molecule has about 32 free SH groups, 16 on each heavy 'meromyosin' fragment, two per globular-actin unit.

Insulin and some other proteins also have this ability to build necklace-fibres from globular molecules (Mercer, 1958). The common form, globular-insulin, spontaneously polymerises in solution, at normal temperatures, though the aggregates remain small. However, if the solution is seeded with fibrous-insulin formed at high temperature, the whole quickly changes to the fibrous form (Waugh, 1957). The seeds of fibrous-insulin can be broken up by repeated freezing and thawing and each fragment will then act as a nucleus for further fibre-formation. This phenomenon is full of interest for a number of different but all relevant reasons. It is an interesting example of autocatalysis which is commonplace at the level of inorganic crystals but is remarkable, even there; at the

level of macromolecules of the present type it is at first glance amazing. On further consideration it may be viewed as just one of the properties which make such a teleonomic phenomenon as life possible. Seeding, growth and fragmentation also have possible relevance to fission and growth at higher levels of magnitude, in organelles and cells; these may be equally spontaneous under appropriate conditions. The question why a chemically active protein such as insulin should form fibres at all may be more parochial: the mode of aggregation may be common, and quite incidental to the function of insulin; on the other hand it could be significant for the aggregation of insulin molecules on the surface of target cells. It seems equally uncertain why actin should so readily assume the globular form, unless this enables it to act as a macromolecular ball-bearing. The metastability of the globular-insulin solution, and the need for seeds, also has its counterpart in solutions of simple inorganic salts; a greater energy requirement for the first than for subsequent bonds seems to be a general phenomenon, not restricted to the formation of peptide bonds (p. 199). This may make the metastable state equally common, in both biological and inorganic systems.

Another feature of interest about the muscle proteins is that myosin also acts as an enzyme, so that the mechanical and chemical functions of proteins are not necessarily sharply segregated. It acts in fact as adenosine triphosphatase and so probably taps the energy required for contraction very directly from ATP. This kind of stream-lining in function encourages the view that biological functions are not as complex as sometimes appears, though it may prove to be a secondary simplification.

Fibrinogen is another protein of the same group as keratin and myosin, therefore known as the keratin–myosin–fibrinogen group, having a fibrous or fibrogenic structure, based on peptides with the α–helix form. In the sense that its normal function involves changes in the molecule it is a dynamic protein like those of the myofibrils, but it has its own unique properties, appropriate to blood-clotting. In the intact animal it is kept inactive by a 'keeper' peptide portion, removal of which by thrombase, itself activated by wounding, exposes four new peptide end-groups (Fox and Foster, 1957, p. 340). The fibrinogen then spontaneously polymerises, via an intermediary with a molecular weight about fifteen times that of fibrinogen itself, i.e. about 5×10^6. Both end-to-end and lateral bonds are formed, so that fibres of indefinite length and width are produced. As usual the units tend to align spontaneously, but at the same time the essential property of the final fibrin is that its fibres interlace in all directions and so form a plug trapping blood corpuscles and staunching the wound. There is some superficial resemblance to the apparently random orientation of collagen fibres in areolar connective tissue, and

the determination of this randomness is as interesting as regular orientations, since collagen *is* very regularly orientated in some tissues (Chapter 29). Both are adaptive in their biological contexts.

At present little is known of the lamellar structure assumed by proteins orientated in relation to lipid interfaces and molecular layers (p. 85). It is believed that hydrophil side chains are orientated towards the aqueous side and hydrophobe side chains towards the lipid side of the interface, and that the polypeptide chains lie in the plane of the interface. Least is known about the arrangement of the peptide chains in the plane of the membrane.

In these structural features the globular, and still more the fibrous, proteins display their great versatility, their ability to produce structures of a great range of sizes and with a great variety of structure, from the completely random to the highly orderly. Further aspects of this will be considered in Chapter 29, and more incidentally elsewhere.

15.5 Solubility and Related Properties of Proteins

The fibrous proteins are relatively insoluble for two main reasons, the formation of stable bonds which, like —S—S—, are less hydrophilic than the free SH groups, and the automatic decrease in solubility with size of solute particle. Biologically insolubility is one of the major virtues of most fibrous proteins. When broken up, however, by almost any method, they pass into solution, like actin and myosin from muscle, gelatin from collagen, fibrin under the action of fibrinolysin and most fibrous proteins under the action of digestive enzymes. Small protein molecules are rarely insoluble in all aqueous media; as already noted both NH and C=O groups of the α-complex are still capable of forming hydrogen bonds after peptidation and so remain as hydrophil as the free ionic groups from which they were derived. If secondary and tertiary structures favour the enclosure of hydrophobe and the exposure of hydrophil side chains, solubility may actually increase with molecular size as in the case of some peptides studied (p. 203). As a group, therefore, proteins are water-soluble, but with an amazing variety in precise solubility properties, which fits them to perform a wide range of functions.

Globulins are mostly insoluble in distilled water, salts being necessary in the medium; as the concentration of the latter is increased, however, the globulins re-precipitate. For ammonium sulphate, one of the most convenient salts, the limiting concentration is 1/3 saturation, 1·4 M. Proteins with a wider solubility-range in both directions, from very dilute to 1/2 saturation with ammonium sulphate, are called pseudoglobulins. Albumins are soluble even in distilled water, and in ammonium sulphate

up to almost full saturation, while at the other extreme fibrinogen is precipitated even by 1/4 saturated ammonium sulphate. Collectively, therefore, they can cover a wide range of requirements, and in addition there is once more the potential basis for a mechanism tending to keep biological systems within a particular physiological range.

Nucleoproteins are soluble in very dilute solutions of sodium chloride (0·02 M), and in distilled water, and again in solutions of 1–2 M, but insoluble in the intermediate concentration of M/7, which in fact is the physiological concentration of sodium chloride. Their solubility properties are almost converse to those of the globulins, therefore. Most proteins are precipitated at their isoelectric point (p. 166) but are soluble on either side of this, throughout the pH range. For most proteins the isoelectric point is on the acid side of neutrality, but that of the histones is on the alkaline side and they are insoluble in dilute ammonia; that of the protamines is at an even higher pH so that they are still soluble in dilute alkalis. Phosphoproteins are precipitated by dilute acetic acid and dissolve in higher concentrations, but vertebrate salivary mucin, a glycoprotein, is insoluble also in the higher concentrations. Glutelins are soluble in both dilute acids and dilute alkalis, but insoluble in neutral salt solutions. The prolamines also are insoluble in neutral salt solutions, and in distilled water, but are soluble in 70–80 per cent ethanol; they are again insoluble in absolute alcohol. Collectively, therefore, proteins have almost every conceivable combination of solubility properties. In most cases the solubility properties are known to be the natural outcome of the composition and structure of the proteins and display their range of potentialities as ably as any other aspect of their physics and chemistry.

In some cases, however, the solubility of the protein is not predictable from its amino acid composition, and perhaps must be considered an emergent property. In virtue of their high proline content the prolamines would be expected to dissolve readily in water, as well as in the ethanol–water mixtures. Some proteins with a high proline content, for instance casein and collagen, however, do not have even the solubility in 70–80 per cent ethanol. Casein, unlike most proteins, and unlike most free amino acids except glycine and proline, has a positive heat of solution and this might be taken to reflect its proline content. However, lactoglobulin also has a positive heat of solution but has no great amount of either glycine or proline. Moreover the heat of solution of a protein can change sign over the physiological temperature range; egg-albumin in ammonium sulphate has a negative heat of solution between 0° and 12°C and a positive value between 20° and 30°C. Its solubility therefore passes through a minimum around room temperature, in the middle of the physiological range. Most proteins show a maximum here, in the

sense that their solubility shows the normal response of most activities to temperature increase but that they become coagulated by heat above the physiological range.

Proteins have the interesting property of a higher density in solution than as solid crystals. This is due to *electrostriction*, a compression of the molecule when in solution, due to ionic electrical forces. Free amino acids show the same phenomenon and differ from most organic substances in this; it is due largely to attraction between oppositely charged groups. There is a parallel in the relative densities of ice and water (p. 14).

The solubility of proteins is affected by inorganic salts in the same general ways as that of the free amino acids, and again is often explicable in terms of the constituent amino acids. In general, however, proteins are more sensitive than free amino acids, and there are emergent properties. Low salt concentrations in general improve ionisation and solubility and are said to *salt in*, while higher concentrations begin to compete with the protein for water and so *salt out*. Salting out is usually induced most easily at the isoelectric point, where + and — charges on the protein are equally balanced, so that dehydration is the explanation, rather than the formation of an insoluble salt between protein and one ion of the inorganic salt. In some cases, however, there is a second point of maximal precipitation, at some other pH value, and here salt formation does occur. Moreover the salting-out potency increases disproportionately with the valency of either ion of the salt and this again indicates salt formation, polyvalent ions complexing with more than one protein molecule and so initiating larger aggregates. Haemoglobin salts in indefinitely as the concentration of sodium chloride is increased, both ions being monovalent in this salt. Typical salting out is reversible but the compounds formed with some of the polyvalent and large cations, such as Hg^{++}, and with the anions having similar properties, picrate, molybdate, etc., are stable and the reaction irreversible. Such compounds are also formed with some monovalent ions, such as Au^{+} and CCl_3COO^{-}, and it is necessary to distinguish this *fixative* effect from reversible salting out by simple dehydration. Salting in and out play a part in normal physiological activities, while fixatives are powerful poisons. Of course the precipitation by simple salting out involves molecular aggregation, the protein molecules sometimes forming salts with each other when they are robbed of water.

Within the salting-out range of concentration of an inorganic salt, proteins are most soluble at low temperatures, like gases in aqueous solution; gases in fact are also sensitive to salting out. Again some proteins are anomalous, however (Cohn and Edsall, 1943, p. 607), and their solubility in salines of this range increases with rise in temperature, like that of most proteins when they are in dilute salines. Others, such as

egg-albumin, show little change with temperature when in salines of the salting-out range, and this is probably associated with the anomalous change in heat of solution of egg-albumin with temperature. Again there is great variety among proteins collectively.

Salting out is of course not peculiar to amino acids and proteins. It is less easily induced in strong inorganic electrolytes, however, than in un-ionised molecules and gases. The amino acids and proteins in this respect have an intermediate status between well-ionised and un-ionised molecules, and show the potentialities and advantages of both.

The solubility of proteins is also affected by the presence of other proteins, as might have been anticipated from the behaviour of amino acids and peptides. *In vivo*, therefore, behaviour may be very different from that of isolated proteins. The interactions are probably in many cases similar to those of simpler electrolytes, but the subtlety and variety of effect are much greater, and are still very poorly understood in detail. When two proteins interact chemically in the manner of protamin with insulin, molecular size is increased, and solubility is decreased for that reason, if not for more specific, ones.

It is by no means unusual for a protein to improve its own solubility —to salt itself in, and to be more soluble in its own solution than in distilled water. The rate of solution when dry protein is actually placed in distilled water, therefore, is at first very low and progressively accelerates; this is perhaps a physical counterpart to autocatalysis and also something reciprocal to crystallisation by seeding (p. 523), but in any case it is another aspect of the versatility of protein properties. The true reciprocal to this self-solution effect is precipitation upon dilution, already noted in connection with peptides (p. 223), a useful safeguard against indefinite dispersion in water.

Proteins also affect the solubility properties of other substances, in ways which are put to biological use. They increase the solubility of calcium carbonate and calcium phosphate and so facilitate the subsequent deposition of inorganic skeletons (Thompson, 1942, p. 652). The solubility of these salts decreases if the protein is degraded to peptone, and this may be a clue to the way the salts are precipitated in the skeleton. In the presence of albumin, calcium carbonate from the reaction between calcium chloride and sodium carbonate is deposited in concentric layers, probably a Liesegang effect (p. 231). There are also radial striations in these 'calcospherites' which strongly resemble natural bodies such as the microscleres of the sponge, *Astrosclera*, the concretions of the parenchyme of cestodes, of the oesophageal glands of earthworms, etc. Moreover, when close-packed these calcospherites become forced into the shape of polygonal prisms, as in the prismatic layer of molluscan

shells. With the additional presence of phosphate, and with the amounts of albumin and gelatin minimal, irregular concretions are formed very similar to the natural sanidasters of some sponges, the warty spicules of Alcyonaria and those of some Holothuria. Other natural forms also have been produced by laboratory models; protein is probably always present in skeletal structures, if only as a foundation.

15.5.1 Colloidal States of Proteins

Because of the large size of their molecules and micelles proteins in solution inevitably differ from solutions of simple *crystalloids*. These differences were first seriously studied by Thomas Graham, over a century ago, and he called them colloid solutions because glue (colla) or gelatin was the type example. Their properties are intermediate between those of crystalloid solutions and those of gross suspensions such as clay, and so also between true solutions and the solid state. Graham was very impressed with the potential importance of these properties as the basis of life and it is interesting to re-read his conclusions in later perspective (Bayliss, 1959, p. 226). They were in effect that colloid is the dynamic state of matter, the crystalloid state being static, that colloids possess energy and may be looked on as the probable primary source of the force appearing in the phenomena of vitality and that the characteristic protraction of chemico-organic changes may be referred to the gradual manner in which colloidal changes take place, for they always demand time as an element. Graham perhaps overemphasised this aspect of the properties of proteins, though not the importance of the properties of the complete range of colloid states, of which his proteins provided a simple example.

In addition to the most important type of colloid solution, the solid-in-liquid, to which the proteins belong, emulsions, or liquid-in-liquid colloids, also are biologically significant, and occasionally foams, or gas-in-liquid colloids (Seifriz, 1936). Most of the other conceivable types occur elsewhere in nature, for instance liquid-in-gas (cloud and fog), solid-in-gas (smoke), gas-in-solid (lava), liquid-in-solid (pearl) and solid-in-solid (gems).

The distinction between crystalloid and colloid solutions is not so sharp as it once appeared, and there is a complete spectrum from crystalloid through colloid to simple suspension; the proteins alone range through much of this spectrum, though no doubt most of them are within the range which has come to be regarded as that of typical colloids.

Perhaps more important than mere size is the distinction between hydrophil colloids such as the proteins and the hydrophobe type characteristic of most inorganic colloids. The former carry little resultant charge and depend on their water shell to keep them in suspension, and to

prevent them from aggregating; this is why neutral dehydrating agents such as ethanol and strong salt solutions can precipitate them. The particles of hydrophobe colloids keep afloat by mutual repulsion due to a substantial charge of the same sign on each; in consequence quite small amounts of ions of the opposite charge neutralise this and precipitate the colloid. This high sensitivity to precipitation is the main reason for the term hydrophobe, and it is evident that this type would have little value in biological systems. The H^+ and ^-OH ions are the most effective in precipitating them, and whereas proteins redissolve on the other side of their isoelectric point, because of their zwitterion character, hydrophobe colloids are precipitated irreversibly. It is perhaps worth emphasising that proteins are precipitated only if the pH is held precisely at their isoelectric point and that in practice they are in solution under most pH conditions whereas hydrophobe colloids are unstable over most of the pH range.

Hydrophobe colloids in general have larger particles than the hydrophil type and so are known alternatively as *suspensoids*. This is sometimes set in opposition to the alternative term, *emulsoid*, for the hydrophil type, depending on the greater stability of emulsions than of solid particles of the same size; the term does not imply that protein colloids are fluid particles, though they are on occasion (p. 233). Perhaps more immediately relevant is the fact that the stability of emulsions depends largely on traces of contaminants at the surface of the droplets and that in biological and commercial emulsions protein is the most usual and efficient stabiliser. In consequence the properties of the emulsion are to some extent the properties of the protein. Further, the amount of hydrocarbon in the side chains of proteins gives some degree of lipid, fluid property to the molecule. In passing it may be noted that proteins can also coat and stabilise suspensoid particles to some extent, and that this general type of stabilisation is by no means the least of their biologically useful functions.

Proteins are molecular colloids whereas the most sensitive hydrophobes, metals, clays, etc., are mere fragments of crystalline material, forcibly broken off. A few inorganic substances, such as silicic acid (hydrated silica), vanadium pentoxide and calcium germanate do form colloids which are molecular aggregates and these are more hydrophil- and protein-like; on the other hand the classical suspensoids, ferric hydroxide and arsenious sulphide, also are chemically produced and presumably are molecular aggregates. Proteins are not absolutely unique as hydrophil or as molecular colloids but they are outstanding for being so as a class.

They possess all the general advantages of colloidal solutions in addition. Their molecules are large enough to be retained by membranes permeable to the small molecules of raw materials, hexoses, amino acids, vitamins and salts, as well as of excretory materials. At the same time their

ratio of surface area to volume, on which chemical activity depends, is not very low; as enzymes the proteins rank among the most reactive of molecules, any disability due to surface/volume ratio being compensated by advantages of specific arrangements of side chains at the surface. In addition the positive advantage of inertia and protracted action, stressed by Graham, must be recognised. The mobility of these molecules also is less than that of crystalloids, but is not of an entirely different order. They diffuse and they are mixed by Brownian movement. Much larger particles, visible under the microscope, are still subject to this enforced mobility, and between this and the protein molecule there is a range of molecular aggregates and micelles sharing this advantage. Viruses and small cocciform bacteria are still within the range but the larger bacteria have elongate, stream-lined forms for autogenous movement, and have crossed an important Rubicon.

One important phenomenon which emerges at the level of systems containing colloidal molecules retained by differentially permeable membranes is the Donnan effect. If the membrane is permeable to small ions but not to proteins, even if it is merely a simple mechanical sieve, then the system tends to be an equilibrium defined by the equation

$$C_1 \times A_1 = C_2 \times A_2$$

where C_1 and C_2 are the concentrations of permeating cations on the two sides of the membrane and A_1 and A_2 the corresponding concentrations of permeating anions. If, to take the simplest possible situation, there is on side 1 a single species of protein molecule, which under physiological conditions will be ionised as an anion, P^-, and no large non-permeating ions on side 2, then electrical equilibrium demands a tendency towards $C_1 = A_1 + P_1$, so that $C_1 > A_1$. If the Donnan relationship holds then A_2 must be greater than C_2 and there will be an excess of negative ions on side 2. Both conditions, Donnan and electrical neutrality, *can* be simultaneously established, but only at the expense of an osmotic imbalance across the membrane; C_2 can equal A_2 and their product can at the same time equal $C_1 \times A_1$ if each equals $\sqrt{(C_1 \times A_1)}$. The general conclusion is that it is impossible to establish simultaneously all three equilibria, Donnan, electrical and osmotic, though *any two* are possible (Fig. 34). In any system containing membranes impermeable to colloids, therefore, there are permanent differentials of one kind or another, potentially the basis for continuous work, which might take many forms. It is probably this situation which has been exploited in biological membranes, so that they usually carry a permanent resting potential, although the components of the membrane itself may be symmetrically distributed across it (p. 86).

It will be appreciated that the protein molecules are often polyanions of

high order, so that the inequality between C_1 and A_1 could be very considerable. Since it is a single osmotic particle whatever its charge, however, proteins provide a large potential variation in ionic/osmotic activity.

Hypothetical situation	One side of membrane		Other side of membrane		Equilibria satisfied	Equilibria not satisfied
	Cations	Anions	Cations	Anions		
1	Na=4*n*	P=2*n* Cl=2*n*	Na=4*n*	Cl=4*n*	Osmotic Electrical neutrality	Donnan
2	Na=4*n*	P=3*n* Cl=*n*	Na=2*n*	Cl=2*n*	Electrical neutrality Donnan	Osmotic
3	Na=4*n*	P=*n* Cl=3*n*	Na=6*n*	Cl=2*n*	Donnan Osmotic	Electrical neutrality

FIG. 34 Three hypothetical distributions of two permeating ions, Na and Cl, and one non-permeating ion, the anion P (protein), on the two sides of a biological membrane to show that the requirements for all of the three relevant equilibria cannot be met simultaneously.

The larger protein molecules in colloidal solution show Tyndall's effect: they visualise a beam of light passed through the solution and viewed at right angles to the beam. This is due to the light being scattered at the surfaces of the molecules, and becomes appreciable as the molecular size reaches the wavelengths of visible rays (380–780 mμ). Violet and blue light, the shortest waves, naturally are the first to be affected and few protein molecules in solution are larger than this. Tyndall blues contribute to the colours of the human iris, feathers, skins, and other biological materials. Not all blues, and not even all 'structural' blues, are due to the Tyndall effect (Fox and Vevers, 1960).

15.5.2 Gels

One important manifestation of the versatile potentialities of hydrophil colloids is the faculty for reversible gelation. Polysaccharides and nucleic acids, and even the silicic acid, vanadium pentoxide and calcium germanate sols already mentioned, also form gels in this way, so that again proteins are not absolutely unique, but they are certainly outstanding for the extent to which their sol–gel changes are used in living organisms. The movement of *Amoeba* and other sarcodines, and protoplasmic

streaming, are classical examples. The change usually involves a volume change which also has potential uses. Sols will sometimes gelate on simply evaporating some of the water and it seems evident that in this case the solute molecules are driven closer together, until eventually they are in contact and form a second continuous phase. On the other hand the change can occur without any loss of water and this probably involves a change in shape of the solute molecules enabling them to form a continuous phase without any increase in relative volume; a change in surface-charge also might be adequate, even without a significant change in molecular shape. Globular molecules may partly uncoil, an incipient denaturation. Slender fibres formed *in situ* can form an effective solid phase at amazingly low concentrations, as low as 0·004 per cent in the case of fibrinogen, and even less in some others. The secretion of the hagfish, *Myxine*, is a dramatic illustration.

Some gels form *thixotropically*, i.e. merely on standing, and they solate on shaking; salad cream is an analogous case among emulsions. Other colloids show the converse phenomenon of *dilatancy*, becoming more solid on mechanical agitation. A simple analogue of this is builder's cement and also sand at a particular degree of waterlogging. The latter is put to use in the locomotion of *Arenicola* (Newell and Chapman, 1947). The two types of colloid have characteristic curves of viscosity and of rate of flow, in relation to the magnitude of a shearing force applied to them (Fig. 35).

Most gels form exothermically, an indication of the truth of the idea that the molecules are more disorderly than in the sol state. They therefore solate on warming and set on cooling. Gelatin among proteins and agar among polysaccharides are typical examples. Polybenzylglutamate (p. 201) behaves similarly though it should be noted that the individual molecules 'melt' when the system as a whole gelates, an interesting example of the complexity of even such a relatively simple system. Methyl cellulose, a laboratory polysaccharide model, shows the converse property of gelating on heating, i.e. endothermically, and solating when cooled, a kind of reversible heat-coagulation. The helices of DNA (p. 361) similarly unwind on heating (Doty, 1960), and the cortex of the egg of the sea-urchin, *Arbacia*, shows a fall in viscosity at low temperatures which may imply a gel–sol change. Thixotropic colloids are those with virtually no entropic change on gelation. Albumin has the orthodox fall in viscosity with increase in temperature, but complicated by a sudden and marked increase restricted to a narrow range between 57·5° and 60°C (Bayliss, 1959, p. 265). Being outside the physiological temperature range this may have little biological significance, but it again illustrates the variety of properties for exploitation among proteins.

Any factor which makes colloidal molecules more elongated, or more branched, or increases the number of their reactive groups, is likely to cause gelation. The molecule of gelatin is C-shaped in the sol state and straightens out when it gelates. Insulin and β–lactoglobulin appear to gelate through increased attractions between molecules, as already suggested. Lactoglobulin effectively gelates when reversibly salted out by ammonium sulphate (p. 223).

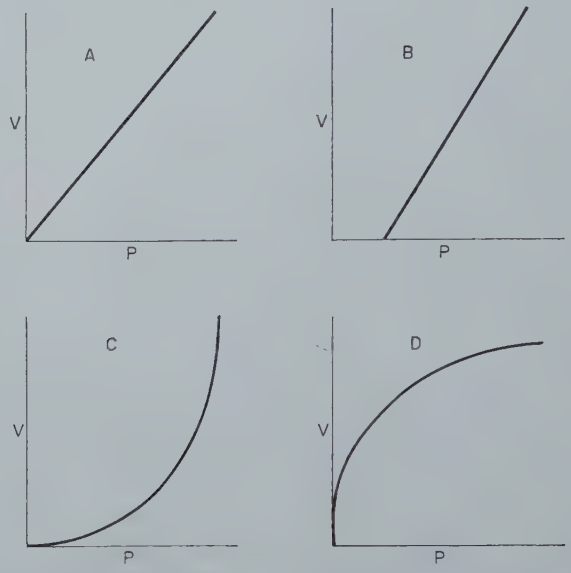


FIG. 35 Diagrams to illustrate the relationship between shear-rate (V) and shearing force (P) for four relevant types of material: A. Pure viscous (Newtonian) fluid, B. Pure plastic material, C. Non-Newtonian liquids and thixotropic gels, D. Elastic materials and dilatant gels.

Gels retain many of the virtues of sols, owing to the persistence of the continuous water phase. They conduct electricity equally well (Bull, 1943) so that the charge on the ionic groups of the protein molecules assists as much as it retards the movement of small, free ions. There is little doubt that the fixed charges should be considered since some gels show *piezoelectric* properties; that is to say they generate an electric potential in response to mechanical distortion, like the crystal detector of the early radio receivers. Piezoelectric potentials may have relevance to the mechanoreceptors of some animals and to other biological situations.

Diffusion also can be almost as rapid in a gel as in a sol. At the same time it does impose new conditions, and if a soluble reagent, A, is predissolved in a gel and a second soluble reagent, B, then allowed to

diffuse into the gel, the insoluble product of reaction may be precipitated in sharp concentric shells around the source of B. These *Liesegang spheres*, or rings when the gel is a thin layer, may be the explanation of some of the periodic formations seen in living organisms, though there are many concentric, lamellar and striped patterns which certainly have not such a simple basis. The alternating spheres of mineral and protein in skeletal structures (p. 224), and of starch and protein in starch grains, conceivably could be simple examples.

15.5.3 Liquid Crystals

Some colloids form gel-like solids even without actual contact between the molecules or micelles, and these are rather common in biological systems. The particles appear to be held rigidly aligned and spaced at the intermolecular distance of minimal coulombic repulsion, and this is a stable, equilibrium state. The nucleoprotein of tobacco mosaic virus (TMV) has this property and it can in fact form a gel even at much lower concentrations than that required to give the uniform spacing just mentioned. It is believed that in this case strands or rather lamellae of aligned micelles form a spongework enclosing lentoid spaces of pure solvent (Fig. 36). The difference from an orthodox gel therefore is simply that the solid particles are not in physical contact.

The apt term *liquid crystal* has been given to such orderly arrays of particles not in actual contact, and they are sometimes regarded as a fourth state of matter, a *mesoform* state between liquids and solids. The soft texture and high water content of the crystals of many proteins indicate that there may be no very rigid distinction between solid and liquid crystals, and it is perhaps significant that the form of the ordinary solid crystals of some proteins varies according to the concentration of the solution in which they are formed (Coleman *et al.*, 1960). Colloids in general might of course be regarded as somewhat in the nature of mesoforms.

There is a great variety of these liquid crystals which again are almost, but not quite, peculiar to biological systems. One outstanding exception is the clay, bentonite (Bateman, 1945); here the particles are discoid, evenly spaced, but randomly orientated, whereas the TMV particles are also orientated in parallel. They are anisotropic or birefringent, therefore, while bentonite is isotropic, though it shows a regular crystal lattice. The axes of the discs of bentonite oscillate about equilibrium positions related to the crystal axes. At certain concentrations the axes of the discs do all become parallel, through mutual interaction, and the gel becomes birefringent, like that of TMV. The degree of order which develops spontaneously even in such simple colloidal systems, and the variety of

structure under different conditions, is very impressive, and indicates that there is nothing supernatural in the proportionately more complex behaviour of biological colloids.

TMV in fact develops other mesoforms than the two already described. If the concentration is further reduced the liquid crystal lamellae break up into lentoids or *tactoids*, isolated liquid crystals suspended in a continuous fluid phase. There has been a reversal of phases, in fact (Fig. 36), and we now have a sol type of liquid in place of a gel type. The tactoids are in osmotic equilibrium with the fluid phase which therefore is called the *equilibrium liquid*. At still lower concentrations the TMV micelles become

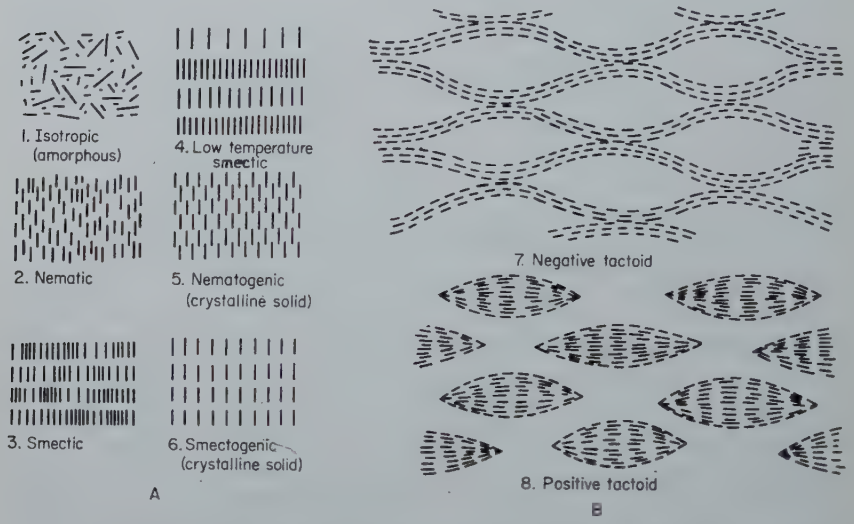


FIG. 36 Diagrams of the biologically relevant types of 'mesoforms' or 'liquid crystals', formed by elongated molecules and micelles in solution, together with two examples of orthodox solid crystals.

A. The grades of orderliness among mesoforms: 1. Completely random, non-crystalline, 2. All molecules aligned but not evenly spaced or in dress, 3. All molecules aligned and in dress, forming a pile of monolayers, but randomly spaced within the layers, 4. In dress and evenly spaced in each layer but the spacing varying from one layer to another, 5, 6. Orthodox crystals with alignment, dress and identical spacing throughout, 5. layers partly imbricating and 6. layers free to shear on each other. Based mainly on J. Needham (1942).

B. Tactoids, or two-phase mesoform systems. In the negative tactoid (7) the mesoform solution is the continuous phase and water with micromolecular solutes fills the lentoid spaces, while in the positive tactoid (8), formed at lower concentrations of the macromolecule, the water is the continuous phase. The lentoids are a compromise between surface forces tending to form a sphere and the elongated molecules tending to produce a parallel, crystalline array. The order of the crystals is nematic to sub-smectic. Based on Bernal and Fankuchen (1941).

disorderly and an isotropic sol results. The micelles are large enough to scatter light and the solutions appear cloudy. They are also large enough to fall under gravity, against repulsion, Brownian forces and so on, and therefore in a TMV preparation there is a vertical concentration gradient. All four colloidal forms may be present, in order from the bottom to the top of a vessel (Bernal and Fankuchen, 1941).

There is probably evidence of liquid crystalline structure in the curious spiral rotation of the plasmodium of a Mycetozoan as it flows (Seifriz, 1952), and in the spiral track of the nucleolus as it falls through the nucleus (Harris, 1939). It may have relevance also to the division-spindle of the cell, and even to the more permanent motor organs, such as the muscle fibril. The actin filaments are probably anchored at one end but free at the other, while the myosin filaments may be effectively free throughout their length (Huxley, 1960), at least during certain phases of the contraction-cycle.

Several degrees of orderliness have been recognised in the arrangement of the micelles of liquid crystals (Fig. 36). When, as in TMV, they are aligned and equally spaced laterally, but not in dress, they are said to be in the nematic state. In others the molecules are aligned and in dress, so that they form successive layers perpendicular to the axes of the individual particles. Within the layer they may or may not be evenly spaced; in the latter case the system is said to be *smectic*. They may be evenly spaced within each layer but unrelated to the spacing in other layers, or finally they may have the complete regularity of solid crystals.

Just as colloids in general combine the virtues of solid and liquid systems so liquid crystals have the mobility of fluids and the structural order of solids. The individual particles may move and be replaced without changing the essential structure, and this is ideal in systems which retain continuity of genetical form notwithstanding the continuous flux of material through the body.

15.5.4 Coacervates

Liquid crystals do not end the list of interesting and biologically significant colloidal forms produced by proteins. One of the most complex and 'vital' is the *coacervate* droplet (Booij and de Jong, 1956), ranging up to 670μ in diameter, covering the size range from cell-organelles up to very large whole cells. There is a separation of two fluid phases, in fact, as in an emulsion, but with both phases essentially aqueous; they are in effective equilibrium, like liquid crystals with their medium. Their properties are the most versatile, teleotropic and eobiological of any system yet considered. Even so, the state is not absolutely peculiar to biological materials, or even to macromolecules, since the bentonite

suspension can assume some such form. The relationship of silicon to carbon (p. 37) may be significant here.

Quite simple organic systems such as phenol in water and methanol in cyclohexanol also produce coacervate droplets, below 65·9° and 49·1°C respectively. These are droplets of solvent-saturated solute in solute-saturated solvent, not simply of solute in solvent, and coacervate droplets are in fact usually polyphasic bodies. They are not necessarily rigidly distinct from true emulsions in this respect, since the solubility of some lipids in water, and reciprocally, is not very high and emulsion droplets are probably saturated with water. Above the critical temperature given, phenol and water are miscible in all proportions, while below this temperature they may be prepared as a simple solution of phenol in water; however, the phenol is so soluble, 500 g or more in 15 ml, that there is no doubt eventually a solution of water in phenol, though there is at no stage anything corresponding to a phase-reversal. Under other conditions a phenol-in-water phase separates in bulk from a water-in-phenol phase, so that the versatility almost compares with that of the more complex TMV nucleoprotein systems. The phenol coacervate in fact, apart from its higher surface-forces, may be compared with the liquid crystal of TMV in water, an intermediary between a true solution of nucleoprotein in water and one of water in nucleoprotein. Coacervate droplets of proteins tend to produce solid precipitates when they separate out, and not a fluid phase like that of the simple polar lipid, phenol.

Another relatively simple example, oleic acid to which is added increasing amounts of potassium chloride, very well illustrates the position of coacervates in the wider spectrum of physical states (Booij and de Jong, 1956). The fatty acid is insoluble in water but dissolves as it is converted to the potassium salt, forming at first a true solution of low viscosity. As the amount of potassium chloride is increased the solution becomes more viscous, but remains for a time plastic. At still higher concentrations it becomes increasingly elastic. Then elasticity and viscosity both suddenly decrease as coacervation occurs; compact globules have replaced a tangle of molecular fibres. Further increase in potassium chloride causes the droplets to fuse to a continuous coacervate layer, separated from a smectic liquid–crystal layer, and a third layer consisting of the equilibrium liquid (p. 232). Finally the coacervate layer progressively disappears, leaving the mesoform system.

Proteins most readily form coacervates consisting of two or more components in addition to the solvent and these have been called *complex coacervates*. Typically two proteins together in solution form common coacervate droplets over a pH range between their respective isoelectric points. Here one is a macro-anion and the other a macro-cation, and

presumably there is some kind of salt-formation since the coacervate breaks at either isoelectric point; the salt may be fully ionised and some such loose association seems demanded: conjugation would tend to cause flocculation and would not be sufficiently reversible. *In vivo* the intracellular pH is one or two pH units on the acid side of neutrality so that the possibilities of coacervate formation are high; cells themselves may have originated in this way and a study of the membrane of coacervate droplets is a leading interest (Booij and de Jong, 1956).

Interaction between oppositely charged proteins must reduce the resultant charge, but the association retains enough of its water-shell not to be precipitated—as would be expected if each protein remains as a free ion. Partial dehydration *is* a factor in coacervation, however, since this can be induced by partial dehydration of gelatin and by partial dealcoholisation of a solution of prolamins in ethanol (Oparin, 1957a). Sugar and other non-electrolytes are effective dehydrating agents for this purpose so that there is no doubt that both water-shell and coulombic charge are important in coacervate formation.

Even polyvalent inorganic ions can produce a coacervate with large ions of opposite charge, for instance Co^{6+} with lecithin (Picken, 1960, p. 192). This example again implies a resemblance to incipient precipitation, the polyvalent ion promoting aggregation (p. 223). It is probably a two-component type, the inorganic ion being one species, and probably all simple coacervates are after all of this type.

Coacervates incorporating three solute components, A, B, C, are not simply droplets of A–B–C in water but of A–B, B–C and A–C separately, in water or in each other. They are therefore called, somewhat cumbrously, multiple complex coacervates. Polyphasic systems of this kind probably play an important part in biological structures. Something very similar is visible under the microscope in the cytoplasm of oocytes and other cells, though with the added complication of lipid phases.

There is good evidence that the limiting membrane of coacervate droplets is vital not only for their initiation and stability but also for their very interesting bioid properties. It is sometimes monomolecular, perhaps in simple systems, but otherwise bimolecular, as in biological membranes (Oparin, 1957a, p. 317; Picken, 1960). Coacervates have been classified as O– and P–types accordingly (Booij and de Jong, 1956). Coacervate droplets swell in sodium solutions and shrink in calcium, as living cells do (Overbeek and Voorn, 1957), and they selectively absorb materials from the equilibrium liquid; in fact they may completely deplete it of some solutes, eventually working against a concentration gradient. This is analogous to selective transport across the gut-wall, the nephric tubule and many other biological membranes, and is usually regarded as one of

the most wonderful and teleonomic of physiological functions. If suitable enzymes of biological origin are incorporated into laboratory coacervates, and the appropriate substrate added to the equilibrium liquid, then the uptake of that substrate is greatly accelerated, in accordance with diffusion and mass-action laws, since the substrate is being changed chemically by the enzyme after transport (Oparin, 1957b, p. 221). This again may be biologically relevant; for instance glucose is converted, no doubt enzymically, to trehalose after absorption across the insect gut (Treherne, 1958); trehalose is also much more impermeant than glucose, so that a second factor improves uptake in this case. Permeases, enzymes in the membrane specifically concerned with transport, deliver the substrate in its original form and so they constitute a different category. As a result of absorption, coacervate droplets grow, like living cells, and they change their composition, i.e. they *differentiate*. They can also ingest oil droplets in a manner similar to *pinocytosis* by living cells.

Vacuoles, and therefore membranes, develop also inside coacervate droplets, in response to heat, electrical fields, and inorganic salts; they may do so reversibly, while the main membrane remains stable. Solid bodies, crystals and fibres also form spontaneously inside, and differentiation is by no means restricted to the limiting membrane. Directed stimuli such as diffusion fields and d.c. electrical fields cause internal flow-patterns similar to cyclosis and protoplasmic streaming, and if free to move the whole droplet moves in amoeboid fashion in the axis of the field. The whole droplet, like liquid crystals, may become birefringent, with all molecules aligned; the form may become non-spherical in consequence and the increased order is shown also by a decrease in viscosity. The contents are often thixotropic, changing between sol and gel with minimal energy change. Clearly the total extent of the resemblances to living cells is very great.

It may seem more difficult to understand how coacervates of gelatin can form from solutions as dilute as 0·001 per cent than to accept the fact of their subsequent absorption of other materials. The droplets come to contain virtually all the gelatin from that dilute solution in fact. This initial segregation of the main component is probably related to the phenomenon of *entmischung*, a tendency for 93 per cent or more of the protein in dilute solutions to accumulate at the air–water interface, the main difference being that the accumulation occurs at spontaneously forming water–water interfaces. However, entmischung is a slow process, requiring 24 hours or so to reach equilibrium, whereas coacervates form instantaneously on mixing appropriate proteins at reasonably *high* concentrations. It seems that there is a strong tendency for proteins to accumulate at interfaces and also to form their own interfaces, and that both

must have contributed to the origin of living cells. These would seem to be inevitable consequences of protein synthesis, which itself seems inevitable under terrestrial conditions.

Fox (1960), who demonstrated the relative spontaneity of peptide formation from a mixture of amino acids (p. 512), also produced coacervate-like spherules, 1·5–3μ in size (Fig. 56), from hot saturated solutions of his proteinoids in saline. The size of these bodies varied with the sodium chloride concentration and they were extremely stable. Somewhat similar bodies, called *protomorphs*, have been prepared from microsomal material (McClure and Roberts, 1958), a natural association of a number of components. In the laboratory nucleic acids, polysaccharides, phosphatides and other lipids have been found to coacervate with protein, and microsomal material may contain all of these and more. It has been suggested (Oparin, 1957a, pp. 317–18) that all organelles are coacervate forms and in a very broad sense this may be true, if the emphasis is on stable segregates, but it is more useful to think of chromosomes, myofibrils and other fibrous structures as solid crystalline forms rather than as liquids. More problematical is the endoplasmic reticulum of the cytoplasm; this may be a coacervate structure, in which there is much membrane and little interior (Scarth, 1942). Coacervates of protein with lipid have very much the properties of lipoproteins, as obtained from biological membranes.

It is believed that coacervates occur free in the ocean depths (Oparin, 1957a, p. 320), which is relevant to the spontaneous origin of living cells. If such bodies appeared spontaneously anywhere else on Earth today they would be rapidly ingested by existing micro-organisms.

15.6 Denaturation of Proteins

The term denaturation is itself a recognition of the uniqueness of proteins. Their biological activity depends on the preservation of a subtle 'natural' structure which is very labile and reactive and therefore all too easily destroyed (Putnam, 1953a). The study of denaturation therefore should be particularly rewarding for the present purpose. It is not an easy study, however; almost every significant change in conditions, physical or chemical, tends to cause denaturation, so that there is little specific relationship between particular causal agents and effects. It is possible to induce partial and reversible denaturation but not easy to control it at this sub-threshold level. Cytochrome *c*, as already noted, is one of the few proteins which shows some recovery of activity even after heat-denaturation at 100°C. In general denaturation is 'all-or-none' and the actual change not very amenable to study. Moreover it is by no means certain that denaturation is always the same phenomenon, just as the

biological function of proteins itself is very variable; destruction of this does not necessarily always involve the same kind of structural change. In some cases the loss of biological activity has been the only available criterion and the structural changes are unknown. The denaturation of globular proteins usually leads to their precipitation, with increase in effective volume and viscosity, whereas the fibrous proteins often become more soluble and less viscous. Further, some globular proteins are inactivated far short of precipitation, whereas others are still active after precipitation. Some are still active after their antigenicity, the hallmark of biological specificity, has been destroyed.

There are other variables, also. Some proteins are denatured by conditions which are normal for others; most proteins are denatured by acids whereas pepsin normally acts in hydrochloric acid, at pH 1–2, and it is denatured in normal fluids, around pH 7·0, even at room temperature. They vary considerably in their susceptibility to denaturation; ribonuclease and lysozyme can be boiled for some time at pH 2–3 without losing much of their activity, but most proteins would be completely denatured by this. The change is probably to some extent specific to each, and the definition of denaturation by Neurath *et al.* (1944), as any change short of opening the peptide bonds, is collective rather than specific. It is not absolutely certain that denaturation is always restricted to secondary and higher grades of structure, but none of the many agents which denature serum albumin releases any new terminal amino or carboxyl groups, and the molecular weight is usually unchanged. The *enthalpy* of denaturation, i.e. the heat liberated at constant pressure, is low (Florkin, 1960) and this indicates that covalent bonds, such as the peptide bond, are not involved.

Heat, acids, alkalis and the poisonous inorganic ions are the best-known denaturing agents. Among the many others which will denature, ionising radiations are biologically interesting because of the regular exposure of living organisms to ultraviolet, and recently to radioactive elements. At the other end of the spectrum, ultrasonic vibrations and sometimes even mechanical shaking are effective. This illustrates the great sensitivity of many proteins, which is further demonstrated by instances of denaturation merely by the protein being spread at a surface; not all proteins are denatured by this process and some antigens and antibodies (p. 270) retain their biological activity. Some dissociate into sub-units when spread, but others associate further. A natural example of sensitivity to mechanical damage is provided by the plasmodium of Mycetozoa; this is killed by being forced through a mesh with pores as large as 200μ. If allowed to flow spontaneously, however, it passes through an 11μ mesh unharmed.

Among denaturant chemicals there are many organic compounds, of which urea is the simplest, the most 'physiological' yet the most potent

in high concentration. This and most organic denaturants appear to act mainly by breaking hydrogen bonds and stealing them. Hydrogen bonds are concerned only in the higher grades of structure so that this further confirms the view that denaturation affects these grades only. Hydrogen bonds are weak enough to be broken by spreading the protein at air–water interfaces. At the same time they are individually rather non-specific and unimportant, and some proteins, such as haemocyanin, can be broken down into sub-units without loss of activity or denaturation; indeed a reversible change in polymer size may be part of their normal physiological behaviour. In some cases new intramolecular bonds are formed in the process of denaturation (Anderson, 1957), giving an artificial tertiary structure, and this is probably due mainly to hydrogen bonds.

Hydrogen bonds are involved also in the structural relationship of protein molecules to those of their aqueous medium, in what may be called their quaternary structure. It is found that dehydrating agents also are among the potent denaturants, so that this quaternary structure is an essential part of the 'nature' of proteins. The undoing of secondary and tertiary structures certainly is a major feature of denaturation in some cases but often mesoform or other colloid properties are affected, depending primarily on the quaternary relationship. The water may be more critical than the protein itself, as one might have concluded in Chapter 2. Ionising radiations denature by two distinct mechanisms, one directly on the protein and the other through ionising the water; as already seen, ionisation affects such properties as hydrogen-bond formation. Sugars and sugar alcohols protect proteins from denaturation by reducing the necessary amount and activity of the water associated with them; presumably some of the hydrogen bonds of water with protein are replaced with sugar–protein bonds of the same type, the larger size of the sugar molecule being the essential protection. This no doubt has its role in the making of preserves. Other proteins also are often protective, for the same reason, and so the mere concentration of a single species of protein is protective (Cohn and Edsall, 1943, p. 571). This sensitivity due to water no doubt provides the rationale for desiccation in the ripening of seeds and spores—as one of the most important, and now dangerous, aspects of the activity of water; reduction in weight by desiccation, and other minor considerations, perhaps also are significant. As already seen, seeds may retain vitality in this condition for centuries and artificially dried mammalian proteins retain full activity for forty years or more (Keilin, 1953a; 1959). Dried protein will then stand heating at 100°C. Of course desiccation must be carefully controlled or the protein is denatured by that alone; this is analogous to the mechanical denaturation of the Mycetozoa, speed being the danger, as Graham emphasised. Vertebrate blood has

enough protein to be very stable *in vitro* without further treatment other than to avoid infection and activation by agents such as heat and light.

Denaturation is accompanied by increased reactivity in virtually all the amino acid side chain groups of the protein and this might be taken to mean that it affects mainly the bonds previously formed by these groups. This is not necessarily true, however, since there are other ways in which the groups could have been masked in the natural protein and it is known that in some cases, at any rate, few of these *prototropic* groups are directly affected in denaturation (Putnam, 1953a). At the same time some ionic groups, whether of end or side chain positions, are involved in the process, to judge from the powerful denaturant action of the long chain anions of the anionic detergents. The action is closely proportional to their cation-binding energy, i.e. their acidity; the potency of acids and alkalis in general confirms this. Somewhat surprisingly, however, the detergents protect proteins from denaturation if they are used in low concentration, and they also protect against denaturation by acids weaker than themselves. A further surprise is that they protect against heat-denaturation also, so that ionic groups may play a part in all forms of denaturation.

In vivo the nucleic acids and the fatty acids may protect in this way. At low protein concentrations the C_{12} fatty acids are the most effective but at higher concentrations C_7 and C_8 members. This may be the significance of the lower fatty acids in milk, and particularly caprylic acid (Table 18); for this acid protection is maximal at a ratio of nine molecules per molecule of protein. Nucleic acids have a particularly strong stabilising effect, which may be attributed to their poly-anion nature. They are able to restore the structure and activity of ribonuclease, denatured previously by urea (Anfinsen, 1959). Apart from the interest in a substrate regenerating its own enzyme in this case there is the further complication that urea is primarily a hydrogen-bond agent. However, urea, and also other similar hydrogen-bonding denaturants, such as guanidine, increase the optical activity of protein solutions in the same way as agents known to unmask side chains, so that they may affect ionic groups also: even *their* action may not be simple and clear-cut.

An astounding feature of the heat-denaturation of proteins is its high temperature coefficient, Q_{10}—the ratio of its speeds at temperatures 10° apart. For most biochemical and chemical processes this is between 2 and 3, over the physiological range, tending towards the lower figure at the upper end of the range. For heat-denaturation, however, it is as high as 700. Consequently at one temperature it is almost infinitesimally slow whereas a few degrees higher it is almost instantaneous, as we know from studying eggs in the frying pan. There are two ways of interpreting

this. In the first place it implies an exceptionally large heat of activation, which in turn indicates that proteins are really extremely stable at physiological temperatures. This is the more surprising because denaturation is strongly exothermic and the change involves a great increase in entropy. It is a particularly striking manifestation of the metastability of biological materials, the activation energy being not so very much less than the energy released in the subsequent process of denaturation. The position is often represented pictorially as a very high hill between two valleys of much the same height, in either of which the system rests with considerable inertia. It is thus evident why denaturation is reversible up to the level of the intervening ridge but so irreversible beyond this point. In addition, and once more paradoxically, the heat of activation of the reverse reaction is negative and under certain conditions renaturation therefore occurs spontaneously. The example of cytochrome *c* has been mentioned; others are trypsin after heat-denaturation, chymotrypsin and a trypsin-inhibitor, all enzyme proteins.

The second way of viewing this high Q_{10} of heat-denaturation is from the biological aspect, of natural selection. It seems possible that there has been selection in favour of proteins which are very heat-stable within the temperature range normally encountered, and which denature abnormally slowly within this range, the discrepancy necessarily increasing towards the upper limit of the range. Selection will not ensure an extension of the property beyond the normal range so that at the upper limit the denaturation response climbs steeply to the value which is spontaneous for that temperature, and so the Q_{10} is fantastically high. In thermophilic organisms, for which the 'normal' temperature range extends to 70° or even 80°C, selection for stability up to those limits has been achieved. The validity of this second view is strengthened by the fact that the temperature coefficient of denaturation by any other agent remains orthodox.

If the view is correct then the interest from the thermodynamic aspect is in the way natural selection has used the energy situation. A high activation energy implies very strong stabilising intermolecular forces. In addition the many intramolecular bonds tend to be mutually sustaining and stabilising (Putnam, 1953a). Another way of saying much the same thing is that large molecules of this kind have many possible configurations of roughly equal energy-content, so that they readily shuttle between these in response to disturbing factors, and are not easily disorganised completely. This is shown for instance in thixotropic, dilatant and other changes which occur, with little change in entropy. The *configurational entropy* of the molecule is therefore said to be large. Kirkwood (1957) visualises the stability-through-variability of the molecule as due to the large number of unoccupied proton-acceptor sites on the typical protein

molecule. It may be compared with stabilisation through molecular resonance.

There is much that is thermodynamically anomalous in heat-denaturation. At moderately high concentrations of pepsin (0·5 per cent) its heat-denaturation behaves as a first-order process, i.e. as though a single reactive group were the pacemaker, but at 0·004 per cent it behaves as a fifth-order process and the heat of activation is three times as high—namely 147 kcal/g-equivalent. At the lower concentration, perhaps, a number of the relevant groups are no longer near enough to react spontaneously.

Those 'all-or-none' changes which are significant physiologically, for instance nerve-conduction, resemble heat-denaturation rather closely in certain important ways. They are graded responses up to a critical threshold strength of stimulus, beyond which they become almost explosive, as heat-denaturation does. The *status quo* is restored by a further sequence of events and not by simple reversal of the impulse-generation, which is as irreversible as denaturation. The perfection of these responses to supraliminal, significant intensities of stimulus is no doubt largely the result of natural selection and it seems equally plausible that there has been selection in favour of proteins which do not denature below the ceiling temperature.

Pressure tends to counteract the effect of heat on denaturation and this might be taken to imply that there is a type of denaturation specific to high temperature, but denaturation by any means can lead to an increase in volume (p. 238), and pressure may be expected to oppose denaturation by all agents.

As already indicated, the Q_{10} for denaturation by agents other than heat is not anomalously high, but of course it is possible to test these agents only below the range over which heat-denaturation shows the anomalous value. For denaturation by surface-spreading the Q_{10} is virtually unity, which implies that it is essentially a simple physical process. That for urea-denaturation however is actually negative over a certain temperature range and so also is anomalous, but in the opposite direction. For the urea-denaturation of TMV the Q_{10} varies curiously with temperature, passing through a minimum as the temperature is increased; for most chemical reactions, by contrast, it is fairly constant over the terrestrial range, or falls steadily with temperature increase.

Denaturation, therefore, is a phenomenon as complex and variable as the structure of the material concerned. Among proteins collectively, if not individually, it probably affects all constructional features except the primary peptide bond itself. It seems certain that the various proteins denature in rather manifold ways because of the varying importance

of the different constructional features, and it is probable that the various denaturing agents have quite distinct actions on the same protein. Further testing of these two generalisations should throw more light on denaturation and on the variety of structure it concerns.

The study to date has shown one more fact of great general importance, namely that proteins are the only group of biological materials which have a 'nature' in this sense: they are the unique living material. Nucleic acids (p. 362), which rank next to the proteins, if not above them, in biological importance, do show certain changes similar to denaturation, but as a group they resemble rather the carbohydrates and lipids, being unaffected by the milder treatments which denature proteins, and being simply hydrolysed at their primary, covalent linkages by stronger agents. Nucleic acids are broken down into nucleotides (p. 355). In the nucleoproteins they have very much the character of coenzymes to a protein apoenzyme.

15.7 Amino Acid Composition of Proteins

The properties of proteins so far considered have been related to general structural features, of which the possession of side chains is one, rather than to the specific properties of the individual side chains. This is natural, since they were mainly general physical properties rather than chemical actions. It has been seen that some structural features in fact do depend on amino acid composition, and then usually on high proportions of particular amino acids such as glycine, alanine and serine in silk-fibroin, glycine and proline in collagen and cystine in keratin. Chemical activity depends much more, but by no means entirely, on amino acid composition, and usually demands a richer variety of the amino acids. An exception to this rule is the high content of the diamino amino acids, and particularly arginine, in protamins: these amino acids can be regarded as chemically balancing, if not actually neutralising, the acidity of the nucleic acids. Most of the chemically active proteins which have been analysed, however (Table 27), are found to contain most of the common amino acids in significant percentage (Synge, 1957), so that it is not easy to relate activity to particular amino acids, or, on the other hand, to say if any of them are irrelevant for particular actions.

One approach to this fundamentally important problem, through the growing body of available data on gross amino acid composition (Table 27), is to find out which amino acids are always relatively abundant in a particular functional group of proteins and reciprocally in which types of protein each amino acid is most abundant. The latter is the most profitable since the absolute amount of some amino acids is always relatively low, or relatively high, and in the first instance it is easiest to

interpret variations in amount than the general level. Not enough proteins in each functional group have been analysed to make good use of the former method. It can be said without much hesitation that leucine is one of the most abundant of the amino acids in hormones; that serine, threonine, glutamic acid and aspartic acid are generally abundant in enzymes; that all three diamino acids are common in the haem-proteins; and that the sulphur-containing amino acids are relatively abundant in albumins, as well as in the keratins which were once classified as 'albuminoid'. Beyond this, however, it is necessary to specify and qualify. There appear to be some amino acids which are abundant only in certain members of a functional group of proteins, and some which are abundant sporadically in many types of protein. Equally, each type of protein may require a number of different amino acids for its chemical activity; this is certainly true of lysozyme and crotoxin, which belong to two different types of defensive protein. Ten years ago the prospects of explaining function in terms of amino acid composition were not very bright (Bailey, 1950; Porter, 1953) and they are still not good. It is worth looking at Table 27 in more detail, however.

It will be seen that apart from hydroxyproline, which is present only in collagen and casein, and cysteine which is often estimated as cystine, no amino acids are at all commonly unrecorded. Moreover each is present in a fairly narrow range of concentrations, the most abundant being the dicarboxylic and diamino amino acids, glycine, alanine and the branched chain group, i.e. all the members of two of the major groups of amino acids recognised in Chapter 13. The sulphur–amino acids and the aromatic group are fairly uniformly present in low percentage. A high percentage of one automatically depresses that of all the others so that the figures can be referred to as 'ratios'.

The average amount of glycine in proteins is 6·3 per cent, so that it is unusually abundant, not only in *silk-fibroin* (43·6 per cent) and collagen (27·2 per cent) but also in another structural protein, *elastin* (28·2 per cent). By contrast its ratio is low in the other main group of structural proteins, the keratin–myosin–fibrinogen group (p. 220). Glycine therefore is associated with the more soluble, glutinous, plastic and elastic of the structural proteins. It is absent from *zein* and some other proteins of plant seeds and scarce in others, as well as in *casein*, an analogous food-protein for animals; this may be merely because glycine is synthesised so easily by the young organism. Its ratio is below average in a large number of chemically active proteins, ribonuclease and most enzymes, protamins, albumins and most hormones. This is to be expected since it has no active side chain. Where it *is* chemically active, as in detoxication, it acts as a free amino acid, and uses the active group of its α–complex.

TABLE 27. AMINO ACID COMPOSITION OF REPRESENTATIVE SELECTION OF PROTEINS

Protein	Amino acids																			
	gly	ala	val	leu	ileu	pro	HOpro	phe	tyr	try	ser	thr	cys *i*	cys *s*	met	arg	his	lys	asp	glu
Salmine	3·0	1·1	3·1	0	1·6	5·8	0	0	0	0	9·1	0	0	0	0	85·2	0	0	0	0
Protamine (bird)	3·7	1·9	1·4	–	–	3·9	0	–	5·0	0	7·1	1·5	0	0	0	58·0	1·4	0	0·7	1·2
Histone (rat liver)	5·8	8·1	6·4	10·1	5·7	2·7	0	2·8	5·4	0	5·4	6·5	0	0	1·6	15·9	2·9	11·5	7·0	10·4
Histone (calf liver)	5·8	7·6	5·5	9·1	4·6	3·4	0	3·5	3·9	0	4·1	6·4	0	0	0·9	14·8	2·3	11·7	5·5	10·3
Histone (calf thymus)	5·1	6·9	3·2	5·2	20·5	4·0	0	4·1	3·3	0	4·7	4·8	0	0	0	17·4	2·7	10·2	5·7	4·3
Histone 1 (thymus)	5·5	8·1	5·6	9·0	4·7	4·2	0	3·2	3·0	0	4·4	5·6	0·8	0	0·9	12·1	2·2	9·4	6·6	11·0
Histone (bird red cell)	5·8	8·6	6·1	10·0	6·4	4·1	0	3·6	3·9	0	6·3	5·3	0	0	0	14·7	2·3	12·1	6·1	11·2
Histone (rat sarcoma)	6·7	5·4	7·8	5·1	17·6	3·6	0	5·1	2·6	0	5·3	6·8	0	0	0	16·3	2·3	11·9	6·8	3·4
Myoglobin (whale)	4·0	8·5	3·7	18·0	0	2·4	0	4·0	2·1	2·6	2·7	4·3	0	0	1·9	2·2	11·1	15·9	6·4	17·4
Myoglobin (horse)	5·9	8·0	4·1	16·8	–	3·3	0	5·1	2·4	2·3	3·5	4·6	0	0	1·7	2·2	8·5	15·5	8·2	16·5
Haemoglobin (horse)	5·6	7·4	9·1	15·4	0	3·9	0	7·7	3·0	1·7	5·8	4·4	0·5	0·6	1·0	3·7	8·7	8·5	10·6	8·5
Serum albumin (ox)	1·8	6·3	5·9	12·3	2·6	4·8	0	6·6	5·1	0·7	4·2	5·8	5·7	0·3	0·8	5·9	4·0	12·8	10·9	16·5
Serum albumin (man)	1·6		7·7	11·0	1·7	5·1	0	7·8	4·7	0·2	3·3	4·6	5·6	0·7	1·3	6·2	3·5	12·3	9·0	17·0
Ovalbumin	3·1	6·7	7·1	9·2	7·0	3·6	0	7·7	3·7	1·2	8·2	4·0	0·5	1·4	5·2	5·7	2·4	6·3	9·3	16·5
Conalbumin (hen)	5·7	4·4	8·2	8·8	5·0	4·9	0	5·7	4·6	3·0	6·3	5·9	3·8	–	2·0	7·6	2·6	10·0	13·3	11·9
γ–Globulin (man)	4·2	–	9·7	9·3	2·7	8·1	0	4·6	6·8	2·9	11·4	8·4	2·4	0·7	1·1	4·8	2·5	8·1	8·8	11·8
β–Lactoglobulin	1·5	7·1	5·7	15·5	5·9	5·3	0	3·9	3·7	1·9	4·1	5·2	2·3	1·1	3·2	2·9	1·6	11·3	11·5	19·1
Ovomucoid	3·8	2·3	6·0	5·1	1·4	2·7	0	2·9	3·2	0·3	4·2	5·5	6·7	–	1·0	3·7	2·2	6·0	13·0	6·5
Edestin (hemp seed)	0	4·3	6·5	7·5	4·7	4·3	0	5·5	4·3	1·5	6·4	3·9	0·9	0·5	2·4	16·7	2·5	2·4	12·0	20·7
Coconut globulin	–	–	5·9	7·2	4·4	–	–	5·1	–	–	–	4·1	–	–	–	16·7	1·5	4·4	–	–
Gourd seed globulin	–	–	6·6	8·2	4·7	–	–	7·1	4·5	1·8	–	2·9		1·1	2·5	16·5	2·2	2·9	–	–
Phaseolin (bean, glob.)	–	–	6·0	10·5	6·7	–	–	8·0	–	–	–	4·2	–	–	–	6·0	2·2	7·2	–	–
Arachin (peanut, glob.)	–	–	4·9	7·6	4·5	–	–	7·0	5·7	0·7	2·3	2·9		1·5	0·7	13·6	2·2	2·7	–	–
Conarachin	–	–	3·7	6·6	4·0	–	–	4·3	2·9	0·9	1·8	2·0		2·9	2·1	16·5	2·1	4·7	–	–
Cotton seed globulin	–	–	6·1	7·1	4·2	–	–	8·1	–	–	–	4·0	–	–	–	14·7	3·4	4·2	–	–
Tobacco globulin	–	–	6·7	10·5	5·3	–	–	5·7	4·1	1·5	–	4·2		1·1	2·2	16·1	2·2	1·6	–	–
Glycinin (soya bean)	–	–	4·6	8·1	5·8	–	–	5·8	3·7	–	–	3·0	1·2	0·3	2·6	7·9	2·0	6·9	–	–
Gliadin (wheat, prolamin)	–	2·1	2·7	11·9		13·6	0	6·4	3·2	6·6	4·9	2·1		2·58	1·7	2·7	1·8	0·7	1·3	45·7
Zein (maize, prolamin)	0	10·5	4·0	21·1	5·0	10·5	0	7·3	5·3	0·2	7·1	3·5	0·8	–	2·4	1·7	1·3	0	4·6	26·9
Gelatin	26·9	9·3	3·3	3·4	1·8	14·8	14·5	2·6	1·0	0	3·2	2·2	0	0	0·9	8·6	0·7	4·6	6·7	11·2
Collagen	27·2	9·5	3·4	5·6		15·1	14·0	2·5	1·0	0	3·4	2·3	0	0	0·8	8·6	0·7	4·5	6·3	11·3
Achilles tendon	24·8	8·7	2·9	3·5	1·9	10·5	–	2·5	0·9	0	3·2	2·5	0·3	0	0·9	8·0	0·9	3·5	6·9	10·4
Yellow tendon	–	–	13·6	7·6	4·3	12·6	–	4·3	1·7	–	–	1·5	0·7	–	0·3	3·1	0·3	1·3	2·9	5·0
Elastin	28·2	6·2	13·8	7·3	3·4	15·6	0	4·8	1·9	0	1·0	1·1	0·6	–	0	1·1	0	0·5	0·6	3·3
Silk-fibroin	43·6	29·7	3·6	0·9	1·1	0·7	0	3·4	12·8	0	16·2	1·6	0	0	0	1·1	0·4	0·7	2·8	2·2
Keratin (hair)	0	0	5·9	8·3	4·7	9·6	0	2·7	3·5	0	7·6	6·3	14·4	0	0·5	10·9	1·1	3·8	8·0	17·9
Keratin (wool)	6·5	4·1	4·6	11·3	0	9·5	0	3·7	4·7	1·8	10·0	6·4	11·9	0	0·7	10·4	1·1	2·8	7·2	14·1
Keratin (feather)	–	–	8·3	8·0	6·0	8·8	0	5·2	2·2	7·5	10·2	4·4	8·2	0	0·5	7·5	0·4	1·3	7·0	9·7
Myosin (rabbit, skel. M)	1·9	6·5	2·6	15·6		1·9	0	4·3	3·4	0·8	4·3	5·1	1·4	0	3·4	7·4	2·4	11·9	8·9	22·1
Tropomyosin (rabbit, M)	0·4	8·8	3·1	15·6		1·3	0	4·6	3·1	0	4·4	2·9	0·8	0	2·8	7·8	0·9	15·7	9·1	32·9
Fibrinogen (man)	5·6	3·7	4·1	7·1	4·8	5·7	0	4·6	5·5	3·3	7·0	6·1	2·3	0·4	2·6	7·8	2·6	9·2	13·1	14·5
Fibrin (man)	–	–	–	7·1	–	–	–	6·0	5·8	3·2	9·8	6·5	2·4	–	2·6	7·9	2·8	8·8	–	–
Ribonuclease	1·3	0	7·3	0	3·1	3·6	0	3·6	7·9	0	12·0	9·0	6·5	0·6	4·4	5·2	4·2	10·4	14·2	13·0
Phosphorylase (rabbit)	3·8	4·9	7·3	10·5	6·5	4·7	0	6·2	5·9	2·0	3·0	4·2	0·4	0	2·7	11·6	3·3	7·2	9·3	13·4
Pepsin	6·4	0	7·1	10·4	10·8	5·0	0	6·4	8·5	2·4	12·2	9·6	1·6	0·5	1·7	1·0	0·9	0·9	16·0	11·9
Chymotrypsinogen	6·3	5·8	10·3	8·9	4·9	3·8	0	3·8	2·7	5·6	10·9	10·7	3·3	1·1	1·1	2·7	1·2	7·7	10·9	7·4
Aldolase (zymohexase)	5·6	8·6	7·4	11·5	7·9	5·7	0	3·1	5·3	2·3	7·3	7·5	1·1	0	1·2	6·3	4·2	9·5	9·7	11·4
Triose-P deHase	6·1	6·7	12·0	6·8	9·1	3·7	0	5·6	4·6	2·1	7·5	7·3	1·1	0	2·7	5·2	5·0	9·4	12·4	4·8
Lysozyme	5·7	5·8	4·8	6·9	5·2	1·4	0	3·1	3·6	10·6	6·7	5·5	6·8	0	2·1	12·7	1·0	5·7	18·2	4·3
Avidin (egg-white)	4·6	–	4·2	4·9	5·5	1·6	0	5·9	0·9	5·4	4·5	10·5	0·5	–	1·4	6·5	1·0	6·2	9·7	6·6
Insulin (ox)	4·3	4·5	7·8	13·2	2·8	2·5	0	8·1	12·5	0	5·2	2·1	12·5	0	0	3·1	5·2	2·5	6·8	18·6
ACTH (sheep)	8·0	–	3·4	7·8	3·1	8·2	0	4·0	2·4	0	6·0	3·2	7·2	0	1·9	8·7	1·3	5·0	6·7	15·6
FSH (sheep)	2·1	–	5·8	9·2	3·3	5·2	0	5·8	3·8	0	3·2	4·7	5·4	0	1·0	5·3	3·7	11·1	9·3	13·4
Prolactin (sheep)	4·0	–	5·9	12·5	7·2	6·2	0	4·1	4·7	1·2	6·5	4·8	3·1	0	3·6	8·6	4·5	5·3	11·6	14·1
APGH (ox)	3·8	–	3·9	12·1	4·0	3·4	0	7·9	5·2	0·8	5·7	6·1	2·5	0	2·9	9·1	2·7	7·1	9·0	13·0
Thyroglobulin (ox)	3·7	7·4	1·5	12·8	–	–	–	6·7	3·1	2·1	10·8	–	3·6	–	1·3	12·7	2·2	3·4	–	–
Casein	2·0	3·2	7·2	9·2	6·1	10·6	–	5·0	6·3	1·2	6·3	4·9	0·3	–	2·8	4·1	3·1	8·2	7·1	22·4
Tobacco mosaic virus	1·4	4·1	7·8	8·0	5·7	4·9	–	7·5	3·4	1·9	6·0	8·4	0·6	–	0	8·8	0·0	1·3	11·7	9·9
Yeast proteins	–	–	5·0	7·3	6·0	–	–	4·4	3·7	1·4	–	5·5	1·0	–	2·0	5·4	2·9	7·6	–	–

The distribution of alanine, also with a short and inactive side chain, is very similar to that of glycine. Its average percentage is 5·1, and it is much more abundant than this in silk-fibroin and collagen (gelatin). It is fairly abundant in elastin, absent from fibrin and ribonuclease, and below average in keratin, protamins, casein, seed-proteins and most hormones. There is considerable justification for considering glycine and alanine as functional analogues, therefore. At the same time alanine has certain peculiarities, the amount being as much as 10 per cent in zein and above average in albumins and haem-proteins. This justifies the view (p. 197) that no two amino acids are simple alternatives. In the haem-proteins glycine is below average and the sum glycine + alanine is about average; this may be significant, i.e. there may be no special requirement for this type of amino acid but a definite disadvantage in glycine.

Serine (6·0 per cent) and threonine (4·5 per cent) form a similar pair, usually distributed together, but sometimes the one acting as sole hydroxyamino acid. There is 16 per cent of serine in silk-fibroin and 13 per cent in *silk-sericin*, and there is 10 per cent in both fibrin and keratin, of the keratin–myosin–fibrinogen group, which contain so little glycine and alanine. In reciprocal contrast to the latter, serine is below average in collagen and elastin. In collagen, of course, hydroxyproline is present, as a special hydroxyamino acid. Serine is readily phosphorylated at its hydroxyl group and this may be the clue to its abundance in *protamins* where the ratios of glycine and alanine are low: it may link the protein to nucleic acids through the phosphate groups of the latter. Phosphoserine is characteristic of secreted proteins, such as milk casein (Anfinsen, 1959, p. 207), and this is probably relevant to the role of phosphatases in secretory activities; it is certainly noteworthy that by contrast there is relatively little serine in the seed-proteins. It is also abundant in triose, phosphate, dehydrogenase, aldolase (zymohexase), ribonuclease, pepsin, chymotrypsin and many other enzymes, in thyroglobulin and in the γ–globulins of the serum, associated with immunological activity. In contrast to glycine and alanine, therefore, serine is a chemically active amino acid, probably as phosphoserine.

Threonine, like serine, is more abundant than average in keratins and fibrins, but there is little in silk-fibroin, which contains so much serine; this confirms the view (p. 216) that the C_1 side chain of serine is critical. The ratio of threonine, like that of serine, is low in collagen, elastin and the seed-proteins. Hydroxyproline therefore appears to have dispensed with the need for either of these two more common hydroxyamino acids, in collagen, and the hydroxyproline relationship to proline may be comparable to that between serine and alanine in silk-fibroin. There is of course a certain amount of various hydroxyamino acids in collagen

(Gross, 1961). Threonine is abundant in most enzymes analysed but also in *avidin*, the antibiotic protein of egg-white, which has little serine. The two amino acids are real analogues, therefore, but not always alternative ones.

The sulphur-containing amino acids are the least abundant of all the common α–amino acids except tryptophan. The average for cysteine + cystine is 3·1 per cent, so that the 14·4 per cent in hair-keratin is very outstanding. The amount is unusually high also in insulin, where again they act mainly in a structural capacity. They are virtually absent from silk and collagen where the —S—S— bonds would probably give too much rigidity or insolubility. Urease, ribonuclease, lysozyme, chymotrypsin, ovomucoid, serum albumin and conalbumin, and the hormones ACTH and FSH, are metabolically active proteins with a high content of these amino acids. There are forty of their residues in the molecule of urease. Either the —S—S— link, or its reversible redox change to ⁻SH, is found to be essential for the action of chymotrypsin, lysozyme, β–amylase, crotoxin, FSH, ICSH, lactogenic hormone, and insulin but not for pancreatic amylase, trypsin, ocytocin and the parathyroid hormone (Table 28). Its significance in insulin therefore is not entirely structural (p. 213). The reduced, SH, state is essential for activity in papain, but not in pancreatic amylase or in ovalbumin and solanine.

Methionine (average 1·6 per cent) is virtually absent from keratin and insulin, although it is the main source of sulphur for the other two sulphur amino acids. In other proteins, however, its distribution does run parallel to that of cysteine. It is relatively abundant in egg-albumin, ribonuclease and some other enzymes, moderately so in fibrinogen and fibrin, tropomyosin, myosin, seed-proteins, β–lactoglobulin, prolactin and APGH, and below average in silk, collagen, protamins and histones. In the globulins and seed-proteins there is more methionine than cystine; here it may be acting as a reserve of both sulphur and methyl groups (p. 178), and similarly in egg-albumin and lactoglobulin. It is evidently a chemically active rather than a structural amino acid.

The distribution of the branched chain aliphatic amino acids bears out the idea that they form a group of analogues, either occurring together with much the same relative abundance or alternatively one being exploited to the complete exclusion of the others. As already noted, their average ratio is high, and leucine (8·1 per cent) is the third most abundant amino acid (second if the protamins be excluded); they are rather more common than glycine and alanine, which they resemble in being mainly structural in function. They are abundant in some enzymes and other globular proteins, but this may still be as structural units; their shape is probably ideal for the complex tangle of the haem-proteins (Kendrew, 1959). The elastic property of the isoprene-like side chain, confirmed by

their abundance in elastin, has been adequately stressed (p. 179); valine is the most common of the three in elastin and it is C_5, like isoprene. In prolamins such as the *gliadin* of wheat and maize the elastic property becomes visco-elastic, in the presence of water, and accounts for the peculiar properties of dough; here leucine is the most common. All are absent from silk and collagen where straight, inextensible fibres are ideal.

Since the general level of all three is high it is probably more true to say that they vary reciprocally from this level than in parallel. Apart from the examples mentioned, valine is the preferred member in chymotrypsin and triosephosphate dehydrogenase, where leucine is in average or lower amount, whereas leucine is particularly abundant in *gluten* (25 per cent), myoglobin (17–18 per cent), myosin (15·6 per cent), lactoglobulin (15·5 per cent), thyroglobulin (12·8 per cent) and wool (11·3 per cent), where valine has only 4·5, 3·9, 5·7, 2·6, 1·5 and 4·6 per cent respectively. Relative to their averages there is more leucine than valine in most hormones and the leucine content is highest in those enzymes which have only moderate amounts of valine.

The amount of isoleucine runs parallel to that of leucine in enzymes but is often low in hormones. In calf *histone* 20·5 per cent of the amino acid residues are isoleucine, compared with 5·2 for leucine and 3·2 for valine, but in other histones leucine is the dominant member of the group. It seems likely, therefore, that a relatively small advantage of one or another can tilt the ratios heavily in favour of that one. The precise significance of isoleucine is not clear but valine seems to be 'preferred' for fibrous, and leucine for globular, proteins.

The group may be abundant in seed-proteins because the plant embryo finds as much difficulty in synthesising them as animals do. However, they are also surprisingly soluble amino acids, considering their hydrocarbon side chain, and in addition leucine has good alcohol-solubility. Proteins containing these amino acids therefore are suitable, if not ideal, for transport; the high leucine content of hormones supports the idea of a positive virtue for this purpose.

Proline, notwithstanding its origin from glutamic acid (p. 181), can be regarded structurally as a ring-closed norvaline, and like the branched chain group themselves it is abundant in elastin, gliadin and other seed-proteins. Its pyrrolidine ring might be expected to complement the lightly tangling effect of the branched chains and so to contribute to elasticity. In collagen, however, it probably gives a much more regular, and therefore rigid, interlocking between the three peptide chains of the triplex (Piez and Gross, 1960). It may confer resilience on peptide chains, through compensatory reversals in sense of coiling (p. 180), and there seems good reason why it is present in elastin but not in fibrin, and in keratin, which is quite

TABLE 28. ESSENTIAL AND NON-ESSENTIAL RADICALS IN VARIOUS PROTEINS

Protein	Radicals found essential	Radicals found not to be essential
Viruses		
TMV	ϕ	NH_2
Toxins		
Diphtheria	NH_2	
Scarlet fever	ϕ	
Crotoxin	NH_2, OH, COOH, $-SS-$, ϕ	
Antigens		
Ovalbumin	ϕ	SH
Serum albumin		ϕ
Antibodies		
Rabbit antiovalbumin		$-SS-$
Horse antiovalbumin	NH_2	
Horse anti-pneumococcus polysaccharide	NH_2	
Enzymes		
Pancreatic amylase	NH_2	SH, $-SS-$, ϕ
Barley β–amylase	$-SS-$, ϕ	NH_2
Chymotrypsin	$-SS-$, ϕ	NH_2
Trypsin	amide NH_2, indolyl	COOH, $-SS-$, ϕ, imidazole, amino
Pepsin	ϕ	NH_2
Papain	SH, ϕ	NH_2
Solanine	ϕ	NH_2, SH
Phosphatase	NH_2	
Ribonuclease	NH_2	
Lysozyme	NH_2, COOH, OH, $-SS-$, guanidyl	
Ovomucoid (trypsin inhibitor)	COOH, ϕ, guanidyl	
Hormones		
Pitocin (ocytocin)		$-SS-$
ACTH	NH_2, $-\phi$	
Lactogenic hormone	NH_2, $-SS-$	
FSH, ICSH	NH_2, $-SS-$	
Pregnant mare serum	NH_2, $-SS-$	
Prolan	ϕ	NH_2
Insulin	COOH, $-SS-$, ϕ	NH_2, OH, guanidyl
Parathormone	COOH, NH_2	$-SS-$

extensible, but not in silk. Its exceptionally high solubility may explain its abundance in casein, the blood γ–globulins, and other soluble transported proteins such as the hormones. By contrast it is not very abundant in enzymes. It probably contributes to the solubility of glue on warming, the swelling of collagen fibres, and the mobilisation for transport of the seed-proteins. Its unusually low melting point (p. 180) may already influence the properties of glue at temperatures exploited commercially.

Hydroxyproline does not appear to be mainly, and certainly not generally, a functional analogue of proline. In collagen it probably bonds one triplex to another (Piez and Gross, 1960) whereas proline forms bonds within each triplex. In casein it may have much the same significance as serine and threonine. Hydroxylysine also is present, just as all the hydroxyamino acids are present in the collagens of animals, collectively. In casein they may be actual precursors of collagen, which is synthesised in quantity by the suckling infant. It is the percentage of hydroxyproline, in particular, which determines the temperature at which collagen becomes denatured (Schmidt, 1959).

The dicarboxylic amino acids, glutamic and aspartic acids, are abundant in most proteins other than the protamins, where they would tend to nullify the effect of the diamino acids. Glutamic acid is the most common of all amino acids (12·9 per cent), while aspartic (7·8 per cent) is as high as fourth in order of abundance. They resemble the branched chain group not only in their high general level of abundance but even in detailed distribution; they are abundant in the cereal proteins, β–lactoglobulin, insulin and myosin. There is a possible correlation also with the distribution of proline, however, since they are abundant in casein as well as in gliadin, and scarce in silk and elastin. In keratin and insulin a possible correlation with cysteine is indicated, but of course they are so widely distributed that some correlation with all other amino acids is almost inevitable. A positive as opposed to a casual correlation of this breadth could be credited in view of the extensive role of the two carboxyamino acids in the biosynthesis of the others (p. 181), and this is no doubt the main significance of their particular abundance in the food-proteins, though they can scarcely be used in this way in the mature, functioning proteins. However, the high percentage in tropomyosin, as compared with myosin, may have this significance, since there is some evidence that tropomyosin is an immature form of myosin (Bailey, 1955). Glutamic acid is most abundant in gliadin, tropomyosin, casein, myosin, edestin, β–lactoglobulin, serum albumin, ovalbumin, myoglobin, insulin and keratin, in that descending order; apart from myosin and the last three all might be considered food- or embryo-proteins, to be used for further biosynthesis.

Relative to its average, aspartic acid is more abundant than glutamic in ribonuclease, pepsin, chymotrypsin, triosephosphate dehydrogenase, fibrinogen, avidin and conalbumin, where glutamic itself is not below average, and in lysozyme and ovomucoid where it is. Aspartic therefore seems more important than glutamic in enzymes; on the other hand, relative to its average again, it is less abundant than glutamic in all hormones, most histones, albumins, globulins, collagens, myosins, keratins and all food-proteins. There is so little aspartic in gliadin, which contains so much glutamic, that the two again (p. 180) appear to be exclusive alternatives on occasion. This asymmetry in gliadin is the more surprising because aspartic is the functional member of the pair in plants (p. 181); perhaps this is a form of division of labour, aspartic being a functional amino acid predominantly, in plants, and glutamic a biosynthetic amino acid. The contrast in their distribution to enzymes and hormones perhaps also fits this picture, with relevance to animals and plants.

Free ω–carboxyl groups were found essential (Table 28) for the activity of five out of six proteins examined (Putnam, 1953b), lysozyme and ovomucoid which have a high aspartic content, insulin with a high glutamic content, and crotoxin and the parathormone. This carboxyl group, therefore, has mainly chemical functions.

The hexone bases, arginine and lysine, also are present in relatively high concentration in nearly all proteins. After glutamic acid, in fact, arginine is the most generally abundant of the amino acids in proteins (9·5 per cent). This seems to stress the importance of ω–carboxyl and amino groups. The third hexone base, histidine, has no free second amino group and is one of the rarer amino acids. In some ways (p. 184) it is more related to the aromatic amino acids, though for some purposes certainly its position is in the present group; in the haem-proteins, in particular, the three hexone bases seem to be interchangeable (p. 182). All three are scarce in silk-fibroin, elastin, and gliadin, where one or more of the glycine, branched chain, dicarboxy, and proline groups are abundant.

Arginine accounts for as much as 88 per cent of some protamins, 15–17 per cent of many seed-proteins, other than the prolamins, 12–16 per cent of histones and about 12 per cent of thyroglobulin and some other globulins, of phosphorylase and of lysozyme. There are moderate amounts in the keratin–myosin–fibrinogen group of the fibrous proteins.

Lysine (6·1 per cent) is absent from protamins and invertebrate haemoglobins, where arginine is so abundant, and reciprocally is abundant in vertebrate myoglobin and haemoglobin where arginine is scanty. This is perhaps the best example of alternative amino acid exploitation within the same group of proteins; at present the decision in favour of the one or the other amino acid seems fortuitous. In other histones the two are

about equally abundant, relative to their respective averages. Lysine is of course absolutely less abundant than arginine in most proteins, including hormones and some enzymes, but has a higher ratio in some enzymes and in a number of albumins and globulins, in ovomucoid, casein, myosin and fibrinogen. There seems to be some parallel therefore to the difference between the distributions of glutamic and aspartic acids, arginine resembling glutamic acid and lysine aspartic acid.

Histidine (average 2·4 per cent) is particularly abundant in the myo- and haemoglobins of vertebrates (8·5–11·1 per cent), where it is directly correlated with lysine and inversely with arginine. In vertebrate haemoglobin, and also in insulin (5·7 per cent), it is absolutely more abundant than lysine, but in the haemoglobin of cyclostomes the emphasis is on lysine alone. Histidine is known to play both a structural and a chemical role in the haemoglobin of gnathostomes; the iron atom is bonded to two imidazole (histidine) rings as well as to the four haem-nitrogen atoms and on the other hand histidine contributes to the buffering of carbon dioxide produced in respiration:

$$RNH_2 + CO_2 \rightarrow R{\cdot}NH{\cdot}COOH$$

When present the other hexone bases may contribute to this action. It is noteworthy that although the *carbamino* compound thus formed accounts for only 6 of the 56 volumes per cent of carbon dioxide carried in venous blood, about a third of it is decomposed in the lungs, so that it contributes about a third of the total carbon dioxide eliminated there, and is a major factor in the exchange. Histidine is also relatively abundant in some enzymes and hormones. The imidazole group is found essential for the activity of chymotrypsin, in which it confers special bonding properties on the hydroxyl of serine. It also accelerates the action of D–amino acid oxidase.

Free amino groups have been shown essential (Table 28) for the activity of pancreatic amylase, lysozyme, phosphatase, ribonuclease, crotoxin and diphtheria toxin, some antibodies, and the hormones ACTH, FSH, ICSH, lactotropin and parathormone. They are not essential for that of β–amylase, trypsin, chymotrypsin, pepsin, papain, solanine, ovomucoid, and the hormones prolan, ocytocin, and insulin. The distribution of arginine and lysine tends to reflect this, but histidine in peptide combination has no free amino group, and shows no clear correlation.

While the phenolic amino acids are among the less abundant they are rarely absent except from some protamins. Phenylalanine (average 4·7 per cent) has usually the higher and more constant ratio, and this is possibly a further indication (p. 186) that it is the precursor of tyrosine (average 4·0 per cent). Only in special cases, e.g. silk-fibroin (12·8 per cent) and insulin (12·5 per cent), has tyrosine a higher ratio than phenyl-

alanine. Both have their highest ratios in relatively highly differentiated proteins, in accordance with Gudernatsch's generalisation (p. 191), but it does not seem to be mainly structural differentiation which concerns these amino acids as might have been anticipated. Silk-fibroin and the *sclerotin* of arthropods are the only common structural proteins with very large ratios. In sclerotin they have been largely converted to quinones, in any case, and this is probably usual wherever they have a structural function. The unmodified amino acids are most abundant in enzymes, hormones, albumins and globulins, mainly chemically active proteins. Phenylalanine has rather high ratios also in a number of food- and seed-proteins, and tyrosine also is abundant in some of these, e.g. casein and gliadin; in such proteins they may be as usual reserve amino acids for the young individual. Except in silk-fibroin and a few other proteins, the two amino acids are distributed in parallel, but phenylalanine has the higher values in seed-proteins and tyrosine in enzymes, which again fits the idea that the latter is the main functional member and phenylalanine the precursor.

Apart from trypsin there are few enzymes containing these amino acids which do not require the phenol ring specifically for their activity (Porter, 1953), and in fact it is essential for the activity of virtually all other types of proteins in which it occurs (Putnam, 1953b)—hormones, antibodies, antigens, inhibitors of enzymes, toxins and virus nucleo-proteins. It is not clear how the ring affects all these different activities so vitally, but this is probably a strong indication that it depends on the general activating action of the resonant structure.

The average amount of tryptophan in proteins, 1·4 per cent, is lower than that of any amino acid except hydroxyproline. It is much more widely distributed than the latter, but even so the low average is partly due to its complete absence, sporadically. At the other extreme, therefore, there is as much as 10·6 per cent in lysozyme, 7·5 per cent in feather keratin, 6·6 per cent in gliadin, 5·6 per cent in chymotrypsin, 5·4 per cent in avidin and 5·9 per cent in casein. Like the phenolic amino acids, it is more abundant in the metabolically active enzymes than in the structural proteins, with the exception of keratin and fibrin. There are as many as twenty-six tryptophan residues in the molecule of lactic dehydrogenase. Also like the phenolic group it is mainly a 'differentiation' amino acid but yet is abundant in many seed-proteins—probably for the same reason. It is absent or scanty in the following structural proteins: collagen, elastin, fibrin and myosin, and also in the enzyme ribonuclease. There is little in most hormones of animals, in contrast to the phenolic group, and little in protamins and in histones other than the haem-proteins.

The indole ring is essential for the activity of trypsin, in contrast to the phenol ring, which is essential for peptic activity. There is good reason to

believe that the indole ring, like that of the phenolic group, is active while in peptide combination. Their contribution to the fluorescence of proteins is one piece of evidence (but see p. 194).

15.7.1 Conclusions

It is certainly true (Porter, 1953), as indicated at the beginning of the section, that at present it is far from possible to explain the properties of all proteins in terms of their gross amino acid composition, but on the other hand there is plenty of promise that the problem is difficult rather than irrational. Some correlations are good and serve a similar purpose to solved clues in a crossword puzzle. There is the correlation of valine with elastic properties and glycine, alanine and serine with the silk-fibroin type of fibre. Equally clear is the dependence of the strength and insolubility of keratin on the cystine —S—S— bond, and it is possible to attribute most of the properties of collagen to their high content of proline and hydroxyproline. Again the strength and hardness of the exoskeleton of insects are largely explained by the conversion of phenolic amino acids to quinones, whether *in situ* or as free amino acids. It is also noteworthy that each different type of skeletal protein has a different preponderant group of amino acids, which is the alternative way of looking at the same question.

Part of the difficulty of the main problem is that each protein and each amino acid may perform several functions, so that the complete tally of amino acids in all known proteins inevitably appears confusing until it is broken down into its separate components. The basis for the structural properties is probably the most evident, and it was to try and reveal more about the chemical properties that the review of this section was undertaken. The basis for the high alkalinity of the protamins and histones has long been evident and also the biological reason for this. It is also clear that the redox properties of cysteine are important for the activity of many enzymes. Serine and threonine, probably as phosphate esters, are essential for the action of many enzymes, and so are the dicarboxylic acids, perhaps because of their ability to build extensive resonant systems of hydrogen-bonded protein (Leach and Lindley, 1952). Tyrosine also is of general importance for enzyme action, and to some extent histidine and tryptophan, all three in virtue of their fully resonant unsaturated ring structures. Amino groups and the guanidyl group are clearly important for the action of some enzymes and other proteins, and in fact every polar side chain is widely used; the non-polar side chains appear equally certainly unimportant for chemical activities. Table 28 illustrates the specific requirements for the polar and resonant side chains in various groups of chemically active proteins.

There are some further possible generalisations. Thus proline and the leucines are abundant in those proteins which must be highly soluble and easily transported, in particular hormones, seed-proteins and serum-proteins, and the solubility properties of these amino acids are a suitable basis for those of the complete proteins. The solubility of these amino acids in ethanol may improve the transport of the proteins through the lipid component of biological membranes. It is doubtful if these amino acids, leucine especially, play a part in the actual function of these proteins. It also seems certain that a number of the amino acids of the food-proteins are significant only as material for definitive proteins, later. The composition of the food-proteins is perplexing if viewed in any other light. This is particularly true of the high content of the 'differentiation' amino acids, in proteins which are anything but differentiated at that time. More light is thrown on the problem of analogous amino acids; for the most part they appear to be alternatives for some purposes but each uniquely specialised for other functions. For instance one is usually the more important in enzymes and its analogue in hormones; this applies to aspartic acid as compared with glutamic acid, lysine compared with arginine, and tyrosine with phenylalanine. In lysozyme there is 18·2 per cent of aspartic acid compared with 4·3 per cent for glutamic, whereas for insulin the ratios are 6·8/18·6. The differentiation affects other classes of protein also and there is 45·7 per cent of glutamic acid in gliadin compared with 1·3 per cent for aspartic. In elastin there is 13·8 per cent of valine and only 7·3 per cent of leucine, whereas in zein there is 21·1 per cent leucine and only 4·0 per cent valine.

The distribution of the hexone bases in the different haem-proteins seems to imply that the only significant differentiation is taxonomic and therefore may be merely fortuitous. In some other cases, also, there is as yet no evident functional basis for differentiation between analogues, but of course there may be such a basis, in the case of the hexone bases and in general. Taxonomic differences usually arise as significant functional differences.

These are the tentative generalisations emerging from the variations in the percentage of each amino acid in the different proteins. The potentially more fundamental problem of the significance of the average percentage of each remains obscure. It may be nothing more than a statistical average, reflecting a general tendency for some amino acids to perform more, or more important, functions than others, but alternatively it is just possible that most proteins are built up of a 'general foundation' of most of the amino acids in their average percentages, together with additional amounts of those amino acids responsible for the special functions of each protein. A few proteins, such as the protamins, have very few amino

acids, in unusual ratios, and here there can be no question of such a foundation, but most other proteins seem to have a far greater variety of amino acids than could possibly be required for one, or even for a number, of special functions. The peptide form, and other structural features, and no doubt certain functions, such as metabolic turnover, are more or less common to all proteins and this might be the rationale for the foundation component. It is significant that the degree of correlation in amino acid composition between the various proteins which have been analysed is surprisingly high (Fox and Homeyer, 1955). In fact the degree of correlation is proportional to the degree of phylogenetic relationship, and all the proteins of one species show a very high degree of correlation, irrespective of special function (Fox, 1953). This raises the further possibility that the common component is inherited from a common, primitive, all-purpose protein; an all-purpose molecule would be expected to contain all the amino acids in their average amounts. The protamins would then be regarded as the most extreme divergents away from the general protein. It is not implied that the general foundation retained by many proteins is in the nature of an evolutionary vestige, though this is possible: if it was once functional it may still be so.

It has been seen that all proteins have much in common in their structural mechanics; no doubt every α–amino acid forms the same peptide bond-unit (with the exception of proline), and the same contribution to the backbone of the peptide, but protein structure also depends on specific side chains, and the same amino acids, in much the same proportions, are likely to be required for the molecular mechanics of all proteins. Glycine may function quite generally as a spacer, or *em* (p. 174), and serine and alanine also act as spacers (p. 216). The cystine —S—S— bridges have a rather similar function, for instance in insulin (Anfinsen, 1959). Other general mechanical functions are known, such as the compensatory reversal in coiling of the peptide helix permitted by proline residues, and no doubt molecular mechanics alone could account for much of the general foundation. It seems equally probable that there are other common requirements, both physical and chemical. Whether the foundation component is a discrete entity or not, however, it is obviously profitable to be able to distinguish between common and specific components of each protein and to relate them to their functions.

The possibility that a foundation component may be scattered throughout the molecule leads to the consideration of specific amino acid sequence in proteins, the acme of uniqueness among proteins, to which almost any problem about them eventually leads. An analysis of the ratios of the various letters in a written message is of small value compared with a knowledge of their sequence, including the spacing into words. A different

arrangement of a group of letters makes an entirely different word and there are known parallels to this in the proteins. Two of the *licheniformin* group of antibiotics are identical in gross amino acid composition, yet they have very different antibiotic activities and toxicities to mice. Again the various antibody proteins of a particular animal may be indistinguishable in gross amino acid composition from the inert γ–globulins of the serum (Porter, 1953).

15.8 The Sequence of Amino Acids in Proteins

Where careful analyses have been repeated the gross amino acid composition of purified proteins has proved constant. This of course could be a purely statistical consistency and the individual molecules might vary considerably. There is evidence that they do (Fox, 1953), to some extent, but the majority of molecules of a purified protein move together in an electrical field, indicating a common size, shape and charge, at least, and they are therefore likely to contain the same amino acids, in the same proportions. Obviously there are various possibilities about this but if consistency of size is established it is very unlikely that there would be variation in composition. By an extension of this argument it is unlikely that the actual arrangement of amino acids would vary in molecules from the same cell-type, with the same gross amino acid complement, and analyses of the actual sequence (Sanger, 1952) confirm this. The sequence is repeatable, and no doubt genetically informed.

This is desirable even if for biosynthetical reasons alone: it is easier to build one pattern than a number—unless the pattern can be completely random. At the same time, however, the probability is that the specific sequence in each protein is also significant functionally or taxonomically. It seems unlikely that precise genetic codes for such large, heterogeneous molecules could be evolved unless the precise amino acid sequence were significant in one or both of these ways. There is the further implication that each protein is as unique in significance as it is in amino acid sequence. There is scarcely a more fundamental problem about proteins, or indeed about life, than to explain the amino acid sequences of the actual proteins which have been evolved.

In proteins which contain very few of the twenty different amino acids it is reasonable to expect some very simple pattern of sequence, and difficult to envisage any other arrangement which would be meaningful. For instance in protamins having 80 per cent arginine the expected arrangement is . . . XAAAAXAAAAXAAAA . . . , where X is one of the other amino acids present, serine, threonine, proline, valine or leucine (Felix *et al.*, 1956). The alternative of an unbroken succession of

arginine residues throughout 80 per cent of the molecule is implausible and would probably be unstable. In silk-fibroin, with 40 per cent glycine and 40 per cent serine and alanine the simplest possible sequence . . . GAGSXGAGSX . . . not only seems the only reasonable one but has definite mechanical considerations in its favour (p. 216). For the crystalline parts of collagen a regular repetition of the sequence glycine, proline and hydroxyproline has been suggested (Crick and Rich, 1955), but for the whole molecule of collagen the pattern must be more complex (Table 27).

At one time it seemed possible that there might be some such simple repeat-pattern in most proteins. Bergmann and Niemann (1937; 1938) believed that the number of amino acids in all proteins conformed to the formula $2^n \times 3^m$, where n and m are integers, variable between proteins. This seemed to imply that all were constructed on some simple pattern of condensation of amino acids in groups of two or three, followed by further polymerisation of these units, and of their products, always in multiples of 2 or 3. In view of the variety of amino acids in most proteins it was evident that the initial dipeptides and tripeptides could not be all alike, but in principle fairly simple and quite regular repeat-patterns would be possible. However, this formula has not proved generally valid and, as already indicated, those proteins which have been analysed show a virtually aperiodic molecule, very few dipeptide sequences, and still fewer longer sequences, recurring in it (Gamow *et al.*, 1956).

An intermediate grade of regularity was detected in some proteins (Sorn, 1957). In *glucagon*, the second hormone of the islets of the pancreas, either aspartic or glutamic acid was found to recur at fairly regular intervals, separated by 6, 6, 5, 4 and 4 other residues, while either serine or threonine recurred with a 3, 3, 3, 5 pattern of separation. In ribonuclease, lysine recurred with a separation of 6, 6, 6, and 5 other amino acids. More complete knowledge of the sequence in these proteins has shown the pattern to be less regular, though there does seem to be a significant deviation from complete aperiodicity and this may help in the elucidation of the plan of more complex sequences. It is reasonable to suppose that the latter do have functional and phylogenetic plans, and that they are not quite 'random' or *aperiodic* in the sense (Gamow *et al.*, 1956) that no dipeptide sequences occur more frequently than any others. Of course in the sense that a very specific sequence is synthesised in every molecule of a protein species, it is the very antithesis of random.

For the purpose of the present thesis the most important question is whether the amino acid sequence is always mainly functional in significance, whatever the degree of aperiodicity in its pattern. The most evident alternative is that it is taxonomically, i.e. phylogenetically, based, and has no necessary functional significance, but, as suggested above, it is also possible

that it has both, since phylogenetic changes must always have been fully functional.

Some evidence for a functional basis has been given in the cases of the licheniformins and mammalian antibodies (p. 257), and there is also the case of human haemoglobin, in which a change of one amino acid in a repeat period of 290, the exchange of valine for glutamic acid, causes the sickle-cell abnormality (Hunt and Ingram, 1957). The abnormal haemoglobin has only 1/100 of the solubility of the normal pigment and only a fraction of its oxygen-carrying power. The haemoglobin of normal mammalian embryos differs from that of adults in a similar way by only one or two amino acids, and again is quite distinct in properties. Of course the change is not really in sequence of amino acids alone: there is an actual replacement of one or two amino acids. The same difficulty complicates the contrary evidence; for instance it is claimed that eleven of the thirty-nine amino acids of the hormone ACTH, a sequence at one end of the molecule in fact, are not essential for its activity and may be removed without effect (Oncley, 1959b). This is such a large fraction of the molecule that it must be taken to mean that even the gross amino acid composition may be unimportant, whatever the significance of sequence. In papain as many as eighty amino acids at the amino end of the molecule are said not to be essential for its proteolytic function. It is of course possible that these are simply carrier portions for a small prosthetic peptide, or that they have other functions, but in any case they provide little evidence concerning sequence alone.

The same difficulty arises in connection with the alternative possibility that sequence may be phylogenetically determined: there are now many known examples of amino acid sequences which vary genetically and taxonomically, but in almost every case both sequences and actual amino acid personnel vary. However, these differences seem more significant when we also know the sequence, and can say that a new amino acid, X, actually occupies the position of Y in the protein of a related organism, or of a related tissue. If we find such differences they must in a sense be more functionally significant than if Y were merely eliminated and X incorporated elsewhere, or anywhere, in the molecule. What is known of these taxonomic variations in homologous proteins tends to indicate that sequences can be changed without affecting biological function very much, since in general homologous proteins remain functionally analogous also. Thus the insulins from different mammals vary in one or more of a particular sequence of three amino acids, where we find:

ox . . . ala–ser–val . . .
sheep . . . ala–gly–val . . .

horse . . . thr–gly–ileu . . .
pig . . . thr–ser–ileu . . .
whale . . . thr–ser–ileu . . .

These differences may also have a functional basis, since the hormone is likely to be working under slightly different conditions in the different mammals, but they could give the impression of ringing all the possible changes on a small group of amino acids simply for the sake of producing variants and so incompatibility with related groups of animals. Under such conditions the sequence in the whale might well by chance come to be identical with that in the pig, which is actually more closely related to the ox and sheep. It will be noted that replacements are partly by analogous amino acids, and in fact each of these insulins is quite highly active in the other mammals.

Similar variations are found in the posterior pituitary hormones, ocytocin and vasopressin, but are more impressive here because there are only nine amino acids in the molecule. In ox vasopressin the sequence is HOOCcys–tyr–phe–glu(NH_2)–asp(NH_2)–cys–pro–arg–glyNH_2, and in the pig arginine is replaced by the analogous lysine. It is interesting to find that the second posterior pituitary hormone, ocytocin, differs from vasopressin again in this position, but also in a second amino acid; the two hormones no doubt have evolved from the same peptide but in this case the two amino acid replacements constitute the only basis for the functional difference between them, and at least it must be concluded that taxonomic and functional differences have the same kind of basis.

Other functional differences seem to involve much greater molecular differences. For instance the pig has two melanocyte-stimulating hormones, α–MSH with 13 amino acids and β–MSH with 18 amino acids, and even so 4 of the 13 differ from those in the corresponding sequence in β–MSH (Oncley, 1959b); the functional difference here is very small, certainly not greater than that between vasopressin and ocytocin. Part of the molecule of MSH is almost identical with a part of another pituitary hormone, ACTH, but here both molecular and functional differences are greater than between the two MSHs. In ACTH all taxonomic variations appear to affect only amino acids in the functionally dispensable part of the molecule, and it is possible that the functional difference between the two posterior pituitary hormones depends only on the second amino acid position, the taxonomically varying position again being in a functionally indifferent part of the molecule.

It remains true that taxonomic and functional differences between proteins appear to be based on very similar differences in amino acid personnel and sequences, and that both are functional in the general

sense of having been naturally selected. This would make it easier to understand how the sequence cysS–A–B–cysS–his–thr–val–glu occurs in the cytochromes of organisms as diverse as mammals, birds, fish, insects and yeasts—the amino acids A, B being the only two which vary: during the long histories of these groups this sequence has remained functionally desirable in all.

There is of course a good *a priori* reason for expecting the amino acid sequence to be functional, since steric and other spatial factors must greatly affect co-operation between amino acids. However, proximity can also be attained by bending and folding of the molecule, in appropriate ways, just as, on a larger scale, chromosomal loci far apart along the chromosome can synergise in the interphase nucleus (Pontecorvo, 1959). For this reason the sequence of amino acids is probably not all-important, and this may explain why there is at present no compelling evidence that it is.

A further question of some importance is whether the general evolution of amino acid sequences has been from the simple to the complex or conversely from the randomly aperiodic to the regularly repeating pattern. A random aperiodic order might be the simplest spontaneous mode of condensation, but a specific, invariable aperiodicity is likely to require the maximal organisation. Today simple repeat patterns are relatively rare, and are largely confined to skeletal proteins, but this does not necessarily decide the issue either way. Taxonomic divergence, for which there is much evidence, is likely to have increased the heterogeneity of proteins rather than conversely and in general it seems probable that the uniqueness of proteins, individually, has progressively increased.

15.9 Proteins as Catalysts

The most numerous and important group of proteins with a chemical action are the enzymes, or biological catalysts. Some reactions, particular steps of various metabolic pathways, have a rapid spontaneous rate *in vivo* and are not catalysed, but the vast majority are, including such as

$$CO_2 + H_2O \rightarrow H_2CO_3$$

which has a high spontaneous rate, but one not high enough for respiratory purposes in a very active animal. Table 29 gives some idea of the variety of reactions speeded by enzymes in living organisms. There could scarcely be a system more important, whether in numbers or in mode of action, and probably every one is a protein, simple or conjugated. From time to time enzymes of great potency but containing very little protein have been obtained from living material, but always the small amount

TABLE 29. CLASSIFICATION OF ENZYMES
(largely based on Hoffmann-Ostenhof, 1953)

1	Enzymes catalysing double-decomposition reactions of the type $A+B \rightleftharpoons C+D$
1.1	Hydrolases: $XY + H_2O \rightleftharpoons XOH + HY$
1.11	Esterases
	Carbonic esterases (lipases, etc.)
	Phosphoesterases
	Sulphuric esterases
	Nucleases
1.12	Glycosidases
	Oligosaccharases (maltase, sucrase, etc.)
	Polysaccharases (amylases, cellulases, chitinases, etc.)
	Thioglycosidases
1.13	Amidases
	Aminases
	Acylamidases
	Cycloamidases
	Amidinases
	Glycosaminases
1.14	Peptidases
	Endopeptidases (pepsin, trypsin, chymotrypsin, papain, cathepsins)
	Exopeptidases (aminopeptidases, carboxypeptidases, dipeptidases)
1.15	Phosphoamidases
1.16	Polyphosphatases (pyrophosphatases, etc.)
1.17	Halogenases
1.18	Carbon–sulphur hydrolases
	Thioetherases
	Thioglycosidases
	Thioesterases
1.19	Desmolases (C–C hydrolases)
1.2	Oxidation–reduction enzymes
1.21	Anaerobic transhydrogenases (dehydrogenases)
1.22	Aerobic transhydrogenases (dehydrogenases)
1.23	Oxidases (tyrosinase, polyphenolase, cytochrome oxidase, ascorbic oxidase, etc.)
1.24	Peroxidases
1.25	Catalases
1.26	Transelectronases (cytochromes)
1.3	Other transferases
1.31	Transcarboxylases
1.32	Transformylases
1.33	Transmethylases
1.34	Transacetylases (and other transacylases)
1.35	Transaminases (and transpeptidases, etc.)
1.36	Transphosphatases
1.37	Transsulphurases (rhodanese)
1.38	Transglycosylases (transglycosidases)
1.39	Transadenylases
2	*Lyases and Syntheases*: catalysing reactions of the type $A \rightleftharpoons B + C$

2.1 Anhydratases and Dehydratases
- Carbonic anhydratase ('anhydrase')
- Fumarase
- P–glyceric dehydratase ('enolase')

2.2 C—C lyases (desmolases) and syntheases
- Decarboxylases and carboxylases (free CO_2 removed or added)
- Zymohexase (aldolase)
- Triose–P lyases
- Carboligases: $R'CHO + R''CHO \rightleftharpoons R'CHOH{\cdot}C{:}OR''$

2.3 C—N Lyases and syntheases
- Aspartic deammoniacase (aspartase)

2.4 C—S Lyases and syntheases
- Desulphydrases and sulphydrases ($-SH$)
- Desulphinases and sulphinases ($-SO_2$)
- Sulphoxide lyases

2.5 Hydrogenases ($H_2 \rightleftharpoons 2H$)

2.6 Hydrogenylases ($X + H_2 \rightleftharpoons XH + H$)

3 *Mutases and Isomerases* (including racemases)
- P–triose isomerase (P–glyceraldehyde $\rightleftharpoons$ P–dihydroxyacetone)
- Other aldo–keto isomerases ('oxoisomerase', ribose–5–P isomerase, etc.)
- Aconitase (citric acid $\rightleftharpoons$ *iso*citric acid)
- Gluco–P–mutase (Glucose–1–P $\rightleftharpoons$ glucose–6–P, ribose–1 $\rightleftharpoons$ 5–P mutase, Glyceric–3 $\rightleftharpoons$ 2–P mutase
- Acetaldehyde mutase (Cannizzaro reaction): $2CH_3CHO \rightleftharpoons C_2H_5OH + CH_3COOH$

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of protein has been found essential and the experiment has merely demonstrated the surprising potency of the protein, *apoenzyme*, moiety. Recently (Binkley, 1954) dipeptidase has been found to be most potent with an RNA content much higher than 99 per cent, and this remains a possibly exceptional case, but if so it only serves to emphasise the rule.

The speeding of reactions by enzymes is illustrated by choline-esterase, one molecule of which hydrolyses 2×10^7 molecules of acetyl choline per minute at 22°C, and by catalase which decomposes $2{\cdot}64 \times 10^6$ molecules of hydrogen peroxide per molecule per minute, even at 0°C. This speed is due largely to the protein moiety, though the non-protein, *coenzyme*, moiety alone usually has some catalytic effect and indeed is probably responsible for the specificity of the reaction accelerated. This is illustrated by a number of analyses. Catalase is a haem-protein, that is an iron-porphyran conjugated with protein, and even inorganic iron has some catalytic action on hydrogen peroxide decomposition, raising the very low spontaneous rate to 10^{-5} moles of hydrogen peroxide per second, per g-equivalent of iron. If the iron is combined with a pyrrole (p. 306) the rate is increased by a further considerable factor, and if it is in porphyran form (p. 315) there is a further 1000 × speeding. Association of the iron-porphyran with the specific protein of catalase however now raises the

activity by a further factor of 10^7 (Stern, 1936). Consequently one mg of iron in catalase is as effective as ten tons of inorganic iron (Oparin, 1957a, p. 246). Haemocyanin, the copper-protein oxygen-carrier of some arthropods and molluscs, also has some catalase activity (Ghiretti, 1956), and even here the copper is 10^5 times as active as its inorganic form (Nicolaev, 1957).

Carboxylase, the enzyme which transfers the carboxyl group in metabolic reactions, is a conjugate of the B-vitamin, thiamine (p. 397), with protein, and again simple analogues of the thiamine have some positive catalytic effect, for instance methylamine, CH_3NH_2. If the carboxyl group is added to this, the activity increases twenty times. The completion of pyrimidine (p. 347) and thiazole rings, the essential moieties of thiamine (p. 397), further increases activity very greatly, but again the addition of the protein component increases it by a further factor of 10^4. Sucrase, the enzyme which hydrolyses cane sugar to glucose and fructose, is 10^7 times as efficient as the H^+ ion used for the purpose in the laboratory.

From these examples it seems that the coenzyme determines the type of reaction which is catalysed while the protein contributes the efficiency of speed. In fact it is also as specific to the reaction as the coenzyme, and no other protein can deputise, but this is to be expected; the conditions for speeding each type of reaction are quite specific and no doubt a protein has been modelled to fit each case. In fact the protein is in a sense more specific than the coenzyme: the pyridine coenzyme, NAD (p. 334), dehydrogenates a large number of substrates but there is a specific protein for each. The protein therefore seems to have two quite distinct functions, one quantitative and the other qualitative. In fact it is the protein also which determines the specificity of reaction to one optical isomer, only, of an optically active substrate. Further, there are a considerable number of enzymes which are purely protein, so that they themselves provide the counterpart of the coenzyme, probably through particular amino acids. It is therefore more correct to regard enzymes as proteins which occasionally have a non-protein coenzyme than as a group of catalysts which have acquired a protein as carrier and enhancer. Enzymes are inactivated by heat, and the Q_{10} of this has the same high value as that for the heat-denaturation of proteins.

This is not to minimise the importance of the non-protein moiety in the conjugated protein-enzymes. Frequently this moiety has a radical, or a larger portion of the molecule, in common with the substrate; this is the part which makes direct contact with the latter. It is an essential part of the classical Michaelis theory of enzyme action that an enzyme–substrate complex is formed and the coenzyme is usually the bonding group. A very simple, instructive case is that of α-glucose 1:6-diphosphate as coenzyme

for the isomerisation or 'mutation' of glucose-1-phosphate to glucose-6-phosphate; in the reaction it donates its 1-phosphate group to position 6 of the substrate molecule, which becomes a new coenzyme for the molecule, the original coenzyme becoming the product of the reaction, glucose-6-phosphate! A similar mechanism may be assumed for 2,3-diphosphoglyceric acid which is coenzyme for the isomerisation of 3-phospho- to 2-phosphoglyceric acid. Other examples are uridine diphosphoglucose as coenzyme for galactowaldenase, the enzyme which isomerises α-galactose-1-phosphate to α-glucose-1-phosphate, bile salts as coenzymes for cholesterol esterase (Long, 1961) and choline pyrophosphate for acid and alkaline phosphatases (Kutscher and Sieg, 1950). In other cases there is at least suggestive evidence for such affinity: β-D-glucopyranosidase contains 3 per cent of carbohydrate, while hepatic phosphorylase, which converts glucose-1-phosphate to glycogen, contains phosphate and is inactivated if this is removed (Stetten, 1959). Nucleohistone has adenosine triphosphatase activity and the lipase of the castor oil plant contains so much lipid itself that it is insoluble in water and must be extracted with lipids. A number of the B-vitamin co-transferases provide examples where a small radical, only, is common to coenzyme and substrate, but with essentially the same functional significance: the pyridine and benzpteridine dehydrogenases become reversibly hydrogenated themselves each time the enzyme acts; pyridoxin becomes aminated during transamination, folic acid becomes formylated during transformylation, and so on. It would therefore be surprising, perhaps, if proteolytic enzymes did not have amino acids as coenzymes, and it may be significant that pepsin, which acts mainly on the bonds between phenolic and dicarboxylic acids, has a high content of tyrosine and glutamic acid.

In some conjugated enzyme-proteins there are no significant groupings common to both coenzyme and substrate but it is still with the coenzyme that the latter makes its bond; the haem-proteins, catalysing the transfer of oxygen or of electrons, are an example. When catalase bonds with its substrate, hydrogen peroxide, there is a change in the absorption spectrum of the haem group, showing that it has been chemically changed.

If this is so general, what is the action of the protein part, to explain its effect on speed, its determination of specificity to substrate, and its invariable presence? The last implies some action common to all enzymes while the other two, or at least the substrate-specificity, imply an action which varies with each enzyme. In support of a generic action there is the demonstration by Leach and Lindley (1952) of very consistent, simple ratios between various groups of amino acids in enzymes, and not in other proteins. Again, there is a common sequence of amino acids, gly–asp–ser–gly at the active site of a number of enzymes of various

functions, trypsin, chymotrypsin, thrombase, choline esterase and phosphoglucomutase. Moreover for chymotrypsin and phosphoglucomutase the next three amino acids in order are glu–val–ala (Frieden, 1959). On the other hand the particular group of side chains essential for the action of different enzymes varies considerably (Table 28), and reflects the specific action of each.

The most probable mode of action, which could explain both generic and specific functions, and also why proteins alone could play this part, is as a resonant network for electron-transfer and activation, a semiconductor or extended oscillator (Evans and Gergely, 1949; Geissman, 1949; Eley, 1962). As already seen (p. 213) proteins can form extensive resonant networks through the numerous peptide bond units, supplemented by hydrogen bonds of secondary and tertiary nature, by resonant rings of the aromatic amino acids, including histidine, and to some extent by other groups, such as the ω–carboxyl of the dicarboxylic amino acids and the guanidine group of arginine. In addition, side chain hydroxyl groups give extra hydrogen bonds (Leach and Lindley, 1952). Molecules as large as this, with resonance throughout large sections, in principle are able to activate virtually all chemical reactions using energy of the low quantal value of visible light and of the chemical bonds formed in photosynthesis; probably only this group of molecules could do so. It is no longer a surprise that the enzyme urease has a molecular weight of 480,000, for, although small itself, urea is a stable molecule and has a high energy of activation. Only three or four active sites on this enzyme molecule are required after activation, for the actual hydrolysis reaction.

The protein might be expected to give actual evidence of the conduction of electrons and energy in some cases, and this is recorded. It might be expected to transfer an electron to or from the coenzyme, permitting it to bond with the substrate, or to this complex to permit its further reaction. In one case in fact it seems to permit the subsequent separation of the substrate (or rather its analogue) once more from the coenzyme. Haemoglobin, which may be considered an enzyme because of its close similarity to the cytochromes, catalase and other enzymes, forms a carbonyl compound, CO·Hb, with carbon monoxide. This complex is decomposed again by ultraviolet light of 280 mμ wavelength, which is the wavelength absorbed by the tryptophan of the globin, the haem itself absorbing at 313–400 mμ. The energy therefore must be absorbed by the tryptophan residue(s) and passed through the *filet* (Winter, 1952) to the CO·Hb link. Catalase, like haemoglobin, has four haem groups per molecule, and if the iron atom of one of these is blocked by poisons, such as cyanide or azide, there is a change of reactivity of all the remaining three. The protein therefore acts as a conductor along many paths, both to and from the pros-

thetic groups. The protein itself can be changed, at points remote from the site of conjugation with the substrate. It is believed that in some cases the coenzyme is completely enclosed in the apoenzyme which therefore must act as intermediary between it and the substrate. In the decomposition of CO·Hb, one molecule of carbon monoxide is released per light quantum absorbed, so that conduction must be 100 per cent efficient and compares with conduction in metals. As in metals, each electron shifts only one step and the one which transfers energy at the end of a filet is not the one which first received energy into the molecule.

It is evident that a hydra-headed protein, one with many prosthetic groups, each specific to a different type of reaction, would be feasible and should have potential advantages, and such enzymes do in fact exist (Dixon, 1951). This may explain the remarkable fact that when all the enzyme-activities of some cells are measured *in vitro*, and the results translated back into total amount of protein, this proves to be much more than the cell is known to contain. The reactions catalysed by a multi-capitate enzyme are sequent members of a particular metabolic pathway and this must greatly speed the whole series. It must also help in solving spatial problems, since if there is a long series of separate enzymes there may be a problem of transporting the substrate from one to the next. Diagrams of metabolic pathways make this transport seem simple, since the arrow symbol for a reaction is taken to imply a movement in space (Needham, 1960c), but in fact spatial problems are serious, as biochemists have long realised (J. Needham, 1936; Peters, 1937). The protein apo-enzyme can simplify this problem even if it is only single-headed, since electron-transfer can deputise for an actual material transport. In the electron-transport sequence of aerobic respiration in living organisms (Fig. 20) the first enzyme receives an electron from hydrogen, converting it to H^+, and the last donates one to oxygen, converting it to O^-. The union of $2H^+$ and O^- to form water is the only other movement necessary. The enzymes themselves are arranged in contact, on the mitochondria (Green, 1962), so that electron transfer can sweep from one to another, along the line.

No doubt there is much more to be discovered about enzymes and spatial problems. For instance myosin is a protein with both structural and enzymic properties, and the enzymic sites are very regularly deployed spatially in the myofilaments. By hydrolysing ATP, and so making energy available at strategic points, this enzyme permits the formation of chemical bonds which are so concerted that they summate into the overt movement of muscular contraction. In this case movement is the teleonomic aim. Somewhat similarly permeases, that is the enzymes which promote the transport of materials across cell membranes, cause an actual movement

of their substrate molecules, and do *not* in fact produce a permanent chemical change in them; a mere electron shift is inadequate here. It has been postulated (Danielli, 1954) that the carrier molecule, which conceivably might be the permease itself, has some kind of cyclic or reciprocating movement across the membrane, carrying the substrate in one direction. This could be compared to the cycle of movements in muscle contraction. Transport of substrate over long distances by overt mechanical means, as in the case of oxygen by carrier pigments, takes this a big step further. In these examples we have an interesting bridge between the biochemical and physiological levels of activity.

While most enzymes speed biological processes it would perhaps be more useful to recognise them as controlling these processes, acceleration being the kind of control which is usually required, but occasionally retardation, and often stabilisation. Some organisms are able to perform the normally explosive reaction $H + H \rightarrow H_2$, but the dissipation of energy is controlled by an enzyme (Bayliss, 1959) and the reaction proceeds at a manageable rate, which also matches that of most other metabolic reactions. The enzyme is therefore a negative catalyst and collectively enzymes are reaction buffers, stabilising metabolism against either extreme in rate. This may be one reason why reactions exploited *in vivo* are not always those which proceed most spontaneously in the test-tube. It is interesting that the ease with which lipases attack a series of substrates is in inverse order to that for non-enzymic catalysts (Scott, 1953).

Enzymes, like other proteins, tend to stabilise substances with which they are conjugated and this is another manifestation of their depressive capacity (Bayliss, 1959, p. 338). In consequence, radicals which would be very unstable if free in solution can be restrained to controllable activity when bound to an enzyme, their excess energy being spread over the whole resonant network. The enzyme-protein can act as a sink of energy, therefore, and reciprocally it can act as a reservoir, giving the whole system inertia and momentum, two outstanding qualities of living systems, as Graham emphasised (p. 225). This tends to reaffirm the view that enzymes are proteins to which non-protein moieties have become attached, rather than vice versa.

There are further properties of enzymes which have the same implication, or at least stress the dominant role of the protein part. Some enzymes, for instance the dehydrogenases and cytochromes of eggs (Bradfield, 1950), are activated merely by stretching the material of the cell. This is reminiscent of piezoelectric changes, the generation of electrical potentials in metals by mechanical forces, and is a further indication of semiconductor properties. Secondly living organisms appear to be able to synthesise an enzyme to deal with any toxic material which may be

inflicted on them (Hoffmann-Ostenhof, 1957), even including newly manufactured molecules. It is perhaps questionable whether these are really new to living organisms, but in any case they do not carry a permanent reserve of constitutive enzymes to deal with every eventuality, and synthesise those for all but the common substrates, only upon actual provocation. Enzyme-induction is a typical piece of epigenesis (Jacob and Monod, 1961) and of protein synthesis, and it would seem almost essential to have a protein for this purpose even if it had not been essential for the subsequent action of the enzyme.

This is perhaps the context in which to consider the phenomenon of *isodynamic* enzymes, that is to say enzymes having the same function but a different amino acid composition. In fact probably every enzyme has its isodynamic counterparts in the different organisms, the difference in amino acid composition increasing with taxonomic distance. If the protein part were required only for the synthesis of the enzyme there would be less cause for surprise in this phenomenon of protein variation than if it is also required for the function of the enzyme; it is difficult to believe that an infinitely variable protein can perform precisely the same catalytic function, in the various organisms, whereas each is known to have its own peculiar genetic codes for biosynthesis. There is another possible explanation of isodynamism, however, namely that only certain portions of the protein are concerned with the catalytic function, the rest having purely taxonomic, genetic significance (p. 259). The example of the cytochromes is relevant here (p. 261): in widely unrelated groups the enzyme has a common short sequence of amino acids, the rest of the molecule being peculiar to each. It is among enzymes, in fact, that the distinction between functional and purely taxonomic amino acid sequences should be most usefully pursued. It may be surmised that in all absolutely isodynamic enzymes, performing the same reaction on the same substrate, there is an amino acid sequence which is identical in all organisms. As already noted (p. 259), large sections of some proteins can be eliminated without affecting their specific action, but one or more small sections may be absolutely specific. Isodynamism does not prove that the whole of the protein part of enzymes is non-specific, or unimportant.

Perhaps the clearest demonstration of the dominant role of proteins in enzymes is the purely protein nature of so many enzymes, in addition to the proteases; in some cases non-protein prosthetic groups may have passed unrecognised as yet, but in others the enzyme has been crystallised and contains only amino acids. Enzymes of this type are mainly hydrolases, and not only of the anhydrides of carboxyl with amino groups. It is now known that the proteases themselves hydrolyse other conjugates, in addition to the peptides (Neurath, 1959), so that it is

possible that a single generic type of bond is attacked by all pure protein enzymes. At the same time hydrolysis and its modifications, such as phosphorolysis, is the most widespread type of reaction in aqueous media, and the various protein enzymes have become further specified as proteases, saccharases, and so on, and some are absolutely specific to a single substrate. The transfer of most radicals other than H^+ and ^-OH involves non-protein prosthetic groups, most of them B-vitamins, but their transfer is usually associated with a hydrolytic type of change and this may be one reason why a protein apoenzyme is always associated with these co-transferases. The hydrolytic action of proteins probably harks back to their fundamental amphoteric, zwitterion nature (p. 165), while the substrate-specificity of each enzyme depends on its particular amino acid sequence and on other constructional features, including its resonance pattern. This does not rule out the probability that proteins can also act as catalysts for non-hydrolytic reactions in virtue of their resonant properties, their amino acid side chain groups and other features. It is probable that the enzymes are at least the most versatile group of proteins.

15.10 Proteins and Immunity Reactions

There could scarcely be more impressive evidence of the uniqueness of proteins than that from the study of immunity reactions, which may be defined as the defence of the body by its own proteins against foreign proteins, *antibodies* against *antigens*. The antibody–antigen reaction is absolutely unique, peculiar to living systems and to proteins among biological materials. Other materials from foreign organisms, including polysaccharides, nucleic acids and other macromolecules, are by comparison relatively harmless and they are dealt with in more orthodox ways, by simple excretion, sometimes preceded by chemical detoxication (Williams, 1959). Only foreign proteins have the violent disturbing effect on metabolism which has come to be called antigenicity, manifested as mysterious protein sicknesses, fevers, allergies, runt diseases and others, and sometimes fatal. Only proteins are used to combat these antigens, probably because they alone are competent. The mode of defence varies with the nature of the antigen and resembles, both in its variety and in the actual types, the denaturation of proteins (p. 237). This also is a process peculiar to proteins and is the simplest way of inactivating them. Free protein antigens are usually precipitated or are chemically inactivated by a type of antibody known as an *antitoxin*, while antigens in organised form, such as living bacteria or foreign cells, are either precipitated by *agglutination* or dissolved by *lysis*. Agglutination usually involves the neutralisation of the surface-charge of the cell, and resembles one type

of denaturation, while lysis is a common mode of denaturation for the fibrous proteins. Antibodies are classified as precipitins, antitoxins, agglutinins and lysins; the last, in particular, work in association with a second component, known as the *complement*, which is also partly, but not entirely, protein.

The whole mechanism therefore is very much a protein 'closed shop', though with certain exceptions and provisos. For instance gelatin and some other proteins are not antigenic, and this may be related to their simple pattern of amino acids (p. 258). Most small peptides and all free amino acids evoke no response: otherwise the heterotrophic mode of nutrition would be impossible. Even some proteins with an aperiodic sequence of amino acids, for instance insulin, are non-antigenic; in this case the reason may be that the molecule is relatively small and varies little between the animals tested. As usual these exceptions provide clues to the understanding of the whole phenomenon.

Reciprocally there is evidence that some non-protein materials *are* antigenic, for instance some lipid–polysaccharide complexes (Boyd, 1947) and even some pure polysaccharides (Howarth and Stacey, 1948). However, as in the case of enzymes, it is difficult to ensure that biological materials are completely freed of protein; on the other hand it would not be conclusive to test, instead, polysaccharides synthesised in the laboratory by simple chemical means, since peptides themselves are non-antigenic when synthesised in this way (Fox, 1960). Certainly these particular non-protein materials rather readily become conjugated with protein, and a host's own protein can become antigenic when in such a conjugated form; this in fact may be the reason for some of the so-called *anaphylactic* responses (Najjar and Robinson, 1959). A similar explanation seems highly probable for the well-established antigenicity of simple phenolic derivatives and even of inorganic iodine. These very readily bond with protein, and form analogues of the aromatic amino acids including the thyronines (p. 186); this is another clue to the general explanation of antigenicity.

Free amino acids are non-antigenic because they can be built into peptide patterns of host-type, to function normally, and it is only completed patterns which are wrong for this purpose which behave antigenically. There appear to be two relevant types of wrong structure: either the amino acids of the protein are in the wrong order or the protein is conjugated with an abnormal prosthetic, or *hapten*, group, whether a simple phenol or iodophenol, or a complex lipopolysaccharide. Most foreign proteins have the wrong amino acid sequence for the host, and as in the case of other groups of metabolic antagonists the most dangerous may be those which differ least, since these compete most effectively with the indigenous metabolite, but only sufficiently to 'jam the works'. It is

therefore not surprising that the proteins of sibs and other near relatives sometimes prove as strongly antigenic as those of distantly related organisms. Only identical twins can safely exchange tissues. Even proteins of another organ of the same body tend to be antigenic and the response to them no doubt helps to prevent one organ from being invaded by the tissues of another, or its function from being disturbed by any vagrant circulating protein originating from another organ. The normal blood-proteins are better tolerated, perhaps, because they circulate throughout life to the whole body. Possibly for the same reason blood can be safely transfused from one individual to another, with only modest precautions concerning a few major antigens of the host's blood cells, and major antibodies of the donor's plasma.

It is probably only under abnormal conditions that one organ is molested by the proteins of another, just as in the case of foreign proteins. Food-proteins are normally broken down far enough to cause no trouble, but occasionally the permeability of the gut-wall is increased and antigenically active peptones enter the circulation. The placenta likewise occasionally has a faulty permeability and the mother is subjected to antigens from her foetus. More common ways of ingress are through the delicate respiratory surfaces, and those of the alimentary canal proximally to the region of digestion, and through wounds.

Living organisms of course may grow and proliferate inside the host they invade, so that the amount of antigenic protein rapidly increases, but even free proteins cause a marked response, if in any significant quantity, and contrast strongly with other materials. The nucleic acids hold an anomalous position among the latter, for they tend to be non-antigenic although they are capable of very important autonomous activity. Since the classical work of Avery *et al.* (1944), it has been known that foreign DNA (p. 347), once inside a host cell, can continue its own genetically specific activity, and even *transform* the host cell towards its own type. The host cell seems to have no defence against this. Virus nucleic acid similarly causes a host cell to produce virus-type protein, as well as more of the specific nucleic acid, and there are other well-known actions of this kind. Bacteriophages invade with virtually pure nucleic acid, leaving their protein outside the host cell, perhaps because it is so antigenic, in contrast to the nucleic acid. The latter carries 'instructions' for the synthesis of more virus-type protein, when the time comes, and by this time there is no longer much to fear from the host's antibody mechanism. This technique of the virus may be the main reason why it has been necessary for the host to develop another defence-mechanism against viruses, involving the production of non-specific agents called *interferons* (Isaacs, 1961), with the added advantage of acting much more promptly than the

typical antibody response. The spermatozoon of higher organisms resembles viruses in invading the ovum with much nucleic acid and little protein.

Virtually everything in the behaviour of antigens and antibodies is relevant to the uniqueness of proteins, and a wealth of further information may be obtained from such works as those of Boyd (1956) and Landsteiner (1962), while the publications of Burnet (1956, 1959, 1960) deal also with recent theory in the subject. For the present purpose interest centres first on the immunological evidence for the uniqueness of proteins among biological materials and secondly on evidence for the uniqueness of each protein among proteins in general. The latter is the highest grade of uniqueness among biological materials, the basis for taxonomic specificity down to the level of the individual organism as well as of functional specificity down to its lowest level. Immunology is unrivalled for the study of this grade of uniqueness and the antigenic criterion of protein specificity is the most useful, if not the most reliable. Antibodies are not merely non-specific protein denaturants: each is highly specific to the particular antigen which evoked it. This opens up a wonderland of speculation, in which there are still very few certain landmarks.

The antibody against a particular protein, *R*, will react similarly but less strongly with a related protein, *R′*, and the intensity of reaction falls progressively with genetic distance between the organisms supplying the test antigens. The same probably applies to degrees of functional difference between antigenic proteins, though this is not so amenable to test; the antibody against fibrinogen cross-reacts with the fibrinogen of various species (Sevag, 1951, p. 50), and this may be quite general. The continuous gradient in strength of reaction with related proteins, as opposed to an 'all-or-none' type of response, is strong evidence that the antibody reacts with a large number of groups on the antigen. These groups might be all alike, and more distantly related proteins possess progressively fewer of them, but since the gradient phenomenon applies whichever antigen is selected to induce the initial antibody it is more likely that they differ qualitatively. Each antigen can be qualitatively unique but not all can be quantitatively outstanding.

It must be pointed out that if an antibody is induced against a laboratory conjugate, *HR*, where *H* is a simple hapten and *R* the protein, then it will likewise react in graded intensity with *H′R*, *H″R*, . . . , where *H′*,*H″*, . . . are progressively more distant analogues of *H* (Tyler, 1955), for instance one of the phenolic series, or the simpler series, phosphonate, arsonate, antimonate, sulphonate, carboxylate. . . . Members of such a series are known to have similar chemical properties, differing mainly quantitatively, so that here the basis for a graded antibody response may be quantitative, in a different sense, but each of the common amino acids is

unique and antigen proteins vary only in their amino acid arrangement. It is features of this arrangement which are likely to be the points of interaction between antigens and antibodies, in the case of the purely protein antigens.

The consequent idea of an antibody which 'fits' extensive parts or even the whole of the antigen molecule, perhaps in somewhat the manner of a die or templet, has been very popular among immunologists, and may be in essence correct, but on the other hand there are indications that in some cases, at least, there is only one point of contact between antigen and antibody. There may be great variation in this respect, but at the present early stage the great need is for working generalisations. In support of the one-point mechanism there is evidence from the actual antibody–antigen reaction (p. 278). In addition it has been found that if two active or *determinant* groups of the antigen are more than 15 Å apart it is necessary to form an antibody against each (Haurowitz, 1960), and if glycyl leucine is made part of an antigen some organisms produce an antibody against the glycine residue only and others against the leucine only (Pressman, 1960). These facts make the immunity reaction appear very similar to a simple detoxication, which is characterised by conjugation or some other type of reaction at a single critically active site on the toxic molecule.

It is difficult to reconcile with the theory of a detailed fitting between antibody and antigen, the discovery that the antibodies produced by a rabbit against antigens from a wide range of sources all contained the same sequence of five amino acids at one end of the molecule, and that this sequence also occurs in the γ–globulins of the plasma (Fox and Foster, 1957, p. 423). It is of course possible that all antibody molecules of a particular individual have a common section of this kind and another section which is highly specific to each antigen, but this work on the rabbit has prompted the alternative suggestion that a single type of protein molecule is used to form all the antibodies of an individual and that only minor changes are necessary to adapt it to each different antigen. This would certainly make the whole phenomenon seem relatively simple.

Also tending to support the one-point, detoxication type of theory is the fact that in conjugated antigens, prepared in the laboratory, there is a close relationship between antigenic power and the chemical reactivity of the hapten. Antigenicity increases with the polarity of the hapten group, electrostatic forces being the most important (Kauzmann, 1959). Arsenic acid is a more potent hapten than the halogens, nitro groups, —CH_3, H_3CO—, and other relatively apolar radicals; these also induce antibodies which are less specific to them, and form weaker bonds with them. The radicals found to be most active antigenically are in general the same as those which determine the normal metabolic activity of the B-vitamins (Marrack, 1950), and probably of metabolites in general.

This shows that antigenicity depends on metabolically active groups and that antibodies are directed particularly against these, but it does not necessarily imply that only one is involved in each case.

When the antigen is a conjugated protein it may well be that the hapten being very active is the main target of the antibody, so giving an approximation to a one-point attack. However, it is striking with what detail the antibody fits this small molecule. It is specific to the optical sense of any asymmetric carbon-atom in the molecule, to any *cis–trans* isomerism and to any *ortho–meta–para* isomerism of phenolic haptens (Pressman, 1960; Haurowitz, 1960). A simple protein antigen possibly has recognition marks more widely distributed through the molecule.

It is important to return to this question of 'recognition' and specificity since this must be considered the essence of the immunity reactions. It is essential to ensure specific destruction of the one foreign protein and to protect the host's own proteins. Occasionally autoimmunisation against its own proteins does occur, but this is probably due to their conjugation with a foreign hapten, which subjects them to a one-point kind of attack against which they have no defence. Any one metabolically active group is useless for immunological recognition since it occurs so widely in all organisms; the minimal requirement for recognition is an array of active groups, such as the side chains of the common amino acids, and this probably demands a many-point relationship between antibody and antigen. Otherwise, of course, a much simpler antibody molecule should suffice.

There are more subtle features of immunological recognition. As already seen the rabbit's fibrinogen will induce it to produce an antibody against the fibrinogens of other mammals, but it does not react with the indigenous fibrinogen, which would amount to autoimmunisation. Again young animals will develop a tolerance to any foreign protein presented before a critical stage of development, and although this protein may be undesirable on functional grounds it is never subsequently attacked; it has ridden in on the bandwagon of the young animal's own proteins which secure immunity in the same mysterious way, at that stage. The animal normally becomes tolerant of all its own and intolerant of all other proteins. Each protein is recognised, challenged or allowed to pass, and where necessary manacled, each by its specific antibody guard. The complete mechanism is much more complex than a simple chemical reaction such as those of detoxication.

A further complication is that the young individual does not begin to produce antibodies until the critical stage of development mentioned, which in fact corresponds to the time when its own proteins begin their definite differentiation and themselves become active antigens. There may be a rational connection here, namely that the body cannot learn to

recognise foreign proteins until it has all its own definitive proteins for comparison. The former may be as deleterious before this time as later, but this is just unfortunate; as already seen they may become permanently tolerated as a result of this misfortune. The rabbit's fibrinogen may be a clue to ways and means in the development of the embryo's immunity mechanism; each of its definitive proteins may induce the formation of an antibody which is active against all possible analogues, the intensity of the reaction being related to the degree of resemblance. At one time the late ontogenetic appearance of the mechanism was thought to be an evolutionary recapitulation, since immunity responses seem less well developed in the lower animals, particularly the lower metazoa. However, it has been known for some time that the Protozoa do have immunity reactions (Reynolds, 1924; Loeb, 1945), and also the sponges (Galtsoff, 1929), and this has recently been shown also for the Coelenterata (Phillips and Yardley, 1960). It is probable that even bacteria produce antibodies and so does the unfertilised egg (Tyler, 1955; Rothschild, 1956). There is merely an ontogenetic hiatus, therefore, which allows the new proteins of the zygote to be differentiated without challenge, to serve as the templets for an appropriate new antibody system.

This possible interpretation of the ontogenetic process could simplify what otherwise appears the most amazing power of the antibody mechanism, that it can produce an effective answer to any foreign protein which might provoke it. If the number and type of antibodies are in fact determined by the individual's own proteins then antibody production is no more (and no less) remarkable than the biosynthesis of the latter. Since each antibody will in fact react in degree with a large group of similar proteins, the total covered by a set of antibodies of this type would be very great. It is known that individuals are unable to produce antibodies against some antigens and at present it would not be possible to assert that any individual effectively copes with more antigens than could be covered by a set of this type.

One reason why some immunologists believe that each individual can produce a separate antibody, 'out of the blue', against any antigen which may appear, is that undoubtedly in many cases there is no great amount of preformed antibody, as would be expected from the above ontogenetic theory; in most cases there is no 'natural' immunity and the antibody is first produced only upon challenge. Most instances of natural immunity are due in fact to early challenge or to passive immunisation by the parent. However, it remains possible that the spontaneous ontogenetic process merely produces the mechanism, and perhaps a small actual sample of each antibody, limited by the need for economy in idle materials. Once challenged, however, a reserve of that antibody may persist through life

and speed up the defence against subsequent encounters, which have now risen from the possible to the probable status.

Those who support the view of a more omnipotent antibody mechanism have tended to mollify its incredibility by attributing to the antigens themselves a suicidally samaritan property of inducing antibodies in their own negative or complementary images, but this antipodism seems biological nonsense, on the grounds that all initiative must come from the host. An alternative possibility, of course, is that since by definition foreign proteins cannot be antigens unless they do react in some way with normal metabolites, there is necessarily a pre-existing complement for every potential antigen. The antibody may be this complement itself, and the surge of new synthesis may follow from a kind of mass action effect, due to the inactivation of existing stocks by the antigen. If the antibody is a simple analogue rather than the complement itself the theory would not be modified very radically. What must be doubted, at all costs, is that an individual can synthesise proteins complementary to virtually every protein of every other individual and species, each of which has evolved along its own lines at each of the innumerable dichotomies of evolution. The contrast between this extreme view and the one-point-detoxication idea needs no further emphasis.

The actual mechanism of antibody induction and production upon challenge is, like that of enzyme induction (p. 269), largely beyond the present terms of reference, since it is essentially a mechanism controlled at the systemic level, an emphatic negation of any possibility that it is simply a direct chemical reaction between foreign and indigenous proteins. Proteins do not become antibodies to foreign proteins simply by mixing them *in vitro*, though it has been claimed that some such direct change does occur if the mixture is heated and allowed to cool (Anon., 1946). The suggested explanation is that the heating causes denaturation and partial unfolding of the potential antibody, which on cooling spontaneously renatures in a form influenced by the foreign protein. The situation is full of difficulties, however; heating is likely to denature the antigen also and this might be indistinguishable from immunological inactivation and there is the further difficulty that under these conditions both proteins are equally antibody and antigen.

In unicellular organisms the systemic mechanisms may be simply the evocation, directly or indirectly by the antigen, of the production of a new nucleic acid code by a relevant gene in the nucleus, but in the Metazoa antibody-production is limited to special organs and cells, the lymphopoetic tissues, and these must be 'informed', wherever the antigen enters the body. Experimentally administered antigens are taken up by phagocytes, liver and kidneys as well as by the spleen and other lymphopoetic

organs (Coons, 1951), so that the simpler defence and excretory mechanisms contribute to antigen destruction, just as they also deal with the eventual product of antibody–antigen interaction. The lymphopoetic tissues are probably stimulated only by those antigen molecules which actually reach them and enter cells of the correct type, the plasma cell. This type of cell probably acquires antigen during its circulation round the body and carries it back to the poetic organs (Gowans, 1959), which produce more cells, containing much antibody, which is then carried to the main sites of antigen accumulation.

The lymphopoetic tissues begin to produce active antibodies as soon as 30 min. after the initial provocation by antigens, but usually they require much longer. Reciprocally the mechanism has momentum, and antibody-production may continue for years after all antigen has been destroyed. This seems very much a systemic mechanism, therefore. The actual synthesis of a molecule of antibody, by contrast, may require no more than 0·1 sec (Haurowitz, 1960), and the long time-lag lies in the preparatory accumulation of samples of antigen in the appropriate sites, in the induction of nucleic acid synthesis, in the consequent antibody synthesis, and in its distribution. The direct action of the antigen is probably limited to specifying which of the host's protein-synthesising genes is evoked; this might be interpreted as suicidal initiative, but it is more logical to maintain that all initiative lies with the host, in its complex systemic mechanism.

The extent to which the mechanism has become systemic in the higher animals is dramatically shown by experiments in which the host's lymphopoetic tissues are destroyed by irradiation or by radiomimetic chemicals. It then no longer reacts to foreign proteins, and all other proteins of its body are immunologically inert. It is then possible to introduce lymphopoetic cells from another individual and to find that they populate the host's bone marrow and other lymphopoetic sites, and proliferate there. The host then becomes tolerant of all proteins of the donor of these cells, and intolerant of all proteins which were antigens to this donor. It may even become intolerant to its own proteins.

The eventual reaction between antibody and antigen is stoichiometric (Marrack, 1950), i.e. there is an exact and simple ratio between the number of molecules of the two which gives maximal immunological effect. This implies that the reaction is a typical chemical one. The ratio is not usually unity, however, and usually favours the antigen so that as many as forty molecules of antibody may be required to deal with one of the antigen. Lower values are more usual, but even so this is not easily reconciled with the idea of an antibody which fits all the essential parts of the antigen molecule. On the other hand there is no evidence that

each of the antibody molecules associated with one antigen is different. If each attacks one of a number of replicated sites on the antigen molecule then these are likely to be relatively small, common features, and the process is more in harmony with the detoxication than the templet type of theory.

When the antigen is in the form of whole cells the tables are effectively turned in favour of the antibody (Boyd, 1947, p. 275). As few as 600 antibody molecules, perhaps as few as thirty, together with their complement (p. 271), may be sufficient to cause the agglutination and lysis of one erythrocyte, containing 10^7–10^8 molecules of haemoglobin. By contrast 5×10^6 molecules of the commercial detergent, sodium tetradecyl sulphate, are necessary to lyse each corpuscle. This high ratio in the reciprocal direction is equally at variance with any theory of a precise templet relationship. Relatively few molecules of antibody adsorbed on to a cell in a relatively non-specific way could be adequate to cause agglutination with others, similarly treated, while the subsequent lysis may be an enzymic type of reaction, with the complement as catalyst. It is known to act after the initial antibody–antigen reaction, and to contain lipopolysaccharide, which could be a suitable coenzyme for lytic reactions.

Immunological processes, therefore, while undoubtedly among the most complex and ingenious of unique biological mechanisms, are also among the most enigmatic. There is evidence on the one hand that antigens are inactivated by relatively non-specific attack on some one or few active sites and on the other that they are completely denatured. Some of the evidence indicates that the active antigenic sites are common metabolic radicals while other facts imply that each antibody is minutely specific to the antigen which induced it. Antibodies are very relevant to the uniqueness of proteins among biological materials, but it is less certain that they are so relevant to the uniqueness of each protein among proteins.

15.11 Conclusions

A concluding section might seem particularly desirable here since proteins represent the zenith of biological synthesis and their uniqueness is manifested in so many ways. For this very reason, however, a short summary would be difficult. To some extent it would also be premature, until the other biological nitrogen compounds, and other biological elements, have been considered, since many of these are very intimately connected with the activity of the proteins. In fact the general conclusions about biological materials will be largely conclusions about proteins, and there are few of these which are better discussed at this stage. We may leave this crucial biochemical group, the polymers of the amino acids, therefore, and return to further classes of monomers.

CHAPTER 16

OTHER OPEN CHAIN NITROGEN COMPOUNDS

In biosynthesis these are probably all derived from amino acids so that there is justification for considering them after the amino acids and also for taking them all together, although some could be considered more, and others less, complex than the amino acids themselves. Four main groups are biologically important, the *amides*, $R{\cdot}C(=O)NH_2$, the *amines*, $R{\cdot}NH_2$, the *amidines*, $R{\cdot}C(=NH)NH_2$, and the *guanidines*, $R{\cdot}NH{\cdot}C(=NH)NH_2$. The last two may be regarded as nitrogen-analogues (p. 154) of the amides and diamides. The amines are taken to include *secondary* (R′, R″):NH, and *tertiary*, (R′, R″, R‴) ⋮ N, compounds, together with their *quaternary* nitrogen compounds, (R′, R″, R‴) ⋮ $N^+{\cdot}R''''$, where up to three of the R groups may be hydrogen atoms. In quaternary compounds proper, all four are alkyl groups, and if one of these can ionise as an anion, it tends to undergo ring closure with the cationic nitrogen, to form *betaines*. As heterocyclic compounds these might be considered later, but they so readily revert to the open chain state, and show the appropriate properties, that they are best included here. They are in fact amino acids of course.

Like the amino acids all these groups are ammonia derivatives, and this dominates their properties. The amidines and guanidines are polyamino compounds and urea is a diamide, but there are many other groups of polyamino compounds, and other ammonia derivatives, which are not common biologically. Hydrazine, $H_2N{-}NH_2$, azides, $R^+N{\equiv}N{=}N^-$, and other series are powerful poisons; like hydroxylamine, $HONH_2$, they are sufficiently active and close analogues of normal nitrogen metabolites to cause metabolic disturbances. For this, if for no other, reason they must be excluded from normal systems; it has already been seen that this is a general principle and that the exploitation of one group of compounds automatically demands the exclusion of nearly related ones. The same principle applies within a group, and altogether this leads to a much more black-and-white differentiation between biological and non-biological materials than would otherwise exist. A particular group

initially may have been merely outstanding, but in the course of evolution it has become unique; it would be reasonable to claim that its uniqueness is biological rather than simply chemical, therefore—unique only in its normal biological context.

16.1 Amides

These differ from the amino acids in having the amino group on the carbonyl carbon itself, replacing the hydroxyl of the carboxyl group and so destroying its acid properties (p. 164). The form $R{\cdot}C(=O)NH_2$ resonates with $R{\cdot}C(=NH)OH$ so that the amides do not behave as strong bases either. They are in fact virtually non-ionic, and so neutral in a stricter sense than the amino acids, which become strong acids if the amino group is covered by formaldehyde (p. 102). The amide structure is of course analogous to that of a peptide, $R_1{\cdot}C(=O)NH{\cdot}R_2$, and has the same resonance stability. In its formation there has been the same kind of condensation reaction, with ammonia acting as an acid (proton-donor) and the carboxyl paradoxically as a base, mutually abolishing their ionising forms. This last effect is put to use in urea formation in animals (p. 284). The amides remain very weakly acidic and basic but the base, for instance, ionises only in the strongly acid region of pH 0 to −1·0!

Like those of the peptide group, the nitrogen and oxygen atoms of an amide readily form hydrogen bonds with other organic molecules and some amides form a kind of hydrogen-bonded polymer. The group forms stronger hydrogen bonds than carboxyl in spite of the fact that in general oxygen forms stronger hydrogen bonds than nitrogen. The amides are more water-soluble than the corresponding fatty acids in consequence, and have higher melting and boiling points.

With these properties amides have potential biological value and some actually are exploited. The simplest members, formamide and *acetamide*, are not of major importance but the mousy smell of acetamide may be significant, as also the side chains of cobalamin (p. 321). The only very important members are in fact the ω–amides of aspartic and glutamic acids, asparagine and glutamine (p. 181). Their names tend to imply that they are amines rather than amides, unfortunately. Much of the significance of the two amino acids is through their amides, which are no longer dicarboxylic acids, and many of the aspartic and glutamic residues of proteins are in this form (Waelsch, 1952). At the same time the α–complex is essential for the biological functions of these amides, and the two polar groups are too far apart in chains longer than C_5; this probably explains why *glutamide* is the longest which is biologically exploited. On the other hand the α–complex tends to inhibit ω–amide formation in the

C_3 homologue, which is the shortest possible dicarboxylic amino acid, so that $aspNH_2$ and $gluNH_2$ are the only members of potential value. For most purposes $gluNH_2$ is the more active, as already noted (p. 181), though $aspNH_2$ is the more common in plants. It is abundant also in some animal materials, for instance the body fluids of the cockchafer, *Melolontha* (Ussing, 1945).

Between 2500 and 4000 cal/mole are required to synthesise $gluNH_2$ from glutamic acid, i.e. about the same as for peptide bonds, not a large amount of energy considering the stability of the bond, but enough to require coupling with the hydrolysis of ATP. Details of the reaction are still uncertain (Florkin, 1960) but it is interesting as one known to be greatly speeded by 'starter' or 'primer' amounts of $gluNH_2$ itself; it is autocatalytic in fact, like that of DNA and starch. This has not been recorded for most other metabolites.

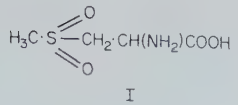
There are large accumulations of $aspNH_2$ in plants and enough $gluNH_2$ in animals to show that they serve as stores of the NH_2 group, mainly for the purpose of aminating the α–keto, and other, precursors of the amino acids:

$$H_2N(O{=})C{\cdot}R_1{\cdot}CH(NH_2)COOH + R_2{\cdot}C(=O){\cdot}COOH \rightarrow$$
$$HO(O{=})C{\cdot}R_1{\cdot}CH(NH_2)COOH + R_2CH(NH_2)COOH$$

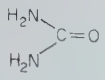
The ω–amino group in fact is much more labile than might be expected, more so than the α–amino of aspartic and glutamic acids, though this group *is* used for transamination in the absence of the ω–amino. The lability of the ω–group in fact probably depends on the implied interaction between the two. The precursors of probably all the common α–amino acids can be aminated by this mechanism and it is also used for the formation of glucosamine (p. 51) from glucose–6–phosphate (Kalckar and Klenow, 1954), of purine bases (p. 347) and of urea (p. 284). Moreover the two dicarboxylic amino acids also form ω–analogues with amines, peptides, etc., and so can mediate transamineation and transpeptidation as well as simple transamination.

The ω–amides are more soluble in water than their free amino acids (Kamin and Handler, 1952), and help in their transport across membranes (Meister, 1956).

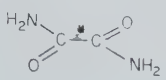
Methionine sulphoxime (Fig. 37, I) is an analogue of the ω–amides, with sulphur replacing the carboxyl carbon; it is strictly the analogue of $aspNH_2$ but in practice it behaves more as the analogue of $gluNH_2$, perhaps because of the additional methyl group on the sulphur atom. It is extremely poisonous, as might have been anticipated (Heathcote and Pace, 1950), and $gluNH_2$ is an effective antidote. It might have been expected that the sulphonamides, $R{\cdot}(O{=})S(=O)NH_2$, also would act as analogues of the



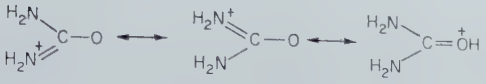
I
Methionine sulphoxime



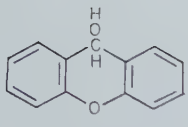
II
Urea



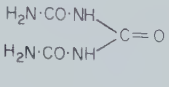
III
Oxamide



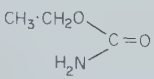
IV Resonant forms of urea



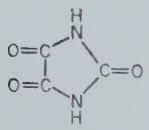
V
Xanthydrol



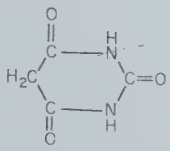
VI
Triuret
(Carbonyl diurea)



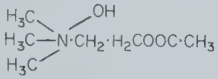
VII
Urethane
(Ethyl carbamate)



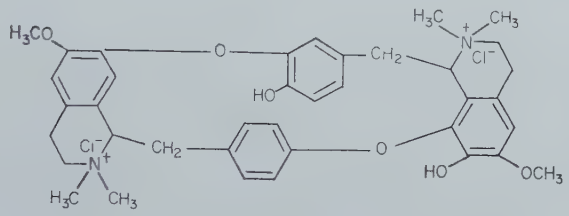
VIII
Parabanic acid



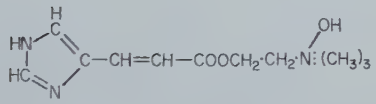
IX
Barbituric acid



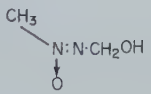
X
Acetyl choline



XI
D-Tubocurarine hydrochloride



XII
Murexine



XIII

FIG. 37 Other types of open chain nitrogen compounds of biological significance

(*continued on pp. 285 and 287*).

amides but in fact they prove functional analogues of carboxylic compounds, O=S=O of C=O, and NH_2 of hydroxyl. This is why the aromatic sulphonamides block the action of *p*-aminobenzoic acid (p. 378), an amino acid and not an amide.

16.1.1 Polyamides: Urea

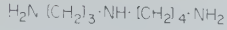
Urea (Fig. 37, II) is structurally the simplest organic diamide, and it is also the only one of major importance, though *oxamide* (Fig. 37, III) and others are known to the organic chemist and occasionally occur in living organisms. Urea is best known as one of the main forms of nitrogenous excreta in animals, though the occurrence of high concentrations of urease in soya beans and other plant materials shows that it is of wider importance. It has some quite unique properties, largely explicable as the combination of those of a typical amide with those peculiar to a diamide which has both amino groups on the same carbon atom, and no side chain—which would tend to depress activity and water-solubility.

In spite of the two amino groups it is neutral in reaction and particularly innocuous in moderate concentration, so that it is an ideal form in which to detoxicate $—NH_2$, without a proportionate sacrifice of carboxylic acids. The source of the C=O group for biological urea synthesis in fact is carbon dioxide, the anhydride of a dibasic but weak acid, so that the neutralisation of two molecules of the strong base by each one of carbon dioxide is a great chemical triumph, probably unsurpassed in metabolism. There is usually an abundance of respiratory carbon dioxide in animals and the great accomplishment is the formation of a neutral amide, in preference to the alternative of the very alkaline and volatile ammonium carbonate. In plants carbon dioxide is more precious, but so is nitrogen, and this is probably the main reason why the vegetative parts, at least, rarely form urea.

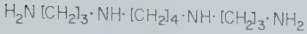
Urea also has only one-third of the osmotic effect of the equivalent amount of ammonium carbonate. In practice indeed it often permeates freely through biological membranes, for instance of the erythrocytes of some species, and exerts no osmotic pressure at all across these; for them it is *isotonic* with distilled water! At the other extreme the membranes of the elasmobranch gills are completely impermeable to urea and these animals use it as additional osmoactive material, because of its low toxicity.

Considerable energy is needed to synthesise urea from its components (Pilgrim, 1954), the energy of two ~P bonds in fact (Fig. 26), but this is a reasonable price to pay for eliminating the very toxic, free $—NH_3^+$ ions.

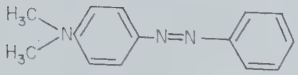
It is well known that urea acts as a weak base, forming monovalent salts with acids. This may not in itself be very important biologically,



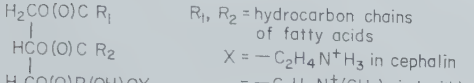
XIV
Spermidine



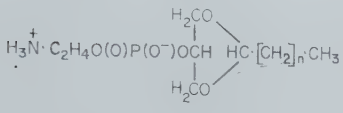
XV
Spermine



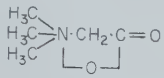
XVI
Butter yellow



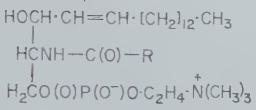
XVII
Phosphatides
(generic formula)



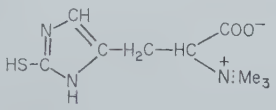
XVIII
β-plasmalogen



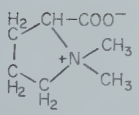
XX
Glycine betaine



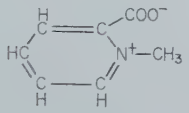
XIX
Sphingomyelin
(R = hydrocarbon chain)



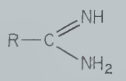
XXI
Ergothioneine



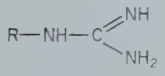
XXII
Stachydrine



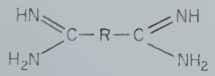
XXIII
Homarine



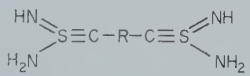
XXIV
Amidines
(generic formula)



XXV
Guanidines
(generic formula)



XXVI
Diguanidines



XXVII
Diisothioureas

FIG. 37 (*continued*).

but it gives a clue to the nature of the resonance in the molecule (Fig. 37, IV). The active form is the third, with the symmetrical, oxonium base. With this structure and resonance forms it is naturally one of the most powerful hydrogen-bonding agents known, and this is why it denatures proteins so effectively. This is the main physiological drawback of urea, but the blood and other proteins of elasmobranchs presumably are adapted to withstand the action of concentrations which can be as high as 1·58 per cent. The denaturation of proteins by urea has a negative temperature coefficient (p. 242) so that the mammals have an advantage over poikilotherms in resisting its action.

Another virtue of urea is that it is extremely soluble in water, 56 per cent w/w at 0°C and 145 per cent at 70°C. This is in turn related to its polar structure and strong hydrogen-bonding power. By contrast oxamide (Fig. 37, III) is virtually insoluble, like most of the oxalates themselves. Like water, urea forms hydrogen bonds with its every atom, though rather weakly with the carbon. It is difficult to find any insoluble urea derivatives to use as the basis for a qualitative test or for its quantitative estimation, but the nitrate and oxalate are only sparingly soluble and urea forms a unique white precipitate with the interesting alcohol xanthydrol (Fig. 37, V). Its monosubstituents are the only other compounds giving this kind of precipitate with xanthydrol.

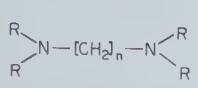
Urea is also very soluble in polar organic solvents such as glycerol, methanol, acetone and dioxane, and in the latter it has a dipole moment (Table 11) as high as $8{\cdot}6 \times 10^{-18}$ e.s.u., the highest value of any substance in such a solvent. In water, urea resembles many of the amino acids in showing a positive dielectric increment, i.e. the dielectric constant increases with concentration. Urea is insoluble in ether, chloroform, benzene and other apolar lipid solvents so that it is not retained in the lipid stores of the body.

In molten form urea itself is a remarkably good solvent, as would be expected from its similarities to water (Clark, 1951), particularly in forming hydrogen bonds. The melting point is as low as 132°C, so that it could be the medium for the thermophilic type of early biological system envisaged by Fox (1960). As a solvent it behaves very much like water and its analogue, ethanol—as might be anticipated. Of 120 water-soluble substances tested sodium silicate was the only one which proved insoluble in molten urea. Its solubility (p. 468) depends on a rather specific ratio of oxygen to silicon and probably urea has too little oxygen. The urea-based thermophiles therefore could not have formed siliceous skeletons from solution. In urea solution the Co^{++} ion has the anhydrous blue and not the hydrated pink colour, but most other ions have the same colour as in water, ureated instead of hydrated. Litmus changes colour

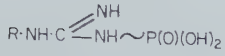
with pH just as it does in water and metals release hydrogen from acids. Precipitation reactions can be performed in molten urea, as in water.

An interesting technical use of molten urea is to improve the water-solubility of materials such as starch. The 'soluble' starch is dissolved in

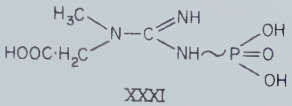
XXVIII
Stilbamidine



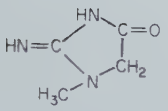
XXIX
Tetraalkyl-diamines



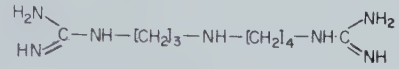
XXX
Phosphagens
(generic formula)



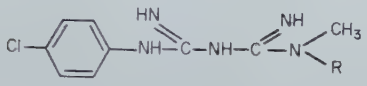
XXXI
Phosphocreatine



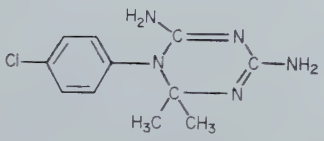
XXXII
Creatinine



XXXIII
Hirudinone



XXXIV
Paludrine



XXXV
Probable active form of
antimalarial diguanidines

FIG. 37 (*continued*).

molten urea, the solution cooled and set, and then powdered. This powder dissolves instantaneously in water, giving a clear solution, and not an opalescent one as the ordinary soluble starch does. The starch is still sensitive to iodine and so forms the ideal indicator for iodine titrations, especially since urea is also a good solvent for iodine. It is in fact a better solvent for

some organic and inorganic substances than water itself. As my colleague, H. K. Pusey, has suggested there may be a significant relationship between the high urea content of elasmobranchs and the absence of a bony skeleton, which was present in more primitive forms. Preliminary experiments have failed to show that bone-phosphate is more soluble in urea than in water but this merits more careful testing under relevant physiological conditions.

As would by now be expected, hydrocarbons are insoluble in urea but all polar lipids are readily soluble (Picken, 1960). Some organic substances are miscible in all proportions, for instance gluten, dextrin, gum arabic, sodium alginate and soap, representatives of most of the major biological groups. Although paraffins are insoluble they form interesting solid *channel* (*canal*) *complexes* with urea as do other relatively long straight chain carbon compounds (Finar, 1959, p. 369). In this type of *inclusion complex* the molecules of the straight chain compound lie parallel, in straight channels between urea crystals. Molecules shorter than C_5–C_7 do not orientate well enough to form these complexes, while branched and cyclic molecules will not fit into the necessary narrow channels. The size of these is presumably dictated by the urea, since the number of molecules included is inversely proportional to their size. The inclusion component can be a mixture of different molecules, provided these are structurally similar. This further curious property of urea conceivably may have biological parallels.

When gently heated to 180°C urea forms a dimer called *biuret*: $NH_2{\cdot}C(=O){\cdot}NH{\cdot}C(=O){\cdot}NH_2$, effectively a series of four peptide bonds. Peptides in fact also form this kind of compound, which can be recognised by the violet colour of its compound with Cu^{++} in alkaline medium. This is the basis of the biuret test for proteins and their initial breakdown products, the proteoses and peptones, as well as for purines and pyrimidines (p. 347), which also have the necessary number of peptide links in series. The minimal requirement is two —NH—C(=O)— groups joined either directly or through another carbon or a nitrogen atom (Hawk *et al.*, 1947, p. 156). The copper is probably bound by nitrogen atoms of two biuret complexes (Rising and Yang, 1933), forming a simple analogue of the metalloporphyrins (p. 315), and it seems very probable that this is the essential type of structure in the copper-proteins (p. 315). Crystals of *triuret*, *carbonyl diurea* (VI), have been identified in *Amoeba*, probably as an excretory product.

Urea is an extremely versatile compound. Among its industrial uses is copolymerisation with formaldehyde to form a hard, colourless, transparent, thermosetting amino-plastic, with properties approximating to those of flint glass, and without the drawback of splintering (Read, 1935,

p. 453). The ethanolic substituent, ethyl carbamate, or *urethane* (VIII), is now well known as a narcotic; it also affects cell division and is mutagenic. It inhibits growth, including that of neoplasms, but, like many growth-inhibitors, it tends to be carcinogenic itself under some conditions (Sexton, 1953, p. 353). Other substituted ureas behave similarly; *phenyl urethane* is antimitotic, inhibiting germination and growth in plants. Urethane inhibits photosynthesis so that an effect on electron-transport may be suspected. It may depress resonance by breaking hydrogen bonds, as urea does. If urease were not so narrowly specific to urea itself it might be suspected of destroying these anti-germination derivatives in the soya bean at the appropriate time; urea itself may have anti-germination activity, perhaps.

Urea condenses with oxalic acid to form *parabanic acid* (Fig. 37, VIII), and with its homologue, malonic acid, to form *barbituric acid* (Fig. 37, IX). These are heterocyclic compounds, belonging to the *pyrazolidine* and *pyrimidine* groups (p. 344), and it is perhaps surprising that the biological members of those groups are not synthesised in this simple way. Even so this further ability of urea may be biologically significant.

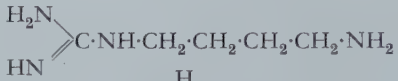
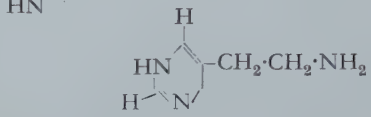
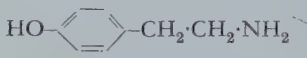
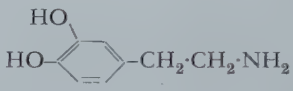
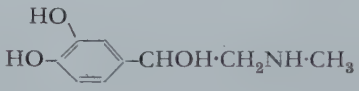
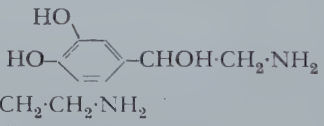
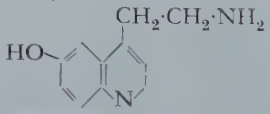
16.2 Amines

Like the parent base, ammonia, and in contrast to the amides, the amines are strongly basic. In fact substitution of the hydrogen atoms in turn by an alkyl group progressively increases the strength, and the quaternary series are the strongest of the monoamino bases. A number of the amines are important biologically, usually for their basicity, combined with their relatively high lipid-solubility. In lipid media they exist mainly as the conjugate base, $(R)_3{:}\,N$, and in aqueous media as the conjugate acid, $(R)_4{:}\,N^+$, where R is either hydrogen or an alkyl group. Consequently they can suddenly produce or lose the active cation, with a corresponding change of pH, which will tend to be localised at aqueous–lipid interfaces. This appears to be particularly effective at the membranes of conducting cells and most *aminium* bases are powerful neurotropic agents. In quantity most of them are powerful neurotoxic agents, and those which act physiologically *in vivo* are present in very low concentration.

Amines are synthesised with difficulty in the laboratory but enzymic α–decarboxylation of the amino acids seems to be a routine method for forming the primary amines *in vivo*. A considerable number of their α–amines are physiologically or at least pharmacologically active (Table 30). Secondary and tertiary amines are quite common as ring compounds, but as open chain biological compounds they are nearly always poly-methyl substituents, and in fact usually derivatives of the one member, *choline*,

or trimethylethanolamine, $(CH_3)_3 : N \cdot {}^{+}C_2H_4OH$. This is synthesised from the natural primary amine, *ethanolamine* (Table 30), itself derived from serine. It is converted to choline by repeated transmethylation from

TABLE 30. PRIMARY AMINES OF BIOLOGICAL IMPORTANCE

Amine	Formula	Source
Ethanolamine	$HOCH_2 \cdot CH_2 \cdot NH_2$	Serine
Taurine	$HO_3S \cdot CH_2 \cdot NH_2$	Cysteine
Putrescine	$H_2N \cdot CH_2 \cdot CH_2 \cdot CH_2 \cdot CH_2 \cdot NH_2$	Ornithine
Cadaverine	$H_2N \cdot CH_2 \cdot CH_2 \cdot CH_2 \cdot CH_2 \cdot CH_2 \cdot NH_2$	Lysine
n-Hexylamine	$H_3C \cdot CH_2 \cdot CH_2 \cdot CH_2 \cdot CH_2 \cdot CH_2 \cdot NH_2$	Norleucine
Agmatine (δ-Guanidyl butylamine)		Arginine
Histamine		Histidine
Tyramine		Tyrosine
Dopamine		Tyrosine
Adrenalin		Tyrosine
Noradrenalin		Tyrosine
Tryptamine		Tryptophan
Serotonin (5-hydroxy-tryptamine)		Tryptophan

methionine (p. 178). The free di- and trimethylamines occur in fish extracts and in beetroot. Their hydroxides are relatively unstable, and this is probably why they are much rarer than the primary and quaternary series; this is another example of stable nodes with unstable intermediary stages.

The two physiologically most important neurotropic amines, *adrenalin* (Table 30) and *acetyl choline* (Fig. 37, X), are thus a number of steps re-

moved from the simple products of decarboxylating amino acids. *Serotonin* also has an extra hydroxyl group, at position 5. At the same time most of the simple α–amines do show neurotropic effects, and some are believed to be normal physiological agents, for instance *tyramine* in cephalopods and coelenterates (Ross, 1945), and *histamine* rather generally. In the vertebrates there is a sharp division of labour, still not fully understood, adrenalin being the mediator at the motor endings of the sympathetic system and acetyl choline at the corresponding parasympathetic endings, as well as at most other motor endings. Within nerve pathways also there is some division of labour. The simpler amines mimic one or other of the two main mediators, but very often only at certain types of synapse, giving further evidence of chemical differentiation—in the synapses themselves. Most of them mimic adrenalin, which may be essentially a synaptic depressor agent, while acetyl choline may be the only normal facilitator (Buday, 1961). Histamine and serotonin are among the few which resemble acetyl choline, and both appear also to be natural agents. The sympathetic system is well known to be concerned with the alarm activities of fright, fight and flight, and the parasympathetic with the 'vegetative' activities of the quiet life; it is therefore interesting to find that serotonin is outstanding for its psychologically tranquillising effect. Dopamine and adrenochrome, which are most closely related to adrenalin, are particularly strong synaptic depressants. At present there appears to be some overlap in function among neurotropic amines, but among those which are normal physiological agents in fact there may be no real duplication; it seems, on the other hand, that acetyl choline acts at a greater variety of junctions than might be expected of a single agent, and that there is theoretically scope for further division of labour. It is very desirable to know which of the amines occurring in animals are normal mediators.

The chemical basis for the action of adrenalin has been analysed in some detail and as in other instances of this kind it shows that a large number of features of the molecule are absolutely essential and that it is a highly unique substance. It also has a number of distinct actions and the various active groups of the molecule therefore must be very nicely balanced. Sympathetic activity includes cardioacceleration, bronchodilation, inhibition of peristalsis in the gut, vasoconstriction in the gut, skin and elsewhere, and vasodilation in lungs, heart and skeletal muscles. The maximal cardioacceleration is given by the hydroxyl group at position 3 of the ring (Table 30), and this is enhanced by the hydroxyl in the side chain, unless there is also a hydroxyl at 4 in the ring, when it potentiates the effect of this group instead. This effect is cardiodepressant! If both of these hydroxyl groups are removed the depressant action is abolished.

However, the bronchodilator effect demands an hydroxyl at 4 and in the side chain, and the second hydroxyl in the ring (at 3) further increases *this* activity (Lands, 1951).

The methyl group on the amine of the side chain depresses the cardio-acceleration and the vasomotor effects, and isopropyl in this position is even more potent, but methyl increases the bronchiodilator effect fifty times, and isopropyl 100 times. The action on the intestine runs parallel to that on the blood vessels, so that there are only two main groups of actions to be balanced in the molecule. Even so, it is understandable that no other amine could deputise precisely for adrenalin in all its actions, and probably that the molecule has been perfected through a long period of evolution.

Acetylcholine (ACh) and the other quaternary amine derivatives are unique in being able to ionise even in lipid media (Sexton, 1953, p. 190), and this is probably vital for the transport of ions across the membranes of nerves and other cells, particularly at synapses. The neurotropic properties of ACh are not due solely to the amine group since the agent is inactivated by the normal lytic enzyme acetylcholine esterase, which removes the acetyl group from the free end of the ethanol side chain. Oxidation of the ethanol to acetaldehyde yields *muscarine*, $(CH_3)_3N^+CH_2CHO$, which occurs biologically, and closely mimics ACh in some of its actions; it completely lacks others, however, so that as in the case of adrenalin the total action of ACh can be fractionated. Other choline derivatives further illustrate this. *Pseudomuscarine*, $ONO{\cdot}CH_2{\cdot}CH_2{\cdot}N^+(CH_3)_3$, has nitrous acid in place of acetic, but, as its name indicates, it resembles muscarine rather closely in action. *Neurine*, in which the ethanol has been desaturated to the vinyl, $H_2C{=}CH{-}$, state still has synaptic transmitter activity (Fearon, 1949, p. 421). *Acetyl β–methylcholine* and *carbamyl choline*, $NH_2{\cdot}C(=O)O{\cdot}CH_2{\cdot}CH_2{\cdot}N^+(CH_3)_3$, behave very similarly to ACh. Even the phosphorus and arsenic analogues, for instance *arsenocholine*, $(CH_3)_3As^+{\cdot}CH_2{\cdot}CH_2OH$, have some activity, and so have all onium compounds, for instance $(R)_3{:}S^+{}^-OH$ and $(R)_3O^+{}^-OH$. Neither side chain nor base therefore is absolutely specific. ACh thus is unique not for any particular one of its effects but for the combination and for the strength of all of them.

The cyclic analogue, *atropine* (p. 338), and *dibutoline* and others have an antagonistic action to ACh, dilating the pupil of the eye, for instance, and the *curarine* group of analogues block transmission by ACh at motor end plates. They are specific to this particular site so that the drug can be used to induce muscular relaxation in the conscious patient, a valuable surgical device on occasion and an interesting psychological contrast to the situation when it was used as an arrow poison. The most satisfactory derivative in present use, D–*tubocurarine chloride* (Fig. 37, XI), is a

rather complex cyclic and dimeric form (Collier, 1950), of great interest chemically. It has that necessary degree of asymmetry (p. 115) for effective resonance activity. Other curariform drugs have a long straight chain forming each bridge between the two condensed rings which contain the base itself, and it is found that their optimal length is C_5–C_6 for synapses in the central nervous system and C_8–C_{11} for peripheral synapses. The simplest and best known are *hexa-* and *deca-methonium iodides*. At peripheral synapses the hexa compound acts as an antidote to the deca compound. In tubocurarine presumably the rings in the bridges count their full value of C_6, if the whole bridge is analogous to the C_8–C_{11} chains.

Murexine (XII), or β–imidazolyl–(4)–acryl choline, from the hypobranchial gland of the dog-whelk, *Murex*, is curariform in action and also blocks the intermediary or 'preganglionic' synapses of the sympathetic system, which are in fact cholinergic, of a special type blocked by nicotine. Presumably the tertiary nitrogen of the imidazole ring acts as the second quaternary group; the two are then separated by six carbon atoms (and an oxygen) passing one way round the ring and by eight, plus one oxygen and nitrogen, the other way round. It is presumably the same nitrogen, rather than the primary amine group, which makes histamine cholinergic, since the α–amino group has been lost in murexine.

Another quaternary cation of biological importance is *tetramethylamine*, $(CH_3)_4 : N^+$. The free conjugate base, trimethylamine, $(CH_3)_3 : N$, has been found in the reproductive tissues of animals and plants, and is associated with the gonadal hormones of vertebrates. The hydroxide, *tetramine*, $(CH_3)_4 \cdot N^+ \ ^-OH$, is a component of the poison of jellyfish and sea anemones, and of the 'salt bush' of Australia, and is a powerful paralysant.

Bacteria are rather immune to the primary, secondary and tertiary amines, as poisons, but in general the secondary and tertiary series are good insecticides (Sexton, 1953, p. 332). They may act as antagonists of the more physiological primary and quaternary series, therefore. Maximal activity is shown by C_{11}–C_{15} compounds, which is fairly near the optimal chain-length for fatty acids and for a number of biocidal amines, and to the length of the bridge chains in the peripherally acting curariform drugs. Among natural antibiotics, aureomycin (Fig. 21, XV) and others are amines of various grades, and (XIII) from a glycoside of *Macrozamia* is an amine oxide (Sexton, 1953, p. 234).

The neurotropic actions of the amines are not the only type with biological significance. As already noted (p. 185) histamine has a strong stimulatory effect on growth processes of various kinds, and choline promotes the growth of mammals. It is usually regarded as a B-vitamin, and *putrescine* (Table 30) is a vitamin for some parasitic bacteria. Two

curious polyamines, *spermidine* (Fig. 37, XIV) and *spermine* (Fig. 37, XV), are essential dietary constituents for the parasitic bacterium, *Haemophilus* (Herbst and Snell, 1948–9). Their structure implies that they may be formed from putrescine, together with its next lower homologue, but such structural resemblances can be very misleading. On the other hand trimethylamine is growth-inhibitory (Wastl, 1942), and most simple nitrogen compounds tend to inhibit this type of activity (Almquist, 1951).

Quaternary amines are alkalis strong enough to dissolve wool (Penn, 1949). Notwithstanding the strength of tetramethylamine hydroxide (tetramine), however, trimethylamine oxide, $(CH_3)_3 : N \rightarrow O$, is neutral in reaction, has other virtues similar to those of urea, and is a major form of excretory nitrogen in teleosts, elasmobranchs and cephalopod molluscs. It is very soluble and little more toxic than urea. Elasmobranchs also retain it for osmotic purposes, as they do urea (Baldwin, 1953). Mammals are able to convert the conjugate base to the oxide, and this appears in the milk of cows feeding on beetroot and other plant sources of the tertiary amine.

Another quite separate property, a special function of choline, is to act as methyl donor for a number of important syntheses in the body; this is in fact its main action as a growth factor. It rather readily exchanges methyl for hydrogen and returns to the ethanolamine stage. It can donate methyl to the precursor of methionine, in those organisms which are able to synthesise this amino acid. Although, as already seen, arsenocholine is a mild neurotropic analogue of choline it cannot *trans*-methylate. This action also requires the ethanol side chain, since homocholine, or trimethylpropanolamine, $(CH_3)_3 \cdot N \cdot CH_2 \cdot CH_2 \cdot CH_2OH$, is inactive. Choline is uniquely active, i.e. quantitatively outstanding for the purpose. It will methylate selenium and tellurium as it does the sulphur of the methionine-precursor. Among other substances which will *trans*-methylate are *glycine betaine* (Fig. 37, XX) and even *monomethyl ethanolamine*. Betaines with a longer chain are inactive, but on the other hand such distantly related bases as *butter yellow* (Fig. 37, XVI) can be effective agents.

16.2.1 Conjugated Amines: Phosphatides and Cerebrosides

A number of the amines form conjugates with other molecules, mainly lipid, and occur in large quantities in the myelin sheath of nerves and elsewhere. The function of some is as obscure as their structure is complex, but one group, the phosphatides, are relatively simple, abundant and fairly well understood. They occur abundantly in the cell-membranes and elsewhere (p. 84), in most tissues of probably all organisms, and are known to help in the transport and chemical mobilisation of lipids, particularly the glycerides to which they are structurally related. They are in fact themselves triglycerides, in which one fatty acid has been replaced

by phosphoric acid, conjugated with a nitrogen base of the amine group. Phosphoryl ethanolamine thus forms *cephalin*, and choline forms *lecithin*, the two best known of the phosphatides (Fig. 37, XVII). Serine, inositol diphosphate (p. 74) and some other non-nitrogenous bases also form this type of phosphatide. The group X is relatively easily removed, leaving a phosphatidic acid. The two fatty acids are nearly always C_{16} or C_{18}, but vary considerably in detail, giving a large number of species of each phosphatide genus. This radiation of limited groups is reminiscent of the steroids and other classes of material. In the biological phosphatides the phosphoryl group is probably always on the end carbon of the glycerol, but in the recently discovered *plasmalogens* (Fig. 37, XVIII) it may be at the α– or β–positions. As already noted (p. 102), in these compounds a fatty aldehyde forms a *gem* diether with both other groups of the glycerol. The function of the plasmalogens is obscure, as yet.

A feature of interest in the phosphatide molecule is the association of strong acid (phosphoric), strong base and virtually neutral alcohol. This association occurs in other amino–lipid conjugates and some other biological molecules, but is most clear in the structure of the nucleic acids (p. 355). They, therefore, are potentially capable of acting as strong zwitterions while keeping the molecule as a whole nearly neutral, in somewhat the same way as the amino acids. In the nucleic acids the acid and base groups are rather widely separated.

The solubility properties of the phosphatides also are very interesting and impressive. Replacement of the one fatty acid of a fat by a phosphoryl-amine makes the resulting phosphatide soluble in water, as an opalescent colloid, and without losing its lipid-solubility. It is, therefore, a most valuable intermediary between the two media. Lecithin no doubt is a zwitterion in both types of media. Perhaps the most interesting feature is their insolubility in acetone, which might be taken as the most typical semipolar lipid. Consequently they are soluble in the more polar media, such as water, acetic acid, methanol and ethanol, and on the other hand in the non-polar solvents such as petroleum, chloroform and even ether, but not in the particular intermediate state represented by acetone. The cephalins are insoluble also in alcohol, which gives some idea of possible division of labour among the phosphatides.

Lecithin has such a high affinity for water that it is hygroscopic, and crystals placed in water rapidly swell. They do not so readily dissolve, however, because the molecules spontaneously orientate at the surface to form a strong membrane. This is an obvious advantage over the fats and fatty acids which need a protein backing to form stable membranes. Since the granules of commercial lecithin are irregular they eventually form amorphous rounded, fluid bodies in water, superficially resembling

amoeboid cells. Small local variations in impurities and other conditions inevitably cause streaming movements inside, local changes in surface forces, and so on. These so-called 'myelin forms' therefore have the appearance of simple models of amoeboid form and movement; to the extent that there is a large phosphatide component of all cell-membranes the model is a rational one.

The spontaneous orientation of phosphatides at interfaces is an expression of their strongly amphiphil nature and a clue to their intermediation in the transport of other materials between the two media. Like all substances found to accumulate here they lower the surface tension, and particularly in association with protein and carbohydrate. The surface activity of the common form of lecithin is greatly increased if its unsaturated, oleic acid, moiety is removed, forming *lysolecithin*. This is the effect of a lecithinase present in the venoms of the cobra and the rattlesnake and the virulence of the venom is due partly to the consequent haemolytic effect on the victim's erythrocytes. Invertebrates without blood cells of this type should be much more resistant to these poisons, but like most venoms they do not depend entirely on the one weapon. Cholesterol (p. 125) gives some protection from the haemolysant action of lysolecithin by forming a complex with it (Fearon, 1949), and this is probably relevant to the general function of steroids on biological membranes (Booij and de Jong, 1956). Some phosphatide derivatives are used as bactericides, probably because of their surface action.

The term myelin form arose because of the particularly large quantities of lecithin in the myelin sheaths of nerves. It has long been thought that the sheath was composed of many concentric cell-membranes and the swiss-roll manner of its formation is now understood (Ben Geren, 1954). It is virtually pure cell-membrane material, and this accounts for the high lecithin content. It is here also that so many related conjugated amines occur in extricable amounts.

Cephalin forms insoluble complexes with proteins having a basic reaction, the protamines and histones, and may be more important than lecithin in the fabric inside the cell (Picken, 1960, p. 347). Lecithin is the more common extracellularly as well as in the membrane and so plays the more important part in fat transport and utilisation. Cephalin appears to have an extracellular function in blood-clotting: conjugated with protein it is an essential component of thromboplastin, the enzyme which activates the first enzyme of the process, thrombase. In fact it is released from damaged cells, and remains inactive in the absence of wounding, a simple, highly adaptive provision. Cephalin also forms insoluble complexes with globulins (Makari, 1960), and so stabilises them. Probably in a similar way it depresses the activity of some proteins and peptides and

in consequence counteracts the inhibitory effect of oligopeptide antibiotic on vital metabolism in the gram-positive bacteria (Sexton, 1953, p. 224).

The phosphatides are much more active chemically than simple glycerides, due no doubt to the active phosphatide–amine group. Any double bonds in the two fatty acid moieties therefore oxidise spontaneously in air, whereas they are stable in the neutral fats. The whole molecule hydrolyses more readily, losing fatty acids and choline, but leaving the powerfully bonded phosphoric residue. This is no doubt relevant to the lipotropic action of lecithin, promoting the catabolism of the triacyl glycerides. Choline is lipotropic, presumably because it is incorporated into lecithin, and this is its second main action as a vitamin. The effect is largely peculiar to this particular base; triethyl choline has some action but the tripropyl analogue is inactive. Another aspect of the reactivity of the phosphatides is the formation of stable compounds with heavy metals as well as with proteins; stable compounds are also formed with carbohydrates and with other lipids.

Phosphoryl choline forms another series of complex lipids called the *sphingomyelins*, also abundant in the myelin of nerve-tissue. The phosphate moiety is bonded to an alcohol group of a long chain amine, known as *sphingosine* (Fig. 37, XIX). The terminal three carbon atoms of this could be regarded as an aminated glycerol residue and there is a fatty acid linked to this amine group by a peptide bond comparable to the acetyl–amino link in acetyl–glucosamine. The significance of the secondary amine group and of the main part of the sphingosine is still unknown but together they are unique features of this group of compounds. The sphingomyelins have much the same solubility properties as the lecithins, but with their own special details. They are soluble in benzene but insoluble in ether, acetone, ethanol and ethylacetate; they dissolve in the last two if heated and they form an emulsion in water. Again these solubility peculiarities may be the basis for a significant division of labour among the conjugated amines. Arsenocholine is an effective substitute for choline in the sphingomyelins (Sexton, 1953).

Glycolipids or *cerebrosides* have precisely the sphingomyelin structure but with galactose substituted for phosphoryl choline. In one way this is not a very great difference since the combination of strong acid and strong base gives a unit which is virtually as neutral as the sugar. The cerebroside group are relevant here not only because of their analogy to the sphingomyelins in this respect: they also have the secondary amine group. The fatty acid bonded to this group varies greatly so that there is a large family of cerebrosides; their function however is still unknown. The galactose depresses the lipid-solubility of the molecule which there-

fore again has its own unique solubility properties; it is soluble in alcohol but insoluble in ether. It seems probable that part, at least, of the significance of the whole group of nitrogen-containing lipids is their varied solubility properties.

16.3 Betaines

The acid and base ionising groups of some of the quaternary amino acids, that is carboxylic acid derivatives of the quaternary amines, are sufficiently far apart for internal salt formation with ring closure, and this group of compounds is known as the betaines (Fig. 37, XX). The phosphoryl amines do not behave thus and compounds like acetyl choline have no free acid group. The simplest betaine, glycine betaine, is formed when the ethanol group of choline is oxidised to acetic acid. It will be noted that betaines close to form the relatively rare heterocyclic ring containing both nitrogen and oxygen (p. 386). The internal salt is a labile structure, and so the betaines retain a number of zwitterion properties as well as the high melting point of strong electrolytes. The betaines of the relatively long chain fatty acids have a dielectric increment (p. 286) greater than that of the α–amino acids, in fact as large as that of oligopeptides (Cohn and Edsall, 1943, p. 147), which have about the same distance between the two free ionic groups as the betaines in their open chain state. The methyl groups of trimethyl glycine are not involved in the ring closure and glycine betaine can act as a methyl-donor; indeed this may be the active form of choline, as a methyl-donor.

Glycine betaine is common in plants but the betaines isolated from animals usually have a longer chain fatty acid. The most common is *carnitine*, $(Me)_3 \cdot N^+CH_2 \cdot CHOH \cdot CH_2 \cdot COO^-$, while *γ–butyrobetaine*, $(Me)_3 \cdot N \cdot {}^+(CH_2)_3 \cdot COO^-$, and *crotonbetaine*, $(Me)_3 \cdot N^+ CH_2 \cdot CH{=}CH \cdot COO^-$, also are fairly common (Baldwin, 1953). These three are interconvertible and probably have a common origin, from glutamic acid, via γ–aminobutyric acid. Carnitine is a very potent B-vitamin for insects, but the nature of its physiological action is still obscure; these long chain betaines do not transmethylate. *Ergothioneine* (XXI), which occurs in red blood cells, is a substituted α–amino betaine, the substituent being a thiol derivative of histidine: in fact ergothioneine is the betaine of thiolhistidine. It occurs also in the fungus ergot, but its function is not yet known.

While the methyl groups are not involved in betaine formation, a ring closure through a hydrocarbon group does occur in a related group of bases, exemplified by *stachydrene* (XXII), isomeric with γ–butyrobetaine. Neither ionic group is affected by this closure, which forms a stable ring but leaves them fully active. Stachydrine may actually be formed by the

methylation of proline (Baldwin, 1953), in which case the initial ring closure is not through a hydrocarbon group. Stachydrine occurs in plants and in the large primitive lamellibranch, *Arca noae*, very appropriately named since it is not only ancient and ark-like but contains a cargo of rare nitrogen compounds (Baldwin, 1953). Among these is another of the present group, *homarine* (Fig. 37, XXIII), occurring also in the lobster. It is hexacyclic and fully unsaturated, so that in fact it is structurally, and probably actually, a derivative of the pyridines to be considered later (Chapter 18). At the same time it has the zwitterion structure of the betaines, and so have some of the biological pyridines.

16.4 Amidines and Guanidines

These are polyamines, with two and three amine groups respectively on the same carbon atom (Figs. 37, XXIV, XXV). They are structures reciprocal to the secondary and tertiary amines, in which two and three alkyl groups are bonded to the same nitrogen atom. They might be regarded also as substituted ureas but this would obscure the fact that they are strong bases, the guanidines being among the strongest nitrogen bases known. They show very clearly that oxygen, as in urea itself, depresses the alkalinity of nitrogen bases, while hydrocarbons enhance it; they are nucleophilic while oxygen is electrophilic.

Amidines are not very common in living organisms, perhaps for the usual reason applicable to compounds of intermediate structure. To some extent this is borne out by the fact that the *diamidines* (Fig. 37, XXVI) synthesised in the laboratory prove strong trypanocidal drugs (Sexton, 1953, p. 309), while the mono-amidines are inactive. However, the activity of this group of compounds depends on the presence of two nitrogen base groups symmetrically disposed at the ends of a link-chain, rather than simply on the number of amine groups in the molecule. Consequently some *diguanidines* and some *diisothioureas* (Fig. 37, XXVII) also have anti-protozoal activity. The general structure recalls the curariform drugs (p. 292) and the optimal length of link is C_{10}–C_{14}. Aromatic diamidines also are active trypanocides, one of the most effective being stilbamidine, *trans*-4:4–diamidinostilbene (Fig. 37, XXVIII); the benzene ring therefore is equivalent to a C_4–C_6 portion of an open chain. The link seems at least as important as the amidine groups themselves, so that this may be taken as another example of the biological actions of the stilbenes. As trypanocides they are most active with —CH_2—CH_2— or —CH_2—O—CH_2— linking the benzene rings, so that the *trans*-configuration is not important. Activity is greatly weakened if —CHOH— or —C(=O)— is substituted for the C_2 link and it is completely destroyed by —S—S,

—SO_2—, —NH—, —N=N—, or —NH—CO—NH—, there. This is another impressive example of great specificity of biological activity to structure. These compounds also show a degree of taxonomic specificity: some diguanidines are trypanocidal whereas others are antimalarial.

The transposition compounds of the diamidines, namely the *tetra-alkyldiamines* (Fig. 37, XXIX), also are anti-protozoal, and this further emphasises the rather low group specificity of the end groups. These compounds tend to be specific to the amoebas. The most effective member has R=C_4H_9 and n=10. It is also acaricidal and carcinogenic, and so is butter-yellow (Fig. 37, XVI), which is a transposition compound of stilbamidine, with the —C=C— link also transposed to —N=N—.

16.4.1 Guanidines

Free guanidine, HN=C:$(NH_2)_2$, may exist in muscle tissue and in some plants, e.g. vetch; it is a base with a p*K* greater than pH 14, i.e. as strong as sodium hydroxide. The substituted guanidines have a lower value, for instance 11·6 to 12·6 for arginine, and are more common as biological compounds. The most important group are the *phosphagens* (Fig. 37, XXX), already noticed in connection with arginine (p. 183); they are widely distributed as a store of ~P, for the rapid synthesis of ATP (p. 404) from ADP in emergency. Their uniqueness lies in the easy reversibility of the Lohmann reaction involved: when ATP is present in high concentration it spontaneously re-phosphorylates the guanidine so that there is a permanent store of phosphagen in normal times. The guanidines are unique in forming a ~P bond through an amino radical.

The phosphagen of vertebrates is *phosphocreatine* (Fig. 37, XXXI), which has a transfer value of 13,000 cal/mole in its ~P bond, whereas that of many invertebrates is *phosphoarginine* (p. 183), with a value of 11,800 cal/mole. The vertebrate phosphagen also has the virtue of a smaller molecule. Vertebrates use arginine to amidinate glycine, as the initial step in creatine synthesis, so that they have probably evolved creatine phosphate to replace arginine phosphate. A number of invertebrates also have exploited creatine, even including the Porifera; these together with the echinoderms and some protochordates have both creatine phosphate and arginine phosphate (Prosser and Brown, 1961). Creatine phosphate occurs also in some coelenterates, sipunculids and errant polychaetes.

A number of other phosphagens occur sporadically and there may be others yet undiscovered. *Nereis* has *phosphoglycocyamine*, P~G·CH_2·COOH, where G is the guanidino group; glycocyamine is an even simpler molecule than creatine, therefore. *Arenicola* has phosphotaurocyamine,

P~G·CH_2·CH_2·SO_3H, an interesting reminder of the co-existence of glycocholic and taurocholic acids in the bile of mammals. The bile acids are, like the phosphagens, very variable (Table 40) and merely group-specific. It is also interesting that free taurine constitutes as much as 5 per cent of the weight of the muscle of *Mytilus*. In *Lumbricus* the phosphagen, *lombricine*, is phosphoseryl glycocyamine phosphate, P~G·CH_2·CH_2·O·P(=O)OH·O·CH_2·CH(NH_2)·COOH; here the trend seems to have been in the opposite direction, towards complexity. A further phosphagen probably exists in the protozoan, *Tetrahymena* (Seaman, 1952). The precise significance of the variety of phosphagens, particularly in annelids, is at present obscure, but they are not group-specific in the sense that any phosphagen will function in any animal.

Creatine is excreted as *creatinine* (Fig. 37, XXXII), a ring-closed structure recalling the betaines. If creatine is fed to mammals they excrete it rapidly as creatinine but their normal output of this excretory form is very constant and implies a steady 'wear' on stocks of creatine in the muscles, the major relevant tissue. In some forms of muscular dystrophy creatine is excreted unchanged, and it would seem to perform some more fundamental function than simply as an emergency store of ~P. It is therefore interesting that free guanidine in quantity causes muscular spasms, similar to those of hypocalcaemic tetany due to deficiency of parathormone. After parathyroidectomy in fact there is an increased excretion of monomethyl guanidine from the muscles (Fearon, 1949, p. 417). The alkali is thought to be a calcium precipitant and in this capacity would have a powerful effect on muscular activity.

All phosphagens possess an acid group which could ring-close in this way with one of the two guanidine nitrogen groups which do not carry the phosphate group. Clark and Warren (1963) suggest that this closure actually occurs and is responsible for the high transphosphorylating power of all of them.

A number of other biological guanidines have been isolated, some perhaps excretory products of phosphagens but mostly of uncertain significance. Dimethyl taurocyamine, or *asterubin*, from the starfish, might be a phosphotaurocyamine derivative; it raises the blood-sugar level of dogs and so probably has a sympathomimetic action. *Octopine*, formed after death in the tissues of the octopus and other molluscs, is a complex of arginine with pyruvic acid, a complex reminiscent of the arginosuccinic acid of the ornithine cycle (p. 183). *Agmatine* (Table 30), the α-amine formed by decarboxylating arginine, occurs in the sponge, *Geodia*, and in cephalopods; naturally it is a very strong base. A compound, *galegine*, G·CH_2·CH:(CH_3)$_2$, occurs in the primates, and is possibly an amidinated α-valamine (isobutylamine).

Guanidines are even stronger hydrogen-bonding agents and denaturants than urea. The halides of guanidinium break up DNA fibres, in which the two strands are held together by hydrogen bonds.

The biological guanidines appear to be all synthesised by transamidination from another guanidine to the amine-precursor, and this is the biological bond between amidines and guanidines. Arginine is probably the initial source of the amidine group and so may be compared with glutamine as an amino-donor and methionine as a methyl-donor. This donation of part of the side chain seems to be a rather general function of the amino acids.

Diguanidines occur naturally, for instance *hirudinone* $G{\cdot}(CH_2)_3{\cdot}NH{\cdot}(CH_2)_4{\cdot}G$, in leeches (Robin, 1962); it is effectively an amidinated spermine (Fig. 37, XV). The pharmacological value of laboratory diguanidines has been mentioned above; for maximal trypanocidal action a C_{10} link is required, as in *synthalin*, α, ω–decamethylene diguanidine, whereas the best antimalarials have the guanidine groups directly connected, as in *paludrine* (Fig. 37, XXXIV). Activity is maximal in this series when $R{=}C_4$, and the nature of the substituents on the aryl ring also is important (Sexton, 1953, p. 295), so that again virtually every component of the molecule is significant. In the active form of these antimalarials the diguanidine unit is probably ring-closed (Fig. 37, XXXV), forming a *triazine* compound (p. 345). The extensive amination and methylation of the ring resemble those of the pyrimidines and purines (p. 347).

16.5 Conclusions

There is a great contrast between the biological functions of the amino acids and those of the other open chain amino compounds. The amino acids are a large and varied group, used in large amounts as fabric material, both for other monomer molecules and for the great group of polymers, the proteins. By contrast few members of each of the other groups of amino compounds are biologically important, and most are used in small amounts, as non-expendable agents (apart from wear and tear, which does sometimes involve destruction after limited use). Compounds such as acetyl choline are absolutely unique and extremely powerful in action, active in concentrations as low as 10^{-12}. Urea and trimethylamine oxide are exceptional in being 'bulk' materials, with a mainly excretory function, but they again are unique members of their group. The pharmacological study of laboratory-synthesised members of the diamidines and diguanidines shows how each feature of a molecule contributes qualitatively or quantitatively to its action, so that for each particular purpose there is a unique, ideal molecule. A number of components other than the amino

groups contribute, though the latter are usually dominant: otherwise it might be questioned whether the complete compound should be considered in this chapter. For instance if stilbamidine had proved similar in properties to the simple stilbenes (p. 142) then it should have been considered there. However, in general it does seem that a nitrogen group dominates the properties of any type of molecule incorporating it, so that the order of treatment adopted in this book is justified also on functional grounds.

The synthetic diamidines and diguanidines also show how minor details of the molecule make it specific to one or another group of animals, and this may prove the explanation of the variety of phosphagens, as of some other groups of metabolite such as the bile acids. The guanidine group is absolutely specific to the main function while the varying side chain adapts it to the conditions in particular groups of organisms; these conditions may depend on the environment rather than on the genetic constitution of the group in question, but in any case they have applied the 'fine adjustment' in the natural selection of the phosphagens.

Compounds such as amides and tertiary amine oxides, which have oxygen or carbonyl directly attached to the amino group, are innocuous, neutral in reaction and very soluble, ideal as excretory forms for nitrogen. By contrast the alkyl amines and the polyamino hydrocarbons are strong to powerful bases, the strength increasing with the number of amino groups attached to one carbon atom and with the number of methyl groups bonded to one nitrogen atom. The former line culminates in the guanidine group of the phosphagens and the latter in choline. Primary amines also become strong bases and good neurotropic agents with the help of suitable aromatic accessory groups, as illustrated by adrenalin, histamine and serotonin. Secondary amines, like many other intermediary series, fall between two functional stools.

A number of the amino compounds have more than one distinct function. Both primary and tertiary monoamines also form the phosphatide type of conjugate, with its own unique properties, while the guanidines also have a function in urea synthesis. Choline has a distinct transmethylating role, as contrasted with the neurotropic action of the acetyl derivative and with the phosphatide function of the phosphoryl derivative. These seem great variations, at present, but eventually it may prove that they are just the differences to be expected from the differences in structure.

Among the interesting features of this general group is the frequent appearance of double-ended or cuff-link molecules; urea itself effectively has this structure, and still more triuret, and it is interesting that xanthydrol, the test reagent for urea, has a double-ended molecule, but doubly linked, like some of the curariform drugs. Spermine and spermidine

have essentially this structure. In addition to the doubly and singly linked curariform compounds there are the diamidines and diguanidines, including the naturally occurring hirudinone. Cuff-link molecules have already been encountered in the stilbenes, in indigo, in the carotinoids, and in other instances, and other examples will be seen later. The terminal groups are usually ring structures, however, whereas apart from tubocurarine most of the present group of substances have bootlace, or tassel, endings.

Some may form betaine-type or other rings, and in several respects the present group is intermediary between the open chain and the heterocyclic nitrogen compounds. Apart from the direct resemblance of the betaines, and stachydrine, homarine, etc., many of the group are agents with similar functions to the heterocylic nitrogen compounds to be considered in the following chapters. It may be noted that the ring-nitrogen atom is *ipso facto* at least a secondary amine, and in fact the important ones are tertiary, with the power of becoming a quaternary base.

CHAPTER 17

HETEROCYCLIC NITROGEN COMPOUNDS: PYRROLES AND PORPHYRINS

As in the carbocyclic (p. 127) and oxycarbocyclic (p. 42) series the only common *nitrogen*-heterocyclic rings in living organisms, as in the laboratory, are the penta- and hexacyclic members; these are the stable rings. There may be one or two, but rarely more, nitrogen atoms in the ring, though three, regularly alternating with carbon, are well known to the organic chemist. A larger number is very improbable for the reason that nitrogen does not form polymer chains. *Oxazoles* and *morpholones*, penta- and hexacyclic rings containing nitrogen and oxygen, are not common biologically, perhaps for the same reason that the amides are rare (p. 281). The betaine ring (p. 298) is of this type, and is rather labile. The two elements occur together more frequently in condensed ring systems, however. Thiazole, the pentacyclic ring containing nitrogen and sulphur, is biologically exploited more than oxazole, as will be seen (Chapter 23).

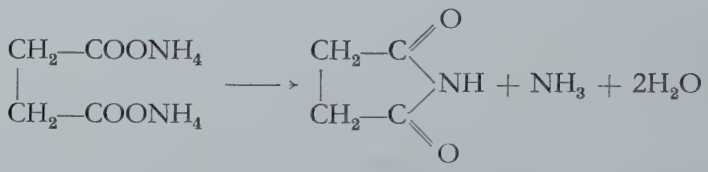
As already indicated, the alkaligenicity and general reactivity of the open chain amines is retained in ring compounds, and indeed is usually enhanced by resonance, because the unsaturated or aromatic series are the most common. They include some of the most powerfully resonant of biological molecules, and very readily show electron-transitions to an excited state (Glass, 1961). The porphyrins, to be considered in this chapter, are the most outstanding example. Unsaturated rings are usually *fully* unsaturated: once more the intermediate conditions are least stable. There is actually some sacrifice of alkalinity in the aromatic ring because as already seen (p. 131) this is always more acid than the corresponding open chain compound, but the open chain quaternary bases are too alkaline for some biological purposes, and the aromatic nitrogen compounds have also unique properties not shown by the aliphatic series.

The heterocyclic nitrogen compounds tend to be special agents in the body, rather than bulk materials like the amino acids and proteins; this might be regarded as a parallel to the contrast between aryl compounds and aliphatic hydrocarbons, such as the fatty acids. They do not polymerise extensively and it is significant that the only bulk materials and

high polymers among the nitrogen-heterocyclic biological compounds, the nucleic acids, polymerise through pentose and phosphoric acid, not through the nitrogen bases. As active agents the nitrogen-heterocyclic compounds are frequently conjugated in this way, in the form of nucleotides, or with proteins.

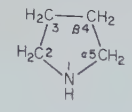
17.1 Pyrrolidines and Pyrroles

Pyrrolidine (Fig. 38, I) is the fully saturated mononitrogen, pentacyclic ring, occurring in proline (p. 179), in one ring of vitamin B_{12}, in *nicotine* (p. 339) and other alkaloids, and in *stachydrine*, already considered. The ring may be formed in the laboratory by heating ammonium succinate, a possible biological compound:

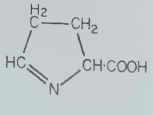


and it can also be synthesised from ammonia and simple hydrocarbons, using ultraviolet light or simpler catalysts (Oparin, 1957a, pp. 181, 202). *In vivo*, however, the proline ring is formed from glutamic acid, as already noted (p. 181), by initial reduction of its ω–carboxyl group to the aldehyde, after which ring closure is spontaneous, as it is in the aldose sugars. The ring compound first formed, pyrroline–5–carboxylic acid (Fig. 38, II), has one double bond, and subsequently moves to the equilibrium position of full saturation and not the other way to pyrrole.

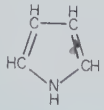
Pyrrolidine is a strong base but *pyrrole* (Fig. 38, III), the fully unsaturated ring, has its alkalinity so much reduced that it can act as a very weak acid, forming salts of the type KNC_4H_4. It might be supposed that this must be the basis for the formation of the metallo-porphyrins to be considered, but in fact in the iron-porphyrans iron acts as an electron-acceptor, i.e. as the acid. Pyrrole remains a stronger base than acid, therefore, and this is true also of the other unsaturated nitrogen-heterocyclic compounds. The charge density of the pyrrole nitrogen is 1·69 e.s.u., compared with the theoretical 2 units for the trivalent nitrogen atom, so that it is nucleophilic only to the extent of 0·31 units and still electrophilic to the extent of the 1·69 units (Finar, 1959, p. 719). The pyrrole molecule resonates between no fewer than five structures, one with no charge and the other four with a positive electrophilic charge on the nitrogen and a small nucleophilic charge on one or other of the four carbon atoms. In all four of these hybrids the nitrogen effectively acquires



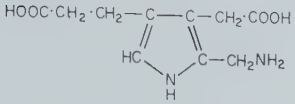
I
Pyrrolidine nucleus



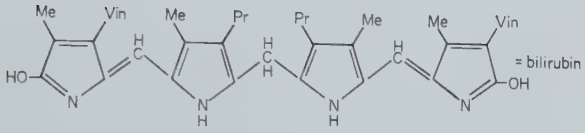
II
Pyrroline–5–carboxylic acid



III
Pyrrole nucleus

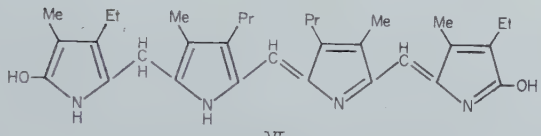


IV
Porphobilinogen

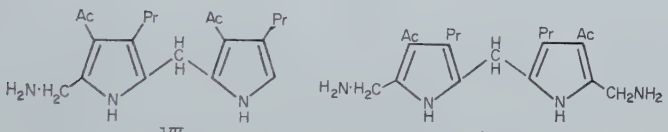


V

Me (methyl) = $-CH_3$
Vin (Vinyl) = $-CH=CH_2$
Pr (Propionyl) = $-CH_2 \cdot CH_2 COOH$
Et (Ethyl) = $-CH_2 \cdot CH_3$



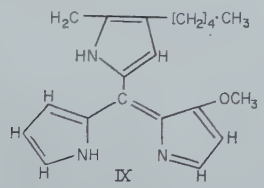
VI
Phycoerythrin



VII
Dipyrryl A

VIII
Dipyrryl B

(Ac = Acetyl $-CH_2COOH$)



IX
Prodigiosin

FIG. 38 Pyrrole derivatives of biological importance (*continued on pp.* 309 *and* 311).

one double bond, by the use of its lone pair of electrons. In consequence it cannot accept a proton to become quaternary and so is never a strong base.

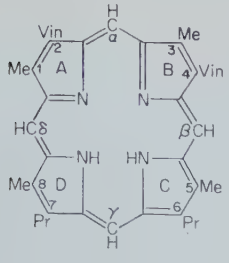
It can be synthesised in the laboratory by heating to 200°C a mixture of ammonium mucate and glycerol, both biological materials:

$$H_4N{\cdot}OOC{\cdot}[CHOH]_4{\cdot}COO{\cdot}NH_4 \rightarrow C_4H_5N + NH_3 + {}_2CO_2 + 4H_2O$$

It is interesting that the oxygen analogue, tetrahydro-furan (p. 144) also can be synthesised in this way, from mucic acid (Finar, 1959, p. 711). If the mucic acid is esterified with amines instead of ammonia then substituted pyrroles are formed, and it is substituted compounds which are important biologically. However the main pathway of pyrrole biosynthesis differs from this laboratory method, as in so many other cases. Further, its condensed ring systems, such as indole (p. 189), are biosynthesised by entirely different routes from that of the simple pyrroles, by ring closure between two side chains of anthranilic acid (p. 187). Not only this but some indoles, such as dopachrome (p. 188) and melanin, form their condensed pyrrole ring by closure through the single side chain of tyrosine. The indoles have been adequately considered in connection with tryptophan (p. 189).

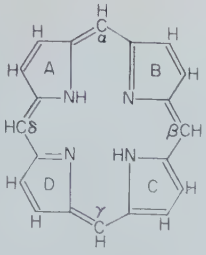
Free pyrroles are not common in living organisms, although the ring is very stable, and has survived hundreds of millions of years in fossils and other biological products. It has survived in an oligopolymeric form, although some free pyrrole can be obtained by heating coal tar or bone-oil to 100–150°C; being volatile it distils over as a vapour. Again, the free monomer, *porphobilinogen* (Fig. 38, IV), occurs in the urine in some pathological states but in normal biosynthesis it polymerises spontaneously. The pyrrole monomer can be synthesised in the laboratory by mixing ammonia and acetylene at red heat, as well as by the method described above, but it will polymerise readily in the laboratory also; concentrated acids convert it into a resin, which indicates a high polymer, so that there are exceptions to the generalisation above (p. 305). By contrast pyrrolidines do not polymerise. The traditional laboratory test for pyrroles is that in vapour form they give a cherry-red colour with pinewood soaked in concentrated hydrochloric acid; this may be due again to a polymer, *pyrrole red.*

The pyrroles are insoluble in water, because of their weak ionisation, their high carbon/nitrogen ratio, and the aromatic character of the ring. They contrast sharply with the pyrrolidine compounds such as proline, therefore (p. 179). However, the pyrroles do resemble proline in being alcohol-soluble; they are also soluble in ether.



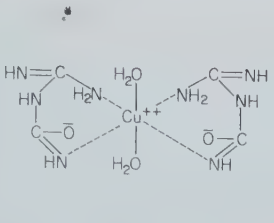
X

Protoporphyrin IX



XI

Porphin nucleus

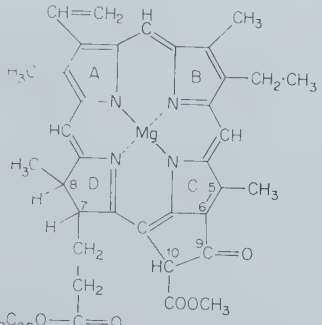


XII

Copper di [monoimino biuret]

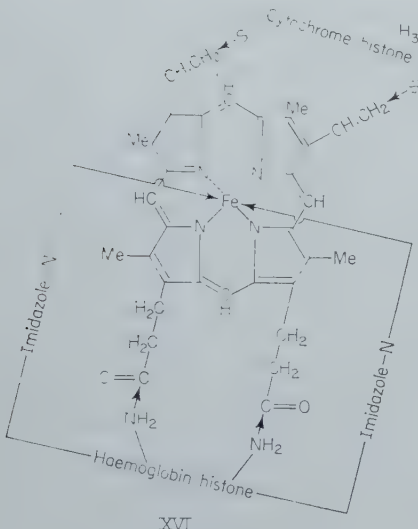
XIII

Octahedral arrangement of six bonds



XIV

Chlorophyll a



XVI

Bonding between haem and protein in cytochromes and haemoglobins

FIG. 38 (*continued*).

L*

17.2 Open Chain Polypyrroles

Open chain *tetrapyrroles* occur in living organisms, but in some cases at least these are secondary products of the porphyrins (below) in which a tetrapyrrole unit has closed to form a second-order ring. *Bilirubin* (Fig. 38, V) and *biliverdin* are reddish and greenish pigments of vertebrate bile, where they are probably primarily excretory products of the blood porphyrins. However, there is some evidence that they have been put to an opportune use here, in connection with digestion. Some molluscs use them as pigments of their shell (Comfort, 1951), and some birds of their egg-shell, that is for purposes of camouflage. They may have active chemical functions also; biliverdin seems to be a normal component of the molecule of catalase (Sexton, 1953, p. 166), an enzyme which already has a porphyrin as prosthetic group. As will be noted, the methylene and single bonds of the middle link of bilirubin break up the molecule into two separately resonant sections, but biliverdin is freely resonant throughout the molecule and should be a more potent activator.

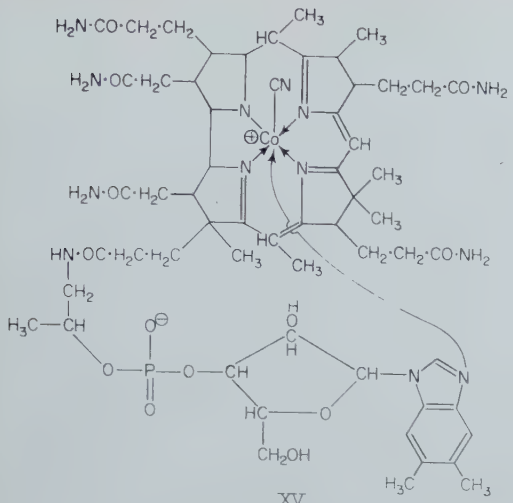
In plants there are open chain tetrapyrrole pigments, the *phycobilins*, which seem to be even more important as agents in their own right, acting as auxiliary pigments in photosynthesis (Florkin, 1960, p. 291). *Phycoerythrin* (Fig. 38, VI) is the pigment in the red algae and *phycocyanin* in the blue–green group; they are clearly related to the bile pigments and have the same near-symmetry. They are conjugated with protein and probably absorb light energy in a region where chlorophyll itself is deficient (p. 117), subsequently passing it on to the latter at a more suitable wavelength.

The tetrapyrroles have characteristic and limited solubility properties; biliverdin is insoluble in water, ether and chloroform. This no doubt facilitates their deposition in shells, but also in the tissues, in jaundice and in bruises. Perhaps for this reason the porphyrin ring is opened while still attached to protein, in some cases.

Tetrapyrrole seems to be a stable stage, and both smaller and larger polymers are rare. Both di- and tripyrroles are formed as intermediary stages in the synthesis of porphyrins and there is some suggestion of a definitive tripyrrole in the blood pigment, *haemerythrin*. A special tripyrryl compound occurs in *Bacterium prodigiosum* (below), but this has probably more affinity with the porphyrins than with the open chain group.

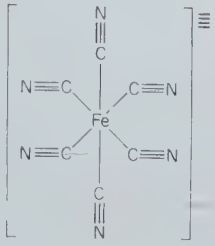
17.3 Porphyrins

These ring-closed tetrapyrroles are the commonest biological pyrrole derivatives, and this depends largely on the great stability of this second-order ring, which is also more compact than the open chain tetrapyrrole.



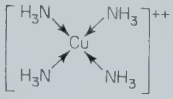
XV

Cyanocobalamin (vitamin B_{12})



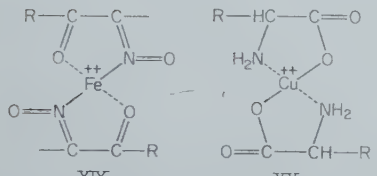
XVII

Ferrocyanide tetranion



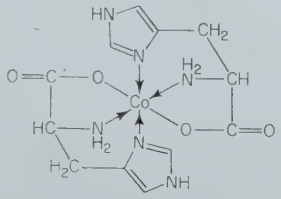
XVIII

Copper tetrammine



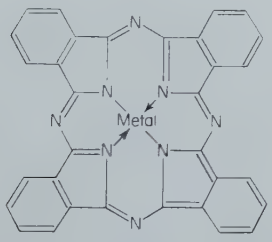
XIX XX

Coordination of metals by N and O



XXI

Synthetic cobalt–histidine O_2–carrier



XXII

Phthalocyanin nucleus

FIG. 38 (*continued*).

New properties also emerge and the porphyrin is able to chelate a metal atom between its four pyrrole nitrogen atoms. The resulting *porphyrans* are compounds with remarkable properties, and a molecule of great geometric beauty (Figs. 38, XIV, XVI).

The biological porphyrins also bear a number of functionally significant side chains on the pyrrole rings, and these are present from the very early stages of biosynthesis, before the completion of the porphyrin ring. Biosynthesis therefore seems to show 'provision' and occupies fewer steps than might have been anticipated from the final complexity of the molecule. This is largely true of other large, complex biological molecules: it is extremely uncommon for a naked nucleus first to be formed and then subsequently to be substituted in a number of separate positions. It is of course possible that many of the side chains are incidental to the starting materials and to the mode of synthesis and that function may have evolved so as to make use of them, retrospectively; on the other hand it is perhaps equally possible that the weight of natural selection has been upon starting materials and on biosynthetic pathways which shall give the functionally ideal final product, so that biosynthesis has often come to show provision. Certainly each side chain and other detail does now seem to be an essential part of each porphyrin. Equally the biosynthesis has become a very stream-lined piece of construction, notwithstanding the complexity of the product, and it is difficult to understand the uniqueness of such compounds without giving some attention to their biosynthesis (Shemin, 1955).

It is interesting to find that a number of the enzymes catalysing steps in the biosynthesis are themselves porphyrino-proteins (Calvin, 1962), so that there is a form of partial and indirect autocatalysis, which is by no means uncommon in biosynthesis, and indeed is inevitable. At the same time it is too partial and indirect to lead to uncontrolled positive feedback. Synthesis begins by the condensation of glycine with active succinic acid, i.e. *succinyl-coenzyme A* (p. 93):

$$H_2N{\cdot}CH_2COOH + HOOC{\cdot}CH_2{\cdot}CH_2COOH \longrightarrow HOOC{\cdot}CH(NH_2)CO{\cdot}CH_2{\cdot}CH_2{\cdot}COOH$$

The product is α–amino–β–keto–adipic acid, and it is noteworthy that adipic acid is an amino acid which occurs free in some organisms. When decarboxylated at the α–position the product is ∂–amino laevulinic acid (the conventional lettering involves an unfortunate reversal here), and two molecules of this condense to form *porphobilinogen* (Fig. 38, IV), already a stable pyrrole. There must be a keto–enol change in the γ–C=O group of each laevulinic unit, to transfer the double bond to the main chain (as in isoprene synthesis, p. 107), and at the same time to provide

hydroxyl groups for the condensation with the other unit. The double bond is transferred to the two alternative positions in the respective chains of the two units and the completed porphobilinogen has several features of asymmetry. The ∂–amino group of one laevulinic unit becomes the pyrrole nitrogen, while that of the other is later used to link the pyrrole to a second porphobilinogen molecule (Fig. 38, VII). The link is formed at the α–position (I) of the second pyrrole ring, since this is more nucleophilic than the β–positions (Finar, 1959, p. 719).

Two types of dipyrryl unit are formed, *A* with both pyrroles facing the same way (Fig. 38, VII) and *B* with them facing in opposite directions (Fig. 38, VIII), and with a second methylamine side chain symmetrically on the second ring. This must have come from a third porphobilinogen unit and it is thought that tripyrryl methane is first formed, as an intermediary. Such a compound does occur as a definitive pigment in *Bacterium prodigiosum* (Fig. 38, IX), a saprophyte able to grow on bread, where its brilliant red pigment is said to have given rise to the legend of the bleeding host (Baldwin, 1964). In porphyrin synthesis, however, the unit which donated its methylamine side chain to form the initial bridge of this tripyrryl intermediary is later removed, leaving dipyrryl *B* in which neither unit has made this sacrifice. Dipyrryl *B* can finally condense with another *B*, or with an *A*, and again two *A* dipyrryl units can condense, forming three possible types of porphyrin. The *A* unit can condense either way round so that in all there are five types of porphyrin. Only *AA* and *AB* combinations are important biologically, and are known as series I and III respectively. In fact most of the known biological porphyrins are derivatives of a single member of series III, *protoporphyrin* (Fig. 38, X).

In the completed porphyrin one pyrrole ring is saturated as far as the intermediary *pyrroline* stage [II], one of the number of minor asymmetries of the molecule, and all the bridges have become desaturated to *methyne*, =CH—, units. The side chains become modified in a number of ways. In protoporphyrin the acetyl side chains at 1:3:5:8 have all been shortened to methyl, while the propionyl chains at 2:4 have been shortened and desaturated to vinyl, —CH=CH_2. Shortening and desaturation occur more easily than the attachment of new chains to a fully formed ring. With its three types of side chain there are 15 possible isomers of protoporphyrin alone, but the biological derivatives probably all come from one of these, known as protoporphyrin IX (Fig. 38, X). The methyne bridges are probably present in all biological porphyrins, though the one between pyrroles *C* and *D* can be replaced by =N— in the enzyme peroxidase without greatly depressing its activity, so that the bridges are not absolutely specific. The nitrogen-bridged porphyrin is known as *azahaemin*.

The plant chlorophylls (Fig. 38, XIV) also are based on a porphyrin of series III, differing from protoporphyrin in having ring *C* partly saturated, as well as *D*, and in the modification of the propionyl side chains at 6:7. However, it may be a secondary modification of protoporphyrin IX, or of a common precursor (Calvin, 1962).

The porphyrin molecule combines complexity with a remarkable geometrical pattern. The four pyrrole rings and the bridges lie virtually in one plane, as in the diagrammatic representation, and as already noted (p. 116) this is ideal for resonance activity. In spite of the asymmetry of the individual porphobilinogen units the molecule as a whole approaches a tetraradial symmetry; there is always some degree of asymmetry, however, probably for the same reason (p. 115), i.e. that this is essential for maximal resonance activity. This may also be the reason why the symmetrical *BB* dipyrryl conjugates do not occur biologically: they may have been excluded by natural selection. Series I porphyrins have a general rotary symmetry and this may be why series III alone has been extensively exploited; at the same time series I derivatives can acquire asymmetry in the pattern of the double bonds and in their side chain arrangements.

With the unsaturation of one bond of each bridge, and never both, the porphyrin become fully resonant, and an =N— bridge preserves this property. In the *porphin* nucleus (Fig. 38, XI) every atom except two of the pyrrole nitrogens has one double bond. Double and single bonds regularly alternate along almost every path through the molecule, except through these two nitrogen atoms. The resonance energy of the molecule, and its ability to activate other molecules, therefore, is exceptionally high, and it is brilliantly coloured, absorbing and using energy of relatively low quantal value, equivalent to that of light in the visible spectrum. The cytochromes transmit yellow and therefore absorb in the violet, but other porphyrins are able to utilise light of longer wavelength and the residual spectrum ranges through the reds and purples of the haemes to the green of chlorophyll. There is usually a particularly strong absorption band around 420 mμ, in the violet, the *Soret band* of their absorption spectra, but additional strong absorption bands in the green–yellow region cause haem to transmit red except when it is in very thin layers, while chlorophyll absorbs most strongly in the red. The brilliance, i.e. unsaturation, of the colours is due to the sharpness of the absorption bands, which in turn show that the molecule has its activating power sharply focused. A number of electrons in the molecule have activation energies of very nearly the same value, so that they act as a team in activating other molecules. This is probably the most unique and important property of the group of compounds, and familiarity with the

absorption spectrum should not blunt our sense of wonder at it. The Soret band is due to the porphyrin ring and disappears when it is opened; the other bands are due partly at least to the metal, and to the protein, conjugated with it in its active biological form.

Like other pigments the porphyrins can handle energy from chemical sources also, and this explains the paradox that haemoglobin and other brightly coloured derivatives may never see the light of day in the bodies of heavily furred or feathered amniotes or in heavily pigmented animals. Indeed their photodynamic power can be troublesome when they are exposed, as in the pathological *erythraemic* responses to excessive insolation. This may be a good reason why the porphyrin ring must be opened for excretory purposes. In chlorophyll, of course, the photodynamic response is the essential physiological action and this may be true also for some other porphyrins.

Notwithstanding its great reactivity the porphyrin nucleus is very stable, as might be anticipated from its compact, flat form and its extensive resonance. Porphyrins therefore are still abundant in petroleum and other ancient and much-processed deposits of biological origin. It is estimated (Abelson, 1957) that the molecule has an average life of billions of years under these conditions. It is not attacked by digestive enzymes and may be heated to 400°C before it begins to decompose.

Nevertheless porphyrins rarely occur free in living organisms, but are usually conjugated with protein. A number of integumental pigments of animals, particularly among molluscs (Comfort, 1951), appear to be free porphyrins (Fox and Vevers, 1960); some of these are derived from the chlorophyll of plant-food and most of them may be secondary products of this type, one implication being that conjugation with protein is no longer necessary when a porphyrin has ceased being metabolically active. However, some of the integumental porphyrins do appear to be conjugated with protein, like some of the open chain tetrapyrroles. The attractive purple pigment of *Aplysia*, the sea hare, and the brown integumental pigment of the earthworm, *Lumbricus*, are examples of such porphyrino-proteins.

17.4 Porphyrans

As already indicated, the most important porphyrins, metabolically, are metallo-porphyrins or *porphyrans* (Fearon, 1949, p. 209), the metal being chelated between the four pyrrole nitrogen atoms, and sometimes also between two groups of a protein conjugant. The outstanding examples are iron-protoporphyrin, or *haem* (Fig. 38, XVI), and magnesium protoporphyrin or *chlorophylline*, the basis of chlorophyll (Fig. 38, XIV). The metal

holds a strategic position symmetrically in the centre of the molecule, and it is also the functional hub of the structure; in view of this and of the general strategy of porphyrin synthesis it might have been expected that the pyrroles would be assembled round the metal atom, but in fact in the biosynthesis of haemoglobin it is probably inserted not only after the porphyrin is complete (Fig. 39) but even after this has been conjugated with the protein, globin (Erikson, 1957). This shows, incidentally, that the bonds which subsequently form between protein and metal, here and in the cytochromes, are not the only bonds formed by the protein (Fig. 38, XVI). Once formed the metal pyrrole bonds are extremely powerful, so strong that they cannot be measured by existing methods (Albert, 1961), and it is possible to open the porphyrin ring with the metal still chelated, as in *verdohaematin* and *verdohaemin* (Fearon, 1949, p. 219). *Verdoglobin* has this structure and also has the protein moiety still attached. It is all the more surprising that the metal is inserted so late in biosynthesis, therefore, but the reason for this may be teleonomic; namely that once the metal *has* been inserted it is difficult to modify the molecule further, because of its stability. In the haem pigment of the bacterium *Haemophilus* the vinyl side chains of the porphyrin appear to play an essential role in the process of inserting the iron into the porphyran.

Copper, manganese, cobalt and other metals can form porphyrans, but only those of iron and magnesium are really stable (Calvin, 1957) and this no doubt accounts for their special exploitation by living organisms. At the other extreme sodium and potassium form such weak bonds that they are spontaneously exchanged for hydrogen (Krasnovsky, 1957). The red pigment, *turacin*, in the feathers of the turaco bird is a copper-porphyran; it is water-soluble, unlike most integumental pigments, and after a storm drips off the feathers like blood. It may therefore have some curious psychological defensive function but it is perhaps significant that there are no copper-porphyrans among more serious metabolites, although there are plenty of copper compounds. A number of these are functional analogues of the iron-porphyrano-proteins so that it is possible that copper forms labile pseudo-porphyrans, of the biuret type (p. 288). It does form this kind of complex (Fig. 38, XII) spontaneously with amino acids and their derivatives (Rising and Yang, 1933), such as monoimino biuret. Vitamin B_{12} (Fig. 38, XV) is a complex cobalt-porphyran derivative (p. 321) and vanadium porphyrans have been found in coal but here the metal may be a replacement for iron (Fox and Vevers, 1960). In the enigmatic vanadium chromogen of the blood of tunicates, vanadium may be associated with an open chain of pyrroles (Webb, 1939).

Interaction between metal and porphyrin is of course fundamental to the action of the porphyrans. A striking demonstration of this is the way

in which iron quenches the brilliant crimson fluorescence of the free porphyrin in ultraviolet light. Energy absorbed by the molecule in that range of high quantal value, and otherwise squandered by re-emission in the far red, therefore is trapped by the metal and made available for useful purposes. Magnesium does not seem to have this effect and chlorophyll retains the fluorescence of the free porphyrin; the main absorption of this pigment is of the longer visible waves, as already remarked, and it conceivably reabsorbs for use its own fluorescent emission due to the action of the small amounts of ultraviolet in sunshine at low altitudes. It may be noted in passing that proteins also can quench the fluorescence of metal-free porphyrins, and that some of the integumental porphyrino-proteins liberate brilliantly fluorescent porphyrins.

Iron-porphyrans are very active as coenzymes and as other types of catalyst, and this may be connected with their suppressed photodynamic action, since the magnesium-porphyrans are strongly photodynamic but on the other hand are poor catalysts for reactions using chemical energy (Krasnovsky, 1957). This is perhaps why the only significant magnesium-porphyrans are the chlorophylls, whereas there is a considerable range of iron-porphyran enzymes. It is possible to prepare porphyrans of Cu, Ni, Mn and Co and to conjugate these with the natural apoenzyme of peroxidase (Sexton, 1953, p. 166), one of the haem-proteins, but only the manganese compound proves at all active; even this has only a fraction of the activity of the natural iron compound. The iron-porphyrans therefore are absolutely unique as enzymes of this type, and they are far outstanding also as oxygen-carriers—their second main catalytic function.

In nearly every case the prosthetic group is *haem*, or iron-protoporphyrin (Fig. 38, IX), and differences are due to the protein moiety. The exceptions are *chlorocruorin* and cytochrome a_3 (cytochrome oxidase) which have *chlorocruorohaem*, differing from haem itself in having the formyl, —HC=O, group in place of vinyl at position 2. Chlorocruorin is the oxygen-carrying pigment of the blood of sabellids and some other polychaetes; it is green in colour and does not change colour on oxygenation, as haemoglobin does, so that this small difference in the porphyran has a considerable effect on its properties.

It has been noticed that in the haemoproteins the iron has a conjugation number, or apparent valency, of 6, much higher than in its simpler inorganic compounds. It has in fact the same valency properties as in the ferro- and ferricyanides (Florkin, 1960, p. 114) which may be considered simple organic substances. They are such powerful poisons largely because of this analogy to key metabolites. Along with other simple models they can be used to illustrate a number of the properties of the iron-porphyrans (Glass, 1961).

This facultative higher valency of iron is a latent property of most of the heavier metals, and particularly those in the so-called transitional group. This is because the electronic complement of the outer shells of their atoms is not rigidly fixed. The energy levels of orbitals near the periphery of the atom are so nearly the same that electrons move very readily between them as the atom receives or loses a small amount of energy. The atom of these metals has already some electrons in a new outermost shell, although the outermost orbitals of the previous shell have not been completely filled. Iron has four of these unfilled orbitals, and it tends to fill them all at the expense of any available electron-donors; at the same time, however, it tends to fill further orbitals in the new outermost shell, up to the stable stage of four pairs. In all atoms the innermost, *K*, shell holds only two electrons, the next, *L*, shell 8 and the third, *M*, shell 18. Iron, with an atomic number of 26, has shells *K* and *L* therefore completely filled, 14 electrons and four 'holes' in the *M* shell and two electrons in the fourth, *N*, shell. These two are the valency electrons of ferrous iron, readily donated to electrophilic groups to form ferrous salts. The four outermost orbitals of the *M* shell have each one unpaired electron, and it is these unpaired electrons which make the atom so strongly paramagnetic—so strongly that ferro-magnetism stands in a class of its own, including in addition only cobalt and nickel, the other two members of the same period in the transitional group of the periodic system (Table 1). When the four orbitals in question of ferrous iron are filled (two electrons each), magnetism disappears, and so acts as a useful indicator of the state of the atom; it becomes in fact weakly diamagnetic, repelling the lines of magnetic force.

Iron becomes the ferric, or tervalent, cation through donating a further electron from the *M* shell, so making a total of five unpaired electrons in that shell, and ferric compounds therefore are more strongly paramagnetic than the ferrous series. Iron is a good redox agent, over a certain range of potential (Table 37), because of the ease with which the third valency electron moves into and out of the atom. Both ferrous, Fe^{++}, and ferric, Fe^{+++}, ions have the same maximal coordination number, however, and accept up to 12 electrons, 4 into the *M* and 8 into the *N* shell. One unpaired electron therefore always remains in Fe^{+++} and this ion is always weakly paramagnetic. The coordination complex of 6 bonds, corresponding to the 12 electrons received, forms a very stable octahedral structure with four bonds in one plane, the equator of the octahedron, and two perpendicular to this (Fig. 38, XIII), the six being mutually at right angles, therefore. They are known as 'hybrid' bonds because they involve electrons in two different shells.

Ferrocyanides are diamagnetic and ferricyanides weakly paramagnetic,

so that they are typical coordination complexes. The iron-porphyrans, however, remain strongly paramagnetic and therefore the six groups bonded must donate their electrons to orbitals in the fourth shell, the bonds being essentially ionic (Coryell, 1954). However, when oxygen and certain other groups replace one of the six ligands the ferrous iron atom is displaced from this stable state to one essentially covalently coordinated. Four unpaired electrons in 3*d* orbitals become paired and the paramagnetism is quenched. The atom and molecule therefore switch over from one relatively stable state to another as they are oxygenated and deoxygenated, and there is no direct resonance between the two. Under appropriate conditions each change is spontaneous, and 'all-or-none'.

Magnetism is also quenched when various nitrogen compounds occupy the two positions not held by the pyrrole nitrogens, as for instance in the so-called haemochromogens (Fig. 39) synthesised in the laboratory. The

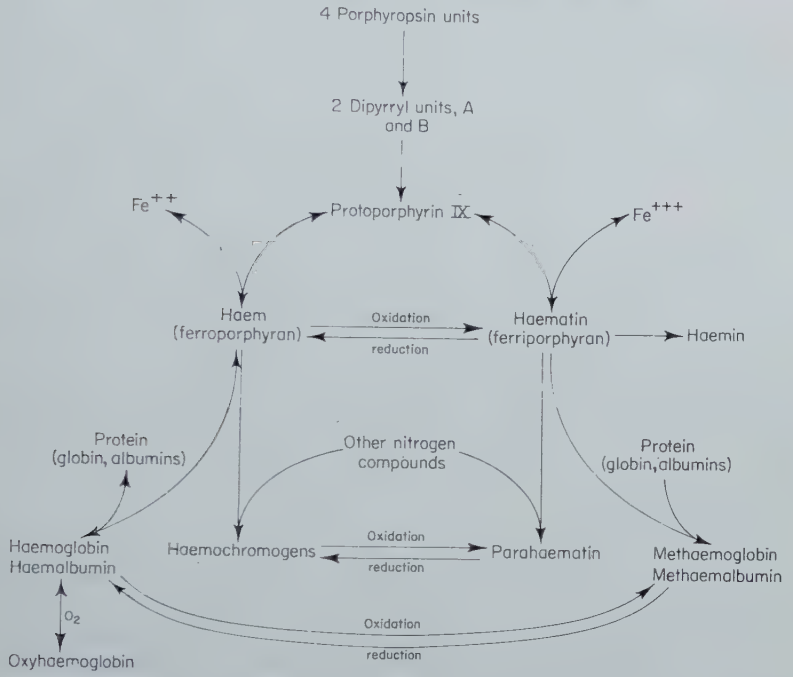


FIG. 39 Diagram of the main steps in the synthesis of haemoglobin and of the more interesting laboratory derivatives of the pathway. The diagram emphasises the parallel between the physiological ferrohaem and the laboratory ferri-haem series and the interconversions between them at all levels. The scheme may incorrectly represent the point at which Fe is added in the biosynthesis of haemoglobin (p. 316), but is correct for its release in laboratory degradation. Based partly on Carter *et al.*, 1959.

natural protein, globin, however, although it holds these two bonds through its histidine imidazole nitrogens, does not quench but preserves the electrostatic type of bonding of the free haem, where the two bonds in question are held by water molecules. Oxygen displaces one of the electrostatically held imidazole nitrogen atoms and immediately the bond with the other becomes coordinate, showing how the electron shifts do indeed affect the whole molecule. The contribution of the protein to this was considered in Chapter 15, in connection with enzyme-action. Rather surprisingly the iron remains ferrous throughout the oxygenation cycle; the oxygen does not become electrophilic and ionic, and the process is not an oxidation.

The change in the imidazole bond upon oxygenation makes the effective p*K* of haemoglobin more acid (Haldane effect) and this drives out the weak acid, carbon dioxide, which is carried back from tissues to respiratory organ partly in combination with the haemoglobin. Reciprocally the loading of carbon dioxide at the tissues is promoted by de-oxygenation and the rise in p*K*, a very neatly automatic piece of biochemical adjustment.

It is possible to oxidise the ferroporphyran, haem, in the laboratory, to the ferriporphyran, *haematin* (Fig. 39), the iron remaining paramagnetic. If the two water molecules are now replaced by nitrogen bases four-fifths of the magnetism is quenched, as in the ferricyanides. If the complete haemoglobin molecule is oxidised in this way, to *methaemoglobin* (MetHb) it will no longer load with oxygen, but remains in the electrostatic, strongly paramagnetic, state. Presumably the globin imidazole residues permanently protect their positions now and prevent the completion of the coordination complex. The extra positive charge on ferric iron gives it less affinity for the nitrogen bases than ferrous iron and so haematin forms salts with such anions as Cl^- (haemin).

17.4.1 Chlorophylls

Chlorophylls, the main pigments of photosynthesis in green plants and some bacteria, are magnesium porphyrans (XIV), the porphyrin being somewhat different from that of haem (p. 314). It will be seen that in fact one propionyl side chain ring-closes with the γ-methyne carbon and the nucleus is therefore known as *phorbin* rather than porphin. The other propionyl side chain is esterified with a C_{20} diterpene alcohol, *phytol*, already considered (p. 112). The vinyl and ethyl side chains are transposed relative to those of haem and the position of the double bonds is different. With magnesium in place of iron, therefore, the complete porphyran differs considerably from haem and chlorocruorohaem. Without the phytol moiety the porphyran is known as *chlorophylline*, and the magnesium-

free porphyrin is *phaeophorbide*. It is possible to remove magnesium without disturbing the phytol, producing *phaeophytin*. Terminology is complicated by the term magnesium chlorin as an alternative to magnesium-phorbin.

With an atomic number of 12, magnesium has its *K* and *L* electronic shells full and two electrons in shell *M*. This is a stable configuration and magnesium does not form coordination complexes of such high value as that of iron. Its partial valencies are fully saturated by the four pyrrole nitrogens and it does not complex further with oxygens or other substrates: as already emphasised, magnesium-porphyrans do not form oxygen-carriers or respiratory enzymes, like the haems. On the other hand they are several thousand times more efficient than the haems in trapping light energy (Calvin, 1962) and the electronic structure of magnesium seems ideal for the separation of electrons from the chlorophyll molecule, an indispensable step in photosynthesis.

Chlorophyll *a* (Fig. 38, XIV) is the main form in all green plants but in addition some have a smaller quantity of another variant, of which several are known. The significance of the variants is not yet known, but in view of the number of other auxiliary pigments associated with chlorophyll they probably have a useful function. The chlorophylls absorb light in eight distinct bands in the visible spectrum, and the position of these differs in the different variants. Chlorophylls *a* and *b*, the two forms in angiosperms, together absorb much of the light of the longer visible wavelengths, and it is thought that the associated carotinoids (p. 117) absorb more of the shorter rays and pass on the energy to chlorophyll. The phycobilins behave similarly. It is interesting to find that the haem-protein, cytochrome, also is involved and may be the initial electron-donor to excited chlorophyll (Calvin, 1962); it is possible that the haems were in fact the original photosynthetic pigments and that the magnesium-porphyrans have evolved from them.

17.4.2 Cyanocobalamin (Vitamin B_{12})

This impressive molecule (Fig. 38, XV) marks the zenith of complexity in monomer molecules known from living organisms. It is larger, more heterogeneous and more chemically active than the aphins (p. 142). Such a structure implies a highly unique function. It is seen to be a cobalt-porphyran analogue of the haemochromogens, one of the two additional nitrogen atoms being that of the cyanide radical! The other is a benzimidazole analogue of the nucleotides (p. 355). The phosphate end of this nucleotide also is bonded to the porphyrin, via one of its side chains. The side chains are another very distinctive feature of the molecule; seven of these are acetamide and propionamide units, and it is one of these which

is bonded, via a hydrocarbon fragment of obscure significance, with the phosphate. Further all four pyrrole rings are extensively saturated so that there is a second side chain, a methyl group, at a number of the β–positions of the rings. There is also a methyl side chain on two of the bridge carbons. Further, pyrroles A and D are connected directly and not via a methyne link. In all, the indication is that its biosynthetic pathway is considerably different from that of the protoporphyrins, though it has the appearance of being a series III type of porphyrin. Conceivably asparagine rather than succinic acid is the initial precursor, so that the amide side chains are present from the outset.

The negative charge on the remaining free hydroxyl group of the phosphoric residue balances the residual positive charge on the cobalt atom and the whole probably resonates between this dipole and the neutral state. The cyanide group is readily exchanged for other radicals of much the same type as those which bond at this position with the iron-porphyrans, for instance OH, NO and SCN. The hydroxy compound, like haematin, acts as a base and forms salts with hydrochloric acid and sulphuric acid (Welch and Nichol, 1952). Vitamin B_{12} is concerned physiologically with the transfer of C_1 units, particularly in the biosynthesis of the purines (p. 347), and it seems possible that the C_1 unit is conjugated with the porphyran in this position.

Elucidation of the structure of B_{12} is a rather recent triumph (Hodgkin, 1955) and little is yet known of the functional significance of its components. The relatively few double bonds in the molecule as a whole is rather surprising, but they appear to permit considerable resonance since its transmission colour is a deep red, absorbing around 550mμ as well as in two bands in the ultraviolet. It is known that the benzimidazole moiety is essential and that antimetabolites of this, such as dimethyldiaminobenzene, are toxic to organisms which synthesise the vitamin (Wooley, 1953a). As might be expected of such a metalloporphyrin it has some redox activity (Huennekens and Osborn, 1959), and there is reason to think that it transfers C_1 units only in a particular state of oxidation. Variants of the vitamin have adenine (p. 347) in place of the benzimidazole (Harrow and Mazur, 1962) and may have an adenine nucleoside at the sixth position on the cobalt. It is a molecule full of interest, and at present of mystery.

As a vitamin, B_{12} is extremely potent, one microgram per day being adequate for blood-cell formation in man. Its uniqueness is further emphasised by the fact that it is the only significant physiological organic cobalt compound known (Kleiner and Orten, 1958). It is known as the haematinic principle because it is required so specifically for haemoglobin synthesis. This might seem in keeping with its elaborate porphyran structure but in fact it does not seem to promote the synthesis of por-

phyrins, and its action on haemoglobin synthesis is thought to be indirect, on the synthesis of the nucleic acid required in turn for the synthesis of the globin moiety. Other plasma proteins (Hsu, 1953), and also general growth, suffer from any deficiency of B_{12}, so that this view is probably correct.

17.5 Porphyranoproteins

In its active form cyanocobalamin is probably conjugated with a protein and chlorophyll is conjugated with a protein of molecular weight 265,000. The bond is labile, however, and contrasts with the strong link in the haem-proteins. This may reflect a greater importance of the protein moiety in the latter, in which a single porphyran has radiated to fill a large number of metabolic niches, due entirely to a variable protein conjugant, therefore. There are relatively few chlorophylls and cobalamins and so there has been less need to distinguish between the functions of prosthetic and protein moieties. The general virtues of conjugation with a protein (Chapter 15) may be assumed to apply here, but there may be also special functions, as there are in the haem-proteins.

Like free pyrroles and other weakly polar nitrogen bases the porphyrins are insoluble in water and in fact free chlorophyll is best extracted with such apolar solvents as petroleum ether; even iron-porphyrans are most soluble in such weakly polar solvents as an acetic–ether mixture, and strong acids or alkalis are necessary to dissolve haem in aqueous medium. The complete haem-protein, cytochrome, however, is very soluble in both lipid and aqueous media, and this is probably quite general. It is thought that in addition the protein associated with chlorophyll may act specifically as an enzyme for transferring inorganic phosphate to bond with ADP (p. 404), in the photosynthesis of ATP (Sexton, 1953, p. 172; Calvin, 1962).

Among the haem-proteins it has already been seen that the protein may speed up the rate of action of the porphyran enormously, and that on the other hand it depresses photodynamic action and fluorescence. It affects the position of the absorption bands and controls the substrate-specificity of those which are enzymes. More specific effects may be considered in connection with the individual members. These fall into two main groups, the oxygen-carrier pigments and the redox enzymes: it is scarcely necessary here to distinguish between those based on haem and the few on chlorocruorhaem. Haemoglobin (Hb) (Fig. 39) is the blood carrier-pigment of vertebrates and various invertebrates. *Myoglobin* of muscle and other tissues differs in having a higher affinity for oxygen, and in fact usually acts as a store of oxygen between blood and tissues.

The most important of the enzymes are the cytochromes, the main terminal oxidases of the cells of all aerobic organisms (Fig. 20). A number of cytochromes are now known, and usually several of them work in series in any one cell; their redox potential-ranges are nicely graded (Table 31), no doubt due to their different protein moieties, so that they

TABLE 31. OXIDATION–REDUCTION POTENTIALS (E_o') OF BIOLOGICAL SYSTEMS (POTENTIAL, RELATIVE TO NORMAL HYDROGEN ELECTRODE, WHEN HALF OXIDISED, AT pH 7·0)

System			E_o (volts)
$O^=$	⇆	$\frac{1}{2}O_2$	+ 0·81
Cytochrome oxidase^{++}	⇆	Cytochrome oxidase^{+++}	+ 0·60
Cytochrome a^{++}	⇆	Cytochrome a^{+++}	+ 0·40
Cytochrome c^{++}	⇆	Cytochrome c^{+++}	+ 0·26
Ascorbic acid	⇆	Dehydrogeno ascorbic acid	+ 0·08
Reduced coenzyme Q	⇆	Oxidised coenzyme Q	
Cytochrome b^{++}	⇆	Cytochrome b^{+++}	+ 0·04
Reduced flavoproteins	⇆	Oxidised flavoproteins	0·0 to − 0·20
Reduced echinochrome	⇆	Oxidised echinochrome	− 0·23
Cysteine	⇆	Cystine	− 0·227
2GSH	⇆	GSSG	− 0·233
Reduced pyridinoproteins	⇆	Oxidised pyridinoproteins	− 0·33
Substrates	⇆	Dehydrogenated substrates	−0·4 to −0·8

minutely control the potential drop between free oxygen and the terminal hydrogen acceptors such as the flavoproteins (benzpteridinoproteins). The peroxidases, which reduce peroxides, are the next group of enzymes and catalase, which specifically reduces hydrogen peroxide, is probably the most widely important. There are also hydrogenases with a haem coenzyme; these catalyse the reaction: $H_2 \rightarrow 2H + 2e$ (Oparin, 1957a, p. 441). *Actiniohaematin* from sea anemones and *helicorubin* from the crop of the snail are haem-proteins possibly with an enzymic function. *Leghaemoglobin*, from the root tubercles of the leguminosae, is unique in being an enzyme for nitrogen fixation and not primarily redox. The biochemical importance of this group of conjugated proteins is therefore very great.

In addition to its two direct links with the iron atom, through histidine

residues (Fig. 38, XVI), the protein of haemoglobin is bonded via two other spare side chain nitrogen groups with the carboxyl groups of the two propionic side chains of the porphyrin. By contrast the protein of cytochrome is bonded via two cysteine residues with the vinyl side chains of pyrrole rings A and B (Fig. 38, XVI); vinyl and ethanol side chains are the only ones which can bond with cysteine (Sexton, 1953, p. 171). The direct bonds with the iron are again through histidine imidazole bases; as already noted these bonds are labile while the other two are strong, permanent links.

In addition to oxygen, similar molecules and radicals, such as CO, H_2S, NO and azide, will displace one of the imidazole nitrogen groups from the iron of haemoglobin (Keilin, 1953b), and they likewise abolish the paramagnetism of the iron. The affinity for carbon monoxide is 300 times greater than that for oxygen; this is now a notorious hazard, negligible before the commercial production of coal gas and so never subjected to natural selection. Pure oxygen in place of air increases the partial pressure sufficiently to drive out carbon monoxide fast enough to save the life of partially asphyxiated cases but it is necessary to add 5 per cent carbon dioxide to stimulate the breathing since what little is being produced by the tissues under these circumstances is being held there by the high partial pressure of oxygen. Methaemoglobin does not complex with carbon monoxide any more than with oxygen but it does bond at this position with —CN, NO, H_2S, H_2O_2, azide, CNO, CNS, and fluoride.

It has been noted that myoglobin has a higher oxygen affinity than haemoglobin, but in addition the haemoglobins differ considerably among themselves. That of *Arenicola* and some other invertebrates is fully loaded with oxygen at a few mm pressure of oxygen so that these yield it to the tissues only under conditions of near-anoxia. The haemoglobin of the frog loads fully at an ambient temperature of 15°C but would be fully dissociated at mammalian body temperature, 37°C; reciprocally that of mammals would remain permanently loaded at 15°C and the tissues would die of asphyxia in the midst of plenty. This is one of the hazards of hypothermia. The reason is that heat promotes the dissociation of oxy-haemoglobin, and that the haemoglobins of the two groups are differently adapted, to their normal body temperatures. The total carrying power and almost every other relevant property likewise vary taxonomically, mainly due to the protein moiety. The mammalian foetus has an haemoglobin with an adaptively higher affinity for oxygen than the maternal haemoglobin. It differs in only one or two amino acids, a similar constitutional difference to that causing the sickle-cell condition (p. 259).

The haemoglobins normally remain in the ferrous state at all times but

the cytochromes shuttle between Fe^{++} and Fe^{+++} and this is the essential component of their redox action. The terminal member, cytochrome a_3, or cytochrome oxidase, is oxidised to ferric by free oxygen and like methaemoglobin does not actually complex with it. The action of cytochrome oxidase is inhibited by —CN, =S and azide, however, so that it probably retains this fundamental property of complexing with a variety of radicals. The other cytochromes become ferric at lower oxidation potentials, and they also retain the property of complex-formation; cytochrome *c* forms a carbonyl compound and when in the ferric state also one with NO, though it does not complex with —CN, =S or F. Each has its unique properties, therefore, which must be explained largely by differences in protein moieties. In view of the great variety of complexes formed it is perhaps not surprising that the cytochromes of some anaerobic bacteria are able to release energy from organic substrates using sulphur instead of oxygen as the oxidant (Kamen, 1958). Cytochromes can also reduce nitrates and free nitrogen under appropriate conditions (see leghaemoglobin), and they are associated with the reduction of carbon dioxide in photosynthesis. It seems likely that there is no rigid distinction between the property of complex-formation and the redox action associated with mere electron-transfer by the iron.

The catalases and peroxidases in fact have their iron permanently in the Fe^{+++} state and act by complexing with the peroxide substrate (ROOH):

$$Fe^{+++}OH + ROOH \rightarrow Fe^{+++}OOR + H_2O$$
$$Fe^{+++}OOR + XH_2 \rightarrow Fe^{+++}OH + ROH + X$$

Catalases are inactivated by —CN, =S, NH_2OH and azide, while peroxidases form reversible compounds with F, CN, =S, and NO. These examples illustrate the great reactivity and versatility of the haemproteins. Their importance is due partly to this and also to the fundamental nature of the processes they catalyse. As essentially catalytic agents, the relatively large bulk of them in the bodies of all organisms is unparalleled—unless the nucleic acids are to be regarded as agents of the same kind.

17.6 Models and Pseudoporphyrins

Because of their importance in metabolism the porphyrins are probably better understood than most groups of effector agents in living bodies, and there is perhaps less need to study models and analogues than in more obscure fields. However, such studies are always useful and interesting and known models cover a wider range of structure and potential function than the common biological porphyrins. Any chelated metal, particularly those held by nitrogen groups, can be considered a simple analogue. The

ease with which such compounds are formed indicates that in evolution the porphyrin was probably built round the metal rather than the latter being added at a late stage, as apparently (p. 316) in present biosynthesis (Oparin, 1957a; Nicolaev, 1957; Calvin, 1962). It is interesting that even metallic platinum has some catalase activity, though metallic iron has more, and no doubt was exploited for that reason. The catalase activity of platinum is inhibited by cyanide and shows other features resembling the more highly evolved catalysts (Sexton, 1953, p. 167). The common redox property of metals of this general group will be considered further in Chapter 26.

As already indicated, the ferro- and ferricyanides are among the closest of the simple analogues (Fig. 38, XVII). It will be seen that the cyanide groups are conjugated through their carbon and not their nitrogen atom, so that the analogy is perhaps not so very close, but rather illustrates the wider scope of the coordination complex. Of course the inorganic cyanides isomerise to isocyanides so readily that the alternative bonding may occur tautomerically in these conjugates. In ferricyanide four of the cyanide groups are coordinated, sharing an electron pair with the iron and retaining the valency electron of the —CN ion for potential reaction with other cations, while the remaining two are electrovalently linked to the iron. In ferricyanide three of the cyanide units are held in this way and the complex as a whole therefore behaves as a tervalent anion only. Both are known to be good redox agents over their redox potential range.

Another simple group are the tetrammines of copper and other metals (Fig. 38, XVIII); these by contrast are complex cations. Each ammonia unit forms a dative (covalent) bond, sharing a pair of electrons with the copper atom, which retains its ability to donate its labile, valency pair of electrons to electrophilic groups and become a divalent cation. The complex has a powerful catalase action (Nyholm, 1957), like the haemproteins, and also like haemocyanin and other copper proteins (Ghiretti, 1956; Needham, 1960b). Here the coordination number of copper seems to be four, and it has no unfilled orbitals in its penultimate electron-shell, but in the copper biurets (Fig. 38, XII) the coordination number seems to be six, and it has even been suggested that the six molecules of water of crystallisation of copper sulphate are held in this way.

In addition to its orthodox porphyrans, cobalt forms such interesting compounds as di-cobalt decammine, $[(NH_3)_5 \cdot Co—O_2—Co \cdot (NH_3)_5]^{4+}$, a simple model for an oxygen-bridged diporphyran. Platinum forms ammines capable of reversible oxygen-transfer (Granick, 1957). The group —C(=NOH)—C(=NOH)— in organic compounds rather readily precipitates nickel and palladium as porphyrin-like structures (Feigl, 1949). Oxygen as well as nitrogen can coordinate metals, as in the

cobalt compound above, and the group —C(=O)—C(=NOH)— forms more distant analogues of the porphyrins, with iron and copper (Figs. 38, XIX and XX). Cobalt forms a complex with histidine which is believed to be of the form (XXI), i.e. the tertiary nitrogen of the imidazole, the α–amino and the carboxyl —OH of two histidine residues, all contribute to satisfy the six coordination bonds of the cobalt atom. The whole complex, again, can act as an oxygen-carrier, and haem is clearly a special selection from a large field of potential carriers (Earnshaw and Larkworthy 1961).

The most impressive analogue of the porphyrans, in its close general resemblance and its differences in detail, is the *phthalocyanine*–metal complex (Fig. 38, XXII). Four benzpyrrole, condensed rings, isomeric with indole (p. 189), act as the four coplanar nitrogen bases, and they are linked into the second-order ring by tertiary nitrogen bridges in place of the methyne links of porphyrans. With this difference and the extra resonant benzene rings it is not surprising that these are brilliant blue pigments (p. 115), compared with the red to purple of haemoglobin. Like haem, they have two of the pyrrole rings partially saturated and the other two with tertiary nitrogen bases, and they are water-insoluble. In spite of this they are used commercially as dyes, since they prove brilliant, fast colours, and with the chelated metal they need no mordant. They are also good catalysts, if iron is the metal chelated (Cook, 1938); their iron-complex catalyses the decomposition of hydrogen peroxide, the oxidation of benzaldehyde and the oxidation of various unsaturated compounds, including the cyclic olefines. Like the haems they are poisoned by cyanide. Structurally and functionally, therefore, they are close analogues and confirm the ability of =N— to deputise for =CH— as a bridge. The benzene ring can perhaps be regarded as an analogue of the side chains at the β–position on the pyrroles of the porphyrins; if so then these side chains are probably not critical for some of the more general properties of the porphyrins, though they are for some of the more specific characteristics.

17.7 Conclusions

As within other classes of organic substances only a few of the pentacyclic 1-nitrogen compounds have been exploited in living organisms, but these have been very fully exploited. Proline and hydroxyproline (p. 179) are the only common pyrrolidine compounds and the porphyrins, with their open chain precursors or derivatives, the only pyrroles of importance, apart from the condensed ring of the indoles. The intermediate, pyrroline ring with a single double bond (Fig. 38, II) does not occur free but it should be emphasised that the porphyrins, and their phthalocyanine analogues,

usually contain both pyrrole and pyrroline rings; in the second-order ring structure they may be stabilised by resonance. This is perhaps adequate even when the porphyrin ring is opened, since some of the bilins contain pyrroline rings.

Apart from a coproporphyrin, which may be secondarily derived, all the biological porphyrins seem to belong to series III (p. 313); moreover, apart from that of cyanocobalamin, and perhaps chlorophyll, all are derived from protoporphyrin (Fig. 313, IX). *A fortiori* most of the biological conjugates contain the one derivative, haem. There is variation in detail, rather in the same way as a single successful stock of organisms radiates to fill a variety of niches, and cyanocobalamin indicates that relatively large modifications are feasible, three of the rings being in the pyrroline and one even in the pyrrolidine state. No doubt there are biological reasons why the variation has not been greater.

The study of simpler coordination compounds shows that functionally as well as structurally they are related to the porphyrins, which therefore are outstanding rather than absolutely unique. By natural selection, however, they have become biologically unique, and this is probably true of many other biological compounds. Another point worth considering in this context is that the porphyrins probably should not be considered primarily as the most useful pyrrole derivatives but as the most useful form of coordination complex for certain useful metals, iron in particular.

Their virtue lies in their extensive resonance, coupled with the ability to chelate the metal by four separate bonds. The resonance property, however, has value apart from any chelation of a metal so that free porphyrins and even tetrapyrroles play certain minor roles in living organisms. The porphyrans themselves are responsible for a large range of indispensable redox properties.

CHAPTER 18

PYRIDINE AND PIPERIDINE COMPOUNDS

THESE are mono-nitrogen, hexacyclic compounds, piperidine (Fig. 40, I) being fully saturated and pyridine maximally unsaturated. Piperidines are even rarer, biologically, than their pentacyclic counterparts (p. 306), though like these they are rather common in the alkaloids of plants; here they may be essentially chemical defensive agents, which often means that they are antagonists of normal metabolites. Most alkaloids are also rather complex molecules, like the antibiotics, so that the analysis of their properties is difficult (p. 338). Piperidine itself is obtained from the alkaloid piperine, and is a strong secondary base, a colourless liquid at ambient temperature. One of the few free biological piperidines is *pipecolic acid* (Fig. 40, II), which could be regarded as a ring-closed norleucine (p. 179). The intermediary stages of desaturation are as usual unstable and rare, but biological examples are *guvacine* (Fig. 40, III) and *baikiaine*, isomeric forms, closely related to pipecolic acid (Florkin, 1960, p. 29). Structurally guvacine is a partially saturated *nicotinic acid* (Fig. 40, V).

18.1 Pyridines

Free *pyridine* (Fig. 40, IV) is one of the many products of coal and other biological materials, but rarely occurs in significant amounts in living organisms. Its strong, unpleasant smell is probably a teleonomic clue to its toxicity, probably as the analogue of its physiological derivatives. It also is a colourless liquid, boiling at 115°C. In this it resembles its counterpart, pyrrole, but unlike the latter it is extremely soluble in water, and indeed mixes with it in all proportions. It is also an excellent solvent for many organic substances, and for some purposes might be a useful deputy for water. The reason why it is so much more soluble than pyrrole is that N being already party to a double bond, resonance activity does not demand the formation of an extra bond, implicating the lone pair of electrons of the nitrogen atom. This pair is available for a change to the strongly polar quaternary state, therefore, by the acquisition of a

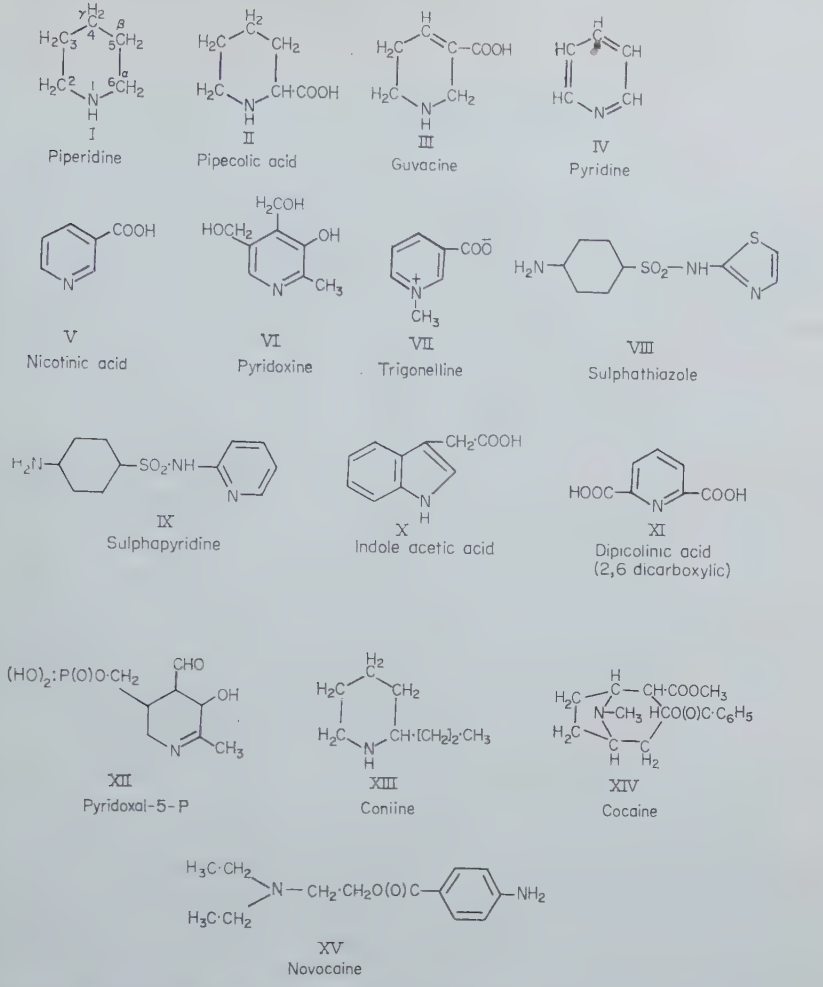


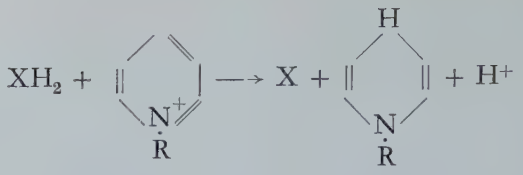
FIG. 40 Pyridine and piperidine derivatives of biological significance (*continued on p.* 332).

proton, H^+:

$$—N= + R^+ \rightleftharpoons —N^+(R)=$$

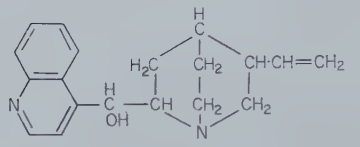
It is also a much stronger base than pyrrole for this reason (Finar, 1959, p. 729).

The electrophilic property of the $\vdots N^+$ radical makes the whole ring more stable, and more resistant to oxidation than benzene and pyrrole: it reduces reversibly to the conjugate base and at the same time the ring becomes hydrogenated elsewhere; this is biologically its most important property:

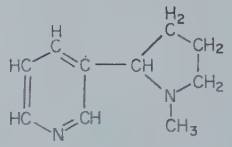


Being electrophilic, it does not form metal coordination-complexes.

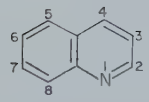
Only two pyridine derivatives are important as normal metabolites, nicotinic acid (Fig. 40, V) and *pyridoxine* (Fig. 40, VI), but these are as fundamentally important as haem. Nicotinic acid is of course an amino acid, and has the stability of the amino acids, subliming without decompo-



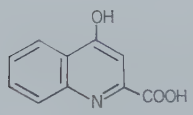
XVI
Quinine



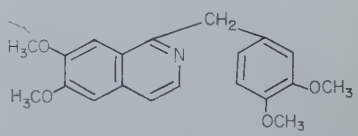
XVII
Nicotine



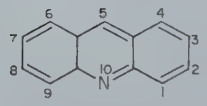
XVIII
Quinoline



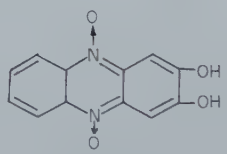
XIX
Kynurenic acid



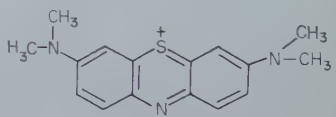
XX
Papaverine



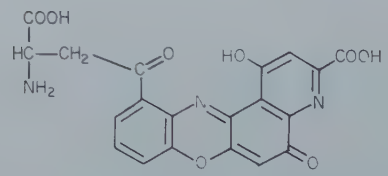
XXI
Acridine



XXII
Iodinin
(Di HO–phenazine
di N–oxide)



XXIII
Methylene blue
(thiazine)



XXIV
Xanthommatin
(phenoxazone)

FIG. 40 (*continued*).

sition, and withstanding boiling at any reasonably physiological pH. It exists *in vivo* also as the amide, which in fact may be the active form. It is more soluble in lipids than the free acid and this may be important because of the extent to which the respiratory reactions of the cell occur in its lipid media (p. 125). The free acid also has neurotropic, vasodilator effects from which the amide is free.

For most animals nicotinic acid or the amide is a vitamin, an essential metabolite demanded in the diet, though mammals and insects certainly synthesise some for themselves, by a degradative pathway from tryptophan (Fig. 41), which also seems to be the main pathway in bacteria and

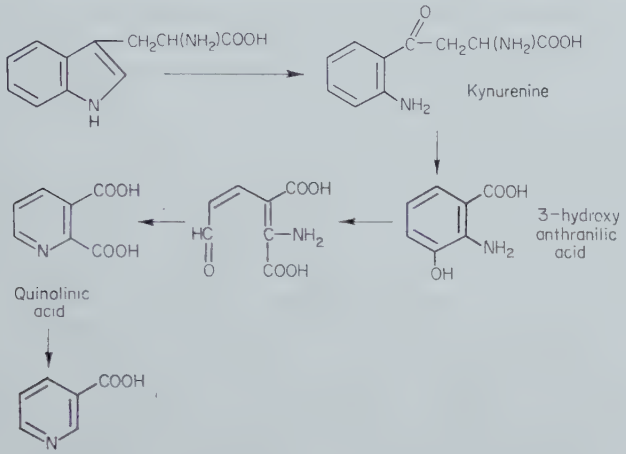


FIG. 41 Pathway of biosynthesis of nicotinic acid from tryptophan.

fungi (Kit, 1960, p. 115). It involves opening both rings of the indole nucleus, in turn, and finally reclosing the benzene ring to include the erstwhile pyrrole nitrogen—altogether an unexpected and intriguing piece of remodelling. Another interesting feature is that the other physiological pyridine, pyridoxine, is coenzyme for one step in the pathway, the deamination of kynurenine; this recalls the role of haem-proteins as enzymes in the synthesis of porphyrins (p. 312). There may be additional pathways for the synthesis of nicotinic acid *in vivo*; in the laboratory it can be obtained by heating glutamic acid, and there is some evidence for a biochemical pathway from this precursor. It is further interesting because glutamic acid is also the precursor of proline (p. 181) and has so many other biological functions (p. 180). Another laboratory method is to heat ethyl allylamine, $CH_2{=}CH{\cdot}CH_2NH{\cdot}CH_2{\cdot}CH_3$, the most 'obvious' precursor, and for that reason, perhaps, unlikely to be a biological one.

Like virtually all the B-vitamins, nicotinamide is biologically active as

part of essential coenzymes, and in this case of coenzymes I, II, and III, which collectively mediate the dehydrogenation of the majority of the common substrates of respiration, as well as transhydrogenation to some of the substrates of biosynthesis. Coenzyme III is a pyridine nucleotide (Fig. 44, XVI, p. 351), i.e. the conjugate nicotinamide–ribose–phosphate, nicotinamide mononucleotide (NMN), while the other two are dinucleotides, the second nucleotide being adenylic acid (p. 355). Coenzyme I is therefore known as nicotinamide–adenine dinucleotide (NAD), Nic–Rib–P–P–Rib–Ad, the nucleotides being joined back to back. Coenzyme II, NADP, differs from this in having a third phosphate residue on the 2′C position of the ribose of the adenine nucleotide. These coenzymes are very loosely bound to a protein apoenzyme, the free energy of bonding ($\Delta F°$) being in fact negative to the extent of 10,000 cal/mole (Klotz, 1953). Coenzymes I and II were formerly known as DPN and TPN, respectively.

As hydrogen-transferring agents these coenzymes react as indicated above, returning to their oxidised state by handing on the hydrogen atom to the next enzyme of the sequence (Fig. 20). Reduction leads to the appearance of a new absorption band at 340 mμ, and this is due specifically to the change in the pyridine moiety; the oxidised form has a single, strong absorption band at 260 mμ, due to both nicotinamide and adenine —a strong indication that there is free energy-transfer between the two bases, by coupled oscillation. This transfer is abruptly arrested by the shift in position of the band of maximal absorption by the nicotinamide, upon reduction (Glass, 1961, p. 836), and thus facilitates the completion of the cycle, the return to the oxidised state. Transfer between the two bases probably depends on a specific steric superimposition of their flat molecules, since there is no provision for resonance from end to end of the dinucleotide. It is probable that serotonin and other indoles also can transfer charge in this way (Szent-Gyorgyi, 1960, p. 86) and the phenomenon offers a potential explanation of a number of molecular associations at present structurally enigmatic.

NAD is ideal as a redox agent, the oxidised form being a particularly good electron-acceptor and the reduced form an even better electron-donor. This is shown by the values of the constant k which characterises the energy of the electrons in the critical orbitals of the molecule, the highest filled and the lowest empty levels. The values of k for these two levels, in order, for oxidised and reduced NAD, are $+1{\cdot}032$, $-0{\cdot}356$ and $+0{\cdot}248$, $-1{\cdot}032$ (Szent-Gyorgyi, 1960). A low positive value for the filled level indicates a good donor and a high value ($> +0{\cdot}5$) a poor donor; reciprocally a low negative value for the empty level indicates a good electron-acceptor. The molecular levels in question of course are not

the same as simple atomic orbitals but refer to a pool of delocalised π-electrons characteristic of these resonant molecules.

Under average conditions in a typical living cell NAD is in the oxidised and NADP in the reduced state (Rutter, 1959), so that with any slight shift in the redox level one or the other comes into its active range. At the average level any NADP will oxidise any NADH to NAD and itself reduce to NADPH. NAD is mainly concerned with the respiratory dehydrogenations and NADP with the biosynthetic transhydrogenations (Talalay and Williams-Ashman, 1960). The latter catalyses, among others, the crucial reduction reaction which permits the Krebs cycle (Fig. 8) to operate in reverse, for synthetic purposes.

In spite of the unique value of nicotinamide for transhydrogenation the activity is not very narrowly tied to its specific structure. The substitution of acetyl, —(O=)C·CH_3, for formamide, —(O=) C·NH_2, at position 3 causes symptoms of nicotinamide-deficiency in mice (Yudkin, 1948), but it has no effect in bacteria and is actually the active group in the coenzyme of the lobster heart, where NAD itself is not active (Kaplan, 1960). For this muscle, *thionicotinamide* and other derivatives also are active. Pyridine–3–sulphonic acid stimulates the growth of staphylococci, and this is a strong indication that it also deputises for nicotinamide, or is a source of it in biosynthesis. Even tetrahydrogen derivatives and piperidines themselves act as growth factors for staphylococci, though curiously enough *trigonelline* (Fig. 40, VII) and *quinolinic acid* (Fig. 41) are inactive. Trigonelline, the methyl betaine of nicotinic acid, is one of the excretory products of nicotinic acid in mammals, so that conversion back is probably difficult. All the forms which do stimulate growth in staphylococci probably are convertible to the vitamin itself.

Even the pyridine ring nitrogen does not seem to be absolutely indispensable. Thiazole (Fig. 48, VI), in which —S— replaces =N—CH— in the ring, can deputise as a growth factor for staphylococci. This ring is needed as a definitive component of the B-vitamin, thiamine (p. 397), in any case, but the deputisation for nicotinic acid seems to be real; moreover the sulphuryl group, —S(=O)$_2$OH, can be substituted for the —$CONH_2$ side chain, again without destroying activity (Erlenmeyer and Kiefer, 1945). It is significant also that sulphathiazole (Fig. 40, VIII), as well as sulphapyridine (Fig. 40, IX), inhibits NAD, presumably competitively, and both are strong bactericides. Other sulphonamide drugs do not have this action, since the *p*–aminobenzene–sulphonic acid moiety itself is not an analogue of this vitamin, though it *is* of *p*–aminobenzoic acid (p. 378). Another surprising analogue of nicotinic acid is *indole acetic* (Fig. 40, X), the common plant *auxin* (p. 192): this causes pellagra, the syndrome of nicotinamide deficiency. Of course it may act as an analogue of tryptophan and inhibit the synthesis of

nicotinic acid, but if it is a direct antimetabolite of the latter it shows that the indole ring will not fully deputise for that of pyridine.

In some ways the specificity of this substance and of its coenzymes is very high. For instance free nicotinamide acts as a competitive inhibitor of NAD and NADP, and it also competes with NAD for the enzyme which breaks down the nucleotide (Zatman *et al.*, 1953). Union of the coenzyme with its apoenzyme is probably immune to this effect since the protein has 2000 times as much affinity for the nucleotide as for the free base (Alivisatos and Denstedt, 1952), so that free base will tend to promote the synthesis and 'cold storage' of the enzyme; with moderate consumption and spaced mealtimes presumably this need not be a serious drawback. The effect is a potential danger wherever, as in this case, the active moiety of the coenzyme combines directly with the substrate of the reaction it catalyses, but not where the apoenzyme first makes this bond (p. 266). In the latter case there is less chance of free analogues of the coenzyme inhibiting its action.

Another pyridine derivative, *dipicolinic acid* (Fig. 40, XI), or pyridine –2:6–dicarboxylic acid, plays a normal role in suppressing developmental activity in dormant bacterial spores (Halvorson, 1959), perhaps by acting as an analogue of nicotinic acid and blocking its transhydrogenation systems. Dipicolinic acid protects the spores against heat and other agents so that it is probably the NAD rather than the NADP systems which are critically suppressed. Immediately germination begins DPA is excreted, which supports the idea that it suppresses metabolic activity. It must be a substance of major importance since it makes up as much as 6–12 per cent of the weight of the spores.

18.1.1 Pyridoxine (Vitamin B_6)

Although, like nicotinamide, *pyridoxine* (Fig. 40, VI) is both a B-vitamin and a coenzyme, it contrasts very strongly in function and in a number of structural features: there is no overlap between the two. Pyridoxine is richly substituted on the ring and this may be related to the fact that its coenzyme catalyses a variety of reactions, in contrast to the restriction of nicotinamide to hydrogen-transfer. All these substituents are at a lower level of oxidation than the carboxyl group of nicotinic acid. The methanolic group at 4 readily oxidises, but only as far as the aldehyde stage, and this pyridoxal then reversibly aminates to pyridoxamine; this is an amine in contrast to the amide of nicotinic acid. All three derivatives of pyridoxine occur *in vivo*, the intermediary aldehyde being perhaps the most common. Unlike nicotinamide, pyridoxine does not form a nucleotide; the complete coenzyme is a simple phosphate (Fig. 40, XII), formed by esterification of the other alcohol group, at 5.

The reactions catalysed by this coenzyme are transaminations, transcarboxylations, amino acid racemation, de-sulphydration of cysteine and dehydroxylation of serine and threonine. The last two are analogous reactions, while amino acid racemation may involve carboxyl and amino transfer around the α–carbon, so that it is related to the first two types of reaction. There are further resemblances of this kind which help to explain why pyridoxine enzymes can catalyse such a variety of reactions, but even so its powers remain impressive.

In transamination the amine form is an actual intermediary in the transfer of that radical. This is in fact a common feature of the transfer reactions of the B-vitamins, virtually all of which are cotransferases in their active form. The aminated form of pyridoxine therefore is comparable to the hydrogenated form of nicotinamide. In transcarboxylation

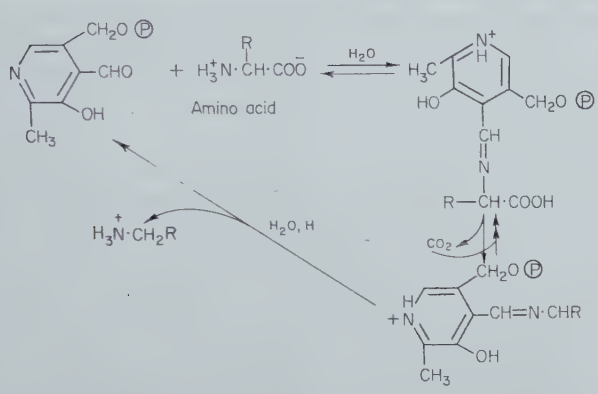


FIG. 42 Action of pyridoxal-5-phosphate as coenzyme for decarboxylation.

from an amino acid this becomes condensed via its α–amino group with the same aldehyde group at 4, forming a Schiff's base which facilitates the subsequent removal of the α–carboxyl group. The relationship between transamination and transcarboxylation is therefore clear. To remove the amine which remains, and to restore the free pyridoxal, transhydrogenation and hydration are involved, and the pyridine–nitrogen becomes a quaternary base in the process. The essential property of the pyridine base therefore *is* put to use (Fig. 42). Cysteine desulphydration and serine dehydroxylation both involve deamination in addition, as a final step, so that the interrelationships between the five types of transfer mediated by this enzyme system are fairly clear. It should be emphasised that other transferase coenzymes also participate in some of these reactions and that pyridoxine is not necessarily the *primum movens* in all.

In spite of the many components of its activity the structure of pyridoxine is not absolutely specific, but resembles nicotinic acid in its grade

of specificity. The 4-carboxy derivative, *pyridoxic acid*, which is a nearer analogue of nicotinic acid, has some of the activity of vitamin B_6, and so has *β–pyracin*, its lactone (Sexton, 1953, p. 151). They promote growth in general and blood-cell production, as the vitamin itself does. Also active is α–pyracin, the lactone of the 5-carboxy analogue. Deoxypyridoxine, which has CH_3 in place of CH_2OH at 4, can deputise for pyridoxine as a growth factor for bacteria, though in the chick it is an antagonist (Yudkin, 1948), and has been used to inhibit the growth of tumours (Sexton, 1953, p. 356). Presumably in bacteria it is converted into the vitamin, and otherwise it acts as an antimetabolite. This could be valid for any analogue differing from the vitamin only in side chains, since these are relatively easily modified.

Substitution of ethyl for the methyl at position 1 does depress activity, a change which was seen to be critical likewise in the open chain quaternary bases (p. 289). Saturation of the ring to the piperidine stage greatly affects activity (Nicolaev, 1957); the greater alkalinity and toxicity of the saturated base are further increased by any substitution of long alkyl chains on the ring. These also increase the lipid-solubility of the molecule.

18.2 Piperidine and Pyridine Alkaloids

The simplest of these is *coniine* (Fig. 40, XIII), the most abundant, though not the most deadly, alkaloid of hemlock to which it gives the mousy odour. It has historical interest, not only in contributing to the death of Socrates but also in being the first alkaloid synthesised in the laboratory. The relative toxicities of piperidine, methyl and ethyl piperidines, and coniine, are in the simple ratios 1:2:3:4 (Read, 1935, p. 634).

The *cocaine–atropine–hyoscyamine* (henbane) group of alkaloids have a nucleus consisting of the piperidine ring condensed with a pyrrolidine ring sharing the same nitrogen atom (Fig. 40, XIV). Cocaine was for long *the* local anaesthetic in dentistry, while atropine was used to block the action of ACh (p. 290) and dilate the pupil; it blocks the muscarine-type action. The nitrogen atom is tertiary and presumably acts as an analogue of the open chain base of ACh. It seems a rather distant analogue, like the tubocurarine group. It is not very surprising therefore to find that the pyrrolidine ring is not very critical for the activity of these alkaloids: the cocaine substitute, *eucaine* (benzamine), has this ring open, while *novocaine*, the best synthetic substitute, as a local anaesthetic, has the piperidine ring also open (Fig. 40, XV). However, it has very low specificity, and its second amine group is a complication.

Hyoscyamine and *hyoscine* (scopolamine) are isomeric with atropine and likewise block the muscarine action of ACh. Like some of the other natural plant alkaloids they have an amazing selectivity for particular

sites in the neuro-effector systems of animals. It is possible that they have equally selective and useful effects in the physiology of the plant itself but the more plausible theory is that they have been naturally selected to deter herbivorous animals in a really practical fashion! L–Hyoscyamine has a greater effect on the iris muscle than the D–isomer while the converse is true for the synapses involved in the spinal reflexes. Hyoscine is an hypnotic with a very selective action within the central nervous system. It is very poisonous in higher concentrations than this.

Quinine (Fig. 40, XVI), the first effective antimalarial drug, extracted from the bark of the cinchona tree (Rubiacieae), has two *N*–heterocyclic nuclei, chain-linked. One is a tripiperidine condensed nucleus, the three piperidine rings sharing an N and a CH group in common, and a $-CH_2$ $-CH_2-$ unit with each neighbour, the three therefore forming a Chinese-lantern structure in three dimensions. Apart from its two side chains this is a symmetrical structure. The other nucleus is a quinoline derivative and this is the more important part (p. 341).

Nicotine (Fig. 40, XVII), best known of all alkaloids, is a pyridine–pyrrolidine complex, with the rings directly linked but not condensed. In the tobacco plant it exists as the salt of malic and citric acids, and so is neutral and relatively inactive as a neurotropic agent. When released the bases of both rings are active, though the pyridine is probably the more important. Nicotine is virtually the only pharmacologically active compound in tobacco (apart from the possible carcinogens) but it is potent enough to need no 'second string'. In low concentration it has a parasympathomimetic effect, and the opposite action at higher concentration. It acts at the intermediary synapse of both components of the autonomic system, a cholinergic synapse in both cases, and above a certain concentration it paralyses both, the parasympathetic first. Applied in high concentration, therefore, there is a brief stimulation followed by paralysis. Inhaled in small quantities it has a sedative action on parts of the central nervous system, and this is probably an attraction to smokers.

It is soluble in both water and lipids, as might have been anticipated from its two components, and this must contribute to its efficacy at nerve-cell membranes. As an insecticide the conjugate base, which is more lipid-soluble, is more active than the water-soluble quaternary nicotinium ion (Sexton, 1953, p. 320), and this illustrates the importance of permeating the cell membrane for action of this type.

The methyl group of the pyrrolidine is not vital for the action of the alkaloid and moreover piperidine can replace the whole pyrrolidine moiety. On the other hand most details of the molecular structure seem very important—more critical perhaps than those of the common pyridine metabolites, nicotinamide and pyridoxine. Neither ring in isolation

is very toxic (Sollman, 1957), so that we have a good example of an 'emergent' property at the higher level of organisation of the molecule. Activity as an insecticide is greatly reduced if the pyrrolidine ring is opened so that in contrast to the findings on cocaine some ring in the normal position is essential. The other, and more important, ring *must* be pyridine and activity is reduced to 1/1000 if a benzene ring is substituted (Sexton, 1953, p. 320); there seems little doubt that the tertiary nitrogen base is the critical atom. The mode of linkage between the two rings also is important, as in most cuff-link molecules. If activity depends on steric superimposition of the ring structures, as in the pyridine dinucleotides (p. 334), this is to be expected, since the length and other properties of the link must be very critical for a precise fitting of the rings.

L–Nicotine is three times as active as the D–isomer. This is a smaller difference than in most optically active agents, for instance adrenalin and hyoscyamine.

Other pyridine and piperidine alkaloids include *gelsemium* (jasmine), *sparteine* (broom), *lobeline* and the bitter alkaloids of lupins and other leguminous seeds. The latter cause lathyrism in animals, manifested primarily in weakness of the leg muscles. Even lentils and peas in quantity can cause lathyrism.

18.3 Quinolines

These are 'benzologues' of pyridine (Albert, 1950), i.e. they have a condensed benzene–pyridine nucleus (Fig. 40, XVIII). They are basic in reaction, rather similar in properties to the pyridines, but for this reason they act as metabolic antagonists of the latter and most of them are very poisonous in quantity, like some of those pyridines which are not themselves normal metabolites. However, some of the quinolines, like some of the pyridines, can be converted into nicotinic acid, at least in the laboratory. On heating their benzene ring is opened and oxidised to *quinolinic acid* (Fig. 41) which can then be decarboxylated to either nicotinic or isonicotinic acid.

One species of bacterium is known to produce three quinolines, all antibiotics, which strengthens the view that quinine (Fig. 40, XVI) and other more complex quinoline alkaloids have an essentially defensive function. Not all biological quinolines are dangerously toxic, however; *kynurenic acid* (Fig. 40, XIX) and its 8-hydroxy derivative, *xanthurenic acid*, are formed from tryptophan as collaterals of the pathway leading to nicotinic acid (Fig. 41). Both pyridines and quinolines are obtained by distilling bone oil, so that quinoline complexes may be normal metabolites. Xanthurenic acid is a component of the molecule of the ommochromes (Fig. 40, XXIII). The angostura quinolinic alkaloids serve as an aperitif, so that they are only mildly toxic, at worst.

Quinolines are less soluble than pyridines, owing to their higher CH/N ratio, and they have some similarity to the pyrroles. For instance 8-hydroxyquinoline chelates metals very readily, and the chelation is of the oxygen–nitrogen type considered at the end of the previous chapter. The benzene ring makes quinolines more extensively resonant systems than the pyridines and the base therefore becomes more active as an energy mediator. At low temperatures quinolines show a dramatic shift in absorption bands towards the blue end of the spectrum, which potentially could help to offset the usual depressant effect of low temperature on chemical processes. Quinolines affect redox reactions, but at a higher oxidation level than those catalysed by nicotinamide. In fact they tend to inhibit those catalysed by the benzpteridines (p. 380) or 'flavins', to which the quinoline molecule has some steric resemblance. Quinine and the related *mepacrine* inhibit two important 'flavo-protein' enzymes, cytochrome *c* reductase and D–amino acid oxidase, while 4–hydroxyquinoline–*N*–oxide inhibits electron transport between cytochromes *b* and *c*, lying within the redox range of the flavo-proteins (Kit, 1960, pp. 23, 24). Reduction of the ring nitrogen atom does not abolish this activity so that it is probably a reversible redox agent as it is in nicotinic acid.

On the other hand oxidation does destroy activity, for instance the antimalarial activity of quinine, which depends mainly on the quinoline moiety. There is in fact little effect of changing the vinyl side chain on the piperidine moiety to ethyl, and laboratory analogues such as *chiniofon* (*yatren*), which completely lack the tripiperidine nucleus, are still good antiamoebiacs. *Bayer–7602*, a commercial cuff-linked diquinoline, also is a good anti-protozoal drug. The link once more seems to be important, however, and virtually any substitution for its CHOH group depresses antimalarial activity and, as a very important practical therapeutic consideration, may increase the toxicity to the host of the parasite (Sexton, 1953, p. 282). Various side chains can be substituted on the quinoline ring without seriously affecting its activity and the maximal antimalarial effect in fact is obtained with a C_4–C_5 aliphatic chain at position 6. The 8-hydroxy compound is an active antibacterial agent, probably because of its chelation-property. However, hydroxyl substituents at any other position are inactive and so is the 5-chloro derivative.

Thus quinolines are rather effective antimetabolites and are used both naturally and commercially for this purpose. Although the ring nitrogen is the focus of their activity various other features of the molecule and its conjugates are functionally important, just as in normal metabolites. The quinolines provide some clue to the functional association between the pyridines and the benzpteridines, as redox catalysts.

Isoquinolines, with the nitrogen atom in the β–position to the benzene

ring, occur in *papaverine* (Fig. 40, XX), one of the opium alkaloids, and in other alkaloids.

18.4 Acridines

These are the next higher benzologues of quinoline (Fig. 40, XXI), and are weaker bases because of the extra benzene ring. The linear condensation of the three rings, with nitrogen in the *para*-position of the middle ring, gives a steric resemblance to isoallozazine (benzpteridine) ring (p. 380). Since even the quinolines act as steric analogues of the latter it is inevitable that the acridines act as powerful antagonists of the flavoprotein enzymes, including cytochrome reductase and D–amino acid oxidase. Some organisms synthesise acridine antibiotics, and the group has been extensively exploited in manufacturing antiseptics (proflavin, acriflavin) and biocides (mepacrine). They are rare as normal metabolites.

Their mode of action has been extensively investigated because of their practical importance and this has thrown considerable light on the properties of aromatic nitrogen bases more generally. The *phenazines* (Fig. 40, XXII), *phenthiazines* (Fig. 40, XXIII) and *phenoxazines* (Fig. 40 XXIV) behave very similarly to the acridines: with the second polar atom in the middle ring they are even closer analogues of benzpteridine than the acridines themselves. The nature of these polar atoms seems relatively unimportant and it would seem possible that the natural phenoxazine derivatives, such as the ommochromes (Fig. 40, XXIV) (Butenandt *et al.*, 1954), may inhibit the benzpteridine dehydrogenases, or even act as redox agents in their own right (Needham and Brunet, 1957). They undergo reversible redox changes *in vitro* and conceivably may do so *in vivo*, under certain conditions. All these analogues stimulate aerobic glycolysis under experimental conditions (Sexton, 1953, p. 211), inhibiting the Pasteur effect, which is the normal switch from glycolysis to complete aerobic respiration when the oxygen-tension reaches an adequate level. This is in keeping with their inhibition of the benzpteridine enzymes which form the link between the proximate pyridine dehydrogenases and the terminal cytochrome system (Fig. 20).

The various azines stimulate aerobic glycolysis, which is the automatic sequel to abolishing the Pasteur effect. A number of phenazines, including *pyocyanin* and *iodinin* (Fig. 40, XXII), are natural biological compounds, but once more they are probably 'deliberate' antimetabolites for the benzpteridine enzymes of would-be predators, rather than deputies for the indigenous enzymes of this group, though some of them do have an appreciable dehydrogenase activity (Sexton, 1953). Iodinin is inhibited by anthraquinones (p. 138) and even by naphthoquinones (p. 135), which are rather distant analogues comparably to the quinolines; this indicates

that the two N→O groups of iodinin are analogous to quinone groups, and take part in the resonance activity of the molecule.

In investigating the significance of the details of the acridine structure it has been shown that at least one basic nitrogen is essential in the middle ring, and most acridines are at least 60 per cent ionised at pH 7·3. Even so there are acidic members, such as the nitroacridines, which are active, and some strongly basic members which are not; this is perhaps for steric reasons. Extra basic nitrogen groups as side chains usually augment activity and bactericidal action is maximal with a basic group on 4 (Fig. 40, XXI) and an acylamino group on 6. Proflavin and acriflavin however have NH_2 at 2 and 8. As in the case of quinolines, alkyl side chains give maximal antimalarial action at a length of C_4. A hydroxyl group in a side chain reduces the chemotherapeutic index (c.t.i.), that is the ratio of the maximal dose tolerated to the minimal therapeutic dose (Sexton, 1953, p. 286). No substituents are completely indifferent. If nitro is substituted for chlorine at 8 then hydroxyl in another side chain increases the c.t.i. The quinolines show that one benzene ring may be removed without greatly reducing activity.

18.5 Conclusions

There are a number of parallels between the piperidine and pyrrolidine series. Relatively few derivatives are exploited but these are used for fundamentally important metabolic purposes. In both cases they are members of the maximally unsaturated sub-series, having maximal resonance activity. In both cases the function is catalytic, with an emphasis on redox activity, which is not the primary function of the nitrogen bases, but is favoured by the more acidic, aromatic ring.

There are also marked differences between the two series, such as the greater alkalinity and water-solubility of the pyridines than of the pyrroles, so that there is a quite distinct biological function for each. Pyridine rings, with their regular alternation of single and double bonds, cannot be linked by the =CH— type of bridge, but only by one single bond or by the sequence —CH=CH—. A chain of the latter type permits the bridge to form the middle ring of a three-ring condensed system and this is more common among pyridine derivatives than the polypyrrole type of structure. These condensed ring systems behave as a steric unit and do not closely resemble the simple pyridines in function. It so happens that they are effective analogues of a structurally different, but functionally associated, group of metabolites, and so they have proved useful in a number of practical and theoretical ways.

CHAPTER 19

COMPOUNDS WITH MORE THAN ONE NITROGEN ATOM PER RING

As USUAL only penta- and hexacyclic compounds are significant. In the laboratory the former have been prepared with as many as four (Fig. 43, V, VI) and the latter with as many as three nitrogen atoms. In living organisms rings with two are common, but there are rarely more. Of the two nitrogen rings, the pyrimidines, purines and pteridines are outstanding and will be considered in subsequent chapters.

There are two possible series of 2N pentacyclic rings, the *pyrazole* series, with the two nitrogens as immediate neighbours (I, II), and the *imidazole* series (III, IV) with one carbon separating them. The 1:3 compound is of course identical with the 1:4 isomer viewed from the opposite side of the molecular plane, so that there is no third series. The pyrazoles have a p*K* of 2·0–3·6 compared with 5·6–7·0 for the imidazoles and are not common biologically. These figures illustrate the importance of the relative positions of the nitrogen atoms.

As usual, the fully unsaturated sub-series are more common than the stronger saturated bases, and the intermediate states are the least common, in living organisms. Although pyrazoles are rare the *pyrazolones*, or keto-pyrazoles, are used as febrifuges. Even among imidazoles, histidine and its derivative histamine (p. 185) are the only ones of universal biological importance. The free base, *glyoxaline*, occurs as *pilocarpine* in the jaborandi plant, and resembles histamine in being a parasympathomimetic agent. It is interesting also because of the bridge across the ring (Fig, 43, VII).

The condensed ring system, benzimidazole, occurs in vitamin B_{12}, as already seen (p. 321). Here it is conjugated with cobalt and, like pyrrole, imidazole rather readily forms coordination-complexes with metals of this group (p. 328); these may show redox properties. Perhaps this is why histidine counteracts the toxicity of inorganic cobalt (Morgan and Parker, 1952). Vitamin B_{12} acts as an anti-histaminic (Traina, 1950) so that the benzologue may be a metabolic antagonist of the simpler, imidazole ring.

Purines (p. 347) are another example of a condensed imidazole ring system.

One component of the *luciferin* of the ostracod, *Cypridina*, the substrate for its bioluminescence, is a partially saturated, *imidazoline* ring (Fig. 43,

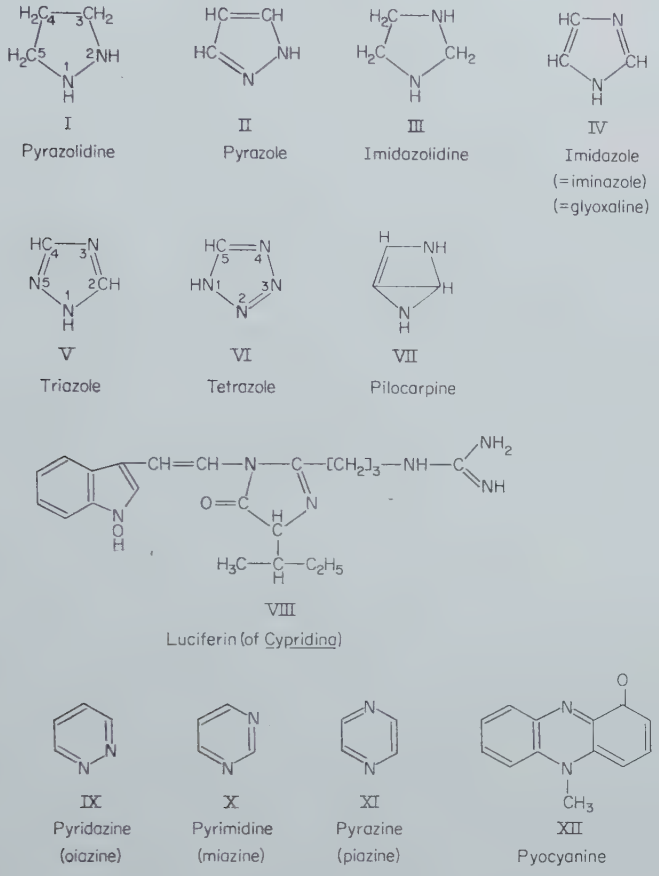


FIG. 43 Polyazole ring-compounds.

VIII). It is a keto-derivative, however, and may resonate with the imidazole state. Its significance in bioluminescence is not yet clear (Glass, 1961) but the other unusual feature of the molecule, the —N—OH group of the indole moiety, is known to prevent fluorescence in the molecule. The bioluminescent emission itself is at a different wavelength from the fluorescence induced by ultraviolet in the absence of the indolic =NOH group, and comes from a different part of the molecule. However, con-

jugation of the luciferin with adenylic acid (p. 355) brings the fluorescence to the same wavelength as the chemiluminescence, so that in the presence of the =NOH it may potentiate the latter.

Of the 2N hexacyclic compounds the *pyrimidines* (Fig. 43, X) are biologically the most important, and the *pyridazines* (Fig. 43, IX) least so; this is a parallel to the imidazole–pyrazole contrast. The *pyrazines* (Fig. 43, XI) occur mainly as condensed ring systems, the most important being the pterins (p. 374), and the benzpteridines or isoalloxazines (p. 380); the former are pyrimidine-condensed pyrazines and the latter simple benzologues of this. Fully saturated sub-series of any of these are biologically rare, and the intermediaries as usual even more so.

The *phenazines* (p. 342) are double benzologues of pyrazine and a number of them occur naturally. Their analogy to the acridines and benzpteridines has been noted. Pyocyanin (Fig. 43, XII), a blue pigment produced by *Bacterium pyocyaneus* (Sexton, 1953, p. 210), may be a normal respiratory coenzyme since it stimulates full aerobic respiration and depresses aerobic glycolysis. However, it does not fully compensate for the lack of terminal oxidases in neoplastic tissues. From its structural resemblance to benzpteridine it would be expected to connect glycolytic with terminal respiration but not to deputise for the latter.

CHAPTER 20

PYRIMIDINES, PURINES, NUCLEOTIDES AND NUCLEIC ACIDS

As already indicated (p. 344) the pyrimidines and their imidazole-condensed structural derivatives, the purines, are the most common and important of biological compounds with 2 nitrogen atoms per ring. Both exist as free bases, and also in simple complexes, but qualitatively, and in bulk, the most important are those incorporated into units called nucleotides (p. 355), which possess the polymerisation property as unrestricted as that of monoses and amino acids, and form fibrous macromolecules, the nucleic acids. A number both of pyrimidines and purines build nucleotides and of these a limited number, varying little among living organisms, take part in nucleic acid formation. Natural polynucleotides are probably always mixed polymers of those pyrimidine and purine nucleotides capable of polymerisation, so that there is good reason for considering the two groups of bases together, although their biosynthetic pathways up to the nucleotide stage are probably quite independent (Fig. 45). A number of free nucleotide monomers, including those which can polymerise, are important metabolites, and one, adenylic acid, is the most critical single metabolite in the body. The free pyrimidine and purine bases also have useful functions, and if the functions of the nucleic acids, in inheritance and protein synthesis, also are considered it is evident that these materials are second in importance to no other group, except perhaps the amino acids and proteins. The amino acids are more varied than the pyrimidine and purine bases, and proteins are more varied, as well as much greater in bulk, than the nucleic acids, but the genetic role of the nucleic acids tends to weigh heavily on the other side.

20.1 Pyrimidine and Purine Bases

Five of these bases are of outstanding importance for building nucleic acids; these are the pyrimidines *cytosine* (Fig. 44, I), *uracil* (Fig. 44, II) and *thymine* (Fig. 44, III), and the two purines *adenine* (Fig. 44, IV) and *guanine* (Fig. 44, V). Adenine, guanine and cytosine occur in both DNA, the

main nucleic acid of the chromosomes, and RNA, which is found in both nucleus and cytoplasm but more abundantly in the latter; uracil and thymine are alternatives, the former occurring in RNA and the latter in DNA. Most samples of the two types of nucleic acid therefore in most organisms contain four bases, two pyrimidines and two purines. Less common bases of both groups also occur in one type of RNA, the so-called 'soluble' or 'transfer' RNA, concerned with transporting amino acids for protein synthesis (Table 32), and viruses have a few unusual ones (Sinsheimer, 1954). The sponge *Cryptotethya* has three unusual bases, including *spongosine*, 2–methoxy–6–amino purine (Bergmann *et al.*, 1957).

Table 32. Rarer Biological Pyrimidine and Purine Bases

Base	Occurrence
Pseudouridine	Soluble RNA (transfer RNA)
Methyl pyrimidines (various)	Soluble RNA (transfer RNA)
5–Methyl cytosine	Soluble RNA (transfer RNA)
Thymine (as a cytoplasmic nucleic acid constituent)	Soluble RNA (transfer RNA)
1–Methyl guanine	Soluble RNA (transfer RNA)
6–Methyl amino purine	Soluble RNA (transfer RNA) and in DNA of *E. coli*
6–Dimethyl amino purine	Soluble RNA (transfer RNA)
2–Dimethyl, 6–hydroxyl purine	Soluble RNA (transfer RNA)
2–Methylamino, 6–hydroxyl purine	Soluble RNA (transfer RNA)
5–Hydroxymethyl cytosine	DNA of T–even bacteriophages
2–Methoxy–6–amino 2–purine (spongosine)	Porifera
Methylated adenine and thymine	RNA of micro-organisms
Methylated guanines	RNA of some plants

A number of purine bases, in particular *hypoxanthine* (Fig. 44, VI) and *xanthine* (Fig. 44, VII), are intermediaries in the metabolic breakdown of nucleotide purines, while *uric acid* (Fig. 44, VIII) is the final excretory form in insects, birds and some other animals. Spiders and some others excrete their purines as guanine. Insects and birds build most of their excretory nitrogen to uric acid, largely in the interests of water economy, but this is also a reminder that the nitrogen for synthesising both types of base in the body comes from amino acids. The pyrimidines are completely broken down for excretion. Some animals use guanine as a white reflecting pigment, for instance to produce the silvering of fish-scales, and other purines are used as white 'pigments' in insects (Fox and Vevers, 1960).

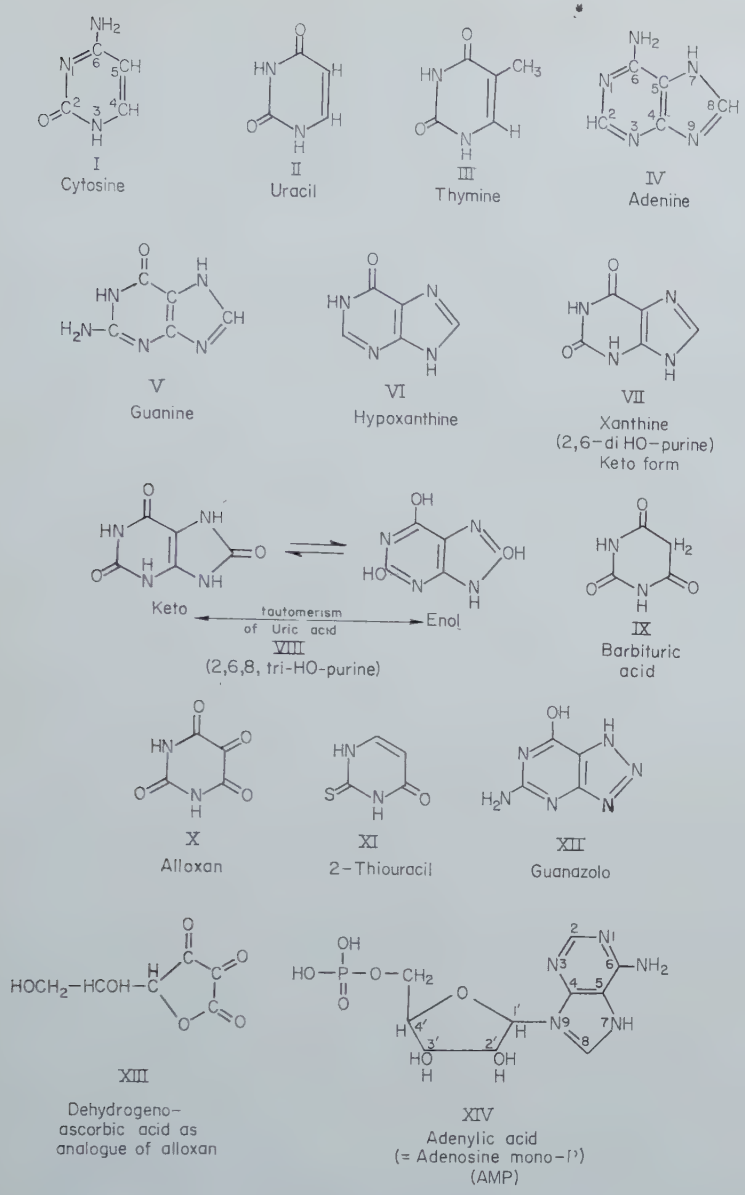


FIG. 44 Biologically important pyrimidines and purines and their derivatives
(*continued on pp. 351 and 353*).

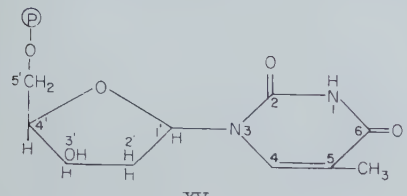
Tea leaves, coffee beans and some other plant tissues contain a variety of methyl-purines: 1:3-dimethyl xanthine (*theophylline*) in tea, 3:7-dimethyl xanthine (*theobromine*) in cocoa and 1:3:7-trimethylxanthine (*caffeine*) in tea and coffee. Once more it is not certain that these bases serve a useful function in the plants themselves, but they have certainly become popular beverages because of their stimulating effect on the nervous system, particularly on the higher association centres. Other plants contain large amounts of a group of pyrimidines, the *barbiturates* (Fig. 44, IX), which have a powerful sedative effect on the nervous system and in quantity are dangerous poisons. *Alloxan* (Fig. 44, X), synthesised in the laboratory by oxidising uric acid, is a powerful diabetogenic agent, antagonising the action of insulin. *Thiouracil* (Fig. 44, XI) and its derivatives are antithyroid agents, inhibiting the secretion of the hormone from the gland. The purine, *paraxanthine*, 1:7-dimethyl xanthine, acts at another point, inhibiting the action of the thryoid hormone on the body. All but the common bases of both groups therefore have powerful effects on the nervous or the endocrine system, poisonous in quantity. The same is true of more distant analogues; for instance *guanazolo* (Fig. 44, XII), a triazolopyrimidine, is a powerful growth-inhibitor (Sexton, 1953). It is important to remember this when considering the normal bases.

A pyrimidine forms part of the molecule of the B-vitamin thiamine (p. 397) and, as already noted, of the pteridine and benzpteridine nuclei, each of which numbers one B-vitamin among its derivatives. Here the action is essentially enzymic, therefore.

Pyrimidines and purines once more are maximally unsaturated rings, resonance being enhanced by the high proportion of nitrogen atoms. Those with OH (enol) substituents on the ring undergo tautomeric change to the keto (=O) form, with one less double bond in the ring (Fig. 44, VIII). An amino side chain similarly can become =NH but this is less common. This extends the scope of resonance in the molecule. The free bases even so have too small a resonant network to be coloured, but their more complex derivatives are, and this is the explanation of the bright colour-reactions used as a test for the two groups, 'murexide' for purines and 'dialurate' for pyrimidines. The resonant system is not extended in nucleotide combination and so the nucleic acids also are uncoloured. The bases absorb maximally well within the ultraviolet range, around 260 mμ; this may be significant for the induction of mutations by ultraviolet, but these molecules are not so easily activated for chemical activity as the coloured cyclic nitrogen compounds. The keto form is less active than the enol, and it is the more common form. They are stable molecules, in fact, the shortness of the side chains (Nimmo-Smith, 1958) as well as resonance contributing to this. The nucleic bases suppress the visible fluorescence

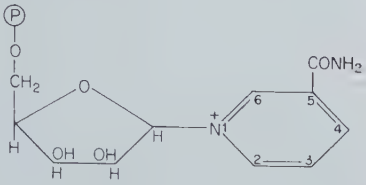
of filter paper (cellulose) when exposed to ultraviolet of 250–290 mμ; this usually means that the energy absorbed is put to use instead of being re-emitted (p. 317), and therefore the bases may help other molecules to harness the energy they absorb. Conceivably they do this to the ribose of nucleotides.

Since these are aromatic rings, keto or enol side chains are acid in reaction. Together with the nitrogen base groups in the rings this makes



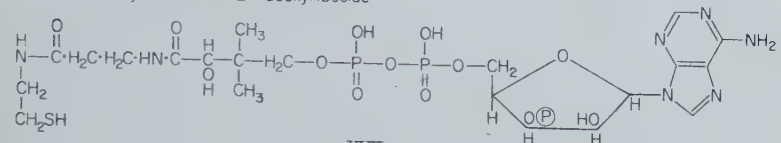
XV

Thymidylic acid
= Thymidine mono-P(TMP)
= Thymine-3-D-2'-deoxyriboside



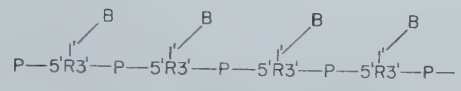
XVI

Nicotinamide mononucleotide (NMN)



XVII

Coenzyme A



XVIII

Polynucleotide skeleton

P = Phosphate
R = Ribose
B = Pyrimidine or purine base

FIG. 44 (*continued*).

for ambivalence, supplemented by the side chain amino groups. Collectively the 'bases' range from quite strong bases to fairly strong acids. Uric acid (Fig. 44, VIII), with three =O (or —OH) side chains, is an acid strong enough to neutralise alkalis for excretory purposes, and so serves a dual purpose in excretion. Only two of the acid groups ionise within the physiological range (pK 5·4 and 10·3). Most of the bases show three acid ionising groups (Table 33) within the significant range so that some of these must be due to the ionisation $\vdots\overset{+}{\mathrm{N}}\mathrm{H} \rightleftharpoons \vdots\mathrm{N} + \mathrm{H}^+$ or to $=\mathrm{NH} \rightleftharpoons =\mathrm{N}^- + \mathrm{H}^+$—all, in fact, in the case of adenine. The three pK values are well

spaced over the significant pH range and the interaction of side chains produces a variety of actual values which collectively cover the whole range rather uniformly. This is reasonably true for the four occurring in each of the nucleic acids and it is probably an important collective property. For this purpose thymine and uracil evidently are interchangeable. The bases are to some extent complementary also in their range of ultraviolet absorption. Adenine absorbs maximally around 260 mμ, whereas

Table 33. Acid Dissociation Constants of Biological Pyrimidines and Purines

Base	Dissociation Constants		
	pK_I	pK_{II}	pK_{III}
Cytosine	4·6	12·16	
Uracil	$\approx$ 0·5	9·43	$>$13·0
Thymine	$\approx$ 0·0	9·94	$>$13·0
Adenine	0	4·15	9·80
Xanthine	0·80	7·44	11·12
Hypoxanthine	1·98	8·94	12·10
Guanine	3·3	9·2	12·3
Uric acid	5·4	10·3	

guanine has two maxima, around 245 and 280 mμ, and a minimum near 260 mμ. Uracil absorbs maximally somewhat below, and the other two pyrimidines somewhat above, 260 mμ.

These bases do not readily form the strong quaternary nitrogen ions, another instance of their chemical restraint, in fact, so that like the pyrroles and some other aromatic compounds they are sparingly soluble. This is put to good use for excretory purposes, particularly in animal groups such as birds, insects and spiders for which water is precious; these animals excrete most of their nitrogen as insoluble purines, whereas other animals excrete most as very soluble forms, ammonia, urea and trimethylamine oxide. Spiders live mainly on the already very desiccated insects, and they excrete guanine, the most insoluble purine possible. The alkali metal salts of uric acid are soluble enough for transport in the body and the free acid is then deposited in the lumen of the Malpighian tubules. In addition uric acid can form a colloidal solution (Höber, 1945, p. 574) which rather readily changes to an unstable suspension, and this also may be exploited for excretory purposes. Combined into nucleotides, by contrast the bases become very soluble, and this may be one reason for the combination; it greatly increases their versatility.

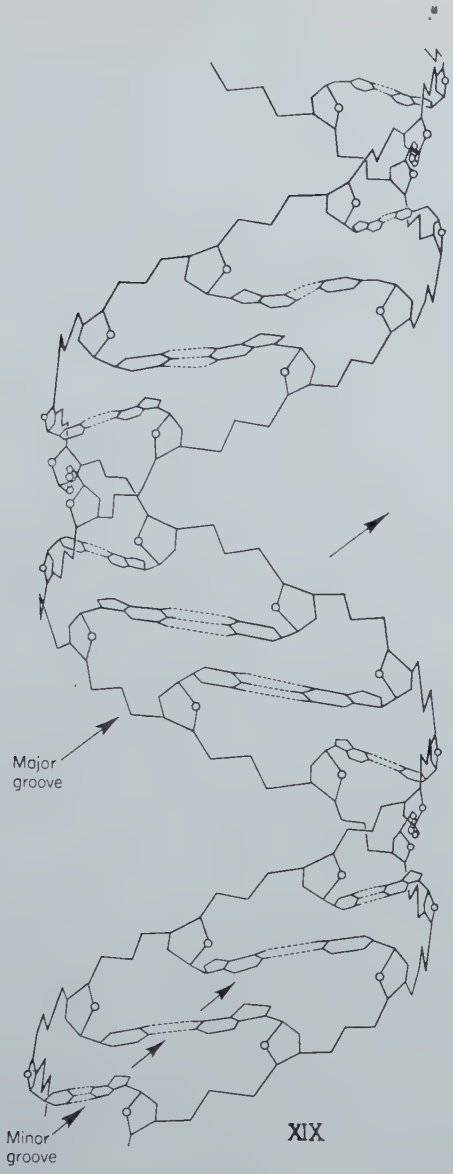


FIG. 44 (*continued*).

The sequence —NH—C (=O)— which occurs so frequently in the rings of these bases is effectively a peptide-bond unit and this may be significant for the close association and interactions between nucleic acids and proteins. The bond of course is present also in urea and therefore it is not surprising that some animals excrete virtually all of their nitrogen as urea and others all as purine. The nitrogen and oxygen atoms take part in hydrogen-bonding as they do in the peptides (p. 215) and the nitrogen of side chain amino groups also contributes to this. It is important for holding pairs of nucleic acid chains together and for bonding nucleic acid to protein, and in quantity it gives the strength of bonding already seen in the proteins.

As side chains on the rings, only —OH (or =O), NH_2 and $—CH_3$ are at all common, the first two as alternatives at the positions 2, 4, 6, 8, and the last usually at 1, 3, 5, 7. These side chains as usual play an important part in the chemical activity of the compounds, in addition to the respects already considered. The pyrimidines found to be most effective as antimalarial agents have amino groups on either 2 or 4 and on one other position, and if the group on 2 is replaced by hydrogen or methyl, or even by another basic group, activity is destroyed. Among its functions, the amino side chain is used to bond nucleotide coenzymes to their apoenzyme (Pullman and Pullman, 1962).

The action of the =O side chain is well illustrated in alloxan (Fig. 44, X), which has this substituent on all four of its ring-carbons (and the ring quite free of double bonds in consequence). It becomes a rather close analogue of dehydrogenoascorbic acid (Fig. 44, XIII), and this also has diabetogenic properties. The $>C=O$ groups cause alloxan to deaminate amino acids, oxidising them to aldehydes. The 5 C=O group can be enolised without causing much change in activity, but any other change, for instance to $>CH_2$, completely inactivates. The barbiturates, which do have this group, are not diabetogenic.

The activity of those members which have methyl side chains—thymine, the beverage purines, and others—lose their activity if the group is removed (Albert, 1957). An alkyl group at 5, or at both 5 and 6, is necessary for the antimalarial action of pyrimidines, and cannot be replaced by halogens, for instance, which are somewhat similar in properties when in organic compounds. On the other hand methyl at position 2 of pyrimidines destroys their antimalarial action: position is all-important. Even the change of such an apparently inert group as methyl can make all the difference to solubility, ionisation, redox level, steric fitting and so on (Albert, 1950). Methylation of the ring-nitrogen atoms of alloxan destroys its diabetogenic activity. It should perhaps be stressed that in many of these examples we are not dealing with the normal physiological function

of natural pyrimidines and purines, but the type of effect is relevant to these; indeed it is relevant to other heterocyclic compounds also. The antimalarial pyrimidines act as antagonists of the benzpteridine coenzymes rather than of normal pyrimidines.

The biological purines and pyrimidines therefore have exploited a limited number of small but very critical side chain groups, permutated according to a rather stereotyped pattern, and this seems a very useful basis for a coding system, the main function now attributed to the nucleic acids. Not all ring positions are necessarily significantly different, but in pyrimidines there are three possible positions for CH_3 and four for $=O$ and $-NH_2$. In purines there are more of both positions. The four bases actually used in most nucleic acids are a small selection of these possibilities, so that coding theory must take them in groups of three in order to code for 20 different amino acids, but transfer nucleic acids offer wider scope (p. 348). From Table 32 it is seen that their bases show a good deal of permutation of the three side chains.

20.2 Nucleotides

Apart from the excretory forms of purine, and their precursors, there are few free pyrimidine and purine bases with major functions in the body, so that more is known about the biological action of the rarer, and the laboratory-synthesised, members than about that of the common bases. These are nearly all active only in the nucleotide form (Fig. 44, XIV, XV). Pyrimidine bases are bonded via their 1–N and purines via their 9–N with position 1′ of a pentose sugar, usually ribose (RNA) or deoxyribose (DNA) but 3–βD–arabofuranose in some nucleotides of sponges (Bergmann and Feeney, 1951), 2′–methoxy ribose in some transfer RNAs (Sinsheimer, 1954; 1962), and the sugar alcohol, ribitol, in the benzpteridine nucleotides. The ribose is bonded via its 5′ position to phosphate. Adenylic acid, or adenosine monophosphate, AMP (Fig. 44, XIV), may be taken as the type for purines and for RNA, and thymidylic acid, TMP (Fig. 44, XV), for pyrimidines and for DNA nucleotides. The base–sugar combination, which can be isolated, is known as a *nucleoside*, and it will be noticed that the complete nucleotide has the trinity of base–neutral–acid already seen in the phosphatides (p. 295), cerebrosides, mucosaccharide units and some other complexes.

As already seen (p. 51), phosphorylation activates sugars, and reciprocally the phosphoric becomes a stronger acid. Pentoses harden gelatin and may have some such stabilising action in nucleic acids also. On the other hand both ribose and deoxyribose are very unstable as free sugars (Baldwin, 1953) and depend on the nucleotide combination to stabilise *them*. Throughout, the structure of nucleic acids seems to be one of balance

and restraint, stability and controlled reactivity. Ribose is the only pentose which could satisfy the constructional requirements for polynucleotides (Todd, 1954), and the D–isomer is essential for steric reasons—to produce a good helical coil (Pullman and Pullman, 1962).

Probably all the five nucleotides which build the common nucleic acids also exist as free nucleotides, with or without extra phosphate groups, and most of these have been found to act as coenzymes for one or more important reactions in the body, mainly connected with biosynthesis. Adenosine triphosphate (ATP) is unique in catalysing almost every reaction in the body which is not spontaneous, whether anabolic or catabolic, including the phosphorylation of the other nucleotides, and many steps in nucleotide synthesis (Fig. 45). In addition to these there are a number of free mono- and dinucleotides which never form nucleic acid polymers but are also coenzymic in function. The pyridine mono- and dinucleotides have already been considered (p. 334), and they are closely paralleled by the benzpteridine mono- and dinucleotides (FMN and FAD), except that ribitol replaces ribose in the benzpteridine nucleotide (though not in the adenylic acid moiety of the dinucleotide). Cyanocobalamin (p. 321) is another nucleotide structure among B-vitamins; the base is benzimidazole in the most common form but adenine in pseudovitamin B_{12} while adenylcobamide has also an extra adenine–pentose unit in the molecule (Harrow and Mazur, 1962).

Pantothenic acid is a B-vitamin which again contains adenylic acid in its completed coenzyme, *co-transacetylase* (XVII) known as *coenzyme A* (CoA). The rest of the molecule, pantatheine-phosphate, is also somewhat analogous to a nucleotide, with a base at one end, a neutral portion in the middle and an acid at the other end. Pantothenic acid is such a unique structure that there has been no occasion to deal with it under any specific heading. It consists of β–alanine (p. 176) peptide-bonded to *pantoic acid* which is unique among biological molecules in being a tertiary alkyl compound. *Cysteamine*, or thioethanolamine, which is peptide-bonded to the other end of the β–alanine in the complete pantatheine, is of course decarboxylated cysteine (Table 30). With an extra phosphate group on the 3′ C of the ribose of the adenylic acid the complete dinucleotide has some resemblance to NADP (p. 334).

Coenzyme A is unique among nucleotides in having an active SH group and this is the critical group of the coenzyme, actually bonding with the acetyl and other acyl radicals transferred. This coenzyme is also unique among sulphur compounds (p. 392) in forming a bond of high transfer value, which can be represented: $CoA{-}S{\sim}C\ (=O){\cdot}CH_3$. An obvious problem for the future is to explain why so much accessory gear is necessary in this coenzyme molecule.

Other B-vitamins, such as pyridoxine (p. 336) and thiamine (p. 397), are not nucleotides in their active form but they are phosphorylated and sometimes pyrophosphorylated, so that they are somewhat analogous to nucleotides. As already noted, adenylic acid becomes a very active metabolite, ATP, through pyrophosphorylation, the bonds between phosphate residues being of high transfer value (p. 402). The other four nucleic acid nucleotides also form pyrophosphates, which are the active forms in which they are incorporated into nucleic acids and in which they act as coenzymes. In nucleic acids, however, they always become simple nucleotides, relatively stable and inactive by comparison with the nucleoside triphosphates. The extra phosphate groups make the latter also more soluble than the simple nucleotides.

The present interest is of course primarily in the pyrimidine and purine nucleotides, rather than in all analogues of the nucleotide structure. The uniqueness of ATP will be considered in more detail later, since it is largely the mouthpiece for the uniqueness of phosphorus itself (p. 404), but there are a number of properties of the pyrimidine and purine nucleotide pyrophosphates which are more relevant here. It has been seen in earlier chapters that ATP is synthesised from its precursor-nucleotide and inorganic phosphorus by the aid of energy trapped in photosynthesis, or subsequently released by reoxidising its products. It is also already clear that ATP then acts as the universal energy mediator as well as a source of phosphate for phosphorylating other metabolites, including the other nucleotides. This last reaction may be represented: NMP + ATP → NTP + AMP.

Guanosine triphosphate, GTP, plays a unique role in protein synthesis, catalysing the final assembly of amino acids which have been brought together, each by its special transfer RNA. No other NTP can deputise for it. Cytidine triphosphate, CTP, is coenzyme for a critical step in the biosynthesis of phosphatides (Kit, 1960). It forms an intermediary pseudodinucleotide with the phosphoryl–choline and phosphoryl–ethanolamine. The resulting CDP–base conjugate then transfers the base to the other moiety, α, β–diglyceride. Uridine is a coenzyme as the disphosphate, which catalyses the Walden isomerisation of galactose–1–phosphate to glucose–1–phosphate. UDP–glucuronate is probably involved in the synthesis of the mucopolysaccharides. Kit (1960, p. 34) gives an impressive list of reactions catalysed by three of the four NTPs: ATP, UTP and GTP. None is recorded for thymidine triphosphate, or any of the other deoxyribose nucleotides. Some of the other NTPs can deputise for ATP in promoting certain phases of the cycle of contraction in muscle, a catalytic function of a rather special kind.

The biosynthesis of the natural pyrimidine and purine bases, which as usual throws light on their unique role, is best left until this point because

they are cast as nucleotides before the definitive base is completed. The pathways (Fig. 45) show other features of interest: notwithstanding the structural relationship between the two series of bases their pathways are quite different, from the beginning. Both are total syntheses, but from different precursors. The earliest precursor of the purines is already conjugated with ribose phosphate but the pyrimidine ring is closed before this conjugation occurs, so that only the final modification of the base is then necessary to give uridylic and cytidylic acids. Neither is built up from triose (or some other C_3 unit) and urea units, as their structure seems to imply, and as early work tried to show, following up such clues as the laboratory synthesis of barbiturates from malonic acid and urea. However, it is interesting that carbamyl phosphate is the starting point for pyrimidine synthesis, as it is for urea synthesis, and that aspartic acid makes a contribution. Once the correct pyrimidine pathway had been discovered it was found that the intermediary, carbamyl aspartic acid, or ureidosuccinic acid, could be prepared in the laboratory (Fox, 1956) by heating together malic acid and urea at 140°C; the biological pathway therefore is probably a spontaneous one. Another interesting feature of the synthesis is the use of the compound, 5′–phosphoribose–1′–pyrophosphate, PRPP, as the agent which adds the phosphoribose unit.

It performs the same function also in the purine pathway, but in this case at the very outset. In this pathway another significant feature is that the imidazole ring is closed first, and this helps to explain why the path is quite distinct from that of the pyrimidines. Perhaps even more surprising, it is quite different from that of histidine (p. 184), the most important biological free imidazole. A formyl group is involved, at positions 8 and 2, in the closing of each ring, which recalls the role of the aldehyde group of glutamic semialdehyde in closing the proline ring (p. 181), in the final closure of the pyridine ring of nicotinic acid in its synthesis from tryptophan, and in other cases, and the role of the keto group in forming the pyrrole ring (p. 312). The carbamyl group used for the purpose in the pyrimidine pathway can be considered analogous. All emphasise the activity of the $\rangle C{=}O$ unit. Another interesting feature is that ATP is necessary for the modification of the initial nucleotide, inosine mononucleotide, IMP, to GMP, while reciprocally GTP is a necessary coenzyme for the conversion of IMP to AMP by a second branch of the pathway; this should maintain a constant ratio between the two.

The acidity of the nucleotides is due to the phosphoric acid, enhanced by conjugation with ribose. The pyrimidines and purines are such weak bases that although a second hydroxyl group of each phosphoric acid residue is involved in polymerising the nucleotides the resulting polynucleotides or nucleic acids are still very strong acids.

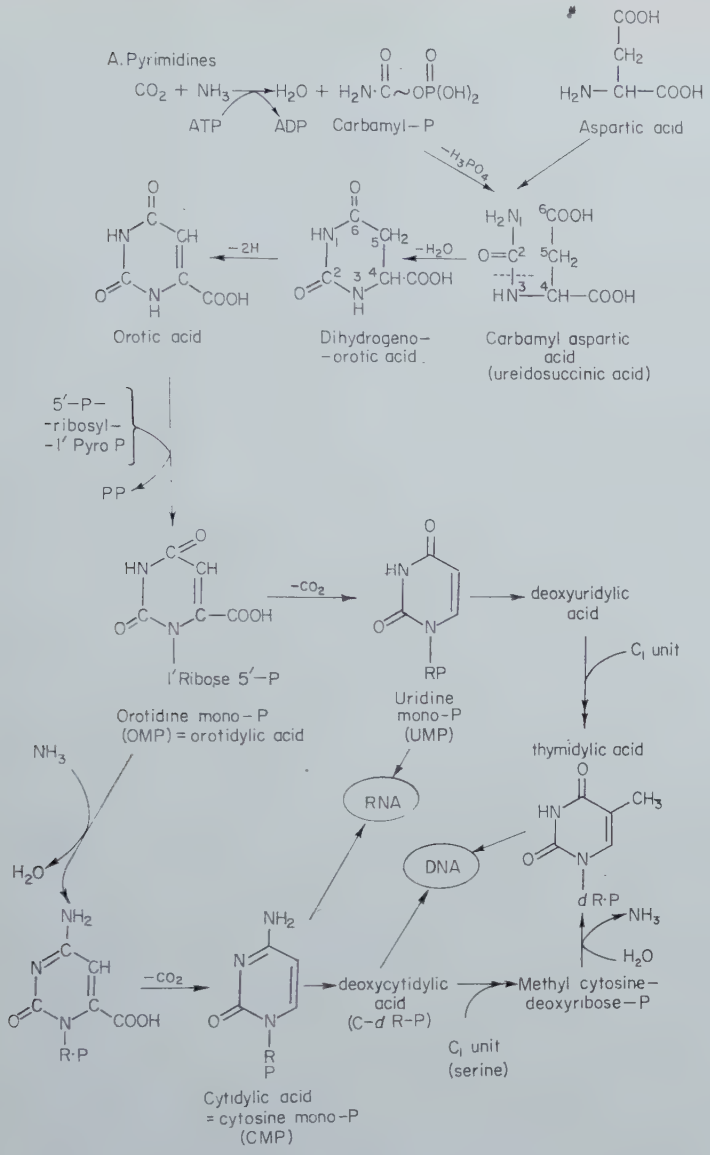


FIG. 45A Pathways of biosynthesis of pyrimidines.

20.3 Polynucleotides: Nucleic Acids

There is less significant distinction between polynucleotides and nucleic acids than between polypeptides and proteins. The coenzymic dinucleotides may be compared with the rare oligopeptides, such as GSH, but there is no other distinct functional category; oligonucleotides are probably rare, and without specific function, though some RNAs have relatively small molecules.

Polymerisation occurs by a second esterification of the phosphate of one nucleotide with the 3′–OH group of the ribose of the next. The result is a linear alternation of phosphate and ribose, forming the backbone of the

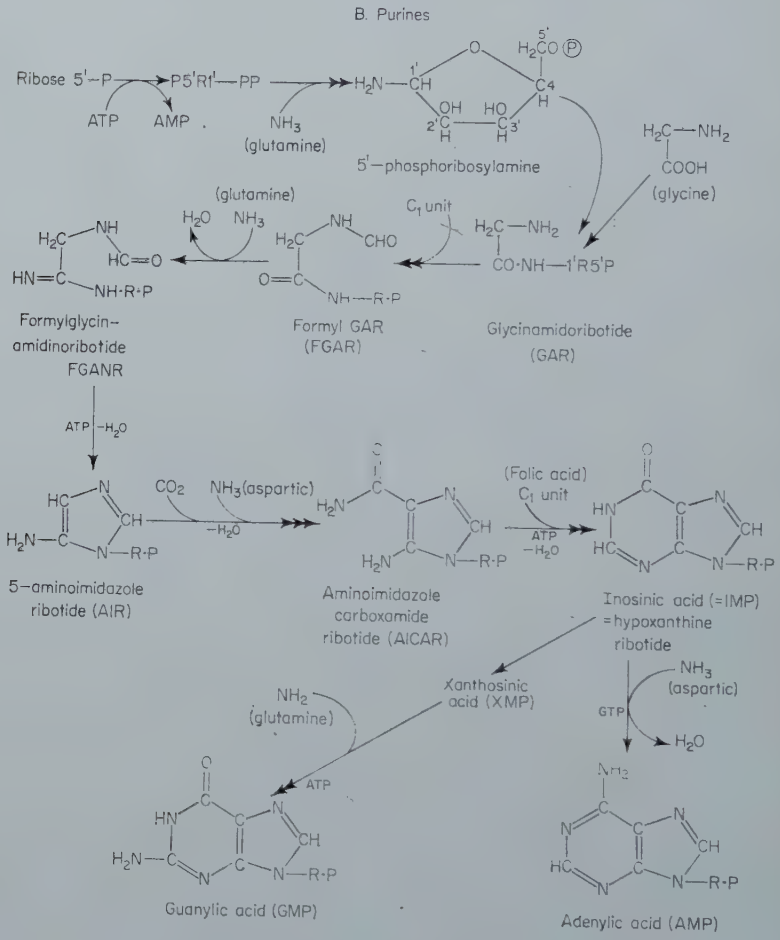


Fig. 45B Pathways of biosynthesis of purines.

nucleic acid, with the bases as side chains (Fig. 44, XVIII). At one time it was thought that there was a regular repetition of the four nucleotides but it is now certain that the order is more irregular and specific to each nucleic acid. There is no theoretical limit in size and some DNAs in fact have molecular weights of over a million daltons.

Asymmetrical monomers automatically build a spiral polymer, so that nucleic acid chains are rather similar helices to those of the fibrous proteins. The association between D–sugar and phosphate gives a helix of minimal strain (Pullman and Pullman, 1962). However, the phosphate residues, with their strong like charges, tend to force apart the turns and single strands tend to be disorderly. If two strands become approximated they may form a stable co-coil because the bases all tend to lie with their planes perpendicular to the axis of the helix (Fig. 44, XIX) and have good powers of hydrogen-bonding, as already noted, with their opposite number of the second chain. The classical triumph of Watson and Crick (1953) was to show that there are strict steric restrictions on which bases can pair in this way: adenine can pair with uracil and with thymine, only, and guanine only with cytosine, among the common five, though hypoxanthine can bond with both adenine and with itself (Rich, 1959b). Hypoxanthine in fact can bond with two others so that in the laboratory triplex helices of inosinic acid have been built, but it is not known that the natural nucleic acids are ever more than duplex, as in most DNAs. The ribose residues lie with their planes in the enveloping cylinder of the helix (Fig. 44, XIX), and contribute to hydrogen-bonding in this plane. The envelope is a remarkably uniform cylinder, the span of the two possible base-pairs differing by less than 3 per cent.

The two strands of a duplex are strictly complementary, and it seems possible (Watson and Crick, 1953) that this also facilitates the autosynthesis of more nucleic acid material. If it is true that DNA carries all genetic information then it must carry that for its own multiplication—by direct or indirect means. A direct and simple replication, in which strand A catalyses the assembly of nucleotides in the specific order to synthesise a new complementary strand, B, while the original B controls the assembly of a second A, has the twin virtues of plausibility and simplicity. The original duplex may unwind as a preliminary, and the new strands be actually laid down as a co-coil, with the existing complement acting as a kind of mould or templet. There is evidence from the replication of the chromosomes, which may be composed of very few strands of DNA, that this type of templeting does occur. It may be unique in biosynthesis, though it has something in common with crystallisation and related phenomena of auto-aggregation into orderly patterns. It is further thought that DNA strands may serve as templets for the synthesis of the

complementary RNA strand, reckoning that uracil ≡ thymine, and that the resulting RNA can act as a simple translation of the information in the DNA. A further theory is that this 'messenger' RNA acts as a code for the assembly of amino acids into genetically specific proteins, and there is growing support for this general idea, as well as for the specific theory that each constellation of three nucleotides is the 'code' for one particular amino acid (Crick, 1963). Whatever the details, it seems clear that here is the most uniquely 'ingenious' property of the nucleic acids, and the most vital of all properties of any biological material.

While the bases determine the even diameter of the helix, the ribose residues tend to ensure even spacing between turns. By their mutual repulsions the phosphate residues contribute to this, as well as to the 'turgidity' of the whole coil. No doubt constant and uniform measurements are very important in molecules with the properties of a spatial templet. The repulsions between phosphates are nicely balanced against the hydrogen-bond attractions and any change in ionic concentration, particularly of H^+, can disturb this and cause the chains to uncoil and separate. Reducing the concentration of cations has this effect and reciprocally an increase in concentration diminishes the electrostatic repulsion between $\equiv PO_4$ groups and makes the coil more compact (Jacobson, 1953). On a somewhat higher scale of magnitude this may control the spiralisation of the chromosomes, prior to nuclear division. The importance of ionic changes in the control of muscular contraction encourages the view that something similar may operate in proteins which are, like myosin and actin, richly bonded by ATP. Here, however, there is the additional factor of the pyrophosphate bond, and no continuous polynucleotide chains.

When two or three polynucleotide strands are co-coiled their forces of attraction are fully satisfied and they do not go on to build cables of higher order, as the proteins do. The forces are not fully satisfied in a duplex and polyuridylic acid co-coiled with polyadenylic acid in the laboratory will incorporate another polyuridylic acid strand after the duplex has been formed. Since biological nucleic acids are usually if not always one- or two-stranded, it has been suggested that a polypeptide chain may occupy the third place (Wilkins, 1956). In fact it is possible that a polypeptide strand may occupy the minor groove between the two nucleic acid strands, as well as the major groove between turns of the duplex (Fig. 44, XIX).

The separation of the strands of a duplex occurs more readily at high temperatures and the process therefore has been called the melting of the DNA crystals (Doty, 1956). It does not occur spontaneously until a temperature of 90–95°C, which helps to explain why nucleic acid does not show the rather low-temperature denaturation characteristic of proteins

(p. 243). Melting is complete over a relatively narrow range of 5°, so that the 'crystals' can be regarded as relatively 'pure', with the same base-composition and molecular size. There is a decrease in viscosity on melting so that the molecules must become more compact, perhaps like a globular protein molecule. Melting occurs spontaneously also at extremes of pH, viz. 2·6 and 11·8; at the higher end presumably the nitrogen bases become active quaternary cations and repel each other.

In principle RNA could form branched molecules because of the third available hydroxyl group on the ribose. Deoxyribose lacks this and DNA can never be branched; it seems very probable that it has been naturally selected for just this reason, since genetic mechanisms demand simple linear structures. No other specific function of the deoxy sugar has become evident: ribothymidylic acid pairs with adenylic acid and also with deoxyadenylic, and the deoxynucleotides in general behave in other respects also very similarly to those of RNA. In practice branching is probably rather exceptional in RNA also. If it were always synthesised by a templet mechanism, on a strand of DNA, or on RNA which had itself been specified in this way, then it would perforce be unbranched. In addition, there is a tendency for the phosphate to form a cyclic diester with *both* of the available OH groups of the ribose of another nucleotide, in the process of polymerisation. The bond in fact resonates between the 3′ and 2′–OH positions through this cyclic intermediary. This makes the polynucleotide bond less stable than that of DNA but at the same time more chemically active (Todd, 1954). In consequence of this cyclic structure, RNA is sensitive to acid and alkaline hydrolysis, a sensitivity very unusual in phosphate-diesters, and not shown by those of DNA. In principle the phosphate residue also has a third group for esterification but this is a very weak acid, ionising at a rather alkaline pH.

RNA is less regularly crystalline than DNA, according to X-ray diffraction pictures, so that it may branch to some extent or develop other irregularities due to the same cause. However it can form duplices like those of DNA, with the same 10 residues per turn. In duplices the amount of A must exactly equal that of T (or U) and G must equal that of C; in natural DNA the ratios in fact are usually near unity but not always in RNA (Leslie, 1961), indicating that it often consists of single strands, without a simple complementary mode of synthesis.

Because of their macromolecular size and the number of polar groups lost by esterification, the nucleic acids are much less soluble in water than the mononucleotides, though alkalis improve solubility by promoting the ionisation of the weak acid groups. Sterols also improve the solubility (Henry and Stacey, 1946), and are used technically for the purpose. Nevertheless nucleic acids have a high affinity for water, because of their

many hydrogen-bonding groups; indeed these no doubt are denied to other strands of nucleic acid largely for this reason. The duplices contract when they are dehydrated by concentrated salt solutions, so that water shares in their crystalline structure. DNA is perfectly crystalline when its ratio to that of water is 60:40. At this proportion an indefinite number of DNA molecules can associate in perfect order; this is borne out by the high dielectric constant of the system. At higher concentrations of water the DNA duplices separate and at higher concentrations of DNA they are packed more amorphously, in paracrystalline arrays (p. 231); this is shown by a narrowing of the magnetic resonance of the DNA. The orderliness of the water itself also is maximal at the 60:40 ratio and its magnetic resonance broadens from 3 mG as distilled water to 15 mG at this ratio (Jacobson *et al.*, 1954). The two components thus form a common, hydrogen-bonded, crystalline continuum, conforming to the fundamental tetrahedral pattern of water, and of the carbon and the quaternary nitrogen atoms. It is thought that there may nevertheless be a polarised transfer of energy, on occasion, along the length of the DNA helix, perpendicular to the base pairs (Glass, 1961), so that the structural organisation may be not only very extensive but also considerably differentiated. Protein also is probably incorporated into the common continuum, *in vivo*, and indeed is a major component, and it is visualised (Jacobson, 1953) that living bioplasm is always such a complex liquid crystalline system, in which everything but mere passive 'inclusions' have an organised place. Such a continuum is amenable to perfect morphogenetic control and seems the best explanation of the remarkable and faithful specificity of form, from the macro level downwards.

Although these liquid-crystals of DNA are best developed at a concentration as high as 60 per cent the salt, sodium deoxyribonucleate, can form a solid gel at a concentration as low as 2 per cent. This remarkable paradox is reminiscent of the boundless variety of structure produced by the proteins. The sodium–DNate gel melts on warming, as gelatin does. Again, other salts prevent gelation, which therefore may occur upon dialysis (Butler and Jones, 1951). Another property reminiscent of the proteins is that DNA binds acidophil dyes more avidly the more dye is already bound (Cavalieri and Stone, 1954); the resemblance is to the greater solubility of some proteins in their own solution than in distilled water (p. 224).

20.4 Nucleic Acids and the Specification of Proteins

Like proteins and unlike most other materials in the body DNA is species—and even individual—specific, although it does not react im-

munologically as the proteins do. The reason is probably that DNA is not metabolically active but expresses its specificity of action only through protein as executive. It may nevertheless be the more important component, like most administrators; if it manages to enter the cell of another species it is able to redirect the whole machinery of biosynthesis in its taxonomically specific direction, as the nucleic acid of viruses does. It is an active and not a passive genetic code. The DNA of one strain of bacterium transferred in this way 'transforms' the host cell to donor type in morphology and activity (Avery *et al.*, 1944). The sperm, introducing little but DNA into the ovum, has a similar effect. Together with the evidence from mitosis and meiosis, and the knowledge that the continuity of the chromosomes is due to the DNA and not to the protein (Callan and MacGregor, 1958), this amounts to strong evidence that DNA carries the genetic information for making specific proteins and other materials.

Exactly how it administers the process is still very uncertain and to some extent may be a systemic matter, beyond the scope of the present theme, but it must be rooted in the innate properties of the nucleic acid and this is the most fascinating aspect of the problem; this problem is rather naturally considered the most important in biology. At the molecular level the only plausible mechanism is considered to be one in which nucleotide groups code for each specific amino acid, as already suggested, or for the link between two specific amino acids. The first alternative is preferred because there are 20 common amino acids but 20^2 different dipeptide linkages. Even so it is not easy to see how combinations from a mere four nucleotides can constitute effective, unambiguous codes for 20 amino acids and determine their peptidation as well as their specific order. Groups of two nucleotides would be inadequate since there are only 16 possible pairs, while a group of four or more would give far too many possibilities. There are 64 possible trios or 'triplets' but precisely 20 if the further requirement be met that overlapping sequences cannot be utilised for fear of confusion—the code having no punctuation marks. If ABC and BCD code for two particular amino acids then BCB and CBC must be 'nonsense' sequences; thus at every junction between two triplets which make sense there must be two which are ruled out, two-thirds of the total number. Some of the 20 amino acids, for instance hydroxyproline and norleucine, are rather casual protein builders, and cystine may be always formed from cysteine so that there may be one or two spare triplets for other rare amino acids in proteins.

The complication of overlap intuitively recognises a spatial problem, that the linear peptide must be specified by a linear code. It is a short jump, if not an inescapable one, from the linear code to the supposition that it acts as a spatial mould or die on which the protein is actually cast.

However, while the idea of a simple templet may be apt for the replication of DNA itself, protein synthesis is more like the translation of a language or deciphering of a coded message, to which such direct spatial contiguity is not so indispensable. If it does occur then the nucleic acid is executive as well as administrator.

Most of the cell's production of protein occurs in the cytoplasm, whereas DNA rarely leaves the nucleus, so that its instructions must be carried out by a deputy or messenger. This is thought to be the role of RNA, also synthesised on the DNA as a templet. There is now strong evidence that RNA synthesised in the nucleus does pass out into the cytoplasm and collects on the ribosomes where most of the protein synthesis of the cell takes place. If, as the base-ratios indicate (p. 363), only one type of strand is synthesised, then we probably have to regard the second strand of each DNA duplex as a dummy or 'keeper'. This is in some ways preferable to the alternative supposition that the adjacent nucleotides on both strands together form the triplets since the restriction of base pairing between the two strands so drastically restricts the number of possible triplets (Levinthal, 1959). For this purpose, in fact, base pairing is a serious handicap.

There appears to be another coding system in connection with the activation, transport and martialling of specific amino acids on the ribosomal sites. It is embodied in the series of soluble or transfer RNAs already mentioned. One RNA of this type is specific to each amino acid. Their variety is increased by various uncommon bases, as already noted, but on the other hand all appear to end in the same sequence of three nucleotides, which of course may be the code for a punctuation mark or for the common peptide bond unit. There is evidence (von Ehrenstein and Lipmann, 1961) that this set of nucleic acids do not code for the order of the amino acids in the eventual peptide but that the messenger RNA does.

It is interesting, but perhaps only generically, that ATP provides the energy for the activation of the amino acids for their union with the transfer RNA. It does so by actually bonding with them:

$$\text{ATP} + \text{amino acid} \rightarrow \text{AMP–amino acid} + \text{PP}$$

The liberation of inorganic pyrophosphate in this way is characteristic of a whole group of 'activation' reactions, not restricted to amino acids (Kit, 1960), and this is only one of the many actions of ATP. Another relatively non-specific component concerned in protein synthesis is the permanent RNA of the ribosomes so that, in all, nucleotides and their polymers play a very major part in protein synthesis. By comparison with the specific coding mechanism, however, the other functions are relatively mundane.

There are alternative possibilities to the literary code, which at first seemed too ingenious even for nature, but evidence in its favour now accumulates rapidly (Crick *et al.*, 1961; Crick, 1963). The most important line of evidence is that one or two deletions in the relatively simple genome of a bacteriophage, each deletion probably one nucleotide in extent, jam the whole code translation, whereas three deletions restore virtual normality. This is in accord with the triplet theory, with its rider that all overlap sequences do not specify an amino acid, but 'read' as nonsense to the biosynthetic mechanism. In this event every triplet from the first one which spans the first deletion to the far end of the code for that particular protein is read as nonsense. Two deletions leave the same situation but three bring the reading back into the normal step beyond the third, with only the short stretch through the three deletions being out of action, corresponding to an unimportant sequence of amino acids, perhaps. The insertion of extra nucleotides shows the same phenomenon, three are relatively harmless but one or two disrupt protein synthesis and produce mutant types.

Another line of evidence comes from the use of simple laboratory polynucleotides to catalyse amino acid incorporation into proteins or rather peptides (p. 198). This has shown that there is indeed a relationship between particular nucleotides and particular amino acids. Polyuridylic acid promotes the incorporation of phenyl alanine particularly, and that of other amino acids in graded, lower rates. This is taken to imply that the code for phenylalanine is UUU and that those amino acids next in the gradient have two Us in their codes. Continuing this analysis to other homopolymers and then to mixed polymers of two different nucleotides has permitted a tentative list of the actual coding triplets and this has shown some corroboration between different groups of workers and conformity with other lines of evidence. Polycytidylic acid has a secondary structure which prevents it acting as a good templet (Crick, 1963); this is not necessarily true of cytosine in natural heteropolymers, but it is recalled (p. 357) that CTP rarely functions as a coenzyme. When it is possible to determine the actual sequence of nucleotides in natural DNAs then another line of attack will be open.

The greatest drawback to the triplet code idea is that it has at present no evident chemical rationale. A purely arbitrary code is satisfactory as a human mathematical exercise but it is not in keeping with our knowledge of biological uniqueness and its evolution. It is difficult to see how two pairs of very similar bases could possibly provide a logical chemical code for 20 much more variable amino acids. They might provide a very neat shorthand representation, for economical storage, and in principle this might have great selective value, but it is difficult to believe in the

possibility of selection of a purely arbitrary code which could also organise the specific polymerisation of the amino acids.

20.5 Biosynthesis of Nucleic Acids

It has already been seen that the biosynthesis of DNA may be autocatalytic, and that this is what might be anticipated if DNA is *the* essential genetic material. There is also positive evidence that a small amount of DNA must be present, as a 'starter' or 'primer', before any DNA synthesis is possible at all. By contrast RNA is not required as a primer for its own synthesis, so that this gives further support for the idea that it is templated on DNA, though in fact it has not been shown that RNA synthesis always needs starter amounts of DNA. A good deal of RNA is synthesised in the cytoplasm and continues for some time in enucleated cells (Brachet, 1960).

The synthesis may be to some extent autocatalytic but it is far from being spontaneous and exergonic. Synthesis of the bases themselves requires about 60 kcal of energy per mole, and a large battery of enzymes, though it does also entail the completion of the nucleotide. The subsequent step of polymerisation has been achieved in the laboratory, but only with the nucleotide triphosphates as starting point and with the help of enzymes of biological origin. The resulting NA is then specific to the species which donated the enzymes. Some of these can synthesise RNA from the nucleotide *di*phosphates as starting point. Both DNA and RNA have been synthesised in this way (Kornberg, 1960; Gruneberg-Manago, 1955).

The enzymes synthesise species-specific nucleic acid if presented with all four nucleotides in adequate amounts, but if given only one they catalyse the synthesis of the homopolymer. They will also catalyse the incorporation of nucleotides such as inosinic acid, which is not normally incorporated into nucleic acids. There is here a curiously paradoxical combination of specificity and non-specificity. There *is* an element of spontaneity in nucleotide polymerisation from their pyrophosphates and it might therefore be supposed that an enzyme is required only to determine the sequence of nucleotides but this does not seem to be true: it is necessary for the polymerisation reaction itself. However, large flat molecules, such as the nucleotide bases, tend to pile up spontaneously at a concentration greater than $M/10^4$ (Calvin, 1957), and this should assist both polymerisation and helix-formation. The co-coiling of completed strands also seems to be spontaneous. If poly-A is mixed with poly-U they form poly-A/poly-U duplices and even AUU triplices, as in the laboratory experiments mentioned earlier. They combine stoichiometri-

cally, so that the process is a true chemical bonding—by hydrogen bonds. The wavelength of maximal absorption shifts when the two combine, which shows that there is resonance across the hydrogen bonds linking the pairs of bases. There may also be charge-transfer between the superimposed bases, along the axis of the helix, as already indicated.

20.6 Nucleoproteins

In the living body nearly all nucleic acid is associated with protein; the possible exceptions are the bacteriophage's invasive material, which may be pure DNA, and the soluble nucleic acid concerned in amino acid transfer. However, even these are associated with, and dependent on, proteins: in the complete infective body of the 'phage' the nucleic acid is surrounded by a shell of protein, and transfer nucleic acid works in close conjunction with activating enzymes, which are protein. Most viruses consist of little but nucleoprotein (NP), which therefore may be considered the minimal constituent of bioplasm; the virus parasitises its host for water and other taxonomically non-specific materials. Nucleoprotein reaches a molecular weight of 6×10^7 in tobacco mosaic virus, and has the largest true molecules in the body.

The structural nature of the association between the two components may vary but a co-coiled helix of the type already mentioned seems to be most common. Protamins follow the minor groove between the two strands of the DNA duplex (Feughelman *et al.*, 1955), not the major groove between successive turns of the pair as might be anticipated, the reason being that the arginine basic groups must be near the acidic groups of the phosphate. They then minimise the disruptive tendency of the latter and stabilise the whole molecule. Other proteins, less basic in reaction, do occupy the major groove of the helix. There is a good deal of both types of protein associated with DNA in the chromosomes, so that both grooves may be occupied simultaneously. In tobacco mosaic virus there is a high proportion (94 per cent) of protein and the structure is effectively reversed, that is to say nucleic acid is wound between successive turns of a protein helix. It seems possible that the structure in bacteriophages, where a core of nucleic acid is enclosed by a protein coat, may be a modification of this co-coiled helix, but that of the icosodecahedral viruses may be different. In addition nucleic acid and protein will form the coacervate type of structure (Oparin, 1957a).

Since DNAase alone, without help from other enzymes, will break up chromosomes it seems probable that both types of protein associated with the DNA have small molecules. The protamines and histones, the basic proteins, certainly do have a particularly low molecular weight

among proteins. That of the ribosomes has a value of 12,000 daltons compared with 120,000 for the RNA. They are combined stoichiometrically in the ratio of 10:1, so that their total weights are equal (Petermann and Hamilton, 1961). The stoichiometric ratio indicates a true chemical compound, but this association between protein and nucleic acid is very loose, so that it can be broken by high concentrations of simple salts such as sodium chloride.

$$Pr^{+}—NA^{-} + NaCl \rightarrow Na\text{-nucleate} + PrCl$$

It may be merely electrostatic (Woodhouse and Sherratt, 1952) and not even ionic since the nucleic acid of the chromosomes is still able to react with basic dyes, one of the most familiar facts of cytology. Other proteins may be more firmly bound, so that it is necessary to denature the protein in order to remove it, and the nucleoprotein moves as a unit in an electrical field. It is worth emphasis that the two components are multiple-bonded, in contrast to those of most conjugated proteins, but probably like the lipoprotein and polysaccharide–protein complexes. Some nucleoproteins have as much as 15–20 per cent of lipid associated with them, so that there may be even less casual 'inclusion' material than previously implied. As much as 30–40 per cent of this associated lipid may be phospholipid.

Nucleoproteins have a number of interesting and biologically important properties. The association sometimes makes for greater stability of both components but sometimes for lability and reactivity. In this connection it is noteworthy that DNA is a relatively stable form, suitable for indefinite genetic transmission, whereas some of the RNAs are particularly labile, chemically active forms. The nucleic acid moiety is often stabilised, by the combination, against irradiation, and against the action of nucleases (Williams, 1956), and the protein is stabilised against heat-coagulation (Greenstein and Hoyer, 1950). Once removed from the nucleic acid it often becomes precipitable by saturated sodium chloride. Virus nucleoprotein is more immune to the host's defences than either component alone (Commoner and Wang, 1956). The λdg bacteriophage is one nucleoprotein which is less stable than its DNA alone (Hogness, 1962). Some deoxynucleoprotein gels are spontaneously unstable (Bernstein, 1956). Nucleic acid tends to depolymerise to some extent before uniting with protein, and globular proteins first unfold. Both of these processes could increase or decrease lability depending on the kind of union subsequently effected between the two components.

The physical properties of the nucleoproteins also are full of interesting paradox and novelty. The viscosity may be less than that of the two components separately, or it may become higher and very anomalous,

in keeping with the complexity of some of the nucleoproteins. A lower viscosity usually means a more spherical molecule, and at this order of magnitude it may mean a larger rather than a smaller molecule, because of the diminished area of contact between solute and solvent molecules which this causes. Deoxynucleoproteins are insoluble in physiological saline but soluble in virtually any other concentration, e.g. in 1–2M NaCl, in 0·02M NaCl and in distilled water. The proximate cause of this is not yet clear, but it may be somewhat analogous to the isoelectric point for proteins. Its final or biological cause may be the advantage of having a sharp difference in state when in physiological media from that in any other concentration. At the same time nucleoproteins do show much the same solubility-properties as simple proteins, being precipitated by high concentrations of ammonium sulphate and calcium chloride, and by acids at a pH between 3 and 5. Nucleoproteins, for instance tobacco mosaic virus, form the interesting liquid-crystals previously described, as tactoids (p. 232), and other forms.

Nucleohistone solutions appear to be capable of incorporating more nucleic acid and protein (Brunish and Luck, 1952), so that some nucleoprotein may be even more directly self-replicating than the nucleoprotein of viruses. This would be consistent with the suggestions (p. 361) that DNA can determine its own replication and the synthesis of a specific protein. In bacteriophages, however, there is an indication of reciprocal and sequential heterocatalysis, the nucleic acid being necessary for protein synthesis and *vice versa*. In either event it is clear that this is the most crucial of biological materials.

20.7 Conclusions

Because of their complexity nucleic acids are less well known than most biological materials, and certainly less well known than their importance merits. Knowledge is already adequate at least to show how unique they are among biological, and among all, materials. Apart from the amino acids the nucleotides are the only nitrogen compounds which polymerise extensively, and this may be a main reason for their association in nucleoproteins. Indeed nucleic acids seem to have little significance divorced from proteins and their biochemistry. This is rather striking even by comparison with free nucleotides and with free pyrimidine and purine bases.

The outstanding feature of nucleotides is their ability to form 'energy-rich' pyrophosphates, which are the essential components of a large number of coenzymes, as mononucleotides, dinucleotides and nucleotides conjugated with other moieties. ATP is unique among the nucleotide

pyrophosphates and among all substances (p. 404) for the qualitative and quantitative extent of its catalytic power. The nucleotide monophosphates and the nucleic acids lack these energetic properties, which depend almost entirely on the pyrophosphate moiety.

Apart from guanine the pyrimidine and purine bases commonly found in nucleic acids rarely have biological functions as the free base, but other bases of both groups have powerful metabolic and physiological actions. This can scarcely be fortuitous: there is considerable evidence that the common nucleic acid bases are under restraint in their normal biological combinations (other than the NTPs). Otherwise perhaps they also would show the kind of properties found in the other pyrimidines and purines —properties generally similar to those of other nitrogen-heterocyclic compounds. The reason for the restraint is far from clear, particularly in the case of RNA. The ultraviolet absorption by the bases is diminished when bound in nucleic acid form, their resonance properties being partly suppressed (Davidson, 1960, p. 33).

To some extent the mysteries concerning the nucleic acids may be due to excessive preoccupation with their role in heredity and protein synthesis. At any rate it is possible to say more about the basis of the uniqueness of many less important metabolites.

CHAPTER 21

PTERIDINES

As already indicated the pteridine nucleus is a pyrazine-condensed pyrimidine (Fig. 46, I). It owes its name to the fact that the first pteridines to attract interest were biological members, the wing pigments of butterflies, in the classical biochemical studies of Gowland Hopkins (1895). Like the purine nucleus it has two nitrogen atoms in each ring, and is indeed a rather close analogue of purine, giving a very similar murexide colour-reaction with alkalis, after oxidation; the main difference is that a stronger oxidising agent is necessary (Ford, 1947). Like purines and pyrimidines, pteridines often carry OH or =O side chains which are again acidic because of the enol transformation. The molecule as a whole therefore is once more amphoteric, and once more the nitrogen bases of the ring are often supplemented by side chain amino groups. The acid and base groups ionise poorly, however, and, like the pyrimidines, purines and pyrroles, these are relatively insoluble molecules in neutral aqueous media. They are more soluble in acids and alkalis and also in alcohol. Some become more soluble by conjugation, but even so folic acid is only 0·1 per cent soluble. Insolubility makes them useful integumental pigments, like the purines. Pteridines resemble the purines also in forming metal-complexes.

There is some evidence (Ziegler-Gunder, 1956; Kit, 1960) that pteridine may actually be formed from purine in the body (p. 384), though from the experience with purines, pyrimidines and other types of molecule this is in need of careful investigation. Certainly xanthine oxidase acts on pteridines as well as on purines, but it is a notoriously nonspecific enzyme, acting even on formaldehyde (Schardinger enzyme). There are further grounds for suspicion in the fact that the integumental pigment-pterins seem to be synthesised via a different path from the vitamin-pteridines (Ziegler-Gunder, 1956). However, on structural grounds alone, even supposing the biosynthetic pathways *are* different, it is regrettable that the conventional numbering of the atoms of the pteridine ring runs the opposite way, and starts with a different nitrogen atom, from that of pyrimidines and purines.

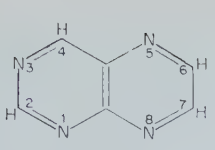
The extra atom, as compared with purines, and the extra double bond, makes the pteridines in general more brightly coloured than the purines and more useful as integumental pigments. They also absorb in the ultraviolet but at a longer wavelength than the purines, namely 340–370 mμ; unlike the purines also they re-emit much of this as visible fluorescence, unless the energy is trapped by conjugation with protein, or with some other materials. The wavelength of the re-emitted fluorescence varies greatly, being yellow in *erythropterin*, blue in *xanthopterin* (Fig. 46, III) and violet in *leucopterin*. Oxidising agents such as potassium permanganate usually quench the fluorescence in acid solution, while those with a blue fluorescence are usually quenched by reducing agents. As a group, therefore, they seem to be potentially very versatile and effective redox agents. Visible light also depresses their fluorescence, and increases their rate of oxidation. It therefore not only activates the molecule itself but also helps to put the ultraviolet absorption to chemical use by the molecule! Equally some pteridines emit light as a result of chemical oxidation, and the substrate for bioluminescence in bacteria is a pteridine derivative (Glass, 1961).

The benzpteridines (p. 380) are redox coenzymes, and other natural pteridines also have redox actions, though not necessarily as their main biological property. Some occasionally may deputise for the benzpteridines as co-dehydrogenases (Ziegler-Gunder, 1956). The pteridines are good electron-acceptors, with a k value for the innermost empty orbital of $-0{\cdot}386$, almost as good as that of nicotinamide (p. 334), therefore. They need help from conjugants in order to become equally good donors of electrons, however ($k = +0{\cdot}864$ for the outermost full orbital). One pteridine, 2–amino–4–hydroxy–6–carboxy–pteridine, is a distant structural analogue of nicotinic acid, by virtue of the carboxyl group.

21.1 Pterins

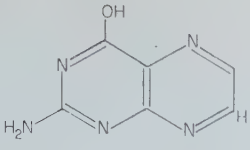
All the simple biological pteridines known have an amino side chain at position 2 and a hydroxyl at 4. This sub-group has been distinguished as the pterins (Fig. 46, II), and is another example of the exploitation of limited sub-groups. The biological pterins include a number of integumental pigments and the *folic acid*, or *pteroyl glutamic acid*, group of B-vitamins; PGA is coenzyme for the transfer of C_1 units in the formyl state and perhaps also in more reduced states. As already indicated, there is some evidence that even these two sections of the same sub-group are not synthesised via the same pathway.

The integumental pterins are interesting in much the same general ways as the other skin pigments (Fox and Vevers, 1960). If it is true that they



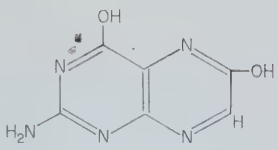
I

Pteridine



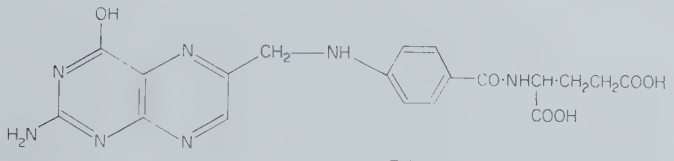
II

Pterin



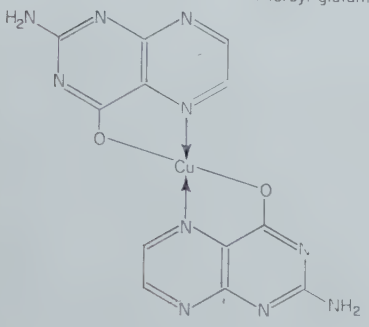
III

Xanthopterin



IV

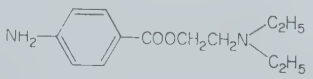
Pteroyl glutamic acid (PGA)(=Folic acid)



V

Pterin

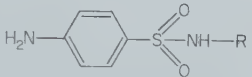
chelate complex



VI

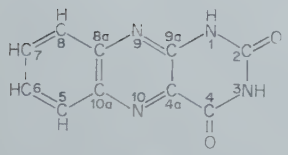
Novocaine

VII



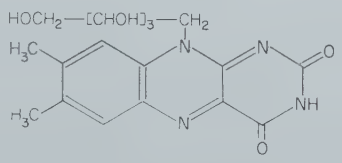
VIII

Sulphonamides (generic formula)



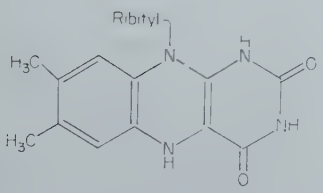
IX

Alloxazine



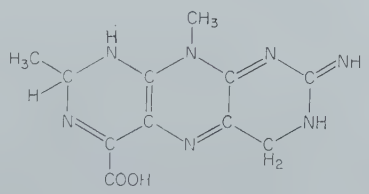
X

Ribityflavin (6, 7,–dimethyl–9–ribityl *iso*–alloxazine)



XI

Leucoflavin



XII

Bacterial luciferin

FIG. 46 Pteridine derivatives of biological significance.

have their own biosynthetic pathway, then like the melanins and ommochromes they may be primary products and not incidental by-products of metabolism, as some carotinoids, phaeophorbides and other porphyrins, the bilins and some purines seem to be. However the benzpteridines, which are primarily redox coenzymes, nevertheless have members which are skin-pigments; these often occur along with the simple pterins which therefore might be by-products of their biosynthesis. This is not necessarily a critical point, however, since it is quite usual for pigments of a variety of unrelated chemical groups to occur together in this way; pterins themselves are almost never found in the absence of melanin, and there are many other examples among integumental pigments, and even among metabolic pigments (p. 321). Camouflage often demands a 'heather mixture' of colours and this is met by several classes of pigment, although many individual classes cover a considerable range of colours. Altogether there is much overlap of ranges and some interesting examples of mimicry between the different classes (Ford, 1937).

The biosynthetic pathways of the different classes of pigment also seem to be much more intermeshed than would be expected on pure chance (Ziegler, 1961), and mutations often affect several simultaneously. There are at least two possible explanations of this; in the first place all pigments have the common property of highly resonant molecules, very active chemically, particularly as redox agents, and this alone could account for much of the intermeshing. Secondly their functional association as integumental pigments may lead to natural selection for their association in biosynthetic pathways, assuming that virtually any two pathways could be brought to intermesh where selection was sufficiently intense. The common redox and activating properties might be expected to facilitate selection in this direction. There is some pterin synthesised in melanophores and it plays a significant part in melanogenesis; this may in fact be an inhibitory control since xanthopterin and pteridine are known to inhibit melanogenesis at the stage of oxidation of dihydroxyindole to melanin (Isaka, 1952). Riboflavin, however, when activated by light of short wavelength promotes the oxidation of dopa to melanin. In turn the oxidation of dopa to dopachrome in the melanin pathway catalyses one step in the biosynthesis of the ommochromes (Butenandt *et al.*, 1956, Johnson, 1961).

21.1.1 Pteroyl Glutamic Acid (PGA)

The pteroyl glutamic acid (PGA) compounds (Fig. 46, IV), as their name indicates, are pterin derivatives of the amino acid type, peptide-bonded to glutamic acid. The pterin, a 9-CH_2OH derivative, is condensed through this position with a special aromatic amino acid, *p*–aminobenzoic

acid (PABA), forming pteroic acid, and it is this which is peptide-bonded to the α–amino of glutamic acid. Like coenzyme-A (p. 356) it may seem, in our present state of knowledge, an unnecessarily baroque piece of architecture to transfer simple C_1 units. PGA and CoA in fact have a similar general plan, with an ω–amino acid as the middle one of three moieties, condensed by its two peptide-forming groups. It is not so clear whether the cysteamine or the pantoic acid of CoA corresponds to the pterin of PGA; the pantoic acid is on the same side of the ω–amino acid but the chemical resemblance, slight in any case, rather favours the alternative comparison.

A number of different forms of PGA have been isolated from living material. Some have two or more glutamic acid residues in series, either α- or γ-linked to each other. One of the most active biologically, the *citrovitrorum* (CF) factor, is formylated at position 5 and this can be regarded as a molecule of the substrate in course of transfer, as in the case of other B-vitamin co-transferases. Other variants of the vitamin have the HC—O group at position 10 also. The latter is part of the PABA moiety, and this helps to explain its association with the pteroic acid. It is probable that ring closure occurs between these two nitrogen atoms via a formyl group (Welch, 1956), which can be transferred from one to the other in this way. Such a labile system also facilitates the subsequent formyl transfer to an acceptor substrate. It may be significant that glutamic acid also forms a formyl derivative—in the synthesis of glutamic itself from histidine, so that all three moieties of PGA can handle the formyl radical. The importance of the formyl group in ring closure in pyrimidines, purines and other cyclic compounds (p. 42) helps to explain the importance of PGA in nucleic acid synthesis and haemoglobin synthesis. It has already been seen (p. 322) that vitamin B_{12} also is associated in the transfer of C_1 units, particularly for this purpose.

The CF factor has two of the double bonds of the pyrazine saturated, and therefore is 5-formyl, 5,6,7,8, tetrahydrogen-PGA. The saturation of course diminishes the resonance of the ring but it appears to be associated with the formylation at 5 and may have functional significance in this connection, as well as in a redox capacity. The essential role of the pteroic moiety seems to be to instigate an electron shift in the substrate donating the C_1 group, and pterins in general readily accept electrons from indoles such as tryptophan and serotonin (p. 192). The anti-PGA action of a range of pterins tested proved proportional to the force with which they accept electrons from tryptophan (Szent-Gyorgyi, 1960). This is another instance where charge-transfer by superposition of molecules is effective; the steric fitting between indole, pterin and purine is surprisingly close.

The 4-hydroxyl group of the pteroic acid seems to be particularly

important, as its only acid group the 4-amino analogue, *aminopterin*, is a powerful antagonist and it is interesting that the power is increased by simultaneous methylation of the *p*-amino group of the PABA moiety, as in *A-methopterin*, and again by simultaneous substitution of aspartic acid for glutamic. This 4-amino aspartic analogue of PGA is also a more powerful antagonist than the simple pteroyl–aspartic acid analogue. It would be useful to know if the two substitutions together have a greater effect than the sum of their independent effects, since this would further prove that the two moieties interact. An extra methyl group at position 9 of PGA itself, or alkyl groups at both 6 and 7, produce metabolic antagonists, so that substitution on any available carbon of the pterin nucleus affects its properties. However, the 4 and 5 positions of the nucleus seem to be the most active, perhaps because of their property of jointly chelating a metal atom (Albert, 1950); two pterin nuclei are necessary and the resulting pseudoporphyrin is closely analogous to that formed by histidine (p. 328), with both nitrogen and oxygen acting as chelating atoms (Fig. 46, V).

The multiple glutamic residues of some variants of PGA further demonstrate the importance of this moiety, already shown in other ways. It may act as the link between the coenzyme and its protein apoenzyme, about which little is yet known. The integumental pterins also are conjugated with protein, but it is not yet known via which amino acid (Ziegler-Gunder, 1956).

21.1.1.1 *para*-Aminobenzoic Acid (PABA)

PABA was known as a vitamin before its PGA compound was recognised (Woods, 1940), and it still seems possible that it may exist as a separate coenzymic agent. Its chemistry and biochemistry have been subjected to intensive study, and rarely with any compulsion to consider it as part of PGA. It is possible that the analogue, *p*-hydroxybenzoic acid, may be a vitamin in its own right, for some bacteria (Davis, 1950). The synthetic local anaesthetic, *novocaine* (Fig. 46, VI), contains a PABA residue so that it may have other physiological conjugates, with neurotropic actions.

Analogues of PABA, in particular the historically famous *sulphonamides*, are certainly powerful antimetabolites, the first class of chemicals found to cope effectively with disease bacteria *in situ* (Sexton, 1953). Their effect is essentially to produce signs of PGA deficiency, which are alleviated by extra PABA alone, so that it is an essential part of the PGA molecule, but probably a very labile one. The benzene ring is fully resonant and the molecule probably resonates between the form shown in PGA (Fig. 46, IV) and a dipole form (Fig. 46, VII), somewhat analogous in structure to the *p*-quinones (p. 132). One of the unshared pair of electrons moves between the *o*- and *p*-positions, the *m*-position being as usual

unstable. This extends the range of resonance, and under the ideal conditions of a fully unsaturated pyrazine ring resonance could be almost continuous from the pyrimidine end of the PGA molecule to the α–complex of the glutamic acid. It also seems probable that the molecule can rotate and fold about the single bonds between the three moieties so as to increase resonance by field effect (Finar, 1959).

The polar amino and carboxyl groups of the PABA certainly are reactive and important. As in other aromatic compounds the group R_2 (Fig. 46, VII) tends to resonate between the two carboxyl oxygen atoms, and so depresses the acidity of the carboxyl group, and this is enhanced by the *p*-amino, again as in other aromatic compounds. The study of variants of the sulphonamide structure (Fig. 46, VIII), as antimetabolites of PABA, is as instructive as attempting to modify the latter itself. The *p*-amino of sulphonamides can be methylated, phosphorylated or acetylated without destroying their anti-PABA activity, and it is probably significant that PABA itself is conjugated with pteroic acid at this position. The amino cannot be replaced by methyl, of course, but in PABA methyl may not even be substituted on the amino; *methopterin*, which is a PGA analogue with this substitution, is an active antimetabolite of PGA. Longer alkyl chains substituted here have less effect. Extra PABA will of course normally counteract any deleterious effect of sulphonamides on animals but substituents on their *p*-amino group protect them from this counteraction (Sexton, 1953); the sulphonamide amino group therefore differs significantly in properties from that of PABA.

In view of this it seems surprising that the $-S(O_2)NH-$ group of the sulphonamides is such a close analogue of the $-C(O)OH$ of PABA. The expectation would be that the sulphonamide group is analogous to the carboxamide group, $-C(=O)NH-$, but since sulphur is much more electronegative than carbon the sulphonamide group is in fact about as acidic as the carboxyl group. The difference between them is further diminished by substituting an electron-attracting group on the amino group of the sulphonamide (Sexton, 1953, p. 129). This is an excellent example of the phenomenon of quantitative uniqueness, since groups which are too electrophilic do not pass on the electron to the SO_2 group; the curve of potency as an anti-PGA drug therefore rises to a maximum and then falls, as the electrophilic power of the substituent R (Fig. 46, VIII) is increased. The analogous group of PABA may be equally sensitive to electrophilic substituents, but it is able to counteract a very large and varied group of sulphonamides substituted in their R-position; it deals with these more widely than with substituents on the *p*-amino group of the sulphonamide. Other *onamide* groups, such as $-SeO_2$,

$-HPO_2$, and $-HAsO_2$ (*atoxyl*) resemble $-SO_2$, in this position, and reciprocally PABA is an antidote to arsenic and antimony poisoning. In passing it may be noted that this implies that these elements are poisonous not merely as analogues of phosphorus.

There is some indication that sulphonamides with substituents on the *p*–amino may act as analogues of other vitamins, such as nicotinamide, and this is further interesting in view of the nicotinamide-like action of some pterins (p. 374). Another interesting complication is that adenine sometimes acts as a more effective antidote to the sulphonamides than PABA itself. One possible explanation of this is that the block is transferred to the pteroic moiety and that the purine is an effective analogue of this, but a rather evident alternative which is not so relevant in the present context is that adenine reduces the demand for purine synthesis and therefore for PGA as an agent in C_1 transfer. At the same time it is generally felt that 2–amino–purine, and probably most purines which act as anti-PGA agents, probably do so as direct analogues of the pteroic acid (Stokstad and Jukes, 1949); there is perhaps here some clue to the fact that as a C_1-cotransferase PGA is concerned more with purine synthesis than with that of other compounds.

21.2 Benzpteridines

These compounds are based on a triple condensed ring, the benzene ring condensed with pteridine, on the pyrazine side. There is one biological group of benzpteridines, the *alloxazines* (Fig. 46, IX), one member of which, *ribitylflavin* (Fig. 46, X), has been mentioned frequently already. It is of major importance as a B-vitamin, part of a mono- and a dinucleotide coenzyme concerned with transhydrogenation, probably in all living organisms. The nucleotides are flavin mononucleotide, FMN, and flavin adenine dinucleotide, FAD, exact parallels to NMN and NAD (p. 334) except that the alcoholic form, ribitol, replaces ribose. There is also a close functional parallel to NAD, except that their redox range is higher, so that while their enzymes do dehydrogenate some respiratory substrates others, the diaphorases, relay hydrogen from the nicotinamide system (Fig. 20). They are rather more concerned with the transhydrogenations of biosynthesis and less with the dehydrogenations of respiration than the nicotinamide systems. Like the latter they form a large number of enzymes, differing in their protein moiety; most of them have the dinucleotide as their coenzyme.

Like the pteridines in general these benzpteridines were first studied from biological sources, and acquired popular names before their chemical nature was known. Their terminology has been beset with confusions in

consequence. The initial name for vitamin B_2, 'riboflavin', refers simply to its yellow colour which earned similar names for a variety of pigments of quite unrelated classes, with the same extent of resonance. In particular confusion with the acridines or flavines (p. 342) and with the nitrogen-free flavones of plants (p. 146) is likely. It was necessary to change the name to ribitylflavin when the sugar proved to be the alcohol and not the aldehyde. Then the name alloxazine was suggested as a chemical name, although this implies that the oxygen is in the ring itself. Ribitylflavin proved to be an isomer of the structure initially suspected, and designated alloxazine, so that it becomes an *isoalloxazine*, and in full: 6, 7–dimethyl–9–D–1′–ribityl–isoalloxazine. The name alloxazine originated from the laboratory method of synthesis, by condensation of alloxan (p. 350) with a diamino-benzene derivative. Alloxazine is benzpteridine–2, 4–quinone, which makes the pyrimidine ring apparently fully saturated. In isoalloxazine a double bond has been secondarily transferred to this ring from the pyrazine ring (Fig. 46, X). Both isomers occur among biological benzpteridines. They are not benzpterins since they lack the 2-amino and 4-hydroxyl side chains of the pterins; in view also of the third ring there is little doubt that their biosynthetic pathway is quite distinct from that of the pterins. They are more closely homologous chemically than biologically.

The benzpteridine nucleus is extensively unsaturated and resonant, coloured and activated by visible light. As already indicated, the alloxazines have this same amount of resonance, similarly extended into quinone side chains. If these show tautomeric change to the enol form the ring system becomes fully resonant. The resonant pathway is longer than in the pterins and other simple pteridines and the colour correspondingly more bathochromic (p. 116). They absorb also in the ultraviolet and re-emit with a yellow, green or blue–green fluorescence, similar in range to that of the pterins. This is quenched by protein and other conjugants, as in the simple pteridines. In fact they resemble the simple pteridines more than might have been anticipated, either from their structures or their modes of synthesis. The implication is that the benzene condensed ring is not of great importance, and further evidence for this will be seen. They form metal chelates as the simpler nucleus does (Fig. 46, V), though the affinity seems more specific, to iron in particular (Albert, 1953).

The biological benzpteridines are probably all related biosynthetically to ribitylflavin, the isoalloxazine form depending merely on the presence of a substituent at 9. *Lumiflavin*, with a methyl group at 9, is formed from ribitylflavin by the action of light, and *lumichrome* is a further product of this action, having no substituent at 9 and therefore having the alloxazine

structure. Most of the properties of the group therefore may be illustrated on ribitylflavin, about which most is known.

Its outstanding feature no doubt is its specialisation as a redox agent, a property rather more incidental in most other pteridines. In this there is some kind of parallel to nicotinamide among pyridines and it might be suggested that ribitylflavin and PGA constitute a pair of B-vitamins rather parallel to nicotinamide and pyridoxin, the one member of each pair being a transhydrogenase, the other primarily transferring a C_1 and other units. The nitrogen atoms concerned occupy corresponding positions in PGA and ribitylflavin.

Like nicotinamide, ribitylflavin becomes hydrogenated in the course of transferring hydrogen, but it takes on two atoms of hydrogen, at 1 and 10 (Fig. 46, XI). The product has one less double bond, therefore, and is colourless (leucoflavin). Also like nicotinamide, ribitylflavin does not reversibly oxidise further: it is a good electron-acceptor but not a good donor. It is rather more specific to particular redox systems than the nicotinamide group (Williams *et al.*, 1950, p. 146).

Considering the degree of specialisation in action it is somewhat surprising that so many rather distant analogues, such as the quinolines (p. 341), act as metabolic antagonists of ribitylflavin. This is the strongest evidence that the benzene ring is not very critical, and in fact some substituents on this ring do not greatly affect activity (Sexton, 1953, p. 110); the 6-ethyl, 7-methyl analogue for instance is still quite active. The 6,7-diethyl derivative will deputise for the vitamin in *Lactobacillus casei*, though it proves to be an antagonist in the rat. In view of the great biosynthetic power of micro-organisms the apparent paradox probably indicates that this and other derivatives are converted to vitamin B_2 itself in those cases where it is apparently a deputy. Some substitutions on the benzene ring have proved to inactivate B_2 very strongly (Knight, 1945); the 7-methyl cannot be moved to 5 and two halogens cannot be substituted for the methyls without destroying activity and inhibiting growth.

The acridine antagonists (p. 342) imply that the central pyrazine ring is most important and the quinolines that only one of its nitrogen atoms is really critical; this is supported by those antagonists which have sulphur or oxygen at the *para*-position in that ring, the thiazines and oxazines (p. 342). The indispensable nitrogen is that at 10, which is directly involved in hydrogen transfer. Presumably the 1-nitrogen of the pyrimidine ring likewise is essential, although there is at present no very clear evidence. It should be appreciated that an antagonist may be very effective if it blocks only one critical group and that this does not prove that it is the only significant group. However, it is interesting that some antipterins are also antipyridines, since pyridines have only one ring nitrogen. One of

the ring nitrogens is involved in the bond between the coenzyme and the apoenzyme, and it may eventually prove that as many features of the pteridine portion are critical as in the pterins.

The ribitol is not absolutely indispensable, that is to say the arabitol analogue shows considerable vitamin B_2 activity; however the galactose analogue is a metabolic antagonist (Yudkin, 1948), so that a pentose is probably essential and, more specifically, the alcohol derivative.

Phosphorylation of ribitylflavin to form a nucleotide is promoted by various metals. Its properties become changed in several important ways; in the first place, like pyrimidines and purines (p. 352), it becomes much more soluble in water, which is important since the dehydrogenases work in the aqueous phase of the cell. It also becomes a good electron-donor (Szent-Gyorgyi, 1960) and so a good reversible redox agent. One molecule can even donate an electron to another, forming a dinucleotide. This may be relevant also to the formation of a mixed dinucleotide with adenylic acid, in FAD. The mononucleotide, FMN, has an intense green ultraviolet fluorescence, which shows that the excited electron squanders its energy if there is not a suitable acceptor of this kind available. This property is put to use in bioluminescent bacteria, however, where FMN may possibly be the actual light-emitting substrate, activated chemically. However, it is now thought (Strehler, 1961) that the interesting analogue, Fig. 46, XII, may be the substrate.

New properties emerge also at the dinucleotide level. The structure seems particularly suitable for charge-transfer by superposition (p. 334), since the two nucleotides are linked tail-to-tail with a maximal length of chain between the two planar ring complexes. The open chain ribityl is much more mobile and suitable for this than ribose (Glass, 1961). The indole ring is rather similar to that of purines, such as adenine, and indole compounds seem to interact with ribitylflavin in this same way, but without necessarily being chemically bound as the adenine nucleotide is. When mixed with FMN the indoles become stronger electron-donors, and the FMN a stronger acceptor, than their individual electronic-energy contents would imply. At low temperatures, in the presence of 5–hydroxy–tryptamine (serotonin), FMN becomes a brown compound which may prove to be the free active radical of the nucleotide. It may be this type of interaction which causes the isoalloxazines to oxidise the plant indole auxins, and inhibit the plant's growth (Galston, 1949).

The nucleotide phosphate group, as well as one of the ring nitrogens, is involved in bonding the nucleotide to its apoenzyme. The protein increases the redox potential of the nucleotide (Williams *et al.*, 1950), making it a better electron-acceptor or oxidising agent, just as the phosphate improves its reducing power in the hydrogenated state. As already

seen (p. 381), the protein quenches the fluorescence of the isoalloxazine, diverting the energy to useful purposes. The phosphate alone enhances the fluorescence, and it is evident that the whole enzyme is specialized for rapid energy transfer and activation, manifested in reversible redox change.

Something is known of the biosynthesis of ribitylflavin (Kit, 1960), and as usual it shows a number of features relevant to the general theme. It does indeed show resemblances (p. 373) to that of the purines (p. 360): for instance carbons 4*a* and 9*a* come from a single glycine molecule, the NH_2 of which becomes one of the adjoining ring nitrogens, presumably that at 10. In addition, purines appear to promote the synthesis, possibly as actual precursors; the use of labelled purines indicates that part of the molecule may be used for benzpteridine synthesis, but not the 8C atom, so that the imidazole ring must be opened, although it is the first ring to be closed in purine biosynthesis (p. 358). Acetic acid residues fill the gap to form the larger pyrazine ring, and further acetic residues build the benzene ring and the methyl side chains of ribitylflavin.

The pattern of synthesis departs sharply from that of the purines themselves, which does not use acetate (Fig. 45). Four acetate units are necessary to complete the dimethylbenzpteridine, their carboxyl carbons providing 8*a*, 10*a*, 6 and 7, and the methyl carbons 5 and 8 and the two methyl side chains. It is a pathway with its own materials, enzymes, and other features, but if it proves that a purine, with relatively little change, is the sole and invariable precursor then it does represent a new class among pathways of biosynthesis, which in general are so 'deliberately' independent, from the earliest precursors. The position of the ribityl group at 9 corresponds to that of the ribose of the purine nucleotides, and this may be converted directly to the alcohol; the reduction may well improve the nucleotide's properties as a redox agent.

If only two acetate residues were added to the purine derivative a 6,7-dimethyl pteridine would be formed and such compounds do occur biologically. It seems even possible that the biological pterins arise in this way after all (p. 376), since they commonly have a long alkyl chain at 9 and their hydroxyl and amino side chains are reminiscent. In view of the suggestions (p. 373) that the pteroic acid of PGA might have a different path of biosynthesis from the other pterins the subject needs further study. With an amino at 2, guanine would be the most appropriate precursor of the pterins; the pterin 4-hydroxyl group is in fact in the same position as the 6 =O group of guanine. In this event the association between white purines and white pterin pigments in the wings of butterflies is not so surprising. It may also be significant that guanine has the same high affinity for iron as the benzpteridines, though the latter of

course have =O in place of the 2-amino group and, again, a high affinity for iron is not rare among heterocyclic nitrogen compounds.

21.3 Conclusions

Two pteridines, PGA and ribitylflavin, have become fundamentally important metabolites, coenzymes of the B-vitamin group. Other biological pteridines, serving as integumental pigments, may be more casual by-products. As in so many classes of chemical compound, therefore, a limited number of members have been exploited biologically, but these have, or have acquired, a uniqueness which is irreplaceable in modern living systems. The molecules of PGA and ribitylflavin in their different ways compare in complexity with the porphyrins, and they illustrate a general tendency among heterocyclic nitrogen compounds. These are the most highly differentiated and specialised catalytic agents in the body and their qualitative complexity contrasts with their quantitative economy. This is the last of the classes of nitrogen compounds of biological importance and it illustrates these features as well as any. It is not yet clear to what extent the pteridine nucleus is unique among nitrogen-heterocyclic nuclei: at present the benzpteridine nucleus seems to have a rather low specificity to its redox function, in the sense that rather remote analogues can affect, or even effect, this function. The precise and efficient adaptation of the flavoproteins to this metabolic niche depends very much on the action of the other components, phosphate, adenine nucleotide and protein, on the innate properties of the isoalloxazine itself a particularly striking example of emergence (p. 1). Certainly pteridine compounds have a biological significance more restricted than that of the porphyrins and nucleic acids; it is more comparable to that of the pyridine group.

CHAPTER 22

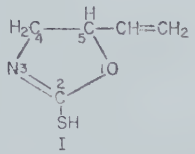
OXAZINES

Compounds with both oxygen and nitrogen in the same ring are uncommon biologically, partly for the same reason that open chain nitrogen oxides are rare, and it is perhaps significant that the betaine ring, one of the few in which the nitrogen and oxygen are directly bonded, is very unstable (p. 298). A kind of N,O–heterocyclic structure is formed in the metal chelates of the pterins (p. 375) and some other pseudoporphyrins (p. 311), where, however, the metal is a member of the ring and some of the bonds are coordinate. One *oxazolidine* compound, 5–vinyl–2–thio–oxazolidine (Fig. 47, I), occurs in brassica plants and has the distinction of being a very potent antithyroid agent (p. 458). With one unsaturated bond it is more correctly an *oxazoline* compound, but the fully saturated ring and the fully unsaturated *oxazole* ring also are rare. By contrast the sulphur analogue of the latter, thiazole, is not uncommon and is metabolically important (p. 398).

The hexacyclic *oxazine* ring occurs as the triple condensed *phenoxazine* nucleus (Fig. 47, II), already noticed as an analogue of alloxazine (p. 380). The nucleus becomes fully resonant through the oxygen becoming the conjugate oxonium cation (p. 152) and its derivatives are often brilliantly coloured and bathochromic, for instance *brilliant cresyl blue* (Fig. 47, III), used as a commercial dye. Living organisms have exploited the group in the *ommochrome* pigments, so called because they were first isolated from the ommatidia of insect eyes. They have redox properties, in some cases quite readily reversible, but at present there is no certain evidence that they serve in any capacity other than as integumental pigments and screening pigments. One group, the *ommatins*, are monomers, probably associated with protein *in situ*, while another group, the *ommines*, are polymeric, more insoluble, less chemically active, and more bathychromic. They often mimic the melanins, in fact, in colour and stability, but can be distinguished by their solubility in acids and particularly in alcoholic solutions of hydrochloric acid, and by their greater chemical reactivity. Even the ommatins occur as dull melanoid pigments, but extract in methanolic hydrochloric acid as brilliant golden pigments re-

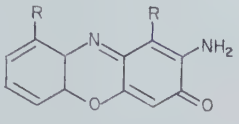
ducing to a crimson form (Butenandt *et al.*, 1954; Needham and Brunet, 1957).

As already seen (p. 193) the ommochromes are synthesised *in vivo* from tryptophan, via kynurenin (p. 333) or aspartyl amino benzene, the pyrrole ring of the indole having been opened (Fig. 47, IV). This is converted to hydroxy aminobenzene, two molecules of which then condense, front to



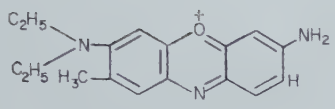
I

5-vinyl-2-thio-oxazolidine



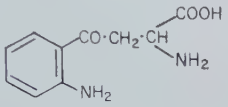
II

Phenoxazines



III

Brilliant cresyl blue



IV

Kynurenin

FIG. 47 Oxazole and oxazine derivatives of biological significance.

back. The intermeshing with the melanin pathway has been mentioned (p. 376). *Xanthommatin* (Fig. 40), the best known of the ommatins, has an extra ring condensed on the phenoxazine nucleus (Butenandt *et al.*, 1954; 1956), like some of the commercial dyes.

There is, in all, a large number of condensed ring systems with these same properties of high resonance, redox properties and brilliant colour. Potentially, and in some cases actually, they are also efficient activating molecules. All have the triple condensed ring with one of the polar non-metals, nitrogen, oxygen or sulphur, at one or both active positions of the middle ring, or substituted on carbon atoms at those positions. The phenazines, acridines, alloxazines and thiazines (Fig. 40) are examples of the first and the anthraquinones (Fig. 21) of the second type. As already seen they behave rather extensively as analogues of each other.

CHAPTER 23

SULPHUR COMPOUNDS

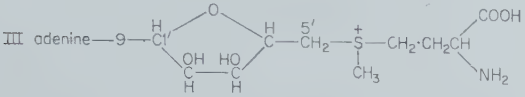
Sulphur is the next higher analogue of oxygen (Table 1) and shares with phosphorus the distinction of being the only analogues of this type to rank among the major biological elements. The most important question about sulphur therefore is whether it is biologically important in its own right, or merely as a deputy for oxygen. There are of course a number of family resemblances between the two elements, and there are situations where they are fairly interchangeable, even in living systems, but in general sulphur is biologically unique and indispensable. The fact that it oxidises so readily, forming compounds with oxygen, is a good indication that it has the opposite, nucleophilic, rather than the same, electrophilic, valency as oxygen.

Some anaerobic bacteria are able to use sulphur instead of oxygen to reoxidise their cytochrome systems, so that it does on occasion act as an electrophilic element. Reciprocally oxygen substituted for sulphur in the biotin molecule (Fig. 48, VII) causes little loss of activity in the vitamin, at least for some organisms (Sexton, 1953, p. 150), whereas most other substituents do destroy activity. The two elements behave very similarly also in the oxazines and thiazines (p. 342), though here any polar non-metal shows considerable similarity.

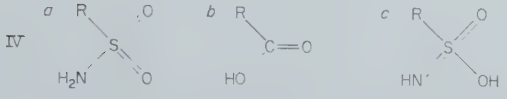
The general resemblances between the two elements are well known in inorganic chemistry and some of these have already been considered in connection with cysteine (p. 177). Hydrogen sulphide resembles water, though it is a considerably stronger acid, and the sulphides resemble the oxides. Organic hydrogen sulphides, RSH, the thioalcohols or mercaptains, and the neutral organic sulphides, or *thioethers* $(R_1.R_2)$:S, correspond to the simple alcohols and ethers. Cysteine is the analogue of serine and is synthesised from it in the body. However, organic sulphur compounds rarely act as substitutes for their oxygen analogues; at most they are similar enough to act as metabolic antagonists, the worst possible situation. Sulphur compounds which behave in this way are not normally synthesised *in vivo*, and in some cases perhaps the oxygen analogues of sulphur-containing metabolites have been similarly excluded, so that the

I Bond formula Electronic formula Sulphuric acid

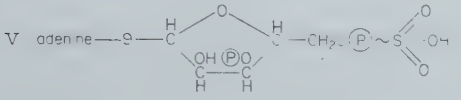
II Thioctanoic acid (α-lipoic acid)



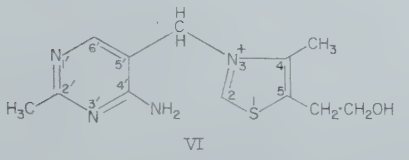
III Active methionine (S-adenosyl methionine)



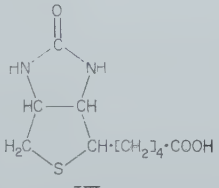
IV a Sulphonamide b Carboxylic acid c Possible resonant form of sulphonamide



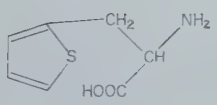
V Active sulphate (adenine–9 ribose–3′, 5′–P–sulphate



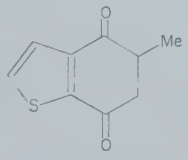
VI Thiamine (vitamin B_1)



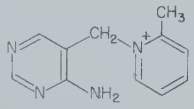
VII Biotin



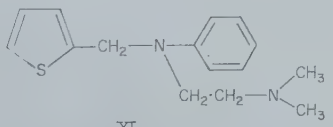
VIII Thienyl alanine



IX analogue of vitamin K



X Pyrithiamine



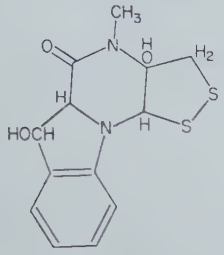
XI Enstamine

FIG. 48 Biologically important sulphur compounds (*continued on p. 391*).

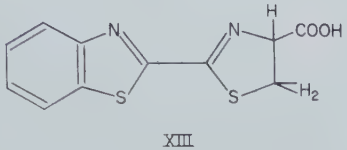
two elements have come to perform more sharply distinct functions than they may have done initially. These do not overlap and still more important they do not interfere. This is an evolutionary development rather similar to the elimination of intermediary compounds with overlapping properties. Differences between sulphur and oxygen have therefore become apparently enhanced biologically.

The intrinsic differences between sulphur and oxygen depend mainly on their difference in atomic size, which puts the nucleus of sulphur farther from the outer shell of reactive electrons. Consequently these are more readily shared with other atoms, including atoms of oxygen itself, than are the outer six electrons of oxygen. Sulphur more readily than oxygen shares a pair datively with a proton or with some other cation to become the tertiary sulphonium ion, $\vdots S^+$, which is also a stronger cation than the oxonium ion (p. 152). Sulphur readily shares electrons datively with two atoms or radicals, as well as covalently with two more, so that in compounds such as sulphuric acid it is effectively hexavalent (Fig. 48, I). Sulphuric compounds are rather common among biological materials, but in most of its other biological forms sulphur is tetra- or divalent. It readily shares electrons covalently with another sulphur atom and this is the basis for the disulphur bridge, RS–SR. Most of the lighter non-metals do this, as in H_2, N_2, O_2, Cl_2, but not when other elements are available, so that peroxide and other such bridges are rare in organic structures, whereas the —S—S— bridge is very strong. The many allotropic forms of sulphur are another reflection of this affinity; they have a varied crystal form and degree of polymerisation. The bonds may be dative or covalent. If sulphur could show a high valency for hydrogen it might have produced macromolecular materials comparing with those of carbon, but hydrogen forms only electro- and covalent bonds and the number with sulphur is therefore restricted to two. They tend to be electrovalent in practice and so the outer shell of sulphur is filled to the stable complement of eight electrons; consequently it no longer tends to polymerise, or to form macromolecules with other elements. The HS bond lacks the stability of the CH bond, and both H atoms of H_2S readily ionise; as already indicated (p. 153), H is the only atom suitable for sealing off 'side-bonds' and building macromolecular threads, and even so only provided it holds the bonds stably and covalently. Since hydrogen sulphide is a stronger acid than water, the HS bond is even weaker than that of hydroxyl. There is an apparent paradox in the fact that the less electrophilic element of the two forms bonds which are more electrovalent, that is in which the one electron of hydrogen becomes more completely the property of the other atom. The reason is probably the greater discrepancy in atomic size: in covalent bonds the two shared electrons move

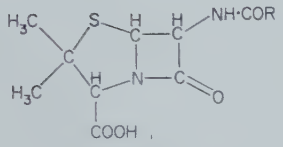
in an orbit round both atoms, but a normal elliptical orbit is not possible if the two atoms are too discrepant in size. Sulphur therefore appears to have more affinity for its electrons, and less for hydrogen itself, than oxygen has, and this is shown also in the weaker hydrogen bonds formed by sulphur. The active electrons of sulphur are in the *M* shell (p. 318) so that some move in the high-energy *d* orbitals, which do not exist in the outer shell of oxygen, and this may be responsible for the electrophilic property,



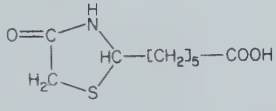
XII
Gliotoxin



XIII
Firefly luciferin



XIV
Penicillins



XV
Actithiazic acid

FIG. 48 (*continued*).

as in iron (p. 318); the high-energy *d* orbitals explain other differences between sulphur and oxygen. The stronger cationic power of sulphonium and of hydrosulphonium, H_3S^+, than of their oxygen counterparts is of course part of the same phenomenon. Even so, hydrogen sulphide is not a strong acid, its p*K* being between 9·1 and 10·8; RSH compounds therefore are a valuable source of H^+ ions under alkaline conditions.

When two RSH form the covalent disulphur bridge, RS–SR, two hydrogen atoms and not two H^+ ions are released and this accounts for the other side of the paradox, that sulphur seems to be more nucleophilic than oxygen, and SH a stronger reducing group than hydroxyl; as already seen, SH is in fact one of the most important of biological reducing radicals, its unique value being that it is active under alkaline conditions whereas most others are acidic reducing agents.

Because of its weaker hydrogen-bonding capacity, sulphur and its compounds are less soluble in water than those of oxygen and nitrogen,

and more soluble in hydrocarbons. This is a reminder that most of its bonds are covalent and not electrovalent. In any case alkyl groups depress the ionisation of the SH bond and RSH can be a weaker acid than water. Consequently the reaction

$$RSNa + H_2O \rightarrow RSH + NaOH$$

will occur. Moreover RSH will be a better reducing agent than hydrogen sulphide to the same extent.

Sulphur may form two anionic electrovalent bonds, or two covalent bonds, or one of each, or it may form three covalent bonds and one cationic electrovalent bond, as in sulphonium, or again it may form two covalent and two dative bonds, as in sulphuric acid. All give stable configurations. Sulphur is in fact stable at all levels of oxidation except the sulphite, which does not occur biologically; it is strongly reducing and illustrates the instability of intermediate states. All levels have much the same energy content, and sulphate reduces with little more expenditure of energy than in the reduction of nitrogen and carbon compounds (Ishimoto and Egami, 1957). Of course this is ignoring any need to reduce the oxygen itself, from such compounds as sulphate. Sulphur is therefore a very valuable element for biological purposes, very versatile and very stable in its various states, yet requiring relatively little energy to change from one to another. In addition it has the uniqueness of properties essential for a major biological element.

There is as much as 2 per cent of sulphur in hair and in some organisms as much as 1 per cent of the dry weight of the whole body is sulphur. In plants the amount increases with age, conceivably an evolutionary recapitulation.

23.1 Open Chain Biological Sulphur Compounds

There are a number of such compounds but it is probably true to say that only cysteine and methionine (p. 178) and their derivatives are of major biological importance, the one mainly for its redox property and the other for its —S— group which can become $\equiv S^+$. In addition there are a number of thioethers, as essential oils, and sulphates, the last often present in bulk. The cysteamine of coenzyme A (p. 356) is probably a cysteine derivative but is important for its ability to form a bond of high transfer energy, $\sim S$, rather than for redox properties, and this is why it has its unique role, unrelated to that of other SH compounds. The formation of this bond depends on the proximity of the keto, C=O, group

$$CoA—SH + HO(O=)C\cdot R \rightarrow CoA—S\sim C(=O)\cdot R + H_2O$$

so that this is another case where organic combination modifies the properties of sulphur.

The main properties of cysteine and other SH compounds have been adequately considered and it is necessary only to emphasise that there are two very different but equally valuable results of the type-reaction: $2RSH \rightarrow RSSR + 2H$, namely the chemical redox effect and the structural function of the disulphur bridge, when R is a peptide chain. In connection with the first it should be added that the RSH/RSSR system is the most active biological system for trapping electrons, apart from the aromatic nitrogen compounds (Reid, 1955); in large molecules, therefore, any labilised electrons tend to be collected at the SH group (Gordy, 1955). No doubt this is why the group is so sensitive to ionising rays. Another point worth re-emphasis is that the tripeptide, glutathione (GSH), is a conjugate of cysteine specially exploiting the redox property. Its redox potential is near that of ascorbic acid and that no doubt is why the two are commonly associated in redox systems (p. 177).

Thioctanoic acid (Fig. 48, II), or α–lipoic acid is an unusual redox form with two SH groups near enough together to form a ring compound in the oxidised form shown. It is functionally associated with thiamine (p. 397), which contains a permanent sulphur ring, and is primarily concerned with transcarboxylation. Thioctanoic acid is thought to play an essential part in the anabolic carboxylations of photosynthesis (Reid, 1955). Cysteine and cystine also are known to promote the polymerisation of a number of C_1 compounds such as CH_3OH, $H_2C{:}O$ and $(NH_4)_2CO_3$ under the influence of light.

The formation of the di-sulphur bridge is used commercially in the hardening of rubber, an interesting analogy to the natural process in keratin. Both remain quite elastic, the single —S—S— bond being pliable and allowing free rotation. There are also a number of small biological molecules with the —S—S— bridge, one of the most notorious being *asafoetida*, $CH_3{\cdot}CH_2{\cdot}CH(CH_3)—S—S—CH{=}CH{\cdot}CH_3$, from the rhizome of the umbelliferous plant, *Ferula*; it is unique in the strength of its unpleasant odour. Among synthetic compounds of this general type are the *thiuram* derivatives, for instance tetraethyl thiuramdisulphide, $(C_2H_5)_2{:}N{\cdot}C{\cdot}S{\cdot}S{\cdot}—S{\cdot}S{\cdot}C{\cdot}N{:}(C_2H_5)_2$, which inhibits the metabolic oxidation of ethanol at the acetaldehyde stage, and so has been used with great success in the cure of alcoholism, under the name *antabuse*. The systemic accumulation of the poisonous aldehyde apparently acts as an effective psychological deterrent! The sodium salt of the monomer, sodium diethyldithiocarbamate $(C_2H_5)_2{\cdot}N{\cdot}C{\cdot}S{\cdot}S{\cdot}Na$, is a good and rather specific colour reagent for iron and copper, and for vitamin K (p. 137). In the thiuram compounds there is a hint of the polymerising tendency of sulphur.

Methionine is the type example of sulphur covalently bonded between

two organic radicals, R_1—S—R_2, and this type of compound is formed also when SH compounds are inactivated by iodoacetate and similar poisons:

$$R_1SH + ICH_2{\cdot}COOR_2 \rightarrow R_1S{\cdot}CH_2{\cdot}COOR_2 + HI$$

Thioethers belong to this group. Probably the best known is *oil of garlic*, $(C_2H_5)_2$:S, the unpleasant smell of which is detectable at a dilution of 4×10^{-8} mg/l. of air. These powerful odours are typical of the divalent sulphur compounds, as illustrated by asafoetida, CS_2, etc. Mustard gas $(Cl{\cdot}C_2H_4)_2$:S, another synthetic war gas, is a powerful vesicant and poison because it goes on to form further addition compounds with almost all biologically active groups, except SH itself (Dixon and Needham, 1946). It may be taken as an indication of the potential reactivity of some biological $R_1{\cdot}S{\cdot}R_2$ compounds.

Heavy metals also poison SH compounds and do so by forming a complex of the type RS—X—SR, where X is the metal. Mercury forms the most stable complexes and this is why the organic thiols came to be called *mercaptans*; reciprocally, of course, thiols are the best antidotes for mercuric poisoning, egg-white being a ready source. It is clear that the linking of two R groups is the special menace of Hg^{++} since mercurous, Hg^{+}, salts are relatively harmless. Since peptides usually have more than one SH side chain mercuric tends to form meshworks and so to denature and precipitate proteins. Its bonds are covalent and not electrovalent as might be anticipated. Equally paradoxically the bond between the protein SH groups and the carotinoid, retinene (p. 112), is believed to be ionic (Reid, 1955), and it is thought that thioctanoic acid may have anionic S^{-} at one phase of the photosynthetic action mentioned above. The arsenic atom in the poisonous war gas, lewisite, $Cl{\cdot}CH{=}CH{\cdot}As{:}Cl_2$, behaves like Hg^{++}, binding two SH groups together, but arsenic is sufficiently non-metallic for this to be the natural type of bond. One of the triumphs of research on this gas was the discovery of a suitable antidote, BAL (British anti-lewisite), $HOCH_2{\cdot}CH(SH){\cdot}CH_2SH$; with two SH groups close together, on a small mobile molecule each of the latter could be expected to inactivate completely one of lewisite, by chelation of the arsenic atom.

Dicovalently bonded sulphur fairly readily changes to the sulphonium form. Among biological metabolites methionine and thiamine are the outstanding examples: in thiamine a ring-sulphur atom is concerned. As a tertiary base, $\vdots S^{+}$ is stronger than quaternary $H_4 \vdots N^{+}$ in the sense that its reduction to the conjugate base :S can be coupled with the formation of $\vdots N^{+}$ from :N (Wooley, 1953b). Its initial formation therefore is more difficult and a special 'active' methionine (Fig. 48, III), *S–adenosyl methionine*, is formed; in this the methionine replaces the phosphate residue of a

nucleotide. The adenosine enhances the electrophilic power of the sulphonium, as it probably does that of the phosphoryl radical; this is the second of many resemblances between sulphur and phosphorus in living systems. It is as an analogue of the quaternary nitrogen compounds that the sulphur of methionine acts as a leading transmethylating agent, the methyl group being already bonded to it. A synthetic analogue, *dimethylthetin*, $(CH_3)_2{:}S^+{\cdot}CH_2{\cdot}COOH$, also is an effective methyl-donor, and is interesting as being a nearer analogue of choline (p. 294): in fact it is known as *sulphocholine*. Trimethylsulphonylchloride, $(CH_3)_3{}^+SCl^-$, also has some of the pharmacological properties of the quaternary nitrogen compounds. *Methionine sulphoxide*, $CH_3{\cdot}S(=O){\cdot}CH_2{\cdot}CH_2{\cdot}CH(NH_2){\cdot}COOH$, synthesised in the laboratory, is an analogue of trimethylamine oxide (p. 282), and proves to be an antagonist of glutamic acid! The explanation would seem to be that $-S(=O)-CH_3$ is analogous to $-C(=O)-OH$. Thus while sulphur is unique in some of its properties it has considerable resemblances not only to oxygen but also to phosphorus, nitrogen, and carbon.

Hexavalent sulphur occurs biologically mainly, if not entirely, as sulphuric acid and its derivatives. There is 3 per cent of free sulphuric acid in the blood corpuscles of tunicates and 4 per cent in the poison from the salivary glands of the snails *Dolium*, *Murex* and *Pleurobranchia*. Here its strong acidity is the significant property, but it also serves other biological functions, in the mucopolysaccharides and other compounds. One function is to increase the water content of the molecules to which it is bonded. Its strong hydrophilia compares with that of phosphoric acid, of which it is a close analogue; the essential difference is that as a member of group V of the periodic system phosphorus forms three covalent bonds with hydroxyl and has only one lone pair of electrons to form dative bonds with oxygen. The so-called sulphonic acid and its derivatives thus correspond to phosphoryl compounds, in which the phosphorus opens a covalent bond, and they would preferably be called sulphuryl compounds to avoid confusion with sulphonium and phosphonium compounds, as well as to stress their derivation. As in phosphoric derivatives it is not always possible to say on which side of the oxygen atom the S—O—H chain was broken in forming a particular organic derivative, since the reaction is usually a condensation with the group C—O—H, which may behave as an acid or as an alcohol group. Chondroitin sulphate and mucoitin sulphate and related substances such as heparin appear to be true salts of sulphuric acid, whereas *taurine*,

$HO{\cdot}S(O_2){\cdot}CH_2{\cdot}CH_2{\cdot}NH_2$, from the bile of mammals, *cysteic acid*, $HO{\cdot}S(O_2){\cdot}CH_2{\cdot}CH(NH_2){\cdot}COOH$, and *asterubin*, $HO{\cdot}S(O_2){\cdot}CH_2{\cdot}CH_2{\cdot}NH{\cdot}C(:NH){\cdot}N:(CH_3)_2$ (p. 301), are equally certainly sulphuryl derivatives. A particularly difficult case is in the detoxication of phenols which are themselves quite strong acids. The products, 'ethereal sulphates', are usually regarded as salts.

Both hydroxyl and =O groups of these acids contribute to the hydrophilia, and the large amount of water in cartilage, connective tissue and mucus therefore is readily understood. With the addition of a further sulphate residue per saccharide monomer, as in heparin, the molecule becomes an anticoagulant, with an amusing contrast to the effect of —S—S— on protein-solubility! Various other sulphuryl compounds also have the anticoagulant property, and some are used to make dyes from insoluble pigments (Baker, 1960). Taurine makes the cholic acid of the bile soluble.

The antimetabolic action of some sulphuryl compounds, as analogues of carboxyl compounds, has already been noticed (p. 378), with the interesting generalisation that $RS(O_2)NH_2$ (Fig. 48, IVa) is the nearest analogue of $RC(=O)OH$ (Fig. 48, IVb), in acidity, perhaps because it resonates with the form (Fig. 48, IVc). In some cases, however, the sulphuryl group itself seems to be an effective analogue, so that cysteic acid acts as an analogue of aspartic acid, inhibiting its conversion to β–alanine, and taurine as an analogue of the latter itself. As in the analogy between methionine sulphoxide and glutamic acid, a rather limited resemblance may be enough for effective antimetabolic action.

Most organisms, including the mammals, are able to reduce sulphate to organic forms (Kit, 1960). Inorganic sulphate is absorbed sparingly from the gut, by comparison with chloride, but of course it is required in much smaller amounts. For this and other purposes thesulphate is handled in the special form of 'active sulphate', adenine–9–ribose–3′, 5′–phosphosulphate (Fig. 48, V). In this the sulphate and phosphate residues are linked by a bond of high transfer energy. Some micro-organisms reduce sulphate primarily as a source of energy, that is by coupling with the oxidation of carbon compounds. This recalls the similar use of nitrate by some bacteria. The oxygen of these compounds is as useful as atmospheric oxygen for suitably adapted organisms, therefore; an alternative view of the phenomenon is that more energy is released in reducing sulphate and nitrate to organic form than is required for that process itself, a view which confirms the ease and relative spontaneity of the biosynthesis of both nitrogen and sulphur organic compounds. Some bacteria can use elemental sulphur to oxidise their respiratory substrates. On the other hand other micro-organisms obtain energy

by oxidising sulphur compounds themselves, with oxygen, using the energy to synthesise their carbon compounds. These variations are matters of supply, demand and biological adaptation: they are not just a case of robbing Peter to pay Paul. *Thiobacillus thiooxidans* oxidises elemental sulphur to sulphuric acid (Postgate, 1954). It can produce and tolerate sulphuric acid stronger than 1N, so that with a little warming, and some evaporation of water from the medium, the organism chars!

Sulphate appears to have a very specific effect in controlling the uptake of molybdenum into the body (Underwood, 1962, p. 107). Sulphur and molybdenum are of course members of the same chemical group, but so are selenium and tungsten, and selenates and tungstates do not have the same effect. Permanganate, phosphate and silicate also are ineffective and chloride, bromide, nitrate, acetate, tartrate, citrate and malonate are all unable to reduce the toxicity of molybdenum (p. 448). Further, molybdenum and sulphate are both necessary to control the copper level in the body; they depress the storage of copper by the liver and neither alone is effective. This is an interesting example of multilateral control which could very well be part of a single 'omnilateral' mechanism. It is known that copper and cobalt are essential for haem synthesis, that iron is necessary for chlorophyll synthesis, and so on.

23.2 Sulphur Heterocyclic Compounds

A number of cyclic sulphur compounds have been identified from living organisms but only three are of general metabolic importance. These in fact are of major importance, all components of B-vitamins and of coenzymes, and all concerned with transcarboxylation primarily. These are thioctanoic acid (Fig. 48, II), *thiamine* (Fig. 48, VI) and *biotin* (Fig. 48, VII). The first two are probably peptide-linked, to form a single active enzyme, *lipothiamide*, which when phosphorylated is the main cocarboxylase of the body. Biotin is concerned with transaminations as well as carboxylations (like pyridoxin) and differs from lipothiamide also in the kind of carboxyl group transferred. It is more concerned with anabolic carboxyl transfers than lipothiamide. At the same time its side chain is so similar to that of thioctanoic acid that it may well conjugate with thiamine in much the same way—a tool with two heads.

These and most other biological S-heterocyclic compounds have pentacyclic rings, however much they vary in other details. The reason seems to be that —S— in a ring is equivalent to the unit, —CH=CH—, so that all pentacyclic sulphur rings are equivalent to a hexacyclic ring with one more double bond. The equivalence was discovered when *thienylalanine* (Fig. 48, VIII) proved a potent antimetabolite of phenylalanine (p. 185),

the compound IX (Fig. 48,) of vitamin K and *pyrithiamine* (Fig. 48, X) of thiamine. The two units cannot be sterically equivalent and therefore —S— must be equivalent in resonance power, and must assume the sulphonium state, $-S^{+}=$. This makes the thiazole ring of thiamine a very active one, but those of thioctanoic acid and biotin are otherwise fully saturated and work in a different way.

The fully saturated ring is unstable and, as already noted, that of thioctanoic acid reversibly opens in connection with a redox cycle. That of biotin also probably opens since it reacts with avidin, a factor in egg-albumin, provided the other ring is intact (Sexton, 1953, p. 149). Even the thiazole ring of thiamine is rather easily opened, and thiamine is one of the most unstable of the B-vitamins. The ring may undergo a regular cycle of fission and reduction (Weil-Malherbe, 1940). The open ring can form a dimer with a second thiamine molecule, through their sulphur atoms, another example of the —S—S— link and of the redox property of SH groups.

The equivalence of —S— and —CH=CH— will make thiazole equivalent to pyridine but these equivalences should perhaps be treated with caution since the *thienyl* ring proves antagonistically analogous to imidazole as well as to benzene, that is to say *enstamine* (Fig. 48, XI) acts as an antihistaminic, implying that the ring sequence, —HC=CH—S—, is equivalent to —N=CH—NH—; however, once more a very partial analogue may be an effective antagonist: only an effective deputy could with certainty be regarded as a close analogue.

What may perhaps be called the *dithiane* ring of thioctanoic acid (Fig. 48, II) occurs also in the antibiotic, *gliotoxin* (Fig. 48, XII). The thiazole ring occurs also in the substrate luciferin (Fig. 48, XIII), of the firefly's bioluminescent system (White *et al.*, 1961); in fact this has both benzthiazole and *thiazoline* rings, directly linked. The antibiotics *penicillin* (Fig. 48, XIV) and *actithiazic* acid (Fig. 48, XV) have the fully saturated counterpart of this, *thiazolidine*, so that these are the only really common series. The structure given for gliotoxin is only tentative and its dithiane ring may be as labile as that of dithioctanoic acid. The thiane ring of biotin may be more stable, since the fairly close analogue, the cyclohexane ring, is an effective metabolic antagonist. In the same light oxybiotin, in which the sulphur atom is replaced by oxygen, can deputise for the vitamin. However, as already seen, the latter still reacts with its specific inactivator, *avidin*, after the thiane ring has been opened.

The thiazole ring, with its quaternary nitrogen and potential tertiary sulphur cation, is both a more active and a more highly differentiated ring. The strong nitrogen base increases the water-solubility of the ring and sulphur in turn strengthens the alkalinity of the nitrogen base. As a result it becomes strong enough to act as a weak curariform agent,

blocking the action of ACh at synapses. This may well be relevant to the normal function of vitamin B_1 in safeguarding the metabolism and health of the nervous system, though this is known to depend also on decarboxylation as an essential step in the respiratory pathways of the tissue. Other features investigated are peculiar to this particular compound, thiamine; for instance the ethanol side chain on position 5 is essential to bind the pyrophosphate of the complete coenzyme. The methyl on 4 is less important and may be replaced by ethyl without serious effect (Sexton, 1953, p. 119). In addition, thiamine fairly readily undergoes a further ring closure in yeast, between the 4′-amino group of the pyrimidine ring and the 2C of the thiazole; a triple condensed ring results with some analogy to benzpteridine (p. 380). It forms a large resonant system and is coloured (*thiochrome*); this compound has no biological activity, however, and in fact the thioctanoic acid moiety is usually attached on 4′, but it shows the potentialities for interaction between the two rings. Both are essential parts of the vitamin and probably interact still in the presence of thioctanoic acid. The bridge between 5′C of the pyrimidine and the 3N of thiazole is essential for activity, which is lost if the CH_2 of the bridge is removed.

A short alkyl group is essential at 2′ and neither a long one nor merely a hydrogen atom. This is the only pyrimidine with methyl at this position (Kleiner and Orten, 1958). The numbering of this ring differs from that usual for pyrimidines (p. 347); the 4′-amino corresponds to the 6-amino of cytosine and the bridge, —CH_2—, group to the 5-methyl of thymine (p. 349). It is probably related biosynthetically to the other pyrimidines (p. 359), but has its own pattern of substituents; its specific association with this particular thiazole and other moieties, at specific positions, is essential for cocarboxylase activity. At present we know little about why or how each component and each relationship between them is essential.

The thioctanoic moiety seems alone to be concerned with the actual transfer of carboxyl, at any rate in the decarboxylation of pyruvic acid, one of the most important in respiration. The terminal SH of the open form of this acid bonds with the keto carbon of pyruvic, and so facilitates the removal of the terminal carboxyl. There remains the structure, —S—C(=O)·CH_3, reminiscent of acetyl–CoA (p. 356), and in fact the acetyl group is then transferred to CoA (Kit, 1960, p. 5). It is probably the sulphur groups of this moiety which are concerned with the oxidation which usually accompanies decarboxylation. On present showing, therefore, this simple facultative ring seems more important than the thiazole, pyrimidine and pyrophosphate moieties, all known to be in a general way very active chemically. By analogy with the behaviour of other cotransferases it may be suspected that the carboxyl group becomes attached to

the coenzyme at some point during the process and this may involve one or more of the other moieties. It will be recalled that PGA, which transfers C_1 at the next level of reduction, can become formylated at two positions, on different moieties, and that PGA has the general resemblance to lipothiamide of two ring moieties, chain-linked, and an open chain portion. In biosynthetic carboxylation a compound $AMP \sim CO_2$, or 'active carbon dioxide', plays a part, so that the pyrophosphate group of lipothiamide pyrophosphate may possibly bond with the carboxyl in decarboxylations, though this is not the usual function of pyrophosphate components of coenzymes in general. There is much to learn of this earliest known of the B-vitamins, and of the other biological compounds with S–heterocyclic rings.

Both dithioctanoic acid and biotin in their ring forms have an alkyl side chain, C_5 long, and this appears ideal for its purpose since both deputies and antagonists of biotin are most effective at a length of C_4 or C_5, but again the purpose is not yet clear. In dithioctanoic acid it might permit appropriate superposition of the ring on the thiazole ring of thiamine.

23.3 Conclusions

Sulphur has no biological significance as an analogue of oxygen. Biological oxygen and sulphur compounds do not even act as metabolic antagonists, that is as partial analogues of each other. Each has been exploited for its unique properties and the precise analogue among the compounds of the other element, when this is also a *functional* analogue, does not occur in living organisms. In fields where the properties of the two elements do overlap it is probable that biological systems have become adapted to synthesise and use the compound of the one or the other and to exclude the analogue which might interfere.

The unique properties of sulphur are quite well known and it is an element of great versatility. It is also clear how these properties are put to use in the open chain biological compounds. The mode of use of the S–heterocyclic compounds is less clear: on the one hand the ring seems to facilitate the formation of the sulphonium base but on the other hand the rings are rather labile, even the fully unsaturated thiazole. Consequently the ring structures may behave very similarly to the open chain compounds. In compounds such as thiamine it is possible that the sulphur ring plays a rather minor role and so at present it is uncertain whether the molecule as a whole should be considered primarily in this context or not.

The versatility of sulphur is shown by its partial analogy to nitrogen, phosphorus, and even carbon, in some of its compounds. Again this probably does not normally lead to antimetabolic effects, but in the laboratory it is possible to synthesise compounds which act in this way.

CHAPTER 24

PHOSPHORUS COMPOUNDS

PHOSPHORUS is related to nitrogen as sulphur is to oxygen. Both are solid non-metals, as contrasted with their gaseous lower analogues. Like sulphur, phosphorus shares with its lower analogue their general group properties, but also has its own unique properties which are the ones exploited biologically. Where their properties overlap either nitrogen or phosphorus is exploited, never both, and the biological compounds of the one are not sufficiently close analogues of those of the other element to interfere with it, though it is not difficult to synthesise such antimetabolites in the laboratory. In some cases overlap and interference are automatically excluded by the intrinsic difference in properties between analogous compounds; for instance ammonia and its derivatives are stable under terrestrial conditions whereas those of phosphine are not. They are powerful poisons and antimetabolites of the amino compounds but fortunately never occur biologically. On the other hand the oxy-compounds of phosphorus are stable and are the only group of its compounds extensively exploited whereas the oxides of nitrogen are rather unstable, excessively strong oxidising agents and rare except in the early stages of nitrogen-assimilation by autotrophes.

24.1 Phosphates

In its oxy-compounds phosphorus is usually pentavalent, forming up to three covalent bonds with hydroxyl or hydrogen groups and one dative bond, P=O, with its one lone pair of electrons. The only group which are common biologically are the derivatives of orthophosphoric acid (Fig. 49, I), the nearest analogue of sulphuric acid (p. 390), so that there is a sharp contrast to the variety of biological sulphur compounds, particularly as the phosphate tends to remain essentially an inorganic radical. It is not in consequence a less important element, however, by any means. Phosphates and phosphoryl compounds have a general resemblance to sulphates and sulphuryl compounds but with one essential difference, which makes phosphorus unique; this is the high transfer

energy of the bonds formed with other phosphate residues and with a number of organic groups, some of which have been noticed already. The transfer energy is the energy released when the bond is transferred to another combination; other elements form such bonds but the uniqueness of phosphorus is in the ability to harness the transfer energy to useful purposes. The *group transfer potential*, as it is called, of these bonds varies between 8 and 16 kcal/g-equivalent *in vivo* (Lehninger, 1959), and the bond is symbolised ~P to distinguish it from those of low transfer potential. The bonds formed by phosphate with some other groups are of this last type, with a transfer potential not greater than 3 kcal/g-equivalent, and these are symbolised —P.

Many reactions which otherwise would be uselessly slow at ordinary temperatures become rapid if coupled with group transfer from a ~P

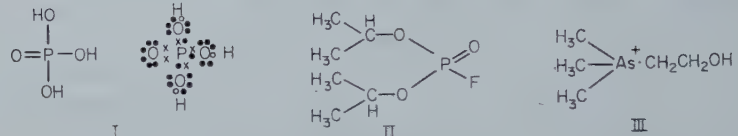


FIG. 49 Phosphorus compounds and analogues with biological significance.

bonding. Relatively few reactions by any molecule yield enough energy to raise an electron from its ground state to its first excited state (Szent-Gyorgyi, 1960, p. 28), but with the co-operation of several molecules of the ~P compound virtually any substrate can be activated. The uniqueness of phosphorus for this purpose lies primarily in its ideal ability both to accept and to donate electrons, a power potentially present in all elements of Group V of the periodic system, and in this group alone. The metallic elements of Groups I–III are good donors but poor recipients and this is true of the heavier, more metallic, elements of other groups, having few electrons in their outermost shell. By contrast elements of Groups VI and VII and the transitional elements have great avidity for electrons but do not donate them readily. The inert gases do neither, of course, because their outer shell is exactly filled, with both this and the penultimate shell in their most stable state. The elements of Group IV have the electron-donating and -accepting tendencies so exactly balanced that all four of their bonds are rigidly covalent.

Within Group V nitrogen is an ineffective donor, because like all members of the first full period it has no *d* electron orbitals (p. 391), from which electrons are transferred most readily, though it does become an active donor when it is part of a heterocyclic aromatic ring. The higher members of this group are too metallic, like those of other groups, and

phosphorus is the ideal element for the purpose, light enough to be mobile and active, and possessing the necessary *d* orbitals. Nitrogen is primarily tricovalent and less readily pentavalent, but phosphorus is usually pentavalent, i.e. tricovalent and di-coordinovalent (dative). The covalent bond is predominantly electron-accepting and the latter -donating; the electrons of a dative bond may become the property of the recipient group almost as completely as in electrovalent bonds, and the pentavalent nitrogen becomes an active cation (p. 151). In phosphorus, accepting and donating tendencies are mild and equally balanced but the accepting tendency is adequate to increase the electrophilia of the oxygen atom of P—OH groups, which therefore ionise as acids, $P—O^- H^+$, and the donating tendency is enough to make the phosphorus slightly positively charged (Finar, 1959), as though it had practically donated the dative pair of electrons. The molecule as a whole therefore is electrostatically polarised.

The further unique feature of the orthophosphate is that it is strongly resonant, each of the four oxygen atoms becoming in turn the doubly bonded one. This is much more restricted in any other compound, for instance in the phosphites where hydrogen groups replace hydroxyl, and in metaphosphoric acid, HPO_3, where the ratio of HO to =O is adverse. This ratio is already too low in sulphuric acid (Fig. 48), although from a knowledge of keto–enol changes two of each group might have been considered ideal; sulphuric acid is no deputy for orthophosphoric acid in this and in the functions which depend on it. It should be noted that in fact this resonance is not similar to a keto–enol change, in which the hydrogen atom does not shift to another oxygen atom; for this to occur the ratio of three OH to one =O in orthophosphoric acid is ideal. The corresponding acid of nitrogen is unstable, and decomposes to nitric acid. In phosphoric acid the four groups and P are virtually coplanar, and this further enhances the resonance (p. 116). The virtues of resonance have been adequately considered already; another aspect of this is that it effectively shortens the bonds between the groups and so facilitates their approach and interaction.

Bonds of high group transfer potential are any in which the resonance becomes suppressed (Bayliss, 1959, p. 357; Picken, 1960, p. 15). The conjugate, P~R, becomes more unstable and reactive by the amount of resonance energy frozen in the bond, and R is readily transferred to effect endergonic or energy-absorbing reactions, the type otherwise most difficult to induce. Alternatively it is transferred to bondage of the type R~R′, in which the transfer potential is still available for further reactions or for useful work, mechanical, electrical, photic, etc. All of these are ultimately driven by ~P energy in living organisms, and it is no exaggeration to say "without phosphorus: no life" (Walsh, 1961).

If there is no change in resonance in the phosphate when it transfers a bond then the transfer potential is in the low, $-P$, range, and there are a considerable number of organic phosphates with this type, for instance the triose monophosphates. It might be supposed that all P—OH bonds would have the high and all PO—H bonds the low transfer potential, since it is the former which are affected by the resonance shifts and are typical covalent bonds whereas the latter are weak, electrovalent bonds; the total energy content of the former is 50–100 kcal/g-equivalent, compared with 5 kcal for the latter. It will be noted that this implies that the transfer potential is only a fraction of the total energy associated with the covalent bond and that an ionic bond could not possibly behave as a ~ bond. A good deal of the evidence supports this interpretation (Lehninger, 1959), but it is not unequivocal. In a link of the type P—O—R neither bond need be ionic. Phosphorylases, which open the ~ bond, in fact appear to act at the point PO—R whereas phosphatases which release only a low transfer potential act at P—OR (Cohn, 1949). The difficulty is greatest in the pyrophosphates, that is two or more orthophosphate residues condensed together in —P—O—P— links, which behave as though they contained a single ~ bond; this *must* be adjacent to one of the phosphorus atoms, unless there is some kind of resonance distribution between both bonds.

24.1.1 Biological Phosphates and Phosphoryl Compounds

Examples of ~P bonds of biological importance are given in Fig. 50. That of pyrophosphates has the lowest transfer potential and it is therefore curious to find that the most important ~P compounds for transfer purposes in the body are nevertheless pyrophosphates, the nucleotide triphosphates (p. 356). Of these, adenosine triphosphate, ATP, is in a class apart since the others are all formed from their monophosphates by transfer of pyrophosphate from ATP itself; this also acts as the universal donor of energy for virtually every reaction and process in the living body, as must be already abundantly clear. In doing so it is degraded to the diphosphate, ADP, or even to AMP, and so must be 'regenerated' by the oxidation of respiratory substrates—and in autotrophes to some extent by light energy: $ADP + P_i \xrightarrow{\text{light}} ATP$.

The reason why these pyrophosphates, and ATP in particular, are the universal mediators of ~P energy is that they have outstandingly high kinetic stability. This is well shown by comparison with acetic anhydride, the carboxyl analogue of the pyrophosphate anhydride, with a bond of the same high transfer potential or thermodynamic instability. In contrast to the pyrophosphates, however, it hydrolyses almost instantaneously on contact with water and is useless for processes of the speed prevalent in

living bodies. ATP, however, hydrolyses very slowly without enzymic help (Lipmann, 1951), and it has the best on both counts, energy to give and the power to control the flow. The nucleoside moiety is invaluable here since inorganic pyrophosphates in fact do hydrolyse in about three minutes at 100°C; this is one more example of the restraining influence of nucleic acids and their components. The inorganic pyrophosphate releases 14,470 cal/mole so that the nucleotide restraint is paid for by some loss of available energy.

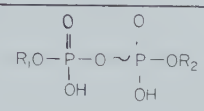
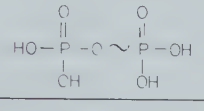
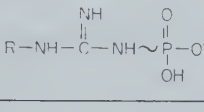
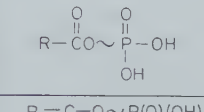
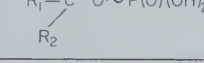
Type of compound	Generic formula	Transfer energy (kg cal/g equiv)
Pyrophosphate		12·0
Inorganic pyrophosphate		14·5
Guanidine phosphate		14·0
Acyl phosphate		16·0
Enol phosphate		16·0
Thiophosphate	$R—S\sim P(O)(OH)_2$	16·0

FIG. 50 Types of compound in which phosphorus has a high group-transfer potential.

Another uniquely important property of ATP is that it is formed from ADP by coupling with respiratory reactions at all redox levels between the initial dehydrogenation of the substrate and the reduction of atmospheric oxygen (Krebs and Kornberg, 1957, p. 271) so that all of the available respiratory energy can be converted to ~bond energy of ATP. The complete oxidation of one molecule of glucose leads to the formation of 38 molecules of ATP from ADP, representing an efficiency of over 66 per cent; there are 686 kcal in a mole of glucose and about 456 kcal are stored in the ATP synthesised at its expense. The redox potential range between the average respiratory substrate and atmospheric oxygen is about 1·2 V and a motive force of about 0·25 V is required to form each

~P bond of ATP, so that if this can be done at any redox level it should be possible to form 4–5 bonds for every pair of electrons which travel the full extent of the electron transfer sequence (Kit, 1960, p. 29). The actual yield is very near 66 per cent of this value. When one molecule of a fatty acid such as palmitic, C_{16}, is fully oxidised 131 ~P are formed (Florkin, 1960, p. 228), which represents about the same efficiency.

A summary of the reactions catalysed in the body by ATP and the other nucleoside triphosphates is given by Kit (1960, p. 34). They fall into a few main classes but collectively these include almost every significant reaction in the body. The types of reaction are given in Table 34,

TABLE 34. TYPES OF REACTION CATALYSED BY ATP (S = SUBSTRATE)

1.	P–transfer (kinase) reactions:	$S + ATP \rightarrow S\text{–}P + ADP$
2.	P~ transfer reactions:	$S + ATP \rightarrow S\sim P + ADP$
3.	P~P–transfers:	$S + ATP \rightarrow S\text{–}P \sim P + AMP$
4.	Activation reactions, with AMP transfer:	$S + ATP \rightarrow S\text{–}AMP + PP_i$
5.	Activation without transfer:	(1) $S + ATP \rightarrow S^* + ADP + P_i$
		(2) $S_1 + S_2 + ATP \rightarrow S_1S_2 + ADP + P_i$
		(3) $S_1 + S_2 + ATP \rightarrow S_1\sim S_2 + ADP + P$
6.	Polymerisations of nucleotide di- and triphosphates:	$n\,(NDP) \rightarrow (NMP)_n + nP_i$

where S is the substrate. In most cases the substrate becomes phosphorylated, and in others this may occur temporarily. A large number of metabolites therefore are isolable as phosphorylated compounds and some, such as the carbohydrates, are rarely unconjugated: again it is an impressive list (Table 35). Where the bond is of the $S\sim P$ type there is transfer energy still available for further use, but if the energy is expended in the reactions associated with the phosphorylation then the bond is of the S–P type. This may subsequently be raised once more to the ~ status by respiratory energy, as in the case of the hexose monophosphates. In this connection it may be noted that the second P of the triose disphosphates is acquired direct from inorganic orthophosphate and is subsequently raised to ~P status without the intermediation of ATP; in fact this is the way in which ~P is initially generated for ATP synthesis. Not all phosphorylations therefore are 'funnelled through' ATP, though certainly most of them are. Equally some S–P compounds perform useful functions without being raised to ~P status again: the phosphate may be used to build phosphate skeletons, as in the vertebrates; it may be used as an acid or as a buffer, as a hydrophilic and solubilising agent (like sulphuric acid, p. 390), and so on. The bond in phosphoproteins, nucleic acids, and phosphatides is of low transfer energy, and compounds between substrates and the mononucleotides are usually not concerned

primarily with energy transfer. Finally, in some cases the bond of high transfer potential is as it were completely separated from P, as in the formation of acyl~coenzyme A (p. 356).

Examples may be given to illustrate the way in which ~P energy is used to make reactions more feasible than they would otherwise be. The polymerisation of glucose to glycogen requires 5000 cal/g-equivalent of bonds formed, but by coupling with ~P transfer it becomes a spontaneous, irreversible process, exergonic to the extent of the difference between 5 kcal and the ~ potential. The ~P compound is more than a mere catalyst, which simply speeds a reaction already thermodynamically possible; ~P makes an improbable reaction spontaneous, and incidentally also rapid. The facilitation of biological activity by ~P may take other forms; for example the redox potential for the change: acetaldehyde → acetic acid has a large negative value but phosphorylation raises the E_o' by 200–400 mV, so bringing it within the range of physiological enzymes (Bayliss, 1959, p. 366). This is associated with a diminution of the free energy change of the reaction. Another example is that conjugation of the luciferin of *Cypridina* with AMP causes the ultraviolet fluorescence emission spectrum to shift, so that it now corresponds to the peak of the bioluminescence emission spectrum (Glass, 1961, p. 848) and supplements the emission (p. 374).

Phosphorylation is sometimes depressant, as for instance on the reducing action of glucose (Walsh, 1961); as a result glucose–6–phosphate has little, and glucose–1–phosphate no, reducing activity. There are situations where this is a valuable safeguard against waste since glucose has a significant spontaneous oxidation rate under aerobic conditions. Phosphate compounds therefore control metabolism in both directions and are general stabilisers. They also stabilise in another way, as buffers (p. 410). For further illustrations of these compounds in detailed action the standard works should be read.

Some bacteria, yeasts and other fungi store large amounts of polyphosphates, that is to say insoluble high polymers of the pyrophosphate type. These may be a reserve of essential phosphate, already ~ bonded, since they reach a maximal concentration just before cell division and disappear rapidly just after (Schmidt, 1961). In some organisms the orthophosphate residues undergo dehydration to the less active metaphosphate, HPO_3, and the whole polymer shows ring closure; this would seem to be the inert storage form *par excellence*. In fact the pyrophosphate form may be rather active, since it promotes amino acid polymerisation in simple laboratory techniques, without the use of enzymes (Fox, 1960; Schramm, 1957). Orthophosphate, the monomer, has some action of this kind, but only at much higher temperatures. No doubt inorganic

TABLE 35. BIOLOGICAL PHOSPHATE COMPOUNDS

(— and ~ bonds are not distinguished)

1. *Inorganic*
   Sodium mono- and dihydrogen phosphates, Na_2HPO_4 and NaH_2PO_4
   Hydroxyapatite, $Ca_{10}(PO_4)_6(OH)_2$
   Sodium triphosphate, Na–P—P—P
   Pyrophosphates
   Polyphosphates
2. *Carbohydrates and derivatives*
   Sedoheptulose–P
   Hexose–1–P, Hexose–6–P, Hexose–1,6–diP
   Glucosamine–1–P
   6–P–gluconate, δ–gluconolactone–6–P
   Ribose–1–P, ribose–5–P, ribose–1,5–P
   Ribulose–5–P, ribulose–1,5–diP, xylulose–5–P
   Deoxyribose–1–P
   5′–P–ribosyl–1′–pyrophosphate (PRPP)
   5–P–ribosylamine
   Erythrose–4–P
   3–P–glyceraldehyde, P–dihydroxyacetone, 1,3–diP–glyceraldehyde
   2–P–glyceric acid, 3–P–glyceric acid, 1,3–diP–glyceric acid
   2–P–enol pyruvic acid
   Imidazole glycerophosphate
3. *Alcohols*
   α– and β–glycerol phosphates
   Phosphoinositides, including phytin (Ca,Mg–inositol–hexaP)
   Ribitol–P
   Propanediol–P
   Indole–3–glycerol–P
   5–P–shikimic acid
   α–Tocopherol(vitamin E)–P
   Calciferol(vitamin D) phosphate
   P–serine, P–threonine, P–HOproline, P–HOlysine, P–HOglutamic acid
4. *Lipids*
   Acetyl and other acyl phosphates
   Phosphatides: lecithin, cephalin, phosphatidyl serine
   Acyl phosphatidic acids
   Plasmalogens
   Sphingomyleins
5. *Carboxylic compounds*
   Acyl phosphates
   Carbamyl phosphate
   Succinyl–P
   β–Aspartyl–P
   γ–Glutamyl–P
6. *Phosphagens: P–guanidines*
   P–creatine
   P–arginine
   P–lombricine

P–taurocyamine
P–glycocyamine

7. *Nucleotides*
DNA, RNA
Mono-, di- and triphosphates of adenosine, guanosine, cytidine, uridine, thymidine, and of rarer bases, and of deoxyribosides of adenine, guanine and cytosine
Inosine mono-, di- and triphosphates
Orotidine–5′–P
Xanthosine–5′–P
Uridine–diP–pentoses and hexoses
Cytidine–diP–choline, CDP–ethanolamine
UDP–glucosamine, galactosamine, glucuronic acid, acetyl glucosamine-SO_4
Guanosine–diP mannose
Nicotinamide mononucleotide, nicotinamide–adenine dinucleotide, NADP
Flavin mononucleotide, flavin–adenine dinucleotide
Cobalamin
Coenzyme A
Adenyl acetate

8. *Other* N–*bases*
Thiamine di-, tri- and tetraP, lipothiamide diP
Pyridoxal–P
P–choline, P–ethanolamine

9. *Sulphur compounds*
Thiophosphates

10. '*Active*' *compounds*
Active carbon dioxide AMP–CO_2
Adenyl-acetate and other acylates
Active amino acids, a.a.–AMP
Active sulphate, adenosine–3′–P–5′–P–sulphate
Adenyl-luciferin

pyrophosphates were the forerunners of the more specialised organic ~P compounds. Some gram-positive bacteria have a lipid–polyphosphate–protein component in their cell wall (Picken, 1960, p. 66), and here a more mechanical function may be implied; the mineral of vertebrate bone and teeth is a somewhat polymeric phosphate (hydroxyapatite).

By comparison with the energetic functions of phosphate in living organisms its other properties are rather dwarfed, but they are real enough, and valuable. Phosphoric acid is almost as strongly hydrophil as sulphuric (p. 395). With three hydroxyl groups it might be expected to surpass sulphuric acid, but only one is a strongly ionising group. This is a main reason why triesters are rarely formed, for instance in a branched RNA (p. 363). The importance of the hydroxyl groups is shown in substituting all three by phenyl radicals, when the product becomes water-insoluble but soluble in lipids (Walsh, 1961, p. 95), like apolar sulphur compounds.

The hydrotropic action is well displayed by the phosphatides (p. 295), though here the strong quaternary nitrogen cation also greatly contributes. The hydrotropic action may be the main biological reason why some organic compounds are phosphorylated *in vivo*; vitamin E (p. 117) becomes much more soluble in consequence, as do the pyrimidine and purine bases. Pyrophosphate improves the absorption of iron from the gut, one of the biggest solubility problems in the body. Sodium triphosphate, Na–P ~ P ~ P, an analogue of the organic triphosphates, is a good solvent for the heavy metal salts in general. Polyphosphates enhance the activity of some ionic detergents: they are called 'builders' and the action is probably hydrotropic. They also have a plasticising action (Picken, 1960, p. 268), which is probably another manifestation of the same property.

On the other hand phosphate can again control in the opposite direction, by depressing the solubility of some organic hydroxy compounds, such as glucose and other hexoses, and this is thought to have a useful biological function in precipitating hexoses after their transport across membrances, so maintaining a positive diffusion gradient, always in the same direction. This action on the monoses is the more interesting in that the latter actually increase the acidity of the phosphate hydroxyl groups. Another aspect of the hydrotropic action is that some organic phosphates, such as that of vitamin D, improve the solubility of inorganic phosphates, including those of magnesium and calcium, which are important biologically, in connection with phosphorus uptake from the gut, its deposition in the bones and its excretion.

The hydrotropic action is of course closely related to the acid properties of orthophosphoric acid. The anion increases the negative charge on starch and so improves its ability to form a coacervate (p. 233) with protein (Oparin, 1957a, p. 309). Phosphoric acid increases the acidity of other acid groups with which it is associated, by as much as 5 pH units (Fearon, 1949, p. 239). Reciprocally, as already noted, organic hydroxy compounds, such as the hexoses, increase the acidity of the phosphate hydroxyl groups; the ionisation of each of the three hydroxyl groups is shifted about one pH unit, compared with free orthophosphoric acid. In the hexose diphosphates the effect is shared between the two phosphate residues (Cohn and Edsall, 1943, p. 134), which implies resonance or induction through the length of the monose molecule.

The three hydroxyl groups of free orthophosphoric acid ionise with pK value at pH 2·0, 7·0 and 12·4 respectively, and with the middle value around neutrality it is an ideal buffer. Under physiological conditions the other two ionisations are rarely used, though they are available, and this is a great additional virtue. The strongly acid group is normally per-

manently esterified and as a buffer it is usually in the form of the sodium salt, NaH_2PO_4; this changes to the disodium–monohydroxy salt, Na_2HPO_4, in the presence of alkali. In the laboratory the mixture NaH_2PO_4 + K_2HPO_4 is used as the best buffer, simply because K_2HPO_4 is more soluble than the disodium salt, whereas the converse is true for the two dihydroxy salts. This phenomenon may well be biologically significant since large Na/K differentials are built up between cells and body fluids; in fact the cell contents are the more acid and the phosphate is not in its most soluble form on either side! It may also be relevant here that organic phosphorylated compounds control the differential uptake of K/Na from the soil, by plants (Sutcliffe, 1952).

This is perhaps a suitable point at which to recognise also the value of bicarbonate as a biological buffer. It is even more innocuous than phosphate and has the special virtue of being formed as a by-product of metabolism, rapidly augmented or eliminated as necessary. The other carboxylic acids are equally effective, and in some cases have a p*K* nearer neutrality, but they are less expendable in quantity, and are less volatile. For plants even carbon dioxide is too precious to be used in this way.

Quantitatively the use of insoluble inorganic phosphate for skeleton-formation is the major use of phosphate in vertebrates, though it may be a relatively recent use in evolution, since apart from the brachiopods invertebrates have a mineral skeleton based on insoluble carbonates or on silicates. The latter are difficult to work (p. 466), but the carbonates are materials very easily obtained and deposited as calcium carbonate. This, however, has the disadvantage of redissolving readily under acid conditions and of being rather brittle compared with phosphate. Vertebrates have solved the problem of obtaining and transporting enough of the relatively rare element, phosphorus, and of depositing it in insoluble form inside the body, where a major problem has always been to absorb it and keep it in solution. In the gut it is mainly associated with calcium, magnesium and other metals, of which even the monohydroxy salt is insoluble. As already seen, its absorption is promoted by vitamin D. Other steroids, such as the bile salts, also assist; the latter are secreted in rather acid media so that the phosphate will be present as the soluble dihydroxy salt.

In the blood soluble $PO_4^{=}$ and Ca^{++} both occur, although the pH is close to 7·0. The relative concentrations of the two ions are very nicely balanced, mainly under the control of vitamin D and the parathyroid hormone. If the level of calcium in the blood falls, output of the parathormone is stimulated and this promotes an increased excretion of phosphate. When the calcium level rises the converse mechanism operates, and the Ca/P ratio therefore remains fairly constant notwithstanding

considerable variations in absolute level. This regulation seems to be primarily concerned with skeleton-formation, but also with buffering, and other functions, of both elements. It is one of the ingenious systemic regulatory mechanisms of the body and strictly speaking beyond the scope of the present book, but its chemical basis may prove to be relevant.

The blood is found to be supersaturated with respect to both ions and this must facilitate their eventual precipitation together, in bone and teeth. They are held in solution until this time, probably in organic combination, e.g. with vitamin D; phosphatases which release phosphate from organic combination certainly are present in skeletogenic tissues and do promote mineralisation (Bourne, 1956; Glimcher, 1959). The organic compound is the link with ATP, which no doubt ultimately provides both phosphorus and energy. The final mineral is hydroxyapatite, a hydrated calcium phosphate with the approximate formula, $Ca_{10}(PO_4)_6(OH)_2$. The ions are not in strictly stoichiometric ratios however (Glimcher, 1959), and are believed to deposit independently from solution, into a crystalline array, but only approximately in the ratios corresponding to apatite. It is a crystalline structure not a chemical molecule, and no molecules of that formula ever exist free in solution. Various other cations and anions, in fact almost any small ions present in the body fluids, are incorporated into the crystal-lattice to some extent, including small organic ions, such as citrate. Ions are exchanged subsequently, also, and continue to do this even after death, so that it is a spontaneous process. Certainly it is not peculiar to living systems but occurs frequently in geological processes. Apatite-deposition is to a large extent spontaneous, particularly in the presence of the organic components of bone (p. 485).

24.2 Other Phosphorus Compounds

As already indicated, biological phosphorus compounds are virtually all ortho-phosphate derivatives, and most other forms, including some of the allotropes of the element itself, are powerful poisons. White phosphorus causes 'phossy-jaw', a bone disease, which perhaps shows how independent the phosphorus atoms of apatite can be of the other ions. The oxides and the other acids of phosphorus also are poisonous, mainly because they are antimetabolites of the orthophosphates, while phosphine and its derivatives are poisonous probably as analogues of amino compounds. Organically substituted phosphorus compounds appear to be sufficiently reduced to act in this way, rather than as analogues of orthophosphoric acid. A group of potent insecticides, of which *di-isopropyl-fluorophosphonate* (Fig. 49, II) is the simplest and best known, act as anti-

cholinesterases (Kilby, 1950), presumably by competition with ACh itself. Some are pyrophosphates, even nearer to the phosphate structure.

Phosphorus acquired its name from its interesting chemiluminescence, producing light as it oxidises spontaneously in air; the light is visible at low ambient light intensities. It differs from the phenomenon now known as phosphorescence since this demands prior illumination and is a photoluminescent phenomenon. The best-known simple examples of this are the sulphides of zinc, calcium and barium, used as auxiliaries in luminous 'paints' for the dials of watches; the average wearer recharges the phosphorescence during the hours of light. The 'phosphorescent' glow of corpses is again different: it is a bioluminescent phenomenon, due to living necrophilic bacteria; it is a type of chemiluminescence and continues indefinitely in the dark. Bioluminescence is always associated with the activity of ~P compounds, but for the usual metabolic reasons and not because of the intrinsic chemiluminescence of elemental phosphorus. At the same time no doubt both are connected with the resonant properties of oxidised phosphorus compounds. A further point of interest in the spontaneous oxidation of elemental phosphorus is that it involves the simultaneous oxidation of oxygen to ozone.

24.3 Conclusions

The biological use of phosphorus shows the usual exploitation of a restricted group of compounds, in this case the orthophosphates and their derivatives. Sulphur in fact is rather exceptional in the variety of its biological forms. The importance of this one form of phosphorus however can scarcely be overemphasised. This is further borne out by the extent to which it is selectively accumulated by living organisms, notwithstanding its great rarity in the environment and its very low solubility. The solubility of phosphate, the common form also in the environment, is only 0·1 ppm (Blum, 1955, p. 178), yet it is as concentrated as 775 ppm in the body (Gulick, 1957). In man, with a bulky phosphate skeleton, as much as 1 per cent of the dry weight is phosphorus, but even in plants it may be as high as 0·8 per cent. It is thought that the percentage has increased progressively during the course of evolution (Fearon, 1949, p. 32), and if this is true the vertebrate skeleton may have been an automatic sequel of chemical allometry (Huxley, 1932; J. Needham, 1933). However, it is necessary to be sure that quite apart from the vertebrates there has been such an evolutionary trend in percentage of phosphorus, and that the evidence for the latter does not rest solely and circularly on the vertebrate skeleton. Terrestrial vertebrates require stronger skeletons than the fishes and so have more phosphate, that is the increase

in phosphorus has continued within this phylum; however, since this has a useful basis the automatic evolutionary increase must be demonstrated and proved from other phyla. Crustacea, which are among the more highly evolved invertebrates, have some phosphate in their skeletons, and the brachiopods also are perhaps fairly highly evolved. A reason for suspicion is that the percentage of phosphorus *decreases* progressively during vertebrate ontogenesis, which can be expected to be directly rather than inversely correlated with evolutionary trends; this again may be misleading, however: the high phosphorus content of the young animal may be a provision for the early rapid growth of the skeleton. The sulphur content of the body *does* increase with age, in plants (p. 392).

24.4 Arsenic and Antimony

These higher analogues of phosphorus are most conveniently considered here, since they are of little physiological importance and are pathologically significant mainly as analogues of phosphorus, or of nitrogen. The other branch of this group, however, containing vanadium and niobium, has its own significance and is better considered in another context (p. 451). Arsenic was for long one of the most notorious poisons, though in fact there are always traces in the body and it has been used therapeutically to strengthen the heartbeat, improve respiration and improve the complexion. As with so many poisons, traces may be beneficial in this stimulatory way, usually perhaps by stimulating the reaction against the poison. This is probably why a surprising degree of habituation to the taking of arsenic is possible, the basis for at least one murder plot of fiction. Some fish normally excrete as much as habituated human beings, so that their food in turn must have a high content.

Arsenate can deputise for phosphate for the *second* phosphorylation of glyceraldehyde, in the glycolytic sequence (Fig. 4), forming 1–arsenic–3–phosphate–glyceraldehyde (Baldwin, 1953, p. 324); this will not subsequently phosphorylate ADP, however, so that it is a typical antimetabolite and not a functional deputy. Arsenites accelerate glycolysis and lactic acid formation, but they block the hydrolysis of fumarate to malate, and so could control the balance between glycolysis and complete oxidation in the Krebs cycle. The normal control of this does depend on phosphorus, but not in the form of phosphite; however, it is noteworthy that some biochemists believe that phosphate was preceded by phosphite in biological evolution (Gulick, 1957). It is more soluble than phosphate but it spontaneously oxidises under present conditions.

The best known of arsenic analogues of nitrogen metabolites is *arsenocholine* (Fig. 49, III), which can replace choline itself for the synthesis

of phospholipids (Sexton, 1953, p. 180). It cannot act as a methyl-donor, however. It has an action on the nervous system resembling that of ACh, but most onium bases have much this same effect, which therefore does not imply a highly specific deputisation (Sexton, 1953). By reacting with mercaptans (p. 394) arsenic acts as a poisonous metallic element, so that it has at least four distinct actions of an antimetabolic nature.

In general, organic arsenic compounds are less toxic than inorganic arsenic (Sexton, p. 303). They are of course less ionic and less water-soluble and in a sense no doubt they are already partially detoxicated. It is probably true to say that the organic forms of most heavy elements are in the same way less toxic than the inorganic compounds.

Antimony and bismuth are progressively heavier, more metallic and less toxic. Insoluble bismuth salts are used to render food radio-opaque for medical X-ray work, and so are relatively harmless. Again, the metal itself is used for dental fillings.

CHAPTER 25

THE PHYSIOLOGICAL INORGANIC IONS

I will go back to the great sweet mother,
Mother and lover of men, the sea. . .
O fair white mother, in days long past . . .
Filled full with life . . .

(SWINBURNE)

As such an one were glad to know the brine
Salt on his lips, and the large air again . . .
Shrill wind beyond the close of heavy flowers. . . .
The surge and thunder of the Odyssey.

(ANDREW LANG, *The Odyssey*)

WITH the exception of sulphur, each of the elements considered so far constitutes 1 per cent or more of the total weight of living organisms (Table 2), and sulphur sometimes approaches 1 per cent (p. 392). They are what Fearon (1949) called the primary elements, on both qualitative and quantitative criteria. The latter alone is not very satisfactory since there is over 2 per cent of calcium in many organisms (Table 2), although it would be difficult to maintain that in consequence it is more important than sodium, potassium, iron, iodine, as well as phosphorus and sulphur. Qualitative criteria are essentially those of specific function. The six major elements already considered are the main components of the 'fabric' of the body, using this word in a broad sense to mean the structural organisation, the agents which do work and the materials they use for the purpose. There are also other elements in the fabric, but they are micro-constituents. Between the primary group and these there is a group, each contributing between 0·1 and 1·0 per cent of the body weight, and present mainly as inorganic ions in the body fluids, intra- and extracellular. These are the *physiological ions*, Na^+, K^+, Mg^{++}, Ca^{++} and Cl^-, and they form a clear group functionally as well as quantitatively. It is almost identical with Fearon's *invariable secondary elements*. There are, of course, other inorganic anions of some importance, $HPO_4^=$, HCO_3^- and

$SO_4^=$ in particular, but these have already been considered. Chloride is the only major elemental anion and in the physiological salts the four elemental cations are associated mainly with it; small, mobile and monovalent, it is ideal for the purpose. The cations have the same general properties, except that two are divalent, and so the physiological ions are also the *major elemental ions* in living organisms. Phosphate and bicarbonate are used mainly for buffering (p. 411) and $SO_4^=$ is a rare ion except in some cells (West and Todd, 1957, p. 646). The uptake of sulphate from the vertebrate gut is severely restricted, in contrast to Cl^-, and there is reason to think that this is not due merely to the lower rate of diffusion and permeation of the larger, divalent ion but results from a 'biological preference' for chlorine. However, this no doubt developed mainly because of intrinsic differences in activity so that the mechanism of selective uptake of chlorine is probably simply an enhancement of these.

The term 'physiological ions' is very apt in the sense that they are greatly concerned with all the more physiological aspects of vital activities, neuromotor, osmoregulatory, etc., and much less with biochemical activities than the primary elements, and some of the microconstituents. The ions control the general state of the aqueous media of the organism, the balance and equilibria between the various 'compartments' of these media, and the transport of materials across their boundaries. A qualification is necessary here, since even Na and Cl to some extent, and K, Mg and Ca in progressively increasing degree, accelerate the rate of action of particular enzymes, and have other special chemical functions such as the building of skeletons. In spirit, however, they do constitute the kind of group suggested above.

They are a real functional group also in the sense that normal physiological activity depends on a precise balance among them, as well as between them collectively and other constituents of the body. Since sodium chloride is so much more abundant than the other chlorides in the body fluids of animals, it can be used as a rough measure of the amount of the group as a whole, and in fact, further, as actually determining the balance between this and other main constituents, such as water, protein, etc.; for the balance among the ions, however, the small amounts of the others are extremely important. Within the cell, moreover, potassium and magnesium are the dominant cations, so that Cl^-, and not Na^+, is the best measure of the group as a whole.

The balance between them and the other constituents is related to a collective function they perform, while the balance among them indicates that each is uniquely significant in particular vital functions. Their individual uniqueness will be considered in more detail later: the common function is to provide a suitable medium for biological materials and

processes. For convenience two requirements may be recognised in the suitable medium, adequate water and adequate salts, the correct total amount relative to the organic components and the correct ratio of water to salts. The maintenance of the latter is usually known as the osmotic function, since inorganic solutes are the most important osmoactive materials, though it is now well known that water-balance or 'osmoregulation' for the body as a whole depends on far more than simple osmosis. Nevertheless salts are essential to keep water in the body, and without them organisms dehydrate, notwithstanding the hydrophilia of the amino acids and proteins. The state of a dehydrated animal is not improved but exacerbated by drinking pure water: this is rapidly excreted, inevitably taking a little more salt with it because no physiological process is 100 per cent efficient. The only satisfactory corrective is adequate salt and water. Very occasionally urea, amino acids, and other organic solutes play a major part in osmoregulation, but this only serves to emphasise the normally predominant role of the physiological salts.

An organism colonising fresh water has no difficulty in obtaining and retaining enough water, and in fact its problem is to prevent waterlogging. Nevertheless, except in a few groups of animals, such as the lamellibranch molluscs, the body accumulates and conserves considerable amounts of salts and this led to Macallum's classical theory that the tissues were adapted to a saline environment approximating to their pristine marine environment, and with minor modifications had carried this in their body fluids ever since (Macallum, 1926). This would be a case of the 'fitness' of the environment (Henderson, 1913), precisely because the living system had fitted itself to the medium. Macallum was well aware that the ratios among the physiological ions did vary considerably among animals but thought that this reflected an evolutionary change in the sea, each group retaining an internal sea approximating to that which its ancestors last experienced. However, there is no good evidence that the ratios in sea water have changed greatly, though the total salt may have increased somewhat; the germ of the idea is correct but there is more fitting of the internal medium to its cellular contents than Macallum supposed.

The outstanding instance is the terrestrial plants: they have virtually no Na^+ or Cl^-, the two main ions of the sea and of the extracellular fluids of animals. The functional reason for this is clear: the rapid transpiration stream, maintained largely by evaporating large quantities of pure water at the leaf surface, would very soon overload the body with any ion not required for anabolic purposes. Again there is little sodium inside the cells of animals, and the inference is that the marine Protista were already selecting their salts ages ago, and that there was never any passive accommodation to everything the medium presented. In accord-

ance with the Chinese maxim they were conciliatory but not accommodating. It seems a reasonable further supposition that the early Metista were little more than colonies of cells, and when they began to evolve completely enclosed fluid spaces in the body these retained approximately the external medium. In animals there may have been no very strong selective pressure to change this except in detail, but the changes which have occurred are full of interest, and show that living systems have evolved the uniqueness of independence.

The independence is particularly clear in the high concentration of K^+ inside most cells of all organisms; it is almost isosmotic with the Na^+ outside whereas in sea water it has only 1/30 the concentration of sodium. Equally impressive is the low Cl^- inside cells, the K^+ and other cations being balanced largely by organic anions. The total diffusible ion-concentration is related to that outside, of course, by Donnan and electrical equilibrium requirements (Fig. 34), but this permits a marked qualitative differential.

It is possible that this differential is more important than the particular ions used to produce it, though this is not to deny the manifest specificity of function of each ion in particular contexts. In nerve and other excitable cells the essential requirement is that two different cations shall be balanced across the membrane, so that when the stability of the membrane is disturbed their electrochemical potentials can be the source of a sharp action potential. Because of their mobility the potassium and sodium ions are the cations of choice, but it is not *de rigueur* which is inside and which on the outside. The cat and dog have high sodium inside and potassium outside their erythrocytes, and it is even claimed that this inversion may be restricted to particular breeds in some species (Kerr, 1937). In general, no doubt, it is more economical to accumulate potassium inside the cells since it is the more mobile ion and since there is such a rapid replacement of the extracellular fluid by external medium, which always has a great preponderance of sodium, even in fresh water. There is further support for the general theme in the fact that the nucleus usually contains sodium and no potassium; this is the simplest way of producing a differential across an intracellular membrane. The intranuclear sodium seems to play the same part in protein synthesis there that potassium plays in the cytoplasm (Allfrey, 1960). Occasionally magnesium and calcium may control the membrane potential of a cell.

The second, and perhaps even more important, general function of the physiological salts is to promote the solution of proteins, nucleic acids and other biological materials. Globulins are insoluble in pure water (p. 221) and the solubility even of such small molecules as the amino acids is very sensitive to one or other of these ions. For the purpose small,

mobile and strong ions are essential and the physiological ions are ideal; the cations belong to chemical Groups I and II, the alkali metals and the alkaline earth metals, donating an electron almost as readily as hydrogen to become cationic, while chlorine belongs to Group VII at the other extreme of the periodic system, very strongly electrophilic, to become anionic. All of the major biological elements already considered belong to the intervening groups, and have less extreme and more ambivalent properties. One consequence of this is that the physiological cations and anions tend to remain associated by their strong reciprocal valencies, and this is why there is the sharp distinction between biological material and medium. The ions do form organic salts, and the elements occasionally form non-ionic biological compounds, but this is a rather secondary function.

25.1 The Four Main Physiological Salts

The importance of NaCl, KCl, $MgCl_2$ and $CaCl_2$ collectively for controlling the water content of the body has been indicated, and water is equally important for maintaining their normal concentration. The plight of those who are shipwrecked or lost in a desert is partly that they become seriously desiccated but also that they do not eliminate salt in proportion. The desert rat and some other mammals can drink sea water and excrete salt differentially but man fails to excrete as high a salt/water ratio as in the sea water itself, and the accumulation of salt in the body quickly causes mental deterioration. Water-excess also is dangerous, causing muscle cramps and nausea and sometimes water-intoxication. Neither condition has been fully analysed into its two components, namely the abnormality in fluid volume relative to organic mass and the abnormality in salt/water ratio, but these two are probably more critical than the third ratio, salt/organic mass: emphasis is always on concentrations, i.e. solute/water, but the third ratio is worth investigation.

Collectively animals tolerate a very wide range in salinity in their body fluids, from the 0·04 per cent of the fluids in freshwater lamellibranchs to the 3·5 per cent of many marine types, a much wider range than that of total organic matter, from 4 per cent in jellyfish to nearly 60 per cent of the body weight in insects, and than that of water, from 45 to 96 per cent. In many species the invididual can tolerate a considerable range, for instance *Mytilus* and *Arenicola* from a normal 3·5 per cent of the body fluids down to 0·6 per cent or less, and again the other two components vary very much less; the water content may change temporarily, when the external salt concentration is changed, but readjusts. These facts all imply that water and organic matter are the more important components.

The ratios among the various ions vary less than the total concen-

tration, though animals acquired the ability to modify these ratios before they managed to cope with large external variations in total concentration (Robertson, 1957); at any rate marine animals with no powers of tolerating external changes already have a strongly differential internal salt composition. The ratios therefore are more important than the total, and individual elements than the total osmotic and total ionic concentrations. The four cations collectively provide an assortment of ratios suitable for a variety of physiological and biochemical purposes. Their actions tend to be antagonistic so that collectively they provide a control over each process, against excess in either direction. In particular the two monovalent cations are usually antagonistic to the two divalent metals, and there is a fine adjustment of antagonism between Na and K, and between Mg and Ca. Any one can become the prime mover, therefore, with the other three antagonistic in graded degree. Analysis usually shows that the antagonisms are based on intrinsic differences in properties between the four ions, and that these have evolved into direct antagonisms by the enhancing tendency so evident in living systems.

Sodium and potassium being the stronger cations depress the concentration of H^+ ions in aqueous solutions more than magnesium and calcium, and so tend to simulate the effect of increasing the concentration of the ^-OH ion (Gray, 1931); the strength of their hydroxides is of course a reflection of this. The hydroxides of calcium and magnesium are slightly alkaline, hence the term alkaline earths, and even the carbonates are alkaline in reaction, but *in vivo* their chlorides have come to simulate acids, and the H^+ ion, by some kind of enhancement of the difference from the alkali salts. Magnesium chloride does hydrolyse slightly, to release free hydrochloric acid, but there seems little doubt that their intrinsic differences from sodium and potassium have become enhanced in their actions on many biological mechanisms. Sodium and potassium tend to increase the solubility and degree of dispersal of proteins, lipids and other components, whereas magnesium and calcium have the opposite effect, driving proteins nearer their isoelectric point, where they are least soluble. This operates also on a larger scale, and outside living organisms in fact, so that sodium and potassium make soil colloidal, sticky and difficult to work whereas calcium makes the grains larger and the soil more crumbly, better aerated, and easy to work. The monovalent ions disperse the molecules of films and membranes, making them gaseous in type (p. 82), while calcium and magnesium make them more solid in properties, with molecules more tightly packed. The former promote permeation through membranes and excite nervous and effector tissues. Calcium and magnesium antagonise the stimulatory effect of sodium on light-production by animals (Nicol, 1960).

Sodium and calcium play the major role in this balance and it so happens that their ratio in the body is about 75:1 which is approximately the ratio at which emulsions spontaneously invert between the oil-in-water and the water-in-oil states (Davson and Danielli, 1952, p. 311). Emulsions are therefore in a labile state in the bodies of all organisms, and this must facilitate traffic between the aqueous and lipid phases of the cell, promoting metabolic activity in various ways. It is a remarkable fact that this should be the ratio of the two ions in sea water and it provides a reason why fresh water and terrestrial organisms should have retained the marine ratio. In the neighbourhood of interfaces the ratio is much more in favour of calcium because of a Donnan relationship:

$$\frac{Na_s}{Na_b} = \frac{\sqrt{Ca_s}}{\sqrt{Ca_b}}$$

where *s* and *b* refer to surface and bulk phases. The calcium, being divalent, has an *s*/*b* concentration ratio which is the square of that of sodium. For monovalent ions this ratio is between 10:1 and 20:1, according to Donnan requirements, so that for calcium it is between 100:1 and 400:1 and the ratio Na_s/Ca_s is only 7·5/1 or less. There is a continuous lipid phase in the membranes of the cell, therefore. Intercellular cement is rich in calcium probably for the same reason, and this further stabilises the membrane. The mucus secreted by planarians has a similar protective function (Pantin, 1931) but here the calcium is an actual secretion. Calcium, and magnesium are in general more extensively bound to the solid components of the cell than sodium and potassium.

On the activity of the membranes of excitable cells magnesium appears to be more depressant than calcium, as shown by its well-known anaesthetic action. There is a conspicuously lower percentage of magnesium in the vertebrates and in the higher Crustacea than in other animals, and this appears to be correlated with their higher rate of physiological activity (Robertson, 1957). The laxative effect of epsom salts depends on paralysing the normal functions of the gut!

Sodium and potassium are antagonistic in several ways. The former usually stimulates excitable tissues but the latter is often depressant. For instance high concentrations of potassium arrest the vertebrate heart in diastole; in this and in other ways potassium resembles ACh (p. 292). For the biochemical transfer of phosphate there is the opposite relationship, potassium promoting and sodium depressing (Florkin, 1960; Long, 1961). This applies also for diuresis, and sodium therefore is in a second way the most important cation for water retention. The excretion of sodium and potassium themselves is bound up with this response, again in reciprocal fashion; the mineralocorticoid hormones of the adrenal

cortex, such as deoxycorticosterone, promote potassium excretion and the retention of sodium and water. The converse occurs in Addison's disease which involves a deficiency of the cortical hormones. Herbivorous mammals of the heart of the large continental masses suffer diuresis and sodium-loss because of the high potassium content of their diet, a serious misfortune where sodium chloride is always, and water frequently, scarce. The animals walk long distances to a water hole and hundreds of miles to a salt-lick. Potassium leaves the cells under rather slight provocation, following trauma and other stresses, and in both acidosis and alkalosis. As potassium leaves the cell sodium enters, much as in nerve-conduction, though in the reverse order.

Calcium and magnesium present a rather similar picture: inside the cell there is more magnesium and less calcium than outside, and each promotes the excretion of the other. If the magnesium content is raised more calcium is necessary to balance it. During hibernation, however, the ratio of Mg/Ca spontaneously increases, perhaps indicating that hibernation is a kind of natural anaesthesia. Calcium is known to antagonise this anaesthetic action of magnesium, which supports the view that the latter is the more strongly depressant. The Ca/Mg balance is particularly important in connection with muscle contraction, and probably with other effector activities. In the contraction cycle of muscle the two ions are associated in turn with a relaxing factor while the other is associated with the actomyosin contractile system; contraction occurs while calcium is associated with the relaxing factor and magnesium with actomyosin, so that here magnesium might seem to be the promoter of activity, but more correctly calcium inhibits the relaxing factor (Weber and Portzehl, 1954). Magnesium strongly antagonises the action of calcium on blood-coagulation.

25.1.1 Sodium

The quantity of sodium salts in the extracellular fluids determines their volumes (West and Todd, 1957, p. 641). This is an adequate demonstration that Na^+ is the ion which primarily determines the water balance of the body. Sodium chloride is also the salt which effectively determines the solubility of proteins, nucleic acids, etc. As already seen, sodium is the most consistently stimulatory of the group for physiological activities, and there is little doubt that it is indispensable for life, and probably irreplaceable by any other ion. The need for salt is well shown by the distressing asthenia and lassitude of sufferers from Addison's disease, by the compulsive craving for it in the animal treks mentioned and by the reactions of those subjected to the no-salt torture of the Chinese. It is less easy to appreciate it in many countries today, where foods are salted

to the verge of nausea, but the past prizing is fossilised in 'salary', 'worth his salt', etc. Virtually every organ of the body is involved, directly or indirectly, in the regulation of the salt balance (Kaunitz, 1956), and it is regulated so well that we do not suffer unduly from the daily insult of oversalting; at the other extreme it is very difficult to produce a real deficiency by short-term laboratory experiments (Orent-Keiles *et al.*, 1937); to induce a retardation of growth in rats it is necessary to reduce their intake to 0·002 per cent of the solid part of their diet. This shows how well it is conserved by the body, when necessary.

Sodium does not form many organic compounds, and there are few enzymes dependent on it for their specific activity; pectinesterase appears to be one of the few. Sodium enhances the action of insulin and, as already mentioned, it promotes the transfer of amino acids to their site of peptide synthesis in the nucleus (Allfrey, 1960), though not in the cytoplasm. More might be known of specific effects if it were easier to induce deficiency. The high concentration maintained in the body leaves no doubt of its great value but this in itself makes it difficult to test whether any other ion can deputise and whether its actions are peculiar to sodium. There is still a strong probability that it is the predominant soluble cation extracellularly in animals simply because it is so in natural waters. Why it should be, is still not clear since it is no more abundant than potassium in the lithosphere, and the chloride is not much more soluble than potassium chloride. It is little more abundant than magnesium and considerably rarer than calcium, both of which have much more soluble chlorides. The K^+ ion is more mobile than Na^+ and in general potassium is the more reactive element, although it has the higher atomic weight. No doubt chlorine has the greatest affinity for the smallest and most electropositive of the four cations, but including its water-shell Na^+ is a larger ion than K^+. However, there is some indication that the other three are more associated with weaker, less soluble anions than sodium is.

Lithium can deputise for sodium in the propagation of a nerve impulse, at any rate *in vitro*. The same is true for other functions in isolation, but there is no indication that it could deputise for sodium generally and permanently. The evidence is in fact to the contrary (p. 430).

25.1.2 Potassium

It is important to remember (Table 2) that potassium is as abundant in the bodies of living organisms as sodium, but is mainly in the cells. It is probably for this reason that there is more positive evidence of specific biological functions for potassium and that a deficiency is more easily induced. Growing and proliferating cells require considerable potassium (Fenn, 1940), as botanists have long appreciated. The growth of rats

suffers if the intake falls below 0·1 per cent as contrasted with the very much lower figure for sodium (Fearon, 1949, p. 16). Permanent deficiency results in fragile bones and sterility. In plants it is essential for the reduction of nitrate to amino compounds. This is only partly related to the K/Na balance across the cell-wall: potassium also acts as a co-factor for a number of enzymes, most of them phosphorus-transferring enzymes.

As a physiological ion it has already been seen to resemble, or to assist in some way, the quaternary $\vdots N^+$ ion of ACh, and so to have a parasympathetic type of action which in its turn favours growth and other vegetative activities. The sympathetic mediator, adrenalin, causes potassium to pass out of the cells and it inhibits growth. After H^+ itself, potassium is the most rapidly diffusing cation: although its atom is larger than those of lithium and sodium, its water-shell, and therefore its total ionic size, is smaller. Further, its speed is almost exactly equal to that of Cl^- so that they set up no diffusion potentials; it is for this reason that potassium chloride is used technically for conductivity bridges, etc., notwithstanding its curious property of evaporating from the solution and depositing on any available surface. Like sodium it exists *in vivo* mainly as the chloride, carbonate and phosphate, but it does also form salts with organic phosphates and with proteins such as haemoglobin (p. 323). By contrast magnesium and calcium in organic combination are held mainly in unionised form (Robertson, 1957).

Potassium is the lightest element to have significant amounts of a naturally occurring radioactive isomer, K^{40}. This is rather abundant in the lithosphere and may have significant mutagenic and other actions on living organisms (Groth and von Weyssenhoff, 1960). It has been suggested that its radiation may have catalysed biosynthesis before photosynthesis became possible (Calvin, 1962), if there was such a time. If, as seems probable, ultraviolet energy was once more abundant in the solar radiation reaching the earth's surface than today, it should have been an alternative agent and it would have the added advantage of a more even distribution than K^{40}. However, the latter also was probably once six times as abundant as now.

The biological specificity of potassium is not very high, possibly lower than that of sodium. The higher analogues, rubidium and caesium, can deputise for it in promoting the growth of fungi, and they often occur with it in the growing points of plants (Branfield, 1950). Even NH_4^+ has the same effect in speeding the action of pyruvate kinase and the methionine-activating enzyme (Long, 1961, p. 391); in two different ways this may be relevant to the association between potassium and choline. Both potassium and magnesium speed the action of fructokinase, and so do a

number of other metal cations. The promotion of amino acid transfer in the cytoplasm by K is paralleled by sodium in the nucleus. No doubt it again is very useful biologically, but it is not for certain irreplaceable.

25.1.3 Magnesium

Magnesium may be a more potent depressant and anaesthetic than calcium because it is a smaller, more mobile, ion, with more soluble salts. The action is not purely deleterious since depression of neuromotor activity differentially favours growth and other vegetative activities, so that it tends to resemble potassium in this. However, excess magnesium depresses even development, and cyclopia may result from differential depression of the most active regions. The magnesium level of the plasma rises in sleep, as it does in hibernation, so that its depressive action has great biological value. Magnesium deficiency can result in convulsions so that the depression is essential in waking hours also.

This ion is much more important than potassium as a co-factor for enzymes. Its unique function in chlorophyll (p. 320) is perhaps in a class apart, but it may also be a clue to the great activity of magnesium in phosphorus-transferring and other enzymes. Virtually every kinase, the enzymes transferring $\sim$P from ATP, are activated by Mg^{++}, at any rate in laboratory experiments (Long, 1961, pp. 389–413), and so are virtually all activating enzymes, which transfer both of the terminal phosphorus residues, together, from ATP. The occurrence of the compound *phytin*, magnesium-calcium-inositol-hexaphosphate, in plants is relevant here and also that of magnesium phosphate in bone (p. 412).

Magnesium forms a number of unique inorganic compounds which may provide clues to its biological significance. The asbestos group are best known; they are hydrated magnesium silicates, in the form of natural fibres up to 11 inches long, long enough to be spun into yarn. One pound will spin to 6 miles of yarn. This is very strong and at a diameter of 0·03 inches will support a weight of 15 lb. (Jones, 1943), comparing quite favourably with metal wire and with biological yarns. Asbestos is a good insulating material, against heat and electricity, like some biological fibres. Soapstone or steatite is another hydrated magnesium silicate mimicking biological material, as its name indicates. It is a solid lubricant, in fact, and is used commercially as French chalk. Silicon may be the more important element in these interesting materials (p. 470), but it forms these particular ones only with magnesium.

In all, magnesium seems to have more unique properties than potassium or sodium and to be more specific in its biological action. Although its concentration in sea water and in body fluids is low, there is almost as much in the body as sodium and potassium (Table 2). It is mainly in the

cells, like potassium; muscle contains as much as 30 mg per cent, functioning as already indicated.

25.1.4 Calcium

The high total calcium in the vertebrate body (Table 2) is due largely to the skeleton; there is only 0·01 per cent in plasma and as little as 0·003 per cent in the erythrocytes. Like sodium it is largely extracellular, in contrast to potassium and magnesium. Some is in ionic form, associated with inorganic and organic anions, but a high percentage is in unionising forms. It is in the third full period of the system of elements, has electrons in the active *d* orbitals, and so it tends to form coordination complexes in which it is chelated by several groups of a large organic molecule, and loses its ionic character. In this and in its biological properties, in fact, it is intermediate between the typical physiological ions and the co-enzymic group of metals to be considered in the next chapter. Magnesium also is rather intermediate, in its wide association with enzymes and related proteins. It might be true to suggest that the biological uniqueness of these two elements depends on this intermediate position, both having rather a variety of functions in consequence. In fact their two main functions become paradoxical—depressant through antagonism to the alkali cations and stimulatory through acting as co-factors for important enzymes: physiological depressants and biochemical excitants.

Hypocalcaemia, for instance, due to deficiency of the parathyroid hormone, is associated with a form of tetany, due to unantagonised excitation by the monovalent cations. Through an interaction between parathyroid and thyroid calcium comes to depress thyroid activity, which is excitant; the high calcium content of the water in limestone districts exacerbates goitre due to iodine deficiency (p. 458). Even its antagonism of the alkali metals presents paradoxes, however, and is not entirely depressant. It promotes the cardioaccelerating action of digitalis (p. 122), which acts by blocking the parasympathetic endings; here it is antagonistic to potassium in *its* depressor capacity. In the absence of calcium, potassium arrests the vertebrate heart in diastole, while calcium alone arrests it in systole.

It has already been seen that the depressant action of calcium is based on its intrinsic properties of forming insoluble salts, of dehydrating and precipitating proteins and of making the lipid phase of membranes continuous. The latter is elegantly demonstrated by placing washed slices of beetroot in pure calcium chloride solution, and others in sodium chloride. Pigment leaks out in the latter but not in the former. Calcium increases the viscosity of phosphatides, such as lecithin, probably through binding in pairs and so causing a closer packing of the interfacial molecules (p. 92);

it prevents the amoeboid movement of the 'myelin forms' (p. 296). It stabilises the histone of intercellular cement (Schmidt, 1955), and the mucus which protects planarians (Pantin, 1931), worms, snails, etc. (Erichsen Jones, 1938). In blood-coagulation it probably acts in its capacity of protein-precipitant, but in fact it acts at three distinct points, each involving a change in a protein, and the first change is in the nature of an activation.

The enzymes activated by calcium include a number also activated by magnesium, mainly phospho-transferases. In addition to these Ca accelerates the action of succinic dehydrogenase, choline esterase, lipase, trypsin, pectase and others. In activating the zymogens, i.e. precursors, of rennase and trypsin it acts in much the same way as on the zymogens of the blood-clotting enzymes.

Thus calcium has a wide and valuable range of functions in addition to its skeleton-building property. Like potassium, however, it is to some extent replaceable by the higher analogues of its group, in this case strontium and barium. Strontium is a major contaminant of calcareous skeletal material, now a great misfortune because of the radioactive Sr^{90} of the fall-out of atomic explosions. Strontium is essential for the calcification of bones and teeth (Underwood, 1962, p. 2), and plants absorb strontium in preference to calcium. The Acantharia have a purely strontium skeleton, while another group of sarcodines have one of barium (Prosser and Brown, 1961, p. 72). As a more distant analogue, barium tends to be antagonistic to calcium; it tends to decalcify and its salts are poisonous. Plants absorb it less readily. Radium, the heaviest and radioactive element of the sub-group, is highly concentrated from its medium by duckweed, algae and other plants. Radium salts placed on sterile gelatin appear to produce microscopic granules which move, grow and divide, like bacteria (Oparin 1957a, p. 81).

25.2 Chloride

The element, chlorine, not only forms chlorides but also covalent chlorocompounds with carbon and other non-metals. These are not common biologically and they are best considered in the general chapter (27) on the halogens. Apart from the chlorides of the physiological cations and other metals there are a number of organic chlorides in living organisms. Most of these are analogous to ammonium chloride, formed by nitrogen bases, and they are usually known as hydrochlorides in the same sense as NH_4Cl is $NH_3 \cdot HCl$, but this should not be allowed to obscure the fact that the base is then in the active, quaternary ionic form. Thiamine (p. 397) is one example of a key metabolite in this form.

There is evidence that Cl^- is essential for living organisms and for their growth (Orent-Keiles *et al.*, 1937), but there is little to indicate that it is more than an incidental anion associated with essential cations. This is still true of the hydrochloric acid formed in the vertebrate stomach and in the salivary glands of the gastropods, *Dolium*, *Murex* and *Pleurobranchia*. As already seen the latter produce mainly sulphuric acid, and hydrochloric acid has the disadvantage of being rather volatile; it may be formed merely because it is the most readily available strong anion. The caterpillar, *Notodonta*, also secretes a concentrated hydrochloric acid. In the vertebrate stomach little will be lost by volatilisation and here it is the main acid produced. It is produced relatively simply, by exchanging Cl^- for HCO_3^-, across the cell-membrane, thus replacing a weak by a strong acid. The $SO_4^=$ divalent ion is less easily transported across. A similar phenomenon underlies the chloride shift associated with the loading and unloading of carbon dioxide in the circulatory cycle. Again carbonic acid is formed, mainly in the erythrocytes, so that HCO_3^- diffuses out into the plasma and Cl^- shifts in to maintain Donnan equilibrium; at the lungs the whole sequence is reversed, and carbon dioxide is eliminated.

Essentially the same mechanism may operate in the chloride cells of the gills and other surfaces of aquatic animals. These cells secrete or absorb Cl^-, according to osmotic requirements, the appropriate cations presumably following in accordance with Donnan and electrical requirements (Fig. 34). On this evidence it might be claimed that Cl^- is the critical ion for osmoregulation, and in a sense this is true, particularly as it is the sole anion (or virtually so), compared with four cations. In nerve-conduction, by contrast, it does seem to be moved passively, following the movement of cations.

Cl^- also seems to be a specific co-factor for salivary amylase but otherwise there is little evidence of any specific biochemical function. Like sodium its significance is mainly physiological, and too general to be at all specific. At present it is easy to underrate its importance and uniqueness. Its uniqueness is that it has remained *the* physiological anion, although the amounts of Na, K, Mg and Ca have been greatly modified in living bodies during evolution. The properties which made it almost the sole anion in sea water determine its retention by living organisms—small size, mobility and ionic strength, so that it forms neutral, stable salts with alkali cations.

25.3 Lithium, Beryllium and Fluorine

While C, N and O are the first members of their respective groups, Na, Mg and Cl are the second members and this raises the important question why the first members, Li, Be and F, respectively, do not form the

physiological ions, or perform any other leading biological function. The first reason, and probably the main one, why they have not become physiological ions, is their rarity in the lithosphere. In the outer 10 miles of the earth's crust, including hydro- and atmospheres, there is 0·005, 0·001 and 0·03 per cent respectively of Li, Be and F, compared with 2·61, 1·94 and 0·23 per cent respectively of Na, Mg and Cl. The particularly low values for Li and Be may be due to their instability at the temperatures at which the earth originated, that is to say their rarity favours a hot origin (p. 454). As hydrogen condenses through helium, lithium, etc., carbon is the first stable stage (Jeans, 1943). Beryllium is easily destroyed by high-energy protons (H^+) (Schubert, 1958), and the intermediate elements are therefore permanently rare on earth.

Lithium is about as abundant in living organisms as in the lithosphere, can deputise for sodium in nerve-conduction (p. 424), and may stimulate the growth of plants (Seifriz, 1936, p. 429). Similarly beryllium can deputise for magnesium in plants, to some extent (Hoagland, 1952). It might therefore be concluded that they would have been used if they had been more abundant, but from phosphorus and other examples, and still more from some to be considered later, it is abundantly clear that this is not an adequate reason why they have not been exploited. It may be adequate in conjunction with the availability of their more abundant analogues; of course silicon is more abundant than carbon, and phosphorus than nitrogen, but carbon and nitrogen have unique properties, very different from those of silicon and phosphorus, and have been exploited for these. By contrast, lithium and beryllium differ from their higher homologues only in minor, biologically useless ways. Moreover they differ mainly in being atypical of their group, in lacking the pronounced electropositive valency for which the physiological cations were selected. Beryllium is the most atypical: it shows a trace of the non-metallic properties more emphatically shown by the lower members of the succeeding groups of the periodic system. It rather closely resembles aluminium, its salts hydrolysing to form the hydroxide and releasing acid. The hydroxide is insoluble at physiological pH, and so there is little possibility of the element being used as a physiological cation. In resembling aluminium, in the next period of the next chemical group it shows 'diagonal affinity' of which other examples will be seen. The metallic character is fully established in progressively earlier periods of the groups of progressively lower number; consequently lithium is more metallic than beryllium, but it is still somewhat abnormal. Its carbonate, phosphate and fluoride are insoluble, and in fact it shows diagonal affinity to calcium, while its urate and picrate are very soluble.

Both lithium and beryllium are sufficiently like their higher analogues

to act as metabolic antagonists in organisms which have now become adapted to the higher members of the groups. Lithium causes serious abnormalities of morphogenesis, generally described as 'vegetalisation' in embryogenesis, depression of activity differentially in the most active regions, rather as in the case of excess magnesium (p. 426). In regenesis it has similar effects, which here seem to be antagonistic to sodium (Kanatani, 1957; Massaro and Schrank, 1959). Potassium tends to increase the effect of lithium, which therefore is not so very different from the normal antagonisms among the four physiological cations.

Beryllium is more toxic than lithium, as might be expected from its more aberrant properties. Like lithium it has powerful effects on morphogenesis (Needham, 1941; Thornton, 1949), probably first as a protein precipitant (Aldridge *et al.*, 1949), secondly as an antagonist of calcium (Kay, 1931) and thirdly as an inhibitor of a number of phosphorus-transferring enzymes, including ATPase and nucleases (Jacobson and Webb, 1952; Dixon and Perkins, 1956). Altogether, therefore, it has a disastrous action, particularly on bone-formation.

Fluorine also is somewhat aberrant, but not in being a weak anion; hydrofluoric acid is extremely reactive, dissolving even glass, and fluorine forms stronger hydrogen bonds than any other element (Table 39). The element is so electrophilic that it will not form compounds with N, O or Cl, the next most electrophilic elements. It is the most reactive element of all, and perhaps this is one reason why it has not been much used in the relatively mild chemical systems of living organisms. As an ion perhaps its greatest drawback is in forming an insoluble calcium salt. This in fact is put to use in hardening teeth, but elsewhere in the body it would be troublesome. The fluoride ion inhibits catalase (Baldwin, 1953, p. 168) and other enzymes, showing that it is an abnormal ion to modern organisms. The position is complicated by the other main halogen property of fluorine, which is best considered in Chapter 27; in this again its action is mainly deleterious and to some extent this may preclude any dealings with the F^- ion also.

25.4 Conclusions

The physiological ions are unusual in a number of ways. They were the main solutes of the medium in which life began and with modifications in proportions have remained essential for the living systems which gradually evolved from this. Cells developed their own special saline composition, designed to maintain useful potentials against the external medium, but in consequence the Metazoa were obliged to enclose, as an internal environment, a saline approximating, at first, to the external

medium. This rather foreign component of the body has persisted in animals, somewhat obscuring the more revolutionary changes inside the cells, which are more intimately related to the evolution of bioplasm. For reasons given, plants were forced to abandon this 'internal sea' and their salt composition is a true intracellular type.

There is no reason why the ions of an external medium should be highly specific and this is perhaps why each of the ions of this group can be replaced to some extent by those of their analogous elements. There is often no serious effect of replacing Cl^- by Br^- or I^-, except a reduction in mobility. As shown, strontium is extensively used as a deputy for calcium and rubidium is a rather good substitute for potassium (Underwood, 1962, p. 343). This and the other analogues of the five ions, particularly the heavier analogues, occur widely in living organisms, in small amounts. Very few other elements show such low specificity: most seem to have acquired an enhanced specificity and uniqueness by their exploitation in living systems.

To some extent extracellularly, but much more intracellularly, the ions have acquired more specific functions, as enzyme co-factors, etc. They have become specified physiologically by a continuation of the process already started in the Protista, a process of balancing them against each other, to establish useful potentials and to effect physiological homoeostasis.

In all three chemical groups the lightest member is not used to any great extent extracellularly, because it is rather rare. It is not used intracellularly, and for the more specific extracellular purposes because it is anomalous in properties. This is in great contrast to the use made of carbon, nitrogen, and oxygen, each the first member of its group—carbon and nitrogen also much rarer than their respective number-twos, but oxygen much more abundant; this shows that the exploitation of elements which are to be constituents of the actual fabric is not determined merely by their abundance in the lithosphere (Table 1, Fig. 1).

CHAPTER 26

THE CATALYTIC METALS

> Life could not exist or even come into being without the mediating action of metals. (J. SCHUBERT, 1954)

IN ADDITION to sodium, potassium, magnesium and calcium there is a large number of other metals which have a biological function, and virtually all are catalysts, active in small or very small quantities and important out of all proportion to these amounts. Probably all are poisonous in large amounts, but so is every element in excess. Some, such as copper, are 'built into' an enzyme, or, like iron, are built into the prosthetic group. Others are more loosely associated and are rather non-committally called co-factors. Magnesium and calcium may come into this category though like most of the metals of this chapter they can form coordination complexes with various organic molecules, resulting in simple analogues of the porphyrans (p. 315).

In the sense that every chemical reaction is an oxidation–reduction (Spiegelman and Sussman, 1952) all these metals catalyse redox reactions, but in the narrower sense also, of energy-yielding reactions in biological respiratory pathways, most are essentially this type of catalyst. Many of them catalyse phosphorus transfer, directly coupled to, or more indirectly related to, respiration, and some catalyse reactions progressively more peripheral to the main respiratory paths, but it is probably true that in essence their actions all centre on respiration, the biochemical lifespring. All lie as a compact set in the middle of the series of all metals, when arranged in order of their electrochemical potentials, i.e all have potentials near the middle of the complete range, another manifestation of the moderation and poise of biological activities. The series (Brody, 1945, p. 106), starting with the most strongly reducing, is: lithium, rubidium, potassium, sodium, strontium, barium, magnesium, aluminium, *manganese*, *zinc*, *chromium*, *iron*, *cadmium*, *cobalt*, *nickel*, tin, lead, hydrogen, antimony, bismuth, arsenic, *copper*, mercury, silver, platinum, gold. Copper is the only isolated member of the present set but in addition molybdenum, selenium, vanadium and titanium are not represented

in this series. Moreover it does not recognise the several redox ranges of some of the metals.

Just as the elements so far considered are all (except potassium and calcium) first or second members of their chemical group, so each of the present set of catalytic metals is the first fully metallic member of its particular group or sub-group (Table 1). Sub-groups, two in each group, arise from the familiar fact that the large electronic shells beyond *M* (p. 318) are filled by two transits or series across the periodic table and the properties of the first or 'even' series of elements are therefore rather different from those of the analogous element of the odd series. The relevant sub-groups are shown in Table 36 and it will be seen that except in the two Cr–Mo–W–U and Fe–Co–Ni only the first member is a catalytic metal; in any case the transitional elements, Fe, Co and Ni, do not constitute a sub-group in the usual sense.

A second noteworthy point is that there is no element of this catalytic set in Group III; the group in fact seems to fall between a number of biological stools. Boron certainly is essential for the growth of plants (Brenchley and Warrington, 1927), but it is a non-metal and might have been expected among the major elements. Reasons why it is not will be seen later (Chapter 28). Aluminium, the first metallic member of the group, is atypical in a number of ways—not metallic enough to be a physiological ion, and with insufficient electronic shells to be a member of the present set; as already seen it resembles beryllium (p. 430) more than any other element.

In fact aluminium may have some small biological significance (Underwood, 1962, p. 325) and this is the most convenient context for its consideration. It promotes the oxidation of succinic dehydrogenase by cytochrome *c*, *in vitro*, though other metals also do this, and there is no reason to think that aluminium is its unique co-factor *in vivo*. However, accumulations of aluminium succinate have been found in the wood of the tree *Orites* (Read, 1935, p. 316). Salts have been used in the laboratory to catalyse the Cannizzaro reaction of acetaldehyde, the synthesis of formaldehyde from carbon dioxide and the synthesis of monoses from formaldehyde. The element is concentrated in onions and some other organisms, and in some neoplasms. It induces a blue colour in hydrangeas (Seifriz, 1936, p. 429), in place of the common pink. There is little to indicate that it is a normal metabolite of importance.

Titanium, the contribution of Group IV to the present set, is biologically the least important, so that a further generalisation is that the groups in the centre of the periodic table are the least important. Their elements have a stable valency, with no power to accept or donate extra electrons under special conditions. Thus the major elements come from the central

groups, the physiological ions only from groups at the two extremes (ignoring the inert gases and the transitional elements), while the catalytic elements come from a wider spread of groups on the two sides of the periodic table.

Another significant generalisation is that there is no important member of the present set from the first two periods of the table: the labile electronic shell is essential. As usual, however, the lightest feasible member is the one exploited, and so with the exception of molybdenum all belong to the third main period. In Group III, therefore, it is scandium, a complete biological nonentity, which should be the most relevant; it belongs to the even series of the third period, along with Ti, V, Cr, Mn and Fe. In fact, however, there is more evidence of a significant biological function in gallium, the first member of the odd series, to which only Cu, Zn and Se of the present set belong. Gallium occurs in living organisms and has even been found essential for the growth of fungi (Fearon, 1949, p. 26) and duckweed (Alexander, 1948). Moreover vitamin D appears to promote its deposition in bone, especially in rachitic animals (Dudley and Friedman, 1952). This resemblance between gallium and calcium is another example of diagonal affinity; it is perhaps also an indication that gallium is too similar to calcium to have a biological function in its own right.

It is interesting that palladium and platinum, which are such outstanding laboratory catalysts for hydrogenation reactions, are so unimportant biologically, and so reinforce the period-3 rule. Their catalytic action is of course by the free metal and depends on 'occlusion' of hydrogen gas by these metals, whereas the present set all operate as compounds, chemically rather than physically.

The halogens also call for comment here. Members of Group VII are the most non-metallic or electron-accepting, and even iodine, the member of period 4, is still a non-metal. The Mn–Ma–Re sub-group are all metallic, however, because of their transitional electronic configuration, and they illustrate most clearly the possible magnitude of the difference between the two sub-groups of the same group. Transitional elements all have much the same electronic activities as iron (p. 318) but the halogens remain always mono-electrophilic and in a milder degree hepta-nucleophilic. They will be considered in the next chapter.

In general the heavier metals in the relevant sub-groups (Table 36) behave similarly to the first member, but more slowly. To some extent they can deputise for the latter but they rarely enter living organisms in sufficient quantity to affect metabolism, whether as synergists or as antagonists of their lighter analogues. In quantity, under laboratory conditions, their soluble salts are toxic, those of mercury being outstanding.

However, soluble salts of the metals of the present set also are toxic in quantity.

Another general feature, which will become more evident when the elements are considered individually, is that there appear to be more than enough of them. Each has a considerable breadth of action so that there appears to be a great deal of overlapping of function. At present it is not possible to say that each is indispensable and biologically unique (Schubert,

Table 36. Position of Biological Catalytic Metals as First Members of their Chemical Sub-group

Metal	Group in periodic table	Sub-group of which it is first member
Copper	I	Cu–Ag–Au
Zinc	II	Zn–Cd–Hg
Titanium	IV	Ti–Zr–Hf–Th
Vanadium	V	V–Cb(Nb)–Ta–Pa
Chromium	VI	Cr–Mo–W–U
Selenium	VI	Se–Te–Po
Manganese	VII	Mn–Tc–Re
Iron	VIII	Fe–Co–Ni

1954). However, there is no doubt of the unique value of magnesium in chlorophyll, of iron in haem, and of others which have been adequately investigated, and judgement on the remainder can reasonably be reserved. Each has its own redox range (Table 37), and in some metabolic pathways it is known that a different metal catalyses each step (Fig. 51), in a regular and gentle gradient of redox potential, and this alone could demand a large number of metals, all told, even granting that a number have several valency shifts, each with its distinct redox potential range (Table 37)

26.1 Iron

This is undoubtedly the most important and irreplaceable of the catalytic metals, demanded by organisms of all kinds. It is required almost entirely for haem synthesis, but also for a more obscure compound in the blood-carrier pigment of echiuroid and some polychaete worms, haemerythrin, and in RNA, where it is often in high concentration (Underwood, 1962, p. 8). In some form it is a co-factor for the synthesis of chlorophyll, and plants suffer from chlorosis if they receive insufficient iron. Deficiency

TABLE 37. OXIDATION–REDUCTION POTENTIALS (E'_o) OF REPRESENTATIVE INORGANIC SYSTEMS[a]

Oxidised form	Reduced form	E'_o(V)	Special media
Co^{3+}	Co^{2+}	+ 1·82	
Co^{2+}	Co	− 0·28	
H_2O_2	H_2O	+ 1·77	
Pb^{4+}	Pb^{2+}	+ 1·75	
Mn^{4+}	Mn^{3+}	+ 1·64	15 N H_2SO_4
Mn^{3+}	Mn^{2+}	+ 1·52	15 N H_2SO_4
Mn^{2+}	Mn	− 1·18	
Ce^{4+}	Ce^{3+}	+ 1·45	acid
Cl	Cl^-	+ 1·36	
Br	Br^-	+ 1·07	
I	I^-	+ 0·54	
$Cr_2O_7^{2-}$	Cr^{3+}	+ 1·30	
Cr^{3+}	Cr^{2+}	− 0·40	
Tl^{3+}	Tl^+	+ 1·24	0·1 N H_2SO_4
Tl^{4+}	Tl^{3+}	− 0·04	
SeO^{2-}	SeO_3^{2-}	+ 1·15	
V^{5+}	V^{4+}	+ 1·02	1 M HCl
		− 0·74	1 M NaOH
H_2O_2	OH^-	+ 1·00	
NO_3^-	HNO_2	+ 0·94	acid
HNO_2	NH_4^+	+ 0·86	
NO_3^-	NH_3	− 0·12	
NH_4^+	NH_2OH	+ 0·56	
Fe^{3+}	Fe^{2+}	+ 0·75	dilute acid
Fe^{2+}	Fe	− 0·44	
AsO_4^{3-}	AsO_3^{3-}	+ 0·57	
$Fe(CN)_6^{3-}$	$Fe(CN)_6^{4-}$	+ 0·49	
$Mo(CN)_6^{3-}$	$Mo(CN)_6^{4-}$	+ 0·73	
Ti^{3+}	Ti^{2+}	+ 0·37	
U^{4+}	U^{2+}	+ 0·36	
Cu^{2+}	Cu	+ 0·34	
Cu^{2+}	Cu^+	+ 0·16	
Sn^{4+}	Sn^{2+}	+ 0·14	
H^+	H	± 0·0	
V^{3+}	V^{2+}	− 0·20	1N H_2SO_4
Mo^{5+}	Mo^{3+}	− 0·34	pH 3
S	S^{2-}	− 0·51	
SO^{2-}	SO^{2-}	− 0·90	
Mg^{2+}	Mg	− 2·37	
Na^+	Na	− 2·71	
Ca^{2+}	Ca	− 2·76	
K^+	K	− 2·92	
Li^+	Li	− 3·05	

also results in retardation of growth in both animals and plants: it is almost self-evident that a deficiency of any essential element must have this effect, which we may take as a general test of the indispensability of an element.

Although it is the second most common metal in the lithosphere, iron is only a microconstituent of living organisms, notwithstanding its conspicuousness in our large volume of blood. It may be noted that aluminium and silicon, the two most abundant mineral elements, are even rarer *in vivo*, and that titanium is another which is very abundant outside but rare in the body. Iron is a microconstituent no doubt because it is required only in small amount, and would be toxic in high concentration, but it does also happen to be very insoluble in most of its compounds so that living organisms have elaborate mechanisms for acquiring and conserving adequate quantities; these are at the same time a further demonstration of its importance. It is salvaged from effete erythrocytes, transported as a special complex, *transferrin* or *siderophilin*, and stored as *ferritin*, a protein complex with $(FeOOH)_8FeOPO_3H_2$. This insoluble complex bears out the view that the body cannot tolerate spare iron in soluble form. For absorption from the gut use is made of such soluble salts as the glutamate. *Phytin* (p. 74) from cereals depresses its uptake and conceivably may be a safeguard in the plant itself. Some diatoms concentrate iron 1000 times as much as they could by diffusion alone (Harvey, 1938). Fortunately iron is relatively soluble in lipid media, the site of terminal oxidation reactions (p. 79).

The unique value of iron as the metal for chelation in a porphyran has been considered (p. 317). It forms chelates with the highest stability constant (Long, 1961, p. 95ff), and among these metals with incompletely filled *d* orbitals, rather readily receiving and donating electrons, iron is unique for the small free-energy change which accompanies any electronic shift, i.e. for the ease and reversibility of the shift (Bernal, 1954). It also has an unusually large number of *d* orbitals containing a single, unpaired electron, and therefore is the paramagnetic element *par excellence*: it is 'ferromagnetic' in fact. Further the redox potential (Table 37) of the $Fe^{++} \rightleftharpoons Fe^{+++}$ change is ideal for terminal oxidation. With an E_o' of +0·75 V it is just capable of reducing atmospheric oxygen ($E_o' = +0{\cdot}80$ V), as the familiar ferric rust on exposed iron shows, and it is potentially capable of oxidising any organic material in living organisms. In organic combination its redox potential can be greatly modified, and it forms enzymes with a large range of E_o' values, some low enough to dehydrogenate rather highly reduced substrates, in controlled, biological fashion. For the most part, however, iron is a specialist on terminal stages of oxidation (p. 324).

Further evidence of its versatility also has been seen. In the cytochromes it reversibly changes between Fe^{++} and Fe^{+++}, but in haemoglobin and chlorocruorin it remains throughout in the ferrous state and in the peroxidases always in the ferric state. Electron movements in the physiological cycle of these last cases affect only the penultimate shell, and there is a change in magnetic properties. In haemerythrin part of the iron changes valency and the rest remains throughout as ferrous.

The biological actions of iron are shown incipiently by its inorganic compounds, which decompose hydrogen peroxide (catalase activity) and catalyse the laboratory reduction of carbon dioxide to formaldehyde (Read, 1935, p. 227; Moore, 1921)—possibly phylogenetically relevant to the role of the cytochromes in carbon fixation in plants (Calvin, 1962). The improvement of catalytic power by biological exploitation, however, is amazing (p. 263).

Some of the hydrated iron oxides exhibit colloidal and other properties similar to those of carbon compounds, so that Francis (1935) went so far as to suggest an iron-based bioplasm as the initial living material. However, its insolubility in other compounds, its inability to polymerise, the small amounts in modern organisms, and other features, speak against any but its present type of function, so dependent for efficiency on its organic combination.

26.2 Cobalt and Nickel

These metals are so similar to iron in their properties, and at the same time are more weakly ferromagnetic and less active, that no independent biological function might be expected for them. This may indeed be true for nickel, though it is rather consistently present in small amounts in all organisms. In plants it has a higher percentage than cobalt, in fact, but the converse is true in animals and some, such as the nudibranch, *Archidoris*, and the tectibranch, *Pleurobranchia*, accumulate large quantities of cobalt. Deficiency of cobalt in domestic artiodactyls results in wasting disease (W. Australia), bush sickness (N.Z.), moor cling (Devon), pining (Scotland) and enzootic marasmus, slow fever and coast disease elsewhere. The large number of names bears witness to the magnitude of the effect as well as the pandemic scarcity of natural cobalt.

Cobalt is now known to be the essential metal of the special porphyran cobalamin or vitamin B_{12} (p. 321), cotransferase for C_1 units. Its function therefore is quite distinct from that of all iron-porphyrans, and there is no overlap. Cobalamin is particularly necessary for haemoglobin-synthesis, but indirectly, for the globin, and not directly for the haem moiety (p. 377). Excess of *erythropoetin*, the active form of B_{12} in the plasma,

causes, polycythaemia, and cobalt is listed as a carcinogen (Heath, 1954), perhaps through the same effect on the synthesis of proteins in general. The action can be mollified by those biological agents which bind this type of metal, in particular the amino acids, methionine, cysteine, cystine and histidine, as well as by laboratory chelating agents such as EDTA. The vitamin is essential for the cockroach, some Protozoa and some green Algae, at least (Prosser and Brown, 1961, p. 93), and therefore may be a general requirement for animals, if not also for plants.

The cobalt ion also accelerates the action of a number of enzymes, *in vitro*: arginase, dipeptidase, pyridoxal kinase, riboflavin kinase and glutamine synthetase (Long, 1961, p. 409). It will be noticed that, like the C_1 transfer of cobalamin, these reactions are not primarily redox, and moreover iron, nickel and other metals also act as co-factors for some at least of them. The E_o' for the change $Co^{++} \leftrightharpoons Co^{+++}$ is + 1·82 V, much higher than for the corresponding change of iron, and far beyond the biological redox range, so that a primarily redox action would not be expected—or any overlap with iron. However, as already seen (p. 328), cobalt compounds, particularly its chelates, can act as oxygen-carriers, as well as in other redox capacities, so that biological conjugates also might suitably shift the E_o' value. The $Co^{++} \leftrightharpoons Co$ change has an E_o' of −0·28 V, which gives some indication of the range within which the change of organic conjugates of cobalt might lie.

Another synthetic chelate which carries oxygen is $[Co(NH_3)_5{-}O_2{-}(NH_3)_5Co]{\cdot}Cl_4$ (Granick, 1957). Cobalt salts also catalyse the laboratory reduction of carbon dioxide to formaldehyde, though this is also a property of nickel and many other metals, some of which have already been noted. More significantly cobalt protects —SH groups from oxidation by copper and iron (Albert, 1957). It is used commercially for the oxidative setting of linseed oil and nickel for the hydrogenation of fats. There is potential redox activity of biological relevance in these metals, therefore, so that there may have been natural selection in favour only of actions such as C_1 transfer, which do not overlap the functions of iron and other metals.

26.3 Copper

This is a universal microconstituent of living organisms and is indispensable for their normal growth and health, though like all other members of this set it is toxic in excess. Deficiency leads to falling disease in cattle, known in various parts of the world as swayback, enzootic ataxia, salt-sick (Florida) and lechtsucht (Holland). The most important symptom in mammals is anaemia, due to an indirect action of copper in

haemoglobin synthesis. It is thought to mobilise stored iron from the liver and so acts in an entirely different way from cobalt. Deficiency of copper also affects bone-growth, hair-pigmentation, reproductive functions and gastrointestinal functions—much the same set of symptoms as many other deficiencies, of metals and of the B-vitamins; these therefore do not give much clue to the other specific functions of the metal.

It has other vital functions, mainly of a redox capacity, and often appearing to be alternative to the action of haem compounds. Copper-proteins in fact act both as oxygen-carriers in the blood and as redox enzymes; as in the case of the haem-proteins this is their sole important function. The resemblance is not so strange as the Group I position of copper might imply, since it is the element next to nickel in atomic size and continues the transitional sequence begun by iron, copper and nickel (Table 1). Like them it has electrons wanting in the *M* shell (p. 318), and is paramagnetic as the Cu^{++} ion.

There are sharp differences between the copper- and iron-proteins, however, which may hold the clue to the division of labour between them, and there are also significant intrinsic differences between the two metals. As already emphasised (p. 316), the only well-known copper-porphyran is a trivial feather pigment, and the chemically active copper-proteins have some more obscure but more direct linkage between metal and amino acids. In view of the affinity of chelating groups for copper, it is almost certainly held in some chelate form. The main copper-proteins are the blood-carrier pigment, haemocyanin, in cephalopods, gastropods, decapod Crustacea and some Chelicerata, and the enzymes polyphenol oxidase, ascorbic acid oxidase, laccase, uricase (Long, 1961), butyryl–Co–enzyme A dehydrogenase, δ-amino laevulinic acid dehydrogenase, flavoproteins concerned in fatty acid oxidation (Kit, 1960), and some other flavoproteins. There are also copper-proteins of less certain function in the blood and tissues, haemocuprein, ceruloplasmin, hepatocuprein, phyllocyanin of plant leaves, and probably the violet connective tissue pigment of chilopods (Needham, 1960b). The enzymes collectively cover a large fraction of the biological redox range, and they certainly overlap the range of the haemoprotein enzymes, but the E_o' values are generally lower, as would be expected from those of the inorganic forms (Table 37). Copper is more suitable for lower rates of metabolism, under more anaerobic conditions. Since women have a lower rate of metabolism than men it is very interesting that the Romans gave the male sign ♂ to ferrum and the female sign ♀ to cuprum. Appropriately enough, also, copper is the metal for which all chelating elements, nitrogen, sulphur and oxygen, are the keenest suitors, while iron-chelates have the highest stability constant; however, those of copper come a good second in this quality.

The most efficient haemocyanins carry less oxygen per unit of copper than the best haemoglobins per unit of iron, as would be expected since the oxygen is not reduced in the process. Animals which evolved haemocyanin 'missed the boat' perhaps, though they use cytochromes and other haem-proteins, and should have been able to adapt these as blood-carrier

FIG. 51 The biological reduction of $—NO_2$ to $—NH_2$, showing the coenzymes and the metal cofactors associated with each step, together with other data. The over-all reduction has a redox potential of —0·12 volts, which is in the flavoprotein range, but the steps at each end of the sequence have high potentials requiring manganese cations and molybdenum anions (Table 37) as cofactors. Based on Nicholas, 1957.

Reduction sequence	NO_3	——→	NO_2	——→	HN:O	——→	NH_2OH	——→	NH_3
Oxidation state of nitrogen	+5		+3		+1		−1		−3
Coenzymes		Flavin		Flavin, NAD or NADP		Flavin, NAD or NADP		Flavin, NAD or NADP	
Metal Cofactor		Mo		Fe Cu		Fe Cu		Mn	
Coupled with phosphorylation		−		+		+		−	

Relevant redox Potentials (V)

$NH_3 + 9\ ^-OH \leftrightarrow NO_3^- + 6H_2O + 8e$ −0·12
$NH_4^+ \leftrightarrow NH_2OH + H^+$ +0·56
$NH_4^+ \leftrightarrow HNO_2$ +0·86
$NO_2^- \leftrightarrow NO_3^-$ 0·0

pigments. Both copper and iron are effective co-factors for some enzymes (Fig. 51) and both seem to be closely associated in cytochrome oxidase, working at the highest redox potential of any enzyme. The question of overlap and synergism between the two needs further elucidation, therefore, but it is clear that copper has its own properties and its unique niche.

It undergoes its valency change, or the alternative electron shifts, under weakly acid conditions (Thompson, 1950), outside the range of metals such as iron, with a much higher redox potential; as already indicated (p. 391), the redox power, measured by rH, is dependent to some extent on pH, and changes with a high E_o' cannot occur at low pH. Moreover, under acid conditions copper has more affinity for oxonium, $^{+}OH_3$, than for ammonium, $^{+}NH_4$, and under alkaline conditions this is reversed (Corwin, 1950), so that other types of reaction may be coupled with its redox activity under appropriate pH conditions.

As already emphasised, copper is outstandingly the most readily chelated of all metals and is even thought to form coordination complexes with its own water of crystallisation, in $CuSO_4 \cdot 5H_2O$ (Partington, 1933, p. 792). In the presence of ultraviolet light it complexes with Cl, O, H_2O and NH_3; the action is selective and does not extend to the analogues, Br, I, S, H_2S, PH_3 and AsH_3. The conjugation with oxygen is a possible precursor of the chemo-activated oxygen-transport of haemocyanin and may help to explain the origin of haemocyanins, while the selectivity for biological radicals may help to explain why copper is an important member of the present set of metals.

Copper is the metal which most readily forms complex cations with ammonia, and these are ions with unique properties. Cuproammine chloride, $CuNH_3Cl \cdot H_2O$ gives colourless solutions in acids and alkalis and both absorb oxygen, changing to the cupric condition (Partington, 1933). It also forms a compound with carbon monoxide, $CuCl \cdot CO \cdot 2H_2O$—a simple model for the behaviour of the respiratory pigments. Cupric salts form coordinates of the kind $Cu(NH_3)_4Cl_2 \cdot 2H_2O$, cupriammine chloride (Fig. 38, XVIII), while the anhydrous salt absorbs ammonia gas and forms $CuCl_2 \cdot 6NH_3$. These are simple but close analogues of the porphyrans.

It is well known that cupriammine solution in ammonium hydroxide, Schweitzer's reagent, dissolves cellulose, forming 'copper-silk', and this kind of property may have relevance *in vivo*. Because of its great reactivity with so many organic compounds copper is the cause of a rather common, and aesthetically objectionable, browning of food products (Thompson, 1950). It is also very soluble in organic solvents, which fits it, like iron, for terminal respiration. In these media its affinity for sulphur decreases and that for oxygen is increased (Corwin, 1950), another potentially very useful property; in aqueous media it might react with SH/SS systems and in lipid media with oxygen.

There is an embarrassment of relevant and possibly relevant properties of copper, and many of them show that it must be a very useful metal biologically. There are some which indicate a unique biological niche not open to iron or to any other element. At the same time there is often close

synergism, if not actual overlap, with iron, and very detailed study will be necessary to clarify their precise roles. As the essential metal in polyphenol oxidase copper cannot be replaced by Fe, Co, Ni, Mn, or Zn, although the simplest copper compounds have some activity (p. 264), so that it appears to be uniquely suitable for this reaction.

26.3.1 Silver and Gold

It is interesting that our early coinage utilised the three metals of this sub-group, and recognised the sequence Cu, Ag and Au as one of increasing value, for stability and wearing qualities. In the present context, however, silver and gold are worth considering precisely as examples of higher analogues which do not merely show the properties of the first member of their sub-group in lesser degree. They do not form a divalent cation and show no transition properties. They have virtually no biological significance, and because most of their salts are insoluble they do not often give trouble as poisons. It is interesting that such a unique and potentially useful property as the spontaneous photodecomposition of the silver halides to the free metal, so well exploited in photography, should not have found application in biological systems, but no doubt the insolubility of the salts and relative irreversibility of the process are serious disadvantages. Copper is unique in the sub-series.

26.4 Manganese

This element has been found indispensable for both animals and plants. It is accumulated in large amounts by the large lamellibranch, *Pinna*, and other molluscs, and by the mid-gut cells of the Hymenoptera (Prosser and Brown, 1961), but more important is the smaller, invariable amount in all organisms. Deficiency leads to retardation of growth in animals and plants, abnormalities in chlorophyll-synthesis, defective bone-formation (perosis) and reproductive failure in mammals. It promotes the synthesis of ascorbic acid (p. 50) and the assimilation of glutamic acid, fat mobilisation and perhaps haemoglobin-synthesis.

Manganese is a component of the molecules of arginase and enolase (Baldwin, 1953, pp. 112, 375) and possibly also of peptidase, and it may be a co-factor for other enzymes, since *in vitro* it greatly speeds the action of phosphorylases, alkaline phosphatase, most phosphokinases, laccase, aminopeptidase, L–malate dehydrogenase and glutamine synthetase. It is probably bound to the flavoprotein which catalyses the reduction of the hydroxylamine stage to ammonia (Fig. 51) and to the enzyme which releases oxygen from water in photosynthesis (Gaffron, 1962).

It is a typical transitional element with a labile valency, and is well known for the oxidising action of permanganates and for the oxygen-

fixation by manganous hydroxide in the Winkler technique. It may be present as a porphyran in erythrocytes (Underwood, 1962) and a number of the enzymes with which it is associated are redox in type. Chemically it is very similar to iron and chromium and a redox action might be anticipated as its main action. However, many of the enzymes above are not primarily concerned with oxidation–reduction and the redox potential of all its relevant changes is much too high for biological purposes (Table 37), though the pigment, pyocyanin (p. 342), is able to reduce manganese compounds as far as the Mn^{+++}/Mn^{++} stage, and various organic substances will do so in the laboratory (Quastel, 1947).

It is difficult to analyse the action of manganese in living organisms, or to pin-point any absolutely unique property, and it probably presents the most formidable problem for the theme of this book. It resembles almost every other metal of the present set, as well as some others, in one or other of its actions. It resembles iron not only in general chemical properties but in its action on chlorophyll-synthesis and in forming a porphyran. It resembles cobalt in a number of the enzymes it assists, and copper in speeding laccase. In new-born animals manganese-deficiency causes ataxia like that due to copper-deficiency, and the element prevents molybdenum from depressing copper-storage. It resembles zinc in being associated with the storage and use of thiamine (p. 397) and with reproductive activities. Selenium and other metals also play some part in these activities. Probably the closest resemblance is to magnesium and calcium, as shown by the large number of phosphate-transferring enzymes it speeds and the osteoporosis resulting from manganese-deficiency.

26.5 Zinc

This is another indispensable metal of the set, rather resembling manganese in biological activities though not very closely in chemical properties. Deficiency retards growth in plants and animals, particularly the growth of bones and hair in mammals. Some lamellibranchs, which seem prone to accumulate metals of this group, contain as much as 300 mg per cent of zinc (Fearon, 1949) and it is concentrated in milk and in the prostate and semen, as well as in many tumours. It is therefore not surprising to find that it plays an essential part in cell-division and also promotes wound-healing. The action on cell-division is concerned with the spindle (Mazia, 1956), but zinc is also associated with the nucleoproteins (Heath, 1949). It promotes protein- and nucleic acid-synthesis (Underwood, 1962), in some way.

These actions may be due mainly to the large number of enzymes containing zinc, or speeded by it. There is 0·33 per cent of zinc in carbonic

anhydratase ('anhydrase'), and it seems to be uniquely essential, among metals, for this enzyme, which catalyses the important reaction:

$$CO_2 + H_2O \longrightarrow H_2CO_3$$

Apart from its respiratory significance this reaction is an essential prelude to the synthesis of hydrochloric acid and other acids in the body. The enzyme is in a sense a carboxylase, which may explain the association of zinc with thiamine. It is also present in the enzymes carboxypeptidase (another link with the carboxyl group), uricase, several pyridine nucleotide enzymes (e.g. alcohol dehydrogenase), some alkaline phosphatases and probably tryptophan desmolase. *In vitro* it promotes the action of phosphorylases and phosphatases rather generally, and the effect on bone-growth no doubt is related to this (Sadasivan, 1951). It also promotes catalase activity, and so has significant redox action, though once more this does not seem to be its main enzymic function.

One of the best-known effects of zinc is not enzymic: it is a stabilising action, by making proteins insoluble. Whereas mercury, its higher analogue, precipitates proteins irreversibly, zinc does not actually denature them. The action of insulin is in consequence greatly prolonged, and zinc is added to the commercial preparations; it is present in some concentration in insulin when initially synthesised, and may have this function naturally. Zinc similarly prolongs the action of the pituitary antidiuretic hormone, and this may be the way in which it enhances the action of prolan (Urbain, 1938). The action on protein-solubility may also cover its effect on the formation of the division-spindle of the cell, and on the shortening of the chromosomes. It also enhances the effects of testosterone and some of the plant auxins (Skoog, 1940), perhaps via proteins with which these are associated.

Chemically zinc is an outstandingly strong reducing metal; it can even release hydrogen from such weak acids as the aliphatic hydrocarbons, and the zinc-alkyls so formed are extensively used in the laboratory for preparing further aliphatic compounds. Because of its very low redox potential, perhaps, it is not extensively used *in vivo* for redox purposes, rather like manganese at the other extreme. Another property of possible significance is the phosphorescence of zinc oxide, since there is a high concentration of zinc in the tapetum and iris of the vertebrate eye. Zinc oxide is known also to catalyse the photo-oxidation of ammonia to nitrites (Ellis and Wells, 1941), a reaction which occurs in the metabolism of some bacteria. The behaviour of zinc oxide is not unique, however, since titanium oxide has rather similar properties.

In carbonic anhydratase and in the effect on protein-solubility we have two unique biological functions of zinc, but in addition there is a wealth

of enzymic action, recalling that of other metals of the set, and at present beyond analysis, mainly because of the amount of overlap and interaction with the others. The association of zinc with manganese has been noted; in addition zinc improves the action of molybdenum in eliminating copper from the body. Again calcium-excess exacerbates zinc-deficiency (Table 38).

26.5.1 Cadmium and Mercury

Mercury is notoriously toxic, as a protein-precipitant, and has no biological significance; cadmium is toxic in excess but in small amounts may have a unique, useful function. It is always present in such quantities in

TABLE 38. SYNERGISM AND ANTAGONISM BETWEEN CATALYTIC METALS

Synergism	Antagonism
1. Cu with Co for Hb-synthesis	1. Mo against Cu
2. Fe with Mn for chlorophyll-synthesis	2. Se against As
3. Ca, Mg, Cu, Mn, Zn, Mo for bone-growth (also non-metals P, F)	3. Co antagonises the promotion of SH-oxidation by Fe, Cu
4. Zn with Mo as anti-Cu metals	4. Mn and V antagonistic on P–lipid oxidation
5. Mg and Mn both promote various enzymes	5. Ca and Mn promote ATPase action, Mg and Ag depress this
6. Mn, Co and Ni all promote arginase	6. Ca-excess exacerbates Zn-deficiency
7. Fe, Co, Cu and Zn all promote zymohexase	7. Mn-excess causes symptoms of Fe-deficiency, and vice versa
8. Cu, Fe, Mn, Mo all co-factors for flavoproteins	8. Ni, Cd (and Pb) counteract the toxicity of excess Na (i.e. interactions not confined to metals of catalytic set)
9. Cu, Mn both promote laccase	9. Mg^{2+} reduces the toxicity of Ni, Co, Cd, Zn, Mn when they are in excess
10. Mo and V both promote enzymes for N-fixation	
11. Zn, Ni both present in insulin	
12. Zn, Mn both promote synthesis of cocarboxylase	
13. Zn, Mn, Se all promote reproductive activities	
14. Co, Zn, Cd, Mo, Cr, V, U and (F) augment effect of Se in antagonising As	

living organisms. The amount is higher in liver and kidney, which may imply that it is largely excreted, but in addition a specific protein, *metallothionein*, containing 2·9 per cent cadmium, has been isolated from the adrenal cortex of the horse. This is one order of magnitude greater than

the metal content of any other protein (Underwood, 1962). Cadmium also occurs in insulin and, like zinc, prolongs its action (Bateman, 1945). *In vitro*, cadmium is found to speed the action of esterases but to inhibit 'dopa'-decarboxylase and peptidase.

26.6 Chromium

Until recently the only known biological association of chromium was as dichromate, in cytological fixation and in the tanning of leather. It is a particularly powerful bonding agent for proteins, in fact, very effective in cross-linking collagen fibres (Gustavson, 1958). The element has now been shown to stimulate the growth of plants (Underwood, 1962, p. 337), and to be an active component of some enzyme systems. It is present in virtually all tissues, up to 0·02 ppm in animals and as high as 1 ppm in plants. It is concentrated to some extent by tunicates (Levine, 1961), and in thyroid and spleen (Fearon, 1949, p. 34). Unlike most other metals its concentration decreases with age. It is particularly abundant in nucleo-protein, up to 1080 ppm in deoxyribonucleo-protein, which represents a concentration up of 2×10^4 times, and fairly high in ribonucleo-protein. It has even been suggested that chromium acts as an alternative to hydrogen bonds for linking the nucleic bases (Underwood, 1962).

Chromium is a component of the 'glucose-tolerance factor', which promotes the uptake of glucose by the tissues from the blood. Here it appears to be in the trivalent cationic form, Cr^{+++}, not in the anionic $CrO_4^{=}$ or $Cr_2O_7^{=}$ forms. These permeate biological membrances rapidly, which improves their value as fixatives, but Cr^{+++} permeates slowly, and this is probably useful in some way for promoting glucose-permeation. Possibly related to this action is its ability to speed the activity of the P–glucomutase system. It also promotes the reaction between cytochrome *c* and succinic dehydrogenase, and this again may involve an action at cell-interfaces, since the dehydrogenase works in aqueous, and the cytochromes in lipid, media (Green, 1962). It inhibits urease even in low, non-toxic concentrations. Commercially, chromous oxide is used to catalyse the reductive synthesis of CH_3OH from CO and H, but *in vivo* the action of the metal, once more, does not seem to be primarily redox.

26.7 Molybdenum

This metal also until recently was known only as a poison, for instance causing 'scours' in cattle feeding on the sour or 'teart' grass of Somerset, on soil with a high molybdenum content (Fergusson, *et al.*, 1938). In lower concentration, however, it is an essential co-factor for a number of enzymes, particularly in plants; it is a permanent component of some of

these enzymes, which are all of the flavoprotein, redox group. The normal content of animal tissues is 1·5 ppm but that of plants is as high as 9 ppm. A concentration of 0·01 ppm is adequate for the growth of tomatoes so that deficiency symptoms are not readily induced.

It is associated with flavoproteins of a particular redox range (Fig. 51), working in series with the Fe, Cu and Mn flavoproteins. This division of labour, depending on relatively small, quantitative differences in properties, may be the essential clue to much of the less-well-analysed biology of this set of metals: every enzyme and therefore every metal in the sequence is indispensable for the operation of the whole pathway. The pathways involving molybdenum flavoproteins are nitrogen-fixation and nitrate-reduction, the two pathways of biological nitrogen-reduction; through these molybdenum stimulates protein-synthesis (Hewitt, 1949). In this connection it is interesting that a simple laboratory system of paraformaldehyde, nitrogen gas and water is converted to amino acids in the presence of light and colloidal molybdenum oxide (Bahadur, 1957). Molybdenum is a co-factor also for xanthine oxidase (aldehyde oxidase), another important flavoprotein.

Molybdenum affects phosphorus-deposition in bone (Comar, 1949), like so many metals of this set, Mn, Cu and Zn, as well as Ca, Sr and Mg. For this reason the action may not involve the formation of phosphomolybdates, a type of complex restricted to molybdenum and tungsten. Their formation would probably inhibit bone-formation, in fact. Another interesting biological action of molybdenum, already mentioned, is its antagonism to copper, which is mutual and also involves sulphate. Molybdenum and sulphate together, but neither alone, prevent copper storage, while copper and sulphate, but also sulphate alone, depress molybdenum-absorption and increase its excretion. This is an active and highly specific phenomenon; as already seen (p. 397), no other anion can deputise for sulphate. In fact it is possible that tungstate may do so and antagonises molybdenum (Underwood, 1962, p. 108). Copper promotes the oxidation of sulphide to sulphate and molybdenum inhibits this, so that the interaction is probably related to redox processes and perhaps to nitrate-reduction (Fig. 51). Molybdenum therefore probably groups with iron and copper as the sub-set with primarily redox action.

26.7.1 Tungsten and Uranium

The next higher analogue of molybdenum, tungsten (wolfram, W), can deputise for molybdenum in the enzymes of nitrogen-fixation (Quastel, 1947), but so can vanadium (p. 451), the 'diagonal analogue' of molybdenum and a member of Group V. As a member of Group V it is an analogue of phosphorus and so forms vanadomolybdates. Tungsten

probably has no normal biological significance, being much too extreme in properties. It has the distinction of being the metal with the highest melting point, 3370°C, and so is used extensively in sparking plugs, electric-lamp filaments and resistance wires. Its alloy with iron cuts steel as easily as a knife cuts lead. The carbide is as hard as boron carbide and almost as hard as diamond.

Uranium is radioactive, which complicates its chemistry and biology. Its salts sensitise other compounds to ultraviolet light. The acetate has been used to induce reluctant proteins to build multilayers (Rothen, 1956) and this 'condensing' effect on membranes depresses fermentation in yeast (Booij and de Jong, 1956). The acetate also promotes the oxidation of sorbitol to sorbose (Alexander, 1948, p. 119). However, it is not certain that any of these actions are either specific to uranium or biologically significant.

26.8 Selenium

Like molybdenum and chromium, selenium was until recently known only for its toxicity, in concentrations greater than 3 ppm of the diet. In moderate concentration it causes alkali disease and in larger amounts 'blind staggers'. Animals are poisoned by feeding on seleniferous plants and this is the basis for the use of the so-called 'systemic insecticides'. These cannot be fed to plants required as food for man and domestic animals, of course, and there is a possible menace to harmless wild animals.

In low concentration, around 0·3 ppm, it is found that selenium has a beneficial effect on metabolism and on specific enzymes, though it is not yet proved that it is indispensable (Underwood, 1962) or a unique co-factor for the enzymes it speeds. It stimulates the growth of the plant, *Astragalus*, but this is unusual in growing preferentially on seleniferous soils. More significantly it promotes the growth of chicks and ruminant mammals, and improves their fertility. It also relieves muscular dystrophy, prevents the oxidation of unsaturated fatty acids, and in fact mimics vitamin E very extensively in its action (p. 117). It is associated with α–tocopherol in promoting the synthesis of ubiquinone, coenzyme Q (p. 132) (Underwood, 1962, p. 319). Like vitamin E, selenium compounds are very good antioxidants, better than the analogous sulphur compounds (p. 391), but Green *et al.* (1961) did not find the action on ubiquinone-synthesis antioxidant in nature. Selenium inhibits a number of enzymes, again mostly redox in their action; these include succinic dehydrogenase, enzymes of alcoholic fermentation and urease.

The toxicity of higher concentrations of selenium is due mainly to its acting as a competitive analogue to sulphur. This therefore may be a good example of a disturbing analogue acting in a quite different and

favourable role at lower concentrations. Compared with most analogues selenium is incorporated in place of sulphur extremely easily, even into the amino acids and proteins. Moreover some of these proteins can deputise for the normal sulphur-proteins, for instance β-galactosidase (Cohen, 1959); most of them 'jam' normal metabolism, however, and this may be how selenium inhibits dehydrogenases. Seleniferous plants of course contain such selenium-proteins, probably functioning satisfactorily, but these, unlike most organometallic compounds, are much more toxic to other organisms than inorganic selenium. As might be expected, sulphur compounds, including sulphate, competitively mollify selenium poisoning; this perhaps indicates that selenium also is involved in the Mo–Cu–SO_4 interaction-complex. Somewhat unexpectedly, perhaps, arsenic also is a selenium-antagonist and at 5 ppm it gives virtually full protection. No other element is known to do this, though a large number augment its toxic effect: F, Mo, Cr, V, Cd, Zn, Co, U.

Relevant to its membership of the present set is the photosensitivity of metallic selenium; it releases electrons on exposure to light and for this reason is used in photocells. By releasing electrons in this way its conductivity is greatly increased, so that the light effect can be measured electrically. The metal is therefore also a good redox catalyst, and both it and selenium dioxide are used to catalyse the conversion of recalcitrant organic nitrogen compounds to ammonium ions in the Kjeldahl process of nitrogen determination. It is a potentially useful and unique element, and the present indications are probably well founded.

26.9 Vanadium

This metal has excited interest since Henze discovered its amazingly high concentration in the blood of many tunicates, up to 6500 ppm, sometimes free in the plasma and sometimes in corpuscles. It is incorporated into a 'vanadium chromogen' so that the superficial resemblance to iron and haemoglobin is evident. However, *haemovanadin* has a low molecular weight, of 900, and could not act as a reversible oxygen-carrier under physiological conditions (Webb, 1939). Its function is more peculiar to it. The redox potential of $V^{+++} \rightleftharpoons V^{++}$ is much too low for it to be reversibly oxidised by free oxygen under physiological conditions (Table 37) and this system could act as a reversible redox agent for organic material only under very acid conditions. In fact 'vanadocytes' do contain as much as 9 per cent sulphuric acid, and in this medium the pigment could reduce terminal oxidases such as the cytochromes (Califano and Boeri, 1950).

Some tunicates have niobium, the next higher analogue of vanadium, instead (Carlisle, 1958), but there are never large amounts of both. Some

have an iron pigment instead (Endean, 1955). The gastropod, *Pleurobranchus*, has a high concentration of vanadium, as well as of cobalt (p. 439); it may prove significant that this gastropod also secretes high concentrations of acid, hydrochloric acid and sulphuric acid, but in the salivary glands. The polyzoan, *Plumatella*, also has unusually large amounts of vanadium, but similar claims for holothurians have not been confirmed (Ciereszko *et al.*, 1962). Some timbers accumulate the element. Most organisms, however, have around 0·1 ppm, or less. Even at this level it appears to have some essential function, since it proves indispensable for plants (Arnon and Wessel, 1953) and is not replaceable by any of the following long list of elements: Ti, Cr, W, As, Pb, Se, Al, Be, Sr, Rb, Cd, Hg, Li, Br, I.

It stimulates the growth of Algae and Protozoa (Bowen, 1938; 1939) and it improves the mineralisation of vertebrate bones and teeth, possibly replacing some of the phosphorus with which it is isomorphous. However, it appears to inhibit the growth of *Mycobacterium* and of the tubercle bacillus, and it is conceivably a relatively recent exploitation, though there are reasons to be sceptical of late exploitations (Needham, 1959).

The most important biochemical function with which it has been found associated is the desulphydration of cysteine, a redox reaction. It is an actual component of the enzyme and so is conjugated with pyridoxal phosphate (p. 336) (Berzel *et al.*, 1958; Snyder and Cornatzer, 1958). Probably in this same capacity, since SH groups are involved, it causes the de-iodination of thyroxin (p. 189) (Anbar and Inbar, 1962) and this is one possible reason for its toxicity in high concentration. It can also to some extent deputise for molybdenum in the fixation of nitrogen, but it is not certain that this is a normal function *in vivo*. It promotes the oxidation of phospho-lipids in the liver, where its concentration is higher than in most tissues (Bernheim, 1939); the concentration of phospholipids in the plasma therefore is depressed. It seems to be essentially a redox agent, therefore, and is used extensively in this capacity commercially, usually in the form of V_2O_5; examples are the oxidation of SO_2 to H_2SO_4, the oxidation of aniline, benzene and naphthalene, and the reduction of fats to mineral oils.

Other interesting properties of possible biological significance include the catalysis by vanadium chloride of polymerisation in some organic monomers (Hinshelwood, 1956); the action is shown at low temperatures which gives it potential biological value. Again, V_2O_5 is unusual among inorganic substances in forming tactoids (p. 232), so that it must readily form macromolecular complexes. Its liquid-crystals are of the smectic type (Pfeiffer, 1948). It should be stressed that in this it is not absolutely unique, since FeO also forms tactoids. In this connection there are a number of

relationships between V and Fe, quite apart from the unique properties of alloys between the two metals. Vanadium tends to accumulate in iron-rich soils and may replace iron in some of its compounds. In coal there is often a large amount of vanadium-porphyran (Fox and Vevers, 1960) probably formed in this way by 'secondary enrichment'. There may be a similar explanation for the very high percentage of vanadium in other deposits of probable biological origin, petroleum, bitumen, asphalt, etc., particularly from the New World, as well as in fossils (Vinogradov, 1943). The ash of Peruvian asphalts is as much as 30 per cent vanadium.

In these deposits vanadium may have replaced other metals also. There is no conviction that it was present in the living organisms in so much higher concentration than today. This would of course be diametrically opposed to the previous hint that the amount used has increased in the course of evolution, but there is no plausibility about this, either. Depression of the growth rate of bacteria by vanadium in fact may be due to interference with the function of other essential metals, since Mn^{++} and $CrO_4^{=}$ counteract the inhibition (Underwood, 1962). There are other instances of competition with molybdenum, which again is very similar to iron chemically; vanadium inhibits mevalonic kinase, a key enzyme in the pathway of biosynthesis of cholesterol (p. 109), while manganese speeds this enzyme.

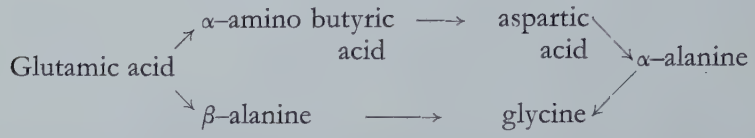
Vanadium therefore is a significant biological redox agent with resemblances to iron and manganese, but also with its own peculiarities, which are probably maximally expressed in tunicates. As already noted its next higher analogue, niobium (columbium, Cb), is sometimes accumulated as an alternative, and to about the same extent. Quite nearly related species may differ in their 'choice', so that it may be a case of almost perfect deputisation. Both elements occur in some petroleum deposits, but along with many other metals, of course.

26.10 Titanium

At present there is not much evidence that titanium has a clear-cut biological function so that it may be true in spirit (p. 434) that no significant metal of the present set occurs in Groups III and IV. However, there is considerable circumstantial evidence that it has a biological role, and it has some outstanding properties. For an element so widespread in the lithosphere it may not be very significant that titanium is present in virtually all tissues, but the lobster accumulates it to the extent of 32·5 mg/kg (Bertrand and Voronca-Spirt, 1929), and it has been found in some fossils in large amounts (Pirie, 1957; Oparin, 1957a). In the latter and in the lignite of Dorset (Hallam and Payne, 1958) it may be a secondary

enrichment, like vanadium. Perhaps most significant is the fact that it is present in all natural xanthopterins (Ziegler-Gunder, 1956).

A property of possible biological value is the photosensitising action of titanium dioxide; this salt will catalyse the photolysis of amino acids, under the influence of sunlight (Giri *et al.*, 1953), and the following changes have been detected:



It also sensitises ammonia to ultraviolet, which catalyses its oxidation to NO_2, a simple analogy to the action of some nitrifying bacteria. However, this action is not unique, zinc oxide and alumina also acting as photosensitisers for the process. Any biological action is likely to be of the redox type and titanium is an outstanding reducing metal. The chloride and even the sulphate are among the most powerful of reducing agents, in acid medium, and are used commercially to bleach textiles. The element has a very great affinity for oxygen, and titanium dioxide is one of the most stable of chemical compounds. It is thought that the oxygen of the sun must be mainly in this form (Osborn, 1918), and also, perhaps, that of the earth in its early period, if the hot-origin theory is correct (Seifriz, 1936, p. 525). It may be noted that the oxide of the next higher analogue of the same sub-group, zirconium, also is very stable, withstanding temperatures up to 2300°C, and the metal zirconium burns in oxygen with a flame 800°C hotter than that of oxy-acetylene.

As a redox agent in acid medium titanium resembles vanadium and it is interesting that tunicates tend to accumulate titanium also (Levine, 1961). Like vanadium chloride, titanium chloride catalyses the low temperature polymerisation of some monomers. Both metals secondarily enrich in coal and other deposits. In this connection it may be noted that germanium, the first metal of the other sub-group of IV, also shows this phenomenon (Hallam and Payne, 1958). It seems that accumulation by tunicates has the same rather low specificity as this enrichment-process, since they also accumulate chromium. This may imply that the whole idea of secondary enrichment should be queried, but in any case it reduces the conviction that any one of the metals in tunicates is biologically critical.

26.11 Conclusions

In some ways this is a critical group of elements for the theme of this book because at present there appears to be an excessive number of them with

very much overlapping, non-specific functions mostly in the same general redox field. There is no doubt, on the other hand, that many of them are indispensable for life and growth, and in many at least one unique function has been discovered. Some of them are accumulated in high concentration sporadically by particular organisms, while most of them are widely distributed in micro-quantities in all organisms, in higher or lower concentration than in the environment, and this is the most certain evidence that they have essential metabolic functions. For some, e.g. molybdenum and selenium, there seems to be a single biological action, clear-cut and unique, but this may be because investigation of them has only recently begun. However, at the other extreme also, where there has been most interest for a long time, each element does seem to have its own distinct biological niche. Iron and copper, although they are both redox agents, working in the same general field, appear to divide the labour in accordance with their intrinsic differences in redox potential and other detailed properties. There is an *a priori* case, therefore, for believing that all of the set of elements have distinct and essential biological functions. The fact that rarely more than one element in any one subgroup is at all active *in vivo* is further indicative evidence.

The problem is greatly complicated by the amount of interaction among the set (Table 38), synergistically or antagonistically, directly or indirectly. Such synergistic interactions are to be expected if they work in series, as in Fig. 51, in particular pathways, or in branches of a common pathway, and antagonistic action, as between copper and molybdenum, or vanadium and manganese, may be connected with the control of a correct balance between them all. When these interactions are all clarified a great deal more will be known about general metabolism, as well as about the present theme. It might be suggested that since nearly all of them belong to different chemical sub-groups the extensive interaction must be a phenomenon at the biological, rather than the chemical, level, but on the other hand they *are* all related by the general nature of their electronic configuration. In this respect they are a natural chemical set, and this is why they have become a definite biological set. Schubert (1954) maintains that all of them can deputise for each other to some extent and that the interaction of each with its biological system is dependent on the concentration of at least two metal ions. Under these circumstances none can be absolutely unique, but it seems that each acquires a high degree of uniqueness for certain reactions, through forming the most stable complex with some one type of biological molecule.

CHAPTER 27

THE HALOGENS

THE term halogen, or salt-former, expresses the very strong anionic property of this main sub-group of Group VII, already seen in the unique role of Cl^- as the one important physiological anion. The halogens have seven electrons in their outermost shell and need only one to complete a state as stable as that of the inert gases. Unlike the manganese sub-group they have no incomplete orbitals in the penultimate shell and show a single-minded tendency to complete the shell, usually electrovalently, but sometimes covalently, as in the gaseous form of the molecules of the free elements, and in their organic compounds. Biologically the ionic property is exploited in chlorine and the covalent property mainly in iodine; more correctly iodine is used for the redox change which accompanies any transition between covalent and ionic states. The halogens do also form inorganic compounds of higher valency, datively, but these have no biological counterparts, and even the covalent organic compounds are very little exploited.

27.1 Chlorine

The role of Cl^- in living organisms has been considered (p. 428). The element also forms some interesting organic compounds, such as chloroform, $CHCl_3$, but these are not used biologically—often for evident reasons. The analogues formed by the other halogens have qualitatively similar properties, and their action is not necessarily weaker than that of Cl compound: for instance CH_2I_2 is more lethal than CH_2Cl_2 (Davenport, 1897). The anaesthetic effect of the chloromethanes increases progressively from CH_3Cl to CCl_4 and is typical of the magnitude of the modification of hydrocarbons by halogen-substitution. Another example is provided by trichloracetic acid, which is 100 times as strong an acid as acetic itself. It is also a powerful protein-precipitant. The compound C_2H_5–S–C_2H_5 is relatively harmless but its monochloride is poisonous and the dichloride is the familiar mustard gas. Covalent chlorine compounds of the hydrocarbons therefore are inimical to modern metabolic

systems, and this is perhaps not surprising: it is bound covalently, like the hydrogen atoms, but the latter are liable to be removed as strong cations whereas the halogen becomes a strong anion if it changes its state. Although some iodine compounds are more poisonous than those of chlorine, in general they are the mildest, and this is probably why this is the halogen most used biologically in covalent form. It is the heaviest of the halogens and so contravenes the rule that only the lightest member of any group is exploited.

27.2 Iodine

As already indicated iodine is exploited mainly for its oxidation–reduction properties and the vertebrate thyroid hormones are the only well-known compounds in this category, though similar compounds do occur in invertebrates (Gorbman, 1959), and may have a similar function. The thyroid principles are iodotyrosine derivatives (Fig. 28, III, IV) (p. 189) and simpler relatives occur in spongin and gorgonin, the organic skeletal materials of the sponges and Anthozoa. Here they have a mechanical function, that of hardening the material, so that once more there is the strong duality of function of amino acids, chemical and physical (p. 170). Other covalent iodine compounds are synthesised sporadically, but only serve to emphasise the uniqueness of the iodotyrosine group. *Balanoglossus*, a hemichordate, probably secretes iodoform, CHI_3, or a related compound, presumably for defence, and some Paussid beetles secrete elemental iodine in vapour form. No doubt bromine would be more irritating, but less easily stored, and iodine is outstanding for the ease with which it is separated from iodides; in contrast to the other halogens it is more stable than its acid, hydriodic acid. Although in general the iodide anion is only an incidental body constituent, seaweeds seem to accumulate large amounts, and are still a main commercial source of the element.

The chemical action of the thyroid hormones is to couple the terminal steps of respiratory oxidation with the crucial phosphorylation of ADP to ATP (p. 405), most of the ATP being synthesised through these terminal stages. Probably through this the thyroid also maintains heat-production in homoiotherms, and may control metabolism in a different way in the poikilothermic vertebrates (Barrington, 1963); without this group of compounds, perhaps, the vertebrates could not have evolved homoiothermy. They also control the supply of energy for growth and in the Amphibia have come to be particularly important for the developmental changes of metamorphosis, both regressive and progressive. Deficiency in man was all too well known, because of the low iodine content

of inland waters, often exacerbated by high calcium content (p. 427); thyroid-deficient adults become myxoedematous, i.e. have puffy connective tissue, but more significantly they have poor temperature-regulation and are metabolically sluggish, physically and mentally. Deficient infants, cretins, are stunted in growth, and mentally deficient or worse. The nervous system in fact depends on the hormone for its growth as well as for its functioning. Occasionally hyperthyroid states develop and the individual then shows the contrasting syndrome of high metabolic rate and temperature, high excitability and emaciation.

It is still far from clear in detail how the thyroid hormone couples phosphorylation with oxidation and whether all its physiological manifestations are through this coupling-control alone, or even through some form of redox action in every case. Dinitrophenol (DNP), an analogue of di-iodotyrosine because of the well-known resemblance between the nitro group and the halogen atom, uncouples phosphorylation from oxidation and is a powerful metabolic poison; it demonstrates very dramatically the importance of the normal coupling. Rather alarmingly, excess of the thyroid hormone itself uncouples the two components, so that the amount of the hormone circulating needs to be very precisely controlled; hyperthyroid patients waste most of the excess energy produced.

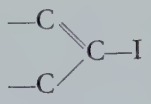
It seems probable that most of the actions of the thyroid hormone are in fact of a redox type since most of its antagonists are reducing agents or 'antioxidants'. Any increase in the amount of the vitamins A, E, and C increases the demand for the thyroid hormone: the vitamins may serve to prevent the amount of thyroid from becoming excessive. Sulphydryl compounds also are rather generally antithyroid. A naturally occurring example which has been of much clinical interest is 5-vinyl-2-thio-oxazolidine, or *goitrin*, which occurs in many species of *Brassica*, and is a member of a biologically rather rare group of ring compounds (p. 386). Experimentally thiouracil (Fig. 44) has proved very potent.

There are other types of antithyroid agent, however; for instance the thiocyanate, ^{-}SCN, and cyanide, ^{-}CN, anions, which are fairly close analogues of the halide anion, inhibit the initial uptake of inorganic iodide by the gland. The sulphydryl compounds on the other hand inhibit the next stage, the critical conversion of I^{-} to the covalent RI form; this is of course the redox change which is believed to recur reversibly in the subsequent functioning of the hormone. It may be significant that the saliva of man usually contains the thiocynate ion, conceivably to control the uptake of iodine from the gut. The thyroid hormone protects the body against the alkyl cyanides (Fearon, 1949, p. 484), which occur in some foods.

In considering the biological uniqueness of iodine in this hormone there are four main features to explain, why a halogen redox agent was used rather than any of the others available, why iodine and not any other halogen, why an aromatic, and not an alicyclic or aliphatic compound, and why only this particular aromatic ring produced active iodo-derivatives of biological importance. Halogen redox agents have the advantage over oxygen, sulphur and their compounds that they require only a single hydrogen or a single electron transfer, which in fact is most convenient metabolically (Geissman, 1949). For this purpose of course the halogen must be in monatomic form and so must be in organic combination. The covalently bonded halogen is an oxidising group whereas the equivalent monovalent HO— group is neutral and HS— is reducing in action. It seems therefore that there is a genuine niche for halogen redox agents. They are of course the strongest oxidising agents, because they are the most electrophilic of the elements.

The uniqueness of iodine among the halogens is that being the weakest of them its bond with the carbon atom is most easily opened and it therefore has the most reactive carbon compounds. The alkyl iodides are not as easily synthesised in the first place as the chlorides and bromides but they are more labile once formed; equilibrium favours their breakdown and they react whenever possible. There is here a parallel to the reactivity of the weak ~P bond contrasted with the stronger —P link (p. 402). This is the explanation of the anomaly that iodine is the heaviest element of its group and incidentally the heaviest atom of biological importance; it is the only biological element beyond the even series of the fourth period of the chemical table.

Aromatic combination is exploited rather than alicyclic or aliphatic because as usual this confers the paradoxical virtues of stability and activity, associated with the conjugate system of double bonds. The halogen is removed much less readily than from saturated, aliphatic combination, and this is an important check on the great intrinsic instability of organic iodine compounds. The vinyl halogen compounds have the same stabilisation, showing that it is certainly due to the double bonds (Finar, 1959, p. 521). The aromatic iodides therefore have the best of both worlds, great reactivity combined with good control. The alkyl iodides are active enough, as shown by the anti-SH behaviour and great toxicity of iodoacetate, $ICH_2 \cdot COOH$, but they lack the stability of the iodotyrosines. The aromatic halogen atom is also more polar than in the primary aliphatic compounds, since it is effectively a secondary compound, approaching the tertiary state in the sense that every other bond on that carbon atom is held by other carbon atoms:



The greater polarity again makes for greater reactivity (Finar, 1959, p. 97), and a slightly ionic character which facilitates reduction to the fully anionic form.

The halogen itself contributes to the biological value of the aryl combination. It increases the acidity of the benzene ring by as much as 3 pH units, about the same as the hydroxyl group does. Di-iodotyrosine therefore has a p*K* near neutrality as contrasted with 9·8–10·4 for tyrosine itself. Iodine and bromine cause a dramatic increase in the electrical conductivity of the aromatic molecule (Szent-Gyorgyi, 1960, p. 68), and some atom develops paramagnetic properties (p. 318), losing one of the electrons from some of its orbitals. Charge-transfer has been induced—the movement of single electrons from one part of the molecule to another. This is the hallmark of activating ability in a molecule, and is commonly associated with strong biological activity.

It is less certain why only tyrosine derivatives have been exploited. In fact histidine is quite readily iodinated (p. 185) and the derivative has appreciable thyroid activity. In the laboratory proteins such as casein are readily iodinated and both tyrosine and histidine residues take up the element, so that there must normally be devices restricting the process to tyrosine. In passing it may be noted that this behaviour of histidine indicates another way in which compounds with the analogous ring, oxazolidine, may act as anti-thyroid agents. Halogenated indoles also occur biologically, for instance dibromoindigotin (p. 193), which resembles thyroxin to the extent of having two aromatic nuclei linked. The virtues of the benzene ring, and more specifically of two oxygen-linked tyrosine residues, in the thyroid hormone have been considered in detail earlier (p. 189). The general structure, the substituents on the rings and the side chain all contribute to its value, and it is clear that in principle only one aromatic ring is likely to be ideal. It seems likely that benzene will prove to be that ring.

The halogenated tyrosines are more insoluble than the amino acid itself, and this is one of their virtues as skeletal materials in spongin and gorgonin. In thyroxin and tri-iodothyronine low solubility may prolong the action, and for transport purposes they are made soluble by protein-conjugation. Iodo- and bromotyrosines, but not the chlorine compounds, affect the hydrogen-bonding within the peptide chain as a whole (Robinson and Bott, 1951), and may further increase the strength and insolubility of skeletal proteins in this way. It is also possible that they promote the

formation of bridges by condensation–oxidation reactions of the form

$$R_1I + R_2H \longrightarrow R_1 - R_2 + HI$$

Spongin and gorgonin are collagen-type proteins, containing little of the sulphur amino acids, so that they cannot form —S—S— bridges. Fluorocarbons might be expected to give even tougher compounds but fluorine does not readily form aromatic compounds (Finar, 1959, p. 520).

The nature of the very sensitive colour-reaction of iodine with starch is still uncertain but it is similar to the colour of iodine vapour and some of its organic solutions, much more bathochromic than that of bromine with glycogen. No doubt it contains an important clue in the biology of iodine. The iodine atoms are thought to form a core down the centre of a helix of the polysaccharide.

27.3 Bromine

Biologically bromine may be the least important of the halogens, because chlorine forms the most useful anion and iodine the most useful covalent compounds, and it has not any unique properties as fluorine has. Most properties show a gradient along the series and bromine is outstanding in none. Even so it is more abundant than most elements of its atomic size. Apart from the special and sporadically distributed tyrosine and indole derivatives (pp. 189 ff.), forms of bromine, probably mainly the inorganic anion, occur widely in all organisms. Moreover it has been found to improve the growth of mice and chicks, the latter even when already receiving adequate iodine (Underwood, 1962). It is often concentrated in fruits and some other parts of plants.

No specific biological function has yet been detected and 6:6′-dibromindigotin (tyrian purple (Fig. 28, X)) is its only unique biological compound. This is secreted by the muricid gastropods as a colourless precursor which is converted to the definitive pigment photochemically in the absence of air. By contrast the halogen-free indigotin must be formed from its leucobase by aerobic oxidation and the difference may depend on the redox properties of the bromine. Unfortunately the function of this pigment *in situ* is not yet clear.

Mono- and dibromotyrosines occur alongside the iodine analogues in spongin and gorgonin, and have the same qualitative significance. The bromine analogues of the thyroid principles similarly prove to have the same qualitative action as the natural hormones, and in the event of iodine deficiency there may be more bromine in the thyroid gland than in the blood (Underwood, 1962). However, normally the thyroid gland is the only organ of the body which *lacks* bromine so that it is quite certain that

it is excluded, probably because its thyroxin and other thyronine analogues are deleterious (if only because they would be competing with the more efficient iodo compounds). This is perhaps the clearest instance of biological exclusion of a chemical analogue, producing absolute uniqueness in place of mere quantitative superiority. The use of bromine when iodine is deficient recalls the use of sodium in muscle cells when potassium is deficient (Krogh, 1946).

Although this may not be a normal biological function, bromides are outstanding among the halides for their sedative effect, but even this may be merely due to antagonism of the Cl^- ion (Sexton, 1953, p. 48), which is more mobile, or of the thyroid hormone, in its action on metabolism. There is of course the complication of the anaesthetic action of alkyl halides and the somewhat related hypnotic effect of chloral hydrate; curiously enough bromal hydrate, $CBr_3 \cdot CH(OH)_2$, is a weaker hypnotic than this.

27.4 Fluorine

Reasons why fluorine is not so important biologically as chlorine have been given (p. 431). Chemists in fact usually regard the elements of the second full period, including Cl, Na and Mg, as the typical elements of their groups, so that the real cause for surprise, perhaps, is that biological systems make so much use of the anomalous first members of Groups IV – VI, and hydrogen, the lightest and most anomalous of elements. In general, no doubt, fluorine is too reactive an element, and too extreme in its properties. It is so strongly electrophilic that it will not form compounds with oxygen or with nitrogen, while its affinity for hydrogen is such that the formation of hydrogen fluoride from the elements is exothermic to the extent of 147·5 kcal/mole. These are all possible reasons why it should not be exploited, whether as an alternative to chlorine or in any other capacity. If it is true that iodine has been developed as the redox halogen of biological choice then there is no reason why fluorine, at the other end of the group, should be used for this purpose; it is the most powerful oxidiser of all. It will form the covalent type of organic compound, and in fact the impressive variety and valuable commercial properties of the fluorocarbons are adequate evidence of this, but they are also proof that fluorine is too effective as an analogue of hydrogen itself to be tolerated in this capacity. Fluoroacetate, $FCH_2 \cdot COOH$, is very toxic, not so much as an oxidant, in the manner of iodoacetate, but through forming the fluorine analogue of citric acid, and so blocking the Krebs cycle. The fluorine homologues of the other C–even fatty acids also are toxic, since they yield fluoroacetic on catabolism, but the C–odd homologues are relatively innoxious, since they are metabolised in a different

way: as oxidants of course all should be equally effective per mole of fluorine.

There are thus at least three main ways in which fluorine might prove an antimetabolite, as an analogue of the Cl^- anion, of the redox agent RI, and of H; it is therefore not surprising to find that it inhibits a large number of enzymes, phosphatases, including ATPase, enolase carboxylase, catalase, lipase, fatty acid oxidase, glucosulphatase, hydrogenylase and choline esterase (Fearon, 1949, p. 37; Baldwin, 1953, pp. 112, 375). In most of the cases analysed it appears to be acting as the hydrogen analogue; for instance it inhibits enolase by forming a Mg, F, PO_4 complex with the metal co-factor of the enzyme and inorganic phosphate. A similar action may be anticipated for choline-esterase inhibition since it is known that various fluorophosphonates are anti-cholinesterases. In the case of lipase, Scott (1953) suggests that fluorine dehydrogenates the enzyme, and a similar action might be relevant to hydrogenylase—or the substrate itself might be involved here. Of course the substitution of fluorine for hydrogen is automatically also an oxidation and a distinction must be made between this process and subsequent action of fluorine as an hydrogen analogue. The inhibitor of fatty acid oxidase may be the F^- anion (Underwood, 1962).

In spite of this fluorine does have a positive biological function. Its insoluble salt, calcium fluoride, hardens vertebrate teeth (Jenkins, 1955) and the bones to a lesser degree. Enamel, the hardest layer of the teeth, is most dependent on this salt which occurs naturally as fluorspar, or 'blue john', and has very interesting fluorescent and other properties. The optimal concentration of fluorine in the diet is very narrowly restricted, as might be anticipated, and excess is deleterious for the teeth themselves, causing a characteristic mottling of the enamel, and a more porous, softer texture; this is found also in the bones. There is calcium fluoride in the otocyst of Mysids and it may have a wide use in skeletal materials. Certainly it occurs widely in plants and animals, between 2 and 60 ppm, compared with 1·4 ppm in sea water and as little as 0·03 ppm in fresh waters. The body wall of *Archidoris* contains as much as 0·57 per cent, about the same amount as in vertebrate teeth (Krogh, 1938, p. 58). The association of calcium fluoride with carboniferous rocks may indicate that the salt was present in the organisms which laid down the rock.

It is readily understood that only an insoluble salt of this kind could be tolerated in the body. At the same time it is clear why the F^- anion could not replace Cl^- in the body fluids; it would be deposited as calcium fluoride, simultaneously depleting the body of soluble calcium. There are also other anomalous and biologically undesirable solubility properties of fluorine compounds. For instance the silver salt, silver fluoride, is

soluble, so that in the presence of fluorine silver and perhaps other metals normally innocuous would become dangerous. Further, the soluble fluorides tend to form complex salts such as KF_2H and $AlF_3 \cdot 3NaF$ (cryolite) which would be deleterious in physiological fluids; they depend on the ability of fluorine to polymerise, even in the form of fluoride, with a full valency shell of electrons.

The biological danger of the fluorocarbons is illustrated by *teflon*, a polymerised tetrafluoroethylene, which is so stable that it resists boiling in *aqua regia*, which is a mixture of concentrated hydrochloric and nitric acids and is the only acid reagent which will dissolve the royal metal, gold. The hydrocarbons depend for biological value on their relative stability, but this is of an entirely different order! Commercially, fluorocarbons promise to 'revolutionise' the wearing properties of materials, but this is not compatible with biological lability.

Fluorine therefore is an extremely interesting but difficult element, biologically. As a light, reactive element the body is exposed to it far more than to the heavy toxic elements, and the control of fluorine in the body is a particularly important aspect of metabolism, practically and theoretically.

27.5 Conclusions

The halogens are of course unique chemically as the most electrophilic, non-metallic of the elements, even iodine being still a true non-metal. It is for this reason, no doubt, that all four of them have biological significance, whereas in most other groups and sub-groups only the first or first two members have been exploited. Because their properties differ only quantitatively, however, they can deputise for each other more than in any other sub-group. For this reason it has been necessary to evolve a biological method for preventing bromine from competing with iodine in the thyroid.

Although potentially so useful the halogens have been used only for restricted, specialist purposes. Their properties are too extreme for exploitation as major biological elements. Monovalency no doubt is one of these, and their proximity to the stable state of the inert gases another.

CHAPTER 28

SILICON AND BORON

THESE two non-metals are the remaining two elements known to have any very appreciable biological significance. They may be taken together because of a general similarity in properties—a diagonal affinity like that already seen between Be and Al and between Ca and Ga (p. 435). They are biologically important more for what they might have been than for what they actually do. Silicon is the next higher analogue of carbon (p. 37), and both it and boron have much of the versatility of carbon, but their properties differ in essentials which rule out both as serious biological alternatives to carbon or even as major synergists with it. In this light their functions in our actual, carbon-based organisms seem trivial in the extreme.

28.1 Silicon

There is no good evidence that silicon is indispensable for the metabolism or the growth of animals or plants (Underwood, 1962, pp. 351–352). It is difficult to believe that grasses or sponges could be normal without silicon as skeletal material, yet maize, which normally has 33 per cent of silicon in its ash, grows quite well in its absence. The siliceous skeleton of the hexactinellid and tetraxonid sponges is a major fraction of their weight and as much as 4 per cent of the dry weight of grasses is silicon (30–40 per cent of the ash). There is as much as 77 per cent of silica in the ash of feathers, and 300 ppm in hair. Siliceous skeletons are present also in the Radiolaria, the silicoflagellates and the diatoms. There is 350 ppm in mammalian connective tissue, again presumably as a skeletal material. It is always present as silica, SiO_2, or as its hydrates, effectively silicic acid derivatives. The latter are analogous to the carbonates, but are harder, and transparent. Silica itself, in its usual form, is a transparent solid, in great contrast to its analogue, carbon dioxide (p. 35). For aquatic animals another advantage of silicates over carbonates is that they are more easily deposited from solution at low temperatures and high pressures, in high latitudes and in the ocean deeps. In grasses silica is used mainly

to give the sharp, serrated edges to the leaves, and here the transparency is a valuable property, since it does not interfere with photosynthesis (Fearon, 1949, pp. 28–29). Silica is harder than apatite and even than calcium fluoride and the wear on the teeth of grazing mammals therefore demands persistently growing types.

There is rarely more than 1·5 mg of silica per l. in natural waters, owing to its low solubility, but sponges are able to extract the element from concentrations down to 0·4 mg/l. Millions of square miles of the ocean floor are now covered with the skeletons of diatoms, and a considerable area with those of the Radiolaria, a total of 3–4 per cent of the sea-bed of the world, and to a great thickness. Freshwater sponges store the mineral in granular form in the young silicoblasts and remobilise it when spicule-formation begins (Jørgensen, 1947). The metabolic handling of silica is an interesting problem: in feathers it is associated with sterols and conceivably may be manipulated in combination with them, or in lipid solution.

The silicates of Cu, Mg and Fe occur in the seeds of cereals, associated with starch (Fearon, 1949), though there is no indication of their significance, except as a store of silica for the young seedling. It is also sometimes firmly bound to cellulose (Underwood, 1962), and its hydroxyl groups therefore may condense with those of the saccharides, as those of phosphate, sulphate, and borate do. It may even be bound to the living fabric (Firsoff, 1962), in which case it may not be such a trivial, inorganic element after all. There is not at present much evidence for organic combination, though silicosis of the lung does show some resemblance to the effect of chemical poisons. However, the silica remains localised in the lung, indicating that the trouble is essentially mechanical.

It is worth studying the properties of silicon in some detail for the reason already given. It is a unique element whatever its biological limitations. The free element is a hard solid, not very strongly reactive but obviously active enough since it never occurs free in nature, whereas silica and the silicates are the most abundant compounds on Earth. Some of the crystalline forms of the element are inert to all reagents except the halogens, so that the Si—Si bond is relatively stable. Again silicon carbide is almost as hard as diamond so that the Si—C bond also is very strong. Silicon can polymerise as carbon does, forming linear chains of silicon atoms with hydrogen holding all spare bonds.

The resulting *silanes*, however, by contrast to the alkanes, are very unstable and could not be the basis of a living fabric. They are very strong reducing agents, in fact, like some of the metal hydrides. The reason is that silicon is more strongly electron-donating than carbon because the atom is larger and the four valency electrons farther from the nucleus; it

will therefore donate one to hydrogen, increasing its reducing power, but the more strongly electrophilic halogens and oxygen rapidly displace hydrogen, forming very stable compounds. Because of the abundance of oxygen on earth the silicates are the most common, and there is also another reason for this. Even sodium hydroxide, which has no effect on the paraffins, will decompose the hydrosilicons:

$$SiH_4 + NaOH \xrightarrow{2H_2O} NaSiO_3 + 4H_2$$

(Wald, 1962). The reason is partly the specifically high affinity between Si and O, the heat of formation of SiO_2 from solid Si and gaseous O_2 being + 191 kcal/mole; it may be more than coincidental that they are the two most common elements on earth, largely in combination. Because of the larger cloud of non-bonding electrons, the Si—Si bond is much weaker than C—C (Table 39), and oxygen therefore is able to insert itself between

Table 39. Variation in Bond-length and Bond-energy with Chemical Group and with Period, among light Non-metals

When X=	First main period				Second main period			
	C	N	O	F	Si	P	S	Cl
Length of bond (Å)								
X—X	1·54	1·48	1·48	1·44	2·34	2·20	2·08	2·00
X—H	1·08	1·00	0·96	0·93	1·50	1·42	1·34	1·28
X—C	1·54	1·47	1·43	1·38	1·88	1·87	1·81	1·77
Energy of bond (kcal/mole)								
X—X	81·5	38·4	33·2	36·6	42·2	51·3	50·9	58·0
X—H	98·8	93·4	110·6	147·5	70·4	76·4	81·1	103·0
X·····H (H-bond)	1+	2–4	3–7	7–10			slight	slight
X—C	81·5	72·8	98·8	121·0	72	77	65	81

the silicon atoms, forming chains of the form —Si—O—Si—O—Si—, and there is no possibility of a silane-based bioplasm.

The chemistry of silicon therefore is largely the chemistry of its oxides, whereas that of carbon is largely that of more reduced compounds. It might be argued that the most stable form of carbon also is the oxide, CO_2, since this is the product of burning and respiration, the hydrocarbons being an end-product of the peculiar activity of living systems, and not the natural state of carbon. This begs the most fundamental question, however, namely: is organic chemistry the outcome of a special phenomenon, life, or conversely is life the outcome of a perfectly spontaneous

synthesis of carbon compounds? Almost certainly the second alternative is the correct one and there is a fundamental difference between carbon and silicon, closely similar to that between nitrogen and phosphorus (p. 401); the higher analogue donates electrons more readily than it accepts them.

Another important difference between carbon and silicon is that the latter does not readily form double bonds, a property which is confined to the lightest elements and is one reason why these are the major biological elements (Wald, 1962). Outside the first full period only sulphur and phosphorus form double bonds at all readily, and they are included among the major elements. The reason again is bound up with atomic size: the larger the atom the greater the repulsion between its non-bonding electrons and those of another atom, and the two form one bond at most. Sulphur and phosphorus, although heavier atoms than silicon, are *smaller* because the increasing number of electrons in their outermost shell compresses the atom as a whole; silicon has the largest atom in its period. Silicon therefore cannot form O=Si=O, corresponding to O=C=O, and silica can only exist in the polymerised *siloxane* form (Fig. 52, I). This has the empirical formula SiO_2 but each silicon atom forms four single bonds. It will be appreciated that the four bonds are directed towards the corners of an imaginary tetrahedron, like those of carbon, so that siloxane is an indefinitely extensible crystalline form and crystals of *quartz*, the most common natural form of silica, have been found up to a ton in weight! Even this is not the size-record for natural crystals; but this is held by silicates, having essentially this same structure; crystals of *muscovite*, an aluminium silicate, may reach 2 tons (Jones, 1943), and one of *spodumene*, $LiAlSi_2O_6$, had a weight greater than 40 tons. This is magnificent but it is not alive.

The silicates in fact must take this form (Fig. 52, II), since otherwise there would be a double bond in the molecule, H_2SiO_3, as in carbonic acid. The free monomer probably never occurs, in fact, and the hard crystalline form of most oxygen compounds of silicon is explained. It might be anticipated that the acid $(HO)_4$⋮Si would be the common form, but this is unstable, like the analogous acid of most non-metals. It will be noticed that in the silicates the atoms of the monovalent metals seal off bonds in the crystal lattice and give a weaker structure than silica; the alkali silicates in fact are water-soluble ('water-glass').

Such a regular crystalline form as quartz, composed entirely of two elements with such high affinity for each other as silicon and oxygen, is inevitably dense (sp. gr. 2·67), inert and so hard that it will cut glass. It is so rigid as to generate an electrical charge if compressed mechanically (piezoelectric response) and reciprocally it oscillates if subjected to an

electrical charge. A mechano–electrical reciprocal transducer of this kind is very valuable technically. Quartz oscillates at a very constant rate of 10^6 per second which makes the quartz clock an extremely accurate measure of very brief time intervals.

Although the individual silicon atoms are immediately surrounded by four identical groups, the oxygen atoms, crystals of quartz are optically

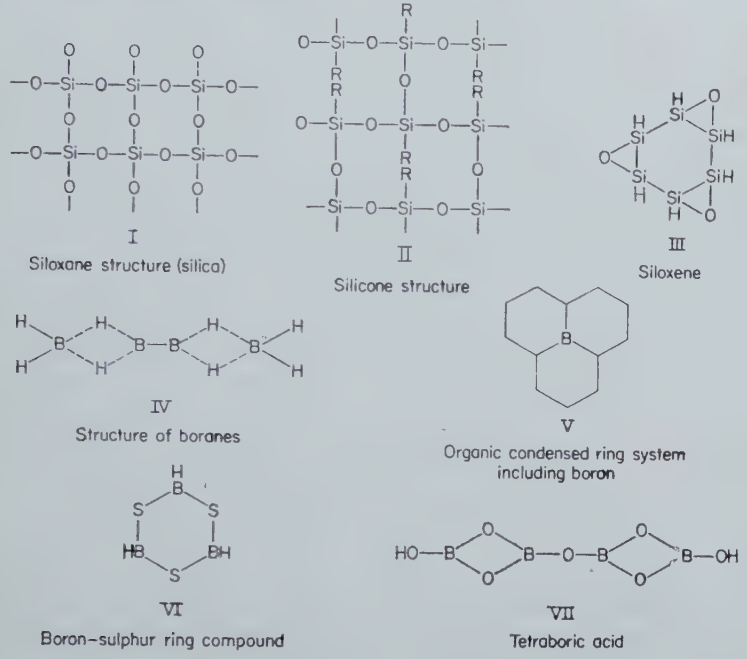


FIG. 52 Silicon and Boron compounds of outstanding interest.

active, and of two optically pure forms, *laevo-* and *dextro*-rotary. The crystals have a corresponding geometrical mirror-image asymmetry, like Pasteur's classical tartrate crystals. It is therefore probably correct to regard the crystal as a continuum in which the relevant four groups on each silicon atom are the chains of atoms indefinitely extended to the limits of the crystal, and therefore are all four different on most of the silicon atoms.

There are forms of silica, such as *opal*, which are more amorphous, due to 3–12 per cent of water entering and disturbing the crystal lattice. Coloured forms, *jasper* and *chalcedony*, are produced by molecules of metallic oxides entering the lattice in the same way. Distortion of the lattice in certain definite ways, by metal oxides in particular, facilitates the formation of glass, which is a supercooled siloxane, produced by melting

together silica and metal oxides, effectively a silicate. The metal oxides act as plasticisers, lowering the melting point because the resulting lattice is less regular. Al_2O_3 and B_2O_3 form tetrahedral structures like those of the SiO_4 group, but they are smaller and so distort the lattice in the necessary way. It will be noted that boron here behaves as a metal.

As a supercooled liquid glass is non-crystalline and can be cast or blown in any form. It has the special virtues among such materials of being transparent and of being reversibly thermoplastic almost indefinitely, reverting to a crystalline form (devitrification) very slowly. It has surprising strength and softens only at a fairly high temperature; it remains the world's most valuable thermoplastic. Recent research is producing an increasing range of carbon-based plastics of remarkable characteristics but none yet excels glass for its combination of properties: chemical resistance, inertness, transparency, toughness, thermoplasticity, etc. Its one drawback is brittleness. Traces of the oxides of the heavy metals confer brilliant colours without destroying the transparency, and this has been one of the glories of the glassmaking industry. The ease with which other atoms are introduced into the molecular lattice recalls apatite (p. 412).

The lenses of the eyes of animals may approach glass in transparency but they are inferior in almost every other respect. Toffee, formed from sugar and other organic ingredients, lacks the colourless transparency, the inertness, etc. Some of the commercial carbon plastics are transparent and colourless but not so hard, durable or inert.

The common 'soft' or soda-glass is a mixed sodium–calcium silicate, while Jena or hard glass is potassium–calcium silicate. Flint glass is the potassium–lead salt, which has lustre and a high refractive index. The inclusion of borax gives glass of the Pyrex type, having a low coefficient of expansion and consequently very resistant to cracking under rapid differential heating or cooling. Quartz itself forms a glass of low coefficient of expansion, as well as with other useful properties, but is not so easily worked as the silicate glasses. The analogous germanium dioxide as well as P_2O_5 and BeF_2 can form glasses but the silica glasses are far outstanding in their combination of useful properties (Lowenstein, 1948).

The general stability, rigidity and covalent bonding of the siloxane lattice makes silica and the silicates good electrical and heat insulators. The typical *micas*, silicates of H, K and Al, are particularly useful as electrical condensers, and in other ways, since they crystallize in layers and split into thin sheets, which can then be stacked with air layers in between for additional insulation. They are transparent and so make suitable doors for lamps and stoves. *Asbestos* (p. 426) also is a valuable insulating silicate, since the fibrous texture traps air, making a lighter and much better insulator than compact silicates. Another interesting natural

material of this group is *vermiculite*, a fibrous mica, modified by natural hydrothermal solutions. When heated it 'exfoliates', or swells, to ten times its initial volume, undergoing lifelike writhing movements and becoming as light as 10 lb/ft^3. It is used for insulation against heat, electricity and sound—as well as for reducing piston-slap in the internal combustion engine. These materials, with their tendency to form linear polymers, show some slight resemblance to carbon polymers.

Because the oxygen atom has only two bonds there is some plasticity and elasticity even in the rigid framework of crystalline silica, but with monovalent metals acting as breakers of some of the bars of the lattice (Fig. 52, II), the pliability is increased, as in the glasses. If monovalent organic groups are introduced instead, the lability is further increased, and a range of technically useful materials, the *silicones*, is based on this. Some are greases and make valuable, non-inflammable lubricants, stable at high temperature, superior to the lipid greases in these respects, yet equally water-repellent. Their main drawback is a tendency to creep indefinitely over all surfaces; this is in sharp contrast to the aggregative tendency of carbon greases and other biological materials, a tendency essential to living systems. It is an exaggeration of the tendency to form monolayers (p. 81).

Another group of silicones have rubbery properties, but these are really due to carbon radicals acting as cross-linking agents between linear polymers of siloxane proper, and could be regarded as complex carbon rubbers. However, the silica component is manifested in other properties, and some are stable at very low temperatures, e.g. phenyl substituents down to −80°C; others are stable at high temperatures, up to 200°C or more (Thorndike, 1958). The latter therefore can be sterilised repeatedly without the rapid deterioration of pure hydrocarbon rubbers. They are also much more inert to ozone and less liable to stick. They are much used for shock absorbers, gaskets and seals, but they are not superior to hydrocarbon rubbers for all purposes. It may be significant that silica gel is used as a catalyst in making butadienes, synthetic hydrocarbon rubbers, from alcohol. One silicone rubber is known as bouncing putty since it flows plastically yet bounces if dropped, both plastic and elastic in properties, therefore, and unequalled by any biological material. Moreover if struck violently it shatters like glass and is too brittle to be drawn rapidly; if drawn slowly, however, it is as ductile as a metal: evidently it is strongly dilatant.

Some silicones are used as high-temperature fluid baths and for other hydraulic purposes and others as anti-foam agents, because of their high surface tension. Some are resinous in properties and can be used for ion-exchange purposes. They are perhaps not so versatile as carbon-based

ion-exchange resins but they add to the galaxy of siloxane derivatives. Much of the variety of course is peculiar to the silicones and therefore largely to the carbon component and there is a parallel to this in the fluorocarbons (p. 462). They emphasise the uniqueness of carbon rather than that of the other component.

As ion-exchange materials the simpler siloxanes, the silicates themselves, are in fact more useful than the silicones. The best are clays, finely divided aluminium silicates, carrying a charge due to previous exchanges between ions of different valency. The *zeolites*, which carry a negative charge, are particularly useful in industry and research. Clays may be important in this way to assist plant nutrition, though the mechanical properties of clay carry certain disadvantages and need to be leavened by coarser materials. It has been suggested that life may have originated on the surface of such ionically active material as clays (Bernal, 1951) but without incorporating any of it into the bioplasm. With clay as catalyst the groups methyl, phosphate, acetyl and peptide can be exchanged without loss of bond energy (McElroy, 1954). Conceivably, also, nucleic bases might be condensed with ribose and phosphate by this means.

The smaller of the charged particles of clay form colloidal solutions and in fact siloxanes in general do this rather readily. Some of the silicones even form emulsions. The silicic acids, themselves, with H^+ in place of metal ions, resemble biological macromolecules in forming the hydrophil type of colloid (p. 226). Metasilicic acid $(H_2SiO_3)_n$ gelates at a concentration of 14 per cent; this is higher than the minimal concentration in most protein colloids but very few other inorganic colloids gelate at all. Polyvalent ions such as Ca^{++} and $PO_4^{\equiv}$ precipitate silica sols but the monovalent H, OH, and Na ions improve their stability, as in the case of the protein sols.

Partially oxidised silicon hydrides can form a hexacyclic ring, *siloxene* (Fig. 52, III), $Si_6H_6O_3$, but this also is unstable, so that the siloxane structure seems to be the only one which is really stable. Siloxene is outstanding as one of the few inorganic substances which is chemiluminescent. One other important difference from carbon is that silicon can show a higher valency than four, like other members of the second main period, and in fact it is more stable as hexavalent than as tetravalent silicon (Edsall and Wyman, 1958, p. 24). The six bonds will be directed mutually at right angles, producing a cubical lattice, more rigid than the tetrahedral. If the six are used in siloxane, then the representation shown in Fig. 52, I and II, will be the actual structure, as seen in one plane.

Silicon shows some family resemblance to carbon, therefore, but differs so sharply in so many of the properties which are most important biologically. It never can have been a serious alternative as a basis for life

reasonably similar to that which we know. It has a certain versatility of properties, even apart from its silicone compounds with carbon itself, and its compounds are outstanding in their own ways, but it seems doubtful if they could have been the basis for any type of life, even remotely different from the existing carbon-based life. Its compounds are too rigid and water-insoluble, and their variety is trivial by comparison with the carbon compounds. In fact silicon admirably emphasises the uniqueness of carbon.

28.1.1 Germanium

This is the next higher analogue of silicon in the main sub-group of Group IV, and it is found in traces in living organisms, as well as abundantly in some coals (p. 454). If it is indeed a secondary enrichment here (Pirie, 1957; Hallam and Payne, 1958) then there is no strong indication that it has biological significance, since most elements occur in traces in living tissues. The glass-forming property of germanium dioxide (p. 470) indicates similarity to silicon, as would be expected, and in that case it is not likely to have any great or unique biological value.

28.2 Boron

In spite of its general chemical resemblance to Si, boron is used biologically for a quite different purpose, in much the same way as Cl and I, O and S, and N and P. The difference is greater than that between Na and K or between Fe and Cu, where the same general field of function is shared between the pair. The discovery that traces of B were essential for the growth of plants (Warrington, 1940; Brenchley and Warrington, 1927) was one of the most dramatic and historical in research on trace elements. Almost equally surprising is the lack of evidence that it is essential for animals (Underwood, 1962), which have much less in their tissues. It is toxic in amounts above 10 ppm and such quantities begin to retard the growth of plants, in fact. In spite of this, however, there are Crustacea and fish living in the borax lakes of Pugh Creek and Lidakh in Kashmir! Dicotyledonous plants require more boron than the monocotyledonous, so that there may be a good deal of taxonomic variation in response, and this probably implies a rather specialised, parochial function.

Its mode of action in plants is not certain but in the form of the borate anion it seems to be concerned with monose mobilisation and transport, since sugar accumulates in boron-deficient plants (O'Kelly, 1959). It is necessary for the growth of pollen tubes, which depends on sugar-absorption. Borates form complexes with sugars, and perhaps have assumed a function elsewhere performed by phosphorylation; this might

permit the use of sugars for special purposes under conditions which demand the suppression of ordinary phosphorylative respiration. It is also perhaps significant that some bacteria and fungi appear to make no use of phosphorus in carbohydrate metabolism (Sexton, 1953, p. 56). Common 'borax', sodium tetraborate or pyroborate, is somewhat analogous to pyrophosphate. The suggestion therefore is that boron is acting as an analogue of phosphorus and not of silicon. Moreover it may be dividing a common field with phosphorus rather as iron and copper do theirs.

There is some indication of the actual division of labour. Boron affects only uptake and anabolism, and does not promote the respiratory catabolism of glucose at all; in fact it inhibits the uptake of oxygen by brain-tissue, which is by no means a minor aspect of its toxicity when in excess, and probably an adequate reason why it has no useful function in animals. Glucose itself protects against this action of boron (Trauter and Messer, 1953), and it is perhaps significant that the dubious use of borax in internal medicine was always in conjunction with honey! Borax inhibits the inductive synthesis of the enzyme β–galactosidase (Spiegelman, 1950), and conceivably this may be related to its inhibition of galactose catabolism by the enzyme, though from what is known of the mechanism of enzyme-induction a more plausible action would be to block the action of galactose as the inducer of its enzyme. There is some evidence of a compound, *borocitrin*, in some micro-organisms (Alexander, 1948, p. 119), which might play a positive role in respiration, and this would be plausible if phosphorus is not used at all by these organisms, but more evidence is needed.

Borate is also a co-factor for the synthesis of inositol (p. 73), which moreover commonly forms a hexaphosphate *in vivo* (p. 74). This requirement for boron could be one reason why inositol is a vitamin for animals, and is not synthesised by them. There is also some accumulation of free amino acids in boron-deficient plants, but at present there is no knowledge that boron controls their transport and mobilisation, also. This might follow from an increased water transport due to the osmotic effect of the sugar, or in some other entailed manner. Boron is not necessary for protein-synthesis in animals, and probably not in plants either, since there is sometimes increased cell-proliferation in boron-deficient plants. From this it also seems probable that boron mainly controls cell-growth through the osmotic effect of sugars in the cell-vacuole.

Like silicon, boron has a number of interesting and potentially useful properties which have not been exploited biologically. There is perhaps a smack of the policy of relegating rivals to junior positions, a policy which is for the best in this case where the rivals are very inferior in calibre and

could only be disruptive elements. The element boron exists in both crystalline and amorphous forms, as carbon and silicon do. The crystals are transparent, and either colourless like diamond or brown like silicon. They are hard, like diamond, and very resistant to heat and acids, though soluble in fused alkali. Moreover, they invariably contain up to 4 per cent of carbon and up to 7 per cent of aluminium (Partington, 1933, p. 720), so that the crystals may not be elemental boron. The carbide, BC, and nitride, BN, are also very hard, like SiC; in fact BN will cut diamond! Ferroboron is used to toughen steel.

Boron is almost unique among elements in reacting directly with nitrogen, in forming the nitride. It probably has the greatest affinity of any element also for oxygen; at least it will take oxygen from the oxides of silicon and carbon. From nitric oxide, however, it takes the nitrogen; this is not surprising, perhaps, since boron and nitrogen have precisely complementary valencies. The free element is also unique in dissolving directly in water, to give a colloidal solution. The particles of this solute form crystal-like bodies which grow in spiral fashion (Simons, 1953) like some true crystals under special conditions. A trivalent element such as boron inevitably forms asymmetrical compounds and like asymmetrical molecules of carbon compounds these must form a spiral structure if they polymerise. This may be the reason why even the 'glass' formed by fusing amorphous boron shows 'conchoidal' fracture, i.e. cleaves along curved planes.

Boron forms boranes, corresponding to the silanes and methanes, and polymers as high as $B_{10}H_{14}$ have been prepared. However they tend to have more hydrogen than would be expected of the linear hydride of a trivalent element and in fact alternate boron atoms are linked not directly but via a hydrogen atom (Fig. 52, IV), by a kind of hydrogen bond known as the three-centre bond (Gerard, 1961). This cannot be strong—less than one-tenth of the strength of covalent bonds, so that large polymers are out of the question. For comparison polyethylene can build chains of 1000 carbon atoms, or more, at −80°C (Robertson, 1947). The boranes are more unstable in the presence of water than the silanes, and contrast strongly with the stability of the methanes and with the curious 'negative' hydrates formed by the latter (Klotz, 1962).

Boron forms compounds with carbon, sulphur and other elements, as well as with nitrogen and oxygen, and collectively its compounds have something of the variety of carbon compounds. It is more catholic than silicon, as might be expected of a first-period element. It forms the greatest number of compounds with carbon, again as might be expected. Chains of the type —B—C—B—C— seem to be the most stable and it would seem that boron bonds with its identity less firmly than with other

elements, even including monovalents, such as hydrogen. This is not surprising for a Group III element with a nucleophilic valency of three. The chains form rings, of which as usual the hexacyclic is the most common. Among its ring compounds are unusual ones, such as Fig. 52, V, where boron is the atom linking three condensed rings, otherwise entirely of carbon. It forms similar cyclic compounds with sulphur and phosphorus (Fig. 52, VI).

Like all elements from beryllium to nitrogen inclusive in the first period, and all from aluminium onwards in the second, boron combines very readily with oxygen, and derivatives of its oxyacids are the commonest forms of boron in nature. The most abundant, orthoboric acid $(HO)_3{:}B$, is tribasic, like phosphoric, but all three acid groups ionise on the alkaline side of neutrality. It is a weak acid, therefore, and provides a useful buffer system over the range pH 7·8 to 10·0, provided its toxicity is not a bar. The familiar borax is the sodium salt of tetraboric acid, an interesting polymer (Fig. 52, VII). This structure differs from that of the boranes in that no two boron atoms are directly linked; it resembles the silicates and the pyrophosphates in the oxygen-links.

It is therefore not surprising that borax forms glasses both in combination with silica (p. 470) and alone. Moreover metal oxides dissolve in it to give coloured glasses; these are used as a test for some metals since borax fuses at a fairly low temperature. Borax is used for glazes and heat-resisting stove enamels. It also dissolves casein to form a suitable glaze for paper; solubility presumably depends mainly on the phosphate groups and the hydroxy-amino acids of casein. Borax is also used for a variety of other commercial purposes, resembling those of the silicates in variety; it is used as a flux for metals in brazing, welding and refining, and as a refractory material, melting only at very high temperatures.

Even so, boron is no more eligible candidate than silicon for the 'lead' in the play of life, its hydrogen compounds being too unstable and too limited in number and molecular size. It forms only low polymers with oxygen, and most of its interesting compounds are strictly carbon compounds. Like silicon it serves mainly to emphasise the uniqueness of carbon.

28.3 Conclusions

Silicon and boron are appropriate members with which to conclude a survey of the elements in relation to biological uniqueness. In the first place they are most nearly related to carbon itself and secondly they illustrate the nature of biological uniqueness as well as any elements. They are poor mimics of carbon in all properties which make that element the

primary element of biological material, but *in vivo* they do not act even as poor mimics; their one biological function depends on a property peculiar to each, and overlapping that of no other. This is equally true for other elements of which the action is adequately known, and will probably prove true for all. A large number of elements perform some one or more useful functions in living systems and each is the most efficient for the purpose. Very often second- and third-best elements exist, but they are not used *in vivo*, though they may be used for some other property peculiar to them. The innate, often merely outstanding, properties of each biological element have become the basis of a function which is now absolutely unique. Nature has become more black and white than she was originally.

The most probable explanation of this is that it is the result of evolution by selection. Elements useful for living, i.e. self-perpetuating, systems have been selected for that use, but at the same time that use, and the system as a whole, have been reciprocally selected to 'fit' each element so that its efficiency relative to that of all rivals has been enhanced. To discover more of this from the only evidence available, namely organisms still living, it is necessary to study the collective functioning of all biological materials, that is to say metabolism and the control of metabolism. This means the study of organization, at progressively higher levels, up to the systemic and genetic. In detail this is beyond the present terms of reference but it is necessary to look at the question briefly before concluding, and the next chapter will be concerned with uniqueness at higher grades of organisation than the molecules so far considered.

CHAPTER 29

HIGHER GRADES OF ORGANISATION

THE aim of this chapter is to see how living bioplasm is constructed from the simpler molecular components already considered, to what extent construction is a spontaneous consequence of the properties of these components and to what extent, on the other hand, it depends on collective controls, culminating in controls at the systemic level, and not amenable to analysis by the present approach. Between the two there seems at present a great gulf; as Brucke said (Thompson, 1942, p. 290): ". . . in addition to the molecular structure of the organic compounds of the cell there is another and in different fashion complicated structure which we call organisation". It is probably only a gulf of ignorance, but this is little comfort at present.

It is a well-accepted fact that new properties 'emerge' at successively higher levels of organisation, the spontaneous outcome of the association between the components, and there is every reason to expect that these will be at least as unique as those of the individual components. The omniscient could perhaps explain all 'emergence' in terms of the properties of the components of the next lower level but at present an emergent is defined as the type of higher-level property which is not predictable merely from a knowledge of the components. It may appear as unrelated to these as the hypothetical systemic control, and the latter should not be invoked unnecessarily.

It has been seen that carbon compounds in general fit into the tetrahedral structure formed by water molecules, and that the most important and active of biological materials, the macromolecules of protein and nucleic acid, do so adequately to form a functional continuum (Eley, 1962; Klotz, 1962). Most other actively functioning materials are associated with protein and so are structurally organised into this continuum. Apart from storage materials the only components outside it are the physiological ions, in part, the monoses and other small substrate molecules, and building units such as the amino acids and nucleotides. Skeletal materials are usually associated with protein and actually deposited on a protein foundation (Needham, 1964). The lipids and other constituents

of the membranes around cells and organelles similarly are organised in relation to protein. It is therefore probable that what is already known of the structure and properties of proteins, together with a study of emergence in larger masses of protein, should go a long way towards explaining bioplasm, the complete structural system. Increase in size in multicellular organisms is largely a matter of replicating cells and at intermediate levels also there is no doubt a good deal of mere replication. Muscles and tendons have the same essential structure and properties at all levels of size, apart from features automatically contingent on size; at lower levels, also, there is much simple increase in bulk, for instance in the endoplasmic membrane system of the cell, the number and size of mitochondria, and the growth of the myelin sheath of nerves. The amount of qualitative emergence between the molecular and the cell level is not so formidable as may sometimes appear, and it is feasible to attempt to estimate it.

29.1 Membranes of Cells and Organelles

These all appear to be based on the 'unit membrane' already described at the molecular level (p. 83), of total thickness 70–80 Å, consisting of a double layer of lipid molecules orientated normally to the plane, and of total thickness 35 Å, sandwiched between single layers of tangentially orientated protein molecules. Intracellular membranes consist of two of these unit membranes; around the mitochondria the inner of the two is folded into the interior as the cristae, which collectively divide up the cavity into spaces each little wider than the membranes themselves. The whole structure therefore is a system of interfaces, with no material more than molecular distance from an organised membrane, and probably under its permanent control, structurally and functionally. This could follow simply from the intrinsic properties of the system. Chloroplasts and some other organelles have a similar structure. In some cells the whole cytoplasm comes to have a rather similar, essentially membraneous, structure, the endoplasmic reticulum; a double membrane-fold permeates the ground cytoplasm proper, by repeated folding or coiling, and the compartments of the ground cytoplasm may be reduced to spaces no wider than those between the two units of the double membrane. In this case the cell interior is one vast system of double membranes, for the structure of the golgi body is essentially similar.

It is now possible to understand the classical observations of Moore (1935), therefore; he found that if Mycetozoa are forced through a fine sieve their structure is disorganised and they die, but if allowed to flow through by their own locomotion they will pass unharmed through a much finer mesh. At that time, when the cytoplasm was believed to be a

single fluid cavity, it was necessary to postulate that some mysterious, invisible structure was disorganised, but the intracytoplasmic membrane basis for this now seems certain (Mercer, 1959b). In Moore's results there is an amusing superficial analogy to the shattering of bouncing putty (p. 471). It would be useful to know how the membrane system is disorganised by sieving and by other agents, and what normally maintains the precise spacing and integrity of the membranes. In all probability they are polarised, like the cell-membrane itself, and folding brings similarly charged surfaces into apposition, so that they are spaced by mutual repulsion, depending on intrinsic properties of the membranes.

Rapidly growing cells, and some mature cells, do not have this extreme sub-division of the cytoplasm (Mercer, 1959b), and it is particularly well developed in gland cells and others having business mainly with their outside worlds. In the simpler cells, therefore, the classical picture of the cytoplasm may still be relevant, structurally organised by the water-protein molecular continuum, and functionally organised by semiconduction (Eley, 1962; Klotz, 1962). These are both intrinsic properties of the material.

Biochemists have for some time felt convinced (Needham, 1936; Peters, 1937) that the control of biochemical activity demanded an organised sub-micro structure, a cytoskeleton, and there is now adequate basis for this in principle, in the ground cytoplasm itself and in the intracytoplasmic membranes. The cristae of the mitochondria and their counterparts in chloroplasts may be regarded as belonging to the latter, and they have been attributed a key function in organising respiration (Green, 1962) and photosynthesis (Calvin, 1962) respectively. A regular spatial deployment of enzymes seems essential, and the closely stacked membranes are ideal for their functioning; what is completely uncertain at present is whether the logical deployment could be a spontaneous process, in evolution. This is perhaps one of the clearest test-cases of the need for an organisation dictated more arbitrarily from higher levels, what we may call rather vaguely systemic control.

A component of the electrical behaviour of membranes may be intrinsic; most, if not all, cells have a standing potential across their surface rather inaptly, perhaps, called the resting potential; it may be as great as 100 mV, the outside usually positive to the inner surface. In some cells it may be an incidental, spontaneous consequence of structure or activity, but in nerve and other excitable cells it has become the basis of their essential function. The surface (unit) membrane may be asymmetrical about its imaginary middle lamella, if the two protein layers differ significantly, but a double membrane formed by the folding of a unit membrane is likely to be quite symmetrical about the space between the two units; any net charge

across such membranes therefore must depend on differences between the bulk phases on the two sides. In fact the Na/K differential (p. 419) across the cell-membrane also is not due to structural asymmetry but is maintained by active processes (sodium pump), requiring respiratory energy. This again may be a systemically dictated control, therefore, in the sense that it does not follow solely from the intrinsic properties of the membrane and its neighbouring media. By contrast, simple asymmetrical membranes, such as monolayers of lipid (Beament, 1961), have a large intrinsic potential and will continue to transfer ions of either charge in the appropriate direction so long as these are not allowed to accumulate on the far side to the level of electrochemical equilibrium. In the insect exoskeleton, and elsewhere perhaps, asymmetrical membranes are deposited and may function in this spontaneous way; equally important, they may be formed in the first place by spontaneous processes.

The maintenance of a potential across the more symmetrical type of membrane could be considered spontaneous if it were shown that respiration is spontaneous. There is no doubt that active living cells are continually respiring and that few can survive the inhibition of all respiration, aerobic and anaerobic, but this only proves that respiration is essential for active life, not that it is an intrinsic property of the system, as assimilation, for instance, has been shown an intrinsic property of coacervate droplets (p. 233). In this connection we may digress for a moment to note that a spontaneously assimilating sphere, such as a living cell, will spontaneously divide when it reaches a mechanically critical size (Rashevsky, 1960), and that the steady-state of assimilation and proliferation may continue indefinitely so long as raw materials are available. Other cells may continuously secrete or do other useful work, as an alternative to growth and proliferation.

If the maintenance of a standing potential *is* spontaneous, then it is equally probable that its restoration after disturbance or 'stimulation' also is spontaneous. Less certain is the spontaneity of the propagation of any disturbance of this kind, as in nerve-conduction, the wave of potential change which sweeps over the surface of the echinoderm egg at fertilisation, and some other cases. Nerve-conduction is decrementless or 'regenerative', and this could be fully spontaneous only if it were 100 per cent efficient; such efficiency in fact is unknown above the electronic level, whereas nerve-conduction is essentially an ionic phenomenon. It would therefore seem that a special, independent boosting mechanism is required and this would be tantamount to a systemic mechanism in the general sense adopted above. The study of Lillie's classical 'model' of nerve-conduction is instructive here (Fig. 53): a piece of iron wire is made

'passive', that is coated with ferric oxide, and is then immersed in dilute nitric acid. If it is now touched at one point with a reducing agent a wave of reduction sweeps along at much the speed of nerve-conduction, and with a number of other properties similar to those of nerve-conduction; it is followed closely by a pulse of reoxidation, similar to the recovery of the 'resting' state in nerve. The local reduction of ferric oxide to ferrous oxide is followed by reoxidation at the expense of nitric acid locally and the resulting nitrous acid reduces ferric oxide farther along the wire.

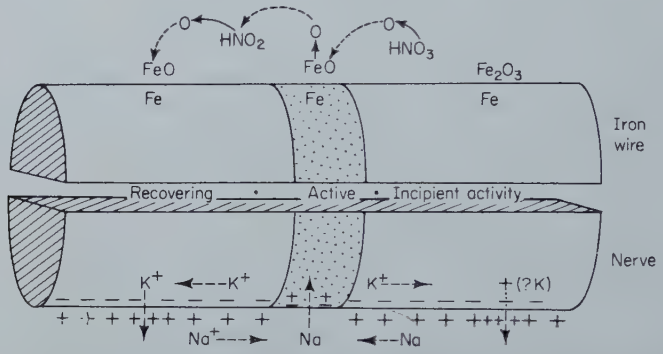


FIG. 53 Lillie's iron wire model of nerve-conduction (above) compared with actual conduction (below). In addition to the more obvious differences it should be emphasised that the real recovery phase in nerve involves the return of Na and K to their *locus quo*. ——→ = donates; - - - → = travels to.

Here the cycle is repeated, and so on. Granted the initial passive coat to the wire, and a stimulus of reduction, the rest of the phenomenon is spontaneous. No doubt the cycle of changes at each point is much simpler than that in nerve-conduction, where the actual cycle ends with sodium inside the cell and potassium outside, so that a separate recovery process is required later. It is difficult to believe that all of this is an automatic sequence, and indeed it is difficult enough to understand the spontaneity of the iron wire model. It is credible that under appropriately different conditions the reaction

$$2FeO + HNO_3 \longrightarrow Fe_2O_3 + HNO_2$$

proceeds either way, but in the model all four reagents are in close proximity in an environment which varies only in the immediate proximity of the active molecules and it is by no means evident what determines the decrementless wave of activity. It is credible that a newly passive area is more resistant or 'refractory' to further reduction than the area ahead of the travelling reduced band, so that it can only travel in the one direction, but the difficulty is to explain why it travels at all. Possibly it depends

on the special electronic properties of iron (p. 318). The model is in a way superior to nerve-conduction in that it is directly driven by redox energy, and that it produces a larger action potential.

If it were shown that nerve-conduction is to any considerable degree a spontaneous response of the materials of the cell and its membrane this would be a very crucial demonstration, since it is largely through the nervous system that animals make responses which eventually counteract or circumvent the deleterious direct effects of external processes. This is a test-case for all mechanisms contributing to self-preservation.

In the rods and cones, the photoreceptor cells of vertebrates, there are dense stacks of unit membranes (Fernandez-Moran, 1959) which may combine the usual electrical properties of the units with semiconduction (electronic) properties (Eley, 1962), and so account for the extraordinary efficiency of these structures. They transduce and amplify into useful nervous information virtually every photon which strikes them. The structure of chloroplasts is very similar and in fact they are using photic energy for a different, but equally efficient, purpose. The light-sensitive pigments, *rhodopsin* and *chloroplastin*, are lipoproteins and may lie in the aquo–lipid pair of monolayers of each half of the unit membrane. The arthropod retinula cells have tubes instead of plates set across the longitudinal axis of the receptor cell.

29.2 Structures Based on Fibrous Proteins

With the exception of keratinous structures most of these are built extracellularly, whereas most of the membrane structures are pericellular or intracellular. In many cases they are a fairly simple multiplication of the constituent macromolecules, but there are some interesting emergents in organisation. The collagen group are the most numerous and important, particularly if the invertebrate collagens be included (Gross, 1961). Their organisation may be illustrated by two extremes, tendon and areolar connective tissue. In the latter collagen fibres of variable diameter criss-cross in all directions, making a loose, pliable but ultimately strong feltwork, whereas in tendon large, uniform bundles of collagen fibres with their axes all parallel are closely bound together, leaving only narrow spaces for the cells and for nerve and blood supplies. Between these two extremes there is a great variety of textures, ideal each for some particular mechanical purpose, and it is not easy to suppose that all are spontaneous products of the collagen macromolecule, even supposing a different condition to evoke each. It is well known that tendon fibres orientate along the lines of experimentally applied tensions but it is not certain that the polypeptide chains or their polymers automatically do this. They

may, since even van der Waals' forces seem to be adequate to orient them in solution (Randall, 1954). As Weiss (1959a and b) has pointed out, any tension partially orientates any existing framework in the ground substance, and definitive fibres may tend to deposit automatically in this as a preferred direction. In connective tissue it may be assumed that tensions chance to occur in all directions and deposition remains throughout random, therefore. Fibres once started in any direction will thicken by spontaneous, stereotactic deposition of more protofibrils, and dense connective tissue becomes a dense mass of bundles of fibres, all parallel within the bundle but random between the bundles. This dense feltwork is very strong against tensions in all directions. In tendon there is an emphatically preferred orientation and conceivably this may be adequate to determine the whole development, automatically.

The texture of bone, which is a specialised connective tissue based on collagen in a ground substance, provides important evidence here, since it seems to be particularly organised in relation to mechanical forces (Murray, 1936; Thompson, 1942; LeGros Clark, 1958;). However, as Murray (1936, p. 110) has emphasised, the material should be laid down as a solid mass at any surface to which a force is applied and not as 'cancellous' bone, an openwork construction of relatively few beams and struts. The cancellous structure in fact combines the virtues of strength with lightness, in the same ingenious way as metal cranes and bridges, and it may be necessary to look for the engineer. D'Arcy Thompson (1942) and other recent biologists have taken to heart Bacon's sagacious remark: ". . . to say bones are for columns and beams is well enquired and collected in metaphysics; but in Physic . . . impertinent . . . the search for physical causes hath been neglected . . ." They have in consequence perhaps been too ready to accept simple physical forces as the sufficient explanation of structure and form. This could scarcely have been put more emphatically than by Haekel, concerning the beautiful arrangement of sponge-spicules to fit the canal system of the animal: "Die geordnete . . . zusammensetzung des skeletal systems ist zum grössen teil unmittelbares Product der Wasserstrommung . . ." (Thompson, 1942, p. 691). Minchin (1898) on the other hand saw this as the result of natural selection, and rightly visualised that the proximate causation might be rather complex. It may be spontaneous only as a complex systemic mechanism.

In the Haversian systems of bone and in the connective tissue of the wall of blood vessels and other tubular structures a much more complexly regular texture is developed, still more difficult to visualise as the spontaneous outcome of the intrinsic properties of the collagen itself. Its fibres are deposited in concentric layers, all the fibres parallel within one layer but the preferred orientation being approximately at right angles to

that in immediately neighbouring layers. The preferred directions are always in the plane of the cylinder, at approximately 45° to longitudinal and transverse axes.

A rather closely analogous texture is found (p. 59) in the cellulose component of plant cell-walls (Preston, 1952; Picken, 1960), with further complications in detail. In the exoskeleton of arthropods also there is a generally similar ply-structure, involving both protein and polysaccharide. In this case if the texture is an expression of the intrinsic properties of the materials these must be very similar in protein and chitin. This is conceivable and it is even more possible that the arrangement of two different proteins should be spontaneously as beautifully patterned as that of myosin and actin in muscle (p. 218), but this remains to be proved.

The mineral components of bone, sponge-spicules and other skeletons are deposited on a primordium of protein fibres and are orientated by them. The crystallites of calcareous sponge-spicules lie all parallel in all the rays (Jones, 1954). The final tri- or quadriradiate star of the completed spicule is not a natural crystalline form of calcite but is determined entirely by the protein primordium. Similarly in vertebrate bone the crystallites of apatite lie all parallel to the collagen fibres in each concentric lamella of the Haversian system described above. These are not spontaneous properties of the mineral, therefore, but are dictated from a higher level, that of the protein macromolecules. In so far as it is safe to argue by analogy it may be suspected that the higher-level organisation of the protein fabric is dictated from still higher, systemic levels. Within the crystallites of calcite and apatite the structure may be normal and intrinsic, for instance the calcite is birefringent, and only the higher-level organisation is dictated from without.

Even so there is probably considerable spontaneity in the whole process of formation of mineral skeletons. The protein spontaneously promotes deposition of mineral, and apatite deposits best on collagen fibres with the natural internal structure (Edds, 1958; Glimcher, 1959). Reciprocally, moreover, calcium promotes collagen-deposition in the skeletal tissues, rather as it does the proteins of blood-clotting. The ground substance, also, consisting mainly of mucoprotein, promotes collagen-deposition in the connective tissues in general, and does so optimally at the concentration actually found in connective tissue. Chondroitin sulphate, one of the essential components of the mucoprotein (p. 54), has been shown *in vitro* to precipitate collagen in the presence of acetate, and a similar function may be suspected for the acetyl glucosamine residues of chitin, in relation to the protein arthropodin. Chitin and the protein of the arthropod skeleton also stabilise each other, and can be dissolved from the exoskeleton and reprecipitated without breaking their links. Similarly

the mucoprotein of vertebrate connective tissue forms a very stable complex with collagen (Picken, 1960, p. 389).

The extent of the spontaneity should not be overestimated, however. The natural periodicity within the structure of the collagen fibres, which is so essential for apatite deposition, occurs also in the collagens of connective tissues and was probably evolved long before calcification. It is not reasonable to suppose that by good fortune this proved ideal for apatite deposition: the more plausible assumption is that a mechanism of deposition was evolved to suit both the existing collagen structure and the mineral. As already seen apatite is a curious crystal and this itself may be a biological adaptation to the rest of the system. This is again implying a systemic mechanism rather than simple intrinsic spontaneity at the molecular level.

There are of course many other structures at this intermediate level of organisation and in general they contribute to the present problem rather than to its solution. Particularly useful examples are the organelles of the Protista since they are discrete, highly differentiated structures, yet are contained within a single cell and so are not very far above the macromolecular level. Cilia and flagella, perhaps the most important and best known, are not restricted to the Protista. The organic capsules and skeletons of the Radiolaria are other examples, and the pseudopodia of the Sarcodina, which are all the more interesting because of their temporary nature. Particularly complex are the cnidocyst of Dinoflagellata, the polar capsule of the Neosporidia, and the nematocyst of the coelenterates (Picken, 1957; 1960), an example outside the group Protista. The mullioned structure of the wall of the nematocyst-thread, the barbs, the eversible mechanism, and the accessory structures of the cell which contains the nematocyst collectively show a complexity which it is difficult to relate to the intrinsic properties of the material, without evoking the help of some organiser of higher status. Some of these structures contrast rather sharply with the simple geometric regularity of most viruses; these do appear to have a structure which could form spontaneously from its components (Caspar and Klug, 1962). Certainly the whole body of many of the Protozoa has one of a number of characteristic geometrical shapes, for which simple proximate causes have been suggested (Thompson, 1942), but at this higher level it seems doubtful if they are spontaneous growth-forms of the materials, as the virus form seems to be.

29.3 Systemic Dictation at the Molecular Level

In the preceding section it has become clear that the higher levels of organisation may be increasingly determined by systemic mechanisms

which override or obscure the intrinsic properties of the materials. This may raise doubts whether even the properties studied at the molecular level are all spontaneous, intrinsic properties of the molecules themselves. There has undoubtedly been some selection of elements and compounds and it is possible that, as in the case of apatite, there may have been dictation of the way in which they should behave. An instructive case is the stronger hydrogen-bonding of the proteins of the flagella of thermophilic bacteria than of those living at normal temperatures (Koffler, 1957). The direct effect of heat is to loosen hydrogen bonds so that this peculiarity of the thermophiles is adaptive, counteracting the direct effect of the external factor. By the same token the adaptation must be indirect and not an intrinsic property of the protein, and this seems to imply a systemic dictation of properties—a systemic means of increasing the hydrogen bond strength in a particular protein. However this is not necessarily the explanation: proteins no doubt vary in their hydrogen-bond strength, depending on the particular amino acids making the bond, and in the thermophiles there has probably been progressive selection for proteins of ever greater hydrogen-bond strength and tolerance of heat. This of course also assumes that the thermophiles have evolved from normal bacteria, and are not relics of a pristine high-temperature flora.

A very similar case, no doubt, is the raised denaturation-temperature of the proteins of animals acclimatised to a raised environmental temperature (Prosser and Brown, 1961, p. 248). Also of the same general type is the increased affinity of the haemoglobin of the frog for oxygen if kept at high temperature; the direct effect of high temperature is to promote dissociation of oxy-haemoglobin so that again the change is adaptive. In these cases the adaptation can occur in the individual animal and it must be assumed that it inherits the ability to switch its protein-synthesis to a variant with modified properties. Another interesting adaptation of this kind (Lundegårth, 1950) is the ability of some organisms, after their cytochrome system has been inhibited by cyanide, to recover the use of the terminal oxidases. Lundegårth suggested that the affinity of iron for the inhibitor decreased but this really would amount to a change in intrinsic properties of an element and it seems more likely that the protein bearing and controlling the iron changes, and that it does so again by a change in the type of protein synthesised. The suggestion that iron might change its properties epitomises the whole problem of this section: do atoms and molecules change their innate properties under the influence of the biological system? From the examples so far considered the answer would seem to be: no; rather, organisms have the plasticity of switching their syntheses to different organic molecules, with appropriately modified properties. The ability is of course an inherited

adaptation, originally evoked by the same kind of hazard in the past. Particular molecules do not change their innate properties, and it is therefore safe to study these even *in vivo*—with a certain amount of care.

The kind of care necessary is illustrated by the haemoproteins at least as well as by any type of molecule. It has been adequately stressed (p. 317) that not merely the same metal ion but also the same haem occurs in a large range of molecules of very varied properties, and so it could be claimed that the different proteins are modifying the innate properties of the iron and of the haem. However, it is more true to say that the properties studied are those of the molecule as a whole and that the strict contributions of iron and haem themselves do not vary.

There is of course no question that systemic mechanisms do affect the molecules which are used and the metabolic reactions which result, but this again is selecting one rather than another and not changing the innate properties of any. Micro-organisms synthesise appreciable amounts of a number of their enzymes only if the appropriate substrate is available. Again the bacterium, *Escherichia coli*, decarboxylates amino acids under acid conditions and deaminates them under alkaline conditions (Gale and Epps, 1942); a volatile acid and alkali respectively is therefore excreted, to evaporate, and the pH is driven towards neutrality in both cases. This is reminiscent of the change in form of the excretory products of man, according to the pH of the diet. There are many other known examples of adaptive mechanisms at the molecular level and, like all adaptive mechanisms, they are essentially systemic in nature; none of them changes the innate properties of molecules. It is only at higher levels of organisation, where the conception of specific molecules no longer has any meaning, that systemic mechanisms may produce such effects, but these cannot any longer be tested against known intrinsic properties of definite chemical entities.

This seems to be an additional difficulty in testing for uniqueness at the higher levels of organisation. There are therefore three main difficulties, that systemic controls complicate the picture, that there are probably emergent properties at each level but that much of the material does not consist of definite molecules with absolutely specific properties which can be used as the yardstick. In the face of these difficulties, and of the apparent chasm already mentioned between the molecular level and the wonderfully organised organism as a whole, there is a tendency to look for simple models which might suggest any possible simplifications of the apparent complexity. There is great danger of regarding purely superficial analogies as significant, and some models have been extraordinarily naïve, but it is worth considering a number of them if

only to show how easily simple systems can show life-like properties, including purposiveness or teleonomy (Pittendrigh, 1958).

29.4 Models of Higher Grades of Organisation

La Nature agit toujours par les moyens les plus simples.
(JANET, 1876, *Les Causes Finales*)

There is no great and no small
To the soul that maketh all,
And where it cometh all things are
And it cometh everywhere.
(EMERSON, *Essays*: 'Epigraph to History')

The essential starting point in the study of models is that living and non-living systems obey the same natural laws so that no analogies, however simple, are completely superficial. Iron filings in the field of a magnet, and Leduc's two droplets of indian ink, resemble the mitotic figure because in each case the pattern of distribution of forces around two neighbouring foci is portrayed. The plant-like sprays which arise from crystals of cobalt salts in a solution of water-glass, sodium silicate, resemble plants perhaps in the nature of the asymmetrical forces which determine their growth. Washing-soda crystals in 5 per cent cupric chloride not only 'grow' in this way but also bend towards the light (Beutner, 1938), probably by differential uptake of water as in the case of the plant tropism.

There is not space to consider all of the many interesting models which have been used; these not only range over the whole field of biological phenomena but over all grades of relationship from the purely superficial to the *in vitro* duplication of what occurs *in vivo*. D'Arcy Thompson's laboratory copies of skeleton-formation (Thompson, 1942, p. 653*ff*) come near this end of the scale. A few of the more dramatic will serve to illustrate the extent of their utility. The most dramatic are those which mimic biological movements, since these are also very relevant to organisation at the level under consideration. Darling (Thompson, 1942) investigated the behaviour of a number of interesting lipids as isolated drops on the surface of water. Drops of aniline pulsate rhythmically, in a life-like manner. The explanation is that the aniline dissolves to some extent and so lowers the surface tension of the water–air interface (1) relative to that of the oil–air (2) and oil–water (3) interfaces (Fig. 54), so that the curvature of the oil drop increases in accordance with the triangle of forces. The rate of solution of the aniline therefore decreases, the surface tension of the water–air interface increases again as the dissolved aniline

disperses, and so the drop is once more pulled out into a flatter disc. The rate of solution now increases again and the whole cycle is repeated indefinitely. It is a typical servo mechanism with a lagging response. The biological structures which it resembles—hearts, contractile vacuoles, etc.—may very well use such servo devices, and the value of the model is to show how easily these can be virtually automatic—but never completely. To continue indefinitely the aniline would need renewal and the water must not become completely saturated with aniline.

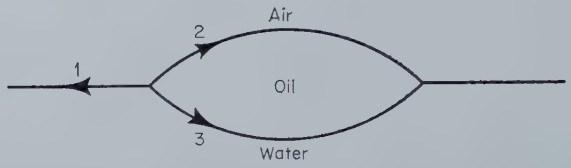


FIG. 54 Vertical section through a drop of oil on the surface of water, to show the three surface-tension forces meeting at the line of junction of the three types of surface, 1. air/water, 2. air/oil and 3. oil/water.

The often-described mercury-drop model (Rogers, 1938) is a rather similar case: the mercury drop in a little 15 per cent sulphuric acid with a trace of potassium dichromide is touched with the point of a needle riding in a floating cork. The needle acquires a negative charge through partial solution in the acid and this discharges the positive charge acquired by incipient oxidation of the surface of the mercury drop. The latter therefore becomes more spherical, breaking contact with the needle; both metals therefore recharge, the drop flattens and the cycle is repeated.

A drop of chloroform will 'ingest' particles of glass if they are coated with shellac and when the shellac has all been dissolved the glass is egested (Beutner, 1938); this is an amusing model of *Amoeba* feeding and egesting the residue. A slender fibre of coated glass is taken in and coiled up inside, in apparently the same ingenious manner as an *Amoeba* ingests a filamentous Alga. The conclusion is that simple surface forces alone may explain the action of *Amoeba* also, but it is a gratuitous conclusion, and needs real evidence.

If a drop of a 10 per cent solution of camphor in benzene is placed on water it shows violent and rapid surface-movements, putting out irregular processes resembling pseudopodia. If olive oil is added to the medium it slows down the movements and inhibits them completely at a concentration of one part of oil to one of water. The camphor–benzene drop still shows streaming movements inside, however, just as some amoeboid cells do when overt movement is prevented. Alcohol induces 'pseudopodia' in the camphor drop, by decreasing its surface tension, and olive

oil no doubt has the opposite effect. It is now known that surface forces do not play a large part in amoeboid movement, but the model does perhaps show that streaming movements are directly related to amoeboid movement and that both may be due largely to simple differential forces in the functional envelope of the amoeboid cell.

One bizarre but very versatile model used by Darling was a large drop of the red oil *ortho-toluidine* (Fig. 55, I). Placed on water the drop indents

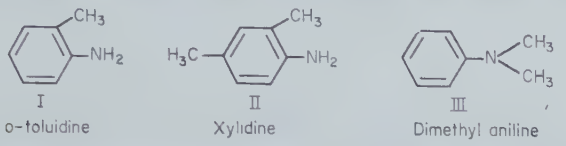


FIG. 55 Formulae of oils used as simple models of biological phenomena.

on one side and then darts across the bowl, often breaking into two as it goes. Then there is a pause and if the drops are still large enough they both indent and dart across simultaneously. It seems likely that the drop causes fountain movements across the surface of the water which split the drop and the daughter drops are then carried across in the return flow, along the sides. *Xylidene* (Fig. 55, II) is even more active. Movement is arrested immediately if the drops are touched with a drop of mineral oil. *o*-Toluidine also ingests drops of *dimethyl aniline* (Fig. 55, III) by 'pseudopodial' action.

Amoeboid movements seem rather characteristic of lipid droplets in water and as already seen (p. 296) biological lipids such as lecithin also are good mimics. The 'myelin forms' of lecithin are more sluggish and life-like than most of the simpler lipids described above, and undoubtedly there are phospho-lipids in the membrane and endoplasmic reticulum of *Amoeba*. Its whole repertoire of movements therefore may be fairly automatic properties of this material. On the other hand some *Amoebae* when in a confined channel will turn on themselves at the end, and retrace their steps (Goldacre, 1957), and will show other equally purposive pieces of behaviour.

The work of Herrera (1942) is probably the most comprehensive of recent endeavours in the field. One of his most interesting models was a mixture of formaldehyde and ammonium cyanate in thin films. This showed many properties reminiscent of biological formations—pseudo cells and tissues of various kinds, figures resembling mitotic division, and so on. These are mainly structural models but they involve non-lipid materials with some slight biological significance. Ammonium cyanide has been used for some interesting laboratory synthesis recently (Table

22), and Herrera's mixture also led to the synthesis of small amounts of amino acids, carbohydrates and pigments. This biochemical model may prove more significant than the morphological ones.

Most of the models are naïve and superficial, no doubt, party tricks more than serious science. With few exceptions they cannot be regarded as close mimics of specific biological structures or processes but they do serve one general purpose, and this is to show how much life-like behaviour can be the automatic result of simple forces. They encourage the more serious work of analysing organisation at the supramolecular levels.

29.5 Conclusions

The problem of higher levels of organisation in relation to biological uniqueness is too difficult and too large for adequate treatment here, but is worth further attention. If the ingredients are unique it follows that in a sense higher levels of organisation must be, also, but it would be useful to show exactly how far this follows automatically from the properties of the constituent materials.

CHAPTER 30

HOW UNIQUE?

The evidence surveyed in the preceding chapters seems to show rather consistently that the materials used in living organisms are outstanding if not absolutely unique among materials available on Earth, and that they are unique in respects which make them ideal for some particular biological function. This is true of the biological elements compared with others and of particular compounds of one element compared with its other compounds. In fact some elements and compounds seem to be more uniquely essential *in vivo* than their general chemical properties would imply, indicating that there may have been selection for a living system which enhances their uniqueness. There are many cases where the evidence is not yet adequate to say if the material is unique, and others where its properties are outstanding but have not so far been shown ideal for a particular biological purpose. Where it is adequate, however, it tends to show that biological materials do not overlap seriously in function and that only the best for any particular purpose is normally used for that purpose. The object of this chapter is to recapitulate some of the more critical evidence bearing on these conclusions. It is not a detailed summary and in fact most weight will be given to difficult and exceptional cases of the type calling for further work. These are the most interesting cases.

There is no doubt that carbon, hydrogen, oxygen, nitrogen, phosphorus, and sulphur are irreplaceable and unique as biological elements, and similarly water as a compound, and as a fluid medium. Even here, however, some qualifications are necessary, particularly for N, P and S. The groups —S— and —CH=CH— can deputise for each other in ring compounds, though the resulting compound is more usually a metabolic antagonist than a deputy. Some organisms can use sulphur instead of oxygen to reoxidise their cytochrome system (Kamen, 1958); in a sense oxygen and sulphur are here respiratory substrates or reagents and there is in general great variation in the substrates used for respiration. The actual 'fabric' of the living system, including the enzymes used in respiration, is less variable. Quaternary compounds of phosphorus and arsenic,

and to some extent the analogous *onium* compounds of sulphur and oxygen, show some of the actions of the key quaternary nitrogen compounds, such as choline (p. 289), but they do not show all its properties, and in particular they do not promote growth. More usually they again act as metabolic antagonists, similar enough to compete but only detrimentally. There is no evidence that any of them are normally synthesised *in vivo*.

One of the most critical examples is that of selenium as a deputy for sulphur. Plants can synthesise large amounts of seleno-proteins, enough to poison stock and to serve as 'systemic' poisons for insect pests, without harming the plant in any detected way. Similarly the bacterium, *Escherichia coli*, can synthesise seleno-proteins which appear to function as well as the normal sulphur-proteins (Cowie and Cohen, 1957) (p. 531). The deputation of germanium for carbon in some coal-depositing organisms, by contrast, is very questionable.

Bromine and iodine seem to be simple alternatives in the skeletal proteins of sponges and Anthozoa (p. 461) and could be so in thyroxin also, but here there is a systemic device for excluding bromine, thus raising iodine from the status of outstanding among halogens to that of absolutely unique among biological elements. Even so thyroid action is not absolutely specific even to the iodo- and bromotyrosine derivatives; the nitrothyronines have a positive action and tetranitrothyronine promotes both the growth of rats and the metamorphosis of tadpoles (Glass, 1951). Some nitro analogues are antithyroid agents (p. 458), but then so is thyroxin in excess. Tetranitrothyronine has only 1/1000 the potency of thyroxin so that iodine is far outstanding, and the body normally excludes nitro-analogues as well as those of bromine. To the organic chemist the nitro group has long been known as an analogue of the halogens, particularly in aromatic combination, and it is interesting to see this rather unusual analogy equally valid physiologically.

Another case where substitution seems a matter of indifference is that of niobium for vanadium in the blood pigment of ascidians (p. 451). Here even the individuals of one species may have either V or Nb, though they never have an equal mixture of the two (Carlisle, 1958). This is perhaps a very minor element, in an isolated group of animals, but it is an outstanding case and deserves close attention. In its turn vanadium can deputise for molybdenum to some extent in catalysing nitrogen fixation by plants, and so can tungsten, the higher analogue of molybdenum (Quastel, 1947).

Some degree of deputation between other catalytic metals also seems very common, at least *in vitro* (Chapter 26), but between the best known of these there appears to be no overlap. This is a very interesting and

critical set of elements; where the metal is only loosely associated with an enzyme it is perhaps not surprising to find that others can deputise quite well *in vitro*, but it is often true also where one metal is a permanent part of the enzyme. For instance arginase is a manganese-protein (p. 444) but cobalt ions greatly speed the release of urea from arginine *in vitro*.

This more facile deputation among ions not firmly bound to the organic fabric is shown also by the physiological ions (Chapter 25). In bone strontium can take the place of calcium, and in fact many ions of either charge can be incorporated into the bone mineral without ill effect —unless in very high concentration, or radioactive! Sr, Ba and Mg can all deputise for Ca in promoting blood-clotting. At the same time it must be stressed that strontium cannot perform all the biological functions of calcium. In the laboratory lithium, but only this cation, can replace sodium as the main cation in the external medium of a conducting nerve (Hodgkin, 1951). In muscle-conduction there is a marked influx of calcium as well as sodium (Prosser and Brown, 1961, p. 436). Calcium and magnesium can replace sodium in the external medium of the brine shrimp, *Artemia* (Thompson, 1942, p. 254), but the internal medium bathing its nerve cells may be maintained near the normal composition and concentration. Phytophagous insects usually have high potassium instead of sodium in the haemolymph but they contrive to maintain an orthodox Na/K ratio near the nerves themselves (Treherne, 1962). However, the Na/K balance is actually reversed in the erythrocytes of some artiodactyls and carnivores (p. 419), and an ionic differential of some sort seems more important than any specific distribution. NH_4^+ can deputise for K^+ in yeast without seriously disturbing its metabolism (Fearon, 1949, p. 19); this is interesting because ammonium is an analogue of the alkali metals in much the same way as nitro is of the halogens (another aspect of the uniqueness of nitrogen). Rubidium, the higher analogue of potassium, is able to deputise for it as much as strontium can for calcium. In very few cases has the extent of the deputisation and the absence of ill effects been tested adequately, or whether the deputisation can continue indefinitely *in vivo*. However, the case of the erythrocytes alone is significant; it indicates that free ions are less critical than components of the living fabric.

Among compounds of the same element deputisation is more common, at least to the extent that different taxonomic groups have exploited alternatives. Perhaps the best example is the use of glutamine by animals and of its homologue asparagine by plants (p. 181). The two amides and their amino acids are far from being complete alternatives, however; the aspartic analogue of PGA (p. 376) is an antagonist of the vitamin itself (Hutchings *et al.*, 1949). The various hydroxyamino acids are extensively interchangeable in collagen (Piez and Gross, 1960). Hydroxy-

proline rarely occurs except in collagen and the hydroxyl side chain seems the essential feature, the amino acid carrier rather immaterial. There are a number of substitutes for ascorbic acid (p. 50), all having the lactone ring system, but none is as effective as ascorbic itself (Kermack and Eggleton, 1938, p. 176). The phosphagens of animals are rather varied (p. 300), though they all have the guanidine group.

Lipids are perhaps the most variable and least specific. The glycerides vary considerably according to the diet of an animal, quite apart from intrinsic taxonomic differences, although in melting point and other properties they tend to be nicely adapted to their position in the body, to the environment of the animal, and so on. For fuel stores molecular details may be relatively unimportant. The essential oils of plants are

Table 40. Bile Acids and Bases of Vertebrates

Acid	Base	Occurrence
Cholic	Glycine Taurine	Most mammals
Deoxycholic		Rabbit
Nutriacholic		*Myocastor* (rodent)
Trihydroxycoprostanic $C_{26}H_{45}O_3COOH$		Crocodilia, Anura
Pythocholic $C_{23}H_{45}O_3COOH$		Boidae (boa constrictor)
Tetrahydroxynorsterocholanic $C_{26}H_{45}O_4COOH$		Teleosts
Sulphuric	Scymnol $C_{26}H_{46}O_5$	Elasmobranchs

probably in a similar position, not permanent components of the functional fabric, and they vary considerably, even among the different species of *Eucalyptus*. Moreover oil of turpentine from European conifers tends to be mainly L–pinene whereas that from America and Burma is mainly the D–isomer.

The bile salts perhaps can be considered much more metabolically active yet they also vary greatly in their acid moiety, and to some extent in their base (Table 40). One of the most striking cases of deputisation is that of the stilboestrols for the natural steroids (p. 122), partly because some of them are even more potent than the latter and partly because the structure seems so distantly analogous; however, it is becoming clear that sterically this is quite a close mimic (Huggins, 1962).

Among more sporadically evolved substances, for instance bioluminescent materials, there is even great variety; the luciferins belong to a

number of quite different chemical groups (McElroy and Glass, 1961; Johnson *et al.*, 1961). This of course emphasises by contrast the uniformity of such key metabolites as the haem-prosthetic group, in all organisms. Antibiotics are another class evolved sporadically, and extremely variable in molecular structure. Here variety has an obvious adaptive value since antibiotics are chemical weapons and variety delays the development of counter-measures by the target organisms.

These are examples of variation and of low specificity among carbon compounds but the great specificity of most of them is emphasised by the exceptions. Specificity extends even to the one optical isomer of a compound. The natural L–nicotine is fatal at concentrations in which the D– form causes only slight shivering. L–Hyoscyamine (henbane) strongly constricts the pupil of the eye, while the D– form has no effect. Adrenalin and other agents show similar differences. Enzymes usually act only on substrate molecules with the correct asymmetry.

One rather impressive example of high specificity is the use of retinene (p. 112) as the photosensitive pigment of the retina of vertebrates, arthropods, molluscs, and possibly other groups. The really impressive feature is that it is the same isomer in all cases and this the sterically hindered, 'improbable' form, 11-*cis*,7, 9,14-*trans*-retinene. It is, however, the most stable and so the most useful (Glass, 1961, p. 904). This seems even more impressive than the identity of the haem in the haemoglobin of so many different phyla, since they all had the same haem already in their cytochromes, from which the haemoglobins probably evolved. In fact most animals also acquire the same carotinoids, as precursors of retinene, so that the two cases may not be very different; in both cases it is impressive that little tendency to exploit alternatives has been found.

The familiar common metabolites are scarcely less impressive, the Krebs cycle substrates, the 20 common α–amino acids, the ribose and the five bases of the nucleic acids, lecithin, the B-vitamins and many others all occur in precisely the same role in all organisms. They may have been inherited from a common ancestor but they have remained unmodified and probably ideal. Even at higher levels of magnitude, in the proteins and nucleic acids, where uniqueness takes a new turn, towards taxonomic specificity, this is equally absolute and the proteins of one individual are incompatible with those of another, even of the same species. In one sense this is the height of biological uniqueness. At a slightly lower level there are functional proteins such as insulin, and the globin of haemoglobin, which are identical in all normal individuals of a species. In the globin it has been possible to show that if a single amino acid of the total of 288 is changed the function is grossly disturbed, as in the sickle-cell genotype. A difference in one or at most two amino acids, again,

Table 41. Length of Aliphatic Chain giving Maximal or Minimal Action to Molecule

Length of chain, C_n. $n=$	Type of molecule	Action
1	Fatty acids	Polar group maximal affinity for other polar molecules
2	Primary amines	Minimally attacked by amine oxidase
2	Amides	Most stable against hydrolysis
2, 3	2-Alkyl-thio-benzthiazoles	Maximal toxicity to micro-organisms
4	Diamines	Most readily attacked by diamine oxidase
4	Quaternary ammonium compounds	Maximal action on nervous system
4	Quinine, side chain	Maximal lethal effect on malarial parasite
4, 5	Quinolines, side chain	Maximal antimalarial action
4, 5	Biotin, side chain	Maximal biological action
4, 5	Dithioctanoic acid, side chain	Maximal biological action
5	Monoses	Second most common size of molecule, biologically
5	Primary alcohols	Maximal toxicity to blowflies
5	Fatty acids	Minimal toxicity
5	Alcohols	Maximal rate of permeation through membranes (partition coefficient, lipid/water, near 1·0)
5	*p*-Alkyl phenols	Maximal bactericidal action
5	Alcohols	Maximal toxicity
5	Various	Near optimum for ring closure
5, 6	Curariform drugs, linking chains	Maximal effect on central nervous system
5, 7	Various	Minimal for forming channel complexes with urea
6	Monoses	Most common size in living organisms
6	Various	Optimal length for ring closure
6	Di-isothioureas, linking chain	Maximal trypanocidal effect
6	Alkyl dihydroxy phenols	Maximal bactericidal action
6	Various	Optimal for conjugation with glucuronic acid
6	Primary alcohols	Maximal toxicity for *Staphylococcus* (detoxication)
6	Fatty acids	Minimal toxicity for potato (slices)
6	n-Alkyl resorcinol	Optimal for anthelminthic action
6, 7	Amides	Optimal for hydrolysis by liver enzymes
7, 8	Various anions	Optimal for protecting proteins of high concentration against denaturation
8–11	Curariform drugs, linking chain	Maximal action on peripheral synapses

Length of chain, C_n. n=	Type of molecule	Action
9	Aldehydes	Maximal promotion of bacterial bioluminescence
9	Fatty acids	Maximal toxicity for potato (slices)
10	Tetra-alkyl diamines	Maximal anti-amoebic action
10	Alkyl thiocyanates	Maximal toxicity for aphids
11	Unsaturated fatty acids	Maximal action against ringworm
12	Alkyl trimethyl quaternary ammonium compounds	Maximal antiseptic action
12	Various anions	Optimal length for protecting proteins of low concentration from denaturation
13	Diamines	Most readily attacked by mono-amine oxidase
10–14	Diamidines	Maximal trypanocidal activity
11–15	Secondary aliphatic amines	Maximal insecticidal activity
16	Fatty acids	Limit of synthesis by malonic mechanism
16, 18	Fatty acids	Most common length in living organisms
36	Various lipids	Limit of stability in monolayers

distinguishes the adult from the embryonic globin which has a higher oxygen-affinity and other distinctions. Of the vast number of different protein molecules which are theoretically possible a very small percentage, but still absolutely a very large number, have been exploited, each for a unique purpose, functional or taxonomic. It would be extremely useful to know if other specific proteins have been tested and discarded by living organisms, or if it has been purely a matter of chance which have been synthesised and what forms and functions have been possible. The student of evolution tends to take this latter view, so that there may be unlimited possibilities for further protein evolution. At lower levels of molecular organisation all variants have been available and it has been a matter of exploiting the useful ones, but at this macromolecular level it may be a matter of exploiting those which could be synthesised.

30.1 Uniqueness of Chain-length

There is an aspect of specificity among carbon compounds which has been mentioned repeatedly throughout the book and which merits a summary here, though it does not at the moment contribute many general principles, perhaps. This is the optimal chain-length of aliphatic compounds and of the side chains of other compounds (Table 41). In each particular case it

illustrates high specificity since very often activity is completely lost if the chain is lengthened or shortened by more than one carbon unit. Collectively, however, the records cover a large range of chain-length which implies an equally large number of different functions, even supposing that a chain of a particular length only performs one function; in fact it probably does many things, depending on its end-group and other features of the molecule in which it occurs. Because of these variants it was not possible to deal with this as a general phenomenon in any other context, but it is clear that alkyl chains play an important part in specific metabolism.

An interesting feature of the available records (Table 41) is that C_5 and C_6 are the most commonly important lengths, and this implies that the chains may form temporary rings. This may not be true of chains of all lengths, or even of all C_5 and C_6 chains, but it probably does indicate that the geometry of the chain is an important aspect of its purpose, permitting the superposition of ring structures of the molecule (p. 334) and so on. In all, no doubt, there are many geometrical and other properties of these chains to be explained. There is a slight indication of another optimal range of chain-lengths between C_{10} and C_{13} but this needs more data.

There is some knowledge of other ways in which chain-length determines properties, for instance the main fact (p. 82) that at short lengths the molecule may be too water-soluble, if the chain has a terminal polar group, and too insoluble at great lengths. Again the degree of ionisation of a carboxylic acid decreases with chain-length but the toxicity of fatty acids for cells of the potato depends on both the H^+ ion and the undissociated molecule; consequently there is an intermediate chain-length which is optimal for maximal effect. Further examples and evidence are given by Sexton (1953).

30.2 Detoxication

A priori it would have seemed impossible to deal with the uniqueness of biological materials without considering the fascinating chemistry and biochemistry of detoxication mechanisms (Williams, 1959). There is a fundamentally important reason, however, why it has been possible to do so and this is that detoxication mechanisms are the very negation of biochemical uniqueness. In sharp contrast to the precise steric fitting of ring structures when they react metabolically, and the critical importance of chain-length, detoxication is effected by an entirely cavalier neglect of everything but an odd chemical grouping, a nose by which to hook the bull. It is a gross chemical rather than a specific biochemical approach.

For this very reason it provides an invaluable foil to biochemistry and its unique processes, and merits brief attention in that light.

To some extent detoxication resembles immunological reactions, but with simpler molecules, to deal with the smaller foreign molecules. The body's proteins seem to be able to inactivate any foreign protein which may enter and similarly almost any foreign molecule in reasonable concentration can be destroyed or inactivated by detoxication. The greatest difference, perhaps, is that antibodies are absolutely specific to their antigens, and the whole mechanism is the acme of biochemical uniqueness, whereas all the simpler toxic chemicals are detoxicated by relatively few stock reagents, acting in a few standard chemical ways, mainly by oxidation, reduction and conjugation (Table 42). Oxidation is more common than reduction since it releases energy while the latter consumes it. The detoxicating agents used by vertebrates are mainly glucuronic acid (p. 50), any of the following amino acids: glycine, glutamic acid, cysteine, serine, ornithine and lysine, other SH compounds and methyl and acetyl compounds. Glucuronic acid forms glucosidic linkages with OH, COOH, NH_2 and SH groups; the others are more restricted in their abilities but they can deal with almost any molecule which contains the appropriate radical for conjugation. The total number of known conjugations is very great but far more toxic substances are destroyed by oxidation and reduction, and the total is very impressive.

Aromatic SH compounds are detoxicated in the same way as phenolic (aromatic OH) compounds, and heterocyclic molecules in the same way as their carbocyclic analogues, for instance the methyl furans, thiophenes and pyridines in the same way as toluene (—O— as well as —S— therefore is equivalent to —CH=CH—), and the dimethyl derivatives of these are treated as xylene. This is a good illustration of the low specificity of detoxication.

Detoxication therefore is rather a blunderbuss type of method but like the blunderbuss there is nothing casual about the mechanism. Insects use much the same devices as the vertebrates (Smith, 1955), though they have exploited glucose rather than glucuronic acid as their main detoxicating agent, and they convert toxic metals to sulphides rather than to organic sulphur compounds. Birds use ornithine where mammals use glycine, and there are some other known variations. In general, however, much the same devices are used, in the same non-specific ways; they are ideal for disposing of a great variety of foreign molecules with a minimum of fuss.

It is very surprising that the oxidation of a single group, or the conjugation of a single group, of a large organic molecule should be adequate to inactivate it, and in fact it does sometimes fail (Williams, 1959), but as

Table 42. Detoxication Mechanisms

1. *Oxidation*
1.1 $-CH_2OH$ to $-HC=O$
1.2 Alkyl to CH_2OH or to COOH
1.3 Oxidative deamination
1.4 Oxidative dealkylation
1.5 Sulphur to sulphoxide and sulphone
1.6 Aryl $-CH=$ to $-COH=$
1.7 Oxidative opening of ring
1.8 Halogenation

2. *Reduction*
2.1 $-HC=O$ to CH_2OH (cf. 1.1)
2.2 $(R_1R_2):C=O \rightarrow (R_1R_2):CHOH$
2.3 $-HC=CH- \rightarrow -CH_2-CH_2-$
2.4 $NO_2 \rightarrow NH_2OH \rightarrow NH_3$
2.5 $R_1N=NR_2 \rightarrow R_1NH_2 + R_2NH_2$
2.6 $R{\cdot}C(=O)NHOH \rightarrow R{\cdot}C(=O){\cdot}NH_2$
2.7 $R_1SSR_2 \rightarrow R_1SH + R_2SH$
2.8 $R{\cdot}AsO(OH)_2 \rightarrow RAsO$

3. *Conjugation*
3.1 Glucuronic acid with
OH of primary, secondary, and tertiary alcohols, including phenols (→ ethers)
COOH of alkyl and aryl compounds (carbocyclic and heterocyclic) (→ esters)
NH_2 of alkyl and aryl compounds
SH of thiophenol compounds
3.2 Glycine with
Benzoic acid, phenyl acetic acid, cholic acid
3.3 Glutamine with
Aromatic acids
3.4 Cysteine with
Aromatic compounds, replacing H, halogens, NO_2, substituents on ring
3.5 Ornithine (both NH_2 groups) with
Aromatic acids
Pyridine derivatives
3.6 Sulphate group with
Phenols
Sterols
Carbohydrates
? Aromatic amines
? Thiophenols
Cyanides (→ thiocyanate)
3.7 Methylation of
Aromatic ring-N groups (not of saturated rings)
Histidine ring-N groups
Phenolic OH groups
3.8 Acetylation of
Aromatic NH_2 groups
Aromatic $-CHO$ groups

usual the exception proves the rule. In explanation of the rule it may be suggested that destruction is so much simpler than construction and maintenance: if the car can be immobilised by removing the rotor-arm why dismantle it completely?

30.3 Conclusions

The uniqueness and specificity of function of biological materials are not absolute, but they are of a very high order in those materials which are part of the actual fabric. From ribose and even simpler molecules, up to the protein molecules which will not function even in an individual's own sibs, there is a highly specific relation between function and one particular atom or molecule. Analogues often show some action of the same kind, but much weaker, and they are rarely able to deputise for all purposes, even qualitatively, so that *in vivo* they upset metabolism. This danger is largely avoided normally by exclusive mechanisms, such as that which prevents bromine forming thyroid hormones, and this in turn heightens the biological uniqueness of the prime mover, iodine.

Where relatively complex molecules such as those of thyroxin, pteroyl glutamic acid and other N–heterocyclic compounds have been tested in detail it is found that almost every feature of the molecule is irreplaceable without seriously depressing activity. The study of side chain length in various classes of compound reinforces this conclusion. Detoxication-mechanisms, a sharply contrasting type to those of normal metabolism, serve to set the uniqueness of the latter in strong relief.

CHAPTER 31

OTHER FORMS OF LIFE?

The main interest through the book has been to decide how far the elements and compounds of living organisms are uniquely ideal for this type of life which we know. There is, however, the theoretical possibility that other forms of life may exist, and might exist even on Earth, under different conditions, and that for them other materials might be ideal. A further theoretical possibility, that two or more types of life might co-exist on earth, probably can be dismissed. Under one set of conditions only one system can be ideal and it will starve any others of material (Needham, 1959). Much of the evidence of the preceding chapters is of absolute uniqueness or superiority, under terrestrial conditions; there are no alternatives which other types of life could usefully exploit. Conditions on Earth no doubt have changed during the last milliard years, since life here was well launched on its evolution, and life itself therefore may have changed its character, but what evidence there is from palaeochemistry indicates that the changes have been small, mainly in the direction of increasing specificity and austerity (Lwoff, 1944; Florkin, 1950, 1960), a process of biochemical stream-lining. Until about 800 million years ago there was probably no free atmospheric oxygen, so that the photosynthetic redox process by which autotrophes fixed solar energy must have been different in detail from that most common today, but it is equally certain that they fixed energy and built energy-rich materials in the same general way as now.

It seems probable that very different conditions would be necessary to produce a really different type of life, and that in consequence there is not much probability of such types anywhere in the universe. The same chemical elements occur everywhere and it seems probable that where conditions permit them to evolve into living systems at all these cannot be very different from that which we know (Wald, 1962). Calvin (1959b) goes so far as to suggest that antimatter is the only possibility of developing an alternative type of life! Short of this, the possibility is that at a different temperature from ours, and with other major differences in environmental conditions, somewhat different forms of self-perpetuating

activity based on the known elements might be possible, particularly if the ratios of the various elements were appropriate.

The most critical factor is a suitable fluid medium as the foundation of a bioplasm: it is difficult to believe that anything with the two essentials of coherence and dynamic activity could survive in any but a fluid medium. Water-based life is possible only between 0° and 100°C, with a little extension in the presence of solutes. Within this range it is doubtful if any other type of life could compete with the type we know, and alternatives therefore are restricted to different solvents, in different ranges. Liquid ammonia is one of the most promising possibilities (Firsoff, 1962), with a temperature range of −33° to −78°C, and there is some evidence that it exists in quantity on some of our solar planets. Temperatures lower than those on Earth are probably more common than higher temperatures, on the planets where life is at all feasible: on the stars themselves most elements are in the gaseous state, and speeds of reaction are incomparably greater than anything remotely resembling our life. Life based on liquid ammonia therefore is the most possible alternative; it would have the further virtue (p. 154) of being fully miscible with aqueous systems.

Ethanol is in some ways a better solvent than liquid ammonia since it is fluid between −112° and +78°C, almost twice the range of water. Methanol, with a range of −95° to +66°C, is not greatly inferior in this respect and is more volatile; its greater toxicity to terrestrial life is not necessarily a serious disadvantage for a different type of life, though its anomalous properties within this series of organic compounds might be, particularly if the life were carbon-based. Ethanol has the advantage of being a fairly close analogue of water, and is perhaps the least toxic of the alcohols for this reason, though even it "at the last biteth like a serpent" (*Proverbs*, xxiii, 32); ethanol conceivably could actually 'take over' from water below 0°C as the medium for a type of life based on them both. At any rate a living system based on ethanol might not be very different from the aqueous one; some salts incorporate alcohol in the same way as water of crystallisation, so that it should form a common molecular fabric with solutes. Its disadvantages of inflammability at high temperatures would be of minor importance below 0°C, particularly if the atmosphere were less strongly oxidising. Its volatility also would be irrelevant in that range. Higher pressures would further improve its suitability.

An aqueous–ethanol mixture also is a feasible medium, and would of course be the only way in which one could take over from the other; its temperature range is almost as good as that of pure ethanol and could equal the complete range of both if the solute system were capable of

existence in either pure solvent. The mixture would be superior to pure ethanol because of the greater chemical and physiochemical activity of water, while at the same time it suppresses the inflammability of ethanol below proof strength. Reciprocally ethanol speeds many activities in our aqueous biological systems.

> . . . the best wine . . . goeth down sweetly, causing the lips of those that are asleep to speak . . . (*Song of Solomon*, vii, 9)

The polyhydric alcohols are other suitable solvents, and are less volatile than ethanol. Ethylene glycol freezes at −11·5°C but optimal mixtures with water freeze at much lower temperatures, as recent winters have shown! Moreover glycol boils only at +197°C so that it could be the basis for life at much higher temperatures than ours. Glycerol could take the temperature up to 290°C and benzyl benzoate to 324°C, near the limit for a carbon-based bioplasm. Pure glycerol melts at +17°C but its aqueous solution is fluid well below 0°C and it is in fact already a functional anti-freeze agent, in the diapausing pupa of Cecropia and other insects. Like the monohydric alcohols the polyhydrics ionise slightly to give the same ions as water. Ethanol is weakly alkaline but glycerol weakly acid, so that their hydroxyl groups ionise on opposite sides of the oxygen atom. Glycerol is a good solvent for copper (Miner and Dalton, 1953), and other metals, and its more lipophil nature than water has other potential advantages (and disadvantages).

Quinoline −19·5° to +238°C, aniline −6·2° to +184°C, pyridine −42° to +115°C, and acetamide +82° to +222°C, are other possible solvents. They are more strongly polar and therefore more active than the alcohols, but have a higher temperature range, and so for the reason given are not such likely candidates for actual bioplasmic media as the latter. There are few suitable organic liquids freezing at a temperature lower than −112°C, but temperatures on planets are often lower than this: the cloud-cover on Jupiter, for instance, stands at −133°C. Isoprene (p. 107) with a freezing point of −146°C is a possible medium and perhaps has the additional virtue that it is formed abundantly in terrestrial biosynthesis. It is non-ionising over the temperature range of its liquid state, however, and would promote redox rather than ionic activities. With the double bond of its molecule it should be chemically active and activating.

Sisler (1961) has given a stimulating discussion of the properties of a number of non-aqueous solvents, all with very useful properties for the chemist, though perhaps not so promising for the cosmobiologist. The most important, in addition to liquid NH_3, are H_2SO_4, N_2O_4, SO_2, hydrazine ($NH_2{=}NH_2$), formamide ($H{\cdot}C(=O){\cdot}NH_2$) and acetic acid. For work at high temperatures urea (p. 284) also might be included

(m.p. 132°C); it has more biologically significant properties than most possible solvents.

These are by no means the only possible media but they are among the most outstanding within the temperature range where a carbon-based solute system seems possible. Above 300°C most carbon compounds char and the macromolecules would probably decompose at much lower temperatures than this, in any solvent. In the downward direction the liquid state of the other common gases and of the lighter elements would be the only feasible media, liquid CH_4, HF, F_2, Cl_2, O_2, N_2 and H_2 in particular. The inert gases would probably be too inert and HF, F_2 and Cl_2 too reactive. The spores of bacteria and the eggs of some higher organisms will survive the temperature of liquid air (−190°C) for some time, so that some form of carbon fabric might be permanently active at those temperatures. At temperatures above 300°C there is of course the possibility of a bioplasm based on silicon, boron, or other elements, with a liquid metal, non-metal or salt as medium, and with properties other than those which make silicon and boron useless as deputies for carbon in our own bioplasm. Their compounds must be capable of existing as discrete, self-reproducing bodies in the medium, and not merely of dissolving in it. The inclusion-complexes of urea (p. 288) may offer an analogy.

It is calculated (Shapley, 1957) that there may be as many as 10^{11} stars with planets in the universe, so that it is very possible that there are some with types of life rather different, if not very different, from ours, in addition to the very large number, perhaps 10^8, with conditions and living organisms not very different from this. Most are likely to be within the first range of analogues, no doubt, i.e. with a medium composed of compounds of H, O, N and C, and a solute system similarly based. There is some reason to think that carbon compounds are being built up and broken down continuously on the colder planets of our solar system, Jupiter, Neptune, Uranus, and Saturn, where a simple aqueous medium cannot exist (Sagan, 1961). A cycle of change, with the day and the season, could of course continue indefinitely in the absence of any fluid medium, and there is the question whether such systems could ever come to qualify for the status of living, i.e. dynamically self-perpetuating: possibly they could. In any case, however, there are considerable possibilities of suitable fluid media on some of the many planets of the universe, and of living organisms which are discrete and mobile without water.

With regard to this first range of analogues of our own life, it is important to remember (Oró, 1961) that C, H, O, and N are the most abundant elements in the universe, apart from the inert gases; indeed the Earth has the misfortune of being unusually poor in C, H, and N. There is a strong probability, therefore, that these four elements are the basis of

life wherever it occurs, and other types are probably not worth further speculation at present. All four elements are abundant in comets and in the carbonaceous meteorites (Bernal, 1961); as much as 8 per cent of the latter is made up of carbon compounds, aliphatic chain molecules of C_{15} to C_{24} length, aromatic molecules, pyrimidines (Calvin, 1961b) and purines (Briggs, 1961), together with a large number of unidentified compounds and bituminous material. These compounds are so like those of terrestrial living organisms that they may have originated from the earth, but otherwise they strongly support the idea of rather similar but independent types of life within the solar system. There is even some evidence of discrete organised bodies, like protistan cells, in some of the carbonaceous meteorites (Claus and Nagy, 1961; Nagy *et al.*, 1963). Supposing that these are truly indigenous and are not subsequent fungal intruders, or concretivorous bacteria (Postgate, 1954), they imply a type of life not very different from ours, and there is so far little evidence for the existence of any type which is very different. Mars almost certainly has a little water, and a form of life effectively the same as ours, and Venus may have a carbon–aqueous system also.

The adaptability of terrestrial organisms indicates that even this one type of life is possible over a wider range of conditions than we normally experience. A few bacteria and Algae live in hot springs at temperatures up to 79°C, where the proteins of most organisms are completely denatured (Fox, 1960). Others live in petroleum wells, in the abyss, in deserts, on mountain peaks, etc. It is likely that longer and shorter days and seasons could be tolerated, and so on. When there was less ozone in the air, and if there were less again, surviving organisms would no doubt tolerate more ultraviolet radiation than they do now. The number of worlds with conditions not completely unsuitable for our type of life may be very great.

This is not to minimise the restrictions in tolerable conditions for our type, the need for mobile H_2O, CO_2, O_2, and N_2 and for solar energy in reasonable supply. It is calculated that if the diameter of the earth were only 20 ft greater gravity would be sufficient to hold all the oxygen in the form of solid oxides, unavailable for respiration; it would then be necessary to use water as an oxidant—or soluble oxides, nitrates, carbonates, etc. If the earth's axis were not tilted relative to its orbit all water would tend to condense as snow at the poles and to become available once more only if the accumulation caused geostatic movements; these probably would be too slow to sustain any conceivable type of water-based life. Life is precarious enough on earth, therefore, but considering the number of planets which exist there may be many biological systems similar to it in the universe.

There even seems a reasonable possibility of types of life in the first range of analogues, using the same very versatile and very abundant major elements, and even in our own solar system there is some slight evidence for this type. For these much of what has been discussed in this book would be relevant. For more distant analogues there is virtually no evidence at present and it is difficult to visualise active, viable systems based on a different set of major elements from those of terrestrial life.

CHAPTER 32

THE ORIGIN AND EVOLUTION OF BIOLOGICAL UNIQUENESS

Soul of the World, inspired by thee
The jarring seeds of matter did agree.
Thou didst the scattered atoms bind,
Which, by thy laws of true proportion joined,
Made up of various parts one perfect harmony.
(PURCELL)

Had I been present at the Creation, I would have given some useful hints for the better ordering of the Universe.
(ALPHONSO THE WISE, 1221–1284. Referring to Ptolemy's astronomy!)

THROUGHOUT this book it has been implicitly assumed that the biological elements and their compounds, apart perhaps from some of the proteins and other complex macromolecules, were available on earth from the earliest origination of life, and that subsequent evolution was mainly a matter of exploiting those which were uniquely ideal for collectively producing a self-perpetuating system of the kind now known as Life. This would probably be generally conceded for the simple inorganic elements and compounds but some biologists prefer to think that there were at first no organic compounds and that their appearance was the first step in the evolution of life itself. They could then be regarded as the fruits of life rather than its source, and it might be argued that life gave them any uniqueness they appear to have, and not that their innate uniqueness made life possible. The antithesis is one of significance since it has already been admitted that the uniqueness even of inorganic elements has been enhanced in and by their biological environment. This is one of the problems which make it imperative to conclude by looking at the uniqueness of biological materials in the light of evolution.

An even more basic question to ask of evolution is whether it has been a completely spontaneous, 'most-probable' process (Needham, 1959), the outcome of the innate properties of the materials and the associations they made. If the answer is in the affirmative then the first question loses

much of its force. Biological uniqueness was latent in the materials from the outset, whenever it actually became manifest.

> Some say that in the origin of things
> When all creation started into birth
> The infant elements received a law
> From which they swerve not since
> That under force of that controlling ordinance they move
> And need not his immediate hand, who first
> Prescribed their course, to regulate it now.
>
> (WILLIAM COWPER, *The Winter Walk at Noon*)

It is an important corollary to the theme of uniqueness that this *is* considered an adequate 'cause' of evolution, and that evolution could only be spontaneous if ideal materials *were* used for each purpose, whereas the alternative possible cause, a fabricator or evolutionary force outside the system itself, might make shift with less exacting requirements in starting materials and in the avoidance of inferior analogues. Another reasonable corollary, which at first glance might seem to be a restatement of the first, is that spontaneous evolution is itself one aspect of the uniqueness of life. In any event the theme of biological uniqueness has much to do with evolution and in particular with the ultra-Darwinian view that evolution was from the outset the survival of the most viable, the most 'probable', of those systems which spontaneously appeared (Needham, 1959).

The outset or 'origin' of life is probably no more than a convenient fiction: as Darwin wrote in a letter to Hooker (Bernal, 1951, p. 17) "... one might as well talk of the origin of matter", and there was never a time when the beginnings of life were not present. Life evolved infinitely slowly and this is why the word 'origination' has been preferred (Needham, 1959), an indefinitely continuous process and not the sudden unique act of a Creator. Such a slow process can only be powered by spontaneity, the spontaneous activities of materials which are synthesised spontaneously.

> Who *is* the Potter, pray, and who the Pot?
>
> (OMAR KHAYYÁM, *Rubáiyát*)

Some of these activities lead sooner or later to 'dead ends', while others continue to operate a flow of energy indefinitely, and are the 'living' systems of the time. They 'survive' until starved of energy by a new and more 'efficient' system through which the spontaneous flow is more rapid. Spontaneity is the driving force throughout, and therefore it is particularly important to enquire how spontaneous is the synthesis of biological materials, and likewise their metabolic reactions.

From a large amount of work in many laboratories (Table 22) it now

seems certain that many of the key metabolites are synthesised fairly easily by methods which could be biologically relevant. In fact rather powerful and artificial sources of energy have been used in many cases, but one, ultraviolet radiation, is natural and was once very abundant in sunlight reaching the earth's surface. Others may be occasional as natural forces but this cannot be considered biologically significant. More important than the sources of energy is the fact that all tend to produce the same products from simple compounds of the major elements, C, H, O, and N, that most of the products are biological compounds and that few are not. The implication is that the production of biological materials is the only thing which could happen to these simple molecules under terrestrial conditions. The variety of compounds strongly resembles that detected in living plants within a few seconds of supplying labelled carbon dioxide: here enzymes now do what ultraviolet may have catalysed in earlier ages. It is worth considering that because of this facile synthesis of so many metabolites the precise significance of the material of the carbonaceous meteorites is the more uncertain, and the nature of the 'organised bodies' is correspondingly more important. The included material would of course be very attractive to terrestrial micro-organisms.

The next main stage in biosynthesis, the polymerisation of amino acids to peptides, also is spontaneous under fairly natural conditions (Fox, 1960), even without special catalysts. Phosphates, and particularly polyphosphates (p. 407), prove effective catalysts, bringing the temperature required down towards the normal biological range; it is still around 70° to 100°C, in fact (Fox, personal note), but again the important fact is that if a mixture of amino acids is made to do anything at all this is liable to be polymerisation. Phosphates, and particularly those with ~bonds, are the key catalysts in living organisms and it is credible that they ensured spontaneous peptide-formation in appropriate environments below 100°C from an early stage in the origination. It is also significant that heteropolymers form more readily than homopeptides of a single amino acid: in fact the latter usually chars on heating whereas a good mixture polymerises and is protected from charring. All natural proteins are heteropolymers. Further, glutamic and aspartic acids catalyse the polymerisation of themselves and of all the other amino acids. Their outstanding biological position among amino acids (p. 250) therefore seems to be based on spontaneous, innate properties. The molecular weight of these laboratory 'proteinoids' was found to increase with temperature. In general molecular weight tends to be inversely related to temperature (Oparin, 1957a), and at very high temperatures there are only free atoms. The proteinoids therefore show that apparently improbable processes may prove spontaneous, that we need to keep an open mind about what is to be expected

spontaneously in any particular case, and that there may be paradoxes in biological processes.

These proteinoids have all the properties of natural proteins except antigenicity, which depends on a specific amino acid sequence (p. 273). Peptides synthesised in the laboratory *are* antigenic if they have the amino acid sequence of a known natural protein (Axelrod, 1963). Not all natural proteins are antigenic, in fact, but it is quite conceivable that proteinoids would prove so if they could be prepared with a consistent amino acid sequence.

When dissolved in hot saline the proteinoids go on to form coacervate-like bodies spontaneously (Fig. 56); in principle, therefore, it is already possible to synthesise simple 'bioid' individuals from the simplest laboratory materials, with the help of natural catalysts only. The bioids formed in this way are about 5 μ in diameter, and their size and form vary with the concentration of salt, the ratios of the various amino acids, and the nature of other components added, such as nucleic acids and starch. They are stable for several weeks, in the absence of infection, and show osmotic properties, so that they must have a differentiated limiting membrane.

Many biochemical reactions are also the reactions which occur most readily in the laboratory, between the same reagents. This might be true of many others if laboratory methods made more use of likely catalysts rather than the all too convenient universal accelerator, heat. Most organic reactions in fact are very sensitive to catalysts in the laboratory (Gilbert, 1950). Laboratory tests should aim at finding the most spontaneous reaction of each reagent under physiological conditions; more parallels to biochemical pathways might then emerge. No doubt some reactions *in vivo* have come to be driven in improbable directions by coupling with stronger reactions spontaneously operating in that direction (Krebs and Kornberg, 1957), so that it may not be possible to copy every one in the laboratory, but it would be worth testing as many as possible.

Examples of biochemical reactions which are driven 'backward' by a coupled reaction are cited among the electron-transfer sequence (p. 133). A simple model (Fig. 57) helps to understand the mechanism (Krebs and Kornberg, 1957). Two electrolytic cells are connected in series, but back to back. The Ce/Fe cell works in the most probable direction, and the ceric ion oxidises ferrous to ferric, but the potential so set up is large enough to drive the Cu/Ti cell in the improbable direction, namely of cupric oxidising titanous to titanic, although Ti^{+++}/Ti^{++} has a higher redox potential than Cu^{++}/Cu^{+}. The parallel *in vivo* is that atmospheric oxygen promotes the *reduction* of NAD (p. 334) which has a redox potential at the lower end of the physiological range.

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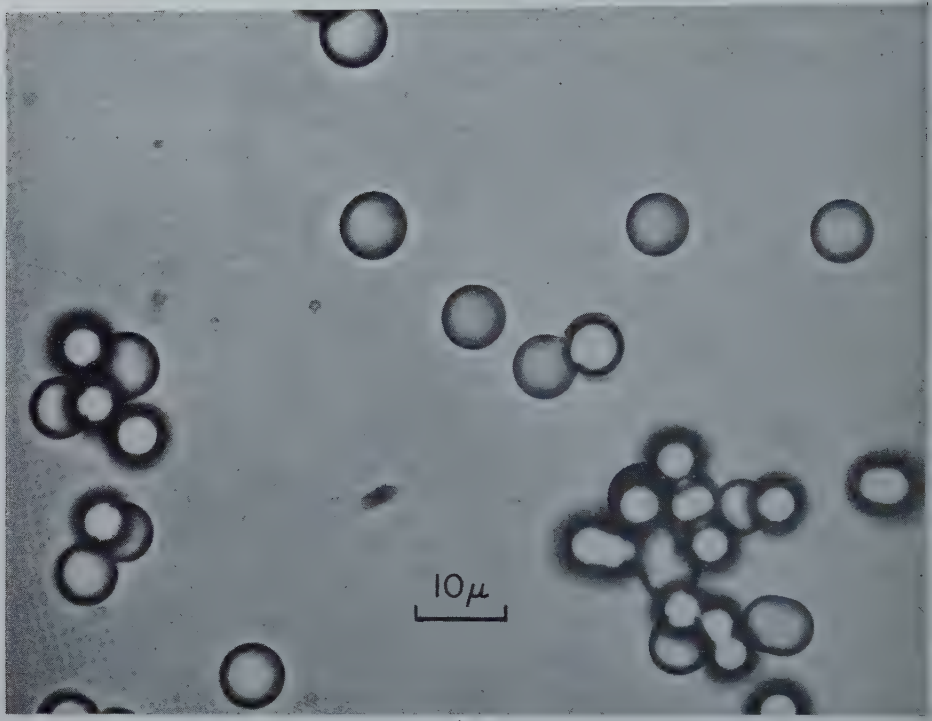


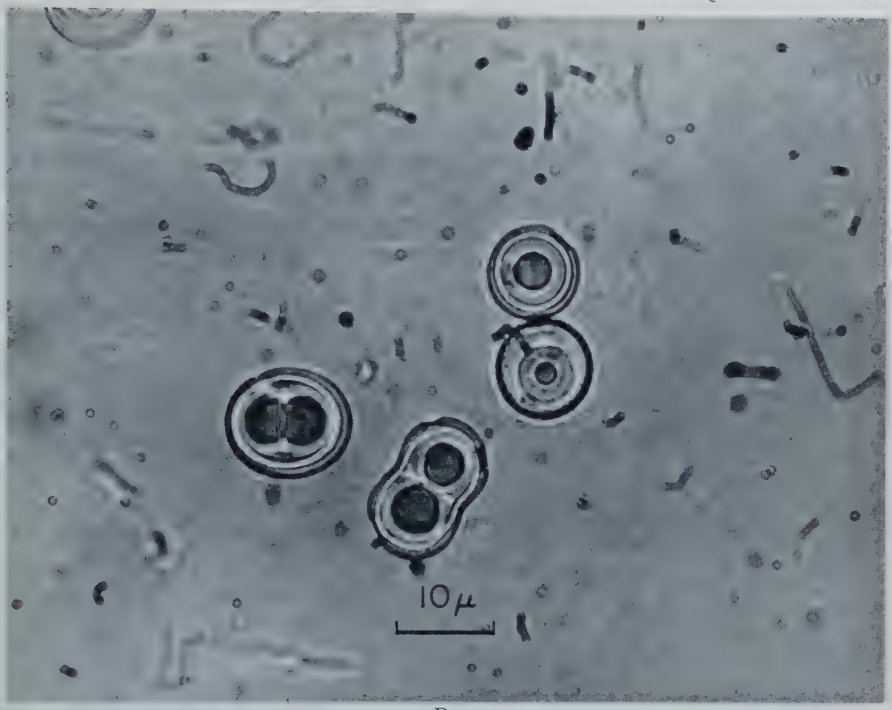
A

FIG. 56 Microscopic spherules, having the dimensions of living cells, formed by heating laboratory-synthesised *proteinoids* in acidified water or saline, A and B as seen under high magnification by the light microscope and C as shown by electron-micrography of a section of spherules embedded in methacrylate and impregnated with OsO_4.

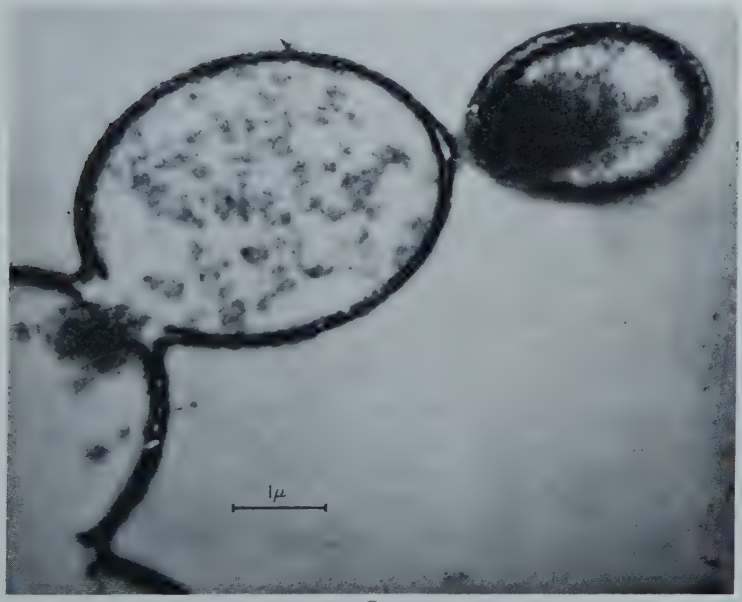
The spherules in B, formed in less acid medium, have double limiting membranes and a denser centre, as in some bacteria, and are dividing in a way resembling septate division in cocci (confirmed by time-lapse photography). The fine filaments are linear aggregates, again with parallels among bacteria, of the population of smaller, 1μ, spherules seen in the picture.

In C the spherules are seen to have a double membrane rather similar to that around the nucleus of living cells and that of other cytological structures, though thicker than most biological double membranes.

Photos kindly supplied by Professor Sidney W. Fox: A and B prepared by Takeshi Fukushima and C by Fukushima and Charlotte Dockery.



B



C

FIG. 56 (*continued*).

The best-known, most simple and fundamental of biological reactions copied by simple methods in the laboratory is the fixation of carbon dioxide under the influence of light. Metal oxides of many kinds have proved effective catalysts and formaldehyde is the usual product. Modern autotrophes also fix carbon dioxide as aldehyde, but by conjugation with existing monose (Fig. 3), and it would be interesting to redesign the laboratory models in this light. After sunny weather there may be considerable amounts of HCHO in rain water (Ellis and Wells, 1941) so that no doubt carbon dioxide fixation is a common spontaneous process.

Formaldehyde further polymerises spontaneously, to paraformaldehyde, which is unstable, but in the presence of suitable catalysts, again metal oxides, it polymerises in quite a different way to the more stable

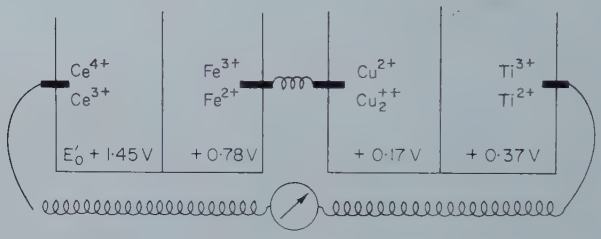


Fig. 57 Arrangement of Ce/Fe electrolytic cell so as to drive a Ti/Cu cell 'backwards'. Based on Krebs and Kornberg, 1957.

monose sugars (Oparin, 1957a, p. 198). The synthesis goes via glyceraldehyde and dihydroxyacetone to fructose, much as in the natural glycolytic sequence (p. 44). It differs from this sequence, when operating in the synthesis direction, in that the fructose is racemic (DL), and that sorbose also is formed. Other products can be obtained by changing the catalyst from $Ca(OH)_2$ to $CaCO_3$; the simplest of the monoses, glycollic aldehyde, $CH_2OH{\cdot}CHO$, is then formed, and it polymerises to tetrose and hexose. It will also unite with glyceraldehyde to form pentose. $Ca(OH)_2$ and $Ba(OH)_2$ catalyse the condensation of two triose molecules to form hexose, a method actually used commercially.

Methane and carbon monoxide, not so common as free reagents, will condense rather readily to form acetaldehyde, the next higher homologue of formaldehyde. This spontaneously polymerises to paraldehyde, an unstable oxy-carbon ring compound similar to paraformaldehyde, in the presence of concentrated sulphuric acid, and under the action of the much milder catalyst, potassium carbonate solution, it polymerises to aldol, $CH_3{\cdot}HCOH{\cdot}CH_2{\cdot}CHO$. Even more significantly, it will form polyenes by crotonic condensation:

$$2\ CH_3CHO \rightarrow CH_3{\cdot}CH_3 = CH{\cdot}CHO + H_2O$$

A number of other biochemical reactions of carbohydrates and related substances occur under simple laboratory conditions. Mild oxidation of glycerol with dilute nitric acid once more yields the two biological trioses, glyceraldehyde and dihydroxyacetone, and if these are warmed they undergo aldol condensation to form DL–fructose (Read, 1935, p. 376). Aldehydes and ketones are readily interconvertible, as they are *in vivo*:

FIG. 58 Household fluff, under low magnification: an example of a fabric which forms spontaneously, in the sense that virtually all the forces to which it is normally subjected promote its fabrication and very few its dispersion. Virtually all ambient fibres are incorporated, though wool, hair and feathers predominate, so that their barbed surfaces, due to microscopical scales, all facing obliquely in the same direction, may play a major part in the fabrication. It would be interesting to compare the rate of fluff formation by smooth-surfaced synthetic fibres, alone. Electrostatic charges are likely to be of the same sign on all fibres of one type, and therefore to oppose fabrication. Simple tangling no doubt plays its part, as in wire-turnings, etc. Photo by J. S. Haywood.

this is particularly true of the two trioses, while glucose and fructose are interconvertible by the osazone method. The latter are also interconvertible via a sugar alcohol, and this is a usual laboratory method. Further, Ruff's laboratory method for converting glucose to pentose goes via gluconic acid, as in the biological 'shunt' pathway (Fig. 5). The laboratory pathway ends in arabinose, not ribose. Mannose is similarly oxidised via mannonic acid to arabinose: as already seen (p. 355) ribose is an unstable pentose.

If sugars are aerated in alkaline solution lactic, pyruvic and acetic acids are formed, as in the glycolytic and peripheral pathways. Once more other products appear, such as glyoxylic and propionic acids (Fearon, 1949, p. 115). *In vivo* there has been a sharpening of specificity, but in fact there may be considerable amounts of by-products in some biochemical reactions, judging by results in isolated preparations. In this case glyoxylic and propionic acids are actually produced by some organisms; rumen micro-organisms produce acetic, propionic and butyric acids from cellulose, while a glyoxylic pathway is operated as a normal one in some micro-organisms (Kornberg, 1961). In stronger alkali, and in the absence of oxygen, the two trioses are the main products. In the laboratory, lactic acid is easily oxidised to pyruvic but it is difficult to prevent the reaction going further. No doubt this is prevented *in vivo* by the small redox-potential change permitted by each enzyme (p. 324). Biological reactions proceed by changes of 3–16 kcal/mole compared with the 300 kcal of an average laboratory reaction (Bernal, 1951); they therefore have less momentum and are more easily controlled, reversed, etc.

A mixture of ascorbic acid (p. 50) and Fe^{+++} ions will aerobically oxidise a number of aromatic drugs (Nimmo-Smith, 1958), and the products are the same as in normal detoxication reactions *in vivo*. Detoxication perhaps should not be taken as a typical biochemical process (p. 500), but in any case the model illustrates the spontaneity and versatility of ascorbic reactions. Ascorbic oxidase is a copper-protein (p. 441) so that in the model iron may be replacing copper functionally. Laboratory models of this kind, involving the catalytic metals, are particularly easily prepared and effective, because of the loose association of the metal and the relatively low specificity of its action. Copper-proteins are a good example (Needham, 1960*b*). The laboratory iodination of proteins (p. 460) may be taken as a parallel case for fairly loosely conjugated non-metals. Much more of metabolism can be duplicated in the laboratory but these examples will perhaps suffice to show that the whole of metabolism is probably spontaneous in its context.

The high degree of spontaneity in so many of these laboratory copies makes it difficult to understand the surprise and incredulity which accompanied the first organic syntheses in the laboratory, although these were indeed by laborious methods, far from spontaneous. At the same time this historical background does help to understand a view still current (p. 523) that the uniqueness of life is precisely that it is not spontaneous but on the contrary most 'improbable'. The first total syntheses in the laboratory were not very impressive by present standards, ethanol by Hennell and urea by Wohler, both in 1828. Only a year before this Berzelius, a contemporary giant of chemistry, had declared that a

special life force was necessary for organic synthesis, which could only occur *in vivo*, therefore; organic compounds originated from life and not *vice versa* as we now believe. Since Hennell and Wohler, however, organic chemistry has been a science of synthesis rather than of analysis and from it biology has received a vast amount of information and suggestions for subsequent application to living organisms. The dyestuffs industry revealed the significance of biological pigments and more recently the plastics industry has taught a great deal about polymers. Initiators are usually necessary to start polymerisation, for instance heat or ultraviolet radiation (Yarsley and Couzens, 1942), but once started the process is often exothermic and therefore spontaneous, like the higher grades of polymerisation in biological materials. The use of starters recalls the *in vitro*, and probably the *in vivo*, synthesis of DNA (p. 368), and the type of agent used is reminiscent of the laboratory synthesis of amino acids, etc., from their simplest precursors. The initiation of industrial polymerisations often depends on free radicals which tend to multiply spontaneously by so-called 'degenerate' chain reactions. This also may have its counterpart in biological systems (Pringle, 1954). The versatility of industrial plastics, collectively, also is support for the view that the great variety of proteins and some other biological polymers may be produced spontaneously.

Laboratory work therefore has in general strengthened the view that biological reactions are the innate, spontaneous properties of materials which are synthesised spontaneously under natural conditions, and that life originated and evolved for this reason. The details of the origination and evolution are not of major importance in the present context, though obviously it would be useful to know how much of the biological material known today (and how much in addition) existed from the earliest days of the cool earth. Molecular size and variety in general may have increased as the temperature fell (even the theory of the 'cold' origin of the earth involves a subsequent hot period before life began (Levin, 1957)) but it has been seen (p. 512) that even this trend is not necessarily universal. In any case the temperature has been suitable for the spontaneous appearance of most of our biological materials for more than a milliard years, and if in fact they evolved only gradually then other reasons for this must be sought.

It is difficult to visualise what could prevent the spontaneous appearance of a vast number of organic compounds, under the influence of solar energy, from the beginning of this long period; it is equally certain that many of them tended to decompose rapidly under the changed conditions of night, winter, etc. Some, however, may have been quite stable and these are of no further consequence for the very dynamic

systems which we must believe life always involved. For these the relevant materials are those which change, but only those which do so as part of a cyclic system; this is the only kind of system which can have the paradoxical essentials of dynamic self-perpetuation. The system is permanent, its personnel come and go: as Shelley wrote in *Adonais*:

> The one remains, the many change and pass.
> Heaven's light forever shines, earth's shadows fly.
> Life, like a dome of many coloured glass,
> Stains the white radiance of Eternity.

Individuals live at the most a few hundred years, and all the oxygen, nitrogen, carbon, etc., in the *biomass*, the total mass of living organisms, circulates through it every few thousand years at most (with the occasional exception of coal, etc.), yet life as a phenomenon, and the circulation of its material, goes on indefinitely. This is the nature of life.

> There is no room for Death,
> Nor atom that his might could render void:
> Thou—Thou art Being and Breath,
> And what Thou art may never be destroyed.
> (EMILY BRONTË, *Last Lines*)

On a smaller scale, the individual may live for 100 years, but during that time his materials have been completely changed many times. Systems survive but the metabolic changes which make this possible may be in very rapid flux. With a chemical flux of this order of speed there seems no reason why all reactions which are spontaneously feasible should not occur very frequently, both now and in the past. This is the material presented to natural selection and with this speed of presentation it is to be expected that a uniquely ideal living system would result.

We are rather in the dark concerning actual metabolic rates at the dawn of life, when less efficient catalysts were available, but it was probably nearer modern metabolic rates than to modern rates of turnover of individuals, and it is important to keep time-scales in mind. In theories on the origin of life it is often suggested that a geological age may have separated the appearance of one biological compound on earth from that of another, but to the chemist it must seem incredible that any geological age could pass without the whole contents of Beilstein appearing in quantity many times. The one possible exception is specific proteins; mammalian proteins differ from those of reptiles and may have been synthesised for the first time in the history of the earth by the mammalian line.

The postulated speed of metabolism is conditional on a continuous supply of solar energy, not incomparably less than now; in fact it is

likely to have been equally great during the last 10^9 years, and at first richer in ultraviolet, so that the variety of reaction may once have been greater, and their speeds always much the same. To handle this energy for biological purposes there must always have been autotrophic systems, continuously fixing it into chemical forms, however soon other systems may have become heterotrophic or parasitic upon them. It would scarcely be necessary to stress this if it were not so generally contended (Oparin, 1957a) that heterotrophes preceded autotrophes. The reason for this view is mainly that there was no free oxygen in the atmosphere until 8×10^8 years ago, and therefore no autotrophes of quite the same type as modern green plants. However, there are many other oxidation–reduction reactions which would permit the conversion of solar into chemical energy. The theory that heterotrophes originated first leads to several absurdities; in the first place it must be assumed that they suddenly appeared fully fledged. They at first fed on organic compounds spontaneously formed by the Sun (the first simple autotrophic systems indeed) until these were exhausted—a matter of a few thousand years at most, without replenishment, even at the slowest imaginable rate of reproduction of the heterotrophes. Finally when the organic compounds were all exhausted some of the heterotrophes quickly metamorphosed into autotrophes, with all the biosynthetic apparatus which heterotrophes lack.

The question of speeds is so important for the evolution of a unique, ideal living system that it may usefully be taken a little further. There is no doubt, of course, that some chemical reactions have an infinitesimally slow rate and possibly no catalyst which is likely to be available naturally would make much difference to this rate. This is the improbable type of reaction, however, which is not considered to be relevant to life; spontaneous reactions are relatively fast reactions, and our interest should be in maximal rather than in minimal recorded speeds. The molecular vibration of water, which is of course relevant to its chemical activity, has a period as short as 10^{-13} seconds (Klotz, 1962), and the molecule of NH_3 reverses the asymmetry of its three hydrogen groups relative to the nitrogen atom 2×10^{10} times per second (Johnson *et al.*, 1954). Water molecules exchange their ions at the rate of 10^{12} per second, and so on. The activated complex which is formed as the first step in a chemical reaction may exist for only 10^{-13} seconds so that energy must change hands twice in that time. It would be reasonable to consider 10^{-8} to 10^{-4} seconds as the maximal life of such complexes (Glass, 1961). Carbon compounds react with speeds which range up to these values; for instance some ethylene derivatives polymerise to chains thousands of carbon atoms long in a fraction of a second, even at −80°C (Robertson, 1947). The maximal rates *in vivo* are almost as impressive: labelled carbon dioxide and other raw

materials are incorporated into the fabric of the organism within a few seconds, and heterotrophes are about as rapid in the syntheses which they do perform. Molecules of the protein *ferritin* (p. 438), of molecular weight 400,000, are synthesised in a few minutes (Loftfield, 1957). Bacteriophages increase their mass a hundredfold in a few minutes. A catalase molecule acts on 4×10^4 molecules of substrate per second, even at 0°C.

Certainly these are processes in modern organisms, the end-products of a long process of selection for efficiency, which almost certainly includes speed. Equally certainly carbon is outstanding for the moderation of its activity, but this is referrable to both ends of the speed range: many of its reactions are moderately fast. The earliest living systems were simpler than those of today and less subject to the braking effect of complexity; indeed, there is plenty of evidence of restraint in the metabolism of living organisms today: potential maximal speeds are much greater. The speed of an army no doubt is the speed of its slowest members—those marching on their stomachs, perhaps—but the speed of an electrical discharge is determined by the best conductor-pathway, and this is true of chemical activity in a reaction-network of an open system (Passynski, 1957), in accordance with what has been called Hinshelwood's Principle (Dean and Hinshelwood, 1955). The most rapid pathway is the most probable. Applied to the eobiological systems the contention is that life has always been precisely this most probable, opportunist exploitation of the most spontaneous pathways. Life is the record of this maximal spontaneity.

In this reaction-network all feasible chemical experiments are likely to have been tried many times, and any collateral pathways which contributed to the cyclic perpetuation of the system to have been permanently incorporated, all within the very early ages of the cool Earth. As Lucretius said (Henderson, 1917):

> The first beginnings of things is to unite in all manner of ways and thoroughly to test every kind of product possible by their mutual combinations.

The more crucial question is whether the *spatial distribution* of materials was always adequate; chemicals cannot react if they do not approach to within molecular dimensions, or at best (Rothen, 1956) to distances infinitesimally small by geographical standards. Solar energy is distributed to the whole surface, with unfailing regularity, while O_2, CO_2, N_2 and H_2O are virtually worldwide in distribution, and very mobile. All other elements of significance are present in mobile form in sea water, and many in fresh water, so that it seems that spatial problems need never have been serious. Life is generally thought to have evolved in the

sea, because of the composition of animal body fluids (p. 418), as well as for other reasons, and it is also generally agreed, with Haldane, that the early seas may have been a veritable soup of organic materials. The average speed of small molecules is 1000 metres a second, and although their free path is very limited this speed ensures chemical reaction between competent pairs once they are near-neighbours. Diffusion, the resultant of molecular jostlings, is rather slow but very rapid by geological standards. In any case, however, grosser forces such as wind, convection, etc., have always been available for distributing and aggregating materials (Fig. 58). They have settled and fossilised in large masses from time to time, and in general spatial problems are not serious.

They have been further eased by spontaneous aggregative tendencies, well shown in coacervate bodies (p. 233). There is a possibility that these tendencies are particularly well developed in biological systems; Fearon (1949, p. 45), for instance, wrote that in regions devoid of life materials in solution disperse but in the presence of life there is the opposite tendency (Fig. 58). However, specific aggregations, usually of pure substances, are well known also to geologists, and are illustrated by the secondary enrichments mentioned frequently in this book. Crystallisation in general, in fact, is this same kind of aggregation with purification. It might be argued that the process is anti-entropic and improbable, but it must be remembered that the earth is an open system and the flow-through of solar energy, which makes life possible, also drives these aggregative processes including those of fossil materials (above). One of the most striking of natural examples is the formation of manganese dioxide nodules free on the ocean floor (Mero, 1960). These initially form, as crystals commonly do, on the nucleus of some foreign body, in this case fragments of pumice, sharks' teeth, etc., and they may grow to a weight of nearly a ton. Their rate of growth is actually very slow, perhaps 1 mm in 1000 years, but manganese is a rare element and its salts not highly soluble.

For many years the apparently anti-entropic, improbable character of aggregation, segregation, reduction-reactions and other aspects of vital activities were taken as evidence that life was not a most probable, spontaneous phenomenon but in fact quite the opposite. Since the work of Onsager, Prigogine and others on open systems, however, it is realised (von Bertalanffy, 1949; Oparin, 1957a) that living organisms may represent the natural steady-state of carbon compounds in such a system—which does exist on earth. Whereas closed systems tend to a state of *maximal* entropy open systems tend to a steady-state of *minimal change* of entropy at an absolute level which may be one of high organisation, i.e. of low entropy. The high organisation is expressed and used in a number of ways, for instance in the storage of compounds which can be

re-oxidised when required, to provide energy for particular reactions which otherwise would indeed be improbable. The important role of ATP (p. 404), as already indicated, is to handle this energy for almost all reactions and processes of the kind. As a further instance of its mode of action the reduction of carboxylic acid to aldehyde may be cited (Bayliss, 1959, p. 398). If an aldehyde is phosphorylated it yields 10 kcal/mole less energy on oxidation to carboxylic acid than does the free aldehyde; in consequence the reverse, improbable direction of reaction is favoured, with the help of NAD coenzymes having a redox potential no lower than −0·3 V.

With 20 per cent of free oxygen in the atmosphere we tend to think that oxidation is the only spontaneous process on Earth, but it must be appreciated that this oxygen is the complement of the various reduced compounds built up by living organisms as part of their steady-state organisation. Every reaction is both an oxidation and a reduction (Spiegelman and Sussman, 1952) and the atmosphere is an integral part of the steady-state system. At the level of maximal entropy from which it may be imagined to have started reductions would be as spontaneous as oxidations; indeed even in the biological steady-state they still are: it is merely that the oxidants and reductants are different. In the existing steady-state ultraviolet energy promotes both oxidations and reductions (Ellis and Wells, 1941); examples of oxidations are those of water, ammonia, —SH, —S—, $—SO_3$ and $—NO_2$, and of reductions those of AgO, HgI_2, $HgSO_4$, CCl_4, $TiCl_4$, $SnCl_4$, and persulphates. In any case, however, all of these reactions are both oxidations and reductions and it is a matter of taste which reagent interests us most. In photosynthesis water is both oxidised to oxygen and reduced to hydrogen.

It is reasonable to single out photoactivation in this way since ultraviolet once reached the Earth's surface in quantity, and is still considerable, while energy in the visible range remains of paramount biological importance. Ultraviolet of 2000 Å wavelength has a quantum value high enough to break most known chemical bonds (Reid, 1957) and to start any reaction which is otherwise feasible; even reactions thermodynamically probable often need considerable energy in a single packet for activation and this is the unique virtue of light-energy. The lower quantum-energy of visible rays is effective for some reactions, and in the presence of photodynamic chemicals, mainly coloured, resonant molecules (p. 114), it will activate many more, so that modern organisms are almost as well provided as eobiological systems were before increasing ozone in the atmosphere filtered out so much of the ultraviolet.

In today's steady-state biosphere most carbon compounds tend to oxidse for the reason given, but it is worth noting that the reduced forms,

and still more the partially reduced forms, are not so highly endothermic (p. 33) or improbable as this tendency might imply, so that those actually synthesised in living organisms may well be the most probable products of solar activation. The most endothermic and least probable form of carbon in fact is not hydrocarbon but the free element, and the change from solid carbon to gaseous methane, CH_4, is exothermic to the extent of 18·6 kcal/mole. If the carbon were in gaseous form indeed the figure would be as high as 362 kcal. The oxides are more exothermic than the hydrocarbons of course, and solid carbon burns in air to carbon dioxide with the release of 94 kcal/mole, but relatively little energy is required at each step of the reduction of carbon dioxide to monose, and further to fat.

It follows from the above considerations that desaturation is endergonic and rehydrogenation exergonic. Acetylene, $CH \equiv CH$, polymerises to benzene with the release of 156 kcal/mole, but in fact this state rather than elemental carbon itself is the most endergonic: acetylene even decomposes to carbon and hydrogen explosively, with the release of 58 kcal/mole.

The combination of methane, carbon monoxide and ammonia to form alanine is exothermic (Pavlovskaya and Passynski, 1957). Starting from CO_2, H_2O and NH_3 no doubt it is endothermic but the example further emphasises the small margin of energy involved, and how easily biological syntheses can become spontaneous, thermodynamically. As already noted (p. 150), nitrogen gas is more stable than nitrogen compounds and the hydrides such as ammonia are more stable, less endothermic in their formation, than the partially oxidised stages:

$$NH_2OH \xrightarrow{-O} NH_3 + 47 \text{ kcal}$$

Nitrogen therefore acts as a useful complement to carbon and a large number of organic nitrogen compounds are virtually isothermic with a mixture of simple carbon and nitrogen precursors. They should form spontaneously on activation under almost any conditions, of temperature, etc.

The relatively low temperature on earth is sometimes emphasised as a factor indicating that life cannot have been spontaneous in origin and continuance. It is argued that but for their specialised enzymes many biological reactions, even in homoiotherms, would be very slow. One reply to this is that life probably originated at higher temperatures and that enzyme-improvement kept pace with the slowing due to a falling temperature. Even if this is not true, however, high temperature is a questionable necessity. Photoactivation is efficient at low temperature and once this energy is trapped organisms can make their own local

'hot-spots' for specific purposes. For some purposes, including structural organisation, heat is a disadvantage and low temperature an actual virtue. Entropy increases with temperature and is of course minimal at 0° Absolute (Mendelsohn, 1949). Large orderly molecules form more easily at low temperature, as in industrial polymerisations (p. 519), though some proteins may be anomalous (p. 202). Increased pressure, for many purposes equivalent to decreased temperature, also facilitates condensation and polymerisation (Bressler *et al.*, 1949), and proteins can be partially renatured in this way. In connection with the apparent anomaly of Fox's proteinoids (p. 512) it must be remembered that these were synthesised dry, and not in aqueous solution. Polymerisation and other reactions (Pringle, 1954) therefore may actually become more rapid as the temperature falls, and some are retarded as the temperature rises (Cullis and Hinshelwood, 1947). Once more it may be emphasised that living systems are likely to have been the result of whatever reactions were most rapidly spontaneous under the conditions prevailing. A number of interesting biochemical complexes form spontaneously on freezing (Szent-Gyorgyi, 1960), and they have a very specific construction and chemical potentialities. Some reactions which are irreversible at high temperatures become reversible at lower temperatures (Freed, 1958); the value of easy reversibility, which is a feature of many biological reactions, needs no emphasis.

Systems do not tend always to the one extreme of their possible range, as the laws of entropy require; many can tend to a relatively stable state at either extreme, because of an intervening 'hill' of activation-energy, or because the intervening stages are the most unstable, as in so many examples cited in earlier chapters:

> Deprivation abates a moderate passion but intensifies a great one—as the wind blows out a candle but fans up a fire.
>
> (LA ROCHEFOUCAULD)

Taking all these facts into consideration there need be no doubt that spontaneous activities *could* account for the origin and evolution of living organisms, and there can be considerable confidence that the origination actually was of this kind. It may be argued that in spite of all this some biological processes seem very improbable, and certainly these call for further attention. When all the circumstances are known they may prove most probable. A good example of a common, indeed, a very common, process which nevertheless seems entirely improbable on almost every individual count is rainfall (Thompson, 1942, p. 59)—an essential process for all but marine life. In the first place raindrops will not form in the absence of the usual nuclei of foreign bodies (Kraus, 1947), and secondly

the molecules of water vapour must deposit on these nuclei more rapidly than they re-evaporate; below a certain size the high surface/volume ratio is strongly in favour of evaporation and there is no stability until the critical size is reached. The droplets then persist, but they still do not form raindrops unless they can coalesce with others. They can only do this if they are of equal size, are falling parallel and near together, at the same speed, and are rotating in opposite directions. In spite of all this, heavy rainfall is one of our most common experiences! There can be very few metabolic reactions which even appear so improbable.

The lesson of this chapter, therefore, is that little, if any, biological chemistry is manifestly improbable in its context, and that most of its reactions in fact are the most spontaneously probable in that context. As Kermack and Eggleton (1938) put it, "Nature is really at heart a confirmed opportunist". She must be so, and has no option. Natural selection is Hobson's choice. Only spontaneous processes can be exploited and these are most probable if they not only proceed with maximal speed but also form part of a system which 'survives', i.e. continues to operate indefinitely, energetically an open system and materially a cyclic system, guaranteed an unfailing replenishment of raw materials. For any situation there can be only one element or one compound which is most probable and this is the origin of the uniqueness of biological materials. Their uniqueness has probably been further enhanced by interaction with each other in the self-perpetuating biological system as a whole. Modern living organisms are wonderfully complex and teleotropic mechanisms, but it seems credible that they could have evolved by natural selection under these conditions, in the time which has been available.

Eobiological systems were cyclically self-perpetuating even before discrete cell-like bodies appeared, and it is still the biomass as a whole, rather than individual species, much less individual organisms, which is perpetual. It is therefore reasonable to consider the uniqueness of the materials somewhat divorced from specific morphological forms. The shortcomings of this book are not due primarily to this divorce but to the incompleteness of our knowledge of the properties of the materials. It is still necessary to agree ruefully with the Abbé Galiani (Thompson, 1942, p. 19):

> La science est . . . plutôt destiné à étudier qu'à connaître—à chercher qu'à trouver la vérité.

On the other hand there is already considerable stimulating knowledge and the outlook is distinctly more hopeful than in the days of Democritus: "But we know nothing really for truth lies deep down." We may reasonably feel content to journey on, hopefully.

God has made everything beautiful in its time:
Also he has set the world in their heart,
Yet so that man cannot find out the work which he has done
From its beginning to its end.

(*Eccles.*, iii, 11)

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